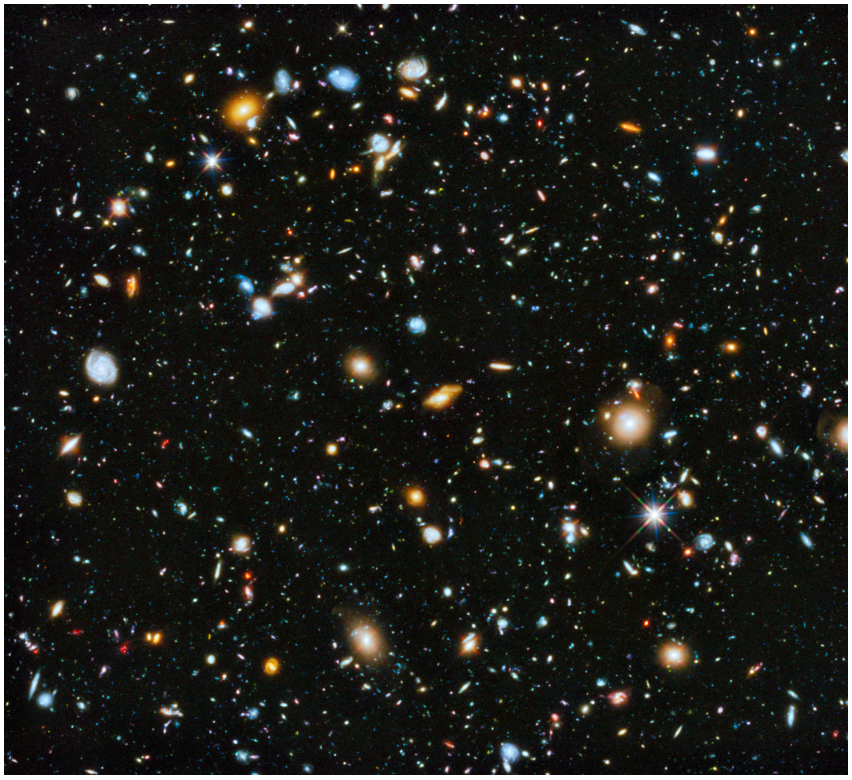
My Mathematical Universe

Haoran Li



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1 Set theory

1.1 Set theory

Definition 1.1 $\{A_i\} \subseteq \mathcal{P}(X)$, $X \xrightarrow{f} Y$ is a map. f **separates** A_i if $\bigcap_i f(A_i) = \varnothing$. f **completely separates** A_i if $f(A_i) = f(a_i)$ for some distinct $a_i \in A_i$. f **perfectly separates** A, B if $A_i = f^{-1}(a_i)$ for some $a_i \in A_i$

Lemma 1.2 (Zorn's lemma) P is a nonempty poset and every chain has an upper bound, then P contains a maximal element

Theorem 1.3 (Schröder–Bernstein theorem) $A \xrightarrow{f} B$ and $B \xrightarrow{g} A$ are injective, then there exists $A \xrightarrow{h} B$ bijective

Theorem 1.4 (Inclusion-exclusion principle) $A_1, \cdots, A_n \subseteq S$ are of finite cardinality, then

$$\left|\bigcup_{i=1}^{n} A_n\right| = \sum_{k=1}^{n} (-1)^k \sum_{i_1 < \cdots < i_k} \left|A_{i_1} \cap \cdots \cap A_{i_k}\right|$$

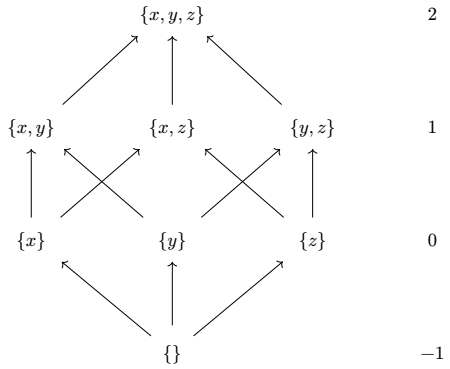
Definition 1.5 A **lattice** is a partially ordered set in which the supremum and infinimum of any two elements exists uniquely

Lemma 1.6 Trees are bipartite

Proof. Take some $v \in T$ as the root, and label the nodes that are even distance away from 2 and odd distance away from 1

□

Definition 1.7 A **Hasse diagram** is a mathematical diagram used to represent a partially ordered set



Definition 1.8 The *Bermoulli numbers* B_n are defined by

$$\frac{t}{e^t - 1} = \sum_{k=0}^{\infty} \frac{B_k t^k}{k!}$$

For instance, $B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_3 = 0$, $B_4 = -\frac{1}{30}$, etc.

2 Category theory

2.1 Category

Definition 2.1 A *semicategory* $\mathcal{C}$ consists of

- A class of *objects* Ob $\mathcal{C}$
- A class of *morphisms* Hom $\mathcal{C}$
- Compositions $\text{Hom}(A, B) \times \text{Hom}(B, C) \to \text{Hom}(A, C)$

Such that

- Compositions are associative: $(hg)f = h(gf)$

Definition 2.2 The *empty category* is the category without any objects nor morphisms

Definition 2.3 A *category* $\mathcal{C}$ consists of

- A class of *objects* Ob $\mathcal{C}$
- A class of *morphisms* Hom $\mathcal{C}$
- Compositions $\text{Hom}(A, B) \times \text{Hom}(B, C) \to \text{Hom}(A, C)$

Such that

- Compositions are associative: $(hg)f = h(gf)$
- $\text{Hom}(A, A)$ contains *identity* 1_A: $1_A f = f$, $g1_A = g$

Definition 2.4 A *subcategory* $\mathcal{S}$ is a category consists of subclasses of objects and morphisms with the same composition map

Definition 2.5 The *dual category* $\mathcal{C}^{op}$ of $\mathcal{C}$ consists of the same objects and morphisms but with morphisms reversed

Definition 2.6 A *functor* $F : \mathcal{C} \to \mathcal{D}$ consists of maps

- Ob $\mathcal{C} \to$ Ob $\mathcal{D}$
- $\text{Hom}_{\mathcal{C}}(A, B) \to \text{Hom}_{\mathcal{D}}(F(A), F(B))$

Such that it

- Preserves identities: $F(1_A) = 1_{F(A)}$
- Preserves compositions: $F(g \circ f) = F(g) \circ F(f)$

Definition 2.7 A *contravariant functor* $F : \mathcal{C} \to \mathcal{D}$ consists of maps

- Ob $\mathcal{C} \to$ Ob $\mathcal{D}$
- $\text{Hom}_{\mathcal{C}}(A, B) \to \text{Hom}_{\mathcal{D}}(F(B), F(A))$

Such that

- $F(1_A) = 1_{F(A)}$
- $F(g \circ f) = F(f) \circ F(g)$

Remark. A contravariant functor $F : \mathcal{C} \to \mathcal{D}$ is a functor $F : \mathcal{C}^{op} \to \mathcal{D}$

Definition 2.8 $A \xrightarrow{f} B$ is a

- *monomorphism* if $fg_1 = fg_2 \Rightarrow g_1 = g_2$
- *epimorphism* if $g_1 f = g_2 f \Rightarrow g_1 = g_2$
- *bimorphism* if it is both a monomorphism and an epimorphism

- *isomorphism* if it is invertible
- *automorphism* if it is an isomorphism and $B = A$
- *constant morphism* if $fg = fh$ for any g, h
- *coconstant morphism* if $gf = hf$ for any g, h
- *zero morphism* if it is both constant and coconstant
- *involution* if $B = A$ and $f^2 = 1_A$

Remark. Isomorphisms are bimorphisms

Example 2.9 A bimorphism is not necessary an isomorphism. In the category of rings, $\mathbb{Z} \hookrightarrow \mathbb{Q}$ is a bimorphism because $\mathbb{Q} = \mathbb{Z}_{(0)}$ is a localization and the universal property of localization

Definition 2.10 A *natural transformation* is a family of morphisms $\eta_A : F(A) \to G(A)$ making the following diagram commute

$$\begin{array}{ccc} & \overset{F(f)}{} & \\ F(A) & \longrightarrow & F(B) \\ {\scriptstyle \eta_A}\downarrow & & \downarrow{\scriptstyle \eta_B} \\ & \underset{G(f)}{} & \\ G(A) & \longrightarrow & G(B) \end{array}$$

For contravariant functors

$$\begin{array}{ccc} & \overset{F(f)}{} & \\ F(B) & \longrightarrow & F(A) \\ {\scriptstyle \eta_B}\downarrow & & \downarrow{\scriptstyle \eta_A} \\ & \underset{G(f)}{} & \\ G(B) & \longrightarrow & G(A) \end{array}$$

η is a *natural isomorphism* if η_A are isomorphisms

Definition 2.11 A category $\mathcal{C}$ is

- *small* if $ob(\mathcal{C})$ and $\mathrm{Hom}(\mathcal{C})$ are sets
- *locally small* if $\mathrm{Hom}(a, b)$ are sets
- **filtered** if it is not empty, and for any two objects $j, j' \in J$, there is an object $k \in J$ and morphisms $f : j \to k$ and $f' : j' \to k$, for any two morphisms $u, v : i \to j$, there is an object $k \in J$ and a morphism $w : j \to k$ such that $w \circ u = w \circ v$

  A filtered colimit is the colimit of a functor $F : J \to \mathcal{C}$ where J is a filtered category, direct limit is a special case of a filtered colimit

  The dual notion is called **cofiltered**
- *pointed* is a category with zero object *connected* if there is a finite sequence of morphisms connecting any two objects

Example 2.12 $C \leftarrow A \to B$ is a connected even there is no morphism between B, C

Definition 2.13 we say categories $\mathcal{C}, \mathcal{D}$ are *isomorphic* if there are functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ such that $G \circ F = 1_{\mathcal{C}}$, $F \circ G = 1_{\mathcal{D}}$ and we say $\mathcal{C}, \mathcal{D}$ are *equivalent* if $G \circ F$ is naturally isomorphic to $1_{\mathcal{C}}$ and $F \circ G$ is naturally isomorphic to $1_{\mathcal{D}}$

Definition 2.14 Suppose $\mathcal{C}, \mathcal{D}$ are categories, define the *functor category* $[\mathcal{C}, \mathcal{D}]$ or $\mathcal{D}^{\mathcal{C}}$ has all functors from $\mathcal{C}$ to $\mathcal{D}$ as objects, and natural transformations as morphisms

Definition 2.15 $\mathcal{C} \times \mathcal{D}$ is the *product category* with $ob(\mathcal{C} \times \mathcal{D}) = ob\mathcal{C} \times ob\mathcal{D}$, $Hom_{\mathcal{C}\times\mathcal{D}}(A \times B, C \times D) = Hom_{\mathcal{C}}(A, C) \times Hom_{\mathcal{D}}(B, D)$

Definition 2.16 A functor $F : \mathcal{C} \to \mathcal{D}$ is

- *faithful* if $\mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ is injective
- *full* if $\mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ is surjective
- *fully faithful* if $\mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ is bijective
- *essentially surjective* if $\forall d \in ob\mathcal{D}$, $\exists c \in ob\mathcal{C}$ such that $Fc \cong d$

Theorem 2.17 $F : \mathcal{C} \to \mathcal{D}$ is an equivalence iff F is fully faithful and essentially surjective

Proof. If F is an equivalence, there exist functor $G : \mathcal{D} \to \mathcal{C}$ and natural isomorphisms $\eta : 1_{\mathcal{C}} \to GF$, $\xi : 1_{\mathcal{D}} \to FG$, $\forall d \in \mathcal{C}$, $\xi_d : d = 1_{\mathcal{D}}(d) \to FG(d) = F(Gd)$ is an isomorphism, i.e. F is essentially surjective, similarly, so is G

The composition of

$$Hom(c, c') \xrightarrow{F} Hom(Fc, Fc') \xrightarrow{G} Hom(GFc, GFc'), \quad f \mapsto Ff \mapsto GFf$$

Is the same as

$$Hom(c, c') \xrightarrow{\eta} Hom(GFc, GFc'), \quad f \mapsto \eta'_c f \eta_c^{-1}$$

By Exercise 35.66, this is bijective, thus $Hom(c, c') \xrightarrow{F} Hom(Fc, Fc')$ is injective, i.e. F is faithful. Similarly, consider the composition

$$Hom(Fc, Fc') \xrightarrow{G} Hom(GFc, GFc') \xrightarrow{F} Hom(FGFc, FGFc')$$

We know $Hom(GFc, GFc') \xrightarrow{F} Hom(FGFc, FGFc')$ is surjective, but we also have the following diagram

$$\begin{array}{ccc} Hom(c, c') & \xrightarrow{\quad F \quad} & Hom(Fc, Fc') \\ {\scriptstyle \eta}\downarrow & & \downarrow{\scriptstyle \xi} \\ Hom(GFc, GFc') & \xrightarrow{\quad F \quad} & Hom(FGFc, FGFc') \end{array}$$

Since η, ξ are bijective, $Hom(c, c') \xrightarrow{F} Hom(Fc, Fc')$ is surjective, i.e. F is full

Conversely, suppose F is fully faithful and essentially surjective, then for any $d \in \mathcal{D}$, there exists c and an isomorphism $d \xrightarrow{\xi_d} Fc$, denote this c as Gd, we can define a functor $G : \mathcal{D} \to \mathcal{C}$, $d \mapsto Gd$(Here we have used the axiom of choice), $d \xrightarrow{f} d' \mapsto c \xrightarrow{Gf} c'$ where $FGf = \xi_d^{-1} f \xi_{d'}$ since F is fully faithful

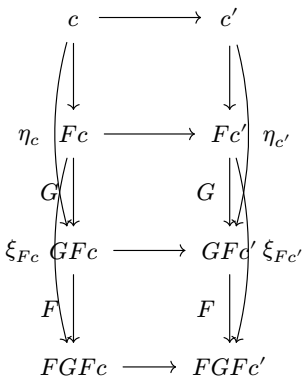
$$\begin{array}{ccc} d & \xrightarrow{\quad f \quad} & d' \\ {\scriptstyle \xi_d}\downarrow & & \downarrow{\scriptstyle \xi_{d'}} \\ FGd & \xrightarrow{\quad FGf \quad} & FGd' \\ {\scriptstyle F}\uparrow & & \uparrow{\scriptstyle F} \\ Gd & \xrightarrow{\quad Gf \quad} & Gd' \end{array}$$

$\xi : 1_{\mathcal{D}} \to FG$ is a natural isomorphism

Since F is fully faithful, there are unique $\eta_c : c \to GFc$, $F(\eta_c) = \xi_{Fc}$

If $f, g : c \to c'$ such that $\eta_{c'} f = \eta_{c'} g$, then $\xi_{Fc'} Ff = \xi_{Fc'} Fg \Rightarrow Ff = Fg \Rightarrow f = g$

If $f, g : c \to c'$ such that $f\eta_c = g\eta_c$, then $Ff\xi_{Fc} = Fg\xi_{Fc} \Rightarrow Ff = Fg \Rightarrow f = g$



$\eta : 1_{\mathcal{C}} \to GF$ is a natural isomorphism

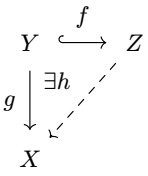
□

Definition 2.18 Suppose $u : S \to A$, $v : T \to A$ are morphisms, v filter through s means there is a morphism $w : T \to S$ such that $v = u \circ w$, then mutually filter defines an equivalence relation on monomorphisms(or equivalent by saying that w is an isomorphism), the equivalence classes are called *subobjects* of A, the dual notion is called *quotient objects*

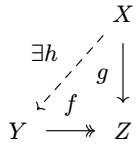
Proposition 2.19 Direct limit is an exact functor

Definition 2.20 An object X is a

- *injective object* if for any monomorphism $f : Y \to Z$ and morphism $g : Y \to X$, there is a morphism $h : Z \to X$ such that $g = h \circ f$



- *projective object* if for any epimorphism $f : Y \to Z$, and morphism $g : X \to Z$, there is a morphism $h : X \to Y$ such that $g = f \circ h$



initial object if for every Y, there is a unique $X \to Y$

final object if for every Y, there is a unique $Y \to X$

zero object is an object which is both initial and final

Remark. Final objects are also called terminal

Lemma 2.21 (Yoneda lemma) $\mathcal{C}$ is locally small

$$\mathrm{Hom}(\mathrm{Hom}(c, -), F) \xrightarrow{\cong} F(c), \eta \mapsto \eta_c(1_c)$$

$$\mathrm{Hom}(\mathrm{Hom}(-, c), F) \xrightarrow{\cong} F(c), \eta \mapsto \eta_c(1_c)$$

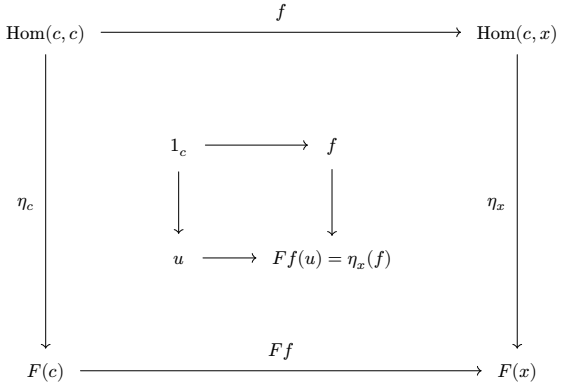
If $F = \mathrm{Hom}(-, d)$ or $\mathrm{Hom}(d, -)$, then

$$\mathrm{Hom}(\mathrm{Hom}(c, -), \mathrm{Hom}(d, -)) \xrightarrow{\cong} \mathrm{Hom}(d, c)$$

$$\mathrm{Hom}(\mathrm{Hom}(-,c),\mathrm{Hom}(-,d)) \xrightarrow{\cong} \mathrm{Hom}(c,d)$$

$\mathrm{Hom}(c,-), \mathrm{Hom}(-,c)$ give fully faithful embeddings

Proof.



The natural transformation η is determined by the element u in $F(c)$

□

Remark. Functor $\mathrm{Hom}(c,-), \mathrm{Hom}(-,c)$ is called **Yoneda embedding**, here embedding in the sense of a fully faithful functor, which is injective on objects up to isomorphism as in Lemma 35.69

Definition 2.22 A functor $F : \mathcal{C} \to Set$ is called a *representable* functor if there is an object A in $\mathcal{C}$ such that $Hom(A,-), F$ are naturally isomorphic, By Yoneda lemma, the isomorphism is determined by an element $a \in A$ called the universal element

Definition 2.23 Let $\mathcal{C}$ be a category, we can define *quotient category* by moding out a congruence relation $\sim$, here $\sim$ is an equivalence relation on $Hom(X,Y)$ for any X, Y and it respects composition, i.e. suppose $f_1 \sim f_2 : X \to Y$, $g_1 \sim g_2 : Y \to Z$, then $g_1 \circ f_1 \sim g_2 \circ f_2$, thus $Hom_{\mathcal{C}/\sim}(X,Y) = Hom_{\mathcal{C}}(X,Y)/\sim$

Definition 2.24 $\mathcal{C}$ is *concretizable* if there is a faithful functor $F : \mathcal{C} \to Set$. A morphism $f : X \to Y$ is an *embedding* if $F(f)$ is injective, and for any $F(Z) \xrightarrow{\phi} F(X)$, $Z \xrightarrow{h} Y$ such that $F(Z) \xrightarrow{F(h)} F(Y)$, $F(h) = F(f) \circ \phi$, $\phi = F(g)$ for some $Z \xrightarrow{g} X$

Note. $\mathcal{C}$ may have different concretization

Definition 2.25 W is a class of morphisms of $\mathcal{C}$, the *localization* of $\mathcal{C}$ with respect to W, denoted $\mathcal{C}[W^{-1}]$, satisfies universal property

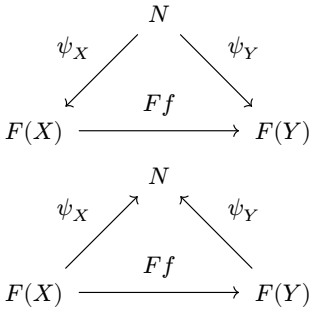
- Any functor $F : \mathcal{C} \to \mathcal{D}$ sending morphisms in W to isomorphisms in $\mathcal{D}$ uniquely factors through the *localization functor* $\mathcal{C} \to \mathcal{C}[W^{-1}]$

Construction 2.26 Add formal inverses of the morphisms in $\overline{W}$

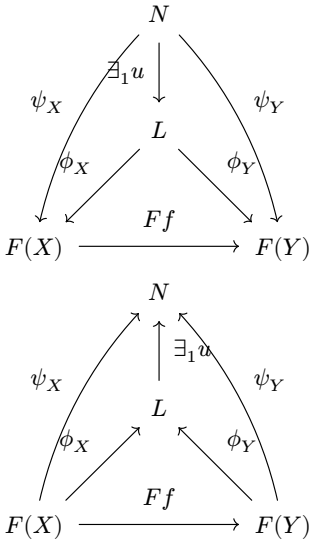
Note. $\mathrm{Hom}_{\mathcal{C}[W^{-1}]}(A,B)$ consists of *roofs* $A \leftarrow C \to B$ and $A \to D \leftarrow B$

Definition 2.27 A **diagram** is a functor $D : J \to \mathcal{C}$, J is called the **indexed category**, the diagram D can be thought of as indexing a collection of objects and morphisms in $\mathcal{C}$ patterned on J, we say D is a diagram in $\mathcal{C}$ shaped J

Let $F : J \to \mathcal{C}$ be a diagram in $\mathcal{C}$, N be an object in $\mathcal{C}$, then a **cone** from N to F is a family of morphisms ψ_X such that the following diagram commutes, a **cocone** from F to N is a family of morphisms ψ_X such that the following diagram commutes



A **limit** of the diagram F is cone (L, ϕ) such that for any other cone (N, ψ) there is a unique $u : N \to L$ such that the following diagram commutes, a **colimit** of the diagram F is cone (L, ϕ) such that for any other cone (N, ψ) there is a unique $u : L \to N$ such that the following diagram commutes



Limits may also be characterized as terminal objects in the category of cones to F, thus unique up to isomorphism, so is colimits, a category contains all limits is called **complete**, and is called **cocomplete** if containing all colimits

The **equaliser** $Eq(f, g)$ is defined to be the limit of the diagram

$$X \overset{f}{\underset{g}{\rightrightarrows}} Y$$

, the **coequaliser** is the colimit

Remark. Direct limit and inverse limit are defined on directed set, thus limit and colimit are more general

Definition 2.28 A **directed set** X is a set with a preorder $\leq$ and any pair of elements has an upper bound, i.e., $\forall x, y \in X, \exists z \in X$ such that $x \leq z, y \leq z$

Definition 2.29 Given a directed set I, we can define a **direct(inductive) system**, with modules A_i, and functions $f_{ij} : A_i \to A_j$, $f_{ii} = 1_{A_i}$, $f_{jk} \circ f_{ij} = f_{ik}, i \leq j \leq k$, we can also define an **inverse system**, with module A_i, and functions $f_{ij} : A_j \to A_i$, $f_{ii} = 1_{A_i}$, $f_{ij} \circ f_{jk} = f_{ik}$

We can define morphism betweem direct and inverse systems and modules

Suppose A_i is a direct system, then a morphism $g_i : A_i \to B$ is such that $g_j \circ f_{ij} = g_i, i \leq j$ or $g_i : B \to A_i$ is such that $g_j = f_{ij} \circ g_i, i \leq j$. Suppose A_i is an inverse system, then a morphism $g_i : A_i \to B$ is such that $g_i \circ f_{ij} = g_j, i \leq j$ or $g_i : B \to A_i$ is such that $g_i = f_{ij} \circ g_j, i \leq j$, We can define morphisms between direct and inverse systems

Suppose A_i, B_i are both direct systems, a morphism $g_i : A_i \to B_i$ is a family of maps such that $g_j \circ f_{ij} = f_{ij} \circ g_i, i \leq j$. Suppose A_i, B_i are both inverse systems, a morphism $g_i : A_i \to B_i$ is a family of maps such that $g_i \circ f_{ij} = f_{ij} \circ g_j, i \leq j$.

Definition 2.30 The **direct limit** of a direct system is a module A_∞ and morphisms $\iota_i : A_i \to A$ with the universal property: given any morphism $g_i : A_i \to B$, it induces a unique $g_\infty : A_\infty \to B$ such that $g \circ \iota_i = g_i$, there is a concrete construction: define the direct limit $\lim_{\to} A_i = \bigsqcup_{i \in I} A_i / \sim$, where $a_i \sim a_j, a_i \in A_i, a_j \in A_j$ if there is an upper bound k such that $f_{ik}(a_i) = f_{jk}(a_k)$, or equivalently, $a_i \sim f_{ij}(a_i), i \leq j$

Definition 2.31 The **inverse limit** of an inverse system is a module A_∞ and morphisms $\pi_i : A \to A_i$ with the universal property: given any morphism $g_i : B \to A_i$, it induces a unique $g_\infty : B \to A_\infty$ such that $\pi_i \circ g = g_i$, there is a concrete construction: define the inverse limit $\lim_{\leftarrow} A_i = \left\{ (a_i) \in \prod_{i \in I} A_i \;\middle|\; a_i = f_{ij}(a_j), i \leq j \right\}$,

Remark. Direct limit and inverse limit are dual to each other in the categorical sense

Definition 2.32 Product, coproduct The **biproducts** $(\bigoplus_i A_i, p_i, \iota_i)$ of A_i is such that $(\bigoplus_i A_i, p_i)$ is the product and $(\bigoplus_i A_i, \iota_i)$ is the coproduct

Remark. The initial and final object are the limit and colimit of empty diagram

In the category of sets, the initial object is $\varnothing$ and a terminal object is $\{*\}$

Lemma 2.33 $\mathrm{Hom}(X, -)$ functor takes limits to limits, dually, $\mathrm{Hom}(-, X)$ functor takes colimits to limits

Proof. Consider $Y \xrightarrow{f_i} \mathrm{Hom}(X, X_i)$, we want to define a map $Y \to \mathrm{Hom}(X, \lim_{\leftarrow} X_i)$, for $y \in Y$, $X \xrightarrow{f_i(y)} X_i$ induces $X \xrightarrow{g} \lim_{\leftarrow} X_i$

□

Remark. In general, $\mathrm{Hom}(X, -)$ functor doesn't take colimits to colimits

Definition 2.34 A *skeleton of a category* $\mathcal{D}$ of $\mathcal{C}$ is a full subcategory such that no two objects in $\mathcal{D}$ are isomorphic and for every object in $\mathcal{C}$ is isomorphic to some object in $\mathcal{D}$, the functor $\mathcal{D} \hookrightarrow \mathcal{C}$ is an equivalence of categories

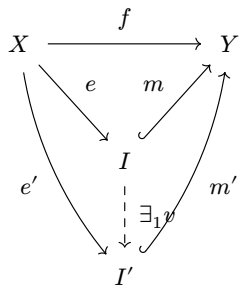
Definition 2.35 Suppose $\mathcal{C}$ is a category, a *filtered object* X is an object with a *filtration* of X, a descending filtration

$$\cdots \to X_{n+1} \to X_n \to X_{n-1} \to \cdots \to X$$

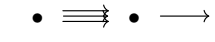
Or an ascending filtration

$$X \to \cdots \to X_{n+1} \to X_n \to X_{n-1} \to \cdots$$

Definition 2.36 Suppose $\mathcal{C}$ is a category, $f : X \to Y$ is a morphism, the *image* of f is a monomorphism $m : I \to Y$ such that there is a morphism $e : X \to I$ such that the following diagram commutes and satisfies the universal property



Definition 2.37 A *quiver* is a functor from



to the category of sets. Equivalently, a directed graph allowing multiple arrows and loops

Definition 2.38 The *free category* generated by quiver Q has objects vertices in Q and morphisms paths in Q with empty path the identity

Definition 2.39 $A \xrightarrow{i} B$ has *left lifting property* or *LLP* and $X \xrightarrow{p} Y$ has *right lifting property* or *RLP* for each other in this diagram if

$$\begin{array}{ccc} A & \longrightarrow & X \\ {\scriptstyle i}\downarrow & \exists \nearrow & \downarrow{\scriptstyle p} \\ B & \longrightarrow & Y \end{array}$$

i, p are *orthogonal* if the lifting is unique

Definition 2.40 A class of morphisms $\mathbf{M}$ of $\mathcal{C}$ satisfies *2 out of 3* if any two of $f, g, f \circ g$ are in $\mathbf{M}$, so is the third. $\mathbf{M}$ is clearly closed under composition

A class of *weak equivalences* is a class of morphisms $\mathbf{W}$ containing isomorphisms and satisfies 2 out of 3. The class of isomorphisms $\mathbf{I}$ is a class of weak equivalences

Definition 2.41 G is a *group object* in category $\mathcal{C}$ (having finite products, thus terminal object 1) if there are multiplication $m : G \times G \to G$, identity $e : 1 \to G$, and inverse $i : G \to G$ satisfying the group law

1. Associativity

$$\begin{array}{ccccc} & & G \times G & & \\ & {\scriptstyle m \times 1} \nearrow & & \searrow {\scriptstyle m} & \\ G \times G \times G & & & & G \\ & {\scriptstyle 1 \times m} \searrow & & \nearrow {\scriptstyle m} & \\ & & G \times G & & \end{array}$$

   commutes

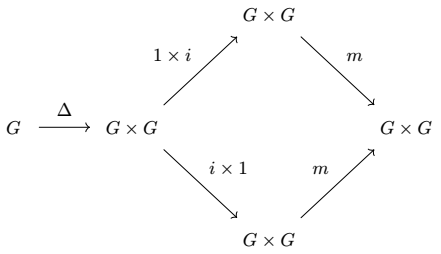
2. Unity

$$\begin{array}{ccc} 1 \times G & \xrightarrow{e \times 1} & G \times G \\ & {\scriptstyle p_2} \searrow & \downarrow {\scriptstyle m} \\ & & G \end{array}$$

$$\begin{array}{ccc} G \times 1 & \xrightarrow{1 \times e} & G \times G \\ & {\scriptstyle p_1} \searrow & \downarrow {\scriptstyle m} \\ & & G \end{array}$$

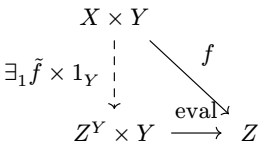
   commute

3. Invertibility



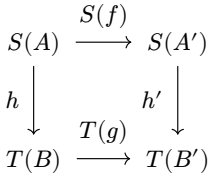
commutes

Definition 2.42 Y is an object such that all binary products $X \times Y$ exist, the **exponential object** is Z^Y together with morphism $Z^Y \times Y \overset{\text{eval}}{\to} Z$ satisfying universal property

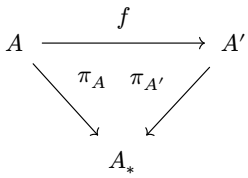


Proposition 2.43 $Hom(X \times Y, Z) \to Hom(X, Z^Y)$ is an adjunction

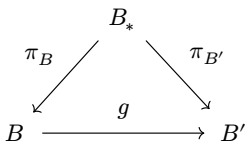
Definition 2.44 Consider functors $S : \mathcal{A} \to \mathcal{C},\ T : \mathcal{B} \to \mathcal{C}$(for source and target), define **comma category** $(S \downarrow T)$ with objects $(A, B, h),\ A \in \mathcal{A}, B \in \mathcal{B}$ are objects, $h : S(A) \to T(B)$ is a morphism, and with morphisms $(f, g) : (A, B, h) \to (A', B', h')$ where $f : A \to A', g : B \to B'$ are morphisms such that the following diagram commutes



Definition 2.45 Consider the comma category of $1_{\mathcal{A}} : \mathcal{A} \to \mathcal{A}, T : * \to \mathcal{A}$ which we call **slice category**, sometimes denoted as $(\mathcal{A} \downarrow A_*)$ where $A_* = T(*)$, the objects of the slice category are $A \overset{\pi_A}{\to} A_*$ and morphisms are



Its dual notion, the comma category of $S : * \to \mathcal{B}\ \ 1_{\mathcal{B}} : \mathcal{B} \to \mathcal{B}$, which we call **coslice category**, sometimes denoted as $(B_* \downarrow \mathcal{B})$ where $B_* = S(*)$, the objects of the slice category are $B_* \overset{\pi_B}{\to} B$ and morphisms are



The comma category of $1_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$, $1_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ which we call **arrow category**, sometimes denoted as $\mathcal{C}^{\to}$ the objects of the arrow category are just the morphisms(arrows) $A \overset{f}{\to} A'$, and morphisms are

$$\begin{array}{ccc} A & \xrightarrow{f} & A' \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle h'} \\ B & \xrightarrow{g} & B' \end{array}$$

Definition 2.46 A right inverse are called a **section**, a left inverse is called a **retraction**

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow^{=} & \downarrow{\scriptstyle g} \\ & & X \end{array}$$

f is a section of g, g is a retraction of f

Definition 2.47 A Galois category is the category of finite G-sets for some profinite group G, the fiber functor Φ is simply to the category finite sets, $G \cong \mathrm{Aut}\ \Phi$

Example 2.48 $\mathsf{Finite\ sets} \xrightarrow{\mathrm{id}} \mathsf{Finite\ sets}$ gives a Galois category, the group being the permutation group

Example 2.49 Covering spaces

Example 2.50 Field extensions

Definition 2.51 (Monoidal category) A category $\mathcal{C}$ is *monoidal* if there are

- A *tensor product(monoidal product)* $\mathcal{C} \times \mathcal{C} \xrightarrow{\otimes} \mathcal{C}$ with a tensor unit 1
- *Associator* $(x \otimes y) \otimes z \xrightarrow{\alpha_{x,y,z}} x \otimes (y \otimes z)$ which are natural isomorphisms
- *Left* and *right unitor* $1 \otimes x \xrightarrow{\lambda_x} x$, $x \otimes 1 \xrightarrow{\rho_x} x$ which are natural isomorphisms

Such that

1. Unital

$$\begin{array}{ccc} (x \otimes 1) \otimes y & \xrightarrow{\alpha} & x \otimes (1 \otimes y) \\ & {\scriptstyle \rho \otimes 1} \searrow \quad \swarrow {\scriptstyle 1 \otimes \lambda} & \\ & x \otimes y & \end{array}$$

2. Associative

$$\begin{array}{ccccc} ((w \otimes x) \otimes y) \otimes z & & \xrightarrow{\alpha} & & (w \otimes x) \otimes (y \otimes z) \\ {\scriptstyle \alpha}\downarrow & & & & \downarrow{\scriptstyle \alpha} \\ (w \otimes (x \otimes y)) \otimes z & \xrightarrow{\alpha} & w \otimes ((x \otimes y) \otimes z) & \xrightarrow{\alpha} & w \otimes (x \otimes (y \otimes z)) \end{array}$$

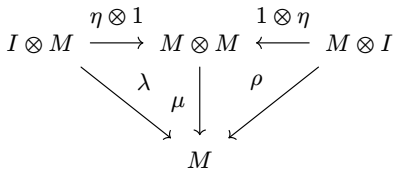
$\mathcal{C}$ is *strictly monoidal* if α, λ, ρ are identities, i.e. $\otimes$ is associative and 1 is an actual unit

Definition 2.52 (Monoids, comonoids and bimonoids) $\mathcal{C}$ is a monoidal category
$M \in \mathcal{C}$ is a *monoid* if there are

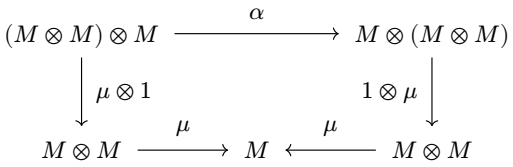
- Multiplication $\mu : M \otimes M \to M$
- Unit $\eta : 1 \to M$

Such that

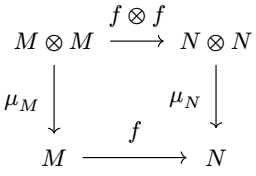
1. unital

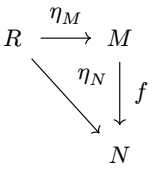


2. associative



A morphism between monoids would be $f : M \to N$ such that the following diagrams commute





R it self is a monoid with

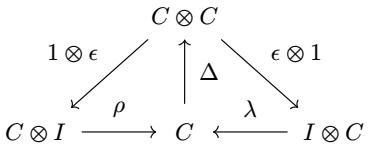
$M \otimes N$ is again a monoid with $\mu_{M\otimes N} : (M \otimes N) \otimes (M \otimes N) \to M \otimes N,\ (m_1 \otimes n_1) \otimes (m_2 \otimes n_2) \mapsto m_1 m_2 \otimes n_1 n_2$, and $\eta_{M\otimes N} : R \to R \otimes R \xrightarrow{\eta_M \otimes \eta_N} M \otimes N$

$C \in \mathcal{C}$ is a *comonoid* if it is a monoid in $\mathcal{C}^{\mathrm{op}}$, i.e. there are

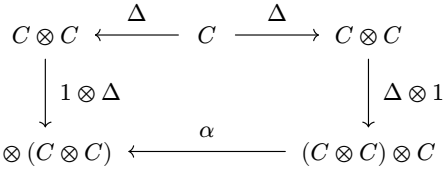
- Comultiplication $\Delta : C \to C \otimes C$
- Counit $\epsilon : C \to I$

Such that

1. counital



2. coassociative



A morphism between comonoids would be $f : M \to N$ such that the following diagrams commute

$$\begin{array}{ccc} M & \xrightarrow{\Delta_M} & M \otimes M \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f \otimes f} \\ N & \xrightarrow{\Delta_N} & N \otimes N \end{array}$$

$$\begin{array}{ccc} R & \xleftarrow{\epsilon_M} & M \\ & \nwarrow^{\epsilon_N} & \downarrow{\scriptstyle f} \\ & & N \end{array}$$

$B \in \mathcal{C}$ is a *bimonoid* if it is both a monoid and a comonoid satisfying following compatibility

1. Multiplication μ and comultiplication Δ

$$\begin{array}{ccccc} B \otimes B & \xrightarrow{\mu} & B & \xrightarrow{\Delta} & B \otimes B \\ {\scriptstyle \Delta \otimes \Delta}\downarrow & & & & \uparrow{\scriptstyle \mu \otimes \mu} \\ (B \otimes B) \otimes (B \otimes B) & & & & (B \otimes B) \otimes (B \otimes B) \\ \downarrow & & & & \uparrow \\ B \otimes (B \otimes B) \otimes B & & \xrightarrow{1 \otimes \tau \otimes 1} & & B \otimes (B \otimes B) \otimes B \end{array}$$

   Here $\tau(x \otimes y) = y \otimes x$

2. Multiplication μ and counit ϵ

$$\begin{array}{ccc} B \otimes B & \xrightarrow{\mu} & B \\ {\scriptstyle \epsilon \otimes \epsilon}\downarrow & & \downarrow{\scriptstyle \epsilon} \\ I \otimes I & \longrightarrow & I \end{array}$$

3. Comultiplication Δ and unit η

$$\begin{array}{ccc} B \otimes B & \xleftarrow{\Delta} & B \\ \uparrow{\scriptstyle \eta \otimes \eta} & & \uparrow{\scriptstyle \eta} \\ I \otimes I & \longrightarrow & I \end{array}$$

4. Unit η and counit ϵ

$$\begin{array}{ccc} I & \xrightarrow{\eta} & B \\ & \searrow^{\epsilon} & \downarrow \\ & & I \end{array}$$

Definition 2.53 Let $L : \mathcal{D} \to \mathcal{C}$, $R : \mathcal{C} \to \mathcal{D}$ be functors, and there is a natural isomorphism $\Phi_{X,Y}$, $X \in \mathcal{C}, Y \in \mathcal{D}$

$$\begin{array}{ccc} Hom_{\mathcal{C}}(LX,Y) & \xrightarrow{\Phi_{X,Y}} & Hom_{\mathcal{D}}(X,RY) \\ {\scriptstyle (Lf,g)}\downarrow & & \downarrow{\scriptstyle (g,Rf)} \\ Hom_{\mathcal{C}}(LX',Y') & \xrightarrow{\Phi_{X',Y'}} & Hom_{\mathcal{D}}(X',RY') \end{array}$$

Here $f : X' \to X$, $g : Y \to Y'$, $Hom_{\mathcal{C}}(Lf, g)(h) = h \circ g \circ Lf$

We say L is the **left adjoint** of R and R is the **right adjoint** of L

Example 2.54 Let $G : Group \to Set$ be the forgetful functor, then the functor $F : Set \to Group$, sending S to $F(S)$ is the left adjoint of G

In the category of R-modules Mod, consider functor $F \coloneqq - \otimes B$ and functor $G \coloneqq Hom(B, -)$, then F is the left adjoint to G, i.e. $Hom(A \otimes B, C) \cong Hom(A, Hom(B, C))$

Theorem 2.55 $\mathrm{Ext}^n(A, B)$ is the equivalent classes of n-extensions

$$0 \to B \to E_1 \to \cdots \to E_n \to A \to 0$$

Proof. First consider $n = 1$, given $0 \to B \to E \to A \to 0$, we have $0 \to \mathrm{Hom}(A, B) \to \mathrm{Hom}(A, E) \to \mathrm{Hom}(A, A) \to \mathrm{Ext}^1(A, B)$, take the image of 1_A. Conversely, Find some $0 \to M \to P \to A \to 0$, where P is projective, then we have $\mathrm{Hom}(M, B) \to \mathrm{Ext}^1(A, B) \to \mathrm{Ext}^1(P, B) = 0$, we get a map $M \to B$, let E be the pushout

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M & \longrightarrow & P & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & B & \longrightarrow & E & \longrightarrow & A & \longrightarrow & 0 \end{array}$$

□

Definition 2.56 An injective resolution $C^\bullet \to I^\bullet$ is a quasi-isomorphism where $I^\bullet \in D^+(\mathcal{A})$ consists of injective objects. A projective resolution $P^\bullet \to C^\bullet$ is a quasi-isomorphism where $P^\bullet$ consists of projective objects

Definition 2.57 $\mathrm{Ext}^i_{\mathcal{A}}(C, D) = \mathrm{Hom}_{D(\mathcal{A})}(C, D[i]) = \mathrm{Hom}_{D(\mathcal{A})}(C[-i], D)$, $\mathrm{Ext}^i_{\mathcal{A}}(C^\bullet, D^\bullet) = \mathrm{Hom}_{D(\mathcal{A})}(C^\bullet, I^\bullet[i]) = \mathrm{Hom}_{D(\mathcal{A})}(P^\bullet[-i], C^\bullet)$, here $D^\bullet \to I^\bullet$ is an injective resolution, $P^\bullet \to C^\bullet$ is a projective resolution

Definition 2.58 $F : \mathcal{A} \to \mathcal{B}$ is a left exact functor, the derived functor $\mathbb{R}F : D^+(\mathcal{A}) \to D^+(\mathcal{B})$ is given as follows

- Take any injective resolution $I^\bullet$ quasi-isomorphic to $C^\bullet \in D^+(\mathcal{A})$, $\mathbb{R}F(C^\bullet) = F(I^\bullet)$

$G : \mathcal{A} \to \mathcal{B}$ is a right exact functor, the derived functor $\mathbb{L}F : D^-(\mathcal{A}) \to D^-(\mathcal{B})$ is given as follows

- Take any projective resolution $P_\bullet$ quasi-isomorphic to $C_\bullet \in D^-(\mathcal{A})$, $\mathbb{L}F(C_\bullet) = F(P_\bullet)$

As cohomology and homology arise from derived functors, hypercohomology and hyperhomology arise from hyper-derived functors

Remark. By definition, $\mathbb{R}F(C^\bullet) = \mathbb{R}F(D^\bullet)$ if $C^\bullet, D^\bullet$ are quasi-isomorphic. Let $A^\bullet$ is the chain complex with only A centered at zero, then R^iF is the composition of the following

$$\mathcal{A} \to D^+(\mathcal{A}) \overset{\mathbb{R}F}{\to} D^+(\mathcal{B}) \overset{H^i}{\to} \mathcal{B}$$

i.e. $H^i(\mathbb{R}F(A^\bullet)) = R^iF(A)$. Denote $H^i \circ \mathbb{R}F$ as $\mathbb{R}^iF$. Since any resolution $0 \to A \to C^\bullet$ is a quasi-isomorphism between $A^\bullet$ and $C^\bullet$, $R^iF(A) = \mathbb{R}^iF(A^\bullet) = \mathbb{R}^iF(C^\bullet)$. Hyper-derived functor gives a way of computing derived functor using any resolution instead of only those "nice" resolutions

Example 2.59 The derived functors of $\Gamma : \mathrm{Sh}(X) \to Ab$, $\mathcal{F} \mapsto \mathcal{F}(X)$ define sheaf cohomology

$$H^i(X, \mathcal{F}) = (R^i\Gamma)(\mathcal{F})$$

The hyper-derived functors of Γ define sheaf hypercohomology

$$\mathbb{H}^i(X, \mathcal{F}^\bullet) = (R^i\Gamma)(\mathcal{F}^\bullet)$$

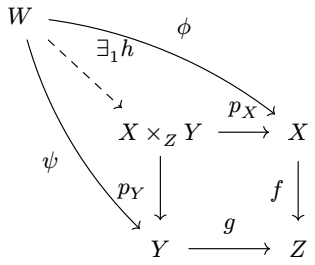
The derived functors of $F : RMod \to Ab$, $M \mapsto M \otimes_{RG} R$ define group homology

$$H_i(G, M) = (L_iF)(M)$$

The hyper-derived functors of F define group hyperhomology

$$\mathbb{H}_i(G, M^\bullet) = (L_iF)(M^\bullet)$$

Definition 2.60 The **pullback** of $f : X \to Z$, $g : Y \to Z$ is $(X \times_Z Y, p_X, p_Y)$ satisfying the universal property

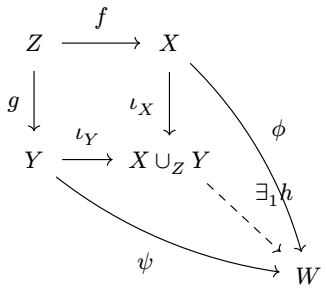


p_X is the **base change** of g along f, p_Y is the base change of f along g

If f is an epimorphism, so is p_X

More generally, we can also define the puullback of $f_i : X \to Y_i$

Definition 2.61 The **pushout** of $f : Z \to X$, $g : Z \to Y$ is $(X \cup_Z Y, \iota_X, \iota_Y)$ satisfying the universal property



ι_X is the **cobase change** of g along f, ι_Y is the cobase change of f along g

If f is a monomorphism, so is ι_X

More generally, we can also define the pushout of $f_i : Z \to X_i$

Proposition 2.62 Pushout preserve epimorphisms and isomorphisms and in the category of sets, pushout preserve injection

Pullback preserve monomorphisms and isomorphisms and in the category of sets, pullback preserve surjection

2.2 Sheaf

Definition 2.63 (Site, also known as Grothendieck topology) Suppose $\mathcal{C}$ is a category. $\{U_i \to U\}_{i \in I}$ is a family of morphisms with a fixed target U. Let's call this an open coverings of U. A *site* is a set of families of open coverings such that

1. $\left\{U \xrightarrow{\text{id}} U\right\}$ is an open covering
2. Suppose $\left\{V_{ij} \to U_i\right\}_{j \in J_i}$ are open coverings, then so is $\left\{V_{ij} \to U_i \to U\right\}_{i,j}$
3. $U_i \times_U V$ exists in the category and $\{U_i \times_U V \to V\}$ is an open covering

$$\begin{array}{ccc} U_i \times_U V & \longrightarrow & V \\ \downarrow & & \downarrow \\ U_i & \longrightarrow & U \end{array}$$

Definition 2.64 The Zariski site has the category of open subsets in Zariski topology. The étale site has the category of étale coverings

Definition 2.65 $\mathcal{A}$ is a category. A *presheaf* is a functor

$$F : \{\texttt{open subsets of } \ X\}^{op} \to \mathcal{A}$$

$F(U \hookrightarrow V) = \text{res}_{UV}$ are *restriction maps*. A *morphism of presheaves* $F \xrightarrow{\phi} G$ is a natural transformation, i.e. the following diagram commutes

$$\begin{array}{ccc} F(V) & \xrightarrow{\text{res}_{UV}} & F(U) \\ {\scriptstyle \phi_V}\downarrow & & \downarrow{\scriptstyle \phi_U} \\ G(V) & \xrightarrow{\text{res}_{UV}} & G(U) \end{array}$$

Definition 2.66 $U \subseteq X$ is an open subset, F is a presheaf over X, the *restricted presheaf* $F|_U$ is given by $F|_U(V) = F(U \cap V)$

Definition 2.67 F is a presheaf, $x \in X$, open subsets containing x is full subcategory $\tau(x)$, the *stalk* F_x is the colimit $\lim\limits_{\substack{\to \\ x \in U}} F(U)$, elements in F_x are called *germs*, denote the germ of f at x as f_x

Lemma 2.68 $B(f, U) = \{f_x | x \in U, f \in F(U)\}$ form a basis on the *étalé space* $|F| = \bigcup F_x$. $|F| \to X$, $f_x \mapsto x$ is a local homeomorphism

Definition 2.69 Presheaf F is a *sheaf* if

$$F(U) \xrightarrow{\text{res}_{U_i,U}} \prod_i F(U_i) \xrightarrow[\text{res}_{U_i \cap U_j, U_j}]{\text{res}_{U_i \cap U_j, U_i}} \prod_{i,j} F(U_i \cap U_j)$$

Is an equaliser. Equivalently, F satisfying

1. If $U = \bigcup_i U_i$, $f, g \in F(U)$, $f|_{U_i} = g|_{U_i}$, then $f = g$
2. If $U = \bigcup_i U_i$, $f_i \in F(U_i)$, $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$, then there exists $f \in F(U)$ such that $f|_{U_i} = f_i$, here f has to be unique because of 1

Proposition 2.70 $F \xrightarrow{\phi} G$ is a monomorphism or a epimorphism iff $F_x \xrightarrow{\phi_x} G_x$ is injective or surjective on each stalk

Definition 2.71 (Sheafification) F is a presheaf over X, the sheaf of sections $X \to |F|$ is the *sheafification*

Proposition 2.72 $\mathcal{F}$ is a presheaf, $\mathcal{G}$ is a sheaf. Then any morphism $\mathcal{F} \to \mathcal{G}$ factors uniquely through $\mathcal{F} \to |\mathcal{F}| \to \mathcal{G}$. i.e. $\text{Hom}(\mathcal{F}, i(\mathcal{G})) = \text{Hom}(|\mathcal{F}|, \mathcal{G})$, the inclusion functor i from the category of sheaves to the category of presheaves is right adjoint to the sheafification functor $|\cdot|$

Definition 2.73 The *constant presheaf* $\underline{A}$ given by $\underline{A}(U) = A$, $\text{res}_{UV} = 1_A$

F is a *locally constant sheaf* if for any $x \in X$, there exists $U \ni x$ such that $F|_U$ is a constant sheaf. $F : \Pi_1 X \to \mathcal{A}$ is a functor. The category of locally constant sheaves is equivalent to the category of covering spaces of X

Definition 2.74 Functor $\Gamma : Sh(X) \to \mathcal{A}$, $F \mapsto F(X)$ is a left exact functor, the *sheaf cohomology* is the right derived functor $R^i\Gamma$, i.e. $R^i\Gamma(F) = H^i(X, F)$

Definition 2.75 $X \xrightarrow{f} Y$ is a continuous map, F is a presheaf/sheaf over X. The *pushforward presheaf/ sheaf* f_*F of F under f is the presheaf/sheaf over Y given by $f_*F(V) = F(f^{-1}(V))$, f_* is the direct image functor

Definition 2.76 A *ringed space* $(X, \mathcal{O})$ is a topological space X and a sheaf of rings over X, $\mathcal{O}$ is the *structure sheaf*. $(X, \mathcal{O})$ is a *locally ringed space* if each stalk is a local ring

Definition 2.77 A *fine sheaf* F is one with "partition of unities", more precisely, for any open cover, there is a family of endomorphisms such that each endomorphism is zero outside some element of the cover

Example 2.78 The de Rham complex is a resolution of the locally constant sheaf $\mathbb{R}$, although not injective but fine sheaves. Thus the de Rham cohomology coincides with the sheaf cohomology

Definition 2.79 An *acyclic sheaf* F if its higher sheaf cohomologies vanishes

Definition 2.80 A *soft sheaf* F is one that any section over a closed subset can be extended to a global section

Definition 2.81 A *flasque sheaf* or *flabby sheaf* F is one that the restriction maps are surjective

Lemma 2.82 Flasque sheaves are acyclic
Fine sheaves are acyclic

Proof.

□

Definition 2.83 (Godemont resolution) A flasque resolution construction

Definition 2.84 A morphism between ringed spaces is $(X, \mathcal{O}_X) \xrightarrow{(f,\phi)} (Y, \mathcal{O}_Y)$, $X \xrightarrow{f} Y$ is a continuous map, $\mathcal{O}_Y \xrightarrow{\phi} f_*\mathcal{O}_X$ is a morphism of sheaves. A morphism between locally ringed spaces require ϕ is a local ring homomorphism between stalks

Definition 2.85 $(X, \mathcal{O}_X)$ is a ringed space, a sheaf of $\mathcal{O}_X$ modules $\mathcal{F}$ is a sheaf of abelian groups such that $\mathcal{F}(U)$ are $\mathcal{O}_X(U)$ modules that respect restriction, i.e. $(rm)|_V = r|_V \; m|_V$

Definition 2.86 The *pullback sheaf* $f^{-1}F$ of F under f is the sheafification of the presheaf $f^{-1}F(U) = \varinjlim_{V \supseteq f(U) \text{ open}} F(V)$. $(f^{-1}F)_x \cong F_{f(x)}$, thus f^{-1} is exact

Definition 2.87 G is a sheaf of $\mathcal{O}_Y$ modules, $f^*G = f^{-1}G \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$, f^{-1} is exact, f^* is right exact is the inverse image functor, f^* is exact iff f is flat, f^{-1} and f_* are adjoints, i.e. $\mathrm{Hom}(f^{-1}G, F) = \mathrm{Hom}(G, f_*F)$. Suppose $f(U) = V$, then $\mathrm{Hom}(f^{-1}\mathcal{G}(U), \mathcal{F}(U)) = \mathrm{Hom}(\mathcal{G}(f(U)), \mathcal{F}(U)) = \mathrm{Hom}(\mathcal{G}(V), \mathcal{F}(f^{-1}(V))) = \mathrm{Hom}(\mathcal{G}(V), f_*\mathcal{F}(V))$

Definition 2.88 $\mathcal{F}, \mathcal{G}$ are $\mathcal{O}$ modules, then $\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G}$ is the sheafification of the presheaf $(\mathcal{F} \otimes_{\mathcal{O}} \mathcal{G})(U) = \mathcal{F}(U) \otimes_{\mathcal{O}(U)} \mathcal{G}(U)$, $\mathcal{F} \oplus_{\mathcal{O}} \mathcal{G}$ is the sheafification of the presheaf $(\mathcal{F} \oplus_{\mathcal{O}} \mathcal{G})(U) = \mathcal{F}(U) \oplus_{\mathcal{O}(U)} \mathcal{G}(U)$

Definition 2.89 A *quasi-coherent sheaf* $\mathcal{F}$ on ringed space X is a sheaf of $\mathcal{O}$ modules that has a local presentation, i.e. for each $x \in X$ there is a neighborhood $U \ni x$ with

$$\mathcal{O}^{\oplus I}|_U \to \mathcal{O}^{\oplus J}|_U \to \mathcal{F}|_U \to 0$$

exact

Note. Quasi-coherent sheaves are being thought of as "genearalized vector bundles"

Definition 2.90 $\mathcal{F}$ is of *finite type* over X if for each $x \in X$ there is a neighborhood $U \ni x$ such that $\mathcal{O}^n|_U \to \mathcal{F}|_U \to 0$ is exact for some n. Quasi-coherent sheaf $\mathcal{F}$ is a *coherent sheaf* if $\mathcal{F}$ is of finite type over X and for any $\mathcal{O}$-module morphism $\varphi : \mathcal{O}^m|_U \to \mathcal{F}|_U$, $\ker\varphi$ is of finite type over X

Definition 2.91 $\mathcal{F}$ is *locally free* if for each $x \in X$ there is a neighborhood $U \ni x$ such that $\mathcal{F}|_U \cong \mathcal{O}^I|_U$

2.3 Abelian category

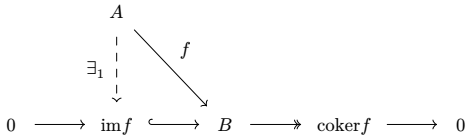
Definition 2.92 $\mathcal{A}$ is an *abelian category* if

1. $\mathrm{Hom}(X, Y)$ are abelian groups, satisfying distribution law
$$f \circ (g + h) = f \circ g + f \circ h, (f + g) \circ h = f \circ h + g \circ h$$
2. Finite biproducts exist
3. Every morphism has a kernel and a cokernel
4. Every monomorphisms is normal and every epimorphisms is conormal

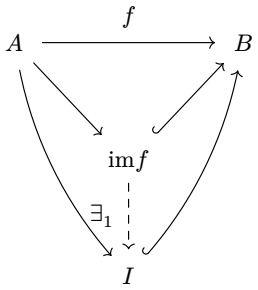
Note. $\mathcal{A}$ is a *preadditive category* or an **Ab**-*category* if it only satisfies 1. An *additive category* if it satisfies 1. 2. A *preabelian category* if it satisfies 1. 2. and 3.
$0 \in \mathrm{Hom}(X, Y)$ is a zero morphism from 1.
$\mathcal{A}$ has a zero object, the empty biproduct
Kernels and cokernels are the equalisers and coequalisers of the morphism $f, 0$ in 3.

Definition 2.93 $F : \mathcal{A} \to \mathcal{B}$ is a functor between abelian cateogories is *additive* if F is are group homomorphisms on $\mathrm{Hom}(F(A), F(B))$. i.e. $F(f + g) = F(f) + F(g)$

Definition 2.94 For a morphism $A \xrightarrow{f} B$, define its image $\mathrm{im}f$ by the following commutative diagram



The image satisfies universal property



Example 2.95 A ring R can be thought of as a preadditive category with a single object and morphisms $r \in R$ The category of left R modules can be thought of as the functor category $[R, Ab]$, where Ab the category of abelian groups

Definition 2.96 Let $\mathcal{A}$ be an abelian category, a ($\mathbb{Z}$-graded) *chain complex* $C_\bullet$ is
$$\cdots \to C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} C_{-1} \to \cdots$$
Such that $\partial_n \circ \partial_{n+1} = 0$, ∂_i are called *boundary maps(differentials)*

We can define chain maps, chain homotopy, boundaries, cycles, and homology groups, and we say the chain complex is exact if each homology groups is zero, the chain complexes form the *category of chain complexes* $Ch\mathcal{A}$

The *homotopy category of chain complexes* often denoted as $K(\mathcal{A})$ is the quotient category with chain maps modulo chain homotopy equivalence as morphisms

a chain map is called a *quasi-isomorphism* if it induces isomorphisms on homology groups

Lemma 2.97 An alternative definition of an exact functor F could be that F preserve exactness, i.e. $F(A) \to F(B) \to F(C)$ is exact for any short exact sequence $A \to B \to C$

Definition 2.98 The *direct sum* $(C \oplus D)_\bullet$ of chain complexes $C_\bullet, D_\bullet$ is

$$\cdots \to C_1 \oplus D_1 \overset{\partial_1^C \oplus \partial_1^D}{\to} C_0 \oplus D_0 \overset{\partial_0^C \oplus \partial_0^D}{\to} C_{-1} \oplus D_{-1} \to \cdots$$

Definition 2.99 A *double complex* $C_{*,*}$ is $\left\{C_{p,q}\right\}_{p,q\in\mathbb{Z}}$ two differentials $\partial' : C_{p,q} \to C_{p-1,q}$, $\partial'' : C_{p,q} \to C_{p,q-1}$ such that $(\partial')^2 = (\partial'')^2 = 0$ and $\partial'\partial'' + \partial''\partial' = 0 (\partial', \partial''$ anticommutes)

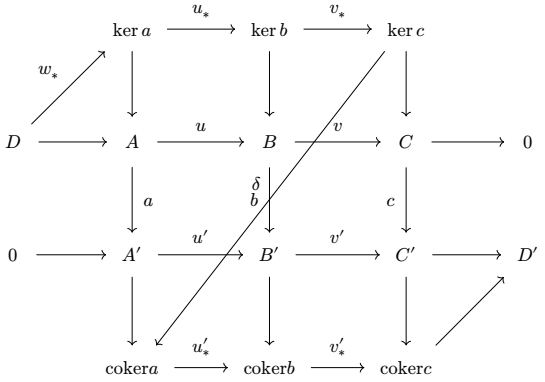
The *total chain complexes* are $(Tot^\oplus)_n = \bigoplus_{p+q=n} C_{p,q}$ and $(Tot^\Pi)_n = \prod_{p+q=n} C_{p,q}$ with $\partial = \partial' + \partial''$

Example 2.100 $C_* \otimes D_*$ is the total complex of double complex $C_{p,q} \coloneqq C_p \otimes D_q$, $\partial' \coloneqq \partial^C \otimes 1$, $\partial'' \coloneqq (-1)^p 1 \otimes \partial^D$

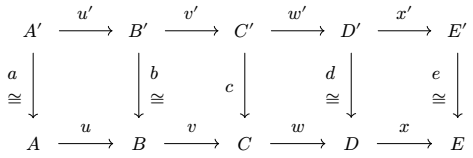
Definition 2.101 A *filtered chain complex* is a filtered object in $Ch\mathcal{A}$

$$\cdots \to F_{p+1}C_\bullet \to F_pC_\bullet \to \cdots \to C_\bullet$$

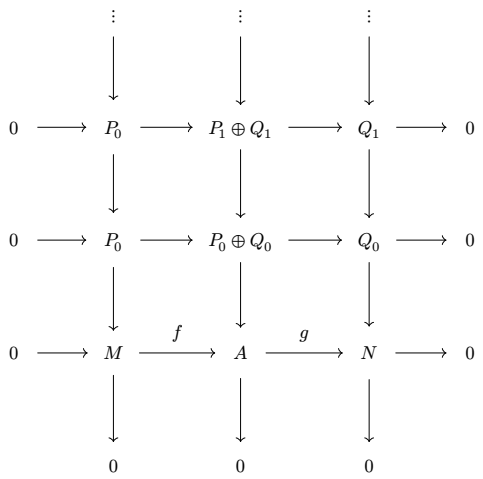
Lemma 2.102 (Snake lemma) Given the following commutative diagram with exact rows, then we have an exact sequence



Lemma 2.103 (Five lemma) If b and d are monic and a is an epi, then c is monic. Dually, if b and d are epis and e is monic, then c is an epi. In particular, if a, b, d and e are iso, then c is also an iso

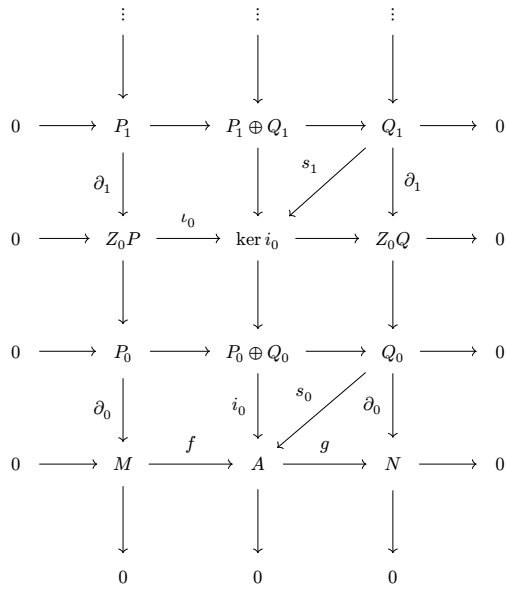


Lemma 2.104 (Horseshoe lemma) Suppose $P_\bullet \overset{\varepsilon}{\to} M$, $Q_\bullet \overset{\eta}{\to} N$ are projective resolutions, then any exact sequence $0 \to M \overset{f}{\to} A \overset{g}{\to} N \to 0$ can be extended into commutative diagram



With $(P \oplus Q)_\bullet$ being a projective resolution, every row and column are exact

Proof. Since $A \xrightarrow{g} N$ is epi and Q_0 is projective, we get $Q_0 \xrightarrow{s_0} A$ such that $gs_0 = \partial_0$ which gives us $P_0 \oplus Q_0 \xrightarrow{(f\partial_0 \ s_0)} A$, by Lemma 2.102, this is epi, and we get an exact sequence $0 \to Z_0P \to \ker i_0 \to Z_0Q \to 0$, similarly, we can construct $Q_1 \xrightarrow{s_1} \ker i_0$, then $P_1 \oplus Q_1 \xrightarrow{(\iota_0\partial_0 \ s_1)} \ker i_0$ is again epi by Lemma 2.102, inductively, we can construct the commutative diagram



□

Lemma 2.105 If $A \xrightarrow{f} B \xrightarrow{g} C$ is a sequence and there is a homomorphism(retraction) $C \xrightarrow{r} B$ such that $rg = 1_B$, then there is an exact sequence $0 \to \mathrm{coker} f \to \mathrm{coker}(gf) \to \mathrm{coker} g \to 0$

Proof. First observe that we have $0 \to \mathrm{im}g/\mathrm{im}(gf) \to C/\mathrm{im}(gf) \to C/\mathrm{im}g \to 0$, $B \to \mathrm{im}g$, $\mathrm{im}f \to \mathrm{im}(gf)$, thus $B/\mathrm{im}f \to \mathrm{im}g/\mathrm{im}(gf)$, since $rg = 1_B$, $B/\mathrm{im}f \cong \mathrm{im}g/\mathrm{im}(gf)$, therefore, $0 \to B/\mathrm{im}f \to C/\mathrm{im}(gf) \to C/\mathrm{im}g \to 0$

□

Lemma 2.106 Suppose $\mathcal{A}$ is abelian category, then $\mathrm{im}f = \ker \mathrm{coker} f = \mathrm{coker} \ker f$

Definition 2.107 Pick $p \in \mathbb{Z}$, define the *translation* of X by p is $X_\bullet[p]$ where $(X_\bullet[p])_n = X_{p+n}$, differential $X_\bullet[p]_n \to X_\bullet[p]_{n-1}$ is given by $(-1)^p\partial$ The *translation functor* $T : Ch(\mathcal{A}) \to Ch(\mathcal{A})$, $X \mapsto X_\bullet[1]$ is an auto morphism of $Ch(\mathcal{A})$

Theorem 2.108 (Acyclic model theorem)[1] *Model* $\mathcal{M} = \{M_j\}$ is a subclass(possibly with repetition) of objects in $\mathcal{C}$, $F, G : \mathcal{C} \to Ch_{\geq 0}$ are functors, $H_n(G(M_j)) = 0$ for any $n \neq 0$, $M_j \in \mathcal{M}$. For any C, there exist $m_j \in F_k M_j$ such that $F_k(C)$ is free with basis $\left\{F_k(\sigma)(m_j) \;\middle|\; M_j \overset{\sigma}{\to} C\right\}$

Theorem 2.109 (Universal coefficient theorem for cohomology) There is an exact sequence

$$0 \to Ext^1(H_{n-1}, A) \to H^n(C; A) \to Hom(H_n, A) \to 0$$

Proof. Since C_n is a free group, so are subgroups B_n, Z_n, exact sequence

$$0 \to Z_n \hookrightarrow C_n \to B_{n-1} \to 0$$

Splits, i.e. we have a splitting homomorphism $B_{n-1} \overset{s}{\to} C_n$, $C_n \cong Z_n \oplus B_{n-1}$, thus exact sequence

$$0 \to H_n = Z_n/B_n \to C_n/B_n \to C_n/Z_n \cong B_{n-1} \to 0$$

Induces exact sequence

$$Hom(B_{n-1}, A) \to Hom(C_n/B_n, A) \to Hom(H_n, A) \to Ext^1(B_{n-1}, A) = 0$$

$Hom(H_n, A)$ is the cokernel of $Hom(B_{n-1}, A) \to Hom(C_n/B_n, A)$

Note that

$$H^n(C; A) = Z^n(C; A)/B^n(C; A) = \frac{\ker(Hom(C_n, A) \to Hom(C_{n+1}, A))}{\mathrm{im}(Hom(C_{n-1}, A) \to Hom(C_n, A))}$$

$C_n \overset{\phi}{\to} A \in Z^n(C; A) \Leftrightarrow \phi\partial = 0 \Leftrightarrow \phi \in Hom(C_n/B_n, A)$, thus $Hom(C_n/B_n, A) \cong Z^n(C; A)$

$$\begin{array}{ccc} C_{n+1} & \xrightarrow{\partial} & C_n \\ & \searrow^{} & \downarrow{\scriptstyle \phi} \\ & & A \end{array}$$

$C_n \overset{\psi}{\to} A \in B^n(C; A) \Leftrightarrow \psi = \phi\partial$ for some $C_{n-1} \overset{\phi}{\to} A \Leftrightarrow \psi = \phi\partial$ for some $Z_{n-1} \overset{\phi}{\to} A$, and since $B^n(C; A) \subseteq Z^n(C; A) \cong Hom(C_n/B_n, A)$, we have $B^n(C; A) \cong \mathrm{im}(Hom(Z_{n-1}, A) \to Hom(C_n/B_n, A))$

$$\begin{array}{ccc} C_n & \xrightarrow{\partial} & C_{n-1} \\ & \searrow^{\psi} & \downarrow{\scriptstyle \phi} \\ & & A \end{array}$$

Exact sequence

$$0 \to B_{n-1} \to Z_{n-1} \to B_{n-1}/Z_{n-1} = H_{n-1} \to 0$$

Induces exact sequence

$$Hom(Z_{n-1}, A) \to Hom(B_{n-1}, A) \to Ext^1(H_{n-1}, A) \to Ext^1(Z_{n-1}, A) = 0$$

$Ext^1(H_{n-1}, A)$ is the cokernel of $Hom(Z_{n-1}, A) \to Hom(B_{n-1}, A)$

Since composition $B_{n-1} \overset{s}{\to} C_n \to C_n/B_n \overset{\partial}{\to} B_{n-1}$ is identity, we have a homomorphism $r : Hom(C_n/B_n, A) \to Hom(B_{n-1}, A)$ induced by $B_{n-1} \to C_n/B_n$ such that composition $Hom(B_{n-1}, A) \to Hom(C_n/B_n, A) \overset{r}{\to} Hom(B_{n-1}, A)$ is identity

Apply Lemma 2.105 to $Hom(Z_{n-1}, A) \to Hom(B_{n-1}, A) \to Hom(C_n/B_n, A)$, we get an exact sequence

[1] Consult Theorem 9.12 of [1] or `https://amathew.wordpress.com/2010/09/11/the-method-of-acyclic-models/`

$$0 \to Ext^1(H_{n-1}, A) \to H^n(C; A) \to Hom(H_n, A) \to 0$$

□

Remark. B_n is not necessarily a direct summand of C_n, a map $B_n \xrightarrow{\phi} A$ may not be possible to extended to $C_n \xrightarrow{\phi} A$, however a map $Z_n \xrightarrow{\phi} A$ can always be extended to $C_n \xrightarrow{\phi} A$

Theorem 2.110 (Algebraic Künneth formula) C, D are free chain complexes, then

$$0 \to \bigoplus_{p+q=n} H_p(C) \otimes H_q(D) \to H_n(C \otimes D) \to \bigoplus_{p+q=n-1} Tor_1(H_p(C), H_q(D)) \to 0$$

Is exact

Proof. If D has trivial differentials, then $H_q(D) = D_q$ is free, hence

$$H_n(C \otimes D) = \bigoplus_{p+q=n} H_p(C \otimes D_q) = \bigoplus_{p+q=n} H_p(C) \otimes D_q = \bigoplus_{p+q=n} H_p(C) \otimes H_q(D)$$

In general, consider exact sequence $0 \to Z \xrightarrow{i} D \xrightarrow{\partial} B[-1] \to 0$, $0 \to B \xrightarrow{i} Z \to H(D) \to 0$, then $0 \to C \otimes Z \to C \otimes D \to C \otimes B[-1] \to 0$ is exact since C_k are free, this gives us long exact sequence

$$\cdots \to H_n(C \otimes Z) \xrightarrow{1\otimes i} H_n(C \otimes D) \xrightarrow{1\otimes\partial} H_n(C \otimes B[-1]) \xrightarrow{1\otimes i} H_{n-1}(C \otimes Z) \to \cdots$$

$Z, B[-1]$ have trivial differentials, hence the connecting homomorphism is just

$$\bigoplus_{p+q=n} H_p(C) \otimes H_q(B[-1]) = \bigoplus_{p+q=n-1} H_p(C) \otimes H_q(B) \xrightarrow{1\otimes i} \bigoplus_{p+q=n-1} H_p(C) \otimes H_q(Z)$$

Then we have

$$0 \to \operatorname{coker}(1 \otimes i) \to H_n(C \otimes D) \to \ker(1 \otimes i) \to 0$$

We also have

$$0 \to Tor_1(H_p(C), H_q(D)) \to H_p(C) \otimes B_q \xrightarrow{1\otimes i} H_p(C) \otimes Z_q \to H_p(C) \otimes H_q(D) \to 0$$

Therefore, we have exact sequence

$$0 \to \bigoplus_{p+q=n} H_p(C) \otimes H_q(D) \to H_n(C \otimes D) \to \bigoplus_{p+q=n-1} Tor_1(H_p(C), H_q(D)) \to 0$$

□

Definition 2.111 A *composition series* of A is a sequence of subobjects

$$A = A_n \supseteq \cdots \supseteq A_1 \supseteq A_0 = 0$$

With *composition factors* H_{i+1}/H_i simple and *composition length* $\ell(A) = n$

Definition 2.112 $F : \mathcal{C} \to \mathcal{D}$ is an additive functor between abelian categories. $C^\bullet$ is *F-acyclic* if $R^i F C^\bullet = 0, \forall i > 0$

Proposition 2.113 F is left exact, $0 \to A \to K^0 \to K^1 \to \cdots$ is a F-acyclic resolution, then $R^i FA = H^i FK^\bullet$

Proof. Consider $0 \to A \to K^0 \to K^1$, get $FA = \ker(FK^0 \to FK^1) = H^0 FK^\bullet$
Consider $0 \to A \to K^1 \to K^0/A \to 0$, get $R^1 FA = \operatorname{coker}(FK^0 \to F(K^0/A))$, and consider $0 \to K^0/A \to K^1 \to K^2$, get $F(K^0/A) = \ker(FK^1 \to FK^2)$, hence $R^1 FA = H^1 FK^\bullet$
For $i > 1$, still have $R^i FA = R^{i-1} F(K^0/A)$, but $0 \to K^0/A \to K^1 \to K^2 \to \cdots$ is now a F-acyclic resolution of K^0/A, thus $R^i FA = R^{i-1} F(K^0/A) = H^i FK^\bullet$

□

Lemma 2.114 $\ell(A)$ is independent of the composition series

Definition 2.115 $f : A^\bullet \to B^\bullet$ is a chain map, the *mapping cone* is $A[1] \oplus B$ with differential

$$\begin{array}{ccccccccc} \longrightarrow & A^{i-1} & \xrightarrow{d} & A^i & \xrightarrow{d} & A^{i+1} & \longrightarrow \\ & \downarrow & & \downarrow & & \downarrow & \\ \longrightarrow & B^{i-1} & \xrightarrow{d} & B^i & \xrightarrow{d} & B^{i+1} & \longrightarrow \end{array}$$

2.4 Spectral sequences

Definition 2.116 Suppose $\mathcal{A}$ is an abelian category, a **spectral sequence** consists of objects $\{E_r\}_{r \geq r_0}$ (r_0 is mostly 0), and morphisms $d_r : E_r \to E_r$ such that $d_r \circ d_r = 0$ and $E_{r+1} \cong H(E_r) = \ker d_r / \mathrm{im} d_r$

Definition 2.117 Suppose $\mathcal{A}$ is an abelian category, an **exact couple** is (D, E, i, j, k)

$$\begin{array}{ccc} D & \xrightarrow{\quad i \quad} & D \\ & {\scriptstyle k} \nwarrow \quad \swarrow {\scriptstyle j} & \\ & E & \end{array}$$

Such that it is exact at each term, define differential $d = jk$, then $d^2 = jkjk = j(kj)k = 0$, we can define the **derived couple**

$$\begin{array}{ccc} D' & \xrightarrow{\quad i' \quad} & D' \\ & {\scriptstyle k'} \nwarrow \quad \swarrow {\scriptstyle j'} & \\ & E' & \end{array}$$

Where $D' = i(D)$, $E' = \ker k / \mathrm{im} j$, $i'(a) = i(a)$, $j'(i(a)) = \overline{j(a)}$, $k'(b) = \overline{k(b)}$, then the derived couple is again an exact couple, thus we can carry this process indefinitely, giving the n-th derived couple $(D^{(n)}, E^{(n)}, i^{(n)}, j^{(n)}, k^{(n)})$

Example 2.118 Suppose $\cdots \subseteq F_{p-1}C_\bullet \subseteq F_pC_\bullet \subseteq \cdots$ is a filteration of chain complex $C_\bullet$(or **filtered chain complex**), exact sequence $0 \to F_{p-1}C_\bullet \to F_pC_\bullet \to (grC_\bullet)_p \to 0$ give a long exact sequence

$$\cdots \to H_n(F_{p-1}C_\bullet) \xrightarrow{i_*} H_n(F_pC_\bullet) \xrightarrow{j_*} H_n(F_pC_\bullet / F_{p-1}C_\bullet) \xrightarrow{k_*} H_{n-1}(F_{p-1}C_\bullet) \to \cdots$$

If we write $D^1_{p,q} \coloneqq H_{p+q}(F_pC_\bullet)$, $E^1_{pq} \coloneqq H_{p+q}(F_pC_\bullet / F_{p-1}C_\bullet)$, then the long exact sequence become

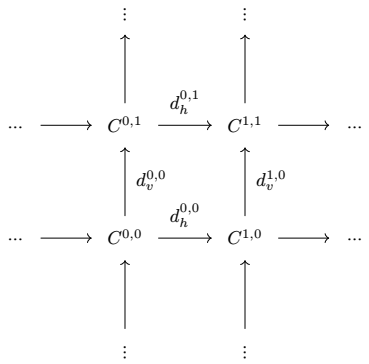
$$\cdots \to D^1_{p,q} \to D^1_{p+1,q-1} \to E^1_{p,q} \to D^1_{p,q-1} \to \cdots$$

Consider $D^1 = \bigoplus D^1_{p,q}$, $E^1 = \bigoplus E^1_{p,q}$, then $(D^1, E^1, i_*, j_*, k_*)$ form an exact couple, note that $d_r : E^r_{p,q} \to E^r_{p-r,q+r-1}$

Remark. $grC_\bullet = \bigoplus_p F_pC_\bullet / F_{p-1}C_\bullet$ is called the **associated graded complex**

If X is a CW complex, we can take $F_pC_\bullet = C_\bullet(X^p)$, here X^p is the p-th skeleton of X

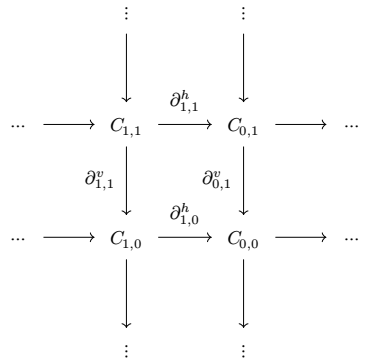
Definition 2.119 A **double cochain complex** $C^{\bullet,\bullet}$ is bigraded with anticommuting differetials d_h, d_v, i.e. $(d_h)^2 = 0$, $(d_v)^2 = 0$, $d_hd_v + d_vd_h = 0$



Define the **total cochain complex** to be $C^n = \bigoplus_{p+q=n} C^{p,q}$, with total differential $d = d_h + d_v$, this is indeed a differential since $d^2 = (d_h + d_v)^2 = (d_h)^2 + d_h d_v + d_v d_h + (d_v)^2 = 0$

We can define the **horizontal filtration** of the total cochain complex $(F_p^h C)^n = \bigoplus_{\substack{k+l=n \\ k \leq p}} C^{k,l}$ and the **vertical filtration** of the total cochain complex $(F_q^h C)^n = \bigoplus_{\substack{k+l=n \\ l \leq q}} C^{k,l}$

A **double chain complex** $C_{\bullet,\bullet}$ is bigraded with anticommuting differetials ∂^h, ∂^v, i.e. $(\partial^h)^2 = 0$, $(\partial^h)^2 = 0$, $\partial^h \partial^v + \partial^v \partial^h = 0$



Define the **total chain complex** to be $C_n = \bigoplus_{p+q=n} C_{p,q}$, with total differential $\partial = \partial_h + \partial_v$

We can define the **horizontal filtration** of the total chain complex $(F_p^h C)_n = \bigoplus_{\substack{k+l=n \\ k \leq p}} C_{k,l}$ and the **vertical filtration** of the total chain complex $(F_q^h C)_n = \bigoplus_{\substack{k+l=n \\ l \leq q}} C_{k,l}$

Remark. If d_h, d_v commutes instead of anticommuting, then $C^{\bullet,\bullet}$ can be viewed as a cochain complex of cochain complexes, the total differential becomes $d^n(c) = d_h^p c + (-1)^p d_v^q c$ for any $c \in C^{p,q}$, this is indeed a differential since

$$\begin{aligned}
d^{n+1} d^n(c) &= d^{n+1}(d_r^p c + (-1)^p d_v^q c) \\
&= d^{n+1} d_h^p c + (-1)^p d^{n+1} d_v^q c \\
&= d_h^{p+1} d_h^p c + (-1)^{p+1} d_v^q d_h^p c + (-1)^p d_h^p d_v^q c + (-1)^{2p} d_v^{q+1} d_v^q c \\
&= (-1)^{p+1} d_v^q d_h^p c + (-1)^p d_h^p d_v^q c \\
&= (-1)^p (d_h^p d_v^q - d_v^q d_h^p) c \\
&= 0
\end{aligned}$$

However, these two types of definitions are equivalent

Proposition 2.120 Let $E_{p,q}^r$ be the spectral sequence corresponds to the horizontal filtration

(1) $E_{p,q}^0 \cong C^{p,q}$

(2) $E_{p,q}^1 \cong H_q(C_{p,\bullet})$

(3) $E^0_{p,q} \cong H_p(H^v_q(C))$

(4) If $C_{p,q}$ vanishes outside the first quadrant, i.e. $C_{p,q} = 0$ for any $p < 0$ or $q < 0$, then the spectral sequence converges to the homology of the total chain complex $E^r_{p,q} \Rightarrow H_{p+q}(C)$, i.e. $E^\infty_{p,q} \cong H_{p+q}(C)$

Proof. **(1)** By definition $E^0_{p,q} \coloneqq (F^h_p C)_{p+q}/(F^h_{p-1}C)_{p+q} \cong C^{p,q}$

(2) $E^1_{p,q} = H_{p+q}(F^h_p C/F^h_{p-1}C) \cong H_{p+q}(C_{p,\bullet})$

(3)

□

Definition 2.121 C is a chain complex, $\cdots \subseteq F_{p-1}C \subseteq F_pC \subseteq F_{p+1}C \subseteq \cdots$ is a filtration of chain complexes. FC is **exhaustive** if $\bigcup F_pC = C$. FC is **Hausdorff** if $\bigcap F_pC = 0$. $\hat{C} = \lim_{\leftarrow} C/F_pC$ is the **completion**. FC is **complete** if $\hat{C} \cong C$, since $C \to \hat{C}$ has kernel $\bigcap F_pC$, hence completeness implies Hausdorff. FC is **bounded below** if $\forall n$, $F_pC_n = 0$ for p small enough. FC is **bounded above** if $\forall n$, $F_pC_n = C_n$ for p big enough. FC is **bounded** if bounded below and above

$F_pH_n(C) = \mathrm{im}(H_n(F_pC) \to H_n(C))$

Definition 2.122 F_nC is a filtered chain complex, $E^0_{pq} = \frac{F_pC_{p+q}}{F_{p-1}C_{p+q}}$ defines a spectral sequence

E^1_{pq} **converges** to H_*C if $E^1_{pq} = H_{p+q}(F_pC/F_{p-1}C) \Rightarrow H_{p+q}C$

Theorem 2.123 $A^r_p = \left\{x \in F_pC \;\middle|\; dx \in F_{p-r}C\right\}$, $Z^r_p = A^r_p + F_{p-1}C$, $B^r_p = dA^{r-1}_{p+r-1} + F_{p-1}C$, $A^r_p \cap F_{p-1}C = A^{r-1}_{p-1}$

$$E^r_p = \frac{Z^r_p}{B^r_p} = \frac{A^r_p + F_{p-1}C}{dA^{r-1}_{p+r-1} + F_{p-1}C} = \frac{\frac{A^r_p + F_{p-1}C}{F_{p-1}C}}{\frac{dA^{r-1}_{p+r-1} + F_{p-1}C}{F_{p-1}C}} = \frac{\frac{A^r_p}{A^{r-1}_{p-1}}}{\frac{dA^{r-1}_{p+r-1}}{dA^{r-1}_{p+r-1} \cap F_{p-1}C}}$$

$$= \frac{\frac{A^r_p}{A^{r-1}_{p-1}}}{\frac{dA^{r-1}_{p+r-1}}{dA^{r-1}_{p+r-1} \cap A^{r-1}_{p-1}}} = \frac{\frac{A^r_p}{A^{r-1}_{p-1}}}{\frac{dA^{r-1}_{p+r-1} + A^{r-1}_{p-1}}{A^{r-1}_{p-1}}} = \frac{A^r_p}{dA^{r-1}_{p+r-1} + A^{r-1}_{p-1}}$$

Lemma 2.124 C and $\hat{C}$ give the same spectral sequence

Theorem 2.125 If F_*C is bounded, then $E^1_{p,q}$ converges to H_*C

If F_*C is bounded below and exhaustive, then $E^1_{p,q}$ converges to H_*C, the convergence is natural

Theorem 2.126 C is a complete filtration, then

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \lim_{\leftarrow}{}^1 H_{n+1}(C/F_pC) & \longrightarrow & H_n(C) & \longrightarrow & H_n(C/F_pC) & \longrightarrow & 0 \\ & & \| & & & & \| & & \\ & & \bigcap F_pH_n(C) & & & & \lim_{\leftarrow} H_n(C)/F_pH_n(C) & & \\ & & & & & & \| & & \\ & & & & & & H_*(C)/\bigcap F_pH_n(C) & & \end{array}$$

Lemma 2.127 F_*C is Hausdorff and exhaustive, then

1. $A^\infty_{pq} = \ker(F_pC_{p+q} \xrightarrow{d} F_pC_{p+q-1})$
2. $F_pH_{p+q}(C) \cong A^\infty/\bigcup dA^r_{p+r,q-r+1}$
3. The subgroup $e^\infty_{pq} = A^\infty_{pq} + B^\infty_{pq}$ is isomorphic to $F_pH_{p+q}(C)/F_{p-1}H_{p+q}(C)$

Proof.

$$\begin{aligned}\frac{F_pH_{p+q}(C)}{F_{p-1}H_{p+q}(C)} &= \frac{\ker(F_pC_{p+q}\to F_pC_{p+q-1})/\mathrm{im}(C_{p+q+1}\to C_{p+q})}{\ker(F_{p-1}C_{p+q}\to F_{p-1}C_{p+q-1})/\mathrm{im}(C_{p+q+1}\to C_{p+q})}\\ &= \frac{A_p^\infty/dC_{p+q}\cap F_pC_{p+q}}{A_{p-1}^\infty/dC_{p+q}\cap F_{p-1}C_{p+q}}\\ &= \frac{A_p^\infty}{A_{p-1}^\infty+\bigcup_{r\geq 0}dA_{p+r}^r}\\ &= \frac{A_p^\infty}{A_{p-1}^\infty+dA_{p+\infty}^\infty}=E_{p,q}^\infty\end{aligned}$$

Note that here $\bigcup_{r\geq 0}A_{p-1+r}^r\subseteq\bigcup_{r\geq 0}A_{p+r}^r$

□

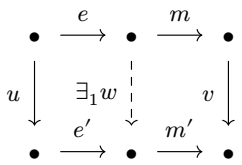
2.5 Factorization system

Definition 2.128 A **factorization system** $(\mathbf{E},\mathbf{M})$ for category $\mathcal{C}$ is two classes of morphisms such that

1. Any morphism f can be decomposed as $f=me$, $m\in\mathbf{M}$, $e\in\mathbf{E}$

2. $\mathbf{E},\mathbf{M}$ are closed under composition and contain all isomorphisms

3. Factorization is functorial, i.e. for any u,v such that $vme=m'e'u$, there exists a unique w such that the following diagram commutes



Example 2.129 $\mathbf{E},\mathbf{M}$ being epi and mono in *Set* is a factorization system

Definition 2.130 A **weak factorization system** $(\mathbf{E},\mathbf{M})$ for category $\mathcal{C}$ is two classes of morphisms such that

1. Any morphism f can be decomposed as $f=me$, $m\in\mathbf{M}$, $e\in\mathbf{E}$

2. $\mathbf{E}$ are exactly those morphisms having left lifting property for all morphisms in $\mathbf{M}$

3. $\mathbf{M}$ are exactly those morphisms having right lifting property for all morphisms in $\mathbf{E}$



3 Linear algebra

3.1 Vector spaces

Definition 3.1 $C \subseteq V$ is **convex** if $tC + (1-t)C \subseteq C$ for $0 \leq t \leq 1$. C is **strictly convex** if $tC + (1-t)C \subsetneq C$ for $0 < t < 1$

Definition 3.2 V is a vector space of dimension n, a q **flag** is

$$0 = V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_q = V$$

A complete flag is an n flag

Definition 3.3 V is an n dimensional $\mathbb{F}$-vector, $\mathcal{V} = \{v_1, \cdots, v_n\}$ is a basis of V, $\{e_1, \cdots, e_n\}$ is the standard basis of $\mathbb{F}^n$, $V \to \mathbb{F}^n, v_i \mapsto e_i$ is an isomorphism, the matrix of $\phi \in \mathrm{End}(V)$ with respect to $\mathcal{V}$ is

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$$

Such that $\phi(\mathcal{V}) = \mathcal{V}A$. If $v = \mathcal{V}c$, then $\phi(v) = \mathcal{V}Ac$

3.2 Matrices

Definition 3.4 $I, J \subseteq \{1, \cdots, n\}$, the *submatrix* A_{IJ} of A is the matrix with entries $\{a_{ij} | i \in I, j \in J\}$. The *principal submatrix* are matrices A_{II}

Definition 3.5 E_{ij} is the matrix with 1 on the (i,j)-th entry and otherwise zeros, then $E_{ij}E_{kl} = \delta_{jk}E_{il}$

Elementary matrices are single row operations, i.e.

$$e_{ij}(r) = \begin{pmatrix} 1 & & & & \\ & \ddots & & r & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}$$

with r on the (i,j)-th entry

$$s_{ij} = \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 0 & 1 & & \\ & & & \ddots & & \\ & & 1 & 0 & & \\ & & & & \ddots & \\ & & & & & 1 \end{pmatrix}$$

and

$$d_i(r) = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & r & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}$$

We have $e_{ij}(-r) = e_{ij}(r)^{-1}$ and

$$
\begin{aligned}
e_{ij}(r)e_{ij}(s) &= e_{ij}(r+s) \\
[e_{ij}(r), e_{kl}(s)] &= I + rs\delta_{jk}E_{il} - sr\delta_{li}E_{kj} + \delta_{jk}\delta_{li}(srsE_{kl} - rsrE_{ij}) + rsrs\delta_{jk}\delta_{li}E_{il} \\
&= \begin{cases} I & i \neq l, j \neq k \\ e_{il}(rs) & i = l, j \neq k \\ e_{kj}(-sr) & i \neq l, j = k \\ * & i = l, j = k \end{cases}
\end{aligned}
$$

Steinberg relations

Definition 3.6 $E(n,R) \subseteq SL(n,G)$ is the subgroup generated by elementary matrices of determinant 1. $E(R) = \bigcup E(n,R)$

Lemma 3.7 $SL(n,F) = E(n,F)$

Lemma 3.8 $[E(n,R), E(n,R)] = E(n,R)$ if $n \geq 3$

Proof. For distinct i,j,k, $e_{ij}(r) = [e_{ik}(r), e_{kj}(1)]$

□

Theorem 3.9 (Whitehead's lemma) $[GL(R), GL(R)] = E(R)$, hence $K_1(R) = GL(R)/E(R)$

Proof. Since

$$
e_{12}(1)e_{21}(-1)e_{12}(1) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}
$$

and

$$
\begin{pmatrix} g & \\ & g^{-1} \end{pmatrix} = \begin{pmatrix} 1 & g \\ & 1 \end{pmatrix} \begin{pmatrix} 1 & \\ -g^{-1} & 1 \end{pmatrix} \begin{pmatrix} 1 & g \\ & 1 \end{pmatrix} \begin{pmatrix} & -1 \\ 1 & \end{pmatrix}
$$

We know

$$
[g,h] = \begin{pmatrix} g & \\ & g^{-1} \end{pmatrix} \begin{pmatrix} h & \\ & h^{-1} \end{pmatrix} \begin{pmatrix} (hg)^{-1} & \\ & hg \end{pmatrix} \in E(R)
$$

□

Definition 3.10 The *Kronecker product* of matrices

$$
A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}, B = \begin{pmatrix} b_{11} & \cdots & b_{1p} \\ \vdots & \ddots & \vdots \\ a_{np} & \cdots & b_{np} \end{pmatrix}
$$

is

$$
A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}
$$

Definition 3.11 $\mathrm{tr}(A^*B)$ defines the *Frobenius inner product* over $M(n,\mathbb{C})$

3.3 Eigenspace decomposition

Proposition 3.12 $T \in \mathrm{Hom}_{\mathbb{F}}(V,V)$ is a linear operator, and $V = \bigoplus_i V_i$, where V_i are T invariant spaces, denote $T|_{V_i}$ as $T - i$, then $ch_T(t) = \prod_i ch_{T_i}(t)$, and $m_T(t) = lcm_i\ m_{T_i}(t)$

Definition 3.13 $T \in \mathrm{Hom}_{\mathbb{F}}(V,V)$. $\lambda \in F$ is an **eigenvalue** if $Tv = \lambda v$ has nontrivial solution, $v \in V$ is a **generalized eigenvector** of rank m of T corresponding to eigenvalue λ if $(T - \lambda 1_V)^m v = 0, (T - \lambda 1_V)^{m-1} v \neq 0$ for some $m \geq 1$, and let V_λ be the subspace of all such generalized eigenvectors, called

generalized eigenspace, notice if V is of finite dimensional, then $V_\lambda = \ker(T - \lambda 1_V)^m$ for some m with m being smallest, suppose $\dim V_\lambda = d$, then the characteristic polynomial of $T|_{V_\lambda}$ is $(t-\lambda)^d$, and the minimal polynomial of $T|_{V_\lambda}$ is $(t-\lambda)^m$

Proposition 3.14 $\overline{F} = F$, finitely dimensional F vector space V can be decomposed into the direct sum of generalized eigenspaces $V = \bigoplus_\lambda V_\lambda$

Definition 3.15 $T \in \mathrm{Hom}_{\mathbb{F}}(V, V)$ give V an $F[x]$ module with $x \cdot v = Tv$, $W \leq V$ be a subspace, W is called T **invariant** if $TW \subseteq W$, or rather W is an $F[x]$ submodule

Definition 3.16 An linear operator $T \in \mathrm{Hom}_{\mathbb{F}}(V, V)$ is called **semisimple** if V is a semisimple $\mathbb{F}[x]$ submodule

Proposition 3.17 Let $T \in \mathrm{Hom}_{\mathbb{F}}(V, V)$ be a linear operator with $\overline{\mathbb{F}} = \mathbb{F}$, then T is semisimple $\Leftrightarrow$ T is diagonalizable

Proof. Since $\overline{\mathbb{F}} = \mathbb{F}$ and T is semisimple, V can be decomposed as a direct sum of eigenspaces of T, thus T is diagonalizable, conversely, if T is diagonalizable, and $TW \subseteq W$, let V_λ be the eigenspaces of T, denote $W_\lambda = W \cap V_\lambda$, and $W' = \bigoplus_\lambda W'_\lambda$, since $T|_{V_\lambda} = \lambda 1_{V_\lambda}$, we can find $W'_\lambda \leq V_\lambda$ such that $V_\lambda = W_\lambda \oplus W'_\lambda$, and of course $TW'_\lambda \subseteq W'_\lambda$ which implies $TW' \subseteq W'$, then we have $V = \bigoplus_\lambda V_\lambda = \bigoplus_\lambda W_\lambda \oplus W'_\lambda = \bigoplus_\lambda W_\lambda \bigoplus \bigoplus_\lambda W'_\lambda = W \oplus W'$

□

Definition 3.18 An linear operator $T \in \mathrm{Hom}_{\mathbb{F}}(V, V)$ is called nilpotent if $T^k = 0$ for some k, T is called unipotent if $T - 1_V$ is nilpotent

Definition 3.19 (Jordan-Chevalley decomposition) Jordan-Chevalley decomposition of a linear operator $T \in \mathrm{Hom}_{\mathbb{F}}(V, V)$ is $T = T_s + T_n$, where T_s is semisimple, T_n is nilpotent and $[T_s, T_n] = 0$

Theorem 3.20 If V is a finite dimensional $\mathbb{F}$ vector space with $\mathbb{F}$ being a perfect field, and $T \in \mathrm{Hom}_{\mathbb{F}}(V, V)$ is a linear operator, then Jordan-Chevalley decomposition always exist, additionally, there exist polynomials $p(t), q(t)$ with no constant terms and $T_s = p(T), T_n = q(T)$, moreover, the decomposition is unique

Proof. First consider $\overline{\mathbb{F}} = \mathbb{F}$, by Proposition 3.14, V can be decomposed into the direct sum of generalized eigenspaces $V = \bigoplus_i V_{\lambda_i}$, where $V_{\lambda_i} = \ker(T - \lambda_i 1_V)^{m_i}$ with being m_i being the least and $\dim V_{\lambda_i} = d_i$, define $T_s \in \mathrm{Hom}_{\mathbb{F}}(V, V)$ such that $T_s|_{V_{\lambda_i}} = \lambda_i 1_{V_{\lambda_i}}$ and $T_n = T - T_s$, thus T_s is diagonalizable(semisimple), T_n is nilpotent, $ch_T(t) = \prod_i (t - \lambda_i)^{d_i}$, by Theorem 7.10, there exists polynomial $p(t)$ such that $p(t) \equiv 0 \mod t$, $p(t) \equiv \lambda_i \mod (t - \lambda_i)^{d_i}$, and let $q(t) = t - p(t)$, then p, q doesn't have constant terms and $T_s = p(T), T_n = q(T)$. For uniqueness, suppose $T = T_s + T_n = T'_s + T'_n$ are two such decompositions, then $T_s - T'_s = T'_n - T_n$ will be nilpotent which implies $T_s - T'_s = 0$

□

3.4 Bilinear form

Definition 3.21 A *symplectic form* ω is bilinear form such that $\omega(u, v) = X^T J Y$, here $J = \begin{pmatrix} & -I \\ I & \end{pmatrix}$, in other words, there are $u_1, \cdots, u_n, v_1, \cdots, v_n$ such that $\omega(u_i, v_j) = -\omega(v_j, u_i) = \delta_{ij}$, $\omega(u_i, u_j) = \omega(v_i, v_j) = 0$

Remark. $\omega(x \oplus \xi, y \oplus \eta) = \eta(x) - \xi(y)$ on $V \oplus V^*$ is a symplectic form. Conversely, such a V is called a Lagrangian subspace, a polarization

4 Abstract algebra

Definition 4.1 (Rings) R is an abelian group with addition $+$ and additive identity 0, a monoid with multiplication $\cdot$ and multiplicative identity 1, and distributive, $a \cdot (b + c) = a \cdot b + a \cdot c$, $(a + b) \cdot c = a \cdot c + b \cdot c$

Definition 4.2 Ring R is **commutative** if $ab = ba$

Definition 4.3 u is a **unit** if there exists $v \in R$ such that $uv = vu = 1$. The set of units $R^\times$ is a multiplicative group

Definition 4.4 A **semiring** or **rig** is a ring without negatives

Definition 4.5 A **rng** is ring without identity

5 Group Theory

Definition 5.1 A **semigroup** is a semicategory with a single object. A **monoid** M is a category with a single object. A **group** is a monoid with all morphisms invertible. A **groupoid** is a category with all morphisms invertible

A G **set** is a functor from G to the category of sets. Equivalently, a **left group action** is $G \times X \to X$, $(g, x) \mapsto g \cdot x$ satisfying $1 \cdot x = x$, $g \cdot (h \cdot x) = (gh \cdot x)$, a right G action is functor G^{op} to the category of sets. A G **space** is a functor from G to the category of topological spaces

An **equivariant map** of G spaces is a natural transformation $f : X \to Y$

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle g}\downarrow & & {\scriptstyle g}\downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

Definition 5.2 The **center** of G is $Z(G) = \{z \in G \mid gz = zg, \forall g \in G\}$. $Z(G_1 \times G_2) = Z(G_1) \times Z(G_2)$

Definition 5.3 Inner automorphism group of G is $Inn(G) \leq Aut(G)$ consists of conjugations $x \mapsto gxg^{-1}$. **Outer automorphism group** is $Out(G) = Aut(G)/Inn(G)$

Definition 5.4 G is **perfect** if $[G, G] = G$

Definition 5.5 A group action is **trivial** if $g \cdot x = x$. A group action is **free** if $g \cdot x = h \cdot x$ for some x implies $g = h$, or equivalently. A group action is **transitive** if $G \cdot x = X$. A free and transitive action is also called **regular**. A group action is **faithful** if $\forall x \in X, G \cdot x \neq x$

A **homogeneous space** is a G space with G acting transitively

Definition 5.6 A **torsor** P is a set with an action $G \times P \to P, (g, p) \mapsto gp$ such that this action is free and transitive

When chosen an element $e \in P$, automatically you are giving it a group structure with e as identity, notice $e : G \to P, g \mapsto ge$ is a bijection, suppose $g_p e = p$, then $g_e = 1$, the group structure could be given as follows, $P \times P \to P, (p, q) \mapsto g_p g_q e$, and the inverse of p is $g_p^{-1} e$. i.e. a torsor is a group forgetting its identity

Definition 5.7 Let X, Y be G sets, the we can give Y^X a G set structure by $(gf)(x) = gf(g^{-1}x)$, it is easy to check that a map $f : X \to Y$ is equivariant iff f is fixed under the G action

Specially, if G acts on Y trivially, we have $(gf)(x) = f(g^{-1}x)$

Definition 5.8 $H \leq G$ is a subgroup, $N \leq G$ is a normal subgroup, $G = NH = N \rtimes H$ or $HN = H \ltimes N$ is the **semidirect product** and every $g \in G$ can be uniquely written as nh or hn

$$(nh)(n'h') = n(hn'h^{-1})hh' \text{ or } (hn)(h'n') = hh'(h'^{-1}n'h')n'$$

Theorem 5.9 (Jordan-Hölder theorem) Composition series is unique up to reordering

Definition 5.10 A **group representation** V is a $\mathbb{F}G$ module

Definition 5.11 Let (ρ, V) be a group representation of finite group G, $W \leq V$ is called G invariant if $GW \subseteq W$, namely, W is $\mathbb{F}G$ submodule, then we get a subrepresentation on $(\rho|_W, W)$ of G, with $\rho|_W(g) \coloneqq \rho(g)|_W$, if the only G invariant subspace of W are 0 and V, we say ρ is irreducible

Definition 5.12 A group representation (ρ, V) is completely reducible(semisimple) if $V = V_1 \oplus \cdots \oplus V_n$, where $(\rho|_{V_i}, V_i)$ are irreducible subrepresentations, namely, V is the direct sum of simple $\mathbb{F}G$ modules

Definition 5.13 Let V be a complex vector space with Hermitian form $(,)$ finite group representation (ρ, V) of G is called unitary if $(\rho(g)v, \rho(g)w) = (v, w), \forall g \in G, v, w \in V$

Proposition 5.14 Let (ρ, V) be a unitary representation of G, (ρ, V) is completely reducible

Proof. W is G invariant $\Rightarrow W^\perp$ is G invariant

□

Proposition 5.15 Let V be a complex vector space, (ρ, V) is a representation, then there exists a positive definite Hermitian form on V such that (ρ, V) become a unitary representation

Corollary 5.16 Let V be a complex vector space, a finite group representation (ρ, V) is always completely reducible

Definition 5.17 Let (ρ, V) be a representation of G, then the dual representation (ρ^*, V^*) is defined as $\rho^*(g) \coloneqq \rho(g^{-1})^{-T}$. The dual representation is chosen so that the following diagram commutes

$$\begin{array}{ccc} & \overset{\rho(g)\otimes\rho^*(g)}{} & \\ V\otimes V^* & \longrightarrow & V\otimes V^* \\ & \searrow & \downarrow \\ & & k \end{array}$$

Definition 5.18 $H \subseteq G$, (V, π) is a H representation, the **induced representation** is $\mathbb{F}G \otimes_{\mathbb{F}H} V$. Suppose $g_1, \cdots, g_n$ is a set of representatives of left cosets in G/H, v_j is a basis for V, then $g_i \otimes v_j$ form a basis for $\mathbb{F}G \otimes_{\mathbb{F}H} V$. If $gg_i = g_j h$ for some $h \in H$, then $g(g_i \otimes v) = gg_i \otimes v = g_j h \otimes v = g_j \otimes hv$

Definition 5.19 If X is a right G space, Y is a left G space, $X \times_G Y$ is $X \times Y/\sim$, $(xg, y) \sim (x, gy)$. Equivalently, $X \times_G Y = X \times Y/G$ with left action $g(x, y) = (xg^{-1}, gy)$ or right action $(x, y)g = (xg, g^{-1}y)$

If X, Y are left G spaces, $X \times_G Y = X \times Y/G$ with left action $g(x, y) = (gx, gy)$

If X, Y are right G spaces, $X \times_G Y = X \times Y/G$ with right action $(x, y)g = (xg, yg)$

Theorem 5.20 (Sylow's theorem) p is a prime, a p-group is a group consists of elements of order p -th power, a maximal one is a **Sylow p-subgroup** P of G which always exists by Zorn's Lemma 1.2

1. If $|G| = p^n m$, $p \nmid m$, then $|P| = p^n$
2. Any subgroup of a Sylow p subgroup is subconjugate to some other Sylow p subgroup
3. $n_p = [G : N_G(P)]$, if the cojugacy class of P is of order $n_p < \infty$, then $n_p \equiv 1 \mod p$

Definition 5.21 An *isogeny* is a morphism that is surjective with finite kernel

Definition 5.22 A *central extension* of G is $1 \to K \to C \to G \to 1$, here $K \leq Z(C)$

Definition 5.23 A *stem extension* of G is $1 \to K \to C \to G \to 1$, here $K \leq Z(C) \cap C'$

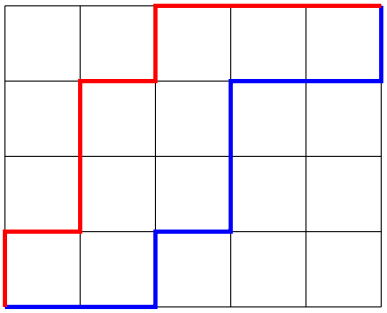
Definition 5.24 Denote $[n] = \{1, \cdots, n\}$, a **permutation** $[n] \overset{\sigma}{\to} [n]$ is a bijection, equivalently write $\sigma = \begin{pmatrix} 1 & \cdots & n \\ \sigma(1) & \cdots & \sigma(n) \end{pmatrix}$. $S_n = \mathrm{Aut}([n])$ is the permutation group

The **length** of $\sigma \in S_n$ is $|\sigma|$ is the number of $\sigma(i) < \sigma(j)$ while $i > j$

Definition 5.25 Just like shuffling a deck of cards, $\sigma \in S_n$ is a $(p_1, \cdots, p_m)$**-shuffle**, $p_1 + \cdots + p_m = n$ if $\sigma(i) < \sigma(i+1)$ for $p_1 + \cdots + p_k + 1 \leq i \leq p_1 + \cdots + p_{k+1}$

A (p, q) shuffle can be represented by a path going only right or up from the lower left corner to the upper right corner in a $(p+1) \times (q+1)$ grid, $|\sigma|$ happen to be the number squares under the path

Example 5.26 ($(5, 4)$ shuffles in S_9) The red one is $\begin{pmatrix} 1&2&3&4&5&6&7&8&9 \\ 2&5&7&8&9&1&3&4&6 \end{pmatrix}$. The blue one is $\begin{pmatrix} 1&2&3&4&5&6&7&8&9 \\ 1&2&4&7&8&3&5&6&9 \end{pmatrix}$



Definition 5.27 G is a *Coxeter group* if it has presentation

$$\left\langle r_1, \cdots, r_n \mid (r_i r_j)^{m_{ij}} = 1 \right\rangle$$

$m_{ii} = 1$(we should think of r_i's as reflections), and $2 \leq m_{ij} \leq \infty, \forall i \neq j (m_{ij} = 2$ means r_i, r_j commutes). $S = \{r_1, \cdots, r_n\}$ is called a *Coxeter system*, G doesn't determine S. The *Coxeter matrix* $M = (m_{ij})$ is symmetric, the corresponding *Schläfli matrix* is $C_{ij} = -2\cos(\pi/m_{ij})$

Definition 5.28 The *Coxeter element* is the product of all simple reflections s_i's, different order in multiplication give conjugation, the *Coxeter number* is the order of the Coxeter element

Definition 5.29 The *Coxeter diagram* consists of n nodes for r_i's, there is an edge numbered m_{ij} between i and j if $m_{ij} \geq 3$(number $m_{ij} = 3$ can just be omitted)

Theorem 5.30 Finite Coxeter groups are classified by Coxeter diagram

Proof.

□

Definition 5.31 (Grothendieck group) The **Grothendieck group** of a commutative monoid M is the abelian group K and $i : M \to K$ satisfying universal property

$$\begin{array}{ccc} M & & \\ {\scriptstyle i}\downarrow & \searrow^{f} & \\ K & \overset{\exists_1 g}{\dashrightarrow} & A \end{array}$$

For any abelian group A

Construction 5.32

$$K = M \times M/\sim, M \xrightarrow{i} K, m \mapsto [m - 0]$$

Abelian group $M \times M \cong \{[m-n] | m, n \in M\}$ is the set of formal differences with addition

$$[m_1 - n_1] + [m_2 - n_2] = [(m_1 + m_2) - (n_1 + n_2)]$$

$[0 - 0]$ is the identity and $[n - m]$ is the inverse to $[m - n]$

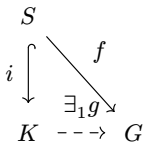
$$[m_1 - n_1] \sim [m_2 - n_2] \text{ if } \ m_1 + n_2 + m = m_2 + n_1 + m \ \text{ for some } \ m \in M$$

Construction 5.33 (Grothedieck completion)

$$K = F(M)/\sim, m +' n \sim (m + n)$$

$F(M)$ is the free abelian group generated by M with addition $+'$

Definition 5.34 The **Grothendieck group** of a semigroup S is a group K and $i : S \to K$ satisfying the universal property



Where G is a group

Construction 5.35

$$K = F(S)/\sim, m *' n \sim (m * n)$$

$F(S)$ is the free group generated by S with multiplication $*'$

Definition 5.36 A is a unital K-algebra. The *standard complex* is

$$\cdots \to A \otimes A \otimes A \to A \otimes A \to A \to 0$$

with differentials

$$d_n(a_0 \otimes \cdots \otimes a_{n+1}) = \sum_{i=0}^{n} (-1)^i a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{n+1}$$

Define $s_i : S^{i+2}(A) \to S^{i+3}(A)$, $a \mapsto 1 \otimes a$, $t_{i+1} : S^{i+3}(A) \to S^{i+2}(A)$, $\lambda \otimes a \mapsto \lambda a$, then $ts = 1$, hence s is a monomorphism. d_n's are uniquely determined by $d_0(\lambda \otimes \mu) = \lambda\mu$, $ds + sd = 1$, i.e. the identity map is null homotopic, hence the standard complex is a free A bimodule resolution of A. If we denote $a_0 \otimes \cdots \otimes a_{n+1}$ as $a_0[a_1|\cdots|a_n]a_{n+1}$, then

$$d_n[a_1|\cdots|a_n] = a_1[a_2|\cdots|a_n] + \sum_{i=1}^{n-1}(-1)^i[a_1|\cdots|a_i a_{i+1}|\cdots|a_n] + (-1)^n[a_1|\cdots|a_{n-1}]a_n$$

Example 5.37 $d[a] = a[] - []a$, $d[a|b] = a[] - [ab] + []b$

Definition 5.38 The *normalized standard complex* is with $A \otimes A/K \otimes \cdots \otimes A/K \otimes A$

Definition 5.39 R is a commutative ring, $C_*(G)$ is the tuple complex, equivalently, $B_*(G)$ is the bar resolution, $\bar{B}_*(G) = B_*(G) \otimes_{R[G]} R$. $B_*(G)$ is a differential graded algebra with shuffled product

Definition 5.40 R is a commutative ring, M is right $R[G]$ module, $C_*(G)$ is the tuple complex, equivalently, $B_*(G)$ is the bar complex, $\bar{B}_*(G) = B_*(G) \otimes_{R[G]} R$, thus

$$M \otimes_{R[G]} B_*(G) \cong M \otimes_R R \otimes_{R[G]} B_*(G) \cong M \otimes_R \bar{B}_*(G)$$

Group homology with coefficients in M is

$$\begin{aligned} H_k(G; M) &= H_k(M \otimes_{R[G]} C_*(G)) \\ &= H_k(M \otimes_{R[G]} B_*(G)) \\ &= H_k(M \otimes_R \bar{B}_*(G)) \\ &= \mathrm{Tor}_k^{R[G]}(M, R) \end{aligned}$$

The differential of $M \otimes_{R[G]} B_*(G)$ is given by

$$\begin{aligned} \partial(m \otimes [g_1|\cdots|g_n]) &= \partial(m \otimes (1, g_1, g_1g_2, \cdots, g_1\cdots g_n)) \\ &= mg_1 \otimes [g_2|\cdots|g_n] + \sum_{i=1}^{n-1}(-1)^i m \otimes [g_1|\cdots|g_i g_{i+1}|\cdots|g_n] \\ &+(-1)^n m \otimes [g_1|\cdots|g_{n-1}] \end{aligned}$$

$H_0(G; M) = M \otimes_{R[G]} R = M_G$. Write $H_k(G)$ for $H_k(G; \mathbb{Z})$

Group cohomology with coefficients in left $R[g]$ module M is

$$\begin{aligned} H^k(G;M) &= H_k(\mathrm{Hom}_{R[G]}(B_*(G),M)) \\ &= \mathrm{Ext}^k_{R[G]}(R,M) \end{aligned}$$

The differential of $\mathrm{Hom}_{R[G]}(B_*(G),M)$ is given by

$$\begin{aligned} (d\phi)[g_1|\cdots|g_{n+1}] &= (d\phi)(1,g_1,g_1g_2,\cdots,g_1\cdots g_{n+1}) \\ &= \phi(\partial(1,g_1,g_1g_2,\cdots,g_1\cdots g_{n+1})) \\ &= g_1\phi[g_2|\cdots|g_{n+1}] + \sum_{i=1}^{n}(-1)^i\phi[g_1|\cdots|g_ig_{i+1}|\cdots|g_{n+1}] \\ &+(-1)^{n+1}\phi[g_1|\cdots|g_n] \end{aligned}$$

$H^0(G;M) = \mathrm{Hom}_{R[G]}(R,M) = M^G$. Write $H^k(G)$ for $H^k(G;\mathbb{Z})$

Remark. A right $R[G]$ module M can be viewed as a left $R[G]$ module and vice versa via $g^{-1}m = mg$

Definition 5.41 The *augmentation ideal* is the kernel of the augmentation map $R[G] \to R$

Definition 5.42 A *derivation* or *crossed homomorphism* is $D : G \to M$ such that $D(gh) = gD(h) + D(g)$, this can be identified with $Z^1(G;M)$. $D_m(g) = gm - m$ are *principal derivations*, this can be identified with $B^1(G;M)$

6 Modules

Definition 6.1 Ring R is a preaddtive categor y with a single object $\bullet$
A *left R-module* an additive functor from R to the category of abelian groups, $\bullet$ is mapped to M which is an abelian group, r are mapped to endomorphisms of M which induces a left group action $R \times M \to M$, $(r, m) \mapsto rm$

1. $1m = m$
2. $(rs)m = r(sm)$
3. $r(m + n) = rm + rn$
4. $(r + s)m = rm + sm$

1 and 2 are given by functoriality, 3 is given by the linearity of r, 4 is given by additivity
A *right R-module* an additive functor from R^{op} to the category of abelian groups, inducing right group action $M \times R \to M$, $(m, r) \mapsto mr$

1. $m = m1$
2. $m(rs) = (mr)s$
3. $(m + n)r = mr + nr$
4. $m(r + s) = mr + ms$

1 and 2 are given by functoriality, 3 is given by the linearity of r, 4 is given by additivity

Definition 6.2 A morphism between left R-modules $\phi : M \to N$ is natural transformation

1. $\phi(m_1 + m_2) = \phi(m_1) + \phi(m_2)$
2. $\phi(rm) = r\phi(m)$

$$\begin{array}{ccc} M & \xrightarrow{r} & M \\ {\scriptstyle\phi}\downarrow & & \downarrow{\scriptstyle\phi} \\ N & \xrightarrow{r} & N \end{array}$$

1 is given by the linearity of ϕ, 2 is given by naturality

Definition 6.3 $X \subseteq M$ is *linearly independent* if for any $x_1, \cdots, x_n \in X$

$$r_1x_1 + \cdots + r_nx_n = 0 \Rightarrow r_i = 0$$

Definition 6.4 The submodule generated by $X \subseteq M$ is Span X, the *span* of X

Definition 6.5 $X \subseteq M$ is a *basis* of M if X is a linearly independent spanning set. M is a free R module on X if X is a basis of M

Note. There is no well-defined dimension for free R modules in general, examplified in Example 34.4

Definition 6.6 M is an right R-module, N is a left R-module and G is an abelian group, a map $\phi : M \times N \to G$ is called an R balanced product if ϕ is bilinear and $\phi(mr, n) = \phi(m, rn)$, we can define tensor product $M \otimes_R N$ is an abelian group satisfying the universal property

$$\begin{array}{ccc} M \times N & \xrightarrow{\otimes} & M \otimes_R N \\ & \searrow^{f} & \downarrow{\scriptstyle\exists_1 \tilde{f}} \\ & & G \end{array}$$

Here f is an R balanced product and $\tilde{f}$ is an abelian group homomorphism

A concrete construction would be $F(M \times N)/\sim$, where $(m+m',n) \sim (m,n)+(m',n)$, $(m,n+n') \sim (m,n)+(m,n')$, $(mr,n) \sim (m,rn)$

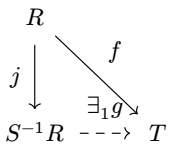
Remark. $r(m \otimes n) = mr \otimes n = m \otimes rn$ is called associativity

Definition 6.7 Module M is *semisimple* or *completely reducible* if it is the direct sum of simple submodules. Ring R is semisimple if it is a semisimple R module

Theorem 6.8 Tensor product is right exact for R modules

Definition 6.9 M is a *flat* R module if $\mathrm{Tor}_1^R(M,-)$ is exact

Definition 6.10 $S \subseteq R$ is *multiplicatively closed* if $1 = s^0 \in S$ and $rs \in S, \forall r,s \in S$, we can define localization $S^{-1}R$ satisfying universal property



Here $f(S) \subseteq T^\times$

Concrete construction: $S^{-1}R \coloneqq R \times S/\sim$, $(r,s) \sim (r',s')$ if there exists $t \in S$ such that $t(rs'-r's)=0$

Let M be an R module, we can define localization, $S^{-1}M \coloneqq M \times S/\sim$, $(m,s) \sim (m',s')$ if there exists $t \in S$ such that $t(s'm - sm') = 0$

Proof. Suppose $0 \to A \overset{f}{\to} B \overset{g}{\to} C \to 0$ is a short exact sequence, then it is obvious that $A \otimes D \overset{f\otimes 1_D}{\to} B \otimes D \overset{g\otimes 1_D}{\to} C \otimes D \to 0$ is a complex and $g \otimes 1_D$ is surjective, now define $\phi : B \otimes D/\ker g \otimes 1_D \to A \otimes D$, $b \otimes d \mapsto a \otimes d$, where a is the unique element in A such that $g(b - f(a)) = 0$

□

Definition 6.11 Let P_i, A, D be R modules, $\cdots P_2 \overset{d_2}{\to} P_1 \overset{d_1}{\to} P_0 \overset{\epsilon}{\to} A \to 0$ be a projective resolution, then we have $0 \to Hom_R(A,D) \overset{\epsilon}{\to} Hom_R(P_0,D) \overset{d_1}{\to} Hom_R(P_1,D) \overset{d_2}{\to} Hom_R(P_2,D)\cdots$, define $Ext_R^n(A,D)$ to be the n-th cohomology group of $0 \to Hom_R(P_0,D) \overset{d_1}{\to} Hom_R(P_1,D) \overset{d_2}{\to} Hom_R(P_2,D)\cdots$, note that $Ext_R^0(A,D) \cong Hom_R(A,D)$

Definition 6.12 Let P_i, B, D be R modules, $\cdots P_2 \overset{d_2}{\to} P_1 \overset{d_1}{\to} P_0 \overset{\epsilon}{\to} B \to 0$ be a projective resolution, then we have $\cdots D \otimes_R P_2 \overset{1\otimes d_2}{\to} D \otimes_R P_1 \overset{1\otimes d_1}{\to} D \otimes_R P_0 \overset{1\otimes\epsilon}{\to} D \otimes_R B \to 0$, define $Tor_n^R(D,B)$ to be the n-th homology group of $\cdots D \otimes_R P_2 \overset{1\otimes d_2}{\to} D \otimes_R P_1 \overset{1\otimes d_1}{\to} D \otimes_R P_0 \to 0$, note that $Tor_0^R(D,B) \cong D \otimes_R B$

Lemma 6.13 (Schur's Lemma) M, N are nonzero simple R modules. A homomorphism $\varphi : M \to N$ is either 0 or an isomorphism. In particular, $\mathrm{End}_R(M)$ is a division ring. Moreover, if F is algebraically closed, $\mathrm{Hom}_F(M,N) = \{\lambda\varphi | \lambda \in F\}$ where $M \overset{\varphi}{\to} N$ is an isomorphism(all isomorphisms are scalar multiple of each other), in particular, $\mathrm{Hom}_F(M,M) = \{\lambda 1_M | \lambda \in F\}$

Theorem 6.14 (Maschke's theorem) G is a finite group, F is a field, $\mathrm{char}F \nmid |G|$, then FG is a semisimple ring

Theorem 6.15 (Artin-Wedderburn theorem) $R = V_1 \oplus \cdots \oplus V_r$ is a semisimple ring, by Schur's lemma 6.13, $D_i = \mathrm{End}_R(V_i)$ are division rings, then

$$R \cong M_{n_1}(D_1) \times \cdots \times M_{n_r}(D_r)$$

where $n_i = \dim_{D_i}(V_i)$. $\sum_{i=1}^r n_i^2 = |G|$, r is the number of conjugacy classes in G

Definition 6.16 F is a field, G is a group, the representation ring $R_F(G)$ is the completion of the set of isomorphic classes of representations

Definition 6.17 The symmetric k algebra is $S^k(V) \subseteq T^k(V)$ consists of k tensors symmetric under the permutation of S_k. The exterior k algebra is $\bigwedge^k(V) \subseteq T^k(V)$ consists of k tensors antisymmetric under the permutation of S_k. We have projections

$$\mathrm{Sym} : T^k(V) \to T^k(V), a_1 \otimes \cdots \otimes a_k \mapsto \frac{1}{k!} \sum_\sigma a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(n)}$$

and

$$\mathrm{Alt} : T^k(V) \to T^k(V), a_1 \otimes \cdots \otimes a_k \mapsto \frac{1}{k!} \sum_\sigma (-1)^{\mathrm{sgn}\ \sigma} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(n)}$$

For $\alpha \in T^k(V)$, $\beta \in T^l(V)$, define

$$\alpha\beta = \alpha \odot \beta = \mathrm{Sym}(\alpha \otimes \beta)$$

$$\alpha \wedge \beta = \frac{(k+l)!}{k!l!}\ \mathrm{Alt}(\alpha \otimes \beta)$$

Which corresponds to determinants

Lemma 6.18 (Nakayama's lemma) M is a finitely generated R module, $I \subseteq R$ is an ideal, $\phi \in \mathrm{End}_R(M)$, if $\phi(M) \subseteq IM$, then

$$\phi^n + a_1\phi^{n-1} + \cdots + a_n = 0$$

for some $a_i \in I$. In particular If $IM = M$, then there exists $r \equiv 1 \mod I$ such that $rM = 0$

Proof. Consider multiplication by scalars as endomorphisms, $1M \subseteq M = IM$ is satisfied, thus $1 + a_1 + \cdots + a_n = 0$

□

Proposition 6.19 Localization is exact. i.e. If $0 \to A \to B \to C \to 0$ is exact, so is $0 \to S^{-1}A \to S^{-1}B \to S^{-1}C \to 0$

Proof.

□

7 Commutative algebra

7.1 Commutative ring

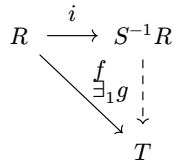
Proposition 7.1 $I, J \subseteq A$ are ideals, $P \subset A$ is a prime ideal, then

1. $IJ \subseteq P \Rightarrow I \subseteq P$ or $J \subseteq P$
2. $I \subseteq \sqrt{J} \Rightarrow \sqrt{I} \subseteq \sqrt{J}$

Proof.

□

Definition 7.2 $S \subseteq R$ is *multiplictive closed*, the localization $S^{-1}R$ of R with respect to S is $R \times S/\sim$, $(r, s) \sim (r', s')$ iff there exists $t \in S$ such that $t(rs' - sr) = 0$. $S^{-1}R$ has the universal property that for any $f : R \to T$ such that maps S to units, then there exists a unique $g : S^{-1}R \to T$ such that $gi = f$



Definition 7.3 Given a ring R and a proper ideal I, we can define an *associated graded ring* $gr_I R \coloneqq \bigoplus_{n=0}^{\infty} I^n/I^{n+1}$, if M is a left R-module, we can define *associated graded module* $gr_I M \coloneqq \bigoplus_{n=0}^{\infty} I^n M/I^{n+1}M$

Definition 7.4 R is a *local ring* if it has a unique maximal ideal m. The *residue field* is $k = R/m$

Definition 7.5 R is a *semilocal ring* if it has only finitely many maximal ideal

Proposition 7.6 Let R be a UFD, f a prime element, then $ht(f) = 1$

Proof. Suppose there exists prime ideal P such that $0 \subsetneq P \subsetneq (f)$, then we can find a prime element in $g \in P$, thus we have $0 \subsetneq (g) \subseteq P \subsetneq (f)$, but then $g = fh$ for some h, but since f is prime, thus $(f) = (g)$ which is a contradiction, such a prime element exists since we can pick any element $0 \neq q = q_1 \cdots q_m \in P$ where q_i's are prime, but then at least one of them has to be in P

□

Theorem 7.7 Let $A \subseteq B$ be finitely generated k-algebras, and A, B are both domains, $0 \neq b \in B \Rightarrow \exists 0 \neq a \in A$ such that for any k-algebra homomorphism $\alpha : A \to k$ with $\alpha(a) \neq 0$ can be extended to k-algebra homomorphism $\beta : B \to k$ with $\beta(b) \neq 0$

Definition 7.8 Suppose R is a commutative ring with identity, a prime element $p \in R$ is an element which is nonzero nor a unit and $p|fg \Rightarrow p|f$ or $p|g$

Definition 7.9 A *graded ring* R is a ring such that $R = \bigoplus_i R_i$ is a direct sum of abelian groups and $R_i R_j \subseteq R_{i+j}$

An ideal is called a *homogeneous ideal* if it consists of only homogeneous elements

Theorem 7.10 (Chinese remainder theorem) Let R be a commutative ring, and $I_1, \cdots, I_n \leq R$ be pairwise coprime ideals, then $R \cong R/I_1 \times \cdots R/I_n, r \mapsto (r \mod I_1, \cdots, r \mod I_n)$

Definition 7.11 An *integral domain* is a commutative ring R such that (0) is a prime ideal. Equivalently, $rs \in R \Rightarrow r \in R$ or $s \in R$

Definition 7.12 Suppose R is a domain, K is the field of fractions, a *fractional ideal* is an R submodule $I \leq K$ such that $rI \subseteq R$ for some nonzero $r \in R$. I is invertible if $IJ = R$ for some other fractional ideal J

Definition 7.13 A *Dedekind domain* is an integral domain such that every proper ideal is a product of prime ideal

Definition 7.14 A *discrete valuation ring(DVR)* is a PID with a unique nonzero prime ideal

Definition 7.15 A local ring homomorphism $\phi : R \to S$ between local rings is such that $\phi(m_R) \subseteq m_S$

Definition 7.16 R is a commutative ring. An *R-linear category* $\mathcal{C}$ is a category enriched over R-modules, i.e. $\mathrm{Hom}(A, B)$ are R-modules, $\mathrm{Hom}(B, C) \otimes_R \mathrm{Hom}(A, B) \to \mathrm{Hom}(A, C)$ is R-bilinear

Definition 7.17 A unital associative R-algebra A is a monoid in the monoidal category of R-modules, coalgebras are comonoids

Definition 7.18 Commutative ring R is a preadditive category with a single object $\bullet$. An *R-algebra* is an additive functor $\phi \in R^{\mathrm{RMod}}$, write $\phi(\bullet) = S$, $\phi(r)s = rs$. A ring A is an R *algebra* is a ring homomorphism $R \xrightarrow{\phi} A$, $ra = \phi(r)a$

Definition 7.19 A *coalgebra* is the categorical dual to a unital associative algebra

Definition 7.20 A is *finite* or ϕ is *finite* if A is a finitely generated R module

ϕ is of *finite type* if A is *finitely generated* R algebra

Definition 7.21 For $p \in \mathrm{Spec}\ A$, $q \in \mathrm{Spec}\ B$, $A \subseteq B$, p *lies under* q or q *lies over* p if $q \cap A = p$ $A \subseteq B$ satisfies *lying over property* if every $p \in \mathrm{Spec}\ A$ lies under some $q \in \mathrm{Spec}\ B$

$A \subseteq B$ satisfies the *incomparability property* if different prime ideals q, q' both lie over p are incomparable, i.e. they don't contain each other

$A \subseteq B$ satisfies *going up property* if for any chain of prime ideals $p_1 \subseteq \cdots \subseteq p_n$, $q_1 \subseteq \cdots \subseteq q_m$ with q_i lies over p_i and $m < n$ can be extended to a chain of prime ideals $q_1 \subseteq \cdots \subseteq q_n$ with q_i lies over p_i

$A \subseteq B$ satisfies *going down property* if for any chain of prime ideals $p_1 \supseteq \cdots \supseteq p_n$, $q_1 \supseteq \cdots \supseteq q_m$ with q_i lies over p_i and $m < n$ can be extended to a chain of prime ideals $q_1 \supseteq \cdots \supseteq q_n$ with q_i lies over p_i

Definition 7.22 $R \subseteq S$ are commutative rings, $a \in S$ is *integral* over R if it is a root of some monic polynomial in $R[x]$. The *integral closure* of R in S are the integral elements of S

Theorem 7.23 B is integral over A, then $A \subseteq B$ satisfies going up property and incomparability property

Lemma 7.24 (Noether's normalization lemma) A is a finitely generated k algebra, then there exists algebraically independent elements $x_1, \cdots, x_d \in A$ such that A is a finite $k[x_1, \cdots, x_d]$ algebra

Definition 7.25 The *height* of prime ideal p is $\mathrm{ht}\ p = \sup_d p_0 \subsetneq \cdots \subsetneq p_d = p$. The *Krull dimension* of a ring R is $\dim R = \sup_d p_0 \subsetneq \cdots \subsetneq p_d = \sup_p \mathrm{ht}\ p$, p_i are prime ideals

Theorem 7.26 (Krull's height theorem) R is Noetherian, I is an ideal which can be generated by n elements, then the minimal prime over I is of height at most n

Definition 7.27 A local ring R is *regular* if $\dim R = \dim_k m/m^2$

Proposition 7.28 A is a integral domain, finitely generated over some subfield k, then $\dim A = \mathrm{trdeg}(\mathrm{Frac}A/k)$

Definition 7.29 A *finitely presented algebra* over R is of the form $R[x_1, \cdots, x_n]/I$, I is a finitely generated ideal

Definition 7.30 R is a commutative ring. Elements in $W(R) = 1 + tR[[t]]$ can be factored

$$\sum_{n=0}^{\infty} A_n t^n = \prod_{n=1}^{\infty} (1 - X_n t^n) = f_X(t)$$

$$A_n = \sum_I (-1)^{|I|} \prod_{i \in I} X_i$$

I runs over subsets of $\{1,\cdots,n\}$ that add up to n. $X=(X_1,X_2,\cdots)$ are *Witt vectors. Witt polynomials*(ghost components) are

$$W_n=X^{(n)}=\sum_{d|n}dX_d^{\frac{n}{d}}$$

The *Witt ring* is the ring of Witt vectors with addtion and multiplication defined by

$$(X+Y)^{(n)}=X^{(n)}+Y^{(n)},(XY)^{(n)}=X^{(n)}Y^{(n)}$$

$1-t$ is the multiplicative identity. Notice

$$\begin{aligned}-t\frac{d}{dt}\log f_X(t)&=\sum_{d\geq 1}\frac{dX_dt^d}{1-X_dt^d}\\&=\sum_{d\geq 1}dX_dt^d\sum_{i\geq 0}X_d^it^{di}\\&=\sum_{d\geq 1}\sum_{i\geq 1}dX_d^it^{di}\\&=\sum_{n\geq 1}X^{(n)}t^n\end{aligned}$$

$Z=X+Y\Leftrightarrow f_Z(t)=f_X(t)f_Y(t)$ since

$$\begin{aligned}\sum_{n\geq 1}Z^{(n)}t^n&=-t\frac{d}{dt}\log f_Z(t)\\&=-t\frac{d}{dt}\log f_X(t)-t\frac{d}{dt}\log f_X(t)\\&=\sum_{n\geq 1}X^{(n)}t^n+\sum_{n\geq 1}Y^{(n)}t^n\end{aligned}$$

Since A_j are polynomials in X_i's, B_j are polynomials in Y_i's, we can show that by induction

$$Z_n=C_j-\sum_{I\neq\{n\}}(-1)^{|I|}\prod_{i\in I}Z_i=\sum_{k+l=j}A_kB_l-\sum_{I\neq\{n\}}(-1)^{|I|}\prod_{i\in I}Z_i$$

are polynomials in X_i,Y_i's
If $Z=XY$, then . In particular, consider $Y=(r,0,\cdots)$, $f_Z(t)=f_X(rt)$

Definition 7.31 A commutative ring R is a λ *ring* if it has λ operations λ^k satisfying

- $\lambda^0(x)=1$
- $\lambda^1(x)=x$
- $\lambda^k(x+y)=\sum_{i=0}^k\lambda^i(x)\lambda^{k-i}(y)$

The last of which is equivalent to a group homomorphism

$$\begin{aligned}\lambda_t:R&\to W(R)=1+tR[[t]]\\x&\mapsto\sum\lambda^k(x)t^k\end{aligned}$$

An λ ideal is an ideal $I\leq R$ such that $\lambda^k(I)\subseteq I$ for any k. A *special* λ *ring* is a λ ring such that

$$\begin{aligned}\lambda^k(1)&=0,k>2\\\lambda^k(xy)&=P_k(\lambda^1(x),\cdots,\lambda^k(x),\lambda^1(y),\cdots,\lambda^k(y))\\\lambda^n(\lambda^k(x))&=P_{n,k}(\lambda^1(x),\cdots,\lambda^{nk}(x))\end{aligned}$$

$P_n,P_{n,k}$ are defined through

$$\sum P_{n,k}(s_1(X),\cdots,s_{nk}(X))t^n=\prod_{1\le X_{i_1}\le\cdots\le X_{i_n}\le nk}(1+tX_{i_1}\cdots X_{i_n})$$

$$\sum P_n(s_1(X),\cdots,s_n(X),s_1(Y),\cdots,s_n(Y))t^n=\prod_{i,j=1}^{n}(1+tX_iX_j)$$

s_i's are elementary symmetric polynomials

Example 7.32 A binomial ring is a $\mathbb{Q}$ algebra R with $\lambda_t(x)=(1+t)^x$, $\lambda^k(x)=\binom{x}{k}=\frac{x(x-1)\cdots(x-k+1)}{k!}$

Example 7.33 R is a semiring. $K_0(R)$ is a λ ring with $\lambda^k(P)=\bigwedge^k P$

Definition 7.34 $R\xrightarrow{\varepsilon}\mathbb{Z}$ is the augmentation. The Adams operation ψ is defined by

$$\psi_t(x)=\sum\psi^k(x)t^k=\varepsilon(x)-t\frac{d}{dt}\log\lambda_{-t}(x)$$

Example 7.35 Vector bundles and representation are λ semirings with augmentation dim

Proposition 7.36 If R satisfies splitting principle, then ψ^k's are endomorphisms of R and $\psi^j\psi^k=\psi^{jk}$

Definition 7.37 Let $s=\frac{t}{1-t}$, then $t=\frac{s}{1+s}$, $R[[t]]=R[[s]]$. The γ operation is defined by

$$\gamma_t(x)=\sum\gamma^k(x)t^k=\sum\lambda^k(x)s^k=\lambda_s(x)$$

Example 7.38 $\gamma^k(x)=\lambda^k(x+k-1)=\binom{x+k-1}{k}=(-1)^k\binom{-x}{k}$

Definition 7.39 The γ dimension $\dim_\gamma x$ is the greatest integer k such that $\gamma^k(x-\varepsilon(x))\neq 0$, $\dim_\gamma R=\sup_x\dim_\gamma x$. The γ filtration is

$$R=F^0_\gamma R\supseteq F^1_\gamma R\supseteq\cdots$$

Here $F^1_\gamma R=\ker\varepsilon$, $F^k_\gamma R$ is the ideal generated by products $\gamma^{i_1}(x)\cdots\gamma^{i_m}(x)$ whereas $\sum i_j\ge k$, $x_j\in F^1_\gamma R$

Definition 7.40 A *discrete valuation* on a field F is $v:F\to\mathbb{Z}\cup\{\infty\}$ satisfying

1. $v(x)=\infty\Leftrightarrow x=0$
2. $v(x+y)\ge\min(v(x),v(y))$, equality holds iff $v(x)\neq v(y)$
3. $v(xy)=v(x)+v(y)$

A valuation is when replacing $\mathbb{Z}$ by some totally ordered abelian group(called the value group). v is trivial if $v(F^\times)=0$

Definition 7.41 A *discrete valuation ring* is an integral domain R with a discretely valued fraction field $F=\mathrm{Frac}(R)$ such that $R=\{x\in F|v(x)\ge 0\}$. In this case $m=\{x\in F|v(x)>0\}$ is the unique prime ideal in R, and $m=(\pi)$, we call π the *uniformizer*, and for $x\in F$, $v(x)$ is the maximal power of π in a.

Definition 7.42 Pick some $0<c<1$, we can define norm $|\cdot|:F\to\mathbb{R}_{\ge 0}$ as $|x|=c^{v(x)}$, this defines a metric topology(which can be shown to be independent of the choice of c)

Example 7.43 $\mathbb{Z}_{(p)}\subseteq\mathbb{Q}$ with the p-adic valuation, $c=p^{-1}$

Definition 7.44 By completing the metric topology, we obtain a complete discretely valued field $\hat{F}$ with extended norm $|\cdot|$, and the closure of R in F is denoted by $\hat{R}$ which can be proven is $\hat{R}=\lim_{\leftarrow}R/m^n$

Example 7.45 $k[t]_{(t)}\subseteq k(t)$ with t-adic valuation, $\widehat{k(t)}=k((t))$ is the field of formal Laurent series, $\widehat{k[t]_{(t)}}=k[[t]]$ is the ring of formal power series

Definition 7.46 $V\subseteq\mathbb{A}^n$ is an algebraic set, $f\in k[V]$

$$D(f)=\{(x_1,\cdots,x_n)\in V|f(x_1,\cdots,x_n)\neq 0\}=V(f)^c$$

form a basis for the Zariski topology on V

$D(f)$ can also be thought of as an algebraic set

$$\{(x_1, \cdots, x_n, z) \mid zf(x_1, \cdots, x_n) = 0\}$$

The coordinate ring can be written as $k[V][\frac{1}{f}] = k[V]_f$, where z is just replaced by $\frac{1}{f}$

Theorem 7.47 $\sqrt{I} = \bigcap_{P \supseteq I \text{ prime}} P$

Theorem 7.48 (Hilbert Nullstellensatz weak form) k is algebraically closed, $m < k[x_1, \cdots, x_n]$ is a maximal ideal, then $k[x]/m \cong k$

Theorem 7.49 (Hilbert Nullstellensatz strong form) k is algebraically closed, $I(V(J)) = \sqrt{J}$

Proof. Since $\sqrt{J} = \bigcap_{P \supseteq J \text{ prime}} P$, suppose $f \notin P$ for some $P \supseteq J$, consider $\varphi : k[x] \to k[x]/P \to A_{\overline{f}} \to A_{\overline{f}}/m$ which is a field, hence $\ker \varphi$ is a maximal ideal, by Theorem 7.48, $B/m \cong k[x]/\ker \varphi \cong k$, then $(\varphi(x_1), \cdots, \varphi(x_n)) \in V(P) \subseteq V(J)$ but $f(\varphi(x_1), \cdots, \varphi(x_n)) = \varphi(f) \neq 0 \Rightarrow f \notin I(V(J))$

□

Proposition 7.50 Morphism $V \xrightarrow{\varphi} W$ induce a ring homomorphism $k[W] \xrightarrow{\varphi^*} k[V], f \mapsto f \circ \varphi$, and if $f(p) = q$, then $(\varphi^*)^{-1}(m_q) = m_p$, thus conversely, if $\alpha : k[W] \to k[V]$ is a ring homomorphism, then $\alpha^{-1} : Spmk[V] \to Spm[W]$ is a morphism which can be identified with $\varphi : V \to W$, and $\varphi^* = \alpha$

Proposition 7.51 A finite morphism $V \xrightarrow{\varphi} W$ between affine varieties is quasifinite

Proof. $\varphi(p) = q \Leftrightarrow (\varphi^*)^{-1}(m_p) = q$, $m_p \supseteq \varphi^*(\varphi^*)^{-1}(m_p) = \varphi^*(m_q)$

$$\varphi^{-1}(q) \leftrightarrow \left\{ \text{maximal ideals of } \ B = \frac{k[W]}{\langle \varphi^*(m_q) \rangle} \right\}$$

Since $k[W]$ is a finite $k[V]$ algerba, so B is finite dimensional over $\frac{k[V]}{m_p} \cong k$ By Chinese Remainder theorem 7.10, $B \to B/m_1 \times \cdots \times B/m_s$ is surjective, $\dim B \geq s$, since $\dim B < \infty$, hence $s < \infty$, thus B has only finitely many maximal ideals

□

Proposition 7.52 $W \xrightarrow{\varphi} V$ is dominant iff $k[V] \xrightarrow{\varphi} k[W]$ is injective

Proof. $f \in \ker \varphi^* \Leftrightarrow f \circ \varphi = 0$, $\text{im}\varphi$ dense $\Rightarrow f = 0$. Conversely, $\overline{\text{im}\varphi} \subsetneq V \Rightarrow 0 \neq f \in I(\overline{\text{im}\varphi})$

□

Proposition 7.53 If $W \xrightarrow{\varphi} V$ is dominant and finite, then φ is surjective

Proof. By Proposition 7.52, $k[W]$ is integral over $k[V]$, by Theorem 7.23, for any $m_q < k[V]$, there exists maximal ideal $n < k[W]$ such that $n \cap k[V] = m_q$

□

Proposition 7.54 Assume A is a k-algebra, $f, g \in A$, then

1. Spec A is irreducible $\Leftrightarrow \sqrt{0}$ is prime
2. f is nilpotent $\Leftrightarrow D(f) = \varnothing$
3. $D(f) \cap D(g) = D(fg)$
4. $D(f) = \text{Spec } A \Leftrightarrow f$ is a unit
5. $\bigcup_{i \in I} D(f_i) = V(\{f_i\}_{i \in I})^c$

Lemma 7.55 There is a bijection

$$\{\text{Idempotents in } \ A\} \to \{\text{Open and closed subsets of } \ \text{Spec } A\}, \quad e \mapsto D(e)$$

Proposition 7.56

1. Localization is exact

2. Localization and quotient commutes. i.e. $S^{-1}(A/I) = S^{-1}A/S^{-1}I$

Proof. Injectivity: Suppose $e_1 \neq e_2$ are both idempotents, denote $e_i' = 1 - e_i$, then

$$0 \neq e_1 - e_2 = e_1(e_2 + e_2') - (e_1 + e_1')e_2 = e_1e_2' - e_1'e_2$$

so one of $e_1e_2', e_1'e_2$ has to be a nonzero idempotent and not a nilpotent, so $\exists p \in \operatorname{Spec} A$ such that one of $e_1e_2', e_1'e_2$ is not in p, this implies that e_1, e_2 cannot both be in or not in p.
Surjectivity: Suppose $U \subseteq \operatorname{Spec} A$ is both open and closed, then both U and U^c are quasi-compact and can be written as $\bigcup_{i=1}^n D(f_i) = V(I)^c, \bigcup_{j=1}^m D(g_j) = V(J)^c$ respectively, denote $I = (f_1, \cdots, f_n), J = (g_1, \cdots, g_m)$, since $\varnothing = D(f_i) \cap D(g_j) = D(f_ig_j)$, f_ig_j is a nilpotent and $(IJ)^N = 0$ for some N. On the other hand, $\operatorname{Spec} A = \bigcup_{i=1}^n D(f_i) \cup \bigcup_{j=1}^m D(g_j) = V(I+J)^c$, so $I + J = A$, raise to the $2N$-th power we get $I^N + J^N = A$, therefore $\exists f \in I^N, g \in J^N$ such that $1 = f + g$ and $0 = fg = f(1-f)$ which implies that both f, g are idempotents. We have

$$p \in D(f) \Leftrightarrow f \notin p \Leftrightarrow I^N \not\subseteq p \Leftrightarrow J^N \subseteq p \Leftrightarrow J \subseteq p \Leftrightarrow p \in V(J)$$

Hence $D(f) = V(J)$. Here $I^N \not\subseteq p \Rightarrow J^N \subseteq p$ because $0 = I^NJ^N \subseteq p$. $J^N \subseteq p \Rightarrow I^N \not\subseteq p$ because $I^N + J^N = A$. $I^N \not\subseteq p \Rightarrow f \notin p$ because otherwise $f \in p \Rightarrow g = 1 - f \notin p \Rightarrow J^N \not\subseteq p$ which is impossible

□

Definition 7.57 Topological space X is an H-*space* if there is a continuous map $\mu : X \times X \to X$ and an identity element e such that $\mu(x, e) = \mu(e, x) = e$

Definition 7.58 A *Hopf algebra* H is a bialgebra with an *antipode* $S : H \to H$ such that the following diagram commutes

$$\begin{array}{ccccc}
H \otimes H & & \xrightarrow{S \otimes 1} & & H \otimes H \\
\uparrow{\scriptstyle \Delta} & & & & \downarrow{\scriptstyle \mu} \\
H & \xrightarrow{\epsilon} & I & \xrightarrow{\eta} & H \\
\downarrow{\scriptstyle \Delta} & & & & \uparrow{\scriptstyle \mu} \\
H \otimes H & & \xrightarrow{1 \otimes S} & & H \otimes H
\end{array}$$

Proposition 7.59 Suppose $\varphi : \operatorname{Spec} B \to \operatorname{Spec} A$ corresponds to $\phi : A \to B$, if $\forall b \in B$, $b = uf(a)$ for some $a \in A, u \in B^\times$, then φ is a homeomorphism

Proof.

□

7.2 Field

Definition 7.60 A **division ring** R is a nonzero ring such that $\mathbb{F}^\times = \mathbb{F} - \{0\}$

A **field** $\mathbb{F}$ is a nonzero commutative ring such that $\mathbb{F}^\times = \mathbb{F} - \{0\}$

Definition 7.61 A **character** is of G is a group homomorphism $G \to \mathbb{F}^\times$, and a **cocharacter** is a group homomorphism $\mathbb{F}^\times \to G$

Lemma 7.62 Characters of G, denoted as $ch(G)$ are linear independent on $\mathbb{F}[G]$

Proof. Suppose not, we can find $c_1\chi_1 + \cdots + c_m\chi_m = 0, c_i \in \mathbb{F}^\times$, with minimal terms, since $\chi_1 \neq \chi_m$, there exists $g_0 \in G$ such that $\chi_1(g_0) \neq \chi_m(g_0)$, on the other hand we have $0 = c_1\chi_1(g) + \cdots + c_m\chi_m(g) = c_1\chi_1(g)\chi_m(g_0) + \cdots + c_m\chi_m(g)\chi_m(g_0), \forall g \in G$ and $0 = c_1\chi_1(gg_0) + \cdots + c_m\chi_m(gg_0) =$

$c_1\chi_1(g)\chi_1(g_0)+\cdots+c_m\chi_m(g)\chi_m(g_0), \forall g\in G$, subtract to get $0=c_1(\chi_m(g_0)-\chi_1(g_0))\chi_1(g)+\cdots+c_{m-1}(\chi_m(g_0)-\chi_{m-1}(g_0))\chi_{m-1}(g)$ with fewer terms which is a contradiction

□

Definition 7.63 E/F is a field extension, $\alpha\in E$ induces an F-linear automorphism $T_\alpha: E\to E$ by multiplication, then the *field trace* is $\mathrm{Tr}_{E/F}(\alpha)=\mathrm{Tr}\ T_\alpha$. The *field norm* is $N_{E/F}(\alpha)=\det T_\alpha$. Suppose

$$f(x)=\prod(x-\sigma_i(\alpha))=x^n+a_1x^{n-1}+\cdots+a_n$$

is the minimal monic polynomial, use $1,\alpha,\cdots,\alpha^{n-1}$ as a basis for $F(\alpha)$, then T_α has the matrix form

$$\begin{bmatrix} 0 & 0 & \cdots & 0 & -a_n \\ 1 & 0 & \cdots & 0 & -a_{n-1} \\ & 1 & \cdots & 0 & -a_{n-2} \\ & & \ddots & \vdots & \vdots \\ & & & 1 & -a_1 \end{bmatrix}$$

Hence $\mathrm{Tr}_{F(\alpha)/F}(\alpha)=-a_1=\sum\sigma_i(\alpha)$, $N_{F(\alpha)/F}(\alpha)=(-1)^na_n=\prod\sigma_i(\alpha)$

Definition 7.64 $\mathbb{F}$ is a **perfect field** if $\mathbb{F}^p=\mathbb{F}$ if $\mathrm{char}\mathbb{F}=p\neq 0$ or $\mathrm{char}\mathbb{F}=0$

Definition 7.65 E/F is a field extension, $\alpha\in E$ is algebraic over F if α is a zero of some polynomial in $F[x]$. The **algebraic closure** of F in E are the algebraic elements of E

Theorem 7.66 (Emil Artin) Any field F has an algebraically closed extension

Theorem 7.67 $F\leq L$ is a field extension, $G=\mathrm{Aut}_F(L)$ as a profinite group with induced topology as a topological group. then there is a one to one correspondence between intermediate fields E and closed subgroups H. E is finite $\Leftrightarrow$ H is open. σE corresponds to $\sigma H\sigma^{-1}$ for any $\sigma\in G$. E is Galois $\Leftrightarrow$ H is normal, in such case, $\mathrm{Aut}_F(E)=G/H$

7.3 Number Field

Lemma 7.68 $K=\mathbb{Q}[\alpha]$ is number field, f is the minimal polynomial of α. Suppose $\sigma: K\hookrightarrow\mathbb{C}$ is an embedding, then $\sigma(\alpha)$ is a root of f, and any such choice gives an embedding

Definition 7.69 E,F are algebraic number fields of finite degree, E/F is finite separable, A,B are corresponding ring of integers, $\{\beta_1,\cdots,\beta_n\}$ is an integral basis of B over A. The **discriminant** of E/F with respect to $\{\beta_1,\cdots,\beta_n\}$ is $D_{E/F}(\beta_1,\cdots,\beta_n)=\det(Tr(\beta_i\beta_j))$

$$\begin{array}{ccc} B & \hookrightarrow & E \\ \uparrow & & \uparrow \\ A & \hookrightarrow & F \end{array}$$

Lemma 7.70 D_K is well defined in $\frac{A}{(A^\times)^2}$

Definition 7.71 E,F are algebraic number fields of finite degree, E/F is finite separable, A,B are corresponding ring of integers which are Dedekind domains

$$\begin{array}{ccc} B & \hookrightarrow & E \\ \uparrow & & \uparrow \\ A & \hookrightarrow & F \end{array}$$

$pB=q_1^{e_1}\cdots q_r^{e_r}$ with $e_i>0$. p is **ramified** if $e_i>1$ for some i, otherwise unramified. p is **inert** if $r=e=1$. p **totally split** if $e_i=f_i=1$

$B/pB\cong\prod_{i=1}^r B/q_i^{e_i}$, $f_i=[k_{q_i}:k_p]$, $[E:F]=\dim_{k_p}(B/pB)=\sum_{i=1}^r e_if_i$

If E/F is Galois, $G = Aut(E/F)$ acts transitively on $\{q_1, \cdots, q_r\}$, then $n = \sum_{i=1}^{r} e_i f_i = ref$

Proof. $B \cong A^n$, $B/pB \cong A^n/pA^n \cong (A/p)^n \cong k_p^n$

□

Example 7.72 $2\mathbb{Z}[i] = (1+i)^2$ is ramified, $3\mathbb{Z}[i]$ is inert, $5\mathbb{Z}[i] = (2+i)(2-i)$ totally split

$$\begin{array}{ccc} \mathbb{Z}[i] & \hookrightarrow & \mathbb{Q}[i] \\ \uparrow & & \uparrow \\ \mathbb{Z} & \hookrightarrow & \mathbb{Q} \end{array}$$

Theorem 7.73 p ramifies in $O_K \Leftrightarrow p \mid \mathrm{disc}(O_K/\mathbb{Z})$

$$\begin{array}{ccc} O_K & \hookrightarrow & K \\ \uparrow & & \uparrow \\ \mathbb{Z} & \hookrightarrow & \mathbb{Q} \end{array}$$

Proof. $pO_K = \beta_1^{e_1} \cdots \beta_r^{e_r}$, $O_K/pO_K \cong O_K/\beta_i^{e_i}$ is an isomorphism of $\mathbb{F}_p$ algebras. $d_i = \mathrm{disc}((O_K/\beta_i^{e_i})/\mathbb{F}_p))$, $d = \mathrm{disc}((O_K/pO_K)/\mathbb{F}_p))$, thus $d = d_1 \cdots d_r$, since discriminant is functorial, $D = \det(Tr_{O_K/\mathbb{Z}}()) \mapsto d$, $p|D \Leftrightarrow d = 0 \Leftrightarrow d_i = 0$ for some i

□

8 Lie algebras

8.1 Lie groups

Definition 8.1 Let G be a Lie group, the *Lie algebra* $\mathrm{Lie}(G)$ of G can be defined in the following equivalent ways

1. The vector space of left invariant vector fields
2. The tangent space at the identity T_eG

Definition 8.2 A *one-parameter subgroup* is a group homomorphism $\phi : \mathbb{R} \to G$

Proposition 8.3

1. Lie groups are parallelizable
2. One parameter subgroups are precisely the maximal integral curves of the left invariant vector fields starting at e. So there is a one to one correspondence between One parameter subgroups of G and $\mathrm{Lie}(G)$

Proof.

1. For any $0 \neq X_e \in T_eG$, we can define a vector field $X_g = g_* X_e$, this is a nonvanishing global section of the tangent bundle
2. Suppose $\phi : \mathbb{R} \to G$ is a one parameter subgroup, let $X_e = \phi'(0)$, then we have a left invariant vector field X on G, think of $\frac{\partial}{\partial t}$ as a left invariant vector field on $\mathbb{R}$, thus ϕ as Lie group homomorphism induces $\phi_* \frac{\partial}{\partial t}$ which is also a left invariant vector field and $\phi'(s) = (d\phi)_s \frac{\partial}{\partial t}|_s = X_{\phi(s)}$. Conversely, if $\phi : \mathbb{R} \to G$ is the maximal integral curve of some left invariant vector field X, suppose the global flow generated by X is $\varphi : G \times \mathbb{R} \to G$, then $\varphi(e,t) = \phi(t)$, $\phi(t+s) = \varphi(e, t+s) = \varphi(\varphi(e,t), s) = \varphi(\phi(t), s)$, since $L_{\phi(t)}$ is an isomorphism, thus $L_{\phi(t)} \circ \phi$ is the maximal integral curve starting at $\phi(t)$, thus $\varphi(\phi(t), s) = \phi(t)\phi(s)$

□

Theorem 8.4 (Lie correspondence)

1. If $\phi : G \to H$ is a homomorphism of Lie groups, then $d\phi : \mathrm{Lie}(G) \to \mathrm{Lie}(H)$ or $(d\phi)_e : T_eG \to T_eH$ is an homomorphism of Lie algebras. Suppose $H \leq G$ is a Lie subgroup, then $\mathrm{Lie}(H) = T_eH \leq T_eG$
2. $d\phi : \mathrm{Lie}(G) \to \mathrm{Lie}(H)$ or $(d\phi)_1 : T_1G \to T_1H$ is an homomorphism of Lie algebras

Proof.

1.
2. Suppose X is a left invariant vector field on G, then $(d\phi)_g X_g = (d\phi)_g (dL_g)_1 X_1(f) = X_1(f \circ \phi \circ L_g) = X_1(f \circ \phi \circ L_g) = (dL_{\phi(g)})_1 (d\phi)_1 X_1(f)$ which gives a left invariant vector field, thus using Lemma 35.81

$$\begin{aligned}(d\phi)[X,Y](f) &= [X,Y](f \circ \phi) \\ &= X(Y(f \circ \phi)) - Y(X(f \circ \phi)) \\ &= X(((d\phi Y)f) \circ \phi) - Y(((d\phi X)f) \circ \phi) \\ &= ((d\phi X)(d\phi Y)f) \circ \phi - ((d\phi Y)(d\phi X)f) \circ \phi \\ &= ([d\phi X, d\phi Y]f) \circ \phi\end{aligned}$$

Therefore $(d\phi)[X,Y] = [(d\phi X), (d\phi Y)]$, $d\phi$ is a Lie algebra homomorphism

□

Proposition 8.5 Suppose (π, V) is a complex representation of a compact Lie group G, then there exists a positive definite Hermitian form such that this representation is unitary

Proof. Choose any positive definite Hermitian form $\langle , \rangle$, define Hermitian form

$$(v, w) \coloneqq \int_G \langle gv, gw \rangle d\mu$$

Here μ is the Haar measure with $\int_G d\mu = 1$, $(,)$ is G left invariant

□

Definition 8.6 Suppose G acts on M, G_p is the stablizer of p. The *isotropy representation* is $G_p \to GL(T_pM)$

Example 8.7 Consider Lie group $\mathrm{SL}(2, \mathbb{C})$ whose Lie algebra is $\mathfrak{sl}(2, \mathbb{C})$, which is generated by $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, thus $D_H = -x_1 \frac{\partial}{\partial x_1} + x_2 \frac{\partial}{\partial x_2}, D_X = -x_2 \frac{\partial}{\partial x_1}, D_Y = -x_1 \frac{\partial}{\partial x_2}$

Definition 8.8 For any $A \in T_1 G$, define the exponential map $\exp A \coloneqq \phi_A(1)$ where $\phi_A : \mathbb{R} \to G$ is the one parameter subgroup corresponding to A, also it is easy to see that $\exp tA \coloneqq \phi_{tA}(1) = \phi_A(t)$ which is a scaling of the integral curve, and $\exp(t + s)A = \exp tA \exp sA$ since $\exp tA$ is a one parameter subgroup, and thus $(\exp A)^{-1} = \exp(-A)$

Proposition 8.9 (*Properties of exponential map*)

Let G, H be Lie groups with Lie algebras $\mathfrak{g}, \mathfrak{h}$

(a) The exponential map is a smooth map

(b) $(d \exp)_0 : \mathfrak{g} \cong T_0\mathfrak{g} \to T_1 G \cong \mathfrak{g}$ is the identity map, which implies that the exponential map is a local diffeomorphism around 0

(c) Suppose $\phi : G \to H$ is a Lie group homomorphism, then the following diagram commutes

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{(d\phi)_1} & \mathfrak{h} \\ {\scriptstyle \exp}\downarrow & & \downarrow{\scriptstyle \exp} \\ G & \xrightarrow{\phi} & H \end{array}$$

Proof. *(a)*

(b) For any $A \in \mathfrak{g}$, consider $\gamma : \mathbb{R} \to \mathfrak{g}, t \mapsto tA$ which is a one parameter subgroup of $\mathfrak{g}$, thus $A = \gamma'(0) \in T_0\mathfrak{g}$, and $\exp A = \gamma(1) = A$

(c) Define $\gamma(t) = \phi(\exp tA)$ which is a one parameter subgroup of H since $\gamma(t + s) = \phi(\exp(t + s)A) = \phi(\exp tA \exp sA) = \phi(\exp tA)\phi(\exp sA) = \gamma(t)\gamma(s)$, then $\gamma'(0) = \frac{\partial}{\partial t}|_{t=0}\, \phi(\exp tA) = (d\phi)_1 \frac{\partial}{\partial t}|_{t=0} \exp tA = (d\phi)_1 A$, on the other hand, $\exp(t(d\phi)_1 A)$ is one parameter subgroup of H corresponds to $(d\phi)_1 A = \gamma'(0)$, thus $\exp(t(d\phi)_1 A) = \gamma(t) = \phi(\exp tA)$

□

Proposition 8.10 Let G be a Lie group and $H \leq G$ a Lie subgroup, then $\mathrm{Lie}(H) = \{A \in \mathrm{Lie}\ (G) \mid \exp tA \in H, \forall t \in \mathbb{R}\}$

8.2 Lie algebra

Definition 8.11 A *nonassociative k-algebra* A is an k vector space with a multiplication $\cdot$ such that it is distributive $(a + b) \cdot c = a \cdot c + b \cdot c$, $a \cdot (b + c) = a \cdot b + a \cdot c$. A

- is *unital* if $1 \in A$
- is *symmetric* if $xy = yx$
- is *antisymmetric* if $xy = -yx$
- satisfies *Jacobi identity* if $(xy)z + (yz)x + (zx)y = 0$

A homomorphism $\phi : A \to B$ is a linear map such that $\phi(xy) = \phi(x)\phi(y)$

Definition 8.12 Suppose $e_1, \cdots, e_n$ is a basis of A, $e_i e_j = \sum_i c_k^{ij} e_k$, c_k^{ij} are called *structure constants* with respect to $e_1, \cdots, e_n$. If A satisfies Jacobi identity, then

$$\sum_l c_m^{il} c_l^{jk} + \sum_l c_m^{jl} c_l^{ki} + \sum_l c_m^{kl} c_l^{ij} = 0$$

Definition 8.13 $B \leq A$ is a *subalgebra* if B is a subspace such that $BB \subseteq B$. $I \leq A$ is a *left ideal* if $AI \subseteq I$. Suppose $I, J \leq A$ are ideals, define *ideal quotients* $(J : I) = \{x \in A | xI \subseteq J\}$ which is an ideal. Homomorphisms preserve ideals

Remark. If A is (anti)symmetric, left ideals are two-sided ideals

Definition 8.14 A is *abelian* if $AA = 0$, A is *simple* if it is not abelian and the only ideals are 0 and A, A is *semisimple* if $A = A_1 \oplus \cdots \oplus A_n$ is the direct sum of simple subalgebras, A is *reductive* if $A = \mathfrak{s} \oplus \mathfrak{a}$ is a direct sum of a semisimple subalgebra $\mathfrak{s}$ and an abelian subalgebra $\mathfrak{a}$

Definition 8.15 A *derivation* is an endomorphism $D : A \to A$ such that $D(ab) = D(a)b + aD(b)$. Derivation algebra $\mathrm{Der}_k(A)$ is the set of all derivations. If $D_1, D_2 \in \mathrm{Der}_k(A)$, then $[D_1, D_2] = D_1 D_2 - D_2 D_1 \in \mathrm{Der}_k(A)$, $\mathrm{Der}_k(A) \leq \mathrm{End}_k(A)$ is a Lie subalgebra

Definition 8.16 $R \xrightarrow{\varphi} S$ is a ring homomorphism between commutative rings. The module of *Kähler differentials* is $\Omega_{S/R}$ satisfies the universal property that any derivation uniquely factors through $d_{S/R} : S \to \Omega_{S/R}$, i.e. $\mathrm{Hom}_S(\Omega_{S/R}, M) \cong \mathrm{Der}_R(S, M)$. $\Omega_{S/R}$ can be constructed as

$$\frac{\{ds : s \in S\}}{dr = 0, d(s+t) = ds + dt, d(st) = sdt + tds}$$

Another construction: define I to be the kernel of $S \otimes_R S \to S$, $s \otimes t \mapsto st$, and $\Omega_{S/R} = I/I^2$, with $ds = 1 \otimes s - s \otimes 1$, the free S module generated by ds consists of $t \otimes s - ts \otimes 1$, which is the kernel of $S \otimes_R S \to S \otimes_R R$, $t \otimes s \mapsto ts \otimes 1$, moding I^2 is precisely the Leibniz rule because $d(st) - sdt - tds = (1 \otimes st - st \otimes 1) - (s \otimes t - st \otimes 1) - (t \otimes s - ts \otimes 1) = (1 \otimes s - s \otimes 1)(1 \otimes t - t \otimes 1) \in I^2$

Definition 8.17 $f : X \to Y$ is a morphism of schemes, consider the diagonal $\Delta : Y \to X \times_Y X$, let $I = \ker \Delta^*$, the *cotangent sheaf* is $\Omega_{X/Y} = I/I^2$, and a derivation $d : \mathcal{O}_X \to \Omega_{X/Y}$

Lemma 8.18 A is an associative algebra. $M \in \mathfrak{gl}_n(A)$, $P \in \mathrm{GL}_n(A)$, then $\mathrm{Tr}(PMP^{-1}) - \mathrm{Tr}(M) \in [A, A]$

Proof.

$$\begin{aligned}
\mathrm{Tr}(PMP^{-1}) - \mathrm{Tr}(M) &= \sum_{i,j,k} p_{ij} m_{jk} p^{ki} - \sum_{k,j} \delta_{kj} m_{jk} \\
&= \sum_{i,j,k} p_{ij} m_{jk} p^{ki} - \sum_{i,j,k} p^{ki} p_{ij} m_{jk} \\
&= \sum_{i,j,k} (p_{ij} m_{jk} p^{ki} - p^{ki} p_{ij} m_{jk}) \\
&= \sum_{i,j,k} [p_{ij} m_{jk}, p^{ki}]
\end{aligned}$$

□

Definition 8.19 $\mathfrak{sl}_n(A)$ is defined by the short exact sequence

$$0 \longrightarrow \mathfrak{sl}_n(A) \longrightarrow \mathfrak{gl}_n(A) \xrightarrow{\mathrm{Tr}} A/[A, A] \longrightarrow 0$$

Definition 8.20 A *Lie algebra* $\mathfrak{g}$ is a anticommutative nonassociative k algebra satisfying Jacobi identity, *Lie bracket* $[,] : \mathfrak{g} \wedge \mathfrak{g} \to \mathfrak{g}$ denotes the multiplication. If $\mathrm{char} k = 2$, we also require $[x, x] = 0$

Definition 8.21 A $\mathfrak{g}$ *module* V is an abelian group with left group action $\mathfrak{g} \times V \to V$ such that $1v = v$, $x(v+w) = xv + xw$, $(x+y)v = xv + yv$, $(xy)v = x(yv) - y(xv)$. Equivalently, a *Lie algebra representation* (π, V) is a Lie algebra homomorphism $\pi : \mathfrak{g} \to \mathfrak{gl}(V)$, where V is an k vector space, $xv \coloneqq \pi(x)v$ give V the $\mathfrak{g}$ module structure

Remark. A $\mathfrak{g}$ module is not a module according to Definition 6.1

Definition 8.22 A $\mathfrak{g}$ *module homomorphism* $\phi : V \to W$ between $\mathfrak{g}$ modules is a group homomorphism such that $\phi(xv) = x\phi(v)$. Equivalently, an intertwine map $\phi : V \to W$ between Lie algebra representations is a linear map such that $\phi(\pi_V(x)v) = \pi_W(x)\phi(v)$, giving the $\mathfrak{g}$ module homomorphism

A subrepresentation (π, W) is a $\mathfrak{g}$ submodule $W \leq V$

Definition 8.23 The *adjoint endomorphism* associated to x is left multiplication by x, i.e. $ad(x)(y) = [x, y]$, Jacobi identity becomes $ad([x, y]) = [ad(x), ad(y)]$, give a Lie algebra representation(adjoint representation) $ad : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$, $ad(x)$ are called *inner derivations* since $ad(z)[x, y] = [ad(z)x, y] + [x, ad(z)y]$. $ad(\mathfrak{g}) \leq \mathrm{Der}(\mathfrak{g}) \leq \mathfrak{gl}(\mathfrak{g})$ are Lie subalgebras

Any Lie algebra homomorphism $\phi : \mathfrak{g} \to \mathfrak{h}$ induce a Lie algebra homomorphism $\phi : ad(\mathfrak{g}) \to ad(\mathfrak{h})$ by $\phi(ad(x)) = ad(\phi(x))$

$$\begin{array}{ccc} & \overset{\phi}{\longrightarrow} & \\ \mathfrak{g} & & \mathfrak{h} \\ ad \downarrow & & ad \downarrow \\ & \overset{\phi}{\longrightarrow} & \\ ad(\mathfrak{g}) & & ad(\mathfrak{h}) \end{array}$$

Definition 8.24 Centralizer of S is defined to be $C_{\mathfrak{g}}(S) \coloneqq \{g \in \mathfrak{g} | [g, S] = 0\}$, in particular, the center $Z(\mathfrak{g}) \coloneqq C_{\mathfrak{g}}(\mathfrak{g})$
Normalizer of S is defined to be $N_{\mathfrak{g}}(S) \coloneqq \{g \in \mathfrak{g} | [g, S] \subseteq S\}$

Definition 8.25

$$\mathfrak{g} \supseteq [\mathfrak{g}, \mathfrak{g}] \supseteq [[\mathfrak{g}, \mathfrak{g}], [\mathfrak{g}, \mathfrak{g}]] \supseteq [[[\mathfrak{g}, \mathfrak{g}], [\mathfrak{g}, \mathfrak{g}]], [[\mathfrak{g}, \mathfrak{g}], [\mathfrak{g}, \mathfrak{g}]]] \supseteq \cdots$$

is called the *derived series*, $\mathfrak{g}$ is *solvable* if derived series terminates

$$\mathfrak{g} \supseteq [\mathfrak{g}, \mathfrak{g}] \supseteq [\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]] \supseteq [\mathfrak{g}, [\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]]] \supseteq \cdots$$

is called the *lower central series*, $\mathfrak{g}$ is *nilpotent* if lower central series terminates

Example 8.26 $[\mathfrak{gl}(V), \mathfrak{gl}(V)] = \mathfrak{sl}(V)$

Proof. Since $\mathrm{Tr}[X, Y] = 0$, thus $[\mathfrak{gl}(V), \mathfrak{gl}(V)] \leq \mathfrak{sl}(V)$, conversely

□

Definition 8.27 Let $\mathfrak{g}$ be a Lie algebra, a Cartan subalgebra $\mathfrak{h} \leq \mathfrak{g}$ is a nilpotent subalgebra such that $N_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$(self normalizing or alternatively $(\mathfrak{h} : \mathfrak{h}) = \mathfrak{h}$)

Definition 8.28 Let $\mathfrak{g} \leq \mathfrak{gl}(V)$ be a Lie algebra, $\mathfrak{g}$ is called *toral* if $\mathfrak{g}$ consists of semisimple elements

Definition 8.29 Let $\mathfrak{g}$ be a Lie algebra, we can show the sum of all solvable ideal is again a solvable ideal, thus $\mathfrak{g}$ has a unique maximal solvable ideal $\mathrm{rad}(\mathfrak{g})$, called the *radical* of $\mathfrak{g}$

Definition 8.30 Let $\mathfrak{g}$ be a complex Lie algebra, $\mathfrak{g}_0$ is called a *real form* of $\mathfrak{g}$ if $\mathfrak{g} \cong \mathfrak{g}_0 \otimes_{\mathbb{R}} \mathbb{C}$

Example 8.31 Let $\mathfrak{g} \leq \mathfrak{gl}(V)$ be Lie subalgebra, the *tautological representation* (τ, V) is defined by $\tau(x) = x$, then $\tau([x, y]) = [x, y] = [\tau(x), \tau(y)]$

Proposition 8.32 Lie algebra $\mathfrak{g}$ is reductive iff its adjoint representation is completely reducible

Lemma 8.33 If $\mathfrak{g}$ is semisimple Lie algebra, and $\varphi : \mathfrak{g} \to \mathfrak{gl}(V)$ is a Lie algebra representation, then $\varphi(\mathfrak{g}) \leq \mathfrak{sl}(V)$

Proof. By Proposition 8.51, $\varphi(\mathfrak{g}) = \varphi([\mathfrak{g}, \mathfrak{g}]) \leq [\mathfrak{gl}(V), \mathfrak{gl}(V)] = \mathfrak{sl}(V)$

□

Proposition 8.34 If $\mathfrak{g}$ is a semisimple Lie algebra, then $ad(\mathfrak{g}) = Der(\mathfrak{g})$

Proof. As an abelian ideal of $\mathfrak{g}$, $Z(\mathfrak{g}) = 0$, thus $\mathfrak{g} \overset{ad}{\to} ad(\mathfrak{g}) \leq \mathfrak{gl}(\mathfrak{g})$ is an embedding. Since $[\delta, ad(x)] = ad(\delta(x)), \delta \in Der(\mathfrak{g})$, thus $[Der(\mathfrak{g}), ad(\mathfrak{g})] \subseteq ad(\mathfrak{g})$, Let $K(,)$ be the Killing form on $Der(\mathfrak{g})$, due to Proposition 8.56(b) and Proposition 8.44(c), $K(,)|_{ad(g)}$ is nondegenerate, denote $I \coloneqq ad(\mathfrak{g})^{\perp}$ under $K(,)$, then $I \cap ad(\mathfrak{g}) = 0$, otherwise $0 \neq I \cap ad(\mathfrak{g}) \subseteq \ker K(,)|_{ad(\mathfrak{g})}$, by Exercise 35.84, $[I, ad(\mathfrak{g})] = 0$, thus for any $\delta \in I$, $0 = [\delta, ad(x)] = ad(\delta(x))$, since ad is an isomorphism, $\delta(x) = 0$, thus $\delta = 0$, $I = 0$, $ad(\mathfrak{g}) = Der(\mathfrak{g})$

□

Remark. When $\mathfrak{g}$ is a semisimple Lie algebra, $\mathfrak{g} \overset{ad}{\to} ad(\mathfrak{g}) \leq \mathfrak{gl}(\mathfrak{g})$ is an embedding, we can identify x with $ad(x)$, by abuse of notations, xy can defined to be the preimage of $ad(x)ad(y) \in \mathfrak{gl}(\mathfrak{g})$

Lemma 8.35 Let G be a compact Lie group, we can pick any nonzero k-form α_I at I, and extend it to a k-form on G by $\alpha_A(Y_1, \cdots, Y_k) = \alpha_I(Y_1 A^{-1}, \cdots, Y_k A^{-1})$, or just $R_A^* \alpha_A = \alpha_I$, then we can define integral $\int_G f(A)\alpha$, then we would have $\int_G f(AB)\alpha = \int_G f(AB)\alpha_A = \int_G f(AB) R_B^* \alpha_{AB} = \int_G f(A) R_B^* \alpha = \int_G f(A)\alpha$, since $R_B^* \alpha = \alpha$, i.e. $(R_B^* \alpha)_A(X_A) = R_B^* \alpha_{AB}(X_A) = \alpha_A(X_A)$, thus this integration is right invariant. Note that this actually gives a right invariant Haar measure

Theorem 8.36 *Weyl's theorem*
Let $\mathfrak{g}$ be a semisimple Lie algebra and $\varphi : \mathfrak{g} \to \mathfrak{gl}(V)$ is a Lie algebra representation, then φ is completely reducible, namely, $\mathfrak{g}$ modules are semisimple(completely reducible), thus any irreducible subrepresentation has to be one of the summand in the decomposition

Proof. Weyl's unitary trick

□

Lemma 8.37 Let $V \neq 0$ be a finite dimensional vector space, suppose $\mathfrak{g} \leq \mathfrak{gl}(V)$ is a Lie subalgebra consists of nilpotent elements, then there exists $0 \neq v \in V$ such that $\mathfrak{g}v = 0$

Theorem 8.38 *Engel's theorem* Consider the adjoint representation $(ad, \mathfrak{g})$ of a finite dimensional Lie algebra $\mathfrak{g}$, then $\mathfrak{g}$ nilpotent iff $ad(X), X \in \mathfrak{g}$ can be strictly upper triangulized simultaneously iff $ad(X)$ is nilpotent for any $X \in \mathfrak{g}$

Lemma 8.39 Let $\mathfrak{g}$ be a nilpotent Lie algebra, $I \leq \mathfrak{g}$ is a nonzero ideal, then $I \cap Z(\mathfrak{g}) \neq 0$, in particular, if $I = \mathfrak{g}$ then $Z(\mathfrak{g}) \neq 0$ which can also easily being shown from the fact that $Z(\mathfrak{g})$ contains the last nonzero term in the lower central series of $\mathfrak{g}$

Proof. Consider adjoint map restrict on I, since $\mathfrak{g}$ is nilpotent, $ad(X)$ is nilpotent for any $X \in \mathfrak{g}$, so is $ad(X)|_I$, i.e. $ad(\mathfrak{g})_I \leq \mathfrak{gl}(I)$ is a Lie subalgebra consists of nilpotent elements, by Lemma 8.37, there exists $0 \neq Y \in I$ such that $[X, Y] = 0, \forall X \in \mathfrak{g}$, thus $Y \in I \cap Z(\mathfrak{g})$

□

Theorem 8.40 *Lie's theorem*
If (π, V) is a finite representation of a finite dimensional Lie algebra $\mathfrak{g}$ with $\overline{k} = k, \mathrm{char} k = 0$, if $\mathfrak{g}$ is solvable, so is $\pi(\mathfrak{g})$, and $\pi(X), X \in \mathfrak{g}$ can be upper triangulized simultaneously

Remark. If (π, V) is a finite representation of a finite dimensional Lie algebra $\mathfrak{g}$ with $\overline{k} = k, \mathrm{char} k = 0$, if $\mathfrak{g}$ is abelian, so is $\pi(\mathfrak{g})$, but it doesn't imply $\pi(X), X \in \mathfrak{g}$ can be diagonalized simultaneously, for example $\left\langle \begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix} \right\rangle \subseteq \mathfrak{gl}(\mathbb{C}^2)$ is abelian, and $\left\langle \begin{pmatrix} 1 & 1 \\ & 1 \end{pmatrix} \right\rangle$ is not diagonalizable at all
However, due to Proposition 35.86, if (π, V) is a finite representation of a finite dimensional Lie algebra

$\mathfrak{g}$ with $\overline{k} = k$, where $\mathfrak{g}$ is abelian, so is $\pi(\mathfrak{g})$, and suppose $\pi(X), X \in \mathfrak{g}$ are diagonalizable($\pi(\mathfrak{g})$ is a toral Lie subalgebra), then they can be diagonalized simultaneously

Definition 8.41 A bilinear form $(,)$ on Lie algebra $\mathfrak{g}$ is *invariant* or *associative* if the Lie derivative is zero, i.e. $(ad_Y X, Z) + (X, ad_Y Z) = 0$, or equivalently, $([X,Y],Z) = (X,[Y,Z])$

Definition 8.42 A *quadratic Lie algebra* is a Lie algebra $\mathfrak{g}$ with an invariant nondegenerate symmetric bilinear form $(,) : \mathfrak{g} \otimes \mathfrak{g} \to k$

Definition 8.43 *Killing form* is the bilinear map $K(,) : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$, $(X,Y) \mapsto \mathrm{Tr}(ad(X)ad(Y))$

Proposition 8.44

1. The Killing form is symmetric and invariant
2. The Killing form on a nilpotent Lie algebra is zero
3. Suppose $I \leq \mathfrak{g}$ is an ideal, then Killing form $K_I(,)$ on I is the same as the restriction of Killing form $K(,)$ to I, i.e. $K_I(,) = K(,)|_I$

Proof.

1.
2. Due to Theorem 8.38
3. By Exercise 35.5, for $X, Y \in I$, we have
$$\begin{aligned} K_I(X,Y) &= Tr(ad(X)|_I \; ad(Y)|_I) \\ &= Tr((ad(X)ad(Y)) \; |_I) \\ &= Tr(ad(X)ad(Y)) \\ &= K(X,Y) \\ &= K(X,Y)|_I \end{aligned}$$

□

Example 8.45 The Killing form is a symmetric, bilinear and invariant form, and it is nondegenerate iff $\mathfrak{g}$ is semisimple due to Proposition 8.56

Lemma 8.46 Any invariant, symmetric and bilinear form on simple Lie algebra $\mathfrak{g}$ is a multiple of the Killing form

Proof. Suppose $(,)$ is an invariant, symmetric and bilinear form, so is $[,]_c = (,) - cK(,)$ for any c. If $(,) \neq 0$, then there exists $x, y \in \mathfrak{g}$ such that $[x,y]_c = 0$ for some c, since the kernel of $[,]_c$ is a nonzero ideal, $[,]_c = 0$

□

Lemma 8.47 Let $\mathfrak{g}$ be a finite dimensional nonassociative $\mathbb{F}$ algebra(including Lie algebra) with $\overline{k} = k$, for any $\delta \in Der(\mathfrak{g}) \leq \mathfrak{gl}(\mathfrak{g})$, let $\delta = \delta_s + \delta_n$ be its Jordan-Chevalley decomposition in $\mathfrak{gl}(\mathfrak{g})$, then $\delta_s, \delta_n \in Der(\mathfrak{g})$

Proof. For any $a \in k$, define $\mathfrak{g}_a$ be the generalized eigenspace of a, then we have $\mathfrak{g} = \bigoplus_{a \in k} \mathfrak{g}_a$, and $[\mathfrak{g}_a, \mathfrak{g}_b] \subseteq \mathfrak{g}_{a+b}$, since for any $x \in \mathfrak{g}_a, y \in \mathfrak{g}_b$, $(\delta - (a+b)1_{\mathfrak{g}})^m([x,y]) = \sum_{k=0}^m \binom{m}{k} \left[(\delta - a1_{\mathfrak{g}})^{m-k} x, (\delta - b1_{\mathfrak{g}})^k y\right]$ which can be easily checked by induction. Then we have $\delta_s(x) = ax, \delta_s(y) = by$, and $\delta_s([x,y]) = (a+b)[x,y] = [ax,y] + [x,by] = [\delta_s(x), y] + [x, \delta_s y]$, thus $\delta_s \in Der(\mathfrak{g})$, so does $\delta_n = \delta - \delta_s$

□

Definition 8.48 *Abstract Jordan-Chevalley decomposition on semisimple Lie algebras*
Because Lemma 8.47 and Proposition 8.34, for any $x \in \mathfrak{g}$, we can identify x with $ad(x)$, and we have Jordan-Chevalley decomposition $ad(x) = ad(x)_s + ad(x)_n = ad(x_s) + ad(x_n)$, where x_s, x_n are defined to be the preimages of $ad(x)_s, ad(x)_n$. Moreover, there exists polynomials $p(t), q(t)$ with no constant

terms such that $ad(x_s) = ad(x)_s = p(ad(x))$, $ad(x_n) = ad(x)_n = q(ad(x))$, by abuse of notations, $x_s = p(x)$ and $x_n = q(x)$

Theorem 8.49 Suppose V is a finite dimensional k vector space with $\overline{k} = k$, $\mathrm{char}k = 0$, $\mathfrak{g} \leq \mathfrak{gl}(V)$ is a semisimple Lie algebra, for any $x \in \mathfrak{g}$, $x = x_s + x_n$ is the Jordan-Chevalley decomposition in $\mathfrak{gl}(V)$, moreover, the abstract and usual Jordan-Chevalley decompositions coincide, i.e. $ad(x_s) = ad(x)_s, ad(x_n) = ad(x)_n$

Proof. Define lie subalgebras $\mathfrak{l}_W \coloneqq \{y \in \mathfrak{gl}(V) \mid yW \subseteq W, Tr(y\mid_W) = 0\}$ with $W \leq V$ being $\mathfrak{g}$ submodules, and define $\mathfrak{l} = \left(\bigcap_W \mathfrak{l}_W\right) \bigcap N_{\mathfrak{gl}(V)}(\mathfrak{g})$, for any $x \in \mathfrak{g}$, due to Proposition 35.5 and Lemma 8.33, $Tr(x|_W) = Tr(x) = 0$, $\mathfrak{g} \leq \mathfrak{l}_W \Rightarrow \mathfrak{g} \leq \mathfrak{l}$, thus $\mathfrak{l}$ is a subalgebra of $N_{\mathfrak{gl}(V)}(\mathfrak{g})$ of containing $\mathfrak{g}$, thus $\mathfrak{l}$ is finite dimensional $\mathfrak{g}$ module, by Theorem 8.36, $\mathfrak{l} = \mathfrak{g} \oplus \mathfrak{h}$ is a direct sum of $\mathfrak{g}$ modules, since $\mathfrak{l} \leq N_{\mathfrak{gl}(V)}(\mathfrak{g})$, $[\mathfrak{g}, \mathfrak{l}] = 0 \Rightarrow [\mathfrak{g}, \mathfrak{h}] = 0$, i.e. $\mathfrak{g}$ acts trivially on $\mathfrak{h}$, fix any irreducible $\mathfrak{g}$ submodule W, for any $y \in \mathfrak{h}$, $x \in \mathfrak{g}$, $xy - yx = [x, y] = 0$, $yxv = xyv$ for $v \in W$, i.e. $y \in \mathrm{Hom}_{\mathfrak{g}}(W, W)$, by Lemma 6.13, y acts on W as a scalar, but $Tr(y|_W) = 0$, thus y acts trivially on W, again by Theorem 8.36, V can written as the direct sum of irreducible $\mathfrak{g}$ submodules, thus y acts trivially on $W \Rightarrow y = 0$, therefore $\mathfrak{h}_j = 0 \Rightarrow \mathfrak{g} = \mathfrak{l}$, for any $x \in \mathfrak{g}$, due to Theorem 3.20, $x = x_s + x_n$ and $x_s = p(x), x_n = q(x)$ for some polynomials $p(x), q(x)$ with no constant terms, thus if $x \in \mathfrak{l}_W$, $xW \subseteq W$ and $Tr(x|_W) = 0$, then $x_sW = p(x)W \subseteq W$, $Tr(x_s|_W) = Tr(p(x|_W)) = 0$, similarly, $x_nW \subseteq W$, $Tr(x_n|_W) = 0$, $x_s, x_n \in \mathfrak{l}_W$, also $x_s = p(x), x_n = q(x) \in N_{\mathfrak{gl}(V)}(\mathfrak{g})$, thus $x_s, x_n \in \mathfrak{l} = \mathfrak{g}$
Since the Jordan-Chevalley decomposition of $ad(x)$ in $\mathfrak{gl}(V)$ is unique and $ad(x_s) + ad(x_n) = ad(x) = ad(x)_s + ad(x)_n$, thus $ad(x_s) = ad(x)_s, ad(x_n) = ad(x)_n$

□

Corollary 8.50 Suppose V is a finite dimensional k vector space with $\overline{k} = k$, $\mathrm{char}k = 0$, $\mathfrak{g}$ is a semisimple Lie algebra, and $\phi : \mathfrak{g} \to \mathfrak{gl}(V)$ is a Lie algebra representation, $x = x_s + x_n$ is the abstract Jordan-Chevalley decomposition, then $\phi(x) = \phi(x_s) + \phi(x_n)$ is the usual Jordan-Chevalley decomposition in $\mathfrak{gl}(V)$

Proof. Due to Proposition 8.52, $\phi(\mathfrak{g})$ is also semisimple
First notice that a linear operator $T \in \mathrm{End}_k(V)$ is diagonalizable(semisimple) iff the dimension of the sum of its eigenspaces is $\dim V$, or equivalently iff all eigenvectors span V
Since $ad(x_s)$ is semisimple in $\mathfrak{gl}(\mathfrak{g})$ as in Proposition 8.34, the eigenvectors y_i's of $ad(x_s)$ spans $\mathfrak{g}$, say $ad(x_s)(y_i) = \lambda_i y_i$, then as in Definition 8.23, $ad(\phi(x_s))(\phi(y_i)) = [\phi(x_s), \phi(y_i)] = \phi([x_s, y_i]) = \phi(ad(x_s)(y_i)) = \phi(\lambda_i y_i) = \lambda_i \phi(y_i)$, thus those $0 \neq \phi(y_i)$ are eigenvectors of $\phi(\mathfrak{g})$, hence $ad(\phi(x_s))$ is semisimple in $\mathfrak{gl}(\mathfrak{gl}(V))$
Since $ad(x_n)$ is semisimple in $\mathfrak{gl}(\mathfrak{g})$ as in Proposition 8.34, $ad(x_n)^m = 0$ for some m, then as in Definition 8.23 $ad(\phi(x_n))^m = \phi(ad(x_n))^m = \phi(ad(x_n)^m) = 0$, thus $ad(\phi(x_n))$ is also nilpotent in $\mathfrak{gl}(\mathfrak{gl}(V))$
Moreover, as in Definition 8.23, $[ad(\phi(x_s)), ad(\phi(x_n))] = [\phi(ad(x_s)), \phi(ad(x_n))] = \phi([ad(x_s), ad(x_n)]) = 0$
Thus $\phi(x) = \phi(x_s) + \phi(x_n)$ is the abstract Jordan-Chevalley decomposition of $\phi(x)$ in $\phi(\mathfrak{g}) \leq \mathfrak{gl}(V)$, by Theorem 8.49, this coincide with the usual Jordan-Chevalley decomposition of $\phi(x) = \phi(x)_s + \phi(x)_n$ in $\mathfrak{gl}(V)$, i.e. $\phi(x_s) = \phi(x)_s$, $\phi(x_n) = \phi(x)_n$

□

Proposition 8.51 Let $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_\mathfrak{n}$ be a semisimple Lie algebra, then any ideal of $\mathfrak{g}$ is certain sum of $\mathfrak{g}_i$'s, and any sum of $\mathfrak{g}_i$'s is an ideal, in particular, $\mathfrak{g}_i$'s are ideals, moreover, $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$

Proof. any ideal $I \leq \mathfrak{g}$ is certain sum of $\mathfrak{g}_i$'s, because $I \cap \mathfrak{g}_i$ is an ideal of $\mathfrak{g}_i$ which is either 0 or $\mathfrak{g}_i$ itself
If $\mathfrak{g}$ is a simple Lie algebra, then it is not abelian, $[\mathfrak{g}, \mathfrak{g}] \neq 0$, thus $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$, generally, $[\mathfrak{g}, \mathfrak{g}] = [\mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_\mathfrak{n}, \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_\mathfrak{n}] = [\mathfrak{g}_1, \mathfrak{g}_1] \oplus \cdots \oplus [\mathfrak{g}_n, \mathfrak{g}_n] = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_\mathfrak{n} = \mathfrak{g}$

□

Proposition 8.52 If $\mathfrak{g}$ is semisimple, $\phi : \mathfrak{g} \to \mathfrak{h}$ is a Lie algebra homomorphism, then $\phi(\mathfrak{g})$ is also semisimple

Proof. Suppose $\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_\mathfrak{n}$ is a direct sum of simple Lie algebras, $\phi(\mathfrak{g}_i)$ are also ideals so $\phi(\mathfrak{g}) = \phi(\mathfrak{g}_1) \oplus \cdots \oplus \phi(\mathfrak{g}_n)$ is a direct sum of ideals, and $[\phi(\mathfrak{g}_i), \phi(\mathfrak{g}_i)] = \phi([\mathfrak{g}_i, \mathfrak{g}_i]) = \phi(\mathfrak{g}_i)$ implies that each $\phi(\mathfrak{g}_i)$ is simple or 0, thus $\phi(\mathfrak{g})$ is also semisimple

□

Proposition 8.53 Let $\mathfrak{g} \leq \mathfrak{gl}(V)$ be a Lie algebra, $X \in \mathfrak{g}$, then X is nilpotent $\Rightarrow ad(X)$ is nilpotent, if in addition V is finite dimensional and $\overline{k} = k$, then X is semisimple $\Rightarrow ad(X)$ is semisimple, or rather diagonalizable

Proof. Let L_X, R_X be left and right multiplications by X, then we have $ad_X = L_X - R_X$ and $[L_X, R_X] = 0$, thus X is nilpotent $\Rightarrow ad(X)$ is nilpotent
Notice that given $A = (a_{ij})$, $D = \text{diag}(d_1, \cdots, d_n)$, $[D, A] = ((d_i - d_j)a_{ij})$, thus $[D, E_{ij}] = (d_i - d_j)E_{ij}$, thus X is diagonalizble $\Rightarrow ad(X)$ is diagonalizable

□

Theorem 8.54 (Cartan's criterion for solvability) Let V be a finite dimensional k vector space with char $k = 0$, $\mathfrak{g} \leq \mathfrak{gl}(V)$ is a Lie subalgebra, then $\mathfrak{g}$ is solvable iff $\text{Tr}(XY) = 0$, $\forall X \in \mathfrak{g}, Y \in [\mathfrak{g}, \mathfrak{g}]$

Corollary 8.55 *Cartan's criterion for semisimplicity*
Let $\mathfrak{g}$ is finite dimensional Lie algebra with char $k = 0$, then $\mathfrak{g}$ is semisimple iff its Killing form is nondegenerate

Proposition 8.56 The following statements are equivalent

(a) $\mathfrak{g}$ is semisimple

(b) The Killing form is nondegenerate

(c) $\mathfrak{g}$ doesn't have have nontrivial abelian ideals

(d) $\mathfrak{g}$ doesn't have have nontrivial solvable ideals

(e) $\text{rad}(\mathfrak{g}) = 0$

Proof. $(a) \Leftrightarrow (b)$ is due to Corollary 8.55

□

Example 8.57 Recall $\mathfrak{sl}(2, k) = \{X \in M(2, k) \mid \text{Tr } (X) = 0\} = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \;\middle|\; a, b, c \in k \right\}$
$\mathfrak{sl}(2, k) = \langle H, X, Y \rangle$, where $X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, and $ad_H X = [H, X] = 2X, ad_H Y = [H, Y] = -2Y, ad_X Y = [X, Y] = H$, this is the adjoint representation of $\mathfrak{sl}(2, k)$

Lemma 8.58 Let (π, V) be a finite dimensional representation of $\mathfrak{sl}(2, k)$, if $V = k^n$, then there are theree $n \times n$ matrices x, y, h such that $[h, x] = 2x, [h, y] = -2y, [x, y] = h$ due to Example 8.57

Lemma 8.59 Let (π, V) be a finite dimensional representation of $\mathfrak{sl}(2, k)$, $V_\lambda \coloneqq \{v \in V \mid \pi(H)v = \lambda v\}$, then $\pi(X)V_\lambda \subseteq V_{\lambda+2}$, $\pi(Y)V_\lambda \subseteq V_{\lambda-2}$ and $\pi(H)V_\lambda \subseteq V_\lambda$

Proof. $\pi(H)V_\lambda \subseteq V_\lambda$ is just by definition, suppose $v \in V_\lambda$, $\pi(H)\pi(X)v = 2\pi(X)v + \pi(X)\pi(H)v = (\lambda + 2)\pi(X)v$, $\pi(H)\pi(Y)v = -2\pi(Y)v + \pi(Y)\pi(H)v = (\lambda - 2)\pi(Y)v$,

□

Remark. $\pi(X), \pi(Y)$ are named *raising and lowering operator*

Theorem 8.60 Suppose $\overline{k} = k$, char $k = 0$, for any integer $m \geq 0$, there is an irreducible representation of $\mathfrak{sl}(2, k)$ with dimension $m + 1$

Proof. Let (π, V) be a finite dimensional irreducible representation of $\mathfrak{sl}(2,k)$, there exists highest $\lambda \in k$ such that $V_\lambda \neq 0$, pick $0 \neq u \in V_\lambda$, let $u_k \coloneqq \pi(Y)^k u \in V_{\lambda-2k}$, then there exists m such that $u_m \neq 0$ but $u_{m+1} = 0$, then $u_0, \cdots, u_m$ are independent since they belong distinct eigenspaces, with $\pi(H)u_k = (\lambda - 2k)u_k$, $\pi(X)u_0 = \pi(X)u = 0$ since $u \in V_\lambda$ is of "highest weight", and $\pi(X)u_k = k(\lambda - k + 1)u_{k-1}, k > 0$ by induction, $\pi(X)u_1 = \pi(X)\pi(Y)u = ([\pi(X), \pi(Y))] + \pi(Y)\pi(X))u = \pi(H)u + \pi(Y)\pi(X)u = \lambda u = 1(\lambda - 1 + 1)u_0$, $\pi(X)u_{k+1} = \pi(X)\pi(Y)u_k = ([\pi(X), \pi(Y))] + \pi(Y)\pi(X))u_k = \pi(H)u_k + \pi(Y)\pi(X)u_k = (\lambda - 2k)u_k + k(\lambda - k + 1)\pi(Y)u_{k-1} = (k+1)(\lambda - k)u_k$
Note that since $0 = Xu_{m+1} = (m+1)(\lambda - m)u_m \Rightarrow \lambda = m$, which implies all possible eigenvalue for $\pi(H)$ has to be integers, when m is even, we call this irreducible representation even, when m is odd, we call this irreducible representation odd
In general, for any finite dimensional representation, we can decompose the representation into irreducible subrepresentations by using this procedure repeatedly
Therefore $0 \neq W \coloneqq \langle u_0, \cdots, u_m \rangle$ is invariant, but π is irreducible, thus $V = W$, and by Lemma 6.13, (π, V) is unique up to isomorphism

□

Example 8.61 $(ad, \mathfrak{sl}(2,k))$ is the unique irreducible 3 dimensional representation of $\mathfrak{sl}(2,k)$ with $V_0 = \langle H \rangle$, $V_{-2} = \langle Y \rangle$ and $V_2 = \langle X \rangle$, it is irreducible because of Lemma 8.64, if we use X, Y, H as basis, then $ad(X), ad(Y), ad(H)$ would have the matrix forms

$$\begin{pmatrix} 0 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Thus

$$K(X,X) = Tr\begin{pmatrix} 0 & -2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0, \quad K(X,Y) = Tr\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} = 4$$

$$K(X,H) = Tr\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -2 & 0 \end{pmatrix} = 0, \quad K(Y,Y) = Tr\begin{pmatrix} 0 & 0 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

$$K(Y,H) = Tr\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -2 & 0 & 0 \end{pmatrix} = 0, \quad K(H,H) = Tr\begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 8$$

Thus its Cartan matrix is $\Phi = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix}$ which is nondegenerate

Example 8.62 The tautological representation (τ, k^2) is the unique irreducible 2 dimensional representation of $\mathfrak{sl}(2,k)$ with $V_1 = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle$, $V_{-1} = \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle$

Example 8.63 Let $S^k(k^2)$ be the k-th symmetric power of k^2 which is isomorphic to the the set of degree k polynomials in $k[x,y]$ generated by
$\langle x^k, x^{k-1}y, \cdots, xy^{k-1}, y^k \rangle$ which is of dimension $k+1$, with this identification, $\left(\pi, S^k(k^2)\right)$ with $\pi(X)(x) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0$, $\pi(X)y = x$, $\pi(Y)x = y$, $\pi(Y)y = 0$, $\pi(H)x = x$, $\pi(H)y = -y$, just as in Example 8.62, and define inductively that $\pi(Z)(fg) = g\pi(Z)f + f\pi(Z)g$, this is the unique $k+1$ dimensional irreducible representation of $\mathfrak{sl}(2,k)$

Lemma 8.64 Let (π, V) be a finite dimensional irreducible representation of $\mathfrak{sl}(2,k)$, and V_k be the k-eigenspace of $\pi(H)$, then the number of irreducible summand of (π, V) is $\dim V_0 + \dim V_1$, whereas $\dim V_0, \dim V_1$ are number of even and odd irreducible summands

Proof. In an even irreducible representation of $\mathfrak{sl}(2,k)$ all the eigenvalues are even, so there is a unique 0-eigenvector, in an odd irreducible representation of $\mathfrak{sl}(2,k)$ all the eigenvalues are odd, so there is a unique 1-eigenvector, thus the number of irreducible summand of (π, V) is $\dim V_0 + \dim V_1$

□

Definition 8.65 $\mathfrak{g}$ is a semisimple Lie algebra, $\mathfrak{h}$ is a maximal toral Lie algebra. For $\alpha \in \mathfrak{h}^* = \mathrm{Hom}_{\mathbb{F}}(\mathfrak{h}, k)$, define *root spaces*

$$\mathfrak{g}_\alpha = \{x \in \mathfrak{g} \mid ad_h(x) = [h, x] = \alpha(h)x, \forall h \in \mathfrak{h}\}$$

α is a *root* if $\alpha \neq 0$ and $\mathfrak{g}_\alpha \neq 0$, denote the set of roots as Δ. $\mathfrak{g}_0 = C_{\mathfrak{g}}(\mathfrak{h})$ is the centralizer of $\mathfrak{h}$

Proposition 8.66

1. $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$
2. $\alpha \in \Delta$, any $X \in \mathfrak{g}_\alpha$ is nilpotent
3. $K(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$ unless $\alpha + \beta = 0$
4. $K(,)|_{\mathfrak{g}_0}$ is nondegenerate

Proof.

1. For $X \in \mathfrak{g}_\alpha$, $Y \in \mathfrak{g}_\beta$, $Z \in \mathfrak{h}$, we have
$$[Z, [X, Y]] = [[Z, X], Y] + [X, [Z, Y]] = \alpha(Z)[X, Y] + \beta(Z)[X, Y] = (\alpha + \beta)(Z)[X, Y]$$
2. For $\beta \in \Delta \cup \{0\}$, $\alpha \in \Delta$, $X \in \mathfrak{g}_\alpha, Y \in \mathfrak{g}_\beta$, $ad(X)^n(Y) \in \mathfrak{g}_{n\alpha+\beta} = 0$ when n is big enough, thus $ad(X)$ is nilpotent
3. Supppose $\alpha + \beta \neq 0$, then there exists $Z \in \mathfrak{h}$ such that $(\alpha + \beta)(Z) \neq 0$, then for any $X \in \mathfrak{g}_\alpha, Y \in \mathfrak{g}_\beta$
$$\begin{aligned}(\alpha + \beta)(Z)K(X, Y) &= \alpha(Z)K(X, Y) + \beta(Z)K(X, Y) \\ &= K(\alpha(Z)X, Y) + K(X, \beta(Z)Y) \\ &= K([Z, X], Y) + K(X, [Z, Y]) \\ &= -K([X, Z], Y) + K(X, [Z, Y]) \\ &= 0\end{aligned}$$
Thus $K(X, Y) = 0$
4. Since $K(\mathfrak{g}_\alpha, \mathfrak{h}) = 0, \forall \alpha \in \Delta$, $\ker K(,)|_{\mathfrak{g}_0} \subseteq \ker K(,) = 0$, thus $K(,)|_{\mathfrak{g}_0}$ is nondegenerate

□

Theorem 8.67 Semisimple Lie algebra $\mathfrak{g}$ has root space decomposition $\mathfrak{g} = \bigoplus_{\alpha \in \Delta \cup \{0\}} \mathfrak{g}_\alpha$

Proof. By Proposition 8.66

□

Lemma 8.68 V is an k vector space, $x, y \in \mathfrak{gl}(V)$ commutes, x is nilpotent, then xy is nilpotent, and $Tr(xy) = 0$

Proof. $x^m = 0 \Rightarrow (xy)^m = x^m y^m = 0$

□

Proposition 8.69 For semisimple Lie algebra $\mathfrak{g}$ with maximal toral Lie subalgebra $\mathfrak{h}$, $C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$

Proof.

1. $C_{\mathfrak{g}}(\mathfrak{h})$ contains semisimple and nilpotent parts of its elements

If $x \in C_{\mathfrak{g}}(\mathfrak{h})$, due to Proposition 8.48, there are polynomials $p(t), q(t)$ with no constant terms such that $ad(x_s) = ad(x)_s = p(ad(x)), ad(x_n) = ad(x)_n = q(ad(x))$, since $x \in C_{\mathfrak{g}}(\mathfrak{h})$, $ad(x)|_{\mathfrak{h}} = 0$, , $ad(x_s)|_{\mathfrak{h}} = p(ad(x))|_{\mathfrak{h}} = 0$, $ad(x_n)|_{\mathfrak{h}} = q(ad(x))|_{\mathfrak{h}} = 0$, thus $x_s, x_n \in C_{\mathfrak{g}}(\mathfrak{h})$

2. $\mathfrak{h}$ contains all semisimple elements of $C_{\mathfrak{g}}(\mathfrak{h})$

   If $s \in C_{\mathfrak{g}}(\mathfrak{h})$ be a semisimple element, use Exercise 35.85, s and elements of $\mathfrak{h}$ are diagonalizable simultaneously, thus $\mathfrak{h} + \langle s \rangle$ is toral in $\mathfrak{g}$, then $s \in \mathfrak{h}$ since $\mathfrak{h}$ is maximal

3. $K(,)|_{\mathfrak{h}}$ is nondegenerate

   Suppose there exists $h \in \mathfrak{h}$ such that $K(h, \mathfrak{h}) = 0$, if $n \in C_{\mathfrak{g}}(\mathfrak{h})$ be a nilpotent element, then $ad(n)$ is nilpotent, and $[n, \mathfrak{h}] = 0$, thus $[ad(n), ad(h)] = ad([n, h]) = 0$, by Lemma 8.68, $Tr(ad(n)ad(h)) = 0$, if $s \in C_{\mathfrak{g}}(\mathfrak{h})$ be a semisimple element, according to Step II, $s \in \mathfrak{h}$, thus $K(s, h) = 0$, and according to Step I, $K(h, C_{\mathfrak{g}}(\mathfrak{h})) = 0$ which contradicts Proposition 8.66(d) that $K(,)|_{C_{\mathfrak{g}}(\mathfrak{h})}$ is nondegenerate

4. $C_{\mathfrak{g}}(\mathfrak{h})$ is nilpotent

   If $n \in C_{\mathfrak{g}}(\mathfrak{h})$ be a nilpotent element, then $ad(n)$ is nilpotent, so is $ad(n)|_{C_{\mathfrak{g}}(\mathfrak{h})}$, if $s \in C_{\mathfrak{g}}(\mathfrak{h})$ be a semisimple element, according to Step II, $s \in \mathfrak{h}$, $ad(s)|_{C_{\mathfrak{g}}(\mathfrak{h})} = 0$, by Theorem 8.38, $C_{\mathfrak{g}}(\mathfrak{h})$ is nilpotent

5. $\mathfrak{h} \cap [C_{\mathfrak{g}}(\mathfrak{h}), C_{\mathfrak{g}}(\mathfrak{h})] = 0$

   Suppose $x \in \mathfrak{h} \cap [C_{\mathfrak{g}}(\mathfrak{h}), C_{\mathfrak{g}}(\mathfrak{h})]$, then $x = \sum [y_i, z_i]$ where $y_i, z_i \in C_{\mathfrak{g}}(\mathfrak{h})$, then $K(x, x) = \sum K(x, [y_i, z_i]) = \sum K([x, y_i], z_i) = 0$, since $K(,)$ is nondegenerate on $\mathfrak{h}$(or $\mathfrak{g}$ or $C_{\mathfrak{g}}(\mathfrak{h})$), thus $x = 0$

6. $C_{\mathfrak{g}}(\mathfrak{h})$ is abelian

   Suppose $[C_{\mathfrak{g}}(\mathfrak{h}), C_{\mathfrak{g}}(\mathfrak{h})] \neq 0$, since $C_{\mathfrak{g}}(\mathfrak{h})$ is nilpotent from Step IV, by Lemma 8.39, there exists $0 \neq z \in Z(C_{\mathfrak{g}}(\mathfrak{h})) \cap [C_{\mathfrak{g}}(\mathfrak{h}), C_{\mathfrak{g}}(\mathfrak{h})]$, then z can't be semisimple, otherwise $z \in \mathfrak{h} \cap [C_{\mathfrak{g}}(\mathfrak{h}), C_{\mathfrak{g}}(\mathfrak{h})]$, contradicting Step V, thus its nilpotent part $n \neq 0$, but its semisimple part $s \in \mathfrak{h} \leq Z(C_{\mathfrak{g}}(\mathfrak{h}))$, so is $n = z - s$, but then $[n, C_{\mathfrak{g}}(\mathfrak{h})] = 0$, by Lemma 8.68, $K(n, C_{\mathfrak{g}}(\mathfrak{h})) = 0$, contradicting Proposition 8.66(d)

7. $\mathfrak{h} = C_{\mathfrak{g}}(\mathfrak{h})$

   Suppose $x \in C_{\mathfrak{g}}(\mathfrak{h}) \setminus \mathfrak{h}$, then it gives a nonzero nilpotent part n, but then since $C_{\mathfrak{g}}(\mathfrak{h})$ is abelian by Step VI, thus $[n, C_{\mathfrak{g}}(\mathfrak{h})] = 0$, by Lemma 8.68, $K(n, C_{\mathfrak{g}}(\mathfrak{h})) = 0$, contradicting Proposition 8.66(d)

□

Remark. $K(,)|_{\mathfrak{h}}$ is nondegenerate is not the same as saying that the Killing form of $\mathfrak{h}$ is nondegenerate which obviously violates Proposition 8.56, it doesn't contradict Proposition 8.44 since $\mathfrak{h} \leq \mathfrak{g}$ is merely a Lie subalgebra but not an ideal, by the nondegeneracy, we can identity $\mathfrak{h}^*$ with $\mathfrak{h}$ by $\mathfrak{h}^* \to \mathfrak{h}, \alpha \mapsto t_\alpha$, where $K(t_\alpha, x) = \alpha(x)$, and here t behaves like the linear isomorphism $t : \mathfrak{h}^* \to \mathfrak{h}, \alpha \mapsto t_\alpha$

Proposition 8.70

1. Δ spans $\mathfrak{h}^*$
2. $\alpha \in \Delta \Rightarrow -\alpha \in \Delta$
3. $\alpha \in \Delta$, $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_{-\alpha}$, then $[x, y] = K(x, y)t_\alpha$
4. $\alpha \in \Delta$, then $0 \neq [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \langle t_\alpha \rangle$
5. If $\alpha \in \Delta$, then $\alpha(t_\alpha) = K(t_\alpha, t_\alpha) \neq 0$
6. If $\alpha \in \Delta$, $0 \neq x_\alpha \in \mathfrak{g}_\alpha$, then there exists $y_\alpha \in \mathfrak{g}_{-\alpha}$ such that $K(x_\alpha, y_\alpha) = \frac{2}{K(t_\alpha, t_\alpha)}$, define $h_\alpha := [x_\alpha, y_\alpha] = \frac{2t_\alpha}{K(t_\alpha, t_\alpha)}$, then $\mathfrak{s}_\alpha := \langle x_\alpha, y_\alpha, h_\alpha \rangle$ is isomorphic to $\mathfrak{sl}(2, k)$ via $x_\alpha \mapsto \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, y_\alpha \mapsto \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, h_\alpha \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

7. Given $\alpha \in \Delta$, $(ad|_{\mathfrak{s}_\alpha}, \mathfrak{g})$ will be a representation of $\mathfrak{s}_\alpha$, thus $\mathfrak{g}$ can be decomposed into irreducible representations of $\mathfrak{s}_\alpha$, and the highest eigenvectors for this representation are also common highest eigenvectors of $ad(\mathfrak{h})$

Proof.

1. Suppose $\langle \Delta \rangle \subsetneq \mathfrak{h}^*$, then there exists $0 \neq h \in \mathfrak{h}$, such that $\forall \alpha \in \Delta, \alpha(h) = 0$, then $\forall x \in \mathfrak{g}_\alpha, [h, x] = \alpha(h)x = 0$, and since $\mathfrak{h}$ is abelian, $[h, \mathfrak{h}] = 0$, thus $[h, \mathfrak{g}] = 0$, but then $h \in Z(\mathfrak{g}) = 0$ which is a contradiction
2. If $\alpha \in \Delta$, and $\mathfrak{g}_{-\alpha} = 0$, then $K(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0, \forall \beta$ by Proposition 8.66(c), then $\mathfrak{g}_\alpha = 0$ which is a contradiction
3. $\forall h \in \mathfrak{h}, K(h, [x, y]) = K([h, x], y) = K(\alpha(h)x, y) = K(t_\alpha, h)K(x, y) = K(h, K(x, y)t_\alpha)$, since $K(,)|_{\mathfrak{h}}$ is nondegenerate, $[x, y] = K(x, y)t_\alpha$
4. Only need to show that $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \neq 0$. There exists $x \in \mathfrak{g}_\alpha, y \in \mathfrak{g}_{-\alpha}$ such that $K(x, y) \neq 0$, otherwise then $K(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0, \forall \beta$ by Proposition 8.66(c), then $\mathfrak{g}_\alpha = 0$ which is a contradiction, thus $[x, y] = K(x, y)t_\alpha \neq 0$ by (c)
5. Suppose instead $\alpha(t_\alpha) = 0$, then $[t_\alpha, x] = [t_\alpha, y] = 0, \forall x \in \mathfrak{g}_\alpha, y \in \mathfrak{g}_{-\alpha}$, by nondegeneracy, we can find $x \in \mathfrak{g}_\alpha, y \in \mathfrak{g}_{-\alpha}$ such that $K(x, y) = 1$, then by (c), $[x, y] = K(x, y)t_\alpha = t_\alpha$, thus $\mathfrak{s} = \langle x, y, t_\alpha \rangle \cong ad(\mathfrak{s}) \leq ad(\mathfrak{g}) \leq \mathfrak{gl}(\mathfrak{g})$ is a 3 dimensional solvable Lie algebra, by Theorem 8.40, for any $s \in [\mathfrak{s}, \mathfrak{s}]$, ad(s) is nilpotent, thus $ad(t_\alpha)$ is both semisimple and nilpotent, hence $ad(t_\alpha) = 0 \Rightarrow t_\alpha = 0$ which is a contradiction
6. $[h_\alpha, x_\alpha] = \frac{2}{K(t_\alpha, t_\alpha)}[t_\alpha, x_\alpha] = \frac{2}{K(t_\alpha, t_\alpha)}\alpha(t_\alpha)x_\alpha = 2x_\alpha \quad [h_\alpha, y_\alpha] = \frac{2}{K(t_\alpha, t_\alpha)}[t_\alpha, y_\alpha] = -\frac{2}{K(t_\alpha, t_\alpha)}\alpha(t_\alpha)y_\alpha = -2y_\alpha$
7. Suppose $x \in \mathfrak{g}$ is a highest eigenvector of representation $(ad|_{\mathfrak{s}_\alpha}, \mathfrak{g})$, then $0 = ad(x_\alpha)(x) = [x_\alpha, x], \forall h \in \mathfrak{h}, [x_\alpha, ad(h)(x)] = [x_\alpha, [h, x]] = [[x_\alpha, h], x] + [h, [x_\alpha, x]] = [-\alpha(h)x_\alpha, x] = 0$

□

Remark. In (f), the choice of x_α is not canonical, however, $h_\alpha = \frac{2t_\alpha}{K(t_\alpha, t_\alpha)}$ is canonical, for example, if we pick $0 \neq x_{-\alpha} \in \mathfrak{g}_{-\alpha}$, $\mathfrak{s}_{-\alpha} = \langle x_{-\alpha}, y_{-\alpha}, h_{-\alpha} \rangle$, then $h_{-\alpha} = h_\alpha = \frac{2t_{-\alpha}}{K(t_{-\alpha}, t_{-\alpha})} = -h_\alpha$, moreover, according to Lemma 8.46, any nondegenerate, symmetric, bilinear and invariant form on $\mathfrak{h}$ is of the form $(,) \coloneqq cK(,)|_{\mathfrak{h}}$ for some $c \neq 0$, then t'_α the dual of $\alpha \in \mathfrak{h}^*$ given by $(t'_\alpha, x) = \alpha(x), \forall x \in \mathfrak{h}$, then we have $K(t_\alpha, x) = \alpha(x) = (t'_\alpha, x) = cK(t'_\alpha, x) = K(ct'_\alpha, x)$, because of the nondegeneracy of $K(,)|_{\mathfrak{h}}$, $t_\alpha = ct'_\alpha \Rightarrow t'_\alpha = \frac{t_\alpha}{c}$, and $\frac{2t'_\alpha}{(t'_\alpha, t'_\alpha)} = \frac{2\frac{t_\alpha}{c}}{cK\left(\frac{t_\alpha}{c}, \frac{t_\alpha}{c}\right)} = \frac{2t_\alpha}{K(t_\alpha, t_\alpha)} = h_\alpha$, thus h_α is even canonical defined regardless of the choice of the nondegenerate, symmetric, bilinear and invariant form on $\mathfrak{h}$, for this reason, we give h_α a new notation $\alpha^\vee$ for later use, whereas $\alpha(\alpha^\vee) = 2$, for any nondegenerate, symmetric, bilinear and invariant form on $\mathfrak{h}$, note that $\alpha \mapsto \alpha^\vee$ is not linear

Also, according to Lemma 8.46, even though $(,)$ is defined up to a scalar, but the orthogonality is always well defined

Due to Proposition 8.70(g), if $0 \neq x \in \mathfrak{g}$ is a highest eigenvector for representation of $(ad|_{\mathfrak{s}_\alpha}, \mathfrak{g})$, then $x \in \mathfrak{g}_\beta$, for some $\beta \in \Delta \cup \{0\}$, when $\beta = 0$, these corresponds to the trivial representations, if $\beta \in \Delta$, we can denote one such highest eigenvector $0 \neq x_\beta \in \mathfrak{g}_\beta$, then $ad(h_\alpha)(x_\beta) = [h_\alpha, x_\beta] = \beta(h_\alpha)x_\beta$, thus x_β is a $\beta(h_\alpha) = \frac{2K(t_\alpha, t_\beta)}{K(t_\alpha, t_\alpha)}$-eigenvector, by Proposition 8.66(a), $ad(y_\alpha)^j(x_\beta) \in \mathfrak{g}_{\beta - j\alpha}$ are all the nonzero eigenvectors corresponds to eigenvalues $(\beta - j\alpha)(h_\alpha) = 2\left(\frac{K(t_\alpha, t_\beta)}{K(t_\alpha, t_\alpha)} - j\right)$, and these roots $\beta - j\alpha, j = 0, \cdots, k = \beta(h_\alpha)$ form an α-string

One of these irreducible representation is $\mathfrak{s}_\alpha$ itself according to Example 8.61

Example 8.71 Consider $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{C})$ which is a semisimple Lie algebra, then

$$\mathfrak{h} = \left\{ \begin{pmatrix} z_1 & & \\ & \ddots & \\ & & z_n \end{pmatrix} \middle| \sum z_i = 0 \right\}$$

is a maximal toral Lie subalgebra, denote $\operatorname{diag}(z_1, \cdots, z_n)$ as h_z, then $[h_z, E_{ij}] = (z_i - z_j)E_{ij}$, and $\mathfrak{sl}(2,\mathbb{C}) = \mathfrak{g} = \mathfrak{h} \bigoplus_{i \neq j} \langle E_{ij} \rangle$, $\Delta = \{e_i - e_j | i \neq j\}$, where $e_i \in \mathfrak{h}^*$ is defined by $e_i(h_z) = z_i$, also

$$\mathfrak{s}_\alpha = \left\{ \begin{pmatrix} 0 & & & & & \\ & \ddots & & & & \\ & & a & b & & \\ & & & \ddots & & \\ & & b & -a & & \\ & & & & \ddots & \\ & & & & & 0 \end{pmatrix} \right\} \cong \mathfrak{sl}(2,\mathbb{C})$$

where $\alpha = e_i - e_j$

Definition 8.72 Define $s_\alpha(\beta) \coloneqq \beta - \frac{2K(t_\alpha, t_\beta)}{K(t_\alpha, t_\alpha)}\alpha, \forall \alpha \in \Delta, \beta \in \mathfrak{h}^*$ is an orthogonal reflection over the hyperplane $H_\alpha = \{\alpha(x) = K(t_\alpha, x) = 0 \mid x \in \mathfrak{h}\} = \ker \alpha$, more precisely, $s_\alpha(\alpha) = -\alpha$, $s_\alpha(\beta) = \beta, \forall \beta \in H_\alpha$, note here any nondegenerate, symmetric, bilinear and invariant form $(,)$ can be used as the definition in place of the Killing form $K(,)$ thanks to Lemma 8.46

Proposition 8.73 *(a)* If $\alpha \in \Delta$, then $\dim \mathfrak{g}_\alpha = 1$
(b) If $\alpha \in \Delta$, then $c\alpha \in \Delta \Leftrightarrow c = \pm 1$
(c) If $\alpha, \beta, \alpha + \beta \in \Delta$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$
(d) If $\alpha, \beta \in \Delta$, the Cartan integer $\beta(h_\alpha) = \frac{2K(\alpha,\beta)}{K(\alpha,\alpha)} \in \mathbb{Z}$ and $\beta - \beta(h_\alpha)\alpha \in \Delta$, if $\beta \neq \alpha, -\alpha$, then $\Delta \bigcap \{\beta + j\alpha \mid j \in \mathbb{Z}\} = \{\beta + j\alpha \mid -r \leq j \leq s, j \in \mathbb{Z}\}$ which is an α string through β, and $\beta(h_\alpha) = r - s$

Proof. (a) Let $\mathfrak{m} = \mathfrak{h} \bigoplus_{c\alpha \in \Delta, c \in k^\times} \mathfrak{g}_{c\alpha}$, then $(ad|_{\mathfrak{s}_\alpha}, \mathfrak{m})$ is a finite representation of $\mathfrak{s}_\alpha$ because of Proppsition 8.66(a), first notice for $x \in \mathfrak{g}_{c\alpha}$, we have $ad(h_\alpha)(x) = [h_\alpha, x] = c\alpha(h_\alpha)x = 2cx$, thus the 0-eigenspace of $ad(h_\alpha)$ is $\mathfrak{h}$, but note that for any $h \in \ker \alpha = H_\alpha \leq \mathfrak{h}$ as in Definition 8.72, $ad(x_\alpha)(h) = [x_\alpha, h] = \alpha(h)x_\alpha = 0$, $ad(y_\alpha)(h) = [y_\alpha, h] = -\alpha(h)y_\alpha = 0$, $ad(h_\alpha)(h) = [h_\alpha, h] = 0$, thus $\mathfrak{s}_\alpha$ acts trivially on $\ker \alpha$ which is of codimension 1 in $\mathfrak{h}$ which gives $\dim \mathfrak{h} - 1$ copies of trivial representation of $(ad|_{\mathfrak{s}_\alpha}, \mathfrak{m})$, since $h_\alpha \notin \ker \alpha$, $\mathfrak{h} = \langle h_\alpha \rangle \oplus \ker \alpha$, by Example 8.61 and Lemma 8.64, $\mathfrak{s}_\alpha = \langle x_\alpha, y_\alpha, h_\alpha \rangle$ is a 3 dimensional irreducible representation of $\mathfrak{s}_\alpha$, and $\mathfrak{s}_\alpha \oplus \ker \alpha$ are the only possible even irreducible representations of $(ad|_{\mathfrak{s}_\alpha}, \mathfrak{m})$, thus $\dim \mathfrak{g}_\alpha = 1$, otherwise, we can choose $0 \neq x'_\alpha \in \mathfrak{g}_\alpha$ linearly independent of x_α, then we have $[h_\alpha, x'_\alpha] = \alpha(h_\alpha)x'_\alpha = 2x'_\alpha$, which gives a contradiction. Moreover, $2\alpha \notin \Delta$, otherwise, we can choose $0 \neq x_{2\alpha} \in \mathfrak{g}_{2\alpha}$, then $[h_\alpha, x_{2\alpha}] = 2\alpha(h_\alpha)x_{2\alpha} = 4x_{2\alpha}$which also gives a contradiction
(b) Suppose $c\alpha \in \Delta$, for $0 \neq x \in \mathfrak{g}_{c\alpha}$, we have $ad(h_\alpha)(x) = [h_\alpha, x] = c\alpha(h_\alpha)x = 2cx$, by Theorem 8.60, we know that $2c \in \mathbb{Z}$, but by symmetry, if we let $\beta = c\alpha$, then $\alpha = \frac{\beta}{c} \in \Delta$ implies $\frac{2}{c} \in \mathbb{Z}$, thus c can only possibly be $\pm 1, \pm 2, \pm \frac{1}{2}$, but from (a), we know $c \neq 2$, thus $c \neq -2$ thanks to Proposition 8.70(b), and by symmetry, $c \neq \pm \frac{1}{2}$, therefore $\mathfrak{m} = \mathfrak{h} \oplus \mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha} = \ker \alpha \oplus \langle h_\alpha \rangle \oplus \mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha} = \ker \alpha \oplus \mathfrak{s}_\alpha$
(c) Obiviously β can't be α or α, otherwise $\alpha + \beta \notin \Delta$, also $\beta + j\alpha \neq 0, \forall j \in k$ by (b), for $\beta \in \Delta \setminus \{\alpha, -\alpha\}$, we can consider $(ad|_{\mathfrak{s}_\alpha}, \mathfrak{m})$ which is a finite representation of $\mathfrak{s}_\alpha$ with $\mathfrak{m} = \bigoplus_{\beta + j\alpha \in \Delta, j \in \mathbb{Z}} \mathfrak{g}_{\beta + j\alpha}$, suppose $\beta + j\alpha \in \Delta$, $\dim \mathfrak{g}_{\beta+j\alpha} = 1$ by (a), choose $0 \neq x_{\beta+j\alpha} \in \mathfrak{g}_{\beta+j\alpha}$, we have $[h_\alpha, x_{\beta+j\alpha}] = (\beta + j\alpha)(h_\alpha)x_{\beta+j\alpha} = (\beta(h_\alpha) + 2j)x_{\beta+j\alpha}$, as j varies in $\mathbb{Z}$, $\beta(h_\alpha) + 2j$ can't take both 0 and 1, thus the sum of dimension of 0-eigenspace and 1-eigenspace of $ad(h_\alpha)$ on $\mathfrak{m}$ is 1, by Lemma 8.64, $(ad|_{\mathfrak{s}_\alpha}, \mathfrak{m})$ is irreducible, according to Theorem 8.60 and Remark 8.70, $\mathfrak{m} = \bigoplus_{-r \leq j \leq s} \mathfrak{g}_{\beta+j\alpha}$, for some $r, s \in \mathbb{Z}_{\geq 0}$, and $\beta + j\alpha \in \Delta, \forall -r \leq j \leq s$ which is the α string through β, note that $ad(x_\alpha)(x_\beta) \neq 0$ as in Theorem 8.60, thus $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$
(d) When $\beta = \alpha, -\alpha$, it is trivial. When $\beta \neq \alpha, -\alpha$, as in (c), we know that when j varies from $-r$ to s, $(\beta(h_\alpha) + 2j)$ are all the possible eigenvalues which are integers symmetric over 0, thus $\beta(h_\alpha) + 2s + \beta(h_\alpha) - 2r = 0 \Rightarrow \beta(h_\alpha) = r - s$ is an integer, then $-r \leq -\beta(h_\alpha) \leq s \Rightarrow \beta - \beta(h_\alpha)\alpha \in \Delta$

□

Proposition 8.74 *(1)* Let $V = \mathbb{Q}\Delta$, then $\mathfrak{h}^* = V \otimes_{\mathbb{Q}} k$
(2) For any $h, k \in \mathfrak{h}$, $K(h,k) = Tr(ad(h)ad(k)) = \sum_{\alpha\in\Delta} \alpha(h)\alpha(k)$
(3) The dual of Killing form restricted on V is rational and positive definite

Proof. Since $K(,)|_{\mathfrak{h}}$ is nondegenerate, we define the dual of $K(,)|_{\mathfrak{h}}$ on $\mathfrak{h}^*$ as $(\alpha,\beta) := K(t_\alpha, t_\beta)$, which is also nondegenerate
(1) Since $\mathfrak{h}^* = k\Delta$ by Proposition 8.70(a), Pick any basis in Δ, say $\alpha_1, \cdots, \alpha_m$, then $\forall \beta \in \Delta$, $\beta = \sum c_j\alpha_j, c_j \in k$, we have $(\beta, \alpha_k) = \sum c_j(\alpha_j, \alpha_k) \Rightarrow \frac{2(\beta,\alpha_k)}{(\alpha_k,\alpha_k)} = \sum c_j \frac{2(\alpha_j,\alpha_k)}{(\alpha_k,\alpha_k)}$ whereas $\frac{2(\beta,\alpha_k)}{(\alpha_k,\alpha_k)}, \frac{2(\alpha_j,\alpha_k)}{(\alpha_k,\alpha_k)}$ are all integers by Proposition 8.73(d), since $\big(\beta - \sum c_j\alpha_j, \alpha_k\big) = 0$ and that $(,)$ is nondegenerate, meaning c_j's are the unique solution, hence matrix $\big((\alpha_j,\alpha_k)\big)$ is nonsingular, so is matrix $\Big(\frac{2(\alpha_j,\alpha_k)}{(\alpha_k,\alpha_k)}\Big)$, thus $c_j \in \mathbb{Q}$, which means $\dim_{\mathbb{Q}} V = \dim_k \mathfrak{h}^*$ and $\mathfrak{h}^* = V \otimes_{\mathbb{Q}} k$
(2) For $0 \neq x_\alpha \in \mathfrak{g}_\alpha$, $ad(h)ad(k)(x_\alpha) = [h,[k,x_\alpha]] = \alpha(k)[h,x_\alpha] = \alpha(h)\alpha(k)x_\alpha$, according to Theorem 8.67, we have

$$K(h,k) = Tr(ad(h)ad(k)) = 0 + \sum_{\alpha\in\Delta} Tr\Big(ad(h)\mid_{\mathfrak{g}_\alpha} ad(k)\mid_{\mathfrak{g}_\alpha}\Big) = \sum_{\alpha\in\Delta} \alpha(h)\alpha(k)$$

Due to Proposition 8.70(b), roots always appears in pairs, if let $\Delta^+ \subseteq \Delta$ consists of exactly one from each pair, then $\sum_{\alpha\in\Delta} \alpha(h)\alpha(k) = 2\sum_{\alpha\in\Delta^+} \alpha(h)\alpha(k)$ *(3)* By (2), for any $\lambda, \mu \in \mathfrak{h}^*$

$$\begin{aligned}(\lambda,\mu) &= K(t_\lambda, t_\mu)\\ &= \sum_{\alpha\in\Delta} \alpha(t_\lambda)\alpha(t_\beta)\\ &= \sum_{\alpha\in\Delta} K(t_\alpha,t_\lambda)K(t_\alpha,t_\mu)\\ &= \sum_{\alpha\in\Delta} (\alpha,\lambda)(\alpha,\mu)\\ &= 2\sum_{\alpha\in\Delta^+} (\alpha,\lambda)(\alpha,\mu)\end{aligned}$$

In particular, for any $\beta \in \Delta$, $(\beta,\beta) = 2\sum_{\alpha\in\Delta^+}(\alpha,\beta)^2 \Rightarrow \frac{2}{(\beta,\beta)} = \sum_{\alpha\in\Delta^+}\Big(\frac{2(\alpha,\beta)}{(\beta,\beta)}\Big)^2$, where $\frac{2(\alpha,\beta)}{(\beta,\beta)} \in \mathbb{Z}$, thus $0 \leq (\beta,\beta) \in \mathbb{Q}$ and equals 0 iff $(\alpha,\beta) = 0, \forall \alpha \in \Delta \Rightarrow \beta = 0$ by the nondegeneracy of $(,)$, thus $(,)|_V$ is rational and positive definite on a basis, which implies it is rational and positive definite

□

Remark. When $k = \mathbb{C}$, note that $\mathbb{Q}\Delta < \mathbb{R}\Delta < \mathbb{C}\Delta$, we can view V embedded in the Euclidean space $V_{\mathbb{R}} := \mathbb{R}\Delta = V \otimes_{\mathbb{Q}} \mathbb{R}$ which helps thinking, then we have a root system

Example 8.75 Let $\Omega = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$, then $\{X \in SL(2n,\mathbb{C}) \mid X^T\Omega X = \Omega\}$ is conjugate to $SO(2n,\mathbb{C})$, thus then also induce isomorphic Lie algebra, hence we can identify $\mathfrak{so}(2n,\mathbb{C})$ with $\{X \in M(2n,\mathbb{C}) \mid \Omega X^T + X\Omega = 0\}$ which is the same as $\left\{\begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} \in M(2n,\mathbb{C}) \;\middle|\; B^T = -B, C^T = -C\right\} =: \mathfrak{g}$, then one Cartan subalgebra of $\mathfrak{g}$ will be $\mathfrak{h} = \left\{\begin{pmatrix} D & 0 \\ 0 & -D \end{pmatrix} \in M(2n,\mathbb{C}) \;\middle|\; D = \mathrm{diag}(d_1,\cdots,d_n)\right\}$, note that

$$\left[\begin{pmatrix} D & 0 \\ 0 & -D \end{pmatrix}, \begin{pmatrix} E_{ij} & 0 \\ 0 & -E_{ji} \end{pmatrix}\right] = (d_i - d_j)\begin{pmatrix} E_{ij} & 0 \\ 0 & -E_{ji} \end{pmatrix}$$

$$\left[\begin{pmatrix} D & 0 \\ 0 & -D \end{pmatrix}, \begin{pmatrix} 0 & E_{ij} \\ 0 & 0 \end{pmatrix}\right] = (d_i + d_j)\begin{pmatrix} 0 & E_{ij} \\ 0 & 0 \end{pmatrix}$$

$$\left[\begin{pmatrix} D & 0 \\ 0 & -D \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ E_{ij} & 0 \end{pmatrix}\right] = -(d_i + d_j)\begin{pmatrix} 0 & 0 \\ E_{ij} & 0 \end{pmatrix}$$

Definition 8.76 E is a vector space, $d : E \to E \wedge E$ can be uniquely extended to $d : \bigwedge^k E \to \bigwedge^{k+1} E$ satisfying the graded Leibniz rule $d(a \wedge b) = da \wedge b + (-1)^{\deg(a) \wedge db}$

Lemma 8.77 $\mathfrak{g}$ is a Lie algebra of dimension n, then $(E = \mathfrak{g}, d = [,]^*)$ is a Lie coalgebra

Proof.

□

Proposition 8.78

$$\{\texttt{Finite Lie algebras}\}^{op} \cong \{\texttt{Finite Lie coalgebras}\}$$

Proof.

□

Example 8.79 R is a noncommutative ring, $\delta(a_1 \cdots a_n) = \sum (a_1 \cdots a_k) \wedge (a_{k+1} \cdots a_n)$ defines a Lie coalgebra

Definition 8.80 The *universal enveloping algebra* algebra of Lie algebra $\mathfrak{g}$ over k is

$$U\mathfrak{g} = T(\mathfrak{g})/\langle x \otimes y - y \otimes x - [x, y] \rangle$$

satisfying universal property

Definition 8.81 A $\mathfrak{g}$ module just means a $U\mathfrak{g}$ module

$$H_n(\mathfrak{g}; M) = \operatorname{Tor}_n^{U\mathfrak{g}}(k, M)$$

is the left derived functor of $M \mapsto k \otimes_{U\mathfrak{g}} \mathfrak{M}$

$$H^n(\mathfrak{g}; M) = \operatorname{Ext}^n_{U\mathfrak{g}}(k, M)$$

is the right derived functor of $M \mapsto \operatorname{Hom}_{U\mathfrak{g}}(k, M)$

8.3 Root system

Lemma 8.82 V is a finite dimensional vector space over a field of characteristic 0, $\Delta \subseteq V$ is a finite set such that $V = \operatorname{Span} \Delta$, then for any $\alpha \in \Delta$, there is at most one linear map $s : V \to V$ such that $s^2 = 1$, $s\alpha = -\alpha, s(\Delta) \subseteq \Delta$

Proof. Suppose s, t both satisfy the condition, $sv = v + (s-1)v$, $s(s-1)v = s^2v - sv = v - sv = -(s-1)v$, thus $(s-1)v \in \langle \alpha \rangle \Rightarrow sv = v + f(v)\alpha$, where $f \in V^*$, similarly, $tv = v + g(v)\alpha$, where $g \in V^*$, thus $stv = s(v + g(v)\alpha) = v + g(v)\alpha + f(v + g(v)\alpha)\alpha = v + g(v)\alpha + f(v)\alpha + f(v)g(v)\alpha$, and since $s\alpha = \alpha + f(\alpha)\alpha = -\alpha \Rightarrow f(\alpha) = -2$, thus check $(st)^n v = v + n(f(v) - g(v))\alpha$, but $(st)^n = 1$ for some n because st is just a permutation of Δ, thus $f = g \Rightarrow s = t$

□

Remark. $s^2 = 1$ and $s(\Delta) \subseteq \Delta$ implies that s is a permutation of Δ, note that this definition doesn't involve inner product on V, you could see this as a more abstract definition of a reflection

Definition 8.83 $V = \mathbb{R}^n$ is the standard Euclidean space with the standard inner product $(,)$, a **root system** Δ is a finite subset of V satisfying

1. $V = \langle \Delta \rangle$
2. If $\alpha \in \Delta$, then the only multiples of α are $\pm\alpha$
3. $s_\alpha(\Delta) \subseteq \Delta$, where

$$s_\alpha(\beta) = \beta - \langle \beta, \alpha \rangle \alpha = \beta - \frac{2(\beta, \alpha)}{(\alpha, \alpha)}\alpha$$

is the reflection across the hyperplane $H_\alpha = \{\beta \in V | (\beta, \alpha) = 0\}$

4. $\langle\beta,\alpha\rangle = \frac{2(\beta,\alpha)}{(\alpha,\alpha)} \in \mathbb{Z}$ are called the **Cartan integers**

Remark. $\langle\beta,\alpha\rangle \in \mathbb{Z}$ is the **integrality** condition, and such roots system is called **crystallographic**

$\langle\alpha,\beta\rangle\langle\beta,\alpha\rangle = 4\frac{(\alpha,\beta)^2}{(\alpha,\alpha)(\beta,\beta)} = 4\cos^2\theta \in \mathbb{Z}$ can only be $0,1,2,3,4$, where θ is the angle between α and β, and it is 4 iff $\alpha = \pm\beta$

$\langle\alpha,\beta\rangle\langle\beta,\alpha\rangle$	0	1	2	3	4
θ	$\pm\frac{\pi}{2}$	$\pm\frac{\pi}{3}$	$\pm\frac{\pi}{4}$	$\pm\frac{\pi}{6}$	0

$a_{ji} = \frac{2(\alpha_i,\alpha_j)}{(\alpha_j,\alpha_j)}$ gives a Cartan matrix A with $D_{ij} = \frac{\delta_{ij}}{(\alpha_i,\alpha_i)}$, $S_{ij} = 2(\alpha_i,\alpha_j)$

Conversely, given a generalized Cartan matrix, we can find the corresponding Lie algebra, see Kac-Moody algebra

Remark. Thanks to Lemma 8.82, s_α is the unique linear map $V \to V$ such that $s_\alpha(\alpha) = -\alpha, s_\alpha(\Delta) \subseteq \Delta$

Definition 8.84 Let (V,Δ) be a root system, define the coroot of $\alpha \in \Delta$ to be $\alpha^\vee = \frac{2}{(\alpha,\alpha)}\alpha$, and let $\Delta^\vee = \{\alpha^\vee \mid \alpha \in \Delta\}$, then $(V,\Delta^\vee)$ is also a root system

Lemma 8.85 $\alpha \neq \pm\beta \in \Delta$. If $\langle\alpha,\beta\rangle > 0$, then $\alpha - \beta \in \Delta$, if $\langle\alpha,\beta\rangle < 0$, then $\alpha + \beta \in \Delta$

Proof. Note that $s_\alpha(\beta) = \beta - \langle\beta,\alpha\rangle\alpha \in \Delta$, $s_\beta(\alpha) = \alpha - \langle\alpha,\beta\rangle\beta \in \Delta$, and $\langle\alpha,\beta\rangle\langle\beta,\alpha\rangle = 4\frac{(\alpha,\beta)^2}{(\alpha,\alpha)(\beta,\beta)} = 4\cos^2\theta \in \mathbb{Z}$ can only be $0,1,2,3$, where θ is the angle between α and β, if $\langle\alpha,\beta\rangle > 0 \Leftrightarrow (\alpha,\beta) > 0$, then $\langle\alpha,\beta\rangle = 1, \langle\beta,\alpha\rangle = 1,2,3$ or $\langle\alpha,\beta\rangle = 2,3, \langle\beta,\alpha\rangle = 1$, hence either $\alpha - \beta$ or $\beta - \alpha$ will be in Δ, but then the other will also be in Δ. Similarly, if $\langle\alpha,\beta\rangle < 0 \Leftrightarrow (\alpha,\beta) < 0$, then $\langle\alpha,\beta\rangle = -1, \langle\beta,\alpha\rangle = -1,-2,-3$ or $\langle\alpha,\beta\rangle = -2,-3, \langle\beta,\alpha\rangle = -1$, hence $\alpha + \beta \in \Delta$

□

Definition 8.86 $\Delta^+ \subset \Delta$ is a set of **positive roots** if $\alpha \in \Delta \Rightarrow$ precisely one of $\alpha, -\alpha$ is in Δ^+, and $\alpha,\beta \in \Delta^+ \Rightarrow \alpha + \beta \in \Delta^+$, thus we can correspondingly define negative roots $\Delta^- = \Delta \setminus \Delta^+ = -\Delta^+$. Any hyperplane H that doesn't intersect Δ separates Δ into Δ^+ and Δ^-. $\gamma \in V$ is called **regular** if $(\gamma,\alpha) \neq 0, \forall\alpha \in \Delta$, then $H_\gamma = \{v \in V \mid (\gamma,v) = 0\}$ separates Δ into $\Delta^+(\gamma) = \{\alpha \in \Delta \mid (\gamma,\alpha) > 0\}$ and $\Delta^-(\gamma) = \{\alpha \in \Delta \mid (\gamma,\alpha) < 0\}$. Conversely, given a hyperplane H that doesn't intersect Δ and separates Δ into Δ^+ and Δ^-, we can find $\gamma \in V$ such that $H = H_\gamma$, $\Delta^+ = \Delta^+(\gamma)$, $\Delta^- = \Delta^-(\gamma)$

Definition 8.87 $\gamma \in V$ is **dominant** if $(\gamma,\alpha) \geq 0, \forall\alpha \in \Delta^+$ and **strictly dominant** if $(\gamma,\alpha) > 0, \forall\alpha \in \Delta^+$

Definition 8.88 $\alpha \in \Delta^+$ is **decomposable** if $\alpha = \alpha_1 + \alpha_2$ for some $\alpha_1,\alpha_2 \in \Delta^+$, α is a **simple root** of Δ^+ if it is not decomposable. $S = \{\alpha_1,\cdots,\alpha_m\}$ is a **base** for Δ^+ if for any $\alpha \in \Delta^+$, there are $c_i \in \mathbb{Z}_{\geq 0}$ such that $\alpha = \sum_i c_i\alpha_i$, which also implies that for any $\alpha \in \Delta^- = -\Delta^+$, there are $c_i \in \mathbb{Z}_{\leq 0}$ such that $\alpha = \sum_i c_i\alpha_i$

Lemma 8.89 $S = \{v_1,\cdots,v_m\}$ are on one side of a hyperplane H, and $(v_i,v_j) < 0, \forall i \neq j$, then S is linearly independent

Proof. Suppose $\sum_{i=1}^m a_iv_i = 0$ and not all a_i's are zero, then we can rewrite as $\sum_{k\in K} a_kv_k = \sum_{l\in L} -a_lv_l$, where $a_k > 0, \forall k \in K, a_l < 0, \forall l \in L$, then we have $0 \leq \left(\sum_{k\in K} a_kv_k, \sum_{l\in L} -a_lv_l\right) = \sum_{k\in K,l\in L} -a_ka_l(v_k,v_l) < 0$ which is a contradiction

□

Lemma 8.90 S is the set of simple roots of Δ^+, then S is a base for Δ^+, and S is linearly independent

Proof. It is obvious that S is a base of Δ^+ by definition. Suppose $\alpha \neq \beta \in S$, $\alpha \neq -\beta$ is obvious, hence $(\alpha,\beta) \neq 0$. If $(\alpha,\beta) > 0$, then by Lemma 8.85, we have $\alpha - \beta \in \Delta$, if $\alpha - \beta \in \Delta^+$, then $\alpha = \beta + (\alpha - \beta)$ gives a contradiction, if $\alpha - \beta \in \Delta^-$, then $\beta - \alpha \in \Delta^+$ and $\beta = \alpha + (\beta - \alpha)$ gives a contradiction, therefore $(\alpha,\beta) < 0$. By lemma 8.89, we know S is linearly independent

□

Remark. Given a set of positive roots, there is precisely one base which is the set of simple roots

Lemma 8.91 $\sigma \in \mathrm{GL}(V)$, $\sigma(\Delta) \subseteq \Delta$, then for any $\alpha \in \Delta$, $\sigma s_\alpha \sigma^{-1} = s_{\sigma\alpha}$, moreover, $\langle \beta, \alpha \rangle = \langle \sigma\beta, \sigma\alpha \rangle$

Proof. $\sigma^m = 1$ for some m since there are only finitely many choices of maps $\Delta \to \Delta$, thus σ is a permutation on Δ, hence $\sigma s_\alpha \sigma^{-1}(\Delta) \subseteq \Delta$, $\left(\sigma s_\alpha \sigma^{-1}\right)^2 = 1$ and $\sigma s_\alpha \sigma^{-1} \sigma\alpha = -\sigma\alpha$ implies $\sigma s_\alpha \sigma^{-1} = s_{\sigma\alpha}$ by Lemma 8.82. Compare $s_{\sigma\alpha}(\sigma\beta) = \sigma\beta - \langle \sigma\beta, \sigma\alpha \rangle \sigma\alpha$, and $\sigma s_\alpha \sigma^{-1}(\sigma\beta) = \sigma s_\alpha \beta = \sigma(\beta - \langle \beta, \alpha \rangle \alpha) = \sigma\beta - \langle \beta, \alpha \rangle \sigma\alpha$, we get $\langle \beta, \alpha \rangle = \langle \sigma\beta, \sigma\alpha \rangle$

□

Definition 8.92 The **Weyl group** W of Δ is $\langle s_\alpha | \alpha \in \Delta \rangle$. $|W| < \infty$ since $s_\alpha(\Delta) \subseteq \Delta$. W is a subgroup of $O(V)$

Remark. W is a finite Coxeter group

Definition 8.93 Weyl chambers are connected components of $V \setminus \bigcup_{\alpha \in \Delta} H_\alpha$, as the intersection of open half spaces, Weyl chambers are open convex, each regular $\gamma \in V$ by definition belongs to precisely one Weyl chamber denote as $C(\gamma)$, $C(\gamma) = C(\gamma')$ iff γ, γ' are on the side for every hyperplane H_α iff $\Delta^+(\gamma) = \Delta^+(\gamma')$, thus Weyl chambers are in one to one correspondence with the sets of positive roots, the fundamental Weyl chamber associated to Δ^+, or rather, associated to S is $\{\gamma \in V | (\gamma, \alpha) > 0, \forall \alpha \in S\}$. For any $\sigma \in W$, since $(\sigma\gamma, \sigma\alpha) = (\gamma, \alpha)$, $\sigma(\Delta^+(\gamma)) = \Delta^+(\sigma\gamma)$, $\sigma(S(\gamma)) = S(\sigma\gamma)$, $\sigma(C(\gamma)) = C(\sigma\gamma)$

Lemma 8.94 If $\alpha \in \Delta^+ \setminus S$,then there exists $\beta \in S$ such that $\alpha - \beta \in \Delta^+$

Proof. It is obvious that $(\alpha, \beta) \neq 0, \forall \beta \in \Delta^+$, suppose $(\alpha, \beta) < 0, \forall \beta \in S$, then by Lemma 8.89, $S \cup \{\alpha\}$ is linearly independent which is impossible, thus $(\alpha, \beta) > 0$ for some $\beta \in \Delta^+$, by Lemma 8.85, $\alpha - \beta \in \Delta$, but since $\alpha = \sum_{\alpha_i \in S} c_i \alpha_i$, and $\beta = \alpha_j$ for some j, since some $c_i > 0$, it necessarily has to be that $\alpha - \beta = (c_j - 1)\beta + \sum_{i \neq j} c_i \alpha_i \in \Delta^+$

□

Lemma 8.95 Each $\alpha \in \Delta^+$ can be written as $\alpha_1 + \cdots + \alpha_k$, $\alpha_i \in S$, here α_i may repeat, such that partial sums are all positive roots, i.e. $\alpha_i + \cdots + \alpha_i \in \Delta^+$

Proof. Each $\alpha \in \Delta^+$ can be written uniquely as a sum of simple roots, by induction on the number of summands and Lemma 8.94

□

Lemma 8.96 If $\alpha \in S$, then s_α permutes $\Delta^+ \setminus \{\alpha\}$

Proof. For any $\beta \in \Delta^+ \setminus \{\alpha\}$, $\beta = \sum_{\alpha_i \in S} c_i \alpha_i$, $c_j > 0$ for some $\alpha_j \neq \alpha$, then $s_\alpha(\beta) = \beta - \langle \beta, \alpha \rangle \alpha$ still has some positive coefficient, thus it has to be positive, and $s_\alpha(\beta) \neq \alpha = s_\alpha(-\alpha)$

□

Corollary 8.97 Let $\delta = \frac{1}{2} \sum_{\beta \in \Delta^+} \beta$, then $s_\alpha(\delta) = \delta - \alpha, \forall \alpha \in S$

Lemma 8.98 Use $\alpha \prec 0$ and $\alpha \succ 0$ to mean positive and negative roots, suppose $\alpha_i \in S, i = 1, \cdots, q$, and denote $s_i \coloneqq s_{\alpha_i}$, if $s_1 \cdots s_{q-1}(\alpha_q) \prec 0$, then $s_1 \cdots s_q = s_1 \cdots s_{p-1} s_{p+1} \cdots s_{q-1}$. In particular, suppose $s = s_1 \cdots s_q$ is of the smallest length, then $0 \prec s_1 \cdots s_{q-1} \alpha_q = -s_1 \cdots s_q \alpha_q \Rightarrow s_1 \cdots s_q \alpha_q \prec 0$

Proof. Write $\beta_i = s_i \cdots s_{q-1} \alpha_q$, then $\beta_q = \alpha_q \succ 0$, $\beta_1 \prec 0$, thus there must be some $1 \leq p < q$ such that $\beta_{p+1} \succ 0$ but $\beta_p = s_p \beta_{p+1} \prec 0$, by lemma 8.96, thus β_{p+1} can only be α_p, hence by Lemma 8.91, we have

$$
\begin{aligned}
& s_p = s_{\alpha_p} = s_{\beta_{p+1}} = s_{s_{p+1} \cdots s_{q-1} \alpha_q} = (s_{p+1} \cdots s_{q-1}) s_q (s_{p+1} \cdots s_{q-1})^{-1} \\
& \Rightarrow s_p \cdots s_{q-1} = s_{p+1} \cdots s_q \\
& \Rightarrow s_1 \cdots s_q = s_1 \cdots s_p s_{p+1} \cdots s_q = s_{p1} \cdots s_p s_p \cdots s_{q-1} = s_1 \cdots s_{p-1} s_{p+1} \cdots s_{q-1}
\end{aligned}
$$

□

Lemma 8.99 Denote $n(\sigma)$ the number of positive roots that σ send to negative. $l(\sigma) = n(\sigma)$. Hence there is a unique element $w_\circ$ such that $w_\circ(S) = -S$ of maximal length, and $w_\circ^2 = 1$

Theorem 8.100

1. Let $\gamma \in V$ be regular, then there exists some $\sigma \in W$ such that $\Delta^+(\sigma\gamma) = \Delta^+$, namely, W acts trasitively on Weyl chambers
2. If S' is a another base, then there exists some $\sigma \in W$ such that $\sigma(S') = S$, namely, W acts trasitively on bases
3. If α is any root, then there exists some $\sigma \in W$ such that $\sigma(\alpha) \in S$
4. W is generated by s_α's for $\alpha \in S$
5. If $\sigma \in W$, then $\sigma(S) \subseteq S \Rightarrow \sigma = 1$, namely, W acts freely and transitively(regularly) on bases(and Weyl chambers)

Proof. Let $W' \leq W$ be the subgroup of $O(n)$ generated by s_α's for $\alpha \in S$

1. Let $\delta = \frac{1}{2}\sum_{\beta\in\Delta^+}\beta$, choose $\sigma \in W'$ such that $(\sigma\gamma, \delta)$ is the biggest possible, then for any $s_\alpha \in W'$, due to Corollary 8.97, we have $(\sigma\gamma, \delta) \geq (s_\alpha\sigma\gamma, \delta) = (\sigma\gamma, s_\alpha\delta) = (\sigma\gamma, \delta - \alpha) = (\sigma\gamma, \delta) - (\sigma\gamma, \alpha) \Rightarrow (\sigma\gamma, \alpha) \geq 0$, since γ is regular, so is $\sigma\gamma$, thus $(\sigma\gamma, \alpha) > 0, \forall\alpha \in S$, $\sigma(C(\gamma)) = C(S)$, i.e. W acts transitively on Weyl chambers
2. Directly from (a)
3. Thanks to (b), it suffices to prove that each root lies in some base, there exists $\gamma \in H_\alpha \setminus \bigcup_{\beta\in\Delta\setminus\{\pm\alpha\}} H_\beta$, and the perturb γ slightly so that $0 < (\gamma, \alpha) < |(\gamma, \beta)|, \beta \in \Delta \setminus \{\pm\alpha\}$, then $\alpha \in S(\gamma)$
4. By (c), if $\alpha \in \Delta$, $\beta \in S$, then there exists $\sigma \in W'$ such that $\sigma\alpha = \beta$, then $s_\beta = s_{\sigma\alpha} = \sigma s_\alpha \sigma^{-1}$, thus $s_\alpha = \sigma^{-1} s_\beta \sigma \in W'$
5. By (d), $\sigma \in W$ can be written as $s = s_{\alpha_1}\cdots s_{\alpha_q}, \alpha_i \in S$ and suppose it is of minimal length, by Lemma 8.98, $\sigma(\alpha_q) \prec 0$ contradicting $\sigma(S) \subseteq S$

□

Proposition 8.101 The root system is irreducible iff the Lie algebra is simple

8.4 Dynkin diagram

Definition 8.102 A *generalized Cartan matrix* A is such that $a_{ii} = 2$, $a_{ij} \leq 0$ for $i \neq j$, if $a_{ij} = 0$, then $a_{ji} = 0$, $A = DS$ for some diagonal matrix D and symmetric matrix S, i.e. A is symmetrizable. Note that D would have nonzero diagonal entries, we can pick positive entries, A is a *Cartan matrix* if S is positive definite A is *decomposable* if $a_{ij} = 0$, $i \in I, j \in J$ for some $\{1, \cdots, n\} = I \sqcup J$, i.e. A can be diagonalized by blocks

An indecomposable matrix A is of *finite type* if all principal minors are positive, *affine type* if all proper principal minors are positive and $\det A = 0$, *indefinite type* otherwise

Definition 8.103 S is a set of positive simple roots, the *Dynkin diagram* is a graph with nodes simple roots, $\langle\alpha, \beta\rangle\langle\beta, \alpha\rangle$ edges between α, β which can only be $0, 1, 2, 3$, and an arrow from α to β if $\langle\alpha, \beta\rangle > 1$. The *Coxter diagram* of the Weyl group W is just the Dynkin diagram without arrows. The *Coxeter graph* of it is the underlying graph

Theorem 8.104 We can recover the root system through Dynkin diagram

Definition 8.105 Type A_n corresponds to Dynkin diagram

Example 8.106 $\mathfrak{sl}_{n+1}$ corresponds to type A_n

Definition 8.107 Type B_n corresponds to Dynkin diagram



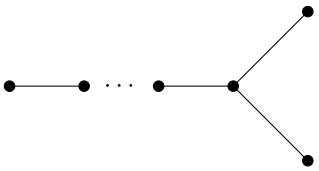
Example 8.108 $\mathfrak{so}_{2n+1}$ corresponds to type B_n

Definition 8.109 Type C_n corresponds to Dynkin diagram



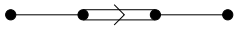
Example 8.110 $\mathfrak{sp}_{2n}$ corresponds to type C_n

Definition 8.111 Type D_n corresponds to Dynkin diagram



Example 8.112 $\mathfrak{so}_{2n}$ corresponds to type D_n

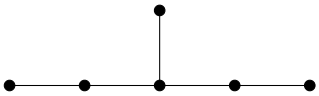
Definition 8.113 Type F_4 corresponds to Dynkin diagram



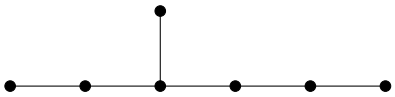
Definition 8.114 Type G_2 corresponds to Dynkin diagram



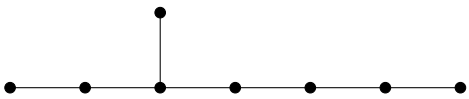
Definition 8.115 Type E_6 corresponds to Dynkin diagram



Definition 8.116 Type E_7 corresponds to Dynkin diagram



Definition 8.117 Type E_8 corresponds to Dynkin diagram



Remark. The number in the subscript is the number of nodes. In particular, we have

$A_1 = B_1 = C_1$

$B_2 = C_2$

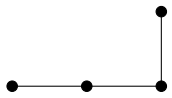


$D_3 = A_3$

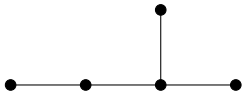
$D_2 = A_1 A_1$

$E_3 = A_2 A_1$

$E_4 = A_4$



$E_5 = D_5$



Theorem 8.118 (Cartan-Killing classification) The above Dynkin diagrams classifies simple Lie algebras

Proof. Consider the **admissible sets** of Euclidean space V, $A = \{v_1, \cdots, v_n\}$ of linearly independent unit vectors with $(v_i, v_j) \leq 0$ and $4(v_i, v_j)^2 \in \{0, 1, 2, 3\}$ if $i \neq j$. Define Coxeter diagram Γ_A for A to have vertices $v_1, \cdots, v_n$, and $d_{ij} = 4(v_i, v_j)^2$ edges between v_i and v_j if $i \neq j$. Assume that Γ_A is connected

1. The number of vertices in Γ_A joined by at least one edge is at most $|A| - 1$

   $v = v_1 + \cdots + v_n \neq 0$ satisfies $(v, v) = n + 2\sum_{i<j}(v_i, v_j) > 0$, thus $n > -\sum_{i<j} 2(v_i, v_j) = \sum_{i<j} \sqrt{d_{ij}} \geq N$, where N is the number of pairs v_i, v_j such that $d_{ij} \geq 1$

2. The graph Γ_A contains no cycles

   The vectors in a cycle of Γ_A form an admissible set which a contradicts a)

3. No vertex in Γ_A has more than 3 edges

   Let w be a vertex of Γ_A with adjacent vertices $w_1, \cdots, w_k$, then $(w_i, w_j) = 0$ for $i \neq j$. Let $U = \mathrm{Span}_{\mathbb{R}}(w_1, \cdots, w_k, w)$, and extend $\{w_1, \cdots, w_k\}$ to an orthonormal basis of U, say by adjoining w_0. Clearly $(w, w_0) \neq 0$ and $w = \sum_{i=0}^{k}(w, w_i)w_i$. Hence $1 = (w, w) = \sum_{i=0}^{k}(w, w_i)^2$, $\sum_{i=1}^{k}(w, w_i)^2 < 1$, w has no more than 3 edges

4. If Γ_A has triple edge, then by c), the Γ_A can only be G_2

5. Assume Γ_A has a subgraph which is a line along $w_1, \cdots, w_n$, if we replace this subgraph with $w = w_1 + \cdots + w_n$, then it is still an admissible set

   $(w, w) = n + 2\sum_{i=1}^{n-1}(w_i, w_{i+1}) = n - (n-1) = 1$, by d) any vertex v has at most edges linked with one such w_i, hence $(v, w) = (v, w_i)$, this gives an admissible set

6. A branch point is a vertex having more than 2 adjacent vertices, in this case, exactly 3. Γ_A has only one double edge, or only one branch point, or neither, but not both. Note that if Γ_A has no brach points and double edges corresponds to A_n

   If Γ_A has two double edges between w_1, w_2 and v_1, v_2, then they can be linked through a line, by e), we can collapse it into a single vertex, but this will contradict c)

7. If Γ_A has a subgraph which is a line through $w_1, \cdots, w_n$, let $w = \sum iw_i$, then $(w, w) = \frac{n(n+1)}{2}$

8. If Γ_A has a double edge, then Γ_A is F_4 or B_n

By f) we know Γ_A is a line through $v_1, \cdots, v_p, w_q, \cdots, w_1$, $q \geq p \geq 1$ with single edges except v_p, w_q, let $v = \sum iv_i$, $w = \sum iw_i$, then

$$(v, w)^2 = (pv_p, qw_q)^2 = \frac{p^2q^2}{2}$$

Since v, w are linearly independent, by Cauchy Schwarz inequality, we have

$$\frac{p^2q^2}{2} = (v, w)^2 < (v, v)(w, w) = \frac{p(p+1)q(q+1)}{4}$$

Which implies $(p-1)(q-1) < 2$, thus if $p = 1$, then q can be any positive interger, giving B_n, if $p = 2$, then $q = 2$, giving F_4

9. If Γ_A has a branch point, then Γ_A is D_n or E_6, E_7, E_8

   Γ_A is has three branch lines $v_1, \cdots, v_p, x$ and $w_1, \cdots, w_q, x$ together with $z_1, \cdots, z_r, x$, $p \geq q \geq r$, let $v = \sum iv_i$, $w = \sum iw_i$, $z = \sum iz_i$ which are pairwise orthogonal, $\hat{v}, \hat{w}, \hat{z}$ be normalized vectors of v, w, z, and consider $U = \text{Span}_{\mathbb{R}}(v, w, z, x) = \text{Span}_{\mathbb{R}}(\hat{v}, \hat{w}, \hat{z}, x_0)$, where x_0 is a unit vector orthogonal to v, w, z, then $(x, x_0) \neq 0$

   $$1 = (x, x) = (x, \hat{v})^2 + (x, \hat{w})^2 + (x, \hat{z})^2 + (x, x_0)^2$$

   Thus by g)

   $$\frac{2p^2}{4p(p+1)} + \frac{2q^2}{4q(q+1)} + \frac{2r^2}{4r(r+1)} < 1$$

   Hence

   $$\frac{1}{1+p} + \frac{1}{1+q} + \frac{1}{1+r} > 1$$

   and we know that

   $$\frac{1}{1+p} \leq \frac{1}{1+q} \leq \frac{1}{1+r} \leq \frac{1}{2}$$

   Hence $r = 1$

   $$\frac{1}{1+p} + \frac{1}{1+q} > \frac{1}{2}$$

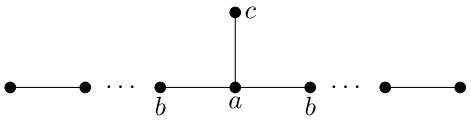
   If $q = 1$, then p can be any positive integer, giving D_n, if $q = 2$, then p can only be $2, 3$ or 4, giving E_6, E_7, E_8

□

Lemma 8.119 (V, Δ) be a irreducible root system, Δ^+ be a set of positive roots and $S = \{\alpha_1, \cdots, \alpha_n\}$ be its base, then there exists unique highest root $\gamma \in \Delta$, meaning $\gamma + \alpha_i \in \Delta, \forall \alpha_i \in S$

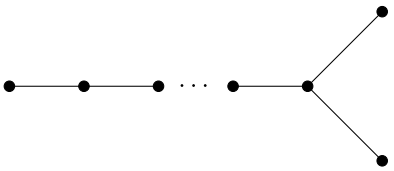
Definition 8.120 Let (V, Δ) be a irreducible root system, Δ^+ be a set of positive roots, $S = \{\alpha_1, \cdots, \alpha_n\}$ be its base, and γ its unique highest root, the extended Dynkin diagram is the usual Dynkin diagram adding $\alpha_0 = -\gamma$, the number of bonds for each two nodes and direction are still defined as before. Finally, suppose $-\alpha_0 = \sum n_i\alpha_i$, then lebel γ with 1, and label node α_i with n_i

Lemma 8.121 If the following part of the extended Dynkin diagram

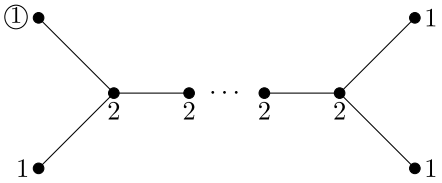


We have $2a = b + c + d$

Example 8.122 Consider the classical root system D_n, $\Delta^+ = \{e_i \pm e_j | i < j\}$, $S = \{e_1 - e_2, \cdots, e_{n-1} - e_n, e_{n-1} + e_n\}$, then $\gamma = e_1 + e_2$, its usual Dynkin diagram is



So its extended Dynkin diagram is



9 Algebraic groups

Definition 9.1

- A *k-group scheme* is a group object in the category of k-schemes
- A *k-algebraic group* is a group object in the category of k-varieties
- A *linear algebraic group* is an affine algebraic group

Definition 9.2 Let k be a commutative ring. A *k-group functor* is

$$G : \{k\text{-}\texttt{algebras}\} \to \{\texttt{Sets}\}$$

With natural transformations

- multiplication $m : G \times G \to G$
- unit $e : \mathrm{Hom}(k, -) \to G$
- inverse $i : G \to G$

such that $m_A : G(A) \times G(A) \to G(A)$, $e_A : \{*\} \to G(A)$, $i_A : G(A) \to G(A)$ give $G(A)$ a group structure. Thus G is actually a functor $\{k\text{-}\texttt{algebras}\} \to \{\texttt{Groups}\}$

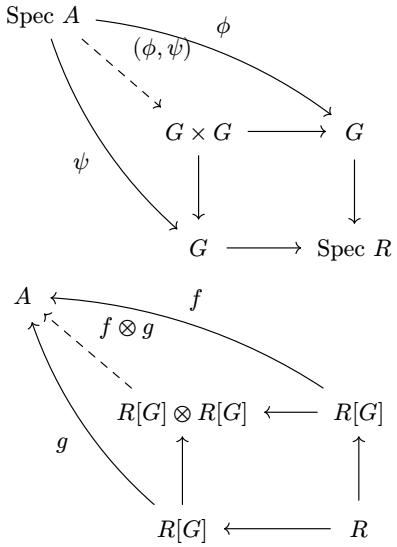
Remark. The functor $G \times G$ is defined to be $(G \times G)(A) = G(A) \times G(A)$ and $(G \times G)(A \overset{\varphi}{\to} B) = G(A) \times G(A) \overset{G(\varphi)\times G(\varphi)}{\to} G(B) \times G(B)$

Proposition 9.3 k-group functor is representable by k-group schemes

Proof. Apply Yoneda lemma

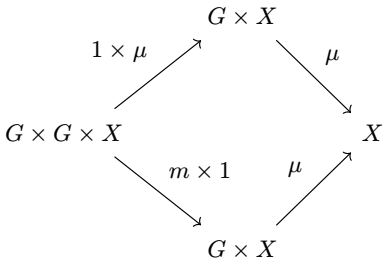
$$G \times G = \mathrm{Hom}(-, G) \times \mathrm{Hom}(-, G) \cong \mathrm{Hom}(-, G \times G) \cong \mathrm{Hom}(-, \mathrm{Spec}\ R[G] \otimes R[G])$$

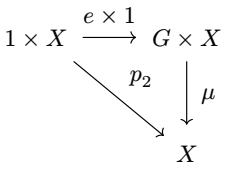
Multiplication $G(A) \times G(A) \to (G \times G)(A)$ is given by



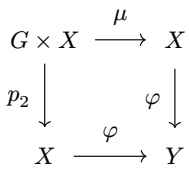
□

Definition 9.4 G is a k-group scheme, X is a k-scheme, then a *group action* is $\mu : G \times_k X \to X$ such that the following diagrams commute

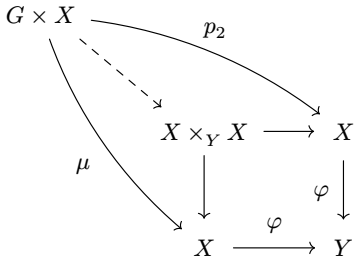




Definition 9.5 G is a k-group scheme, X, Y are k-group schemes. $\varphi : X \to Y$ is a G-torsor if the following diagram commutes



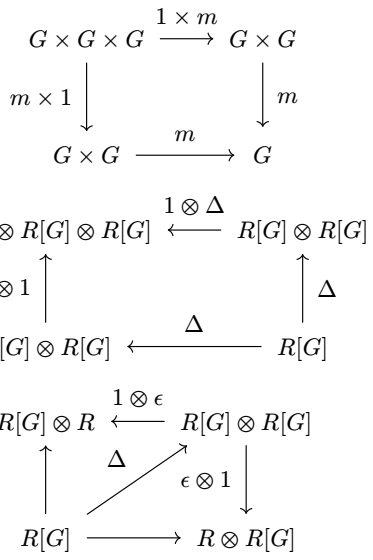
And $G \times X \to X \times_Y X$ induced by

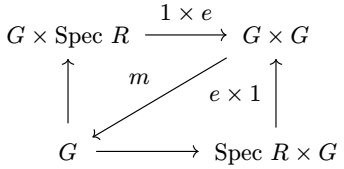


is an isomorphism

Proposition 9.6 The ring of regular functions $R[G]$ of a linear algebraic group G form a Hopf algebra

Proof.





If $g \in G(A) = \operatorname{Hom}_R(R[G], A)$, $f \in R[G]$, then write $f(g) \in A$. There are coproduct $\Delta : R[G] \to R[G] \otimes R[G]$, counit $\epsilon : R[G] \to R$, antipode(coinverse) $S : R[G] \to R$. Suppose $\Delta f = \sum f_i \otimes f_i'$, then for any $g_1, g_2 \in G(A)$, $f(g_1 g_2) = \sum f_i(g_1) f_i'(g_2)$. The counit is given by $\epsilon(f) = f(1)$. The antipode is given by $S(f)(g) = f(g^{-1})$

□

Example 9.7 The category of $\mathbb{Z}$-graded vector spaces $(V^n)_{n \in \mathbb{Z}}$ is equivalent to $\operatorname{Rep}(\mathbb{G}_m)$

Proof. Suppose the coaction is $\rho : V \to V \otimes k[t, t^{-1}], \rho(v) = \sum_n \rho_n(v) \otimes t^n$, $(\rho \otimes 1)\rho(v) = \sum_{m,n} \rho_m(\rho_n(v)) \otimes t^m \otimes t^n$. On the other hand $(1 \otimes \Delta)\rho(v) = \sum_n \rho_n(v) \otimes t^n \otimes t^n$ also $(1 \otimes \epsilon)\rho(v) = \sum_n \rho_n \otimes 1$, which would imply $\rho_n \rho_m = \delta_{nm} \rho_n, 1 = \sum_n \rho_n$, i.e. ρ_n are orthogonal idempotents. The inverse would be given $\sum_n \rho_{-n} \otimes t^n$. Thus $V = \bigoplus_n V^n$ where V^n corresponds to the subrepresentation $\rho_n : V^n \to V^n \otimes k[t, t^{-1}]$

□

Example 9.8 (Deligne torus) $\mathbb{S} = \operatorname{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$ is the Deligne torus, with Hopf algebra $\mathbb{R}[x, y, (x^2 + y^2)^{-1}]$, The coproduct is $\Delta x = x \otimes x - y \otimes y$, $\Delta y = x \otimes y + y \otimes x$, or $\mathbb{C}[z^{\pm 1}, \bar{z}^{\pm 1}]^{\mathbb{Z}/2}$, The coproduct is $\Delta x = x \otimes x - y \otimes y$, $\Delta z = z \otimes z$, $\Delta \bar{z} = \bar{z} \otimes \bar{z}$. $\mathbb{S}(\mathbb{R}) = \mathbb{C}^\times$, $\mathbb{S}(\mathbb{C}) = \mathbb{C}^\times \times \mathbb{C}^\times$

10 Cluster algebras

10.1 Cluster algebra

Lemma 10.1 $\mathbb{P}$ is a torsion free abelian group written multiplicatively, then the group ring $\mathbb{Z}\mathbb{P}$ is a UFD

Proof. Finitely generated torsion free abelian groups are free

□

Definition 10.2 (Exchange pattern) $I = \{1, \cdots, n\}$, $\mathbb{T}_n$ is the regular n tree, the coefficient group $\mathbb{P}$ is a torsion free abelian group under multiplication, thus the group ring $\mathbb{Z}\mathbb{P}$ is a domain. Cluster variables are $\mathbf{x}(t) = \{x_i(t)\}_{i\in I}$ for $t \in \mathbb{T}_n$ such that for $\neq j$ and $t -^j t'$

$$x_i(t) = x_i(t')$$

$\mathcal{M} = \{M_j(t)\}$ are monomials such that

$$M_j(t)(\mathbf{x}) = p_j(t)\prod_i x_i^{b_i}, p_j(t) \in \mathbb{P}, b_i \geq 0$$

and for $t -^j t'$, b_i's depend on j and t

$$x_j(t)x_j(t') = M_j(t)(\mathbf{x}(t)) + M_j(t')(\mathbf{x}(t'))$$

satisfying **exchange pattern**

1. $x_j \nmid M_j(t)$
2. $x_i \mid M_j(t) \Rightarrow x_i \nmid M_j(t')$ for $t -^j t'$
3. $x_j \mid M_i(t) \Leftrightarrow x_i \mid M_j(t')$ for $t -^i t' -^j t_1$
4. $\frac{M_i(t_3)}{M_i(t_4)} = \frac{M_i(t_2)}{M_i(t_1)}|_{x_j \leftarrow \frac{M_0}{x_j}}$ for $t_1 -^i t_2 -^j t_3 -^i t_4$, $M_0 = (M_j(t_2) + M_j(t_3))|_{x_i=0}$

Remark. The substitution $x_j \leftarrow \frac{M_0}{x_j}$ is effectively a monomial. Since if $M_j(t_2)$ nor $M_j(t_3)$ contain x_i, then $M_i(t_2)$ nor $M_i(t_3)$ contain x_j which it substitute for nothing

$$\frac{M_i(t_2)}{M_i(t_1)} = \left(\frac{M_i(t_2)}{M_i(t_1)}|_{x_j\leftarrow\frac{M_0}{x_j}}\right)|_{x_j\leftarrow\frac{M_0}{x_j}} = \frac{M_i(t_3)}{M_i(t_4)}|_{x_j\leftarrow\frac{M_0}{x_j}}$$

Definition 10.3 There is an involution between $(\mathbf{x}, \mathcal{M})$ and $(\mathbf{x}', \mathcal{M}')$ where $x'_j(t) = x_j(t')$, $M'_j(t) = M_j(t')$ for every $t -^j t'$

Definition 10.4 Suppose $J \subseteq I$ is a subset of size m, delete sides labeled in $I - J$ in $\mathbb{T}_n$ and choose choose a connected component which would be $\mathbb{T}_m$, add to the coefficient group x_k's $k \in I - J$. This is called a restriction

Definition 10.5 Exchange pattern on exponents is a family of $B(t)$ such that for each $t -\quad t'$

$$\frac{M_j(t)}{M_j(t')} = \frac{p_j(t)}{p_j(t')}\prod_i x_i^{b_{ij}(t)}$$

Thus

$$M_j(t) = p_j(t)\prod_i x_i^{[b_{ij}(t)]_+}, M_j(t') = p_j(t')\prod_i x_i^{[-b_{ij}(t)]_+}$$

Definition 10.6 An $n \times n$ matrix B is **sign-skew-symmetric** if $b_{ii} = 0$ and for $i \neq j$, b_{ij}, b_{ji} are both zeros or of opposite signs. B is **skew-symmetrizable** if there is a diagonal matrix D such that DB is skew symmetric, i.e. $d_i b_{ij} = -d_j b_{ji}$. Skew-symmetrizable matrices are obviously sign-skew-symmetric

Lemma 10.7

$$\frac{|a|b+a|b|}{2}=\begin{cases}ab & a,b>0\\ -ab & a,b<0\\ 0 & ab<0\end{cases}=\operatorname{sgn}(a)[ab]_+=\operatorname{sgn}(b)[ab]_+$$

Note. $|a|=[a]_++[-a]_+$

Definition 10.8 A **mutation** on a $m\times n$ $(m>n)$ matrix B in direction k denoted by μ_k is given by

$$b'_{ij}=\begin{cases}-b_{ij} & i=k \text{ or } j=k\\ b_{ij}+\frac{|b_{ik}|b_{kj}+b_{ik}|b_{kj}|}{2}=b_{ij}+\operatorname{sgn}(b_{ik})[b_{ik}b_{kj}]_+ & \text{otherwise}\end{cases}$$

Here $\mu_k(B)=B'$. μ_k is involutive

Theorem 10.9 If $B(t)$ are sign-skew-symmetric and $\mu_k(B(t))=B(t')$ for each $t-^k t'$, then it gives a exchange pattern

Proof. Suppose $B(t)$ is an exchange pattern, then $B(t)$ is obviously sign-skew-symmetric. For $t-^k t'$, we have

$$\frac{M_k(t)}{M_k(t')}=\frac{p_k(t)}{p_k(t')}\prod_i x_i^{b_{ik}},\frac{M_k(t')}{M_k(t)}=\frac{p_k(t')}{p_k(t)}\prod_i x_i^{b'_{ik}}$$

Hence $b'_{ik}=-b_{ik}$. Consider $t_1-^j t'-^k t-^j t_2$

$$\frac{M_j(t')}{M_j(t_1)}=\frac{M_j(t)}{M_j(t_2)}|_{x_k\leftarrow\frac{M_0}{x_k}}$$

becomes

$$\frac{p_j(t')}{p_j(t_1)}\prod_i x_i^{b'_{ij}}=\frac{p_j(t)}{p_j(t_2)}\prod_i x_i^{b_{ij}}|_{x_k\leftarrow\frac{M_0}{x_k}}$$

Where

$$M_0=\left(p_k(t)\prod_i x_i^{[b_{ik}]_+}+p_k(t')\prod_i x_i^{[-b_{ik}]_+}\right)|_{x_j=0}$$

1. $b_{jk}>0\Leftrightarrow b_{kj}<0$, then $M_0=p_k(t')\prod_{i\neq j}x_i^{[-b_{ik}]_+}$, thus

   $\prod_{i\neq j}x_i^{b'_{ij}}=\prod_{i\neq j,k}x_i^{b_{ij}}\cdot\left(x_k^{-1}\prod_{i\neq j,k}x_i^{[-b_{ik}]_+}\right)^{b_{kj}}=\prod_{i\neq j,k}x_i^{b_{ij}+b_{kj}[-b_{ik}]_+}x_k^{-b_{kj}}$
2. $b_{jk}<0\Leftrightarrow b_{kj}>0$, then $M_0=p_k(t')\prod_{i\neq j}x_i^{[-b_{ik}]_+}$, thus

   $\prod_{i\neq j}x_i^{b'_{ij}}=\prod_{i\neq j,k}x_i^{b_{ij}}\cdot\left(x_k^{-1}\prod_{i\neq j,k}x_i^{[b_{ik}]_+}\right)^{b_{kj}}=\prod_{i\neq j,k}x_i^{b_{ij}+b_{kj}[b_{ik}]_+}x_k^{-b_{kj}}$
3. $b_{jk}=0\Leftrightarrow b_{kj}=0$, then

$$\prod_{i\neq j,k}x_i^{b'_{ij}}=\prod_{i\neq j,k}x_i^{b_{ij}}$$

Therefore $b'_{kj}=-b_{kj}$ and $b'_{ij}=b_{ij}+\operatorname{sgn}(b_{ik})[b_{ik}b_{kj}]_+$

Conversely, if $B(t)$ are sign-skew-symmetric and $\mu_k(B(t))=B(t')$ for each $t-^k t'$, take

$$M_k(t)=\prod_i x_i^{[b_{ik}(t)]_+},M_k(t')=\prod_i x_i^{[-b_{ik}(t)]_+}$$

for $t-^k t'$, then obviously $x_k\nmid M_k(t)$ since $b_{kk}=0$ and

$$x_j\mid M_k(t)\Leftrightarrow b_{jk}>0\Leftrightarrow -b_{jk}<0\Rightarrow x_j\nmid M_k(t')$$

For $t -^k t' -^j t_1$

$$x_j \mid M_k(t) \Leftrightarrow b_{jk} > 0 \Leftrightarrow b'_{kj} = -b_{kj} > 0 \Leftrightarrow x_k \mid M_j(t')$$

For $t_1 -^j t' -^k t -^j t_2$, it is the exact argument above by taking $p_j(t) \equiv 1$

□

Proposition 10.10 Given a skew-symmetrizable matrix B, the all possible mutations $B(t)$ in $\mathbb{T}_n$ are skew-symmetrizable with the same skew-symmetrizing matrix D

Proof. True for each mutation μ_k

□

Remark. For cluster algebra of rank $n \leq 2$, the exchange pattern is skew-symmetrizable. If $n = 1$, $B(t) \equiv 0$. If $n = 2$, $B(t_n) = (-1)^n \begin{pmatrix} 0 & b \\ -c & 0 \end{pmatrix}$

Definition 10.11 Denote $2n$ tuple $\mathbf{p}(t)$ the coefficients $p_j(t), p_j(t')$ for $t -^j t'$. $\Sigma(t) = (\mathbf{x}(t), \mathbf{p}(t), B(t))$ is a **seed**, $\mathbf{x}(t)$ is the **cluster** of the seed. If we assume $\mathbf{x}(t_0)$ are algebraically independent ($\mathbf{x}(t_0)$ is a cluster of rank n), then so are $\mathbf{x}(t)$ since they are all mutationally equivalent. Denote the collection of all cluster variables $\mathcal{X}$, the collection of all coefficients $\mathcal{P}$, the collection of exchange matrices $\mathcal{B}$, the collection of $M_j(t)$'s $\mathcal{M}$, the collection of seeds $\mathcal{S}$

We can take $\mathcal{F} = \mathbb{ZP}(x_1, \cdots, x_n)$ to be the **ambient field**, $\mathbf{x}$ can be some cluster $\mathbf{x}(t_0)$. The **cluster algebra** is the subalgebra $\mathbb{Z}\mathcal{P}[\mathcal{X}]$

Proposition 10.12 Given $B(t)$ that give rise to exchange pattern, the coefficients must satisfy

$$p_i(t_1)p_i(t_3)p_i(t_3)^{[b_{ji}(t_3)]_+} = p_i(t_2)p_i(t_4)p_i(t_2)^{[b_{ji}(t_2)]_+} \tag{10.1}$$

Proof. For $t_1 -^i t_2 -^j t_3 -^i t_4$

$$\frac{p_i(t_3)}{p_i(t_4)} \prod_k x_k^{b_{ki}(t_3)} = \frac{M_i(t_3)}{M_i(t_4)} = \frac{M_i(t_3)}{M_i(t_4)}\Big|_{x_j \leftarrow \frac{M_0}{x_j}} = \frac{p_i(t_2)}{p_i(t_1)} \prod_k x_k^{b_{ki}(t_2)}\Big|_{x_j \leftarrow \frac{M_0}{x_j}}$$

Here

$$M_0 = \big(M_j(t_2) + M_j(t_3)\big)|_{x_i=0} = \left(p_j(t_2) \prod_k x_k^{[b_{kj}(t_2)]_+} + p_j(t_3) \prod_k x_k^{[b_{kj}(t_3)]_+}\right)|_{x_i=0}$$

Take $x_k = 1$ for $k \neq j$, and use the fact that $B(t)$ are sign-skew-symmetric, we get

$$p_i(t_1)p_i(t_3) = p_i(t_2)p_i(t_4)M_0^{b_{ji}(t_2)}$$

With

1. $b_{ij}(t_2) > 0$. Then $b_{ij}(t_3) < 0$, $M_0 = p_j(t_3)$ and
$$p_i(t_1)p_i(t_3)p_i(t_3)^{b_{ji}(t_3)} = p_i(t_2)p_i(t_4)$$
2. $b_{ij}(t_2) < 0$. Then $b_{ij}(t_3) > 0$, $M_0 = p_j(t_2)$ and
$$p_i(t_1)p_i(t_3) = p_i(t_2)p_i(t_4)p_i(t_2)^{b_{ji}(t_2)}$$
3. $b_{ij}(t_2) = 0$. Then $b_{ij}(t_3) = 0$, $M_0 = p_j(t_2) + p_j(t_3)$, but $b_{ji}(t_2) = b_{ji}(t_3) = 0$, hence
$$p_i(t_1)p_i(t_3) = p_i(t_2)p_i(t_4)$$

□

Note. A trivial solution of 10.1 is $p_j(t) = 1$

Proposition 10.13 The **universal coefficient group** $\mathcal{P}$ of $\mathbb{P}$ is the free abelian group generated by $p_i(t)$ modulo 10.1. $\mathcal{P}$ is torsion free, more precisely, it is the free abelian group generated by $p_i(t_0), p_i(t)$ for every $t_0 -^i t$ and exactly one of $p_i(t), p_i(t')$ for every $t -^i t'$ where $t, t' \neq t_0$

Definition 10.14 Take the field of rational functions of cluster variables $\mathbf{x}(t_0)$ with coefficients in $\mathbb{Z}\mathcal{P}$ to be the ambient field $\mathcal{F}$, all other cluster variables $\mathbf{x}(t)$ are also in $\mathcal{F}$ by Theorem 10.18. The **universal cluster algebra** $\mathcal{A}$ is the subalgebra generated by all cluster variables with coefficients in $\mathbb{Z}\mathcal{P}$

Definition 10.15 $t, t' \in \mathbb{T}_n$ are $\mathcal{M}$**-equivalent** if there is a permutation σ of I such that

- $x_{\sigma(i)}(t) = x_i(t')$
- $M_{\sigma(j)}(t)(\mathbf{x}(t)) = M_j(t')(\mathbf{x}(t'))$ and $M_{\sigma(j)}(t_1)(\mathbf{x}(t)) = M_j(t_1')(\mathbf{x}(t'))$ for $t -^{\sigma(j)} t_1$ and $t' -^j t_1'$

10.2 Laurent phenomenon

Lemma 10.16 (Caterpillar lemma) Define the caterpillar tree $\mathbb{T}_{n,m}$ consists of a spine of $m+2$ nodes, with an orientation from t_{tail} to t_{head} with t_{base} connected to t_{tail}, as illustrated in Figure 1

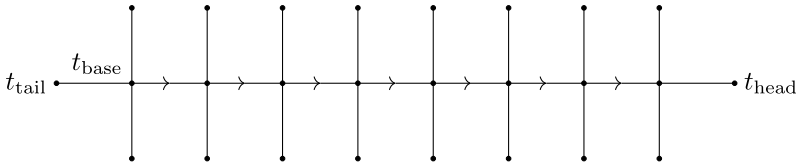


Figure 1: $\mathbb{T}_{4,8}$

Let $\mathbb{A}$ be a UFD, exchange polynomial $P \in \mathbb{A}[x_1, \cdots, x_n]$ for each edge $t -^j t'$, denoted $x -_P^j t'$ satisfying the generalized exchange pattern

- P doesn't depend on x_j and x_i doesn't divide P
- For $t_0 -_P^i t_1 \overset{j}{\underset{Q}{\to}} t_2$, P, Q_0 are coprime in $\mathbb{A}[x_1, \cdots, x_n]$, where $Q_0 = Q|_{x_i=0}$
- For $t_0 -_P^i t_1 \overset{j}{\underset{Q}{\to}} t_2 -_R^i t_3$, $LQ_0^b P = R|_{x_j \leftarrow \frac{Q_0}{x_j}}$ for some $b \geq 0$ and some Laurent monomial L with coefficients in $\mathbb{A}$ coprime with P

Cluster variables $\mathbf{x}(t) = \{x_i(t)\}$ for $t \in \mathbb{T}_{n,m}$ satisfying for each $t -_P^i t'$

- $x_i(t) = x_i(t')$ for any $i \neq j$
- $x_j(t)x_j(t') = P(t)(\mathbf{x}(t))$

Then $\mathbf{x}(t_{\text{head}})$ are Laurent polynomials in $\mathbf{x}(t_0)$ with coefficients in $\mathbb{A}$

Proof. Write the subring of Laurent polynomials generated by $\mathbf{x}(t)$ as

$$\mathcal{L}(t) = \mathbb{A}[x_1(t)^{\pm 1}, \cdots, x_n(t)^{\pm 1}]$$

Make induction on m. If $m = 1$, consider $t_{\text{tail}} = t -_P^i t_{\text{base}} = t' -_Q^j t_{\text{head}} = t_1$, we have for $k \neq i, j$

$$x_k(t_1) = x_k(t') = x_k(t)$$

$$x_i(t_1) = x_i(t') = \frac{P(\mathbf{x}(t))}{x_i(t)}$$

$$x_j(t_1) = \frac{Q(\mathbf{x}(t'))}{x_j(t')} = \frac{Q(\mathbf{x}(t'))}{x_j(t)}$$

Now suppose $m \geq 2$, let's show that $X = x_k(t_{\text{head}}) \in \mathcal{L}(t_0)$, by induction, $X \in \mathcal{L}(t_1) \cap \mathcal{L}(t_3)$. Since $X, x_i(t_1) = \frac{P(\mathbf{x}(t_0))}{x_i(t_0)} \in \mathcal{L}(t_0)$, $X = \frac{f_0}{x_i(t_1)^a}$ for some $f_0 \in \mathcal{L}(t_0)$ and $a \geq 0$, similarly, $X = \frac{g_0}{x_j(t_2)^b x_i(t_3)^c}$ for some $g_0 \in \mathcal{L}(t_0)$ and $b, c \geq 0$, thanks to Lemma 10.17, $X \in \mathcal{L}(t_0)$

□

Lemma 10.17 For $t_0 \overset{i}{-}_P t_1 \overset{j}{\underset{Q}{\to}} t_2 \overset{i}{-}_R t_3$, $\mathbf{x}(t_1), \mathbf{x}(t_2), \mathbf{x}(t_3) \in \mathcal{L}(t_0)$, and

$$\gcd(x_i(t_1), x_i(t_3)) = \gcd(x_j(t_2), x_i(t_1)) = 1$$

in $\mathcal{L}(t_0)$

Note. $\mathcal{L}(t_0)$ is a UFD, $\mathcal{L}(t_0)^\times$ consists of Laurent monomials with coefficients $\mathbb{A}^\times$

Proof. Denote $x = x_i(t_0)$, $y = x_j(t_0) = x_j(t_1)$, $z = x_i(t_1) = x_i(t_2)$, $u = x_j(t_2) = x_j(t_3)$, $v = x_i(t_3)$, think of P, Q, R as functions of x_j, x_i, x_j respectively, then

$$z = \frac{P(y)}{x}$$

$$u = \frac{Q(z)}{y} = \frac{Q\left(\frac{P(y)}{x}\right)}{y}$$

$$v = \frac{R(u)}{z} = \frac{R\left(\frac{Q(z)}{y}\right)}{z} = \frac{R\left(\frac{Q(z)}{y}\right) - R\left(\frac{Q(0)}{y}\right)}{z} + \frac{R\left(\frac{Q(0)}{y}\right)}{z}$$

$$\frac{R\left(\frac{Q(z)}{y}\right) - R\left(\frac{Q(0)}{y}\right)}{z} = R'\left(\frac{Q_0}{y}\right)\frac{Q'(0)}{y} + \frac{1}{2}R\left(\frac{Q(z)}{y}\right)''|_{z=0}\ z + \cdots \equiv R'\left(\frac{Q_0}{y}\right)\frac{Q'(0)}{y} \mod z$$

$$\frac{R\left(\frac{Q_0}{y}\right)}{z} = \frac{L(y)Q_0(y)^b P(y)}{z} = L(y)Q_0(y)^b x$$

Thus $v \in \mathcal{L}(t_0)$

Since $\gcd(P, Q_0) = \gcd(P, L) = 1$

$$\gcd(z, v) = \gcd\left(\frac{P(y)}{x}, L(y)Q_0(y)^b x\right) = \gcd\left(P(y), L(y)Q_0(y)^b\right) = 1$$

Since $\frac{Q(z)}{y} \equiv \frac{Q_0}{y} \mod z$

$$\gcd(z, u) = \gcd\left(z, \frac{Q_0}{y}\right) = \gcd(P(y), Q_0) = 1$$

□

Theorem 10.18 Catepillar lemma 10.16 implies that in a cluster algebra, any cluster variable can be expressed as a Laurent polynomial in a given $\mathbf{x}(t_0)$ with coefficients in $\mathbb{Z}_{\geq 0}\mathbb{P}$ since there is no subtraction involved

Proof. $\mathbb{T}_{n,m}$ can be embedded in $\mathbb{T}_n$. $M_j(t) + M_j(t')$ doesn't depend on x_j and not divisible by x_i for $t \overset{j}{-} t'$ and any $i \neq j$

For $t_0 \overset{i}{-}_P t_1 \overset{j}{\underset{Q}{\to}} t_2 \overset{i}{-}_R t_3$, we have

$$\frac{P}{M_i(t_0)} = 1 + \frac{M_i(t_1)}{M_i(t_0)} = 1 + \frac{M_i(t_2)}{M_i(t_3)}|_{x_j \leftarrow \frac{M_0}{x_j}} = \frac{R}{M_i(t_3)}|_{x_j \leftarrow \frac{M_0}{x_j}}$$

Where $M_0 = (M_j(t_1) + M_j(t_2))|_{x_i=0} = Q_0$, thus

$$\frac{R|_{x_j \leftarrow \frac{Q_0}{x_j}}}{P} = \frac{M_i(t_3)|_{x_j \leftarrow \frac{Q_0}{x_j}}}{M_i(t_0)}$$

Note that $M_i(t_0) = p_i(t_0)\prod_k x_k^{[b_{ki}(t_0)]_+}$ and

$$\begin{aligned}
M_i(t_3)|_{x_j\leftarrow\frac{Q_0}{x_j}} &= p_i(t_3)\prod_k x_k^{[b_{ki}(t_3)]_+}|_{x_j\leftarrow\frac{Q_0}{x_j}} \\
&= p_i(t_3)\left(\frac{Q_0}{x_j}\right)^{[b_{ji}(t_3)]_+}\prod_{k\neq i,j} x_k^{[b_{ki}(t_3)]_+} \\
&= p_i(t_3)Q_0^{[b_{ji}(t_3)]_+}x_j^{-[b_{ji}(t_3)]_+}\prod_{k\neq i,j} x_k^{[b_{ki}(t_3)]_+}
\end{aligned}$$

Hence

$$R|_{x_j\leftarrow\frac{Q_0}{x_j}} = \frac{p_i(t_3)}{p_i(t_0)}x_j^{-[b_{ji}(t_3)]_+-[b_{ji}(t_0)]_+}\prod_{k\neq i,j} x_k^{[b_{ki}(t_3)]_+-[b_{ki}(t_0)]_+}Q_0^{[b_{ji}(t_3)]_+}P = LQ_0^bP$$

Since the sum of two monomials P doesn't depend on x_i and is not divisible by any x_k for $k\neq i$, Q_0 is a monomial, L is a Laurent monomial, Q_0, P are coprime in $\mathbb{A}[\mathbf{x}]$ and L, P are coprime in $\mathcal{L}[\mathbf{x}]$

□

10.3 Y-system

A is the Cartan matrix of root system Φ with simple system Π, denote $[\alpha:\alpha_i]$ as the coefficients of $\alpha\in\Phi$, write $\Phi_{\geq -1}=\Phi_+\cup(-\Pi)$. Since the Coxeter graph is a tree, it is bipartite, up to renaming $I=\{1,\cdots,n\}=I_-\sqcup I_+$, $\varepsilon(i)=\varepsilon$ for $i\in I_\varepsilon$ be the indicator. Let $t_\varepsilon=\prod_{i\in I_\varepsilon}s_i$, $t=t_-t_+$ is a Coxeter element, h is the Coxeter number. $s_{\mathbf{i}_-}, s_{\mathbf{i}_+}$ are reduced words of t_-, t_+, then

$$w_\circ = \underbrace{s_{\mathbf{i}_-}s_{\mathbf{i}_+}\cdots s_{\mathbf{i}_\pm}}_{h \text{ times}} = s_{\mathbf{i}_\circ}$$

is the element of longest length

Definition 10.19 Suppose Φ is irreducible, then

$$[s_i(\alpha):\alpha_k] = \begin{cases} -[\alpha:\alpha_i]-\sum_{j\neq i}a_{ij}[\alpha:\alpha_j] & k=i \\ [\alpha:\alpha_k] & k\neq i \end{cases}$$

Define a piecewise linear modification

$$[\sigma_i(\alpha):\alpha_k] = \begin{cases} -[\alpha:\alpha_i]-\sum_{j\neq i}a_{ij}[\alpha:\alpha_j]_+ & k=i \\ [\alpha:\alpha_k] & k\neq i \end{cases}$$

Proposition 10.20

1. σ_i are involutions
2. σ_i, σ_j commutes if i, j are not adjacent in the Coxeter graph
3. σ_i preserves $\Phi_{\geq -1}$

Proposition 10.21 Let $\tau_\varepsilon=\prod_{i\in I_\varepsilon}\sigma_i$

1. τ_ε are involutions that preserve $\Phi_{\geq -1}$
2. $\tau_\varepsilon\alpha = t_\varepsilon\alpha$ for $\alpha\in\mathbb{Z}_{\geq 0}\Pi$
3. $\Phi_{\geq -1}\to\Phi^\vee_{\geq -1}$, $\alpha\mapsto\alpha^\vee$ are τ_ε equivariant, i.e. $(\tau_\varepsilon\alpha)^\vee=\tau_\varepsilon\alpha^\vee$

Definition 10.22 A **Y-system** is a family of commuting variables $Y_i(t)$, $i\in I=\{1,\cdots,n\}$, $t\in\mathbb{Z}$ such that

$$Y_i(t+1)Y_i(t-1)=\prod_{j\neq i}(1+Y_i(t))^{-a_{ij}} \tag{10.2}$$

Remark. 10.2 only really involve those $Y_j(k)$ with $\varepsilon(j)\cdot(-1)^k = \text{const}$, assume $Y_j(k) = Y_j(k+1)$ for $\varepsilon(i) = (-1)^k$, then we have

$$Y_i(k+1) = \begin{cases} \frac{\prod_{j\neq i}(1+Y_i(k))^{-a_{ij}}}{Y_i(k)} & \varepsilon(i) = (-1)^k + 1 \\ Y_i(k) & \varepsilon(i) = (-1)^k \end{cases}$$

Denote $\mathcal{Y}$ as the collection of all $Y_j(k)$'s, $u_i = Y_i(0)$, define

$$\tau_\varepsilon(u_i) = \begin{cases} \frac{\prod_{j\neq i}(1+u_i)^{-a_{ij}}}{u_i} & \varepsilon(i) = \varepsilon \\ u_i & \text{otherwise} \end{cases}$$

Theorem 10.23 (Zamolodichikov) $Y_i(t)$'s are $2(h+2)$ periodic, i.e. $Y_i(t+2(h+2)) = Y_i(t)$

Theorem 10.24 There is a unique family $\{F[\alpha]\}_{\alpha\in\Phi_{\geq -1}}$ of polynomials in u_i such that $F[-\alpha_i] = -1$ and

$$\tau_\varepsilon(F[\alpha]) = \frac{\prod_{\varepsilon(i)=-\varepsilon}(u_i+1)^{[\alpha^\vee:\alpha_i^\vee]}}{\prod_{\varepsilon(i)=\varepsilon} u_i^{[\alpha^\vee:\alpha_i^\vee]_+}} F[\tau_{-\varepsilon}(\alpha)]$$

Furthermore, $F[\alpha] \in \mathbb{Z}_{\geq 0}[\mathbf{u}]$ has constant term 1. Call $F[\alpha]$ Fibonacci polynomials

Any $\alpha \in \Phi_{\geq -1}$ can be written as $\alpha(k,i) = (\tau_-\tau_+)^k(-\alpha_i)$, denote $N[\alpha] = \prod_{j\neq i} F[\alpha(-k,i)]^{-a_{ij}}$, note that $N[\alpha] \in \mathbb{Z}_{\geq 0}[\mathbf{u}]$ also has constant term 1

Theorem 10.25 There is a unique bijection $\Phi_{\geq} - 1 \to \mathcal{Y},\ \alpha \mapsto Y[\alpha] = \frac{N[\alpha]}{\mathbf{u}^{\alpha^\vee}}$ such that $Y[-\alpha_i] = u_i$, $\tau_\varepsilon(Y[\alpha]) = Y[\tau_\varepsilon(\alpha)]$

Definition 10.26 A **tropical specialization** r_{trop} of a rational expression r is changing the addition $+$ and multiplication $\cdot$ into $\oplus$ and $\odot$ where $a \oplus b = \max(a,b)$, $a \odot b = a + b$

The **compatibility degree** for $\alpha, \beta \in \Phi_{\geq -1}$ is

$$(\alpha||\beta) = (Y[\alpha]+1)(\beta)_{\text{trop}}$$

Here $(Y[\alpha]+1)(\beta)$ is evaluation at $\{u_i = [\beta:\alpha_i]\}$. α, β are **compatible** if $(\alpha||\beta) = 0$

$\Delta(\Phi)$ is a simplicial complex with $\Phi_{\geq -1}$ as vertices and mutually compatible subsets of $\Delta(\Phi)$ are simplices, the maximal simplices are **clusters**. The **exchange graph** $E(\Phi)$ is an unoriented graph with clusters as vertices and an edge between clusters which has intersection of cardinality $n-1$

Remark. $(||)$ is uniquely characterized by

$$(-\alpha_i||\beta) = (Y[-\alpha_i]+1)(\beta)_{\text{trop}} = (u_i+1)(\beta)_{\text{trop}} = [\beta:\alpha_i]_+$$

$$(\tau_\varepsilon(\alpha)||\tau_\varepsilon(\beta)) = (Y[\tau_\varepsilon(\alpha)]+1)(\tau_\varepsilon(\beta))_{\text{trop}} = (\tau_\varepsilon(Y[\alpha])+1)(\tau_\varepsilon(\beta))_{\text{trop}} = (\alpha||\beta)$$

Proposition 10.27 Consider perfect bilinear pairing

$$\begin{aligned} \mathbb{Z}\Pi^\vee \times \mathbb{Z}\Pi &\to \mathbb{Z} \\ (\xi,\gamma) &\mapsto \{\xi,\gamma\} \end{aligned}$$

Where $\{\xi,\gamma\} = \sum \varepsilon(i)[\xi:\alpha_i^\vee][\gamma:\alpha_i]$. Then

$$(\alpha||\beta) = \max(\{\tau_+\alpha^\vee,\beta\},\{\alpha^\vee,\tau_+\beta\},0) = \max(-\{\tau_-\alpha^\vee,\beta\},-\{\alpha^\vee,\tau_-\beta\},0)$$

Note. $(||)$ doesn't depend on the choice of the indicator ε

Proposition 10.28

1. $(\alpha||\beta) = (\beta^\vee||\alpha^\vee)$, in particular, if Φ is simply laced, then $(\alpha||\beta) = (\beta||\alpha)$
2. If $(\alpha||\beta) = 0$, then $(\beta||\alpha) = 0$

3. $J \subseteq I$, $\Phi(J) \subseteq \Phi$ is a root subsystem, $(||)$ on $\Phi(J)$ is the same as the restriction

Theorem 10.29 $\Delta(\Phi)$ is pure of dimension $n-1$, and each facet forms a $\mathbb{Z}$-basis for the root lattice

Theorem 10.30 The simplicial cones of all clusters form a complete simplicial fan

Corollary 10.31 The geometric realization of $\Delta(\Phi)$ is $\mathbb{S}^{n-1}$

Conjecture 10.32 The simplicial fan of $\Delta(\Phi)$ is the normal fan of some convex polytope $P(\Phi)$

Theorem 10.33 $E(\Phi)$ is a regular n tree

Example 10.34

10.4 Associahedron

Definition 10.35 Any n regular polygon has $\binom{n}{2} - n = \frac{n(n-3)}{2}$ diagonals, with these as vertices, noncrossing subsets as simplexes, we have given it a abstract simplicial complex structure

10.5 Cluster algebra of geometric type

Lemma 10.36 Semifield $\mathbb{P}$ is multiplicative torison free

Proof. Suppose $p^m = 1$, then

$$p = \frac{p^m \oplus p^{m-1} \oplus \cdots + p}{p^{m-1} \oplus p^{m-2} \oplus \cdots + 1} = \frac{1 \oplus p^{m-1} \oplus \cdots + p}{p^{m-1} \oplus p^{m-2} \oplus \cdots \oplus 1} = 1$$

□

Definition 10.37 Exchange pattern is **normalized** if $\mathbb{P}$ is a semifield and $p_j(t) \oplus p_j(t') = 1$ for any $t -^j t'$

Proposition 10.38 Given p_j, r_j in a semifield $\mathbb{P}$ such that $p_j \oplus r_j = 1$, and exchange matrix $B(t)$ on $\mathbb{T}_n$, define $p_j(t_0) = p_j$, $p_j(t) = r_j$ for each $t_0 -^i t$, this completely determines the cluster algebra

Proof. Define $u_j(t) = \frac{p_j(t)}{p_j(t')}$ for $t -^j t'$, then

$$p_j(t) = \frac{u_j(t)}{1 \oplus u_j(t)}, p_j(t') = \frac{1}{1 \oplus u_j(t)}$$

Then 10.1 becomes

$$u_i(t_3) p_j(t_3)^{[b_{ji}(t_3)]_+} = u_i(t_2) p_j(t_2)^{[b_{ji}(t_2)]_+}$$

1. $u_i(t_3) p_j(t_3)^{b_{ji}(t_3)} = u_i(t_2) \Rightarrow u_i(t_3) = u_i(t_2)(1 \oplus u_j(t_2))^{b_{ji}(t_2)}$
2. $u_i(t_3) = u_i(t_2) p_j(t_2)^{b_{ji}(t_2)} = u_i(t_2)\left(\frac{u_j(t_2)}{1 \oplus u_j(t_2)}\right)^{b_{ji}(t_2)}$

Thus for $t -^j t'$, we have

$$u_i(t') = u_i(t) u_j(t)^{[b_{ji}(t)]_+} (1 \oplus u_j(t))^{-b_{ji}(t)}$$

□

Remark. $\mathbf{p}$ determines $\mathbf{u}$ which in turn determines $\mathbf{p}$

Fix semifield $\mathbb{P}$, B is skew-symmetrizable, then $(B, \mathbf{p})$ determines the cluster algebra $\mathcal{A} = \mathcal{A}(B, \mathbf{p})$ up to isomorphism

Corollary 10.39 The exchange graph of a normalized cluster algebra is n-regular

Definition 10.40 The **tropical semifield** $(\mathbb{R}, \oplus, \odot)$ is a semifield with multiplication as $\odot$, min or max as $\oplus$

The tropical semifield generated by p is the free abelian group generated multiplicatively by p with $p^a \oplus p^b = p^{\min(a,b)}$

Definition 10.41 A normalized cluster algebra is of geometric type if $\mathbb{P}$ is the tropical semifield generated by $\{p_i\}_{i\in I'}$ and each $p_j(t)$ is a monomial with nonnegative exponents

Remark. In this particular case, normality just means that for $t -^j t'$, $p_j(t), p_j(t')$ doesn't have a common variable, or the support doesn't intersect

Proposition 10.42 $\mathbb{P}$ is the tropical semifield generated by $p_i, i \in I'$, $B(t)$ is the exchange pattern of exponents, $p_j(t)$ give rise to a cluster algebra of geometric type iff $C(t)$ satisfies the exchange pattern of coefficients, i.e. $p_j(t) = \prod_i p_i^{[c_{ij}(t)]_+}$ and

$$c'_{ij} = \begin{cases} -c_{ij} & j = k \\ c_{ij} + \frac{|c_{ij}|b_{jk}+c_{ij}|b_{jk}|}{2} & \text{otherwise} \end{cases}$$

Here the mutation is in direction k

Proof. Suppose $p_j(t)$ give rise to a cluster algebra of geometric type. Define $u_j(t) = \frac{p_j(t)}{p_j(t')} = \prod_{i\in I'} p_i^{c_{ij}(t)}$ for each $t -^j t'$, then according to Proposition 10.38

$$p_j(t) = \frac{u_j(t)}{1 \oplus u_j(t)} = \frac{\prod_i p_i^{c_{ij}}}{1 \oplus \prod_i p_i^{c_{ij}}} = \frac{\prod_i p_i^{c_{ij}}}{\prod_i p_i^{-[-c_{ij}]_+}} = \prod_i p_i^{[c_{ij}]_+}$$

$$1 = u_k(t)u_k(t') = \prod_i p_i^{c_{ik}+c'_{ik}} \Rightarrow c'_{ik} = -c_{ik}$$

And

$$\begin{aligned} \prod_i p_i^{c'_{ij}} &= \prod_i p_i^{c_{ij}} \left(\prod_i p_i^{c_{ik}}\right)^{[b_{kj}]_+} \left(1 \oplus \prod_i p_i^{c_{ik}}\right)^{-b_{kj}} \\ &= \prod_i p_i^{c_{ij}} \prod_i p_i^{c_{ik}[b_{kj}]_+} \prod_i p_i^{b_{kj}[-c_{ik}]_+} \\ &= \prod_i p_i^{c_{ij}+\frac{|c_{ij}|b_{jk}+c_{ij}|b_{jk}|}{2}} \end{aligned}$$

□

Remark. Note if we take $\tilde{B}(t) = (\tilde{b}_{ij})_{i\in I\cup I', j\in I}$ where $\tilde{b}_{ij} = b_{ij}$ for $i, j \in I$ is the principal part of $\tilde{B}$, $\tilde{b}_{ij} = c_{ij}$ for $i \in I', j \in I$

Corollary 10.43 Given $\tilde{B}_0$ with a skew-symmetrizable principal part B_0, then there exists a unique exchange pattern of geometric type such that $\tilde{B}(t_0) = \tilde{B}_0$ for $t_0 \in \mathbb{T}_n$

Proof. By Proposition 10.10

□

Remark. The class of exchange patterns of geometric type is stable under restriction and direct product

10.6 Rank two case

Example 10.44 If $n = 2$, consider $\mathbb{T}_2$

$$-^1 t_0 -^2 t_1 -^1 t_2 -^2 t_3 -^1 t_4 -^2 t_5 -^1$$

The cluster variables are y_i, y_{i+1} for t_i

$$y_{2k+1} = x_1(t_{2k}) = x_1(t_{2k+1}), y_{2k} = x_2(t_{2k-1}) = x_2(t_{2k})$$

$M_2(t_0)$ and $M_2(t_1)$ don't have x_1 and can't both have x_2

If both of them don't have x_2, then $M_2(t_0), M_2(t_1) \in \mathbb{P}$, thus

$$\cdots x_2 \nmid M_1(t_{-1}) \Leftrightarrow x_1 \nmid M_2(t_0) \Leftrightarrow x_2 \nmid M_1(t_1) \Leftrightarrow x_1 \nmid M_2(t_2) \cdots$$

$$\cdots x_2 \nmid M_1(t_0) \Leftrightarrow x_1 \nmid M_2(t_1) \Leftrightarrow x_2 \nmid M_1(t_2) \Leftrightarrow x_1 \nmid M_2(t_3) \cdots$$

So is every $M_*(t_*) \in \mathbb{P}$, write q_m, r_m as the two monomials of $t_{m-1} - \quad t_m$, then we have

$$y_{m-1}y_{m+1} = q_m + r_m$$

And for $t_{m-2} - \quad t_{m-1} - \quad t_m - \quad t_{m+1}$ we have

$$\frac{q_{m+1}}{r_{m+1}} = \frac{r_{m-1}}{q_{m-1}} \Leftrightarrow q_{m-1}q_{m+1} = r_{m-1}r_{m+1}$$

If $M_2(t_0) = q_1 x_1^b$, $M_2(t_1) = r_1$(the other case corresponds to the involution) for some $b > 0$, then $M_1(t_1) = q_2 x_2^c$, $M_1(t_2) = r_2$ for some $c > 0$, we have

$$\frac{M_2(t_2)}{M_2(t_3)} = \frac{M_2(t_1)}{M_2(t_0)}|_{x_1 \leftarrow \frac{M_0}{x_1}} = \frac{r_1}{q_1 x_1^b}|_{x_1 \leftarrow \frac{r_2}{x_1}} = \frac{r_1 x_1^b}{q_1 r_2^b}$$

Since $x_1 \mid M_2(t_3) \Rightarrow x_2 \nmid M_2(t_2)$ gives a contradiction, $x_1 \nmid M_2(t_3) \Rightarrow M_2(t_3) = r_3$, thus $M_2(t_2) = q_3 x_1^b$, periodically, we can conclude

$$\begin{aligned} M_2(t_{2k}) &= q_{2k+1}x_1^b \\ M_2(t_{2k+1}) &= r_{2k+1} \\ M_1(t_{2k-1}) &= q_{2k}x_2^c \\ M_1(t_{2k}) &= r_{2k} \end{aligned}$$

Therefore we have

$$\begin{aligned} y_{2k-1}y_{2k+1} &= q_{2k}y_{2k}^c + r_{2k} \\ y_{2k}y_{2k+2} &= q_{2k+1}y_{2k+1}^b + r_{2k+1} \end{aligned}$$

For $t_{2k-1} -^1 t_{2k} -^2 t_{2k+1} -^1 t_{2k+2}$ we have

$$q_{2k}q_{2k+2}r_{2k+1}^c = r_{2k}r_{2k+2}$$

For $t_{2k-2} -^2 t_{2k-1} -^1 t_{2k} -^2 t_{2k+1}$ we have

$$q_{2k-1}q_{2k+1}r_{2k}^b = r_{2k-1}r_{2k+1}$$

The exchange matrices are

$$B(t_m) = (-1)^m \begin{pmatrix} 0 & b \\ -c & 0 \end{pmatrix}$$

Conversely, given such relation, we can always find a corresponding cluster algebra

In particular, consider the coordinate ring of $\mathrm{Gr}_2(5)$, let

$$\begin{aligned} y_m &= [\overline{2m-1}, \overline{2m+1}] \\ q_m &= [\overline{2m-2}, \overline{2m+2}] \\ r_m &= [\overline{2m-2}, \overline{2m-1}][\overline{2m+1}, \overline{2m+2}] \\ b &= c = 1 \end{aligned}$$

Note. If we denote $m \mod 2$ as $\langle m \rangle$, then

$$p_{\langle m\rangle}(t_m) = q_{m+1}$$
$$p_{\langle m+1\rangle}(t_m) = r_m$$
$$x_{\langle m\rangle}(t_m) = q_m$$
$$x_{\langle m+1\rangle}(t_m) = y_{m+1}$$

Example 10.45 As in Example 10.44, by Theorem 10.18, we know that

$$y_m = \frac{N_m(y_1, y_2)}{y_1^{d_1(m)} y_2^{d_2(m)}}$$

Where $N_m(y_1, y_2) \in \mathbb{ZP}[y_1, y_2]$ not divisible by y_1 or y_2

10.7 Cluster algebra of finite type

Definition 10.46 Seeds $\Sigma(t), \Sigma(t')$ are **equivalent** if t, t' are $\mathcal{M}$-equivalent, i.e. there is a permutation σ of I such that $x_{\sigma(i)(t)} = x_i(t')$, $b_{\sigma(i)\sigma(j)}(t') = b_{ij}(t)$, $p_{\sigma(j)}(t) = p_j(t')$. For geometric type, $c_{i\sigma(j)}(t) = c_{ij}(t')$, or rather, $\tilde{b}_{\sigma(i)\sigma(j)}(t') = \tilde{b}_{ij}(t)$ since σ as only a permutation of I fixes I'. By Proposition 10.38, if t, t' are equivalent, and $t -^{\sigma(j)} t_1$ and $t' -^j t_1'$, then t_1, t_1' are equivalent

Cluster algebras $\mathcal{A}(\mathcal{S})$, $\mathcal{A}'(\mathcal{S}')$ are strongly isomorphic if there is a field isomorphism $\mathcal{F} \to \mathcal{F}'$ that sends seeds in $\mathcal{S}$ to seeds $\mathcal{S}'$, thus inducing bijection $\mathcal{S} \to \mathcal{S}'$ and an isomorphism $\mathcal{A} \to \mathcal{A}'$

$\mathcal{A}(B, -)$ are all the possible normalized cluster algebras. $\mathcal{A}(B), \mathcal{A}(B')$ are strongly isomorphic if there is a one-to-one correspondence between $\mathcal{A}(B, \mathbf{p})$ and $\mathcal{A}(B', \mathbf{p}')$, this is true iff B, B' are mutationally equivalent modulo relabelling rows and columns

$\mathcal{A}$ is of **finite type** if it has finitely many seeds up to equivalences

Definition 10.47 The **Cartan counterpart** of B is the **generalized Cartan matrix** $A(B) = (a_{ij})$, $a_{ii} = 2$, $a_{ij} = -|b_{ij}|$ for $i \neq j$, with the same symmetrizing matrix D, i.e. $d_i a_{ij} = d_j a_{ji}$

Theorem 10.48 A is a Cartan matrix of finite type, there is a sign-skew symmetric $B_\circ$ such that $A(B_\circ) = A$ and $b_{ij} b_{ik} > 0$ for all i, j, k, and $\mathbf{p}_\circ$ is normalized, then $\mathcal{A}(B_\circ, \mathbf{p}_\circ)$ is of finite type. Any cluster algebra of finite type is strongly isomorphic to one such data

Remark. Since the Coxeter graph of A is a tree which is bipartite, thus we can certainly divide them into sinks and sources. Since $b_{ij} > 0$ would be there is a directed edge from i to j, thus we can always find such a $B_\circ$

Theorem 10.49 B, B' sign-skew symmetric, $\mathcal{A}(B), \mathcal{A}(B')$ iff $A(B), A(B')$ are of the same Cartan-Killing type

Theorem 10.50 $\mathcal{A}$ is a cluster algebra, the following are equivalent

1. $\mathcal{A}$ is of finite type
2. $|\mathcal{X}| < \infty$
3. For every seed $(\mathbf{x}, \mathbf{p}, B)$, $|b_{xy} b_{yx}| < 3$ for $x, y \in \mathbf{x}$
4. $\mathcal{A} = \mathcal{A}(B_\circ, p_\circ)$ as in Theorem 10.48

Theorem 10.51 $\mathcal{A}(B)$ consists of cluster algebras all simultaneously of finite type or of infinite type. There is a bijective correspondence between generalized Cartan matrices of finite type and strong isomorphic classes of normalized cluster algebras, through $B \to A(B)$

Theorem 10.52 There is a unique bijection $\Phi_{\geq -1} \to \mathcal{X}$, $\alpha \mapsto x[\alpha] = \frac{P_\alpha(\mathbf{x}_\circ)}{\mathbf{x}^\alpha}$, $P_\alpha \in \mathbb{Z}_{\geq 0}\mathcal{P}$ with nonzero constant term such that $X[-\alpha_i] = x_i$

Theorem 10.53 Every seed $(\mathbf{x}, \mathbf{p}, B)$ in $\mathcal{A}$ is uniquely determined by the cluster $\mathbf{x}$, and for any $x \in \mathbf{x}$, there is a unique cluster $\mathbf{x}'$ such that $\mathbf{x} \cap \mathbf{x}' = \mathbf{x} - \{x\}$. The the cluster complex $\Delta(\mathcal{A})$ encodes the combinatorics of seed mutations

Theorem 10.54 The bijection in Theorem 10.52 identifies $\Delta(\mathcal{A})$ and $\Delta(\Phi)$, in particular, the cluster complex doesn't depend on $\mathbb{P}$ nor $\mathbf{p}_\circ$

11 General topology

Definition 11.1 A *topological space* X is a set with *topology* $\tau \subseteq \mathcal{P}(X)$, such that $\varnothing, X \in \tau$, $U_i \in \tau \Rightarrow \bigcup_i U_i \in \tau$, $U, V \in \tau \Rightarrow U \cap V \in \tau$, elements in τ are *open sets*, complements of open sets are *closed sets*

N is a *neighborhood* of $A \subseteq X$ if $A \subseteq U \subseteq N \subseteq X$ for some open set U

x is a *limit point* of A if any neighborhood of x intersects A. x is a *limit* of $\{x_n\}$ if for any neighborhood U of x, all but finitely many lies in U

A *subspace* is $A \subseteq X$ with *subspace topology* given by $\{U \cap A | U \in \tau\}$

Definition 11.2 $X \xrightarrow{f} Y$ is *continuous* at x if for any neighborhood V of $y = f(x)$, there exists a neighborhood U of x such that $f(U) \subseteq V$. Then f is continuous iff $f^{-1}(V)$ is open for any open set $V \subseteq Y$

Definition 11.3 A *base* for τ is $B \subseteq \tau$ such that B covers X and for any $U_1, U_2 \in B$ such that $U_1 \cap U_2 \neq \varnothing$, there exists $U_3 \in B$ such that $U_3 \subseteq U_1 \cap U_2$

A *local base* for τ at x is a collection of neighborhoods $B(x)$ of x such that any neighborhood of x contain an element of $B(x)$

A *subbase* for τ is $B \subseteq \tau$ such that B generates τ, i.e. by arbitrary union of finite intersections, equivalently, τ is the smallest topology containing B. Here empty union and empty intersection are $\varnothing$ and X

Definition 11.4 X is *first countable* if each point has a countable local base. X is *second countable* if it has a countable base

Definition 11.5 X is *regular* if any point and a disjoint closed set have disjoint neighborhoods. X is *normal* if disjoint closed sets have disjoint neighborhoods

Definition 11.6 $\{A_i\}$ can be *completely separated* if $\{A_i\}$ can be completely separated by a continuous function $X \xrightarrow{f} \mathbb{R}$. Closed subsets $\{A_i\}$ can be *perfectly separated* if $\{A_i\}$ can be perfectly separated by a continuous function $X \xrightarrow{f} \mathbb{R}$. $\mathbb{R}$ can be replaced with I considering $\mathbb{R} \to I$, $x \mapsto \begin{cases} \frac{x}{x-1} & x \leq 0 \\ x & 0 \leq x \leq 1 \\ \frac{2}{x+1} & x \geq 1 \end{cases}$ and $I \hookrightarrow \mathbb{R}$

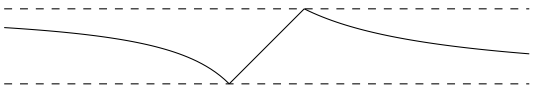


Figure 2:

Definition 11.7 (Kolmogorov classification of topological spaces) X is a T_0 *space* if for any two distinct points in X, at least one of them has a neighborhood which doesn't intersect the other piont, i.e. they are *topologically distinguishable*

X is a T_1 *space* if for any two distinct points in X, each of them has a neighborhood which doesn't intersect the other point. $T_1 \Leftrightarrow$ points are closed

X is a T_2 *space* or *Hausdorff space* if any two distinct points have disjoint neighborhoods. Then the limit of $\{x_n\}$ is unique, denotes the limit $x = \lim x_n$

X is a $T_{2^1/_2}$ *space* or *Urysohn space* if any two distinct points have disjoint closed neighborhoods

X is a T_3 *space* if X is regular Hausdorff

X is a $T_{3^1/_2}$ *space* if X is completely regular Hausdorff

X is a T_4 *space* if X is normal T_1 space $\Leftrightarrow$ normal Hausdorff

X is a T_5 *space* if X is completely normal Hausdorff

X is a T_6 *space* if X is perfectly normal $\Leftrightarrow$ perfectly normal Hausdorff

Definition 11.8 $A \subseteq X$ is *nowhere dense* if the interior of $\overline{A}$ is $\varnothing$

Definition 11.9 X is *sequential compact* if every sequence has a convergent subsequence

Definition 11.10 The *box topology* on $\prod_{i\in I} X_i$ has base $\left\{\prod_{i\in I} U_i \;\middle|\; U_i \subseteq X_i \text{ open}\right\}$

Lemma 11.11 X is Hausdorff iff the diagonal $\{(x,x)|x \in X\}$ is closed

Proof.

□

Definition 11.12 $X \times I \xrightarrow{F} Y$ is a *homotopy* between $X \overset{f_0,f_1}{\to} Y$ if $F(x,0) = f_0(x)$, $F(x,1) = f_1(x)$, write $f_t = F(\cdot,t)$. $X \xrightarrow{f} Y$ is a *homotopy equivalence* if there is $Y \xrightarrow{g} X$ such that $gf \simeq 1_X$, $fg \simeq 1_Y$

Definition 11.13 $X \xrightarrow{f} Y$ is a *topological embedding* if $f : X \to f(X)$ is a homeomorphism

Definition 11.14 $K \subseteq X$ is *compact* if any open cover has a finite subcover. Equivalently, K is disjoint from the intersection of a family of closed sets, then K is disjoint from the intersection of finitely many of them

X is *locally compact* if there is a compact neighborhood for each point

$Y \subseteq X$ is *precompact* if $\overline{Y}$ is compact

Definition 11.15 $A \subseteq X$ is *dense* if $\overline{A} = X$

X is *separable* if X has a countable dense subset

Definition 11.16 $X_\alpha \subseteq X$, $\{X_\alpha\}$ is *locally finite* if for any $x \in X$, there is a neighborhood of x intersecting only finitely many X_α's

$\mathcal{U} = \{U_\alpha\}$, $\mathcal{V} = \{V_\beta\}$ are covers of X, $\mathcal{V}$ is a *refinement* of $\mathcal{U}$ if for any V_β, there exists U_α containing V_β

X is *paracompact* if every open cover has a locally finite open refinement

Lemma 11.17 Closed subsets of compact space are closed

The image of a compact set is compact

Compact subsets of a Hausdorff space are closed

Lemma 11.18 X is compact, Y is Hausdorff, an injective map $X \xrightarrow{f} Y$ is a topological embedding

Proof. $f : X \to f(X)$ is a continuous bijection. If $K \subseteq X$ is closed, K is also compact since X is compact, thus $f(K)$ is compact, $f(K)$ is also closed since Y is Hausdorff

□

Definition 11.19 X is called *connected* if it can be written as the union of two open subsets

X is called *locally connected* if for any $x \in X$, there is a local basis that are connected

Proposition 11.20 Connected components are closed

Connectedness and local path connectedness implies path connectedness

Remark. Connected components may not be open

Definition 11.21 $E \xrightarrow{p} B$ has *lift extension property* for (X, A) if for any $X \xrightarrow{f} B$, a lift $A \xrightarrow{\tilde{f}} E$ can be extended to $\tilde{f} : X \to E$

$$\begin{array}{ccc} A & \xrightarrow{\tilde{f}} & E \\ \downarrow & \overset{\tilde{f}}{\nearrow} & \downarrow p \\ X & \xrightarrow{f} & B \end{array}$$

$E \xrightarrow{p} B$ has *homotopy lifting property* for (X, A) if it has lift extension property for $(X \times I, X \times \{0\} \cup A \times I)$

Proposition 11.22 If (X, A) satisfies homotopy extension property, and A is contractible, then the quotient map $X \xrightarrow{q} X/A$ is a homotopy equivalence

Proof. Consider $X \times \{0\} \cup A \times I \to A \hookrightarrow X$, where $(x, 0) \mapsto x$, $(a, 1) \mapsto *$ can be extended to $f : X \times I \to X$, $f_0 = 1_X$, $f_1(A) = \{*\}$, thus f_1 induces $r : X/A \to X$, $f_1 = rq$, $X \times I \xrightarrow{f} X \xrightarrow{q} X/A$ also induce $g : X/A \times I \to X/A$, where $qf_t = g_t q$, and $g_0 = 1_{X/A}$, $g_1 = qr$ thus $qr \simeq 1_{X/A}$

□

Definition 11.23 $U \subseteq X$ is open if $U \cap K$ is open for any compact subspace $K \subseteq X$ defines a topology. Equivalently, $F \subseteq X$ is closed if $F \cap K$ is closed for any compact subspace $K \subseteq X$. X is *compactly generated* if X has this topology

Definition 11.24 A map is *proper* if the preimage of a compact set is compact

A map is *discrete* if the preimage of a discrete set is discrete

Definition 11.25 X has *discrete topology* if $\tau = \mathcal{P}(X)$. X has *trivial topology* if $\tau = \{\varnothing, X\}$

Proposition 11.26 Suppose X has discrete topology

(a) Any map $f : Y \to X$ is continuous iff $f^{-1}(x)$ is open for all $x \in X$

(b) If continuous maps $f, g : X \to X$ are homotopic, then they are actually the same

Proof. *(a)* For any subset $U \subseteq X$, $f^{-1}(U) = \bigcup_{x \in U} f^{-1}(x)$ is open

(b) If $F : X \times I \to X$ is a homotopy, then the restriction on $\{x\} \times I$ is gives a continuous map $I \to X$, the image has to be connected, thus the restriction is a constant, thus $f(x) = F(x, 0) = F(x, 1) = g(x)$

□

Lemma 11.27 $F_i \subseteq X$ are closed, $\bigcup_i F_i = X$, $f|_{F_i}$ are continuous, then f is continuous

Lemma 11.28 If X is compact, Y is Hausdorff, a surjective continuous map $f : X \to Y$ is a quotient map

Proof. Let's use the universal property of quotient space, consider a continuous map $g : X \to Z$ such that g maps fibers of f to points, thus we have a map $\tilde{g} : Y \to Z$, $\tilde{g}f = g$, for any closed set F in Z, so is $K = g^{-1}(F) = f^{-1}(\tilde{g}^{-1}(F))$, since X is compact, so is K, hence $f(K) = \tilde{g}^{-1}(F)$ is compact, and since Y is Hausdorff, $\tilde{g}^{-1}(F)$ is closed

□

Lemma 11.29 X is locally compact, Hausdorff, $F \subseteq X$ is closed iff $F \cap K$ is compact for any compact subset $K \subseteq X$

Proof. F closed $\Rightarrow F \cap K$ closed. Conversely, suppose $F \cap K$ is compact for any compact subsets $K \subseteq X$, for any $x \notin F$, there is a compact set K containing an open neighborhood U of x, $F \cap K$ is compact thus closed, hence $G = U - F \cap K$ is an open neighborhood of x which is disjoint of F, hence F is closed

□

Lemma 11.30 X, Y are locally compact, Hausdorff, $p : X \to Y$ is continuous, proper, then p is closed

Proof. Suppose $F \subseteq X$ is closed, since $p(F \cap p^{-1}(K)) = p(F) \cap K$, by Lemma 11.29, we can take any $K \subseteq Y$ compact, hence F is closed

□

Definition 11.31 X is noncompact, the *Alexandorff extension* of X is $X^* = X \cup \{\infty\}$ with open sets $\varnothing, X^*$, open sets in X and complements of closed compact sets of X

$X \hookrightarrow X^*$ is an open topological embedding

If X is also locally compact Hausdorff, X^* is the *one point compactification* of X which is Hausdorff

Lemma 11.32 X, Y are locally compact Hausdorff, $X \xrightarrow{f} Y$ is proper, then f sends discrete sets to discrete sets

Proof. Suppose $A \subseteq X$ is discrete, $x_0 \in A$, $y_0 = f(x_0) \in Y$, K is a compact neighborhood of y_0, then $f^{-1}(K)$ is a compact neighborhood of x_0, thus $f^{-1}(K) \cap A$ is finite, so is $K \cap f(A)$, since Y is Hausdorff, there is a neighborhood U of y_0 such that $U \cap f(A) = y_0$

□

Lemma 11.33 X, Y are locally compact, $X \xrightarrow{p} Y$ is proper and discrete, then $p^{-1}(y)$ is finite, and for any neighborhood V of $p^{-1}(y)$, there is a neighborhood U of y such that $p^{-1}(U) \subseteq V$

Lemma 11.34 X, Y are locally compact Hausdorff, $X \xrightarrow{p} Y$ is a proper local homeomorphism, then p is a finite sheeted covering

Definition 11.35 The *compact-open topology* on Y^X is given by a subbase $V(K, U) \coloneqq \{f \in Y^X \mid f(K) \subseteq U\}$, with $K \subseteq X$ compact and $U \subseteq Y$ open

A *normal family* $\{f_i\}$ is a precompact subset of Y^X

Lemma 11.36 $\{f_n\}$ converges pointwise on X iff $\{f_n\}$ converges in Y^X with the product topology $\prod_{x \in X} Y$. Hence we call the product topology the *topology of pointwise convergence*

Proof. If f_n converges pointwise on X to f, then for any neighborhood V_i of $f(x_i)$, $i = 1, \cdots, k$, V_k contains all but finitely many $f_n(x_i)$, thus for n big enough, $f_n \in V_1 \cap \cdots \cap V_k \cap \prod_{x \neq x_0} Y$, i.e. $\{f_n\}$ converges to f in Y^X

□

Theorem 11.37 X is compact, Y is a complete metric space, then the topology induced by metric $d(f, g) = \sup_{x \in X} d(f(x), g(x))$ is the same as the compact-open topology on Y^X

Theorem 11.38 $Y^* \cong Y$

Theorem 11.39 The composition $Z^Y \times Y^X \to Z^X, (g, f) \mapsto g \circ f$ is continuous, in particular, if $X = *$, then this becomes the evaluation map $\mathrm{eval} : Z^Y \times Y, (f, y) \mapsto f(y)$

Theorem 11.40 $Z^{X \times Y} \cong \left(Z^Y\right)^X$

Definition 11.41 A topological space X is reducible if $X = X_1 \cup X_2$, X_1, X_2 are proper nonempty closed subsets, $X_1 \not\subseteq X_2$, $X_2 \not\subseteq X_1$, X is *irreducible* if not reducible

Definition 11.42 A topological space X is *Noetherian* if $X \supsetneq X_1 \supsetneq X_2 \supsetneq \cdots$ terminates, $\dim V = \sup_d(X_0 \supsetneq X_1 \supsetneq \cdots \supsetneq X_d)$, V_i's are closed and irreducible

Theorem 11.43 (Tychonoff's theorem) $\{K_i\}_{i \in I}$ are compact, so is $\prod_{i \in I} K_i$

Proposition 11.44 Connected sets of $\mathbb{R}$ are intervals (a, b), $[a, b)$, $(a, b]$ or $[a, b]$

Theorem 11.45 (Jordan curve theorem) $S^n \xrightarrow{i} \mathbb{R}^{n+1}$ is injective thus an open embedding by Lemma 11.18, denote $X = i(S^n)$, then $Y = \mathbb{R}^{n+1} \setminus X$ consists of exactly two connected components, the interior

U which is bounded, and the exterior V which is not. When $n = 1$, U and V are homeomorphic to D and $\mathbb{R}^2 \setminus D$

Definition 11.46 $W \subseteq X$ is a *locally closed set* if for any point $p \in W$, there is an open neighborhood $U \ni p$ such that $U \cap W$ is closed in U. Equivalently, W is the intersection of an open subset and a closed subset, or equivalently, W is relatively open in $\overline{W}$. A *constructible set* is a finite union of locally closed sets

Theorem 11.47 (Lefschetz fixed point theorem) X is a compact triangulable space of dimension n, the *Lefschetz number* of f is $\sum_{k=0}^{n} \operatorname{tr}(f_*|_{H_k(X;\mathbb{Q})})$. If the Lefschetz number of f is nonzero, then f has fixed points. The converse is not true, i.e. even if the Lefschetz number is zero, then could be fixed points

If $f = \mathrm{id}_X$, then the Lefschetz number is the Euler characteristic χ

Definition 11.48 The *join* of X, Y is

$$X * Y = \frac{X \times Y \times I}{(x, y_1, 0) \sim (x, y_2, 0), (x_1, y, 1) \sim (x_2, y, 1)}$$

We can also interpret it as all possible paths from X to Y. In general, $*_i X_i$ can be thought of as finite sum $\sum_i t_i x_i$, $t_i \in I$, $x_i \in X_i$

Definition 11.49 X is *Noetherian* if every descending chain of subspaces is finite

Proposition 11.50

1. Subspaces of Noetherian spaces are Noetherian
2. X is Noetherian iff X is locally Noetherian and compact

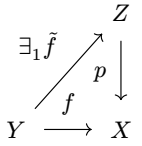
Proof.

1. A descending chain of a subspace is also a descending chain
2.

□

Definition 11.51 A covering space is a fiber bundle with discrete fibers

Proposition 11.52 $Z \xrightarrow{p} X$ is a covering, $f(y_0) = p(z_0)$. f lifts $\tilde{f} : Y \to Z$ with $f(y_0) = z_0$ iff $f_*\pi_1(Y, y_0) \leq p_*\pi_1(Z, z_0)$



Proposition 11.53 Covering $Y \xrightarrow{p} X$ is regular if $\mathrm{Aut}(Y/X)$ is a normal subgroup of $\pi_1(X, x_0)$

Proof. Assume $p(y_1) = p(y_2) = x_0$, by Proposition 11.52, $p_*\pi_1(Y, y_1) = p_*\pi_1(Y, y_2)$ are conjugate, hence normal

□

Definition 11.54 $A \xrightarrow{i} X$ is inclusion. A **deformation** of A into $B \subseteq X$ in X is a homotopy $A \xrightarrow{f_t} X$ such that $f_0 = i$ and $f_1(A) \subseteq B$, onto if equality holds. $X \xrightarrow{r} A$ is a **retraction** if $ri = 1_A$. r is a **weak retraction** if inclusion $A \xrightarrow{i} X$ has a left homotopy inverse, i.e. $ri \simeq 1_A$. A **deformation retraction** is a deformation $X \xrightarrow{f_t} X$ such that $f_1 = ri$ for some retraction $X \xrightarrow{r} A$. Deformation retraction f_t is **strong** if $f_t|_A = 1_A$

X is **contractible** if X deformation retracts onto a point

(X, A) is a **good pair** if A is a strong neighborhood deformation retract of X

Lemma 11.55 $A \overset{i}{\to} X$ is inclusion

1. X is deformable into A iff i is a **weak section**, namely i has a right homotopy inverse, i.e. $ir \simeq 1_X$
2. i is a homotopy equivalence iff A is a weak retract of X and X is deformable into A
3. If X is deformable into a retract A, then A is a deformation retract of X
4. If (X, A) is cofibered, then A is a weak retract of X iff A is a retract of X

Proof.

1. If $X \times I \overset{H}{\to} X$ is a homotopy from 1_X to ir, then H is a deformation of X into A since $H_0 = 1_X$, $H_1(X) \subseteq A$. If H is a deformation of X into A, since $H_1(X) \subseteq A$, define $X \overset{r}{\to} A$ such that $ir = H_1$, then r is a right homotopy inverse of i
2. i is a homotopy equivalence $\Leftrightarrow$ there exists $X \overset{r}{\to} A$ such that $ri \simeq 1_A$, $1_X \overset{H}{\simeq} ir \Leftrightarrow r$ is a weak retract, H is a deformation of X into A
3. $X \overset{r}{\to} A$ is a retraction, $X \times I \overset{H}{\to} X$ is a deformation of X, then $1_X \simeq ir'$ for some $X \overset{r'}{\to} A$, hence $r \simeq rir' = r' \Rightarrow 1_X \simeq ir \simeq ir$ giving a deformation retract
4. $A \times I \overset{H}{\to} A$ is a homotopy from ri to 1_A, since $r(a) = H_0(a)$ and (X, A) is cofibered, we have $X \times I \overset{F}{\to} A$, then $F_0 = r$, $F_1 i = 1_A$, i.e. r is homotopic to retraction F_1

□

Definition 11.56 $\mathcal{C}$ is a class of topological spaces closed under homeomorphism and closed subsets. X is an **absolute retract** for $\mathcal{C}$ if for $Y \in \mathcal{C}$, embedding $X \hookrightarrow Y$ is closed $\Rightarrow$ X is a retract of Y. X is an **absolute neighborhood retract** for $\mathcal{C}$ if for $Y \in \mathcal{C}$, embedding $X \hookrightarrow Y$ is closed $\Rightarrow$ X is a neighborhood retract of Y

12 Topological groups

Definition 12.1 G is a **topological group** if it is a group and a topological space so that the group multiplication $G \times G \to G$ and the inverse map $G \to G$ are continuous maps

Definition 12.2 $f : G \to \mathbb{R}/\mathbb{C}$ is a continuous function, $L_y f(x) = f(y^{-1}x)$, $R_y f(x) = f(xy)$, $L_{yz} = L_y L_z$, $R_{yz} = R_y R_z$, f is called left/right uniformly continuous if $\forall \varepsilon > 0$, $\exists V \ni e$ such that $\| L_y f - f \| < \varepsilon / \| R_y f - f \| < \varepsilon$, $\forall y \in V$, $\| \cdot \|$ is the supremum norm

Proposition 12.3 If $f \in C_c(G)$, then f is both left and right uniformly continuous

Proof. Easy proof by a very standard analysis argument

□

Definition 12.4 If f is a Borel measurable function on G, then f factor through G/H, otherwise suppose $f(y) \neq f(z)$, $y, z \in xH$, $f^{-1}(f(y)) \cap xH \subsetneq xH$ is a Borel set which is impossible, because then $x^{-1}f^{-1}(f(y)) \cap H \subsetneq H$ will also be a Borel set, consider $\Gamma = \{S \in \mathcal{P} \mid H \subseteq H \text{ or } H \cap S = \emptyset\}$, then Γ is a sigma algebra containing all open sets hence Borel algebra, we reached a contradiction

Thus for most purposes one may as well work with G/H which is Hausdorff(L^p spaces for instance, mod almost everywhere vanishing function)

For a locally compact Hausdorff group, A Borel measure μ on G is called left/right invariant if $\mu(xE) = \mu(E)/\mu(Ex) = \mu(E)$, $x \in G, E \in \mathcal{B}(G)$

A linear functional I is left/right invariant if $I(L_x f) = I(f)/I(R_x f) = I(f)$

A left/right Haar measure on G is a left/right invariant Radon measure μ on G, for example, Lebesgue measure on $\mathbb{R}^n$, counting measure on G with discrete topology

Example 12.5 Continuous bijective group homomorphism doesn't imply homeomorphism, which is really obvious, by taking the identity map and a discrete topology on the topological group G

Definition 12.6 Let G be a topological group, then a 1-parameter subgroup means a continuous group homomorphism $\varphi : \mathbb{R} \to G$, $\varphi(s+t) = \varphi(s)\varphi(t)$, in the case of a Lie group, φ is required to be smooth

Definition 12.7 Suppose G is a connected, locally pathconnected and (semi-)locally simply connected topological space, then it has a universal cover $\tilde{G}$ which is unique up to an isomorphism, a connected Lie group certain satisfies this

Proposition 12.8 Denote $\pi : \tilde{G} \to G$ as the covering map, let $\bar{G}$ be the set of maps $T : \tilde{G} \to \tilde{G}$, such that $\pi(Tx) = g(\pi x)$ for some $g \in G$, i.e. the following diagram commutes

$$\begin{array}{ccc} \tilde{G} & \xrightarrow{T} & \tilde{G} \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ G & \xrightarrow{g} & G \end{array}$$

Then $\bar{G}$ which is a group acts transitively and freely on $\tilde{G}$, thus we can think of the universal cover $\tilde{G}$ also as a topological group

Proof. Given $x, y \in \tilde{G}$, there is a unique $g \in G$ such that $g(\pi x) = \pi y$, since $\tilde{G}$ is the universal cover, there is a unique lift such that $T(x) = y$, thus the action is free and transitive

□

13 Cellular structure

Definition 13.1 An **abstract simplicial complex** is $K \subseteq \mathcal{P}(S) \setminus \varnothing$ such that $X \in K \Rightarrow \mathcal{P}(X) \setminus \varnothing \subseteq K$. Finite elements of K are called **faces**. The **dimension** of a face X is $\dim X = |X| - 1$. The d skeleton K^d is the union of faces of dimension no more than d. $\dim K = \sup_{X \in K} \dim X$. K^0 are **vertices**. Maximal elements are **facets**. K is **pure** if all facets have dimension $\dim K$. A **simplex** is a subcomplex which contains all its nonempty subsets, for $X \in K$, $\overline{X}$ is the corresponding simplicial complex

Definition 13.2 The **closure** $\overline{L}$ of $L \subseteq K$ is smallest subcomplex of K containing L. The **star** of $Y \in K$ is st $L = \{X | X \cap Y \neq 0\}$, the star of $L \subseteq K$ is st $L = \bigcup_{Y \in L}$ st Y. The **link** of a face $Y \in K$ is or lk $Y = \{X | X \cap Y = \varnothing, X \cup Y \in X\}$. Equivalently, lk $Y = \overline{\text{st } Y} - \text{st } \overline{Y}$

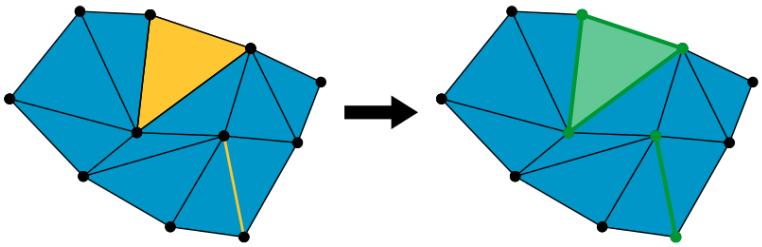


Figure 3: Two simplices and their closure

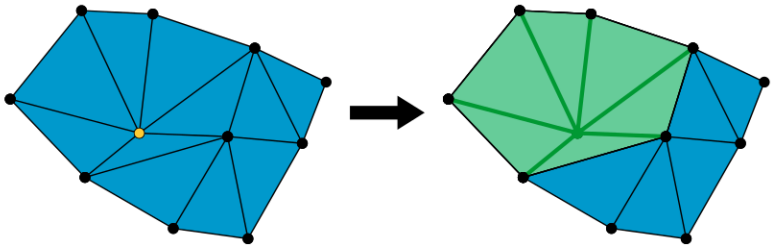


Figure 4: A vertex and its star

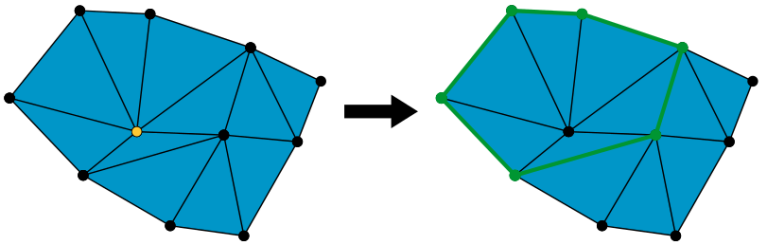


Figure 5: A vertex and its link

Note. lk $\varnothing = K$

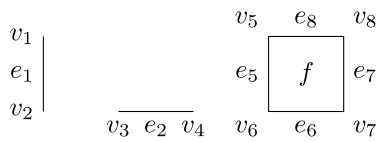
Definition 13.3 If K, L has disjoint sets of vertices, then $K * L = \{X \sqcup Y | X \in K, Y \in L\}$

Definition 13.4 $L \subseteq K$, the **deletion** $K \setminus L$ consists of those sets which don't contain sets in L as subsets. The stellar subdivision of $X \in K$ is by introducing a new vertex x, and form $K \setminus X \cup (\overline{x} * \partial \overline{X} * \text{lk } X)$

Definition 13.5 A simplicial map $f : K \to L$ is such that $f(K^d) \subseteq L^d$

The Eilenberg-Zilber map $C_*(X) \otimes C^*(Y) \to C^*(X \times Y)$ is a quasi-isomorphism

Take $X, Y = I$



The map sends $v_1 \otimes v_3 \mapsto v_5, v_1 \otimes v_4 \mapsto v_8, v_2 \otimes v_3 \mapsto v_6, v_2 \otimes v_4 \mapsto v_7, e_1 \otimes v_3 \mapsto e_5, e_1 \otimes v_4 \mapsto e_7, v_1 \otimes e_2 \mapsto e_8, v_2 \otimes e_2 \mapsto e_6, e_1 \otimes e_2 \mapsto f$. The we get the quasi-isomorphism

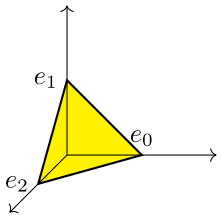
$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Z}[\{e_1 \otimes e_2\}] & \longrightarrow & \mathbb{Z}[\{e_1 \otimes v_3, e_1 \otimes v_4, v_1 \otimes e_2, v_2 \otimes e_2\}] & \longrightarrow & \mathbb{Z}[\{v_1 \otimes v_3, v_1 \otimes v_4, v_2 \otimes v_3, v_2 \otimes v_4\}] & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathbb{Z}[\{f\}] & \longrightarrow & \mathbb{Z}[\{e_5, e_7, e_8, e_6\}] & \longrightarrow & \mathbb{Z}[\{v_5, v_8, v_6, v_7\}] & \longrightarrow & 0 \end{array}$$

Definition 13.6 With the standard basis $\{e_i\}$ for $\mathbb{R}^\infty$ as vertices, the **standard n-simplice** is

$$\Delta^n = \left\{(t_0, \cdots, t_n) \in \mathbb{R}^{n+1} \subseteq \mathbb{R}^\infty \;\middle|\; \sum t_i = 1, 0 \le t_i \le 1\right\}$$

The **i-th face** of Δ^n is the face oppose to the i-th vertex, i.e. $\{t_i = 0\} \cap \Delta^n$

The **boundary** of Δ^n to be $\partial\Delta^n$ is the union of faces. $\partial\Delta^0 = \varnothing$



$\Delta^{n-1} \overset{d_{n,i}}{\to} \Delta^n$, $e_j \mapsto \begin{cases} e_j & j<i \\ e_{j+1} & j \ge i \end{cases}$ is i-th **face map** attaching Δ^{n-1} to the i-th face of Δ^n

$\Delta^{n+1} \overset{s_{n,i}}{\to} \Delta^n$, $e_j \mapsto \begin{cases} e_j & j \le i \\ e_{j-1} & j > i \end{cases}$ is the i-th **degeneracy map** which is a projection

Definition 13.7 X has a **cell decomposition** if X can be written as the disjoint union of open n cells, i.e. $X = \bigcup_{n,\alpha} e^n_\alpha$, where cells e^n_α with subspace topology are homeomorphic to open n disks or open n simplices and disjoint, $X^n = \bigsqcup_{k \le n,\alpha} e^k_\alpha$ is called the **n-skeleton**, define $X^{-1} = \emptyset$

Suppose X, Y have cell decomposition $X = \bigcup_{n,\alpha} e^n_\alpha$, $Y = \bigcup_{m,\beta} e^m_\beta$, then $X \times Y$ also has a cell decomposition $X \times Y = \bigcup_k \bigcup_{\substack{n+m=k \\ \alpha,\beta}} e^n_\alpha \times e^m_\beta$, note that $e^n_\alpha \times e^m_\beta \cong e^{n+m}$

Every topological space has a cell decomposition into points

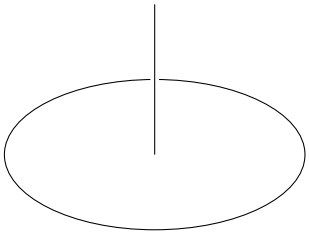
Definition 13.8 A **cellular map** is a map $f : X \to Y$ between topological spaces with cell decompositions such that $f(X^n) \subseteq Y^n$

Definition 13.9 X is called a **cell complex** if X is a Hausdorff space with cell decomposition $X = \bigcup_{n,\alpha} e^n_\alpha$ and a cell complex structure: a family of characteristic maps $\Phi^n_\alpha : \Delta^n \to X$ such that Φ^n_α restricted on $\Delta^n \setminus \partial\Delta^n$ is a homeomorphism onto e^n_α and $\Phi^n_\alpha(\partial\Delta^n) \subseteq X^{n-1}$

Note that in the definition, we could also replace Δ^n with D^n

Remark. Since Δ^n is compact Hausdorff and X is Hausdorff, $\overline{e^n_\alpha} \subseteq \Phi^n_\alpha(\Delta^n)$, $\partial e^n_\alpha \subseteq \Phi^n_\alpha(\partial\Delta^n)$ for $n > 0$, on the other hand, if $\partial e^n_\alpha \subsetneq \Phi^n_\alpha(\partial\Delta^n)$, then there exists $x \in \partial\Delta^n$ such that $y = \Phi^n_\alpha(x) \notin \overline{e^n_\alpha}$, this means there is an open neighborhood U of y disjoint from $\overline{e^n_\alpha}$, but then the preimage of U under Φ^n_α would be a nonempty open subset of Δ^n which intersects $\Delta^n \setminus \partial\Delta^n$ which is impossible, hence $\Phi^n_\alpha(\Delta^n) = \overline{e^n_\alpha}$, $\Phi^n_\alpha(\partial\Delta^n) = \partial e^n_\alpha$ for $n > 0$

A Hausdorff space X with a cell decomposition doesn't immediately give a cell complex structure, for example, consider an open disk union with an open segment right in the middle



Cell complex X can be seen as Δ^n is compact Hausdorff and X is Hausdorff and Lemma 11.28

Definition 13.10 A cell complex X is called **regular** if all characteristic maps are embeddings

Definition 13.11 Let X be a cell complex, it is **closure finite** if $\overline{e_\alpha^n}$ is contained in the union of finitely many cells, and we say X has the **weak topology**, meaning $F \subseteq X$ is closed iff $F \cap \overline{e_\alpha^n}$ is closed in $\overline{e_\alpha^n}$ for any cell, if a cell complex is both closure finite and has the weak topology, we say it is a **CW complex**

Closure finiteness is equivalent of saying $\partial e_\alpha^n \subseteq \bigcup_{k<n,\alpha} e_\alpha^k$ a finite union of cells

Example 13.12 Consider $X = D^2$ with a cell complex structure $D^2 \to D^2$ and $* \to x$ for each $x \in \partial D^2$, this doesn't satisfy closure finiteness since $\overline{e^2} = X$, but the weak topology is the same as the original one, since if $F \subseteq D^2$ is closed in the weak topology, then $F \cap \overline{e^2} = F$ is closed

Consider $X = S^1$ with a cell complex structure $* \to x$ for each $x \in S^1$, the weak topology is the discrete topology on S^1 which doesn't match with the original topology on S^1, but it does satisfy closure finiteness

Remark. Suppose X is a CW complex

X^n is obviously closed due to the weak topology

Since $\overline{e_\alpha^n}$ is contained in the union of finitely many cells, $\overline{e_\alpha^n}$ contains at most finitely many 0 cells, thus any union of 0 cells F is closed because $F \cap \overline{e_\alpha^n}$ is finitely many points which is closed given that X is Hausdorff, therefore X^0 is discrete

Suppose $K \subseteq X$ is a compact subset, then $K \subseteq X = \bigcup e_\alpha^n \subseteq \bigcup \overline{e_\alpha^n} \setminus \partial e_\alpha^n$

contained in finitely many cells, since $K \subseteq \bigcup \overline{e_\alpha^n} \setminus \partial e_\alpha^n$

if $K \cap e_\alpha^n \neq \emptyset$,

, otherwise K intersects infinitely many cells,

Theorem 13.13 Another description of CW complexes is as follows:

These two definitions coincides

Proposition 13.14 Any compact set of a CW complex is contained in finitely many cells

Proposition 13.15 CW complexes are locally contractible, thus they are locally path connected, hence connectedness and path connectedness are equivalent for CW complexes

Theorem 13.16 CW complexes are normal, satisfies T_4 axiom

Proposition 13.17 If $A \subseteq X$ is a CW subcomplex, then (X, A) is a good pair

Theorem 13.18 CW complexes have partitions of unity

Proposition 13.19 Covering space of CW complexes are CW complexes

Proposition 13.20 The product of two countable CW complexes is again a CW complex

Theorem 13.21 (CW approximation) For any topological space X, there is a CW complex Z and a weak homotopy equivalence $f : Z \to X$, this is called a CW approximation

Theorem 13.22 (Whitehead's theorem) Suppose $f : X \to Y$ is weak homotopy equivalence between CW complexes, then it is a homotopy equivalence

Definition 13.23 A **pseudomanifold** M is a pure triangulated space of dimension n such that M is not branching, i.e. any two n simplices have precisely one $(n-1)$ common face, M is strongly connected, i.e. any two n simplices can be linked with a sequence of simplices having common $(n-1)$ face pairwise

Note. The dual graph Γ of M is connected and n-regular

Example 13.24 The 0 dimensional pseudomanifold is the disjoint union of two points, since the empty set has to be the common face two point. The dual graph is a two points joined by an edge, this example is weird

A 1 dimensional pseudomanifold is a infinite chain or a loop, its dual graph is the same

Definition 13.25 D is a nonmaximal simplex, then $\text{lk } D$ is a $n - |D|$ pure dimensional simplicial complex. A pseudomanifold such that $\text{lk } D$ is also pseudomanifolds for any nonmaximal simplex is an **abstract polytope**

14 Homotopy theory

14.1 Homotopy

Definition 14.1 $\pi_n(X) \coloneqq [S^n, X]$ are the homotopy groups, the relative homotopy groups $\pi_n(X, A)$ are defined to be all homotopy classes of maps $(I^n, \partial I^n) \to (X, A)$ or equivalently all homotopy classes of maps $(S^n, s_0) \to (X, A)$, in particular, if $A = \{x_0\}$, we have $\pi_n(X, x_0) \coloneqq \langle S^n, X\rangle$ with basepoints x_0, s_0, furthermore, we also define $\pi_n(X, A, x_0)$ to be all homotopy classes of maps $(I^n, \partial I^n, J^{n-1}) \to (X, A, x_0)$ or equivalently all homotopy classes of maps $(D^n, S^{n-1}, s_0) \to (X, A, x_0)$, here $J^{n-1} = \partial I^n - I^{n-1}$

$\pi_1(X, x_0)$ is called the fundamental group, note that $\pi_n(X, x, x) = \pi_n(X, x)$

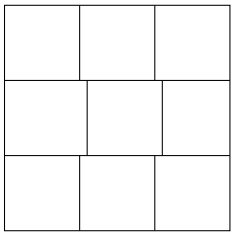
Definition 14.2 All homotopy classes of paths form the *fundamental groupoid* $\Pi_1(X)$ of X, suppose $A \subseteq X$, we can also define $\Pi_1(X, A)$ to be the full subcategory with objects $x \in A$ and morphisms $\mathrm{Hom}(x, y), x, y \in A$, $\Pi_1(X, x) = \pi_1(X, x)$, $\Pi_1(X)$ is a connected category if X is path-connected since there is a morphism connecting any two objects, thus $\pi_1(X, x)$ is a skeleton of $\Pi_1(X)$, $\pi_1(X, x) \hookrightarrow \Pi_1(X)$ is an equivalence of categories

Proposition 14.3 $\pi_n(X, x_0)$ are groups, abelian if $n > 1$

Theorem 14.4 (Van Kampen's theorem) Suppose $X = \bigcup_\alpha A_\alpha$, interiors of A_α cover X, where $X, A_\alpha, A_\alpha \cap A_\beta$ are path connected and $x_0 \in A_\alpha$, then the map induced by inclusion $*_\alpha \pi_1(A_\alpha, x_0) \to \pi_1(X, x_0)$ is surjective. Moreover, if $A_\alpha \cap A_\beta \cap A_\gamma$ are path connected, then then kernel is generated by $i_\alpha(w) i_\beta(w)^{-1}, w \in \pi_1(A_\alpha \cap A_\beta, x_0)$, where $i_\alpha : A_\alpha \to X$ are the inclusions

Proof. Since $A_\alpha \cap A_\beta$ are path connected, we can cut a loop in X into pieces such that each intermediate point is in $A_\alpha \cap A_\beta$ for some α, β, thus the map is surjective

Suppose $f_1 \cdots f_n$, $g_1 \cdots g_m$ are homotopic as loops, suppose F is the homotopy, consider the following diagram, each rectangle is so small that it is inside some A_α, then homotopy the path across a cube one at a time, and since $A_\alpha \cap A_\beta \cap A_\gamma$ are path connected, each vertex lies in $A_\alpha \cap A_\beta \cap A_\gamma$ for some α, β, γ, then we can connected it to x_0 through a path in $A_\alpha \cap A_\beta \cap A_\gamma$



□

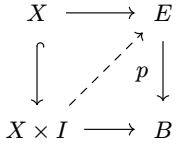
Theorem 14.5 X is a path-connected, attach cells $\{e^2_\alpha\}$ along loops

Definition 14.6 $X \xrightarrow{f} Y$ is a **weak homotopy equivalence** if $\pi_0(X, x_0) \xrightarrow{f_*} \pi_0(Y, f(x_0))$ is bijective, and on each path connected component, $\pi_n(X, x_0) \xrightarrow{f_*} \pi_n(Y, f(x_0))$ are isomorphisms

Theorem 14.7 (Hurewicz theorem) Choose a generator $e \in H_n(S^n)$, define $\phi : \pi_n(X) \to H_n(X)$, $f \mapsto f_*(e)$, here $f : S^n \to X$ is a map. If X is $(n-1)$-connected, then ϕ is an isomorphism

14.2 Hurewicz fibration

Definition 14.8 $E \xrightarrow{p} B$ is a **Hurewicz fibration** if it has homotopy lifting property for any space X



$p^{-1}(A) \overset{p}{\to} A$ is also a fibration for any $A \subseteq B$

Definition 14.9 $A \overset{i}{\to} X$ is a **Hurewicz cofibration** if for any $X \overset{f_0}{\to} Y$, $A \times I \overset{g}{\to} Y$ such that $g_0 = f_0 i$, there there exists $X \times I \overset{f}{\to} Y$ extends f_0 such that $g = fi$

$$\begin{array}{ccc} & & f_0 \\ Y & \longleftarrow & X \\ \uparrow & \overset{f}{\nearrow} & \uparrow i \\ & \underset{g}{\swarrow} & \\ Y^I & \longleftarrow & A \end{array}$$

By Lemma 14.10, $A \hookrightarrow X$ is an embedding, (X, A) is a **cofibered pair** satisfying the **homotopy extension property**

Remark. The fiber of a fibration is the kernel in Top. The cofiber X/A of a cofibration is the cokernel in Top

$A \times I$ can be thought of as the "continuous" coproduct $\coprod_i A$, and A^I can be thought of as the "continuous" product $\prod_i A$

Lemma 14.10 Cofibration $A \overset{i}{\to} X$ is a topological embedding

Proof. M is the mapping cylinder of $A \overset{i}{\to} X$, $X \times \{0\} \sqcup A \times I \overset{[]}{\to} M$ denote the quotient map, $f_0(x) = [(x, 0)]$, $g_t(a) = [(a, t)]$, then $g_0(a) = [(a, 0)] = [(i(a), 0)] = f_0 i(a)$

$$\begin{array}{ccc} & & f_0 \\ M & \longleftarrow & X \\ \uparrow & \overset{f}{\nearrow} & \\ & & i \uparrow \\ & \underset{g}{\swarrow} & \\ M^I & \longleftarrow & A \end{array}$$

$M|_{A\times\{1\}} \overset{k}{\to} A$, $[(a, 1)] \mapsto a$ is a homeomorphism, $kf_1|_{i(A)}$ is the inverse of i because $kf_1 i(a) = kg_1(a) = k([(a, 1)]) = a$, $ikf_1(i(a)) = i(a)$

□

Lemma 14.11 A surjective fibration $E \overset{p}{\to} B$ with B locally path connected is a quotient map

Lemma 14.12 If $X \times \{0\} \cup A \times I$ is a retract of $X \times I$, then the topology of the mapping cylinder of the inclusion $A \hookrightarrow X$ is the the same as the subspace topology on $X \times \{0\} \cup A \times I$ induced from $X \times I$. In particular, f is continuous on $X \times \{0\} \cup A \times I$ iff f is continuous on both $X \times \{0\}$ and $A \times I$

Proof. This is trivial if A is closed due to Lemma 11.27

Write $Y = X \times \{0\} \cup A \times I$. If $O \subseteq Y$ is open in Y, then obviously $O \cap A \times I$ is open in $A \times I$, $O \cap X$ is open in X

Suppose $O \subseteq Y$ is such that $O \cap A \times I$ is open in $A \times I$, $O \cap X$ is open in X. Define $U_n = \bigcup_V \left\{ V \overset{\text{open}}{\subseteq} X \;\middle|\; (V \cap A) \times [0, \frac{1}{n}) \subseteq O \right\}$, i.e. U_n is the largest such open subset of X, $U = \bigcup_{n=1}^{\infty} U_n$, then $O \cap A \subseteq U$ since for any $(x, 0) \in O \cap A \subseteq O \cap A \times I$, because $O \cap A \times I$ is open $A \times I$, there exists open subset $V \subseteq X$ containing x such that $(V \cap A) \times [0, \frac{1}{n}) \subseteq O \cap A \times I$ for some n, hence $x \in V \subseteq U_n \subseteq U$

$O \cap A \times (0, 1]$ is open in Y, $O \cap (X \setminus \overline{A})$ is open in Y, we only need to show that for any $x \in \overline{A}$, there exists an open neighborhood of $(x, 0)$ contained in O, and it suffices to show that $x \in U$, then $x \in U_n$ for some n, $(U_n \cap O) \times [0, \frac{1}{n}) \cap Y$ is open in Y. Now fix $x \in \overline{A}$

Write the retraction r as (r_1, r_2), for $t > 0$, $r(x, t) = (r_1(x, t), r_2(x, t))$, since $x \in \overline{A}$, $r(a, t) = (a, t)$, we know $r_1(x, t) \in A$, $r_2(x, t) = t$. We claim: if $r_1(x, t) \in U_n$, then $x \in U_n$. Since U_n is open, there exists

open neighborhood V of x such that $r_1(V \times (t-\varepsilon, t+\varepsilon)) \subseteq U_n$ for some $\varepsilon > 0$, in particular $r_1((V \cap A) \times \{t\}) \subseteq U_n$, thus $V \cap A \subseteq U_n \cap A$, by maximality of U_n, $V \subseteq U_n$

Suppose $x \notin U$, then by the claim, $r_1(x,t) \in A \setminus U$ for $t > 0$, then $r_1(x,t) \in A \setminus O$ since $A \cap O \subseteq U$, thus $x = r_1(x,0) \in \overline{A} \setminus O$ which contradicts the fact $(x,0) \in A$

□

Proposition 14.13 (X, A) is cofibered iff $X \times \{0\} \cup A \times I$ is a retract of $X \times I$ iff $X \times \{0\} \cup A \times I$ is a strong deformation retract of $X \times I$

Proof. If $A \xrightarrow{i} X$ is a cofibration, then $X \times \{0\} \cup A \times I \xrightarrow{1} X \times \{0\} \cup A \times I$ induces a retraction. Conversely, by Lemma 14.12, $A \times I \xrightarrow{g} Y$, $X \xrightarrow{f_0} Y$ with $g_0 = f_0|_A$ gives a map $X \times \{0\} \cup A \times I \to Y$, composing with retraction gives $X \times I \to Y$

A strong deformation retraction is given by $H((x,t),s) = (\mathrm{Pr}_X\, r(x,st), s\,\mathrm{Pr}_I\, r(x,t) + (1-s)t)$

□

Lemma 14.14 If (X, A) is cofibered, so is $(X, \overline{A})$

Proof. Define $\phi(x) = \inf_{t \in I}\{\mathrm{Pr}_I\, r(x,t) \neq 0\}$, then there is a retraction $X \times I \xrightarrow{r'} X \times \{0\} \cup \overline{A} \times I$

$$r'(x,t) = \begin{cases} (\mathrm{Pr}_X\, r(x,t), 0) & t \leq \phi(x) \\ (\mathrm{Pr}_X\, r(x, \phi(x)), t - \phi(x)) & t \geq \phi(x) \end{cases}$$

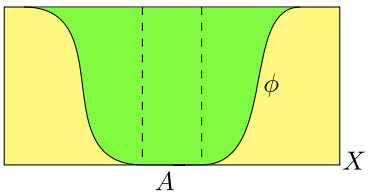
□

Lemma 14.15 $E \xrightarrow{p} B$ is a fibration, A is a strong deformation retract of X and A can be perfectly separated, then

$$\begin{array}{ccc} A & \xrightarrow{f''} & E \\ {\scriptstyle i}\downarrow & \overset{f}{\nearrow} & \downarrow{\scriptstyle p} \\ X & \xrightarrow{f'} & B \end{array}$$

f is unique up to homotopy rel$A$

Proof. $X \xrightarrow{\phi} \mathbb{R}$ is a function such that $\phi^{-1}(0) = A$. $X \xrightarrow{r} A$ is the retract. $X \times I \xrightarrow{D} X$ is a homotopy from ir to 1_X. Define $H(x,t) = \begin{cases} D(x, t/\phi(x)) & t \leq \phi(x) \\ D(x,1) & t \geq \phi(x) \end{cases}$

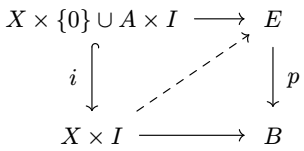


Since $E \xrightarrow{p} B$ is a fibration, we have a lift $X \times I \xrightarrow{F} E$ such that $pF = f'H$ and $F_0 = f''r$ because $pF_0 = pf''r = f'ir = f'H_0$. Define f to be the composition $X \xrightarrow{1\times\phi} X \times I \xrightarrow{F} E$, then we have $fi = F(1 \times \phi) = F_0 i = f''$ and $pf = pF(1 \times \phi) = f'H(1 \times \phi) = f'D_1 = f'$

$$\begin{array}{ccccc} X & \underset{i}{\overset{r}{\rightleftarrows}} & A & \xrightarrow{f''} & E \\ \downarrow & & & \overset{F}{\nearrow} & \downarrow{\scriptstyle p} \\ X \times I & \underset{1\times\phi}{\overset{H}{\rightleftarrows}} & X & \xrightarrow{f'} & B \end{array}$$

□

Proposition 14.16 Fibrations have homotopy lifting property for closed cofibrations



Proposition 14.17 If (X, A), (Y, B) are cofibered, $A \subseteq X$ is closed, then $(X \times Y, X \times B \cup A \times Y)$ is also cofibered. If in addition A or B is a strong deformation retract of X or Y, then $X \times B \cup A \times Y$ is a strong deformation retract of $X \times Y$

Proposition 14.18 $E \overset{p}{\to} B$ is a fibration, then the fibers over connected components of B are homotopic. More over, for any path $\gamma : I \to B$, we can get a lifting $g_t : F_{\gamma(0)} \to F_{\gamma(t)}$ of $F_{\gamma(0)} \hookrightarrow E$, define $L_\gamma : F_{\gamma(0)} \to F_{\gamma(1)}$ to be g_1, if $\gamma \simeq \eta$ rel $\partial I : I \to B$, then $L_\gamma \simeq L_\eta$, and for any $\gamma, \eta : I \to B$, $\eta(0) = \gamma(1)$, $L_{\gamma\eta} \simeq L_\gamma L_\eta$

Proof. According to homotopy lifting property, lifting up $A \times F_{\gamma(0)} \to E$ is homeomorphic to $B \times F_{\gamma(0)} \to E$

□

Remark. We can think of this as an action of $\pi_1(B)$ on $H_*(F)$

Definition 14.19 $E_1 \overset{p_1}{\to} B_1$, $E_2 \overset{p_2}{\to} B_2$ are fibrations, $p_1 \overset{f_0, f_1}{\to} p_2$ are **fiber homotopic** if there exists $p_1 \overset{f_t}{\to} P_2$ varying from f_0 to f_1. p_1, p_2 are **fiber homotopy equivalent** if there are fiber homotopies $p_0 \overset{f}{\to} p_1$ and $p_1 \overset{g}{\to} p_0$ such that fg, gf fiber homotopic to 1

Lemma 14.20 Cofibration $A \overset{i}{\to} X$ is a homotopy equivalence iff A is a strong deformation retract of X. Fibration $E \overset{p}{\to} B$ is a homotopy equivalence iff there exists a section $B \overset{s}{\to} E$ such that sp is fiber homotopic to 1

Proof. If i is a homotopy equivalence, then there exists $X \overset{r'}{\to} A$ such that $ir' \simeq 1_X$, $r'i \simeq 1_A$, by Lemma 11.55, since (X, A) is cofibered, $r' \simeq r$ is a retract and then A is a deformation retract of X. Suppose $X \times I \overset{F}{\to} X$ is a homotopy from 1_X to ir

$\Gamma = X \times \{0\} \cup A \times I \cup X \times \{1\} = X \times \{0\} \cup A \times [0, \frac{1}{2}] \cup A \times [\frac{1}{2}, 1] \cup X \times \{1\}$ is a retract of $X \times I$. Construct $\Gamma \times I \overset{G}{\to} X$

$$G((x,t),s) = \begin{cases} F(x,(1-s)t) & (x,t) \in X \times \{0\} \cup A \times I \\ F(r(x), 1-s) & (x,t) \in X \times \{1\} \end{cases}$$

G_1 can be extends to $X \times I \overset{H}{\to} X$, then $H_0(x) = G_1(x,0) = F(x,0) = x$, $H_1(x) = G_1(x,1) = F(r(x),0) = r(x)$, $H_t(a) = G_1(a,t) = F(a,0) = a$, i.e. H is a strong deformation retraction

□

Lemma 14.21 $E \overset{p}{\to} B$ is a fibration, $A \overset{f}{\to} B$ is a map, $A \times_B E = f^*(E)$ is the **pullback fibration**, suppose $f_t : A \to B$ is a homotopy, then pullback fibrations $f_0^*(E) \to A$, $f_1^*(E) \to A$ are fiber homotopy equivalent. In particular, a morphism $p \overset{f}{\to} q$ between two fibrations is a homotopy equivalence iff f is a fiber homotopy equivalence

Proof. $A \times I \overset{F}{\to} B$ is a homotopy, we have the pullback fibration $F^*(E)$, it suffices to show that for any fibration $E \overset{p}{\to} B \times I$, $E_0 \coloneqq p^{-1}(B \times \{0\}) \simeq p^{-1}(B \times \{1\}) =: E_1$ are fiber homotopy equivalent

Consider the following diagrams

$$\begin{array}{ccc} E_0 & \hookrightarrow & E \\ \downarrow & \nearrow_{(p,t)} & \downarrow p \\ E_0 \times I & \longrightarrow & B \times I \end{array}$$

$$\begin{array}{ccc} E_1 & \hookrightarrow & E \\ \downarrow & \nearrow_{(p,1-t)} & \downarrow p \\ E_1 \times I & \longrightarrow & B \times I \end{array}$$

Then we get fiber preserving maps $f : E_0 \to E_1$ and $g : E_1 \to E_0$, and restricts them to each fiber

$$\begin{array}{ccc} p^{-1}(b,0) & \hookrightarrow & E \\ \downarrow & \nearrow_{(p,t)} & \downarrow p \\ p^{-1}(b,0) \times I & \longrightarrow & \{b\} \times I \end{array}$$

$$\begin{array}{ccc} p^{-1}(b,1) & \hookrightarrow & E \\ \downarrow & \nearrow_{(p,1-t)} & \downarrow p \\ p^{-1}(b,1) \times I & \longrightarrow & \{b\} \times I \end{array}$$

We get maps $f|_{p^{-1}(b,0)} : p^{-1}(b,0) \to p^{-1}(b,1)$, $g|_{p^{-1}(b,1)} : p^{-1}(b,1) \to p^{-1}(b,0)$, according to Proposition 14.18 they are homotopy equivalence and inverses to each other, hence f, g are fiber homotopy equivalences and inverses to each other

□

Corollary 14.22 $E \xrightarrow{p} B$ is a fibration, B contractible, then p is fiber homotopy equivalent to $B \times F \to B$. If B is locally contractible, the fibration is locally homotopy equivalent to a product

Proof. Since B is contractible, identity map is homotopic to a constant map, and the pullback of E under the identity map is E itself, the pullback of E under a constant map is fiber bundle $B \times F$

□

Definition 14.23 (X, x_0) is a pointed space. The **loop space** ΩX consists of all the loops on X starting and ending at x_0, the constant loop being the basepoint. The **path space** PX consists of all the paths starting at x_0. $\Omega X \subseteq PX \subseteq X^I$ endowed with the subspace topology

Proposition 14.24 $\langle \Sigma X, Y \rangle = \langle X, \Omega Y \rangle$ is an adjunction

Definition 14.25 The **mapping path space** P_f is the pullback

$$\begin{array}{ccc} P_f & \longrightarrow & Y^I \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

P_f deformation retracts onto X by shrinking paths. The **homotopy fiber** F_f of f over y is the pullback

$$\begin{array}{ccc} F_f & \longrightarrow & Y^I \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \times Y \end{array}$$

The **mapping cylinder** M_f is the pushout

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & & \downarrow \\ X \times I & \longrightarrow & M_f \end{array}$$

M_f deformation retracts onto Y by sliding the cylinder The mapping cone $M_f/A \times \{0\} \cong C_f$ is the **homotopy cofiber** of f

Proposition 14.26 $X \xrightarrow{f} Y$ can be factorized as $X \hookrightarrow P_f \to Y$ or $X \hookrightarrow M_f \to Y$. $P_f \to Y$, $(x, \gamma) \mapsto \gamma(1)$, $M_f \to Y$ are Hurewicz fibrations. $X \hookrightarrow P_f$, $X \hookrightarrow M_f$ are closed Hurewicz cofibrations

Proof.

$$\begin{array}{ccc} X & \longrightarrow & M_f \\ \downarrow & \nearrow & \downarrow \\ X \times I & \longrightarrow & Y \end{array}$$

□

Example 14.27 PX is the mapping path space of $* \to X$, $\Omega X \to PX \to X$ is a Hurewicz fibration

Proposition 14.28 Suppose $E \xrightarrow{p} X$ is a fibration, then $E \hookrightarrow E_p$ is a fiber homotopy equivalence, and the restriction on each fiber to the homotopy fiber of p is a homotopy equivalence

Proposition 14.29 If $F \to E \to B$ is a fibration, and E is contractible, then F is weakly homotopic to ΩB

Theorem 14.30 $F \to E \to B$ is a fibration, the **fibration sequence** is

$$\to \Omega^2 B \to \Omega F \to \Omega E \to \Omega B \to F \to E \to B$$

$A \to X \to X/A$ is a cofibration, the **cofibration(Puppe) sequence** is

$$A \to X \to X/A \to \Sigma A \to \Sigma X \to \Sigma(X/A) \to \Sigma^2 A \to$$

Theorem 14.31 $E \xrightarrow{p} B$ is Serre fibration, fix $x_0 \in p^{-1}(b_0) = F$, $\pi_n(E, F, x_0) \xrightarrow{p^*} \pi_n(B, b_0)$ is an isomorphism, and we have long exact sequence

$$\cdots \to \pi_n(F, x_0) \to \pi_n(E, x_0) \xrightarrow{p^*} \pi_n(B, b_0) \to \pi_{n-1}(F, x_0) \to \cdots \to \pi_0(E, x_0) \to 0$$

Theorem 14.32 $\mathbf{W}$ consists of all homotopy equivalences, $\mathbf{F}$ consists of all Hurewicz fibrations, $\mathbf{C}$ consists of all closed Hurewicz cofibrations, $(\mathbf{W}, \mathbf{F}, \mathbf{C})$ defines a closed model structure on Top

Proof. 2 out of 3 is obvious

By Lemma 14.15 and Lemma 14.20, $i \in \mathbf{C} \cap \mathbf{W} \Rightarrow i$ has LLP for any $p \in \mathbf{F}$. Suppose i has LLP for any $p \in \mathbf{F}$, since $Y^I \to Y \in \mathbf{F}$, $i \in \mathbf{C}$, $A \to * \in \mathbf{F}$, we get a retraction $X \xrightarrow{r} A$ by

$$\begin{array}{ccc} A & = & A \\ {\scriptstyle i}\downarrow & \nearrow_{r} & \downarrow \\ X & \longrightarrow & * \end{array}$$

$X^I \to X \times X \in F$, $\gamma \mapsto (\gamma(0), \gamma(1))$, then we get a strong deformation retraction

$$\begin{array}{ccc} A & \longrightarrow & X^I \\ {\scriptstyle i}\downarrow & \nearrow_{F} & \downarrow \\ X & \longrightarrow & X \times X \end{array}$$

Hence $i \in \mathbf{C} \cap \mathbf{W}$

□

14.3 Serre fibration

Definition 14.33 $E \xrightarrow{p} B$ is a **Serre fibration** if it has homotopy lifting property for all $(D^n, \partial D^n)$

15 Bundles

Definition 15.1 A **bundle** is $E \xrightarrow{p} B$, where E is the **total space**, B is the **base space**, and p is the projection, $p^{-1}(b)$ is the **fiber** over b. A **cross section** is $s : X \to E$, such that $ps = 1_X$. The restriction $p^{-1}(A) \xrightarrow{\pi} A$, $A \subseteq B$ is also a bundle

Definition 15.2 Suppose $E \xrightarrow{p} B$ is a bundle, $f : A \to B$ is a map, then the pullback $f^*(E) = A \times_p E \to A$ is the **pullback bundle**, the pullback of a section $s : B \to E$ is defined as $f^*s := s \circ f$, notice $p(f^*s(y)) = p(s(f(y))) = f(y)$

Definition 15.3 A **fiber bundle** is a bundle $E \xrightarrow{p} B$ such that there exists an open neighborhood U of b and a homeomorphism ϕ making the following diagram commute

$$\begin{array}{ccc} p^{-1}(U) & \xrightarrow{\phi} & U \times F \\ {\scriptstyle p}\downarrow & \swarrow_{pr_1} & \\ U & & \end{array}$$

Definition 15.4 G is a topological group, a G **fiber bundle** $E \xrightarrow{p} B$ is a fiber bundle and also a morphism of G spaces

Lemma 15.5 A fiber bundle is a Serre fibration

Definition 15.6 $\mathbb{F}$ is a topological field, a **vector bundle** is a fiber bundle $E \xrightarrow{p} X$ with fiber being $\mathbb{F}^n$ and ϕ restricts on each fiber is an $\mathbb{F}$ isomorphism

Definition 15.7 G is a topological group, a **principal G bundle** $p : P \to B$ is a morphism of G spaces, B with the trivial G action, and for each $b \in B$, there is a local trivialization

$$\begin{array}{ccc} p^{-1}(U) & \xrightarrow{\phi} & U \times G \\ {\scriptstyle p}\downarrow & \swarrow_{pr_1} & \\ U & & \end{array}$$

ϕ is an isomorphism

Remark. G action on P preserves fibers, and the action on fiber is free and transitive, each fiber is a G torsor. A morphism of principal G bundles is always an isomorphism. A principal G bundle is trivial iff it has a global section

Proposition 15.8 Suppose $P \to B$ is a principal G bundle, $G \to G/H$ is a principal H bundle, then $P \to P/H$ is a principal H bundle

Proof. $P \cong P \times_G G \to P \times_G (G/H) \cong P/H$

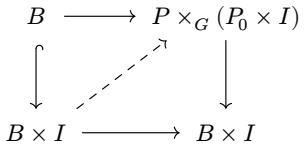
□

Proposition 15.9 Suppose $P \to B$ is a principal G bundle, F is a left G space, $P \times_G F \to P \times_G * \cong B$ is a G fiber bundle. $X \xrightarrow{f} Y$ is a map, $f^*(P \times_G F) \to f^*(P) \times_G F$ is a natural homeomorphism

Proposition 15.10 $P \xrightarrow{p} B$ is a principal G bundle, X is a right G space, a morphism $P \xrightarrow{f} X$ induce $P \xrightarrow{\binom{1}{f}} P \times X$, $B \cong P/G \to P \times X/G \cong P \times_G X$ which is a section s_f of $P \times_G X \to B$, this is a natural bijection

Proposition 15.11 $P \to B \times I$ is principal G bundle, then P and $P_0 \times I$ is an isomorphism, here P_0 is the restriction of P over $B \times \{0\}$

Proof.



□

Definition 15.12 Let $E \xrightarrow{p} X$ be a vector bundle, consider two trivializations $\varphi_U : E_U \to U \times \mathbb{R}^n$ and $\varphi_V : E_V \to V \times \mathbb{R}^n$ around $x \in X$, then $\varphi_V \circ \varphi_U^{-1}$ restricted on $U \cap V \times \mathbb{R}^n$ is a local isomorphism with inverse $\varphi_U \circ \varphi_V^{-1}$ restricted on $U \cap V \times \mathbb{R}^n$, it is also called a transition function and it can also be regard as a continuous map $g_{VU} : U \cap V \to GL(n, \mathbb{R})$ or $g_{VU} \in GL(n, C(U \cap V))$, such that $\varphi_V \circ \varphi_U(x, v) = (x, g_{VU}(x)v)$, notice then $g_{UV} = g_{VU}^{-1}$, and g_{VU}'s satisfy the cocycle relation $g_{WV}g_{VU} = g_{WU}$ on $U \cap V \cap W$

Conversely, given $\bigsqcup_{\alpha \in A} U_\alpha \times \mathbb{R}^n \times A$ transition functions $g_{\alpha\beta} : U_\alpha \cap U_\beta \to GL(n, \mathbb{R})$ that satisfying cocycle relation $g_{\gamma\beta}g_{\beta\alpha} = g_{\gamma\alpha}$ on $U_\alpha \cap U_\beta \cap U_\gamma$, mod equivalence relation $(x, v, \alpha) \sim (x, g_{\beta\alpha}(v), \beta), x \in U_\alpha \cap U_\beta$, you will get back the vector bundle

Suppose $s : X \to E$, is a section, denote $\varphi_i \circ s|_{U_i}(x) = (x, f_i(x))$ over U_i, then $(x, f_j(x)) = \varphi_j \circ s|_{U_j}(x) = \varphi_j \circ s|_{U_i}(x) = \varphi_j \circ \varphi_i^{-1} \circ \varphi_i \circ s|_{U_i}(x) = \varphi_j \circ \varphi_i^{-1}(x, f_i(x)) = (x, g_{ji}(x)f_i(x)), \forall x \in U_i \cap U_j$, thus $f_j = g_{ji}f_i$, conversely, this relation also defines a section

Definition 15.13 The pullback of a transition function is defined to be $f^*g_{ij} \coloneqq g_{ij} \circ f$

Definition 15.14 A morphism between vector bundles $\varphi : E \to F$ is map such that the following diagram commutes

$$\begin{array}{ccc} E & \xrightarrow{\varphi} & F \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ X & \xrightarrow{f} & Y \end{array}$$

and $\varphi_x : E_x \to F_{f(x)}$ is a homomorphism between vector spaces

Definition 15.15 Let $E \xrightarrow{p} X$ and $F \xrightarrow{q} Y$ be vector bundles, then direct sum $E \times F \xrightarrow{p \times q} X \times Y$ is also a vector bundle, suppose $\varphi_U : U \to U \times \mathbb{R}^n$, $\psi_V : V \to V \times \mathbb{R}^m$ are trivializations, then $\varphi_U \times \psi_V : U \times V \to U \times \mathbb{R}^n \times V \times \mathbb{R}^m \cong U \times V \times \mathbb{R}^{n+m}$ is also a trivialization

Proposition 15.16 Let $E \xrightarrow{p} X$ is a vector bundle, and $f : X \to Y$ is a homeomorphism, then $E \xrightarrow{f \circ p} Y$ is a vector bundle, suppose $\varphi_U : E_U \to U \times \mathbb{F}^n$ is a trivialization, then $(f \times 1) \circ \varphi_U =: \psi_{f(U)} : E_U \to U \times \mathbb{F}^n \to f(U) \times \mathbb{F}^n$ is a trivialization

Proposition 15.17 $p : \Gamma_f \to X, (x, f(x)) \mapsto x$ is homeomorphism

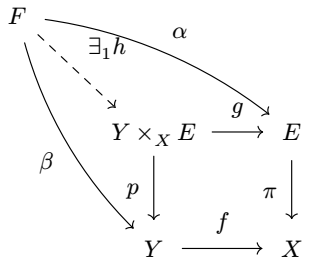
Proof. p as a restriction on Γ_f of $X \times Y$ projecting to X is continuous, and define $q : X \to \Gamma_f, x \to (x, f(x))$, since the composition $X \xrightarrow{q} \Gamma_f \hookrightarrow X \times Y$ which is continuous because $X \xrightarrow{f} Y$, $X \xrightarrow{id} X$ are continuous, q is continuous, and p, q are inverses to each other

□

Definition 15.18 $E \xrightarrow[p]{\pi} X$ ia vector bundle, $f : Y \to X$ is a continuous map, then we can construct the pullback bundle $f^*E \xrightarrow{} Y$

$$\begin{array}{ccc} f^*E & \xrightarrow{g} & E \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle \pi} \\ Y & \xrightarrow{f} & X \end{array}$$

satisfying universal property



Concrete construction: let $f^*E = Y \times_X E \subseteq Y \times X$ with subspace topology, where $Y \times_X E = \{(y, v) \in Y \times E \mid f(y) = \pi(v)\}$, let's check it is a vector bundle over Y, notice that $Y \times_X E \to Y$ factor through $Y \times_X E \to \Gamma_f \to Y$, $(y, v) \mapsto (y, \pi(v)) = (y, f(y)) \mapsto y$, where Γ_f is the graph of f which is homeomorphic to Y due to Proposition 15.17, notice that $Y \times_X E \to \Gamma_f$ is the restriction of vector bundle $Y \times E \overset{1\times\pi}{\to} Y \times X$ over Γ_f, thus $Y \times_X E \to Y$ is a vector bundle, suppose F as in the commutative diagram, then h is simply defined as $h(w) \coloneqq (\beta(w), \alpha(w))$

Remark. In general, this is a pullback, but it has a vector bundle structure such that it induces an isomorphism on each fiber, now suppose $F \overset{q}{\to} Y$ is a another vector bundle such that not only the diagram commutes but also induce isomorphism on each fiber, then $F \cong f^*E$

Use this we have $(fg)^*E \cong g^*(f^*E)$, $f^*(E \oplus F) \cong f^*E \oplus f^*F$, $f^*(E \otimes F) \cong f^*E \otimes f^*F$, $1^*E \cong E$

Definition 15.19 Suppose E, F are vector bundles both trivialized over $\{U_\alpha\}$(this can easily be achieved, just take intersections), suppose the transition functions are $g_{\alpha\beta}, h_{\alpha\beta}$, then define the tensor product of vector bundles $E \otimes F$ by letting its transition functions be $g_{\alpha\beta} \otimes h_{\alpha\beta}$

Similarly, we can define symmetric power and exterior power of vector bundles by specifying its transition function

Does it have universal property also?

Definition 15.20 Let $E \overset{p}{\to} X$, $E \overset{p}{\to} X$ be vector bundles, then the direct sum $E \oplus F \overset{p}{\to} X$ is defined by transition functions $g_{\alpha\beta} \oplus h_{\alpha\beta}$, where $g_{\alpha\beta}, h_{\alpha\beta}$ are transition functions of E, F

Definition 15.21 Let $E \to X$ be a vector bundle, define its dual bundle as follows, if $g_{\alpha\beta}$ is a transition function, the transition function for E^* would be $\left(g_{\alpha\beta}^{-1}\right)^T$

Definition 15.22 quotient bundle, exterior and symmetric power of vector bundle

Proposition 15.23 $E \overset{p}{\to} X$ is a vector bundle with X being a paracompact space, then there exists a continuous map $\langle , \rangle : E \oplus E \to \mathbb{R}$ with $\langle , \rangle|_{E_x}$ defines an inner product

Definition 15.24 $F \subseteq E$ is called a vector subbundle if F is a subspace of E and $F \overset{p}{\to} X$ is also a vector space

Proposition 15.25 $E \overset{p}{\to} X$ is a vector bundle with X being a paracompact space and $F \subseteq E$ is a vector subbundle, then there exists a vector subbundle $F^\perp \subseteq E$ such that $F_x \oplus F^\perp|_x = E|_x$ and $(F\ |_x)^\perp = F^\perp|_x$

Proof.

□

Theorem 15.26 If $E \overset{p}{\to} X$ is vector bundle over a compact Hausdorff space X, then there exists a vector bundle $E' \overset{p'}{\to} X$ such $E \oplus E'$ is a trivial bundle

Proposition 15.27 Every Lie group G is parallelizable

Proof. Pick an arbitrary basis $e_1, \cdots, e_n$ of T_1G, then $L_g^*(e_i)$ will be a basis of $T_{g^{-1}}G$ since L_g^* is an isomorphism, they form independent global sections of the tangent bundle

□

Definition 15.28 Tautological bundle

Definition 15.29 Let X be a smooth manifold of dimension n(depending on the field), Ω denote the cotangent bundle, then $\omega \coloneqq \bigwedge^n \Omega$ is called the canonical bundle

Definition 15.30 Universal bundle

Theorem 15.31 Let X be a paracompact Hausdoff space, there is a bijection $[X, \lim_{\to} Gr_{\mathbb{C}}(n, N)] \to \mathrm{Vect}^n_{\mathbb{C}}(X), [f] \mapsto [f^*(E)]$

Definition 15.32 Let $E \xrightarrow{p} X$ is a vector bundle, an inner product is a continuous map $\langle , \rangle : E \oplus E \to \mathbb{R}$ with $\langle , \rangle|_{E_x}$ defines an inner product on E_x

Proposition 15.33 Let $E \xrightarrow{p} X$ is a vector bundle with an inner product $\langle , \rangle$, then we can local trivialization to be isometry on each fiber, i.e. $\langle v, w \rangle = (\varphi_U(v), \varphi_U(w)), v, w \in E_x$, where $(,)$ is the standard inner product on $U \times \mathbb{R}^n$

Proposition 15.34 $E \xrightarrow{p} X$ is a vector bundle with X being a paracompact space, then there exists a continuous map $\langle , \rangle : E \oplus E \to \mathbb{R}$ with $\langle , \rangle|_{E_x}$ defines an inner product

Definition 15.35 let G be a topological group, E, X be G-spaces, then $E \xrightarrow{p} X$ is a G-vector bundle if it is a vector bundle, p is a G map, and for any $x \in X$, $g : E_x \to E_{gx}$ is a linear map

Definition 15.36 Let G be a topological group, H be a closed subgroup, a G vector bundle $\pi : E \to G/H$ is called a homogeneous vector bundle

Lemma 15.37 Let $Y \xrightarrow{f} X, Z \xrightarrow{g} X$ be open surjective continuous maps, then the projection $p_Y : Y \times_X Z \to Y$ is open surjective

Proof. For surjectivity, if $y \in Y$, since g is surjective, $\exists z \in Z$ such that $g(z) = f(y)$, then $(z, y) \in Y \times_X Z$

To prove p_Y is open, suppose $(z_0, y_0) \in Y \times_X Z$ is in some open set, then $(z_0, y_0) \in U \times V \cap Y \times_X Z$ for some $y_0 \in U, z_0 \in V$ open, since f, g are open, $U' \coloneqq f(U) \cap g(V)$ is open, let $V' \coloneqq V \cap f^{-1}(U')$, then we can show V' is in the image of $U \times V \cap Y \times_X Z$, since $\forall y \in V'$, $f(y) \in U' \subseteq g(V)$, thus $f(y) = g(z)$ for some $z \in V$, hence $(y, z) \in U \times V \cap Y \times_X Z$

□

Proposition 15.38 Let $\pi : E \to G/H$ be a homogeneous vector bundle, E_H be the fiber over the coset H, action $G \times E_H \to E$ can be regard as $\alpha : G \times_H E_H \to E$ which is an isomorphism of G vector bundles. Moreover, if H is locally compact, then for a given $\mathbb{R}H$ module E_H, $G \times_H E_H \to G/H$ is indeed a G vector bundle, hence G vector bundle E is in one to one correspondence with representations of H on E_H, so $K_G(G/H) \cong R(H)$

Proof. E_H is an $\mathbb{R}H$ module, let $G \times_H E_H$ denote the space of orbits of $G \times E_H$ under H by $h \cdot (g, \xi) = (gh^{-1}, h\xi)$, $G \times_H E_H$ is a G space with G action $g \cdot (g', \xi) \mapsto (gg', \xi)$, then the group action can be regarded as $\alpha : G \times_H E_H \to E, (g, \xi) \mapsto g\xi$, we can find its inverse $\beta : E \to G \times_H E_H, E_{gH} \ni \xi \mapsto (g, g^{-1}\xi)$, to show that this is continuous, consider $\gamma : G \times E \to G \times E, (g, \xi) \mapsto (g, g^{-1}\xi)$, then the preimage of $G \times E_H$ will be the pullback $G \times_{G/H} E \coloneqq \{(g, \xi) \in G \times E \mid gH = \pi\xi\}$, then $G \times_{G/H} E \to G \times E_H \to G \times_H E_H, (g, \xi) \mapsto (g, g^{-1}\xi)$ factors as $G \times_{G/H} E \to E \xrightarrow{\beta} G \times_H E, (g, \xi) \mapsto \xi \mapsto (g, g^{-1}\xi)$ which open surjective, therefore β is continuous due to the previous Lemma

□

Definition 15.39 A clutching function for S^k is $f : S^{k-1} \to GL(n, \mathbb{C})$, then we can define vector bundle E_f with f being the transition function, conversely, if E is a vector bundle over S^k, since its upper and lower hemispheres are both contractible, $E = E_f$, where f is the transition function, denoting the corresonding matrix T_f

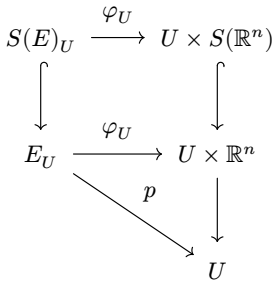
Theorem 15.40 $[S^{k-1}, GL(n,\mathbb{C})] \to \mathrm{Vect}^n_{\mathbb{C}}(S^k), f \mapsto E_f$ is a bijection

Lemma 15.41 Suppose $f, g : S^{k-1} \to GL(n,\mathbb{C})$, then $(E_f \otimes E_g) \oplus \varepsilon^n \cong E_{fg} \oplus \varepsilon^n \cong E_f \oplus E_g$

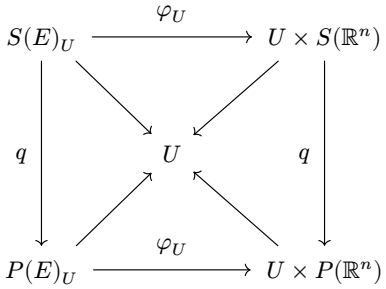
Proof. Since $GL(n,\mathbb{C})$ is path connected, there is a path $A_t \in GL(2n,\mathbb{C})$ that $A_0 = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, A_1 = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$, then $\begin{pmatrix} T_f & \\ & I \end{pmatrix} A_t \begin{pmatrix} I & \\ & T_g \end{pmatrix} A_t$ is $\begin{pmatrix} T_f & \\ & T_g \end{pmatrix}$ when $t = 0$ and $\begin{pmatrix} T_f T_g & \\ & I \end{pmatrix} = \begin{pmatrix} T_{fg} & \\ & I \end{pmatrix}$ when $t = 1$

□

Definition 15.42 Let $E \xrightarrow{p} X$ be vector bundle of rank n, and there is a inner product over E, we can define the sphere bundle $S(E)$ associated to E to be $S(E) = \bigcup_{x\in X} S(E_x)$ with the subspace topology, this is a fiber bundle, suppose φ_U is a local trivialization, since we can choose φ_U to be isometry over each fiber, thus the following diagram commutes



Definition 15.43 Let $E \xrightarrow{p} X$ be vector bundle of rank n, and there is a inner product over E, we can define the projective bundle $P(E)$ associated to E to be $P(E) = \bigcup_{x\in X} P(E_x)$ with the quotient topology, this is a fiber bundle, suppose φ_U is a local trivialization, since we can choose φ_U to be isometry over each fiber, thus the following diagram commutes



Definition 15.44 Let $E \xrightarrow{p} X$ be vector bundle of rank n, and there is a inner product over E, we can define the flag bundle $F(E)$ associated to E to be $F(E) = \bigcup_{x\in X} F(E_x)$ with the subspace topology in $P(E) \times \cdots \times P(E)$

Remark. Consider the pullback of $\pi : F(E) \to X$, $\pi^*(E) \subseteq F(E) \times E$, consider its subbundles $L_1, \cdots, L_n$, where L_i is the subbundle that over a point in $F(E)$, it is the i-th factor, then $\pi^*(E) \cong L_1 \oplus \cdots \oplus L_n$

Definition 15.45 Let X be a paracompact and Hausdorff space, there exist unique functions $w_1, w_2, \cdots$, $w_i : \mathrm{Vect}_{\mathbb{R}}(\mathrm{X}) \to H^i(X, \mathbb{Z}_2)$, $E \to w_i(E)$, and they only depend on the isomorphism classes of E, satisfying

1. $w_i(f^*(E)) = f^*(w_i(E))$, for pullback bundle $f^*(E)$

2. $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$ where $w = 1 + w_1 + w_2 + \cdots \in H^*(X, \mathbb{Z}_2)$

3. $w_i(E) = 0, \forall i > \dim E$

4. If $E \to \mathbb{R}P^\infty$ is the canonical line bundle, then $w_1(E)$ is the generator of $H^*(\mathbb{R}P^\infty, \mathbb{Z}_2) \cong \mathbb{Z}_2[x]$

$w_i(E)$ are called the Stiefel-Whitney classes of E

Definition 15.46 Let X be a paracompact and Hausdorff space, there exist unique functions $c_1, c_2, \cdots$, $c_i : \mathrm{Vect}_{\mathbb{C}}(\mathrm{X}) \to H^{2i}(X;\mathbb{Z})$, $E \to c_i(E)$, and they only depend on the isomorphism classes of E, satisfying

1. $c_i(f^*(E)) = f^*(c_i(E))$, for pullback bundle $f^*(E)$

2. $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$ where $c = 1 + c_1 + c_2 + \cdots \in H^*(X;\mathbb{Z})$

3. $c_i(E) = 0, \forall i > \dim E$

4. If $E \to \mathbb{C}P^\infty$ is the canonical line bundle, then $c_1(E)$ is a generator of $H^*(\mathbb{C}P^\infty;\mathbb{Z}) \cong \mathbb{Z}[x]$, specify a generator in advance

$c_i(E)$ are called the Chern classes of E, also we define the Chern polynomial to be $c_t = 1 + c_1 t + c_2 t^2 + \cdots$ where t is just a formal variable used to keep tracking of the degree

Lemma 15.47 Let L_1, L_2 be line bundles, then $c_1(L_1 \otimes L_2) = c_1(L_1) + c_1(L_2)$

Definition 15.48 Suppose L is a line bundle, define the Chern character $ch(L) = e^{c_1(L)} = 1 + c_1(L) + \frac{c_1(L)^2}{2!} + \cdots \in H^*(X;\mathbb{Q})$, then we have $ch(L_1 \otimes L_2) = e^{c_1(L_1 \otimes L_2)} = e^{c_1(L_1)+c_1(L_2)} = e^{c_1(L_1)} e^{c_1(L_2)} = ch(L_1)ch(L_2)$, If we assume $ch(L_1 \oplus L_2) = ch(L_1) + ch(L_2)$, then for $E = L_1 \oplus \cdots \oplus L_n$, $ch(E) = ch(L_1) + \cdots + ch(L_n) = n + (c_1(L_1) + \cdots + c_1(L_n)) + (c_1(L_1)^2 + \cdots + c_1(L_n)^2)/2! + \cdots$, on the other hand, we have $c(E) = c(L_1) \smile \cdots \smile c(L_n) = (1 + c_1(L_1)) \smile \cdots \smile (1 + c_1(L_n)) = 1 + c_1(E) + \cdots + c_n(E)$, where $c_i(E)$ would just be the i-th elementary symmetric polynomial of $c_1(L_1), \cdots, c_1(L_n)$, i.e. $c_i(E) = \sigma_i(c_1(L_1), \cdots, c_1(L_n))$, so we can express $c_1(L_1)^k + \cdots + c_1(L_n)^k$ in terms of $c_i(E)$, i.e. $c_1(L_1)^k + \cdots + c_1(L_n)^k = s_k(c_1(E), \cdots, c_n(E))$, thus we have an abstract definition of Chern character, $ch(E) = \dim E + s_1(c_1(E), \cdots, c_n(E)) + s_2(c_1(E), \cdots, c_n(E))/2! + \cdots$

Proposition 15.49 $ch(E_1 \oplus E_2) = ch(E_1) + ch(E_2)$, $ch(E_1 \otimes E_2) = ch(E_1)ch(E_2)$

Theorem 15.50 (Thom isomorphism) Λ is a ring, $p : E \to B$ is an oriented vector bundle of rank r, then there exists the *Thom class* $u \in H^r(E, E \setminus B, \Lambda)(B \subseteq E$ as the zero section) such that its restriction to any fiber F, $H^r(F, F \setminus 0, \Lambda)$ is the local orientation. And $H^r(E, E \setminus B, \mathbb{Z}) \to H^r(E, \mathbb{Z}) \to H^r(B, \mathbb{Z})$ defines the *Euler class* $e(E)$

For the tangent bundle of a smooth manifold, the Euler class evaluated at the fundamental class of B is the Euler characteristic

Lemma 15.51 V is a vector space of dimension n, $\mathcal{V}$ is the set of frames(bases) of V, then $\mathrm{GL}(\mathbb{F}, n)$ acts transitively and freely on the right of $\mathcal{V}$

Definition 15.52 If G is a topological group, then a principal G-bundle P is a fiber bundle with a continuous right G action $P \times G \to P$, and the action is free and transitive(thus regular), which imply each fiber is a G-torsor, also, $g \mapsto yg$ is a homeomorphism

Definition 15.53 (Vector bundle(Koszul) connection)
$E \to B$ is a vector bundle. E valued k-forms are

$$\Omega^k(E) = \Gamma\left(\bigwedge^k T^*B \otimes E\right) = \Omega^k(B) \otimes \Gamma(E)$$

A *vector bundle(Koszul) connection* is

$$\nabla : \Omega^0(E) \to \Omega^1(E) = \mathrm{Hom}(TB, \mathbb{F}) \otimes \Gamma(E)$$

Satisfying the Leibniz rule

$$\nabla(f \otimes s) = df \otimes s + f \wedge \nabla s \tag{15.1}$$

By feeding $X \in TB$, $\nabla_X : \Gamma(E) \to \Gamma(E)$ is the covariant derivative along X. This extends to exterior covariant derivative $d_\nabla : \Omega^k(E) \to \Omega^{k+1}(E)$ through 15.1

Definition 15.54 (Connection form)
Over a local trivialization, $s = \sum s_i \otimes e_i$, $\nabla e_j = \sum_i \omega_{ij} \otimes e_i$, $\nabla s = \sum_i (ds_i + \sum_j \omega_{ij} \wedge s_j) \otimes e_i$, in short, $\nabla s = ds + \omega \wedge s$ in local coordinates, here $\omega = (\omega_{ij})$ is the *connection form*

Definition 15.55 (Curvature) In general d_∇ may not give a chain complex. The curvature tensor is

$$d_\nabla^2 : \Omega^0(E) \to \Omega^2(E)$$

Over a local trivialization, $d_\nabla^2 s = d_\nabla(ds + \omega \wedge s) = (d\omega + \omega \wedge \omega) \wedge s$, $d\omega + \omega \wedge \omega$ is the *curvature form.* A connection is *flat(completely integrable)* if its curvature is zero

Definition 15.56 Two vector bundles $E \to X$, $F \to X$ are stably isomorphic if $E \oplus \varepsilon^n \cong F \oplus \varepsilon^n$, denoted as $E \approx F$, we also denote $E \sim F$ if $E \oplus \varepsilon^n \cong F \oplus \varepsilon^m$ for some n, m

Remark. Here stably isomorphic does not imply isomorphic, for example, $TS^2 \overset{s}{\approx} \varepsilon^2$, since $\varepsilon^3 \approx T^2 \oplus NS^2 \approx T^2 \oplus \varepsilon^1$ whereas TS^2 is not trivial by the hairy ball theorem, and $NS^2 \approx \varepsilon^1$ is trivial because it is very easy to find a nonvanishing global section

Definition 15.57 Define the reduced K group to be $\tilde{K}(X)$ which consists of $\sim$-equivalent classes, and define K group to be the formal difference of isomorphic classes $E - F$, and $E - F = E' - F'$ if $E \oplus F' \oplus G \cong E' \oplus F \oplus G$ for some vector bundle G

Remark. When X is compact Hausdorff, $E \oplus F' \oplus G \cong E' \oplus F \oplus G$ is equivalent to $E \oplus F' \oplus \varepsilon^m \cong E' \oplus F \oplus \varepsilon^m$, since we can find G' such that $G \oplus G' \cong \varepsilon^m$ due to Theorem 15.26

$K(*) = \{\varepsilon^m - \varepsilon^n\} \cong \mathbb{Z}$, $\tilde{K}(*) = 0$, and when X compact Hausdorff we have an exact sequence $0 \to K(*) \to K(X) \to \tilde{K}(X) \to 0$, where $K(*) \to K(X)$ is simply given by $\varepsilon^m - \varepsilon^n \mapsto \varepsilon^m - \varepsilon^n$, $K(X) \to \tilde{K}(X)$ is defined as follows, given $E - F \in K(X)$, $E - F = E \oplus F' - F \oplus F' = E' - \varepsilon^m$ is mapped to E', this exact sequence splits since we have map $K(X) \to K(*)$ given by restriction

Conjecture 15.58 Let M be the Möbius line bundle over S^1, since $M \oplus M \cong \varepsilon^2$, and $M \otimes M \cong \varepsilon^1$, thus real K-theory of S^1 is isomorphic to $\mathbb{Z}[M]/(M^2 - 1, 2M - 2)$

Example 15.59 Let $S^n \subset \mathbb{R}^{n+1}$ be the unit sphere, TS^n, NS^n be the tangent bundle and normal bundle, then $TS^n \oplus NS^n$ can be seen as the restriction of the trivial bundle $\mathbb{R}^{n+1} \times \mathbb{R}^{n+1}$ on S^n, thus $TS^n \oplus NS^n$ is trivial

Definition 15.60 Define external product $K(X) \otimes K(Y) \to K(X \times Y), a \otimes b \mapsto p_1^*(a)p_2^*(b) =: a \times b$, this is a ring homomorphism

Definition 15.61 Suppose G is a topological group, P_G is the contravariant functor from the category of CW complexes to the category of sets, mapping X to all the principal G bundles over X, a **classifying space** BG is a topological space such that $[-, BG] \to P_G(-)$ is a natural isomorphism

Lemma 15.62 BG is unique up to weak homotopy equivalence

Proof. Suppose $B'G$ is also a classifying space, then $[-, BG] \cong P_G(-) \cong [-, B'G]$ are natural isomorphic, by Theorem 13.21, we may assume BG, $B'G$ are both CW complexes, and by Lemma 2.21, $X \to Hom(-, X)$ is fully faithful functor, thus $BG, B'G$ are homotopic

□

Theorem 15.63 (Milnor's contruction for classifying space) Define E^nG to be $\overbrace{G * \cdots * G}^{n+1}$ are formal sums $t_0g_0 + t_1g_1 + \cdots + t_ng_n$, with $\sum t_i = 1$. $EG := \lim_{\to} E^nG$ are finite formal sums $\sum t_ig_i$ with $\sum t_i = 1$. $E^nG \to E^nG/G$, $EG \to EG/G =: BG$ are principal G bundles, any principal G bundle over X is a pullback bundle of $EG \overset{p}{\to} BG$

Proof. Define G right action on E^nG, EG

$$E^nG \times G \to E^nG, \left(\sum t_ig_i, g\right) \mapsto \sum t_ig_ig$$

$$EG \times G \to EG, \left(\sum t_i g_i, g\right) \mapsto \sum t_i g_i g$$

Let $U_i = \{p(\sum t_i g_i) \mid t_i \neq 0\}$, then we would have a equivariant homeomorphism $p^{-1}(U_i) \to U_i \times G, \sum t_i g_i \mapsto (p(\sum t_i g_i), g_i)$ with inverse $U_i \times G \to p^{-1}(U_i), (p(\sum t_i g_i), g) \mapsto \sum t_j g_j g_i^{-1} g$, this is well defined since $(p(\sum t_i g_i h), g) \mapsto \sum t_j g_j h h^{-1} g_i^{-1} g = \sum t_j g_j g_i^{-1} g$

□

Definition 15.64 A **topological category** $\mathcal{C}$ is a small category where $ob\mathcal{C}$, $mor\mathcal{C}$ are topological spaces and $i : ob\mathcal{C} \to mor\mathcal{C}, c \mapsto 1_c$, $s : mor\mathcal{C} \to ob\mathcal{C}, c \xrightarrow{f} d \mapsto c$, $t : mor\mathcal{C} \to ob\mathcal{C}, c \xrightarrow{f} d \mapsto d$, $\circ : mor\mathcal{C} \times mor\mathcal{C} \to mor\mathcal{C}$ are continuous. A **continuous functor** between topological categories is a functor that are continuous on both objects and morphisms

Definition 15.65 Define **nerve** $N\mathcal{C}$ on category $\mathcal{C}$ which is also a simplicial set, $N\mathcal{C}([n]) \coloneqq Hom([n], \mathcal{C})$, the set of all functors from $[n]$ to $\mathcal{C}$, viewing $[n] = 0 \to 1 \to \cdots \to n$ as a category

Definition 15.66 (Segal's contruction for classifying space) Define the classifying space of $\mathcal{C}$ to be $B\mathcal{C} \coloneqq |N\mathcal{C}|$ as in Definition 15.65

16 Differential geometry

16.1 Manifold

Definition 16.1 A *submanifold* N is a inclusion and an immersion $i : N \hookrightarrow M$

Definition 16.2 The kernel of $C_p^\infty(M) \to \mathbb{R}, f \mapsto f(p)$ is a maximal ideal m_p, define the *cotangent space* $T_p^*M := \frac{m_p}{m_p^2}$, for $f \in C^\infty(M)$, define $(df)_p = f - f(p) \mod m_p$, $(dx_1)_p, \cdots, (dx_n)_p$ form a basis of T_p^*M locally, $(df)_p = \frac{\partial f}{\partial x_1}(p)(dx_1)_p + \cdots + \frac{\partial f}{\partial x_n}(p)(dx_n)_p$

Definition 16.3 The *tangent space* T_pM at p are the derivations $Der(C_p^\infty(M))$

$\left(\frac{\partial}{\partial x_1}\right)_p, \cdots, \left(\frac{\partial}{\partial x_n}\right)_p$ form a basis of T_pM locally

Definition 16.4 The *pushforward(differential)* of smooth map $\phi : M \to N$ is $\phi_p : T_pM \to T_pN$, $\phi_p(X_p)(f) = X_p(f \circ \phi)$

Definition 16.5 The *pullback* of smooth map $\phi : M \to N$ is $\phi^* : C^\infty(N) \to C^\infty(N)$, $\phi^*(f) = f \circ \phi$

Definition 16.6 $(\varphi^*\alpha)_x(X) = \alpha_{\varphi(x)}(d\varphi_x(X)) = \alpha_{\varphi(x)}((\varphi_*)_x(X))$, or in short $\varphi^*\alpha(X) = \alpha(\varphi_* X)$, similarly, for k forms, $\varphi^*\alpha(X_1, \cdots, X_k) = \alpha(\varphi_* X_1, \cdots, \varphi_* X_k)$

In particular, $\varphi^*(dx) = d(x \circ \varphi)$, pullback is compatible with wedge product, $\varphi^*(\alpha \wedge \beta) = \varphi^*\alpha \wedge \varphi^*\beta$, and pullback is compatible with exterior derivative, $\varphi^*(d\alpha) = d(\varphi^*\alpha)$ The exterior multiplication by α is $\beta \mapsto \alpha \wedge \beta$ The interior multiplication by $v \in TM$ is $v \lrcorner : \omega(-) \mapsto \omega(v, -)$

Definition 16.7 $\pi : T^*M \to M$ is the cotangent bundle, the *tautological one form* or *canonical one form* is $\pi^*\omega$

Definition 16.8 Let M be a smooth manifold, $X, Y \in C^\infty(M, TM)$ are vector fields, define Lie bracket $[X, Y] \in C^\infty(M, TM)$, $[X, Y](f) := (XY - YX)(f) = X(Y(f)) - Y(X(f))$

Remark. Check from local coordinates, $X(Y(f))$ is not well defined

Definition 16.9 Let M, N be smooth manifolds, $f : M \to N$ is a smooth map, it is called an immersion if df is injective at any point, it is called submersion if df is surjective at any point

Theorem 16.10 Suppose M, N are smooth manifolds with dimension m, n, $f : M \to N$ is a smooth map with constant rank r, then for any $p \in M$, denote $f(p) = q$, there are local charts (p, U), (q, V) such that $\chi_V \circ f \circ \chi_U(x_1, \cdots, x_m) = (x_1, \cdots, x_r, 0, \cdots, 0)$. Moreover, suppose M is second countable, if f is injective, then f is a immersion, if f is surjective, then f is a submersion, if f is bijective, then f is a diffeomorphism

Proof. If f is surjective but not a submersion, then $r < n$, but then by Theorem [Unresolved source reference: Baire category theorem], f can't be surjective which is a contradiction

□

Theorem 16.11 Suppose M, N are smooth manifolds with dimension m, n, $f : M \to N$ is a smooth map with constant rank r, then a level set $S = f^{-1}(c)$ is an embedded submanifold in M of codimension r with $f|_S$ being a proper map

Proposition 16.12 Let G be a Lie group, M, N be smooth manifolds with a G action, and G acts transitively on M, for any equivariant map $f : M \to N$, f has constant rank

Proof. For any $x \in M$, denote $y = f(x)$, it suffices to show $\operatorname{rank}(df)_x = \operatorname{rank}(df)_{gx}$ since G acts transitively on M, note that $f(gx) = gf(x)$, thus $fL_g = L_g f$, $(df)_{gx}(dL_g)_x = d(L_g)_y(df)_x$, and group actions are isomorphisms, we have $\operatorname{rank}(df)_x = \operatorname{rank}(df)_{gx}$

□

Theorem 16.13 (Stokes' theorem) $\langle \partial\Omega, \omega \rangle = \langle \Omega, d\omega \rangle$, here $\langle \Omega, \omega \rangle = \int_\Omega \omega$

Theorem 16.14 (de Rham's theorem) M is a smooth manifold. $H^p_{\mathrm{dR}}(X;\mathbb{R}) \xrightarrow{\cong} H^p(X;\mathbb{R})$ is an isomorphism

Proof. Since $\mathbb{R}$ is a divisible abelian group, thus an injective $\mathbb{Z}$ module, hence $\mathrm{Ext}^1_{\mathbb{Z}}(A,\mathbb{R})$, thus universal coefficient theorem gives exact sequence

$$0 = \mathrm{Ext}^1_{\mathbb{Z}}(H_{p-1}(X;\mathbb{Z}),\mathbb{R}) \to H^p(M;\mathbb{R}) \to \mathrm{Hom}(H_p(X;\mathbb{Z}),\mathbb{R}) \to 0$$

The isomorphism is given by $H^p_{\mathrm{dR}}(X;\mathbb{R}) \to \mathrm{Hom}(H_p(X),\mathbb{R})$, $\omega \mapsto \int_{_} \omega$

□

Definition 16.15 $E \to X$ is a vector bundle, a connection on E is an $\mathbb{R}$ linear map

$$\nabla : \Gamma(E) \to \Gamma(T^*X \otimes E)$$

satisfying Leibniz rule $\nabla(f\sigma) = f \otimes \nabla\sigma + df \otimes \sigma$

Lemma 16.16 $\nabla_X(\sigma) = (\nabla\sigma)(X)$ defines a covariant derivative, conversely, every covariant derivative is defined this way

16.2 Differential geometry of surfaces

Definition 16.17 A **differentiable surface** is an embedding $S \hookrightarrow \mathbb{R}^3$

Lemma 16.18 $\gamma(t)$ is a geodesic iff $\ddot{\gamma}$ is parallel to the normal $\vec{n}$, meaning no acceleration in S

A geodesic γ on S has constant speed

The geodesic curvature of a curve γ is the curvature of the projection onto tangent plane, γ is a geodesic iff the geodesic curvature of γ is zero

Proof. $\frac{d}{dt}|\dot{\gamma}|^2 = 2\ddot{\gamma}\cdot\dot{\gamma} = 0$

□

16.3 Connection and curvature

Definition 16.19 The *volume form* is $\sqrt{|\det g|}dx_i \wedge dx_j$, which happen to be $\star 1$

Definition 16.20 *Hodge star* is defined to be $\eta \wedge \star\,\xi = \langle \eta, \xi\rangle\omega$, ω is the volume form. Consider $(\alpha,\beta) = \int_X \alpha \wedge \star\,\beta$, $d^* = (-1)^{kl+1} \star d \star$ is the *codifferential* that $(d\alpha,\beta) = (\alpha, d^*\beta)$, $\Delta = dd^* + d^*d$ is the Laplacian

Definition 16.21 An **affine connection** is

$$\begin{aligned}\nabla : \Gamma(TM) \otimes \Gamma(TM) &\to \Gamma(TM)\\ (X,Y) &\mapsto \nabla_X Y\end{aligned}$$

satisfying

- $\nabla_{fX}Y = f\nabla_X Y$, i.e. ∇ is $C^\infty(M,\mathbb{R})$ linear in the first variable
- $\nabla_X(fY) = XfY + f\nabla_X Y$, i.e. ∇ satisfies Leibniz rule in the second variable

From this we can define covariant derivative ∇, $\nabla_X f = X_f$, $(\nabla_X\alpha)(Y) = \nabla_X(\alpha(Y)) - \alpha(\nabla_X Y)$, here α is a covector, similarly for any tensor, Write contraction $(\nabla T)(\alpha_1,\cdots,\alpha_m,X_1,\cdots,X_n,X) = (\nabla_X T)(\alpha_1,\cdots,\alpha_m,X_1,\cdots,X_n)$, T is a tensor

Note. $\nabla_X(\alpha(Y)) = \nabla_X(\alpha)(Y) + \alpha(\nabla_X Y)$

Definition 16.22 ∇ is an affine connection, the **torsion tensor** is

$$T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$$

Definition 16.23 The Levi-Civita connection ∇ is the one satisfying

- $\nabla_Z g(X,Y) = g(\nabla_Z X, Y) + g(X, \nabla_Z Y)$, i.e. $\nabla g = 0$
- $\nabla_X Y - \nabla_Y X = [X,Y]$, i.e. ∇ is torsion free

Definition 16.24 ∇ is the Levi-Civita connection, the Riemannian curvature tensor is $R_{XY} = [\nabla_X, \nabla_Y] - \nabla_{[X,Y]}$

Remark. X, Y are commuting vector fields around x_0, then $\frac{d}{ds}\frac{d}{dt}\tau_{sX}^{-1}\tau_{tY}^{-1}\tau_{sX}\tau_{tY}Z = R_{XY}Z$, τ is the parallel transport

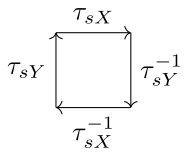


Figure 6:

Definition 16.25 ∇ is a affine connection, denote $\partial_i = \frac{\partial}{\partial x_i}$, $g_{ij} = \langle \partial_i, \partial_j \rangle$, $\nabla_i = \nabla_{\partial_i}$. The *Christoffel symbols* Γ_{ij}^k is defined such that $\nabla_i \partial_j = \sum_k \Gamma_{ij}^k \partial_k$, then $\nabla_i \partial_j - \nabla_j \partial_i = \sum_k (\Gamma_{ij}^k - \Gamma_{ji}^k)\partial_k$, if ∇ is torsion free, then $\nabla_i \partial_j - \nabla_j \partial_i = [\partial_i, \partial_j] = 0 \Rightarrow \Gamma_{ij}^k = \Gamma_{ji}^k$. Since

$$\frac{\partial g_{ij}}{\partial x_k} = \nabla_k g_{ij} = \sum_l \Gamma_{ki}^l g_{jl} + \sum_l \Gamma_{kj}^l g_{il}$$

Switch j, k, we have

$$\frac{\partial g_{ik}}{\partial x_j} = \sum_l \Gamma_{ji}^l g_{kl} + \sum_l \Gamma_{kj}^l g_{il}$$

Then switch i, j, we have

$$\frac{\partial g_{jk}}{\partial x_i} = \sum_l \Gamma_{ij}^l g_{kl} + \sum_l \Gamma_{ki}^l g_{jl}$$

Thus

$$\frac{\partial g_{jk}}{\partial x_i} + \frac{\partial g_{ik}}{\partial x_j} - \frac{\partial g_{ij}}{\partial x_k} = 2\sum_l \Gamma_{ij}^l g_{kl}$$

Therefore

$$\Gamma_{ij}^k = \frac{1}{2}\sum_l g^{kl}\left(\frac{\partial g_{jk}}{\partial x_i} + \frac{\partial g_{ik}}{\partial x_j} - \frac{\partial g_{ij}}{\partial x_k}\right)$$

Proposition 16.26

1. $R_{YX} = -R_{XY}$
2. $(R_{XY}Z, W) = -(R_{XY}W, Z)$
3. $R_{XY}Z + R_{YZ}X + R_{ZX}Y = 0$
4.

The **second Bianchi identity** follows

$$\nabla_X R_{YZ} + \nabla_Y R_{ZX} + \nabla_Z R_{XY} = 0$$

Remark. If write $(R_{XY}Z, W) = R(X, Y, Z, W)$, then R is antisymmetric about the first two variables and the last two variables, R satisfies Jacobi identity, the first two and the last two variables can switch place

Proof.

1.

2.

3.

4. Follow from above

□

Definition 16.27 There is a unique way to define the connection on all (r, s) tensors that are compatible with tensor products(Leibniz rule) and tensor contraction. In particular we would have

$$\nabla_X(\omega(Y)) = (\nabla_X\omega)(Y) + \omega(\nabla_X Y)$$

Hence we have

$$\nabla dx_i = -\sum_{j,k} \Gamma^i_{kj} dx_k \otimes dx_j$$

Definition 16.28 $\{e_i\}$ is an orthonomal basis, the **Ricci curvature** is $\mathrm{Ric}(X) = \sum R_{X,e_i}e_i$. The **scalar curvature** is $S = \mathrm{Tr}\ \mathrm{Ric} = \sum(\mathrm{Ric}(e_j), e_j) = \sum(R_{e_j,e_i}e_i, e_j)$. The **Einstein curvature** is $G = R - \frac{1}{2}gS$

Definition 16.29 The gradient grad f of a function f is defined so that $g(\text{grad } f, X) = Xf$. Assume grad $f = Y$, $\sum_i g_{ik}Y_i = \frac{\partial f}{\partial x_k}$, suppose (g^{ij}) is the inverse of (g_{ij}), then $Y_j = \sum_i \frac{\partial f}{\partial x_i} g^{ij}$, grad $f = \sum_j \left(\sum_i \frac{\partial f}{\partial x_i} g^{ij}\right) \frac{\partial}{\partial x_j}$

Definition 16.30 The Hessian Hess f of a function f is $\nabla^2 f$, or defined through $\text{Hess } f(X, Y) = g(\nabla_X \text{ grad } f, Y)$. In local coordinates, we have

$$\begin{aligned}
\nabla^2 f = \nabla(df) &= \sum_i \nabla\left(\frac{\partial f}{\partial x_i} dx_i\right) \\
&= \sum_i \nabla\left(\frac{\partial f}{\partial x_i}\right) \otimes dx_i + \sum_i \frac{\partial f}{\partial x_i} \nabla dx_i \\
&= \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j} dx_j \otimes dx_i - \sum_{i,j,k} \frac{\partial f}{\partial x_i} \Gamma^i_{kj} dx_k \otimes dx_j \\
&= \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j} dx_i \otimes dx_j - \sum_{i,j,k} \frac{\partial f}{\partial x_k} \Gamma^k_{ij} dx_i \otimes dx_j \\
&= \sum_{i,j} \left(\frac{\partial^2 f}{\partial x_i \partial x_j} - \sum_k \frac{\partial f}{\partial x_k} \Gamma^k_{ij}\right) dx_i \otimes dx_j
\end{aligned}$$

In other words, $\text{Hess } f(X, Y) = X(Yf) - (\nabla_X Y)f$

Definition 16.31 (Musical isomorphisms) $\sharp : T^*M \to TM, \flat : TM \to T^*M$. $(\omega^\sharp, X) = \omega(Y)$, $(X^\flat, \alpha) = \alpha(X)$. For example, $dx_i^\sharp = \sum_k g^{ik} \frac{\partial}{\partial x_k}$

Riemann metric induces an inner product on T^*M through musical isomorphisms. For example, $(dx_i, dx_j) = (dx_i^\sharp, dx_j^\sharp) = \sum_{kl} g^i k g^{jl} \left(\frac{\partial}{\partial x_k}, \frac{\partial}{\partial x_l}\right) = \sum_{kl} g^{ik} g^{lj} g^{kl} = g^{ij}$. This also gives an inner product on T^k_M

Definition 16.32 The Laplacian $\Delta f = \text{tr Hess } f$ is the trace of the Hessian. Here the trace is the trace contraction taken with respect to the metric. In local coordinates

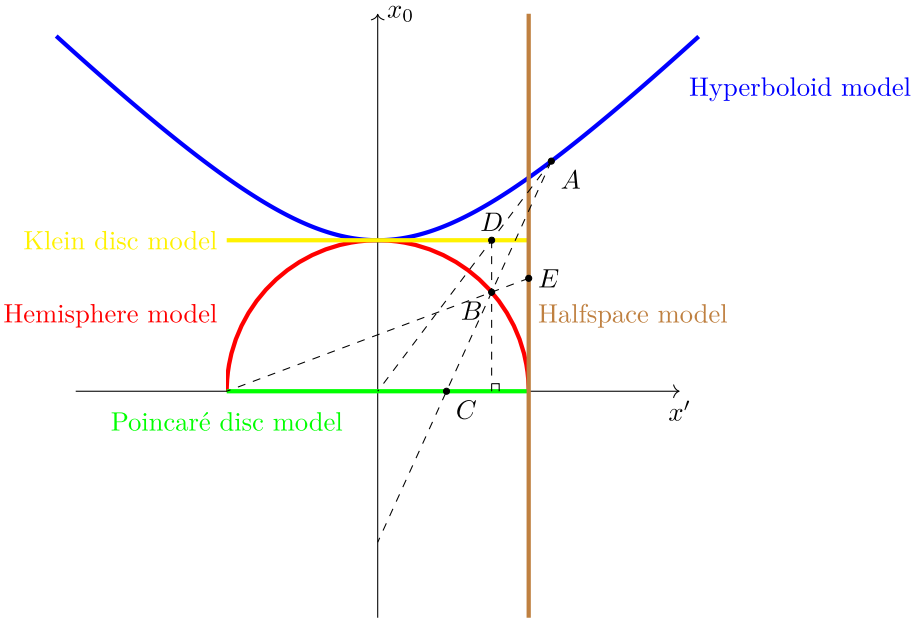
$$\Delta f = \sum_{i,j} g^{ij} \left(\frac{\partial^2 f}{\partial x_i \partial x_j} - \sum_k \frac{\partial f}{\partial x_k} \Gamma^k_{ij}\right)$$

16.4 Hyperbolic geometry

Definition 16.33 $\mathbb{R}^{n+1}$ with metric $ds^2 = dx_1^2 + \cdots + dx_n^2 - dx_0^2$ is the *Minkowski space*

The **hyperboloid model** is $\mathbb{H} = \{x_1^2 + \cdots + x_n^2 - x_0^2 = -1, x_0 > 0\}$. The Riemannian metric is the pullback metric $ds^2 = dx_1^2 + \cdots + dx_n^2 - dx_0^2$

The geodesics are intersections of $\mathbb{H}$ and two dimensional subspaces of $\mathbb{R}^{n+1}$ $(d\sinh s)^2 + (d\cosh s)^2 = \cosh^2 sds^2 - \sinh^2 sds^2 = ds^2$, thus $\mathbb{H}^1$ is isomorphic to $\mathbb{E}^1$



$(x', x_n) \mapsto \left(\frac{2x'}{1+x_n}, 1\right)$, $x' = (x_0, \cdots, x_{n-1})$ is the isometry from the hemisphere to the halfspace

$(x', 1) \mapsto \left(\frac{4x'}{4+|x'|^2}, \frac{4-|x'|^2}{4+|x'|^2}\right)$, $x' = (x_0, \cdots, x_{n-1})$ is the isometry from the halfspace to the hemisphere

$x \mapsto \left(\frac{x'}{1+x_0}\right)$, $x' = (x_1, \cdots, x_n)$ is the isometry from the hemisphere to Poincaré disc

$x \mapsto \left(\frac{2x}{1-|x|^2}, \frac{1+|x|^2}{1-|x|^2}\right)$ is the isometry from Poincaré disc to the hyperboloid

$x \mapsto (1, x')$, $x' = (x_1, \cdots, x_n)$ is the isometry from the hemisphere to Klein disc

$x \mapsto \left(\frac{x'}{x_0}, \frac{1}{x_0}\right)$, $x' = (x_1, \cdots, x_n)$ is the isometry from the hyperboloid to the hemisphere

$x \mapsto \left(\frac{x'}{x_0}, \frac{1}{x_0}\right)$, $x' = (x_1, \cdots, x_n)$ is the isometry from the hemisphere to the hyperboloid

$(x_0, x') \mapsto \left(1, \frac{x'}{x_0}\right)$ is the isometry from the hyperboloid to the Klein model

$(x, 1) \mapsto \left(\frac{1}{\sqrt{1-|x|^2}}, \frac{x}{\sqrt{1-|x|^2}}\right)$ is the isometry from the Klein model to the hyperboloid

The **hemisphere model** is $\mathbb{H} = \{x_0 > 0\} \cap S^n$. The Riemannian metric is pullback metric

$$\begin{aligned}
\sum_{i=0}^{n-1}\left[d\left(\frac{x_i}{x_n}\right)\right]^2-\left[d\left(\frac{1}{x_n}\right)\right]^2 &= \sum_{i=0}^{n-1}\left(\frac{x_0dx_i-x_idx_0}{x_0^2}\right)^2-\left(-\frac{dx_0}{x_0^2}\right)^2 \\
&= \sum_{i=0}^{n-1}\frac{x_0^2dx_i^2-2x_ix_0dx_idx_0+x_i^2dx_0^2}{x_0^4}-\frac{dx_0^2}{x_0^4} \\
&= \frac{dx'^2}{x_0^2}-\frac{d(|x'|^2)d(x_0^2)}{2x_0^4}+\frac{|x'|^2\ dx_0^2-dx_0^2}{x_0^4} \\
&= \frac{dx'^2}{x_0^2}-\frac{d(1-x_0^2)d(x_0^2)}{2x_0^4}-\frac{dx_0^2}{x_0^2} \\
&= \frac{dx'^2}{x_0^2}+\frac{2dx_0^2}{x_0^2}-\frac{dx_0^2}{x_0^2} \\
&= \frac{dx'^2+dx_0^2}{x_0^2}
\end{aligned}$$

The **half space model** is $\mathbb{H}=\{x_0>0\}\cap\{x_n=1\}$. The Riemannian metric is pullback metric

$$\begin{aligned}
\frac{\sum_{i=0}^{n-1}d\left(\frac{4x_i}{4+|x'|^2}\right)^2+d\left(\frac{4-|x'|^2}{4+|x'|^2}\right)^2}{\left(\frac{4x_0}{4+|x'|^2}\right)^2} &\overset{X=4+|x'|^2}{=} \frac{\sum_{i=0}^{n-1}d\left(\frac{4x_i}{X}\right)^2+d\left(\frac{8}{X}-1\right)^2}{\left(\frac{4x_0}{X}\right)^2} \\
&= \frac{X^2}{x_0^2}\left(\sum_{i=0}^{n-1}d\left(\frac{x_i}{X}\right)^2+4d\left(\frac{1}{X}\right)^2\right) \\
&= \frac{X^2}{x_0^2}\left(\sum_{i=0}^{n-1}\left(\frac{Xdx_i-x_idX}{X^2}\right)^2+4\frac{dX^2}{X^4}\right) \\
&= \frac{1}{x_0^2}\left(\sum_{i=0}^{n-1}\frac{X^2dx_i^2+x_i^2dX^2-2Xx_idXdx_i}{X^2}+4\frac{dX^2}{X^2}\right) \\
&= \frac{1}{x_0^2}\left(dx'^2+\frac{|x'|^2\ dX^2}{X^2}+\frac{4dX^2}{X^2}-\frac{dXd(|x'|^2)}{X}\right) \\
&= \frac{1}{x_0^2}\left(dx'^2+\frac{XdX^2}{X^2}-\frac{dXd(X-4)}{X}\right) \\
&= \frac{dx'^2}{x_0^2}
\end{aligned}$$

The **Poincaré disc model** is $\mathbb{H}=D^n$. The Riemannian metric is pullback metric

$$\sum_{i=1}^{n} d\left(\frac{2x_i}{1-|x|^2}\right)^2 - d\left(\frac{1+|x|^2}{1-|x|^2}\right)^2 \overset{X=1-|x'|^2}{=} \sum_{i=1}^{n} d\left(\frac{2x_i}{X}\right)^2 - d\left(\frac{2}{X}-1\right)^2$$

$$= 4\sum_{i=1}^{n}\left(\frac{Xdx_i + x_i dX}{X^2}\right)^2 - 4\left(-\frac{dX}{X^2}\right)^2$$

$$= 4\sum_{i=1}^{n}\frac{X^2dx_i^2 + x_i^2dX^2 - Xx_i dXdx_i}{X^4} - 4\frac{dX^2}{X^4}$$

$$= 4\left(\frac{dx^2}{X^2} + \frac{|x|^2\, dX^2}{X^4} - \frac{dX^2}{X^4} - \frac{d(|x|^2)dX}{X^3}\right)$$

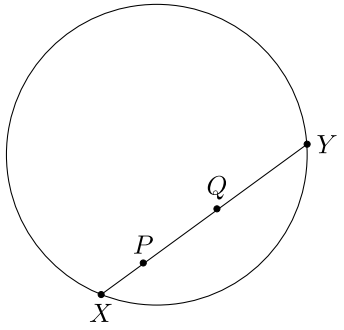
$$= 4\left(\frac{dx^2}{X^2} - \frac{XdX^2}{X^4} - \frac{d(1-X)dX}{X^3}\right)$$

$$= \frac{4dx^2}{X^2} = \frac{4dx^2}{(1-|x|^2)^2}$$

The **Klein disc model** is $\mathbb{H} = D^n$. The Riemannian metric is

$$\left|d\left(\frac{x}{\sqrt{1-|x|^2}}\right)\right|^2 - d\left(\frac{1}{\sqrt{1-|x|^2}}\right)^2 = \frac{|dx|^2}{1-|x|^2} + \frac{(x\cdot dx)^2}{(1-|x|^2)^2} = \sum \frac{(1-|x|^2+x_i^2)}{(1-|x|^2)^2}dx_i^2$$

The distance between P, Q is $\frac{1}{2}\ln\left(\frac{|XQ||PY|}{|XY||PQ|}\right) = \frac{1}{2}\ln(X,P;Q,Y)$, $(X,P;Q,Y)$ is the cross ratio



Theorem 16.34 $\mathrm{Isom}(\mathbb{H}^2) = PSL(2,\mathbb{R})$

Proof. An isometry sends half circles and orthogonal lines to half circles or orthogonal lines, by Schwarz reflection principle 19.33, it is can be regard as an isometry on $\mathbb{C}P^1$ sending $\mathbb{R}P^1$ to $\mathbb{R}P^1$, then it necessarily has to be in $PSL(2,\mathbb{R})$

□

Theorem 16.35 $\mathrm{Isom}(\mathbb{H}^3) = PSL(2,\mathbb{C}) \ltimes \mathbb{Z}/2\mathbb{Z} \cong SL(2,\mathbb{C})$

Proof. Since $\partial\mathbb{H}^3$ is the Riemann sphere, every isometry on $\mathbb{H}^3$ restricts to a conformal map on $\partial\mathbb{H}^3$ because it sends hemispheres and orthogonal planes to hemispheres or orthogonal planes, hence it is a Möbius transformation. On the other hand, Möbius transformations which can all be extended to an isometry on $\mathbb{H}^3$, translations $z \mapsto z+\lambda$ can be extended to $(z,x_3) \mapsto (z+\lambda, x_3)$, dilations $z \mapsto \lambda z$ can be extended to $(z,x_3) \mapsto (\lambda z, |\lambda| x_3)$, inversions $z \mapsto -\frac{1}{z}$ can be extended to $(z,x_3) \mapsto \left(\frac{-\bar{z}}{|z|^2+x_3^2}, \frac{x_3}{|z|^2+x_3^2}\right)$. Therefore the isometry group for $\mathbb{H}^3$ is $PSL(2,\mathbb{C}) \ltimes \mathbb{Z}/2\mathbb{Z} \cong SL(2,\mathbb{C})$

□

16.5 Complex manifold

Suppose $\langle , \rangle$ is a Hermitian inner product on $\mathbb{C}^n$, then Re $\langle , \rangle$ is a real inner product on $\mathbb{R}^{2n}$, i.e. Re $\langle X_1 + iX_2, Y_1 + iY_2\rangle = \text{Re}\,\langle X_1, Y_1\rangle + \text{Re}\,\langle X_2, Y_2\rangle$. On the other hand, a inner product on $\mathbb{R}^n$ can be extended to a Hermitian inner product on the complexification $\mathbb{C}^n$ by complex linearality

Definition 16.36 M is an even dimensional real manifold, an *almost complex structure* is $J : T_{\mathbb{R}}M \to T_{\mathbb{R}}M$ such that $J^2 = -1_{T_{\mathbb{R}}M}$. A complex manifold has an almost complex structure

$$J\frac{\partial}{\partial x_i} = \frac{\partial}{\partial y_i}, J\frac{\partial}{\partial y_i} = -\frac{\partial}{\partial x_i}$$

Example 16.37 S^4 cannot be given an almost complex structure. S^6 can be given an almost complex structure but not a complex structure

Definition 16.38 A is a $(1,1)$ form, the Nijenhuis tensor is

$$N_A(X,Y) = -A^2[X,Y] + A([AX,Y] + [X,AY]) - [AX,AY]$$

Theorem 16.39 (Newslander-Nirenberg theorem) J is *integrable* iff $N_J = 0$. Meaning there is a unique complex structure which will give J

Proposition 16.40 Given an almost complex structure, we can find coordinate charts $(z_1, \cdots, z_n, \bar{z}_1, \cdots, \bar{z}_n)$ such that Span $\left\{\frac{\partial}{\partial z_i}\right\}$, Span $\left\{\frac{\partial}{\partial \bar{z}_i}\right\}$ to be the i and $-i$ eigenspaces of J

Definition 16.41 A *Hermitian manifold* M is a complex manifold with a *Hermitian metric* $h = \sum h_{\alpha\bar{\beta}} dz_\alpha \otimes d\bar{z}_\beta$ on $TM \otimes \mathbb{C}$, where $h_{\alpha\bar{\beta}}$ is a positive definite Hermitian matrix. The real part gives a Riemannian metric

$$\begin{aligned} g &= \frac{1}{2}(h + \bar{h}) \\ &= \frac{1}{2}\Big(\sum h_{\alpha\bar{\beta}} dz_\alpha \otimes d\bar{z}_\beta + \sum h_{\beta\bar{\alpha}} d\bar{z}_\alpha \otimes dz_\beta\Big) \\ &= \frac{1}{2}\sum h_{\alpha\bar{\beta}}(dz_\alpha \otimes d\bar{z}_\beta + d\bar{z}_\beta \otimes dz_\alpha) \\ &= \sum h_{\alpha\bar{\beta}} dz_\alpha d\bar{z}_\beta \end{aligned}$$

Also gives *associate* $(1,1)$ *form*

$$\omega = -\frac{h - \bar{h}}{2i} = \frac{i}{2}(h - \bar{h}) = \frac{i}{2}\sum h_{\alpha\bar{\beta}}(dz_\alpha \otimes d\bar{z}_\beta - d\bar{z}_\beta \otimes dz_\alpha) = \frac{i}{2}\sum h_{\alpha\bar{\beta}} dz_\alpha \wedge d\bar{z}_\beta$$

Note that the volume form is $\text{vol}_M = \frac{\omega^{\wedge n}}{n!}$

Remark. $\omega(u,v) = g(Ju,v)$, $h = g - i\omega$, $g(u,v) = \omega(u,Jv)$. Any one determines the other two. ω corresponds to *Kähler class* in $H^2(M,\mathbb{R})$

Definition 16.42 A Hermitian manifold M is a *Kähler manifold* if $d\omega = 0$, ω is called the *Kähler form.* A symplectic manifold (M,ω) with an almost complex structrue J is a Kahler manifold if $\omega(Ju,Jv) = \omega(u,v)$ and $g(u,v) = \omega(u,Jv)$ is a Riemannian metric. An even dimensional Riemannian manifold (M,g) with an almost complex structure J is a Kahler manifold if $g(Ju,Jv) = g(u,v)$ and J is preserved by parallel transport, i.e. $\nabla J = 0$

Since ω is a real form, thus by Poincare's lemma, $\omega = d\alpha$ for some $\alpha = \beta + \bar{\beta}$, where β is a $(1,0)$ form, then we have $\partial\beta = \bar{\partial}\bar{\beta} = 0$, hence $\omega = \partial\bar{\beta} + \bar{\partial}\beta$, $\beta = \partial\phi$, $\omega = \partial\bar{\partial}(\bar{\phi} - \phi)$. We have $\partial_\gamma h_{\alpha\bar{\beta}} = \partial_\alpha h_{\gamma\bar{\beta}}$, $\partial_{\bar{\gamma}} h_{\alpha\bar{\beta}} = \partial_{\bar{\beta}} h_{\alpha\bar{\gamma}}$, this implies at least locally $h_{\alpha\bar{\beta}} = \partial_\alpha f_{\bar{\beta}}$, and then $\partial_\alpha \partial_{\bar{\gamma}} f_{\bar{\beta}} = \partial_{\bar{\gamma}}\partial_\alpha f_{\bar{\beta}} = \partial_{\bar{\beta}}\partial_\alpha f_{\bar{\gamma}} = \partial_\alpha \partial_{\bar{\beta}} f_{\bar{\gamma}}$, hence $h_{\alpha\bar{\beta}} = \partial_\alpha \partial_{\bar{\beta}}\rho$, ρ is called the local *Kähler potential*, ρ is a Kähler potential if $\omega = \frac{i}{2}\partial\bar{\partial}\rho$

Definition 16.43 Consider $L = \omega \wedge - : H^k(M) \to H^{k+2}(M)$, the *primitive cohomology* is

$$P^{n-k}(M) = \ker\Big(H^{n-k}(M) \overset{L^{k+1}}{\to} H^{n+k+2}(M)\Big)$$

The *hard Lefschetz theorem* says

$$H^n(M) = \bigoplus L^k P^{n-2k}(M)$$

Theorem 16.44 (Serre duality) X is complex manifold of complex dimension n, $E \to X$ is a holomorphic vector bundle, then we have

$$H^i(X, E) \cong H^{n-i}(X, K \otimes E^*)^*$$

Where $K \coloneqq \bigwedge^n T^*X$ is the canonical bundle

For example, if X is a Riemann surface, $E = \mathcal{O}$, then $H^1(X, \mathcal{O}) \cong H^0(X, K \otimes \mathcal{O})^* \cong H^0(X, \Omega)^* = \Omega(X)^*$

16.6 Symplectic manifold

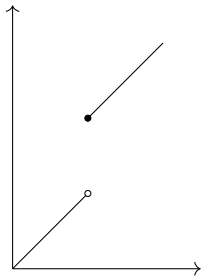
Definition 16.45 M is a smooth manifold, a *symplectic structure* on M is a 2 form ω that is nondegenerate and anti-symmetric on T_pM

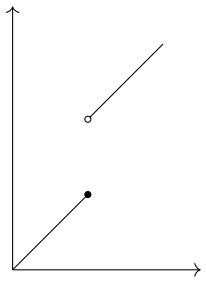
17 Real analysis

Definition 17.1 (Hyperbolic functions) $\sin z = \frac{e^{iz}-e^{-iz}}{2i}$, $\cos z = \frac{e^{iz}+e^{-iz}}{2}$ $\sinh z = -i\sin(iz) = \frac{e^{z}-e^{-z}}{2}$, $\cosh z = \cos(iz) = \frac{e^{z}+e^{-z}}{2}$

Definition 17.2 X is a convex set, $X \xrightarrow{f} \mathbb{R}$ is **convex** if $f(tx+(1-t)y) \le tf(x)+(1-t)f(y)$ for $0 \le t \le 1$ and $x, y \in X$, f is **strictly convex** if $f(tx+(1-t)y) < tf(x)+(1-t)f(y)$ for $0 < t < 1$ and $x \ne y \in X$. f is **concave** if $-f$ is convex

Definition 17.3 $X \xrightarrow{f} [-\infty, \infty]$ is **upper semicontinuous** at x if for any $y > f(x)$, there exists a neighborhood U of x such that $f(U) < y$, i.e. f can only jump down at x. Thus $X \xrightarrow{f} [-\infty, \infty]$ is upper semicontinuous if $\{f < a\}$ are open. $X \xrightarrow{f} [-\infty, \infty]$ is **lower semicontinuous** if $-f$ is upper semicontinuous, i.e. f can only jump up





Lemma 17.4 $\{f_\alpha\}_{\alpha \in A}$ is a family of upper semicontinuous functions, $f = \inf_{\alpha \in A} f_\alpha$ is also upper semicontinuous

Proof.

$$\{f < a\} = \bigcup_{\alpha \in A} \{f_\alpha < a\}$$

□

Lemma 17.5 f is upper semicontinuous, K is compact, then f attains maximum over K

Definition 17.6 $\Omega \xrightarrow{u} [-\infty, \infty)$ is **harmonic** at $x \in \Omega$ if u is continuous at x and for any ball $B(x,r)$, $u(x) = \frac{1}{|B(x,r)|}\int_{B(x,r)} u(y)dy$. $\Omega \xrightarrow{u} [-\infty, \infty)$ is **subharmonic** at $x \in \Omega$ if u is upper semicontinuous at x and for any ball $B(x,r)$, any continuous v harmonic on $B(x,r)$, $u \le v$ on $\partial B(x,r) \Rightarrow u \le v$ on $\overline{B(x,r)}$. $\Omega \xrightarrow{u} [-\infty, \infty)$ is **superharmonic** if $-u$ is subharmonic

Harmonic $\Leftrightarrow$ subharmonic and superharmonic

Lemma 17.7 $\Omega \xrightarrow{u} \mathbb{R}$ is subharmonic, $\mathbb{R} \xrightarrow{f} \mathbb{R}$ is convex, then $f \circ u$ is also subharmonic

Example 17.8 If f is holomorphic, then $\log|f|$ is subharmonic

Lemma 17.9 (Vitali covering lemma) $\{B_i\}_{i \in I}$ is a collection of balls in $\mathbb{R}^d$ such that $\sup_i B_i < \infty$, then there exists a disjoint subcollection $\{B_j\}_{j \in J}$, $J \subseteq I$ such that $\bigcup_{i \in I} B_i \subseteq \bigcup_{j \in J} 5B_j$

Proof.

□

Theorem 17.10 (Vitali covering theorem)

Theorem 17.11 (Interpolation) If $f_n \to f$ in $L^\lambda \cap L^\mu$, $\lambda < \mu$, then $f_n \to f$ in L^p for $\lambda < p < \mu$

Proof. Holder's inequality

□

18 Measure theory

Definition 18.1 μ is a finite measure on X if $\mu(X) < \infty$

Definition 18.2 μ is inner regular if $\mu(A) = \sup_{K \subseteq A \text{ compact}} \mu(K)$. μ is outer regular if $\mu(A) = \inf_{U \supseteq A \text{ open}} \mu(U)$

Theorem 18.3 (Riesz–Markov–Kakutani representation theorem) X is a locally compact Hausdorff space, for each linear functional $\phi : C_c(X) \to \mathbb{R}$, there exists a unique regular Borel measure μ such that

$$\int_X f d\mu = \phi(f)$$

Note. The converse that a regular Borel measure defines a linear functional is obvious. ϕ is nonnegative iff μ is nonnegative. The norm of ϕ is the total variation of μ

Definition 18.4 Suppose X_t, Y_t are random processes with $E(|Y_t|) < \infty$. Y_t is called a *martingale* with respect to X_t if $E(Y_t | \{X_\tau, \tau \leq s\}) = Y_\tau$

19 Complex analysis

Definition 19.1 A **polydisc** $D(z,r) \subseteq \mathbb{C}^n$ is $D(z_1,r_1) \times \cdots \times D(z_n,r_n)$

Definition 19.2 (Wirtinger derivatives)

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right), \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$$

Note.

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial x}\frac{\partial x}{\partial z} + \frac{\partial}{\partial y}\frac{\partial y}{\partial z}, \frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial x}\frac{\partial x}{\partial \bar{z}} + \frac{\partial}{\partial y}\frac{\partial y}{\partial \bar{z}}$$

$$dz \wedge d\bar{z} = -2i dx \wedge dy$$

Definition 19.3 $f : \Omega \to \mathbb{C}$ is **holomorphic** at $z_0 \in \Omega$ if $f'(z)$ exists around z_0. f is **univalent** if f is injective

Theorem 19.4 (Cauchy-Riemann equations) If we write $z = x + iy$, $f(z) = u(x,y) + iv(x,y)$, then the existence of $f'(z)$ implies that $\frac{\partial f}{\partial x} = \frac{1}{i}\frac{\partial f}{\partial y}$ which give the **Cauchy-Riemann equations**

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \end{cases}$$

If f satisfies Cauchy-Riemann equations around z_0, then f is holomorphic at z_0

Lemma 19.5 A univalent map is a biholomorphism to its image

Theorem 19.6 (Goursat) If f is holomorphic on $\Omega \subseteq \mathbb{C}$, $\overline{T} \subseteq \Omega$ is a triangle, then $\oint_T f(z)dz = 0$

Theorem 19.7 (Cauchy's integral theorem) If f is holomorphic on $\Omega \subseteq \mathbb{C}$, $\gamma \subseteq \Omega$ is a piecewise C^1 curve, then $\oint_\gamma f(z)dz = 0$

Theorem 19.8 (Morera's theorem) $U \subseteq \mathbb{C}$ is open, if $\oint_T f(z)dz = 0$ for any triangle $T \subseteq U$, then f is holomorphic on D

Theorem 19.9 (Cauchy-Pompeiu formula) f is a complex valued C^1 function on a disc $D \subseteq \mathbb{C}$, then

$$f(\zeta) = \frac{1}{2\pi i}\int_{\partial D} \frac{f(z)dz}{z-\zeta} - \frac{1}{\pi}\iint_D \frac{\partial f(z)}{\partial \bar{z}}\frac{dx \wedge dy}{z-\zeta}$$

In particular, if f is holomorphic, then

$$f(\zeta) = \frac{1}{2\pi i}\int_{\partial D} \frac{f(z)}{z-\zeta}dz$$

Proof. Denote $D_\epsilon = D - B(0,\epsilon)$, consider

$$\eta = \frac{f(w)dw}{w-z}, d\eta = \frac{\partial f(w)}{\partial \bar{w}}\frac{d\bar{w} \wedge dw}{w-z}$$

By Stokes' theorem

$$\frac{1}{2\pi i}\int_{\partial D_\epsilon} \eta = \frac{1}{2\pi i}\int_{D_\epsilon} d\eta$$

As $\epsilon \searrow 0$

$$f(\zeta) = \frac{1}{2\pi i}\int_{\partial D} \frac{f(w)dw}{w-z} + \frac{1}{2\pi i}\iint_D \frac{\partial f(w)}{\partial \bar{w}}\frac{dw \wedge d\bar{w}}{w-z}$$

□

Lemma 19.10 (Osgood's lemma) f is continuous on an open subset $\Omega \subseteq \mathbb{C}^n$ and holomorphic on each variable, then f is holomorphic

Proof. For each $a \in \Omega$, pick $P = D(a, r) \subseteq \Omega$, since $\frac{\partial f}{\partial \bar{z}_j} \equiv 0$ on Ω, fix $z_2, \cdots, z_n$, then

$$f(w_1, z_2, \cdots, z_n) = \frac{1}{2\pi i} \int_{|z_1 - a_1| = r_1} \frac{f(z_1, \cdots, z_n)}{z_1 - w_1} dz_1$$

For $w_1 \in D(a_1, r_1)$, iterate and we get

$$f(w_1, \cdots, w_n) = \frac{1}{(2\pi i)^n} \int_{|z_1 - a_1| = r_1} \cdots \int_{|z_n - a_n| = r_n} \frac{f(z_1, \cdots, z_n)}{\prod (z_j - w_j)} dz_1 \cdots dz_n$$

For $w \in P$. Since f is continuous, it is bounded on $\overline{P}$, $\frac{1}{z_j - w_j} = \sum_{m=0}^{\infty} \frac{(w_j - a_j)^m}{(z_j - a_j)^{m+1}}$ converges uniformly on compact subsets of $D(a_j, r_j)$. Hence $f(w) = \sum c_\alpha (w - a)^\alpha$, where

$$c_\alpha = \frac{1}{(2\pi i)^n} \int_{|z_1 - a_1| = r_1} \cdots \int_{|z_n - a_n| = r_n} \frac{f(z)}{\prod (z_j - a_j)^{\alpha_j + 1}} dz_1 \cdots dz_n$$

□

Corollary 19.11 (Cauchy inequality)

Theorem 19.12 (Maximum principle)

Theorem 19.13 $\{f_n\}$ are holomorphic on $\Omega \subseteq \mathbb{C}^n$, f_n are uniformly convergent on each compact subset, then f_n converges to a holomorphic function f, and $D^\alpha f_n \to D^\alpha f$ on each compact subset

Theorem 19.14 (Montel's theorem) $\mathcal{F} = \{f_n\}$ are holomorphic on $\Omega \subseteq \mathbb{C}^n$ and locally uniformly bounded, i.e. for any $z_0 \in \Omega$, there exists a neighborhood U and M such that $\sup_{z \in K} |f_n| \leq M$, then $\mathcal{F}$ is normal

Lemma 19.15 (Schwarz lemma) f is holomorphic on the unit disc $D \subseteq \mathbb{C}$, $f(0) = 0$ and $|f| \leq 1$ on D, then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$, if $|f(z)| = |z|$ for some nonzero z or $|f'(0)| = 1$, then $f(z) = az$, $a = f'(0)$

Proof. Define $g(z) = \frac{f(z)}{z}$, since $f(0) = 0$, 0 is a removable singularity, since $|f(z)| \leq 1$, $|g(z)| \leq 1$ on ∂D, by maximum principle 19.12, $|g(z)| \leq 1$ on D, thus $|f(z)| \leq |z|$ on D and $|f'(0)| = |g(0)| \leq 1$, if $|f(z)| = |z|$ for some nonzero z or $|f'(0)| = 1$, then g attains maximum within D, then $g \equiv a$ for some $|a| = 1$, thus $f(z) = az$

□

Corollary 19.16 $D \xrightarrow{f} D$ is a biholomorphic, then $f = e^{i\phi} \frac{z - a}{1 - \bar{a}z}$ for some ϕ and $a \in D$

Proof. Denote $\psi_a(z) = \frac{z - a}{1 - \bar{a}z}$, ψ_{-a} is the inverse of ψ_a

Assume $f(a) = 0$, consider $g(z) = f \circ \psi_{-a}$, then $g(0) = 0$, by Schwarz lemma 19.15, $g = e^{i\phi}$, $f = g \circ \phi_a = e^{i\psi} \frac{z - a}{1 - \bar{a}z}$

□

Lemma 19.17 Suppose $0 \in U \subsetneq D$ is a simply connected open set, there exists $U \xrightarrow{f} D$ univalent such that $f(0) = 0$, $|f'(0)| > 1$. Note that this is impossible if $U = D$ due to Schwarz lemma 19.15

Proof. Denote $\psi_a(z) = \frac{z - a}{1 - \bar{a}z}$, $\psi_a'(z) = \frac{1 - |a|^2}{(1 - \bar{a}z)^2}$. Consider $f = \psi_{g(a)} \circ g \circ \psi_{-a}$ with some $\psi_{-a}(U) \xrightarrow{g} D$ univalent, then $f(0) = 0$

$$f'(0) = \frac{1-|g(a)|^2}{(1-|g(a)|^2)^2} g'(a)(1-|a|^2) = \frac{1-|a|^2}{1-|g(a)|^2} g'(a)$$

Since U is simply connected, so is $\psi_{-a}(U)$ given $-a \in D \setminus U$, we can take $g(z) = \sqrt{z}$ to be one branch, since $|a| < 1$, we get

$$|f'(0)| = \frac{1-|a|^2}{1-|a|} \frac{1}{2\sqrt{|a|}} = \frac{1+|a|}{2\sqrt{|a|}} > 1$$

□

Lemma 19.18 φ is holomorphic on D, f is meromorphic on D and $f \neq 0$ on ∂D, $a_1, \cdots, a_m$ and $b_1, \cdots, b_n$ are the zeros and poles of order $k_1, \cdots, k_m$ and $l_1, \cdots, l_n$ of f in D, then

$$\frac{1}{2\pi i} \int_{\partial D} \varphi(z) \frac{f'(z)}{f(z)} dz = \sum_{i=1}^{m} k_i \varphi(a_i) - \sum_{i=1}^{n} l_i \varphi(b_i)$$

Proof. $f(z) = g(z) \prod_{i=1}^{m} (z - z_i)^{q_i}$ with $g \neq 0$ on $\overline{D}$, z_i, q_i could be a_i, k_i or $b_i, -l_i$ depending on whether it is a zero or a pole, hence

$$\begin{aligned}
\frac{1}{2\pi i} \int_{\partial D} \varphi(z) \frac{f'(z)}{f(z)} dz &= \frac{1}{2\pi i} \int_{\partial D} \varphi(z) \frac{g'(z) \prod_{i=1}^{m} (z-z_i) + g(z) \sum_{i=1}^{m} \prod_{j \neq i} (z - z_j)}{g(z) \prod_{i=1}^{m} (z - z_i)} dz \\
&= \frac{1}{2\pi i} \int_{\partial D} \left[\frac{\varphi(z) g'(z)}{g(z)} + \sum_{i=1}^{m} \frac{\varphi(z)}{z - z_i} \right] dz \\
&= \sum_{i=1}^{m} k_i \varphi(a_i) - \sum_{i=1}^{n} l_i \varphi(b_i)
\end{aligned}$$

□

Theorem 19.19 (Rouché's theorem)

Theorem 19.20 (Hurwitz's theorem) $U \subseteq \mathbb{C}$ is open connected, holomorphic functions $\{f_n\}$ converges uniformly to f on compact subsets of U and $f \not\equiv 0$, f has order m at z_0, for r small enough, there exists K such that for any $k \geq K$, f_k has precisely m zeros in $B(z_0, r)$, counting multiplicities, and these zeros converge to z_0 as $k \to \infty$

Remark. $B(z_0, r)$ can't be arbitrarily large. For example, $f_n(z) = z - 1 + \frac{1}{n}$ converges uniformly to $f(z) = z - 1$ on compact subsets, f has no zeros in the unit disc D, but f_n all have zeros in D

Proof. For r small enough, f doesn't vanish on $\partial B(z_0, r)$ on which $|f|$ attains minimum, then apply Rouché's theorem 19.19

□

Corollary 19.21 U is open connected, univalent maps $\{f_n\}$ converges to f on compact subsets, then f is either univalent or constant

Proof. If f is not a constant and $f(z_0) = f(w_0) = \zeta$, then $f(z) - \xi$ has z_0, w_0 as zeros, by Hurwitz's theorem 19.20, there exist $\{z_k\}, \{w_k\}$ converging to z_0, w_0 such that $f_{n_k}(z_k) = f_{n_k}(w_k) = \xi$, but f_n's are univalent, hence $z_k = w_k \Rightarrow z_0 = w_0$, i.e. f is univalent

□

Theorem 19.22 (Riemann mapping theorem) $U \subsetneq \mathbb{C}$ is a nonempty simply connected open subset, $z_0 \in U$,then there is a unique biholomorphism f from U to the unit disc such that $f(z_0) = 0$, $f'(z_0) > 0$

Proof of uniqueness. Suppose $U \overset{f_1, f_2}{\to} D$ are biholomorphisms such that $f_i(z_0) = 0$, $f_i'(z_0) > 0$, consider $g = f_2 f_1^{-1}$, $g(0) = 0$, $|g| \leq 1$ on D and $g'(0) = \frac{f_2'(z_0)}{f_1'(z_0)} > 0$, by Schwarz lemma 19.15, $g(z) = z$, i.e. $f_1 = f_2$

□

Proof of existence. Fix $a \notin U$, $z_0 \in U$. Define

$$\mathcal{F} = \{f \text{ univalent on } U \mid \mid f \mid \leq 1, f(z_0) = 0\}$$

Since U is simply connected, we can pick one branch $h(z) = \sqrt{z-a}$, then $h(U) \cap -h(U) = \varnothing$, $\frac{h(z)-h(z_0)}{h(z)+h(z_0)}$ is univalent and bounded, scale to get some $f_0 \in \mathcal{F} \Rightarrow \mathcal{F}$ is nonempty

Let $A = \sup_{f\in\mathcal{F}} |f'(z_0)| > 0$, $f_n'(z_0) \to A$ for some $\{f_n\} \subseteq \mathcal{F}$, by Montel's theorem 19.14, f_{n_k} converges to g uniformly on compact subsets, then $|g| \leq 1$, $g(z_0) = 0$ and $0 < A = |g'(z_0)| < \infty$, according to Hurwitz's theorem 19.20, g is also univalent, i.e. $g \in \mathcal{F}$ attains maximal derivative at z_0

Suppose $0 \in g(U) \subsetneq D$, if not, by Lemma 19.17, there exists univalent map $g(U) \xrightarrow{f} D$ such that $f(0) = 0$, $|f'(0)| > 1$, then $f \circ g \in \mathcal{F}$, but $|(f \circ g)'(z_0)| = |f'(0)g'(z_0)| > |g'(z_0)|$ which is a contradiction

□

Remark. Suppose $f_1, f_2 \in \mathcal{F}$ and f_1 is biholomorphic, then $g = f_2 f_1^{-1}$ is a map $D \to D$, with $g(0) = 0$, according to Schwarz lemma 19.15, $\frac{|f_2'(z_0)|}{|f_1'(z_0)|} = |g'(0)| \leq 1$, and if $|f_2'(z)| = |f_1'(z)|$, $g = e^{i\phi}$, f_2 is also biholomorphic

Example 19.23 $U = \mathbb{C} - \{z \geq 0\}$, then $h(z) = \sqrt{z}$ maps U to the upper half plane

Theorem 19.24 (Runge's theorem) $K \subseteq \mathbb{C}$ is compact, then $\mathbb{C} \setminus K$ is the union of its connected components whereas the components are either bounded or not, denote

Theorem 19.25 (Hartogs's extension theorem) An isolated singularity is always a removable singularity when $n \geq 2$

Proof. It suffices to consider the case $P = \{|z_1| \leq 1, |z_2| \leq 1\}$ is a polydisc, f is holomorphic on ∂P, then f is holomorphic on P

□

Lemma 19.26 $\Omega \subseteq \mathbb{C}^n$ is connected, $\Omega \xrightarrow{f} \partial B^n$ is holomorphic, then $f \equiv \text{const}$

Proof. If h is holomorphic, then $\frac{\partial^2}{\partial z \partial \bar{z}} |h|^2 = |h'|^2$, hence

$$0 = \frac{\partial^2}{\partial z \partial \bar{z}} |f|^2 = \sum_{i=1}^{n} \frac{\partial^2}{\partial z \partial \bar{z}} |f_i|^2 = \sum_{i=1}^{n} |f_i'(z)|^2 \Rightarrow f_i'(z) = 0 \Rightarrow f \equiv \text{const}$$

□

Theorem 19.27 (Remmert-Stein) $U_1 \subseteq \mathbb{C}^{n_1}, U_2 \subseteq \mathbb{C}^{n_2}$ are nonempty connected open subsets, $B = \{|z| < 1\} \subseteq \mathbb{C}^n$, then there is no proper holomorphic map $U_1 \times U_2 \to B$

Proof. Suppose $f : U_1 \times U_2 \to B$ is a proper holomorphic map. For any $(x, y) \in U_1 \times \partial U_2$, there is a discrete sequence $\{y_\nu\} \subseteq U_2$ converging to y as in Exercise 35.33, apply Lemma 11.32 to $f(x, y) : \{x\} \times U_2 \to B$, $\{f(x, y_\nu)\}$ is discrete, thus there exists a subsequence $\{y_\mu\} \subseteq \{y_\nu\}$ such that $f(x, y_\mu)$ such that $f(x, y) = \lim f(x, y_\mu) \in \partial B$. Then $f(x, y) : U_1 \times \{y\} \to \partial B$ is a holomorphic, by Lemma 19.26, $f(x, y)$ is constant on $U_1 \times \{y\}$, hence $U_1 \times \{y\} \subseteq f^{-1}(f(x, y))$ which is noncompact since it has noncompact image under projection to U_1. This contradicts the fact that f is proper

□

Corollary 19.28 (Poincaré) The 2 polydisc $P = \{|z_1| < 1, |z_2| < 1\}$ and the 2 ball $B = \{|z_1|^2 + |z_2|^2 < 1\}$ are not biholomorphic

Theorem 19.29 (Weierstrass preparation theorem) f is analytic near 0, $f(0) = 0$, $f(z)$ written as power series around 0 has terms only involve z_1 which can always be achieved by a change of variables as

in Exercise 35.34, then $f = wh$, where $w(z) = z_1^k + g_{k-1}z^{k-1} + \cdots + g_0$ is a **Weierstrass polynomial**, i.e. $g_i(z)$ are analytic around 0 and $g_i(0) = 0$, $h(z)$ is analytic around 0 and $h(0) \neq 0$

Theorem 19.30 (Weierstrass division theorem) Suppose f, g are analytic near 0, g is a Weierstrass polynomial of degree k, then there exist unique h, r such that $f = gh + r$, where r is a polynomial of degree less than k

Definition 19.31 A conformal mapping is a map preserves angles and orientation

Note. Antiholomorphic map preserves angles but changes orientation

Definition 19.32 Möbius transformations are $f(z) = \frac{az+b}{cz+d}$, $\begin{vmatrix} a & b \\ c & d \end{vmatrix} \neq 0$, **Möbius group** acts regularly on $\mathbb{C}P^1$ and preserves cross ratio $(z_0, z_1; z_2, z_3) = \frac{(z_2-z_0)(z_3-z_1)}{(z_3-z_0)(z_2-z_1)}$

Lemma 19.33 (Schwarz reflection principle) If f is holomorphic on $\{\mathrm{Im} z > 0\}$ and continuous on $\{\mathrm{Im} z \geq 0\}$ with real values on $\mathrm{Im} z = 0$, then it can be extended to $\mathbb{C}$ with $f(\bar{z}) = \overline{f(z)}$ for z < 0

Definition 19.34 $\Lambda \subseteq \mathbb{C}$ is a lattice. *Weierstrass sigma function* associated to lattice Λ is

$$\sigma(z) = z \prod_{\omega \in \Lambda^*} \left(1 - \frac{z}{\omega}\right) e^{\frac{z}{\omega} + \frac{1}{2}(\frac{z}{\omega})^2}$$

Weierstrass zeta function is the logarithmic derivative of σ

$$\zeta(z) = \frac{\sigma'(z)}{\sigma(z)} = \frac{1}{z} + \sum_{\omega \in \Lambda^*} \frac{1}{z-w} + \frac{1}{w} + \frac{z}{w^2}$$

Weierstrass eta function is

$$\eta(w) = \zeta(z+w) - \zeta(z), w \in \Lambda$$

This is independent of choice of z

Weierstrass elliptic function is

$$\wp(z) = -\zeta'(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda^*} \left(\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} \right)$$

$$\wp'(z) = -\sum_{\omega \in \Lambda} \frac{2}{(z+\omega)^3}$$

20 Harmonic analysis

Definition 20.1 Group G is a *topological group* if the multiplication and inverse maps are continuous

Definition 20.2 k is a topological field, V is a *topological vector space* over a k if

- V is a topological abelian group
- Scalar multiplication $k \times V \to V$ is continuous

Definition 20.3 k is a field. A *semi-norm* $\| \cdot \|$ on a

- Abelian group A is $A \xrightarrow{\|\cdot\|} k_{\geq 0}$ such that $\| -a \| = \| a \|$, $\| a + b \| \leq \| a \| + \| b \|$
- Rng R is $R \xrightarrow{\|\cdot\|} k_{\geq 0}$ such that $\| rs \| \leq \| r \| \| s \|$
- Vector space V is is $V \xrightarrow{\|\cdot\|} k_{\geq 0}$ such that $\| kv \| = |k| \| v \|$

$\| \cdot \|$ is a *norm* if $\| x \| = 0 \Leftrightarrow x$ is trivial

Remark. A norm $\| \cdot \|$ induces a metric $d(x, y) = \| x - y \|$

Definition 20.4 A topological vector space V is

- *locally convex* if 0 has a neighborhood basis consists of convex subsets. Such neighborhood basis exists iff it is induced by a family of seminorms p_i as $\{p_i < c\}$
- *Banach* if it is normed and complete with respect to the metric induced by that norm
- *Fréchet* space is a locally convex and completely metrizable

Definition 20.5 Y is a topological vector space, T is a set, $\mathcal{G} \subseteq \mathcal{P}(T)$ is a directed set by inclusion, $\mathcal{N}$ is a local base around $0 \in Y$. The *topology of uniform convergence* on sets in $\mathcal{G}$ or $\mathcal{G}$ *topology* is the unique translation invariant topology given by basis

$$U(G, N) = \left\{ f \in Y^T \;\middle|\; G \in \mathcal{G}, N \in \mathcal{N}, f(G) \subseteq N \right\}$$

Example 20.6 $\mathcal{G}$ is the set of compact subspaces, Y is a metric space

Definition 20.7 Let X, Y be a topological spaces, a family of continuous functions $A \subseteq Y^X$ is *equicontinuous* at $x \in X$, if for any open neighborhood V of $y = f(x)$, there is an open neighborhood U of x such that $f(U) \subseteq V, \forall f \in A$

Definition 20.8 A topological space X is called separable if X has a countable dense subset

Theorem 20.9 (Arzela-Ascoli theorem) Let X be a topological space and Y be a complete metric space, $A \subseteq Y^X$ be a family of equicontinuous functions(meaning pointwise equicontinuous). If X is compact, and $A_x \coloneqq \{f(x) | f \in A\} \subseteq Y$ is relatively compact for any $x \in A$, then A is relatively compact in Y^X. If X is separable with S being a countable dense subset, and A_x is relatively compact for any $x \in S$, then any sequence $\{f_n\}$ has a subsequence $\left\{f_{n_k}\right\}$ converges uniformly on any compact subset of X

Definition 20.10 A topological space X is a *Baire space* if for any countable open dense subsets $\{U_i\}$, $\bigcap_{i=1}^{\infty} U_i$ is also dense

Theorem 20.11 (Baire's category theorem) Every complete metric space X is a Baire space

Proof. Let $\{U_i\}$ be a countable open dense subsets, suppose $\bigcap_{i=1}^{\infty} U_i$ is not dense, then the complement of its closure is open nonempty, suppose $B(x, r)$ is in the complement of the closure, since U_1 is dense, $U_1 \cap B(x, r) \neq \emptyset$, then there exists $\overline{B(x_1, r_1)} \subseteq U_1 \cap B(x, r)$, similarly, we can find $\overline{B(x_n, r_n)} \subseteq U_n \cap B(x_{n-1}, r_{n-1})$, and we can also assume $r_i \to 0$, thus $x_i \to y \in X$ since X is complete, but $y \in B(x, r) \bigcap \bigcap_{i=1}^{\infty} U_i = \emptyset$ which is a contradiction

□

Definition 20.12 (Test function spaces) Suppose $U \subseteq \mathbb{R}^n$ is an open subset, a *test function space* $\mathcal{D}$ could be

1. $\mathcal{D} = C_c^\infty(U)$, $\phi_i \to \phi$ in $\mathcal{D}$ if there exists compact $K \subseteq U$ such that $\bigcup_i \operatorname{supp} \phi_i \subseteq K$ and $\partial^\alpha \phi_i \to \partial^\alpha \phi$ uniformly. Or a neighborhood basis given by W such that $W \cap \mathcal{D}_K$ is open
2.

Theorem 20.13 $\mathcal{D} = C_c^\infty(U)$ is sequential compact

Theorem 20.14 $\mathcal{D} = C_c^\infty(U)$ is not metrizable

Proof. Let $\mathcal{D}_K = \{f \in \mathcal{D} | \operatorname{supp} f \subseteq K\}$ for $K \subseteq U$ compact, they are Frechet with seminorms $\sup_K |\partial^\alpha f|$, they are closed subspaces of $\mathcal{D}$. It is easy to see that the interior of $\mathcal{D}$ in $\mathcal{D}$ is empty, so it is nowhere dense, By Baire's category theorem 20.11 and $\mathcal{D}$ is sequential compact, $\mathcal{D} = \mathcal{D}_K$ is not metrizable

□

Definition 20.15 Suppose $\mathcal{D}$ is a test function space. Distributions are defined to be the continuous functionals

Definition 20.16 A *Banach algebra* is an associative algebra A which is a complete normed rng such that $\| rs \| \leq \| r \| \| s \|$. A is *unital* if A is a ring with identity element having norm 1

Definition 20.17 A **-algebra* is a Banach algebra over $\mathbb{C}$ such that there is an antilinear involution $* : A \to A$, such that $(xy)^* = y^* x^*$. A is a *C^*-algebra* if $\| x^* x \| = \| x^* \| \| x \|$

Example 20.18 X is locally compact, $C_0(X)$ are the continuous functions vanishes at infinity, then $C_0(X)$ is a Banach algebra with the supremum norm, $C_0(X)$ is unital if X is compact with 1 being the identity. $C_0(X)$ is a C^*-algebra with complex conjugation as the involution

Definition 20.19 A is a unital Banach algebra over $\mathbb{R}, \mathbb{C}$, $e^x = \sum_{k=0}^\infty \frac{x^k}{k!}$ defines the *exponential*

$$\| e^x \| = \| \sum_{k=0}^\infty \frac{x^k}{k!} \| \leq \sum_{k=0}^\infty \| \frac{x^k}{k!} \| \leq \sum_{k=0}^\infty \frac{\| x \|^k}{k!} = e^{\|x\|}$$

The *logarithm* $\log x = \sum_{k=1}^\infty \frac{(-1)^{x+1}(x-1)^k}{k}$ is defined on $\| x - 1 \| < 1$

Lemma 20.20 e^x and $\log x$ are inverses to each other locally

Proposition 20.21 Let A be a k-Banach algebra, V is a Banach A-module

1. $v'(t) = av(t)$ with $v(0) = x$ has the unique solution $v(t) = e^{ta} x$
2. A linear map $D : A \to A$ is a derivation iff e^{tD} are automorphisms for all $t \in k$
3. For any $a \in A$, $t \in k$, a-invariant subspaces are precisely e^{ta}-invariant subspaces
4. Consider the representation $A^\times \to \mathrm{GL}(V)$, $(af)(x) := f(a^{-1}x)$. This induces another representation $A \to \mathfrak{gl}(V)$ via $(af)(x) = \frac{d}{dt}|_{t=0}\ f(e^{-ta}x)$

   Suppose $V = k[x_1, \cdots, x_n]$, $A = \mathfrak{gl}_n(k)$, $A^\times = \mathrm{GL}_n(k)$

   consider $v(t) = e^{tA} \cdot f := f(e^{-tA}x)$, then $v'(t) = \frac{d}{dt}|_{t=0}\ f(e^{-tA}x) =: D_A f$, where D_A is a linear differential operator $V_m \to V_m$ by Lemma [Unresolved source reference: AV<=V <=> e^tAV<=V], then we should have $f(e^{-tA}x) = v(t) = e^{tD_A} f$, therefore we would get $D_A = -A^T$, and it will be easy to check that $D_{[A,B]} = [D_A, D_B]$
5. If $[X, [X, Y]] = 0$, then $e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]}$

Proof.

1.
2.

3. If $aV \subseteq V$, then $e^{ta}V = \sum_{k=0}^{\infty} t^k \frac{a^k}{k!} V \subseteq V$. If $e^{ta}V \subseteq V, \forall t$, since V is closed, $\frac{d}{dt}|_{t=0}\ e^{ta}V = aV \subseteq V$
4. If we denote $g = (g_{ij}) \in GL(n, \mathbb{C})$, $f(x) = \sum_{i_1,\cdots,i_n} C_{i_1,\cdots,i_n} x_1^{i_1} \cdots x_n^{i_n}$, then $f(g^{-1}x) = \sum_{i_1,\cdots,i_n} C_{i_1,\cdots,i_n} (g_{11}x_1 + \cdots + g_{1n}x_n)^{i_1} \cdots (g_{n1}x_1 + \cdots + g_{nn}x_n)^{i_n}$ is still a homogeneous polynomial in n variables of degree m

   Denote $A = (a_{ij})$,

$$\begin{aligned}
\frac{d}{dt}|_{t=0}\ f(e^{-tA}x) &= \nabla f(x) \cdot \frac{d}{dt}|_{t=0}\ e^{-tA}x \\
&= -\nabla f(x) \cdot Ax \\
&= -\sum_{i,j} a_{ij} x_j \frac{\partial f}{\partial x_i} \\
&= \left(-\sum_{i,j} a_{ij} x_j \frac{\partial}{\partial x_i} \right) f \\
&= (-\nabla^T Ax) f \\
&= D_A f
\end{aligned}$$

   In particular, $D_A x_i = -\sum_{j=1}^n a_{ij} x_j$, thus D_A has matrix $-A^T$ with respect to $x_1, \cdots, x_n$, basis of V_1
5. Let $A(t) = e^{tX}e^{tY}e^{-\frac{t^2}{2}[X,Y]}$, $B(t) = e^{t(X+Y)}$, then $A(0) = B(0)$, $B'(t) = B(t)(X+Y)$ and

$$A'(t) = e^{tX}Xe^{tY}e^{-\frac{t^2}{2}[X,Y]} + e^{tX}e^{tY}Ye^{-\frac{t^2}{2}[X,Y]} - e^{tX}e^{tY}t[X,Y]e^{-\frac{t^2}{2}[X,Y]}$$

   Since $[X,[X,Y]] = 0$, $[Y,[X,Y]] = -[Y,[Y,X]] = 0$

$$e^{-tY}Xe^{tY} = Ad_{e^{-tY}}(X) = e^{ad_{-tY}}(X) = X + t[X,Y]$$

$$A'(t) = e^{tX}e^{tY}(X+Y)e^{-\frac{t^2}{2}[X,Y]} = e^{tX}e^{tY}e^{-\frac{t^2}{2}[X,Y]}(X+Y) = A(t)(X+Y)$$

   Thus $A(t), B(t)$ satisfies the same ODE and initial condition, $A(t) = B(t) \Rightarrow e^X e^Y = A(1) = B(1) = e^{X+Y+\frac{1}{2}[X,Y]}$

□

Theorem 20.22 (Lie product formula) $e^{A+B} = \lim_{n\to\infty} \left(e^{\frac{A}{n}} e^{\frac{B}{n}}\right)^n$

Theorem 20.23 (Lie commutator formula) $e^{[A,B]} = \lim_{n\to\infty} \left[e^{\frac{A}{n}}, e^{\frac{B}{n}}\right]^{n^2}$, the left and right $[,]$ are Lie bracket and commutator

Theorem 20.24 (Backer-Campbell-Hausdorff formula) $e^X e^Y = e^Z$ around 0, where $Z = X + \int_0^1 \psi(e^{ad_X} e^{tad_Y}) dt(Y)$ and

$$\psi(x) = \frac{x\log x}{x-1} \overset{y=1-x}{=} = \sum_{n=1}^{\infty} \frac{y^{n-1}}{n} - \sum_{n=1}^{\infty} \frac{y^n}{n} = 1 + \sum_{n=1}^{\infty} \left(\frac{y^n}{n+1} - \frac{y^n}{n}\right) = 1 - \sum_{n=1}^{\infty} \frac{(1-x)^n}{n(n+1)}$$

The first few terms are

$$e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]+\frac{1}{12}[X,[X,Y]]+\frac{1}{12}[Y,[Y,X]]+\cdots}$$

Proof. Note that $\lim_{m\to\infty} \sum_{k=0}^{m-1} \frac{1}{m} e^{-\frac{kx}{m}} = \int_0^1 e^{-tx} dt = \frac{1-e^{-x}}{x}$, we have

$$\begin{aligned}
\frac{d}{dt}|_{t=0}\ e^{X+tY} &= \frac{d}{dt}|_{t=0} \left(e^{\frac{X}{m}} e^{\frac{tY}{m}}\right)^m = \lim_{m\to\infty} \frac{d}{dt}|_{t=0} \left(e^{\frac{X}{m}} e^{\frac{tY}{m}}\right)^m = \lim_{m\to\infty} \sum_{k=0}^{m-1} e^{\frac{kX}{m}} \frac{Y}{m} e^{\frac{(m-k)X}{m}} \\
&= \lim_{m\to\infty} \sum_{k=0}^{m-1} \frac{1}{m} e^{\frac{kX}{m}} Y e^{-\frac{kX}{m}} e^X = \left(\lim_{m\to\infty} \sum_{k=0}^{m-1} \frac{1}{m} e^{\frac{kad_X}{m}}\right)(Y) e^X = \frac{e^{ad_X}-1}{ad_X}(Y) e^X
\end{aligned}$$

Let $e^{Z(t)} = e^X e^{tY}$, $\frac{d}{dt} e^{Z(t)} = \frac{d}{dt}(e^X e^{tY}) = e^X e^{tY} Y = e^{Z(t)} Y$, but $\frac{d}{dt} e^{Z(t)} = \frac{d}{ds}|_{s=t}\ e^{Z(s)} = \frac{d}{ds}|_{s=t}\ e^{Z(t)+Z'(t)(s-t)} = \frac{e^{ad_{Z(t)}}-1}{ad_{Z(t)}}(Z'(t)) e^{Z(t)}$, hence $\frac{e^{ad_{Z(t)}}-1}{ad_{Z(t)}}(Z'(t)) = e^{Z(t)} Y e^{-Z(t)} = Ad_{e^{Z(t)}}(Y) = e^{ad_{Z(t)}}(Y)$, $Z'(t) = \frac{ad_{Z(t)} e^{ad_{Z(t)}}}{e^{ad_{Z(t)}}-1}(Y)$, since $e^{ad_{Z(t)}} = Ad_{e^{Z(t)}} = Ad_{e^X e^{tY}} = e^{ad_X} e^{tad_Y}$

$$Z = Z(1) = Z(0) + \int_0^1 \frac{ad_{Z(t)} e^{ad_{Z(t)}}}{1 - e^{-ad_{Z(t)}}}(Y)dt = X + \int_0^1 \frac{e^{ad_X} e^{tad_Y} \log(e^{ad_X} e^{tad_Y})}{e^{ad_X} e^{tad_Y} - 1}(Y)dt$$

□

Theorem 20.25 (Stone-Weierstrass) X is compact Hausdorff, $A \subseteq C(X, \mathbb{R})$ is a unital subalgebra. A is dense in $C(X, \mathbb{R})$ with the topology of uniform convergence iff A separates points

$S \subseteq C(X, \mathbb{C})$ is a unital *-algebra that separating points, then S is dense in $C(X, \mathbb{C})$

21 Partial differential equations

21.1 Laplace's equation

21.2 Heat equation

Definition 21.1 The fundamental solution to solution to **heat equation** $u_t - \Delta u = 0$ is

$$E(x,t) = \begin{cases} \frac{1}{(4\pi t)^{\frac{n}{2}}} e^{-\frac{|x|^2}{4t}} & t > 0 \\ 0 & t \leq 0 \end{cases}$$

Theorem 21.2 $U \subseteq \mathbb{R}^n$ is open and bounded, $f \in C_c^1(U \times (0,T])$, then

$$u(x,t) = \int_{\mathbb{R}^{n+1}} E(x-y, t-s) f(s,y) ds dy$$

Satisfies

$$\left(\frac{\partial}{\partial t} - \Delta\right) u(x,t) = f(x,t)$$

Where u is C^1 in t and C^2 in x

Proof. $E(x,t)$ is supported in $t \geq 0$ and $\int_{\mathbb{R}^n} |\nabla_x E(x,t)| dx \leq \frac{C}{\sqrt{t}}$ if $t > 0$, so $\nabla_x E(x,t)$ is integrable near $(0,0)$

$$\begin{aligned} \nabla_x \int_{\mathbb{R}^{n+1}} E(y,s) f(x-y,t-s) ds dy &= \int_{\mathbb{R}^{n+1}} E(y,s) \nabla_x f(x-y,t-s) ds dy \\ &= \lim_{\varepsilon \to 0} \int_\varepsilon^\infty \int_{\mathbb{R}^n} E(y,s) \nabla_x f(x-y,t-s) ds dy \\ &= \lim_{\varepsilon \to 0} \int_\varepsilon^\infty \int_{\mathbb{R}^n} (\nabla E)(y,s) f(x-y,t-s) ds dy \\ &= \lim_{\varepsilon \to 0} \int_{\mathbb{R}^{n+1}} (\nabla E)(y,s) f(x-y,t-s) ds dy \end{aligned}$$

And

$$\begin{aligned} \Delta \int_{\mathbb{R}^{n+1}} E(y,s) f(x-y,t-s) ds dy &= \int_{\mathbb{R}^{n+1}} (\nabla E)(y,s) \cdot (\nabla f)(x-y,t-s) ds dy \\ &= \lim_{\varepsilon \to 0} \int_\varepsilon^\infty (\nabla E)(y,s) \cdot (\nabla f)(x-y,t-s) ds dy \end{aligned}$$

And

$$\begin{aligned} \left(\tfrac{\partial}{\partial t} - \Delta\right) \int_{\mathbb{R}^{n+1}} E(y,s) f(x-y,t-s) ds dy &= -\lim_{\varepsilon \to 0} \int_\varepsilon^\infty \int_{\mathbb{R}^n} (\nabla E)(y,s) \cdot (\nabla f)(x-y,t-s) ds dy \\ &\quad + \lim_{\varepsilon \to 0} \int_\varepsilon^\infty \int_{\mathbb{R}^n} E(y,s) \tfrac{\partial f}{\partial x}(x-y,t-s) ds dy \\ &= \lim_{\varepsilon \to 0} \int_\varepsilon^\infty \int_{\mathbb{R}^n} \left(\tfrac{\partial}{\partial s} - \Delta_y\right) E(y,s) f(x-y,t-s) ds dy \\ &\quad + \lim_{\varepsilon \to 0} \int_{\mathbb{R}^n} E(y,\varepsilon) f(x-y,t-\varepsilon) ds dy \\ &= f(x,t) \end{aligned}$$

Next, let $u \in C^2(U \times (0,T])$ and $u_t - \Delta u = 0$, $\chi \in C^\infty$, $\chi(x,t) = 1$ if $d((x,t), \Gamma_U) \geq 2$, $\chi(x,t) = 0$ if $d((x,t), \Gamma_U) \leq \varepsilon$ and $(x,t) \in U \times (0,T]$, apply the previous argument to $f(x,t) = \left(\frac{\partial}{\partial t} - \Delta\right)(\chi(x,t)u(x,t)) = \left(\left(\frac{\partial}{\partial t} - \Delta\right)\chi(x,t)\right)u - 2\nabla\chi \cdot \nabla u \in C_c^1(U \times (0,T])$, we get

$$\left(\frac{\partial}{\partial t}-\Delta\right)\left(\chi(x,t)u(x,t)-\int_{-\infty}^{t}\int_{\mathbb{R}^n}E(x-y,t-s)f(y,s)\right)=0$$

And

$$u(x,t)\chi(x,t)-\int_{-\infty}^{t}E(x-y,t-s)f(y,s)dsdy=0$$

if $t=0$, so if $0\le t\le T$

$$\chi(x,t)u(x,t)=\int_{-\infty}^{t}\int_{\mathbb{R}^n}E(x-y,t-s)\left(\frac{\partial}{\partial t}-\Delta\right)(\chi(y,s)u(y,s))dsdy$$

□

21.3 Wave equation

Definition 21.3 The fundamental solution to **wave equation** $\Box u=\left(\frac{\partial^2}{\partial t^2}-\Delta\right)u=0$ is

$$E(x,t)=\begin{cases}\frac{1}{2\pi^{\frac{n-1}{2}}}\chi_+^{\frac{1-n}{2}}(t^2-|x|^2) & t>0\\ 0 & t<0\end{cases}$$

Theorem 21.4 $f\in C^2(\mathbb{R}^3)$, $u(x,t)=\frac{1}{4\pi t}\int_{\partial B(x,t)}f(y)dS_y=\frac{t}{4\pi}\int_{S^2}f(x+tw)dS_w$, then $u\in C^2\big(\mathbb{R}^3\times[0,\infty)\big)$, $u(x,0)=0$, $\frac{\partial}{\partial t}|_{t=0}\ u(x,t)=f(x)$ and $\Box u=0$ for $t>0$

Proof.

$$\begin{aligned}\frac{\partial}{\partial t}u(x,t)&=\frac{1}{4\pi}\int_{S^2}f(x+tw)dS_w+\frac{t}{4\pi}\int_{S^2}(w\cdot\nabla)f(x+tw)dS_w\\&=\frac{1}{4\pi}\int_{S^2}f(x+tw)dS_w+\frac{1}{4\pi t}\int_{\partial B(x,t)}n\cdot\nabla f(y)dS_y\\&=\frac{1}{4\pi}\int_{S^2}f(x+tw)dS_w+\frac{1}{4\pi t}\int_{B(x,t)}\Delta f(y)dy\end{aligned}$$

Thus

$$\begin{aligned}\frac{\partial^2}{\partial t^2}u(x,t)&=\frac{1}{4\pi}\int_{S^2}(w\cdot\nabla)f(x+tw)dS_w-\frac{1}{4\pi t^2}\int_{B(x,t)}\Delta f(y)dy\\&\quad+\frac{1}{4\pi t}\frac{d}{dt}\int_0^t\int_{S^2}\lambda^2\Delta f(x+\lambda w)dS_w d\lambda\\&=\frac{1}{4\pi t^2}\int_{B(x,t)}\Delta f(y)dy-\frac{1}{4\pi t^2}\int_{B(x,t)}\Delta f(y)dy\\&\quad+\frac{t}{4\pi}\int_{S^2}\Delta f(x+\lambda w)dS_w\\&=\frac{1}{4\pi t}\int_{\partial B(x,t)}\Delta f(y)dS_y\\&=\Delta u(x,t)\end{aligned}$$

□

Theorem 21.5 $f\in C^2(\mathbb{R}^2)$, then $u(x,t)=\frac{1}{2\pi}\int_{|y|<t}\frac{1}{\sqrt{t^2-|y|^2}}f(x-y)dy$ solves $\Box u=0$ for $t>0$, $u(x,0)=0$, $u_t(x,0)=f$

Proof. Consider $f:\mathbb{R}^3\to\mathbb{R}$, $f(x_1,x_2,x_3)=f(x_1,x_2)$ is independent of x_3, then $u(x,t)=\frac{1}{4\pi t}\int_{\partial B(x,t)}f(y)dy=\frac{1}{4\pi t}\int_{\partial B(0,t)}f(x-y)dS_y$

$$y_3=\pm\sqrt{t^2-y_1^2-y_2^2}=\gamma(y), ds=\sqrt{1+|\nabla\gamma(y)|^2}dy_1dy_2=\frac{t}{t^2-y_1^2-y_2^2}, \text{upper + lower hemisphere}$$

$=\frac{2}{4\pi t}\int_{|(y_1,y_2)|<t}f(x-y)\frac{tdy_1dy_2}{\sqrt{t^2-|(y_1,y_2)|^2}}=\frac{1}{2\pi}\int_{|y|<t}\frac{1}{\sqrt{t^2-|y|^2}}f(x-y)dy$

□

Theorem 21.6 $f\in C^\infty(\mathbb{R}^n\times[0,\infty))$, $u(x,t)=\int_0^t E(\cdot,t-s)*f(\cdot,s)ds$, then $\Box u=f$, $u(x,0)=u_t(x,0)=0$

Proof. Define $u(x,t,s)=E(\cdot,t-s)*f(\cdot,s)\in C^\infty$ for $t>s$

$$\frac{\partial}{\partial t}u(x,t)=u(x,t,t)+\int_0^t\frac{\partial}{\partial t}u(x,t,s)ds$$

$$=\int_0^t\frac{\partial}{\partial t}u(x,t,s)ds$$

$$\frac{\partial^2}{\partial t^2}u(x,t)=\int_0^t\frac{\partial^2}{\partial t^2}u(x,t,s)dx+\frac{\partial}{\partial t}|_{t=s}\ u(x,t,s)$$

$$=f(x,t)+\int_0^t\frac{\partial^2}{\partial t^2}u(x,t,s)dx$$

Thus $\left(\frac{\partial^2}{\partial t^2}-\Delta\right)u(x,t)=f(x,t)+\int_0^t\left(\frac{\partial^2}{\partial t^2}-\Delta\right)u(x,t,s)dx$, the second term is zero for $s<t$

By the same argument, $\Box\int_{-\infty}^t E(\cdot,t-s)*f(\cdot,s)ds=f(\cdot,t)$, thus $\Delta E=\delta_{(x,t)}$ is the fundamental solution

□

Lemma 21.7 The solution to $\Box u=0$ in $t>0,x>0$ with $u(0,t)$ for all $t>0$, $u(x,0)=0$, $u_t(x,0)=f(x)$, $f\in C^1([0,\infty))$, $f(0)=0$ is

$$u(x,t)=\frac{1}{2}\int_{|t-x|}^{t+x}f(\lambda)d\lambda$$

Proof. Define $\tilde{f}:\mathbb{R}\to\mathbb{R}$, $\tilde{f}(x)=\begin{cases}f(x) & x\geq 0\\ -f(-x) & x<0\end{cases}$ which solves $\Box\tilde{u}=0$ for $t>0,x\in\mathbb{R}$, $\tilde{u}(x,0)=0$, $\tilde{u}_t(x,0)=\tilde{f}$, hence

$$\tilde{u}(x,t)=\frac{1}{2}\int_{x-t}^{x+t}\tilde{f}(\lambda)d\lambda=\frac{1}{2}\int_{|x-t|}^{x+t}f(\lambda)d\lambda$$

□

Lemma 21.8 $f(x)=f(|x|)$ is spherical symmetric in $\mathbb{R}^n$, then $(\Delta f)(x)=\left(\frac{\partial^2}{\partial r^2}+\frac{n-1}{r}\frac{\partial}{\partial r}\right)f$

Proof. Δu is characterized by

$$\int_{\mathbb{R}^n}\nabla u\cdot\nabla v dx=-\int_{\mathbb{R}^n}v\Delta u, \forall v\in C_c^\infty(\mathbb{R}^n)$$

If $u(x)=u(|x|)$, $v(x)=v(|x|)$

$$\begin{aligned}\int_{\mathbb{R}^n}\nabla u\cdot\nabla v dx &= \int_{S^{n-1}}\int_0^\infty \frac{\partial u}{\partial r}\frac{\partial v}{\partial r}drdS_w\\ &= -\int_{S^{n-1}}\int_0^\infty \frac{1}{r^{n-1}}\frac{\partial}{\partial r}\left(r^{n-1}\frac{\partial u}{\partial r}\right)v(r)r^{n-1}drdS_w\\ &= -\int_{\mathbb{R}^n}\frac{1}{r^{n-}}\frac{\partial}{\partial r}\left(r^{n-1}\frac{\partial u}{\partial r}\right)v(r)dx\\ &= -\int_{\mathbb{R}^n}\left(\frac{n-1}{r}\frac{\partial}{\partial r}+\frac{\partial^2}{\partial r^2}\right)uv(r)dx\end{aligned}$$

Note that $\frac{1}{r^{n-1}}\frac{\partial}{\partial r}\big(r^{n-1}\frac{\partial u}{\partial r}\big)=\frac{n-1}{r}\frac{\partial u}{\partial r}+\frac{\partial^2 u}{\partial r^2}$

□

Theorem 21.9 The solution to $\Box u=0$ in $\mathbb{R}^{3+1}$ with $u(x,0)=0$, $u_t(x,0)=f(x)=f(|x|)$, $f\in C^\infty(\mathbb{R}^3)$ is

$$u(x,t)=\frac{1}{2|x|}\int_{t-|x|}^{t+|x|}\lambda f(\lambda)d\lambda$$

Proof. By Lemma 21.8, when $n=3$, $\big(\frac{\partial^2}{\partial r^2}+\frac{2}{r}\frac{\partial}{\partial r}\big)u=\frac{1}{\partial r}\frac{\partial^2}{\partial r^2}(ru)$, thus if $\Box u=0$ in $\mathbb{R}^{3+1}$, $u(x,t)=u(|x|,t)$, then $\big(\frac{\partial^2}{\partial t^2}-\frac{\partial^2}{\partial r^2}\big)(ru(r,t))=0$ and $ru(r,t)=0$ if $r=0$, $\frac{\partial}{\partial t}|_{t=0}(ru(r,t))=rf(r)$, by Lemma 21.7, $ru(r,t)=\frac{1}{2}\int_{|t-r|}^{t+r}\lambda f(\lambda)d\lambda$. We can check $u\in C^1$

□

Theorem 21.10 (Energy estimate version 1) $\Box u=0$ for $t>0$, then the energy $\frac{1}{2}\int_{\mathbb{R}^n}|u_t|^2+|\nabla u|^2\,dx$ is a constant

Theorem 21.11 (Energy estimate version 2) $\Box u=0$ in $U_T=U\times(0,T]$, $u=0$ on Γ_U, $u_t(x,0)=0$, implicitly, $u_t=0$ on $\partial U\times[0,T]$, then $\frac{1}{2}\int_{\mathbb{U}}|u_t|^2+|\nabla u|^2\,dx$ is a constant

Theorem 21.12 (Energy estimate version 3) $C=\big\{(x,t)\in\mathbb{R}^{n+1}\,\big|\,|\,x-x_0\,|\le|\,t-t_0\,|\big\}$ is the cone, $D_t=\{x\in\mathbb{R}^n\,|\,|\,x-x_0\,|\le|\,t-t_0\,|\}$ is the section at time t, considet the case $t<t_0$, then $\frac{1}{2}\int_{\mathbb{D}_t}|u_t|^2+|\nabla u|^2\,dx$ is decreasing on $0\le t\le t_0$

21.4 Euler-Lagrange equation

21.5 Energy momentum tensor

Definition 21.13 ∇ is the gradient, write $\nabla^T\nabla=\nabla\cdot\nabla=\Delta$ is the laplacian, $\nabla\cdot 1=\mathrm{div}$ is the divergence, $\nabla\nabla^T=D^2$ is the Hessian

Definition 21.14 $L(z,q):\mathbb{R}\times\mathbb{R}^n\to\mathbb{R}$ is C^∞, u satisfies Euler-Langrange equation, then

$$\begin{aligned}\nabla_x L(u,\nabla u)&=\frac{\partial L}{\partial z}\nabla u+(\nabla\nabla^T u)(\nabla_q L)\\ &=(\nabla_x\cdot\nabla_q L)(\nabla u)+(\nabla\nabla^T u)(\nabla_q L)\\ &=(\nabla^T u\nabla_q L)\nabla_x\end{aligned}$$

Energy-momentum tensor $T_{\alpha\beta}=\frac{\partial u}{\partial x^\alpha}\frac{\partial L}{\partial q_\beta}-\delta_{\alpha\beta}L$, $T=\nabla^T u\nabla_q L-L1$, then $T\nabla_x=(\nabla^T\nabla_q L)\nabla_x-\nabla_x L=0$

Example 21.15 $u_{tt}-\Delta u+u^3=0$, $L(u,\nabla_{x,t}u)=\frac{1}{2}(u_t^2-|\nabla_x u|^2)-\frac{1}{4}u^4$, $T_{00}=u_t^2-\big[\frac{1}{2}(u_t^2-|\,\nabla u\,|^2)-\frac{1}{4}u^4\big]=\frac{1}{2}(u_t^2+|\nabla u|^2)+\frac{1}{4}u^4$, $T_{0i}=-u_t\frac{\partial u}{\partial x^i}$, thus $0=(T_{00},\cdots,T_{0n})\nabla_x=\mathrm{div}(T_{00},\cdots,T_{0n})$

22 Riemann surface

Definition 22.1 A **Riemann surface** is a one dimensional complex manifold

Theorem 22.2 (Riemann's removable singularity theorem) f is holomorphic on $X \setminus \{a\}$ and bounded near a, then f is holomophic on X

Theorem 22.3 (Principle of analytic continuation) X is connected, $X \xrightarrow{f} Y$ is holomorphic and $f \equiv c$ on some nondiscrete subset of X, then $f \equiv c$ on X

Remark. This does not apply to higher dimensions, for example, $f(z, w) = z$, but in higher dimensions, we have Theorem [Unresolved source reference: Identity principle]

Theorem 22.4 (Local behaviour of holomorphic maps) $X \xrightarrow{f} Y$ is a nonconstant holomorphic map, $a \in X$, $f(a) = b \in Y$. There are local charts $U \xrightarrow{\phi} \mathbb{C}$, $V \xrightarrow{\psi} \mathbb{C}$ of a, b such that $\psi f \phi^{-1} = z^k$ for some $k \geq 1$

Remark. If the **multiplicity** $k > 1$, a is a **branch point**

Theorem 22.5 $X \xrightarrow{f} Y$ is a proper nonconstant holomorphic map between Riemann surfaces, there exists some n such that f take every value $c \in Y$, counting multiplicities, n times

Theorem 22.6 (Radó's theorem) A connected Riemann surface is second countable

Theorem 22.7 (Uniformization theorem) A simply connected Riemann surface is either $\mathbb{C}$, $\mathbb{P}$ or $\mathbb{H}^2$

23 Schemes

Definition 23.1 A *subscheme* Y of X is a topological subspace of $Y \subseteq X$

Proposition 23.2 (Gluing schemes) $\{X_i\}_{i\in I}$ is a family of schemes, suppose that

- $V_{ij} \subseteq X_i$ are open subschemes with $V_{ii} = X_i$
- $\varphi_{ij} : V_{ij} \to V_{ji}$ are isomorphisms with $\varphi_{ii} = \mathrm{id}$
- $\varphi_{ik} = \varphi_{ij} \circ \varphi_{jk} : V_{ki} \cap V_{kj} \to V_{ik} \cap V_{ij}$
- $\varphi_{ij}|_{V_{ij}\cap V_{ik}} : V_{ij} \cap V_{ik} \to V_{ji} \cap V_{jk}$ are isomorphisms

Then there is a scheme X (up to isomorphism) with an open covering $\{U_i\}_{i\in I}$ with open immersions $\psi_i : X_i \to X$ with $\psi(X_i) = U_i$ such that d

Proof.

□

Lemma 23.3 (Functor of points) X is a scheme, its *functor of points* is

$$\mathrm{Hom}(-, X) : \{\texttt{Affine schemes}\}^{\mathrm{op}} \to \{\texttt{Sets}\}$$

Conversely, a functor is the functor of points of some scheme if it is a Zariski sheaf covered by affine schemes

Proof.

□

Definition 23.4 Suppose $(X, \mathcal{O})$ is a ringed space. $\mathcal{F}$ is a sheaf of $\mathcal{O}$-module. $\mathcal{F}$ is quasi-coherent if for any $x \in X$, there is an open neighborhood U of x such that $\mathcal{F}_U$ is the cokernel of $\bigoplus_{i\in I} \mathcal{O}_U \to \bigoplus_{j\in J} \mathcal{O}_U$

Proposition 23.5 Quasi-coherent sheaves over Spec R are R-modules.

Proof.

□

Definition 23.6 $\varphi : X \to Y$ is *dominant* if $\varphi(X)$ is dense

Definition 23.7 $\varphi : X \to Y$ is *quasi-finite* if $\varphi^{-1}(x)$ is finite for any $x \in X$

Definition 23.8 A *variety* is a separated, finite type and integral scheme X over k.

Proposition 23.9 An affine variety X is of the form Spec $k[x_1, \cdots, x_n]/(f_1, \cdots, f_m)$

Definition 23.10 $\varphi : X \to Y$ is *affine* if $\varphi^{-1}(U)$ is affine for any affine open U

Definition 23.11 An affine morphism $\varphi : X \to Y$ is *finite* if $k[U]$ is finite $k[V]$-algebra for any $\varphi^{-1}(V) = U$ with U, V affine.

Proposition 23.12 A finite morphism φ is quasi-finite

Definition 23.13 An affine morphism $\varphi : X \to Y$ is *integral* if $k[U]$ is integral over $k[V]$ for any $\varphi^{-1}(V) = U$ with U, V affine.

Definition 23.14 The *canonical bundle* of an algebraic variety X of dimension n is $K = \bigwedge^n \Omega$, Ω is the cotangent bundle

Definition 23.15 The *Picard group* is $H^1(X, \mathcal{O}^*)$

Definition 23.16 X is *complete* if for any variety Y, the projection $X \times Y \to Y$ is closed

Proposition 23.17 A closed subvariety of a complete variety is complete. The image of a complete variety is complete. A complex variety is complete if and only if it is compact as a complex-analytic variety

Definition 23.18 A *rational map* between algebraic varieties $f : X \to Y$ is a morphism from a nonempty open subseteq $U \subseteq X$ to Y, if $g : Y \to Z$ is also a rational map, then we can consider $g \circ f : X \to Z$, f is a *birational map* if it has an inverse

Definition 23.19 A rational curve is an algebraic variety birational equivalent to $\mathbb{A}^1$, since birational equivalence preserve function field, we may assume it's $F(t)$

Definition 23.20 X is *integral* if $\mathcal{O}(U)$ is an integral domain for any open U

Lemma 23.21 Scheme X is integral $\Leftrightarrow$ X is irreducible and reduced

Proof. Suppose X is integral, then obviously it is reduced, if X is not irreducible, then there exist disjoint nonempty open subsets U, V(just take irreducible component minus the others), but then $\mathcal{O}(U \cup V) \cong \mathcal{O}(U) \times \mathcal{O}(V)$ is not integral. Conversely, suppose X is irreducible and reduced, note that if $U \subseteq X$ is affine integral $\Leftrightarrow$ $\mathcal{O}(U)$ is the spectrum of an integral domain. $\mathcal{O}(U)$ is a domain for any open subset U, just show that $\mathcal{O}(U) \to \mathcal{O}(V)$ restriction is injective for any affine $V \subseteq U$

□

Definition 23.22 X is *reduced* if $\mathcal{O}(U)$ is reduced for any open U

Definition 23.23 $X \to S$ is *separated* if $\Delta(X)$ is closed, $\Delta : X \to X \times_S X$ is a closed immersion

Definition 23.24 $f : X \to Y$ is locally P if for each affine chart $V \subseteq Y$, $f^{-1}(V)$ is covered by affine charts U_i such that $f|_{U_i} : U_i \to V$ between affine schemes are of P. Such examples include

1. Locally of finite type
2. Locally finitely presented

Definition 23.25 $f : X \to S$ is a flat morphism iff for each $x \in X$, $\mathcal{O}_{X,x}$ is a flat $\mathcal{O}_{S,f(x)}$ module. Suppose $\mathcal{F}$ is a quasi-coherent sheaf on X, then $\mathcal{F}$ is flat iff $\mathcal{F}_x$ is a flat $\mathcal{O}_{S,f(x)}$ module.

Definition 23.26 $f : X \to Y$ is universally P if $f_Z : X \times_Y Z \to Z$ is P for any $Z \to Y$, i.e. under base change

Definition 23.27 X is an integral scheme, $X = \bigcup \text{Spec } A_i$, A_i are integral domains, let B_i be the integral closure of A_i, the *normalization* of X is $Y = \bigcup \text{Spec } B_i$ with the induced finite morphism $Y \to X$

Lemma 23.28 Normalizations of dimension 1 schemes are regular, normalizations of dimension 2 schemes only have isolated singularities

Example 23.29 $k[x, y]/(x^2 - y^3) \cong k[t^2, t^3]$ with field of fractions $k(t)$ and integral closure $k[t]$, thus the normalization of curve $\text{Spec } \left(\frac{k[x,y]}{(x^2-y^3)}\right)$ is

$$\text{Spec } k[t] \to \text{Spec } \left(\frac{k[x, y]}{(x^2 - y^3)}\right)$$
$$t \mapsto (t^3, t^2)$$

Definition 23.30 $f : X \to Y$ is *flat* if $f : \mathcal{O}_{X,f(x)} \to \mathcal{O}_{Y,x}$ is flat for all $x \in X$

Definition 23.31 X is a *regular* scheme $\Leftrightarrow$ $\mathcal{O}_{X,x}$ is a regular local ring for any $x \in X$

Definition 23.32 X is a *normal* scheme $\Leftrightarrow$ $\mathcal{O}_{X,x}$ is a normal domain for any $x \in X \Leftrightarrow \mathcal{O}_X(U)$ is a normal domain for any open set $U \subseteq X \Leftrightarrow \forall x \in X$, there exists $U \ni x$ such that $\mathcal{O}_X(U)$ is a normal domain

Lemma 23.33 If X is an integral normal scheme, then $\Gamma(X, \mathcal{O}_X)$ is a normal domain.

Proof.

□

Lemma 23.34 Regularity implies normality.

Proof.

□

Definition 23.35 $f : X \to S$ is a smooth if it is smooth on each affine patch, or equivalently, locally of finite presentation and flat

Definition 23.36 $f : X \to Y$ is *unramified* if

- f is locally finitely presented
- For $f(x) = y$, $k(x)/k(y)$ is a separable algebraic extension and $m_y \mathcal{O}_{X,x} = m_x$

Definition 23.37 $f : X \to Y$ is an *étale morphism* if f is flat and unramified. Étale morphism should be thought of as a local isomorphism.

Definition 23.38 The *direct image functor* of $f : X \to Y$ is $f_* : \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$, $f_*(F)(V) = F(f^{-1}V)$ The f_* is left exact, $R^q f_* F$ is the associated sheaf to the presheaf $U \mapsto H^q(f^{-1}(U), F)$

Definition 23.39 The *inverse image functor* of $f : X \to Y$ is $f^{-1} : \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$, the sheafification of

$$U \mapsto \varinjlim_{V \supseteq f(U)} G(V)$$

The pullback sheaf of $y \hookrightarrow Y$ is the stalk $O_{Y,y}$

For $\mathcal{O}_Y$ modules $\mathcal{V}$, we have $f^*\mathcal{V} = f^{-1}\mathcal{V} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$

Remark. Given a vector bundle, its sheaf of sections is locally free. Conversely, if we have a locally free sheaf then it's the sheaf of sections of a vector bundle which we can build by taking sheaf Spec of the symmetric algebra of the locally free sheaf

Definition 23.40 $f : Z \to X$ is a closed immersion if $f(Z)$ is closed and locally regular functions on Z can be extended to regular functions on X, i.e. $f^\# : \mathcal{O}_X \to f_* \mathcal{O}_Z$ is surjective

Definition 23.41 $f : X \to Y$ is proper if it is separated, universally closed and of finite type

Proposition 23.42 Composition of proper morphisms is proper. Closed immersions are proper

Definition 23.43 L/k is a field extension, *Weil restriction* is the right adjoint of fiber product, i.e. $\mathrm{Res}_{L/k} X(S) = X(S \times_k L)$, or $\mathrm{Hom}(-, \mathrm{Res}_{L/k} x(-)) = \mathrm{Hom}(- \times_k L, -)$

Proposition 23.44 Suppose $X = \mathrm{Spec}\ A$, $Y = \mathrm{Spec}\ B$, $f : B \to A$, M, N are modules over A, B that corresponds to $\mathcal{O}$-modules $\mathcal{F}, \mathcal{G}$ over X, Y, then $f_*\mathcal{F}$ corresponds to M regarded as a B module, $f^{-1}\mathcal{G}$ corresponds to

Definition 23.45 Let X be a locally Noetherian integral scheme, a prime divisor is an integral subscheme $Z \subseteq U$, a *Weil divisor* is $\sum_{Z \text{ prime divisors}} n_Z Z$, where the sum is locally finite

Definition 23.46 Let X be a integral Noetherian scheme, a Cartier divisor is a family $\{U_i, f_i\}$, where $\{U_i\}$ is an open cover of X, and $f_i \in \Gamma(U_i, \mathcal{M}_X^\times)$ such that $f_i/f_j \in \Gamma(U_i \cap U_j, \mathcal{O}_X^\times)$, i.e. a Cartier divisor is a global section of $\mathcal{M}_X^\times / \mathcal{O}_X^\times$

Lemma 23.47 There is a map $\mathrm{Div}(X) \to \mathrm{Pic}(X)$ that sends a Cartier divisor to its associated line bundle, which is given by $\mathcal{L}|_{U_i} = \mathcal{O}_X \frac{1}{f_i}$, hence the transition map is f_i/f_j on $U_i \cap U_j$ from U_i to U_j

Example 23.48 (Vector bundles over $\mathbb{A}_k^n$) Suppose $E = \mathbb{A}_k^n \times_k \mathbb{A}_k^r \xrightarrow{\pi} \mathbb{A}_k^n$ is a vector bundle, then a section would be determined by $(t_1, \cdots, t_r) \mapsto (f_1(x_1, \cdots, x_n), \cdots, f_r(x_1, \cdots, x_n))$

Definition 23.49 (Gluing construction of $\mathbb{P}^n$) Consider $U_i = k[\frac{X_0}{X_i}, \cdots, \widehat{\frac{X_i}{X_i}}, \cdots, \frac{X_n}{X_i}]$, then $U_i - [0 : \cdots : 1_j : \cdots : 0]$ would be $k[\frac{X_0}{X_i}, \cdots, \widehat{\frac{X_i}{X_i}}, \cdots, (\frac{X_j}{X_i})^{\pm 1}, \cdots, \frac{X_n}{X_i}]$, and the transition map would be $\varphi_{ij} : U_i \xrightarrow{\frac{X_i}{X_j}} U_j$

Definition 23.50 ($\mathcal{O}(n)$ bundles over $\mathbb{P}^n$) Suppose $L \xrightarrow{\pi} \mathbb{P}^n$ is a line bundle, and $U_i \times_k \mathbb{A}^1_k \cong \operatorname{Spec} k[\frac{X_0}{X_i}, \cdots, \widehat{\frac{X_i}{X_i}}, \cdots, \frac{X_n}{X_i}] \otimes_k k[t_i]$, and consider map

$$U_i \times \mathbb{A}^1_k \overset{\frac{X_i}{X_j} \otimes (\frac{X_i}{X_j})^d}{\to} U_j \times \mathbb{A}^1_k$$

Hence a global section of $H^0(\mathbb{P}^n, \mathcal{O}(d))$ would necessarily looks like a homogeneous polynomial f of degree d, so that

$$(\frac{X_i}{X_j})^d f(\frac{X_0}{X_i}, \cdots, \widehat{\frac{X_i}{X_i}}, \cdots, \frac{X_n}{X_i}) = f(\frac{X_0}{X_j}, \cdots, \widehat{\frac{X_j}{X_j}}, \cdots, \frac{X_n}{X_j})$$

Example 23.51 (Tautological bundle and Hyperplane bundle(Serre's twisting sheaf))

Definition 23.52 Dimension

Proposition 23.53 $\dim(\mathcal{O}_{Y,y}) = \operatorname{codim}(X, Y)$

Classically, ideal $I \subseteq k[x_1, \cdots, x_n]$ corresponds to an algebraic set $V(I) \subseteq \mathbb{A}^n$, $p = (a_1, \cdots, a_n) \in \mathbb{A}^n$, then tangent space $T_p V$(view as embedded in $\mathbb{A}^n$) will just be

$$\left\{ (y_1, \cdots, y_n) \in \mathbb{A}^n \;\middle|\; \sum_{i=1}^n (y_i - a_i) \frac{\partial f}{\partial x_i}(p) = 0, \forall f \in I \right\}$$

Definition 23.54 $k[\epsilon] = k[x]/(x^2)$ is a local k algebra with maximal ideal (ϵ), $p \in X(k)$, then $T_p V \cong \operatorname{Hom}_{\text{loc}}(\mathcal{O}_{V,p}, k[\epsilon])$, i.e. $T_p X$ is the fiber over p of $X(k[\epsilon]) \to X(k)$ induced by $k[\epsilon] \to k$, $\epsilon \mapsto 0$, i.e. $\alpha : A \to k[\epsilon]$ such that $A \xrightarrow{\alpha} k[\epsilon] \to k$ has kernel p(which is a maximal ideal), i.e. $\alpha^{-1}((\epsilon)) = p$. For $\phi : V \to W$, $(d\phi)_p : \operatorname{Hom}_{\text{loc}}(\mathcal{O}_{V,p}, k[\epsilon]) \to \operatorname{Hom}_{\text{loc}}(\mathcal{O}_{W,\phi(p)}, k[\epsilon])$ is induced by $\mathcal{O}_{W,\phi(p)} \to \mathcal{O}_{V,p}$

Definition 23.55 A is a k algebra, M is an A module, a derivation is a k linear map $A \to M$ such that $D(ab) = aD(b) + D(a)b$. Denote the k vector space of derivations as $\operatorname{Der}_k(A, M)$

Example 23.56 R is a local k algebra with maximal ideal m and $R/m = k$, then $R \cong k \oplus m$ as a k vector space since the exact sequence split

$$0 \longrightarrow m \longrightarrow R \underset{\longleftarrow}{\longrightarrow} R/m \cong k \longrightarrow 0$$

Here $k \to R$ maps 1 to 1, making $k \to R \to R/m \cong k$ a k linear isomorphism. Thus $d : R \to m/m^2$, $f \mapsto f - f(m) \mod m^2$ is a k derivation

Proposition 23.57 (R, m) is a local ring, there is a canonical isomorphism $\operatorname{Hom}_{\text{loc}}(R, k[\epsilon]) \to \operatorname{Der}_k(R, k) \to \operatorname{Hom}_k(m/m^2, k)$

Proof. Given a local homomorhpim $\alpha : R \to k[\epsilon]$, $\alpha(f) = f(m) + D_\alpha(f)\epsilon$, here D_α would be a derivation. Given a derivation D, D vanishes on k as well as m^2, thus induces a linear map $m/m^2 \to k$. Given a linear map $m/m^2 \to k$, define derivation $R \to m/m^2$, $f \mapsto df$

□

Definition 23.58 The blow up of the origin in $\mathbb{A}^n$ is

$$Bl_0 \mathbb{A}^n = \left\{ (x_1, \cdots, x_n) \times [y_1, \cdots, y_n] \in \mathbb{A}^n \times \mathbb{P}^{n-1} \;\middle|\; x_i y_j = x_j y_i \right\}$$

Let $\varphi : Bl_0 \mathbb{A}^n \to \mathbb{A}^n$ be the projection to the first factor, then $Bl_0 \mathbb{A}^n$ is covered by n open affine charts $U_i = \{y_i \neq 0\} \cap Bl_0 \mathbb{A}^n$, where $k[U_i] = k\left[x_i, \frac{y_1}{y_i}, \cdots, \frac{y_n}{y_i}\right]$, so $U_i \cong \mathbb{A}^n$, with $\varphi|_{U_i} : U_i \to \mathbb{A}^n$ given by $k[x_1, \cdots, x_n] \xrightarrow{\alpha_i} k\left[x_i, \frac{y_1}{y_i}, \cdots, \frac{y_n}{y_i}\right], x_j \mapsto x_i \frac{y_j}{y_i}$

$\forall i$, $\varphi|_{U_i}|_{D(x_i)} : D(x_i) \to D(x_i)$ is an isomorphism, $\varphi|_{U_i}^{-1}(0) = V(\alpha_i(x_1, \cdots, x_n)) = V(x_i) \cong Spmk\left[\frac{y_1}{y_i}, \cdots, \frac{y_n}{y_i}\right] \cong \mathbb{A}^{n-1}$, and these $V(x_i)$'s glue to give $\varphi^{-1}(0) \cong \mathbb{P}^{n-1}$ which called the exceptional divisor

Proposition 23.59 There is a bijection between points on $\varphi^{-1}(0)$ and the line in $\mathbb{A}^n$ passing 0

Proof. Let $\underline{L = \bigcap_{i=1}^n \{x_i = a_i t\}}$, not all a_i's are zero be a line, then $\varphi|_{U_i}^{-1}(L \setminus 0) = \{x_i = a_i t, t \neq 0, a_i y_j = a_j y_i\}$, and $\underline{\varphi|_{U_i}^{-1}(L \setminus 0) = \{x_i = a_i t, a_i y_j = a_j y_i\}}$, so this line corresponds to $[a_1, \cdots, a_n] \in \mathbb{P}^{n-1} \cong \varphi^{-1}(0)$, thus if $L' \neq L$, $\varphi|_{U_i}^{-1}(L \setminus 0) \cap \varphi|_{U_i}^{-1}(L' \setminus 0) = \emptyset$

$Bl_0\mathbb{A}^n$ is nonsingular since it is covered by affine spaces $\mathbb{A}^n$, $Bl_0\mathbb{A}^n$ is irreducible since $Bl_0\mathbb{A}^n \setminus \varphi^{-1}(0) \cong \mathbb{A} \setminus 0$ is irreducible, and each point of $\varphi^{-1}(0)$ is in the closure of some line L in $Bl_0\mathbb{A}^n \setminus \varphi^{-1}(0)$, so $Bl_0\mathbb{A}^n \setminus \varphi^{-1}(0)$ is dense in $Bl_0\mathbb{A}^n$

□

Definition 23.60 If $V \subseteq \mathbb{A}^n$ is a closed subvariety containing 0, then the blow up of the origin $Bl_0V := \overline{\varphi^{-1}(V \setminus 0)}$, from this, we get an birational isomorphism $\varphi : Bl_0V \to V$ which is an isomorphism away from 0

Lemma 23.61 Suppose X, Y are both S-scheme, $x \in X, y \in Y$ are both over s, then the set of points in $X \times_S Y$ whose projections to X, Y are x, y are in bijection with the prime ideals of $k(x) \otimes_{k(s)} k(y)$, i.e. Spec $k(x) \times_{\text{Spec } k(s)}$ Spec $k(y)$

Definition 23.62 $f : X \to Y$ is a morphism between k-schemes, $y \in Y$ is a point, then the *scheme-theoretic fiber* over y of f is

$$\begin{array}{ccc} X \times_Y \text{Spec } k(y) & \longrightarrow & X \\ \downarrow & & f \downarrow \\ \text{Spec } k(y) & \hookrightarrow & Y \end{array}$$

Proposition 23.63 $p_2 : \text{Spec } k(y) \times_Y X \to X$ is homeomorphism onto the set-theoretic fiber $f^{-1}(y)$

Proof.

- Surjectivity: $\forall x \in f^{-1}(y)$ there is a unique point in Spec $k(y) \times_Y X$ mapping to x under p_2 (by Lemma 23.61, Spec $k(y) \otimes_{k(y)} k(x) =$ Spec $k(x)$)
- Injectivity: Image of p_2 lies in $f^{-1}(y)$
- Homeomorphism: The assertion about homeomorphism is local, so we can assume $Y =$ Spec A, $X =$ Spec B, then p_2 corresponds to $B \to k(y) \otimes_A B$, $b \mapsto 1 \otimes b$. Apply Lemma 7.59, any element in $k(y) \otimes_A B$ can be written as $\sum \frac{\overline{a_i}}{\overline{s}} \otimes b_i = (\frac{1}{\overline{s}} \otimes 1)(1 \otimes (\sum a_i b_i))$, here $y \in$ Spec A corresponds to prime $p < A$ and so $k(y) = \frac{A_p}{pA_p}$

□

Lemma 23.64 X scheme, $\mathcal{J}$ is a quasi-coherent sheaf of ideals in $\mathcal{O}_X$, let $Y := \text{supp}(\frac{\mathcal{O}_X}{\mathcal{J}})$. Then $Y \overset{i}{\hookrightarrow} X$ is closed and $(Y, \mathcal{O}_Y)$ is a scheme with $\mathcal{O}_Y := i^{-1}(\frac{\mathcal{O}_X}{\mathcal{J}})$

Proof.

□

Definition 23.65 $(Y, \mathcal{O}_Y)$ is a *subscheme* of $(X, \mathcal{O}_X)$ if $Y \subseteq X$ is locally closed and if $U = X - (\overline{Y} - Y)$ (largest open subseteq of X in which Y is closed), then $Y = \text{supp } \frac{\mathcal{O}_{X|U}}{\mathcal{J}}$ for some quasi-coherent sheaf of ideals in $\mathcal{O}_{X|U}$. In the case $U = X$ we say Y is a *closed subscheme* of X.

Proposition 23.66 $\exists$ bijection between closed subschemes of X and quasi-coherent ideal sheaves of $\mathcal{O}_X$

Proof. This follows from the fact that if $\mathcal{J}, \mathcal{J}'$ are tow quasi-coherent sheaves of ideals with the same support Y and the restriction of $\mathcal{O}_X/\mathcal{J}$, $\mathcal{O}_X/\mathcal{J}'$ to Y are the same, then $\mathcal{J} = \mathcal{J}'$ (follows from $I, I' < A$ two ideals then $I = I' \Leftrightarrow A/I = A/I'$)

□

Proposition 23.67 There is a equivalence of categories between A modules and quasi-coherent sheaves over affine scheme Spec A, sending A module M to the constant sheaf $\underline{M}$, and quasi-coherent sheaf $\mathcal{F}$ to A module of global sections $\mathcal{F}(\text{Spec } A)$

Theorem 23.68 Quasi-coherent sheaves over a scheme forms an abelian category

24 Stack

Definition 24.1 site

Definition 24.2 Topos is the category of sheaves over a site

Definition 24.3 Fibered category

Definition 24.4 stack

Definition 24.5 Algebraic stack

25 Hodge theory

Definition 25.1 An abelian group $H_{\mathbb{Z}}$(write $H_R = H_{\mathbb{Z}} \otimes R$) has a *pure Hodge structure* of weight n if

$$H_{\mathbb{C}} = \bigoplus_{p+q=n} H_{\mathbb{C}}^{p,q}, \quad \overline{H_{\mathbb{C}}^{p,q}} = H_{\mathbb{C}}^{q,p}$$

Or equivalently, if there is a decreasing *Hodge filtration* $F^p H_{\mathbb{C}}$ on $H_{\mathbb{C}}$ such that $H_{\mathbb{C}} = F^p H_{\mathbb{C}} \oplus \overline{F^{n+1-p} H_{\mathbb{C}}}$. The equivalence of the two definitions is given by $(p+q=n)$

$$F^p H_{\mathbb{C}} = \bigoplus_{i \geq p} H_{\mathbb{C}}^{i,n-i}, \quad \overline{F^q H_{\mathbb{C}}} = \bigoplus_{j \leq p} H_{\mathbb{C}}^{j,n-j}$$

$$H_{\mathbb{C}}^{p,q} = F^p H_{\mathbb{C}} \cap \overline{F^q H_{\mathbb{C}}}, \quad F^p H_{\mathbb{C}} \cap \overline{F^{n+1-p} H_{\mathbb{C}}} = 0$$

Example 25.2 X is a complex manifold, $H_{\mathbb{Z}} = H^n(X;\mathbb{Z})$, then

$$H_{\mathbb{C}}^n(X;\mathbb{C}) = \bigoplus_{p+q=n} H^{p,q} = \bigoplus_{p+q=n} H^p(X;\mathbb{C}) \wedge \overline{H^q(X;\mathbb{C})}$$

Example 25.3 (*Hodge-Tate structure*) The first ???? , $\mathbb{Z}(1) = (2\pi i)\mathbb{Z}$ has a pure Hodge structure of weight -2. Here $\mathbb{Z}(1)_{\mathbb{C}} = H^{-1,-1}$ is the unique one dimensional weight -2 pure Hodge structure. Uniqueness is due to the fact that $H^{-2,0} = \overline{H^{0,-2}}$ must be 0 or nonzero at the same time.

Definition 25.4 The tensor product of two Hodge structures H_1 and H_2 is given by $H^{p,q} =$

Example 25.5 (Hodege-Tate n) The previous example generalizes to $\mathbb{Z}(n) \coloneqq \mathbb{Z}(1)^{\otimes n}$. This has a one dimensional pure Hodge structure of weight n

Definition 25.6 A *polarization* over $\mathbb{Q}$ of a Hodge structure over $\mathbb{Q}$ of weight k is a $(-1)^k$ symmetric nondegenerate flat bilinear map $\beta : \mathbb{V}_{\mathbb{Q}} \times \mathbb{V}_{\mathbb{Q}} \to \mathbb{Q}$ such that the Hermitian form $\beta_x(C_x v, \bar{w})$ on each fiber $\mathcal{V}_x$ is positive definite, here C_x is the *Weil operator*, given as the direct sum of multiplication i^{p-q} on $\mathcal{V}_x^{p,q}$

Definition 25.7 A *mixed Hodge structure* on $H_{\mathbb{Z}}$ consists of an increasing *weight filtration* $W_\bullet$ on $H_{\mathbb{Q}}$ and a decreasing filtration $F^\bullet$ on $H_{\mathbb{C}}$ that are compatible, i.e.

$$F^p(\mathrm{gr}_k W)_{\mathbb{C}} = \frac{F^p \cap W_{k+1}(\mathbb{C}) + W_k(\mathbb{C})}{W_k(\mathbb{C})}$$

is a pure Hodge structure of weight k of $\mathrm{gr}_k W$

Definition 25.8 A *variation* of Hodge structure of weight k over $\mathbb{Q}$ and a complex manifold X is $(\mathbb{V}_{\mathbb{Q}}, \mathcal{F}^\bullet)$, $\mathbb{V}_{\mathbb{Q}}$ is a locally constant sheaf of $\mathbb{Q}$ vector spaces, $\mathcal{F}^\bullet$ is a decreasing filtration of holomorphic subbundles of the locally free sheaf $\mathcal{V} = \mathcal{O}_X \otimes \mathbb{V}_{\mathbb{Q}}$ such that

- $(\mathcal{V}_x, \mathcal{F}_x^\bullet)$ has a pure Hodge structure of weight k, i.e. $\mathcal{V}_x = \mathcal{F}^p \oplus \overline{\mathcal{F}^{k+1-p}}$
- (Griffiths transversality) $\nabla \mathcal{F}^p \subseteq \Omega_X^1 \otimes_{\mathcal{O}_X} \mathcal{F}^{p-1}$

Definition 25.9 A variation of mixed Hodge structure over $\mathbb{Q}$ and a complex manifold X is $(\mathbb{V}_{\mathbb{Q}}, \mathcal{W}_\bullet, \mathcal{F}^\bullet)$, $\mathcal{W}_\bullet$ is an increasing filtration of $\mathbb{V}_{\mathbb{Q}}$ by locally constant subsheaves such that

- $(\mathcal{V}_x, (\mathcal{W}_\bullet)_x, \mathcal{F}_x^\bullet)$ has a mixed Hodge structure, i.e. () is a pure Hodge structure of weight k
- $\nabla \mathcal{F}^p \subseteq \Omega_X^1 \otimes_{\mathcal{O}_X} \mathcal{F}^{p-1}$

26 Tannakian categories-Milne,Deligne

Remark. It is easy to see that $\underline{\mathrm{Hom}}(\mathbb{1}, Y) \cong Y$ since $\mathrm{Hom}(-, Y) \cong \mathrm{Hom}(-, \underline{\mathrm{Hom}}(\mathbb{1}, Y))$ is the representable functor

Remark. [Definition of ${}^t f : Y^\vee \to X^\vee$]

$$\mathrm{Hom}(Y^\vee, X^\vee) \cong \mathrm{Hom}(Y^\vee \otimes X, \mathbb{1}) \cong \mathrm{Hom}(X, \underline{\mathrm{Hom}}(Y^\vee, \mathbb{1})) \cong \mathrm{Hom}(X, Y^{\vee\vee})$$

And there is a canonical choice of $X \to X^{\vee\vee}$ given by ev in $\mathrm{Hom}(X^\vee \otimes X, \mathbb{1}) \cong \mathrm{Hom}(X, X^{\vee\vee})$

27 Plucker embedding

Definition 27.1 Consider Grassmannian $W \in Gr_k(n)$, the *Plücker coordinates* $W_{i_1,\cdots,i_k}$ to be the minor of $i_1,\cdots,i_k$-th columns. For $1 \le i_1 < \cdots < i_{k-1} \le n$, $1 \le j_1 < \cdots < j_k \le n$, $r \le k$, the **Plücker relations** is

$$W_{i_1,\cdots,i_k} W_{j_1,\cdots,j_k} = \sum W_{i'_1,\cdots,i'_k} W_{j'_1,\cdots,j'_k}$$

The summation is over all swaps of a size r order set of $\{i_1,\cdots,i_k\}$ with $w_1,\cdots,w_r$, respectively

Proof. If $r = k$, it is trivial. So we may assume $r < k$. For $v_1,\cdots,v_k,w_1,\cdots,w_k \in \mathbb{C}^k$, consider multilinear function

$$f(v_1,\cdots,v_k,w_1,\cdots,w_k) = |v_1\cdots v_k||w_1\cdots w_k| - \sum |v'_1\cdots v'_k||w'_1\cdots w'_k| = \text{LHS} - \text{RHS}$$

Let's first show that f is skew-symmetric, it is suffices to prove if $v_i = v_{i+1}$ or $v_k = w_k$, then $f = 0$

1. If $v_i = v_{i+1}$, LHS $= 0$, RHS consists of terms $|\cdots v_i\cdots||\cdots v_{i+1}\cdots|$ or $|\cdots v_{i+1}\cdots||\cdots v_i\cdots|$, and each pair will cancel out in summation
2. If $v_k = w_k$, through a linear transformation, $v_k = w_k$ can be taken to be $(0,\cdots,0,1)^T$, and then it reduces to a lower case

Since w_k, v_k can be move to any column up to a sign, we know f is indeed skew-symmetric

□

Example 27.2 Consider $\text{Gr}_2(4)$, the only Plücker relation is

$$W_{12}W_{34} - W_{13}W_{24} + W_{14}W_{23} = 0$$

Theorem 27.3 The *Plücker embedding* is

$$\text{Gr}_k(n) \to \mathbb{P}(\bigwedge^k \mathbb{C})$$
$$\text{Span}(v_1,\cdots,v_k) \mapsto [v_1 \wedge \cdots \wedge v_k]$$

The image is an irreducible projective algebraic variety defined exactly by Plücker relations on Plücker coordinates

28 Teichmuller space

Let $S_{g,b,n,m}$ be the surface with genus g, b boundaries, n punctures inside and m punctures on the boundaries. Then

$$\chi(S_{g,b,n,m}) = (1+b) - (2g+2b+n+m) + 1 = 2 - 2g - b - n - m$$

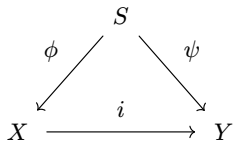
Definition 28.1 Suppose $\mathrm{Aut}(X)$ has a natural topology, the mapping class group is $\mathrm{Aut}(X)/\mathrm{Aut}_0(X)$, where $\mathrm{Aut}_0(X)$ is the path connected component of the identity, hence we have exact sequence

$$0 \to \mathrm{Aut}_0(X) \to \mathrm{Aut}(X) \to \mathrm{MCG}(X) \to 0$$

If X is a space, then a path connecting $f, g \in \mathrm{Aut}(X)$ is an isotopy

Example 28.2 $\mathrm{MCG}(S^2) = \mathbb{Z}/2\mathbb{Z}$

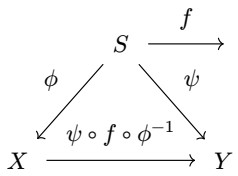
Definition 28.3 Let S be a compact surface with finitely many holes, X be a surface with a complete, finite area hyperbolic metric. A *hyperbolic structure* on S is a diffeomorphism $\phi : S \to X$, ϕ is called a *marking*, (X, ϕ) is a marked hyperbolic surface. (X, ϕ), (Y, ψ) are equivalent if there is an isometry $i : X \to Y$ such that $i \circ \phi$ and ψ are homotopic



The Teichmuller space of S is

$$T(S) = \{(X, \phi)\}/\sim$$

Definition 28.4 (Change of marking) $f : S \to S$ is a homeomorphism



When $f = 1_S$, $\psi \circ \phi^{-1}$ is the *change of marking* The mapping class group left acts on $T(S)$ by $h \cdot (X, f) = (X, fh^{-1})$, then $T(S)$ mod the action is just S

Example 28.5 By Uniformization theorem 22.7, $T(\mathbb{S}^2)$ is a point corresponds to the Riemann sphere, $T(\mathbb{R}^2)$ is two points corresponds the complex plane and the unit disc. $T(A) = [0, 1)$, where A is the open annulus, and $\lambda \in [0, 1)$ corresponds to $\{\lambda < |z| < \lambda^{-1}\}$ according to Exercise 35.53

By Gauss-Bonnet theorem, it is necessary that a closed hyperbolic surface X has area $\mathrm{Area}(X) = -\int_X K dS = -2\pi\chi(X)$ since the Gaussian curvature K is -1. Similarly, by Gauss-Bonnet theorem, it is reasonable to consider flat structures on the torus T^2, by modulo homothety, we may just assume it has unit area. Thus let's define $T(T^2)$ as the isotopy classes of unit-area flat structures on T^2, i.e. markings $T^2 \to \mathbb{T}^2$. Similarly, $T(S^2)$ should be defined to be the unique induced metric on the unit sphere $\mathbb{S}^2$

A marking on a lattice Λ in $\mathbb{R}^2$ is an ordered pair of generators, two marked lattices are equivalent if they transitive under $\mathrm{Isom}(\mathbb{R}^2)$. Marked lattices in $\mathbb{R}^2$ are in bijection with the upper half plane $\mathbb{H}^2$ as follows: $\mathbb{Z} + \mathbb{Z}\tau \leftrightarrow \tau$. note that $\mathbb{Z}\lambda + \mathbb{Z}\mu \sim \mathbb{Z} + \mathbb{Z}\frac{\mu}{\lambda}$ by homothety, $\mathbb{Z} + \mathbb{Z}\tau \sim \mathbb{Z} + \mathbb{Z}\bar{\tau}$ by reflection

Proposition 28.6 $T(T^2)$ is in bijection $\mathbb{H}^2$, this induces a hyperbolic metric on $T(T^2)$ so that $T(T^2) \cong \mathbb{H}^2$

Proof. It suffices to show that $T(T^2)$ is in bijection with equivalence classes of marked lattices in $\mathbb{R}^2$. $\mathbb{R}^2$ is the metric universal cover of $\mathbb{T}^2$

Given a marked lattice $\mathbb{Z}+\mathbb{Z}\tau$, $\tau\in\mathbb{H}^2$, using homothety, we get an equivalent lattice $\mathbb{Z}\lambda+\mathbb{Z}\mu$ with unit area, we can simply take the marking to be the map induced by the linear map $\mathbb{R}^2\to\mathbb{R}^2$, taking $\mathbb{Z}+\mathbb{Z}i$ to $\mathbb{Z}\lambda+\mathbb{Z}\mu$
For any marking $\phi: T^2\to\mathbb{T}^2$, we have the following diagram

$$\begin{array}{ccc} & \overset{\tilde{\phi}}{\dashrightarrow} & \\ \mathbb{R}^2 & & \mathbb{R}^2 \\ \pi\downarrow & & \downarrow\pi \\ & \overset{\phi}{\longrightarrow} & \\ T^2 & & \mathbb{T}^2 \end{array}$$

Hence $\tilde{\phi}\in\mathrm{Isom}(\mathbb{R}^2)$, the image of the standard lattice gives us the desired marked lattice

□

Since $\mathbb{H}^2$ is the universal cover of X, for any marking $\phi: S_g\to X$, we have

$$\begin{array}{ccc} & \overset{\tilde{\phi}}{\dashrightarrow} & \\ \mathbb{H}^2 & & \mathbb{H}^2 \\ \pi\downarrow & & \downarrow\pi \\ & \overset{\phi}{\longrightarrow} & \\ S_g & & X \end{array}$$

$$\begin{array}{ccccc} & \overset{\tilde{\psi}}{\longleftarrow} & \overset{\tilde{\imath}}{\leftrightarrows} & \overset{\tilde{\phi}}{\dashrightarrow} & \\ \mathbb{H}^2 & & \mathbb{H}^2 & & \mathbb{H}^2 \\ \pi\downarrow & & \downarrow\pi & & \downarrow\pi \\ & \overset{\psi}{\longleftarrow} & & \overset{\phi}{\longrightarrow} & \\ Y & & S_g & & X \\ & & \underset{i}{\leftrightarrows} & & \end{array}$$

$\tilde{\phi}\in\mathrm{Isom}(\mathbb{H}^2)\cong\mathrm{PGL}(2,\mathbb{R})$

Proposition 28.7 Let $DF(\pi_1(S_g),\mathrm{PSL}(2,\mathbb{R}))$ be the subset of discrete and faithful representations in $\mathrm{Hom}(\pi_1(S_g),\mathrm{PSL}(2,\mathbb{R}))$, there is a natural bijection

$$T(S_g)\leftrightarrow DF(\pi_1(S_g),\mathrm{PSL}(2,\mathbb{R}))/\mathrm{PGL}(2,\mathbb{R})$$

Proof. Consider map $T(S_g)\to\mathrm{Hom}(\pi_1(S_g),\mathrm{Isom}(\mathbb{H}^2))$ by $\pi_1(S_g)\overset{\phi_*}{\to}\pi_1(X)\overset{\cong}{\to}\mathrm{Aut}(\mathbb{H}^2/X)\hookrightarrow\mathrm{Aut}(\mathbb{H}^2)\cong\mathrm{Isom}(\mathbb{H}^2)$, if $(X,\phi)\sim(Y,\psi)$

□

Definition 28.8 Use the discrete topology on $T(S_g)$, and Lie group topology on $\mathrm{PSL}(2,\mathbb{R})$, and then use compact-open topology on $\mathrm{Hom}(\pi_1(S_g),\mathrm{PSL}(2,\mathbb{R}))$ which can be embedded in $\mathrm{PSL}(2,\mathbb{R})^{2g}$ (this is well defined regardless of the choice the generator), called the algebraic topology on $T(S_g)$

Proposition 28.9 Let c be an isotopy class of simple closed curves, then the map $T(S_g)\to\mathbb{R}$, $\mathcal{X}\to\ell_{\mathcal{X}}(c)$ is continuous

29 Graph Theory

Theorem 29.1 For every group G, there is a connected two dimensional CW complex X with $\pi_1(X) = G$

Proof. We can always find a surjection from a free group F to G, suppose F is generated by g_α's, and the kernel K is generated by r_β's, i.e. F has a group presentation $\langle g_\alpha | r_\beta \rangle$, then define X to be $\bigvee_\alpha S^1_\alpha$ attached with cells e^2_β's along each word r_β

□

Definition 29.2 Cayley graphs, Cayley complexes

Definition 29.3 A **graph** G is a one dimensional CW complex, a **tree** T is a contractible graph, $T \subseteq G$ is maximal if T contains all vertices, note that in a tree there is a unique path between two vertices

Proposition 29.4 Let X be a connected graph, any tree in X is contained in a maximal tree, in particular, X has a maximal tree

Proof. Let's prove more generally any subgraph X_0 is the deformation retraction of subgraph Y which contains all the vertices

Construct $X_0 \subseteq X_1 \subseteq X_2 \subseteq \cdots$ as follows, X_{i+1} is obtained by adding the closures of all the edges that connected to X_i, $X = \bigcup_i X_i$, since X is path connected, let $Y_0 = X_0$, and construct $Y_0 \subseteq Y_1 \subseteq Y_2 \subseteq \cdots$ as follows, for any vertex in $X_{i+1} - X_i$, choose one edge that connects to Y_i, and add the closure, so we have Y_{i+1} from Y_i, it is easy to see the Y_{i+1} deformation retracts onto Y_i, so $Y = \bigcup_i Y_i$ deformation retracts onto $Y_0 = X_0$

If $X_0 = T$ is a tree, so is Y since Y deformation retracts onto T which is contractible

□

Proposition 29.5 (Free basis for connected graphs) For a connected graph X with maximal tree T, for any edge $e_\alpha \in X - T$, there is a corresponding loop f_α goes from x_0 to one endpoint of e_α, across e_α to the other, and go back to x_0, $\pi_1(X, x_0)$ is a free group generated by f_α

Proof. Consider $X \to X/T$ which is a homotopy equivalence

□

Theorem 29.6 Any subgroup of a free group is also free

Proof. Let F be a free group, there exists a graph X such that $\pi_1(X) = F$ by just taking the wedge sum of circles at x_0, let $G \leq F$ be a subgroup, then there exists a covering $Y \xrightarrow{p} X$ such that $p_*(\pi_1(Y, y_0)) = G$, thus $\pi_1(Y, y_0) \cong G$, and since Y is a covering of X, Y is also a graph, by Proposition 29.5, $G \cong \pi_1(Y)$ is free

□

Definition 29.7 A **complete graph** K_n consists of n vertices and all possible edges

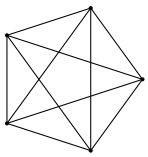


Figure 7: K_5

Complete bipartite graphs are $K_{n,m}$ with n, m vertices on each side and all possible edges between them

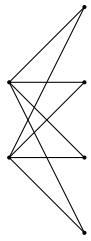


Figure 8: $K_{2,4}$

Definition 29.8 G is k vertex connected if G has more than k vertices and remain connected when removing less than k vertices

Proposition 29.9 Every convex polytope can be represented by a 3 vertex connected planar graph

Remark. A graph embedded in the plane through different ways may have different dual graphs

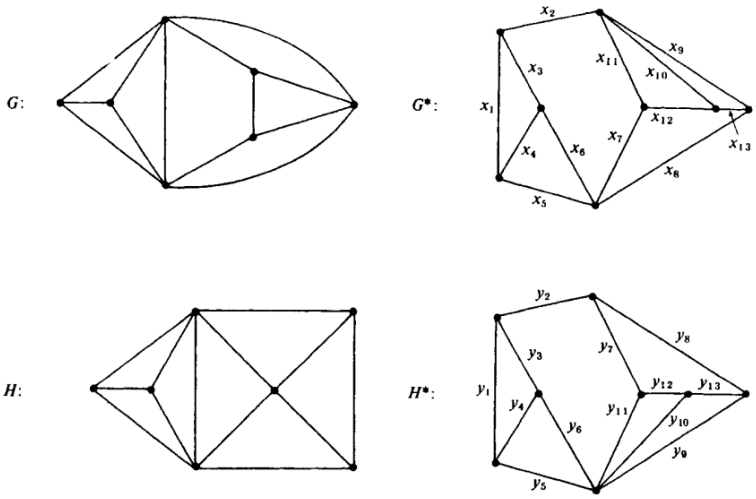


Figure 9: Different dual graphs for different planar embeddings of the same graph

However, if the graph is 3 vertex connected, then the dual graph will be canonical

30 Zeta function

Proposition 30.1

$$\begin{aligned}\int_0^\infty \frac{x^{s-1}}{e^x-1}dx &= \int_0^\infty x^{s-1}e^{-x}\sum_{n=0}^\infty e^{-nx}dx \\ &= \sum_{n=1}^\infty \int_0^\infty x^{s-1}e^{-nx}dx \\ &= \sum_{n=1}^\infty \frac{1}{n^s}\int_0^\infty x^{s-1}e^{-x}dx \\ &= \sum_{n=1}^\infty \frac{1}{n^s}\Gamma(s) \\ &= \zeta(s)\Gamma(s)\end{aligned}$$

Theorem 30.2 (Euler's reflection formula) $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$, $z \notin \mathbb{Z}$

31 p-adic numbers

Definition 31.1 The **p-adic intergers** are

$$\begin{aligned}\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z} &= \left\{(a_0, a_1, a_2, \cdots) \in \prod \mathbb{Z}/p^n\mathbb{Z} \;\middle|\; a_n \equiv a_m \mod p^m, n \geq m\right\} \\ &= \{b_0 + b_1 p + b_2 p^2 + \cdots\}\end{aligned}$$

Here $a_n = b_0 + b_1 p + \cdots + b_n p^n$

Example 31.2 If $p = 7$, we can write $3 + 6 \cdot 7 + 7^2 + 4 \cdot 7^3 + 2 \cdot 7^4 + \cdots$ as

$$\cdots 24163$$

with base 7

Definition 31.3 Given $a_0 \in \mathbb{F}_p$, there is a unique solution to $x^p - x = 0$ in $\mathbb{Z}_p$ such that $\mathbb{Z}_p \to \mathbb{Z}_p/p\mathbb{Z}_p$, $x \mapsto a_0$, giving the unique section the $\omega : \mathbb{F}_p \to \mathbb{Z}_p$, the restriction $\omega : \mathbb{F}_p^\times \to \mathbb{Z}_p^\times$ is the *Teichmuller character*, the image is called the *Teichmüller representative*, the coefficients in the p power series are Teichmuller coefficients

Definition 31.4 To compute additions and multiplications, we introduce *Witt vectors* $(X_0, X_1, \cdots, X_n, \cdots)$, define ghost components or Witt polynomials

$$W_n = X^{(n)} = \sum_{i=0}^{n} p^i X_i^{p^{n-i}}$$

Define addition and multiplication using $X^{(n)}$ by

$$(X + Y)^{(n)} = X^{(n)} + Y^{(n)}, (XY)^{(n)} = X^{(n)} Y^{(n)}$$

Thus we have

$$\begin{aligned}(X_0, X_1, \cdots) + (Y_0, Y_1, \cdots) &= \left(X_0 + Y_0, X_1 + Y_1 + \frac{X_0^p + Y_0^p - (X_0 + Y_0)^p}{p}, \cdots\right) \\ (X_0, X_1, \cdots) \times (Y_0, Y_1, \cdots) &= (X_0 Y_0, X_0^p Y_1 + X_1 Y_0^p + p X_1 Y_1, \cdots)\end{aligned}$$

32 Special relativity

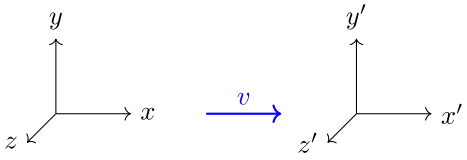
Definition 32.1 (Galilean group) The **Galilean group** is the group of **Galilean transformations** generated by rotations in $\mathbb{R}^n$, translations in $\mathbb{R}^{n+1}$ and **Galilean boosts** $(x,t) \mapsto (x+tv,t)$

$$\begin{pmatrix} R & v & w \\ 0 & 1 & s \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ t \\ 1 \end{pmatrix} = \begin{pmatrix} Rx+tv+w \\ t+s \\ 1 \end{pmatrix}$$

Definition 32.2 (Lorentz group) The **Lorentz group** is the group of **Lorentz transformations** generated by rotations in $\mathbb{R}^n$ and **Lorentz boosts** $(x,t) \mapsto (\sinh sx - \cosh st, \sinh st - \cosh sx)$

Definition 32.3 (Poincaré group) The **Galilean group** is the isometry group of the Minkowski space $\mathbb{R}^{n+1}$

Definition 32.4 $(x,ct) \mapsto \left(\gamma(x-vt), \gamma\left(t-\frac{vx}{c^2}\right)\right)$ $\beta = \frac{v}{c}$, $\alpha = \sqrt{1-\beta^2}$. $\gamma = \frac{1}{\sqrt{1-\beta^2}}$ is the **Lorentz factor** $\begin{cases} t' = \gamma\left(t - \frac{vx}{c^2}\right) \\ x' = \gamma(x-vt) \end{cases}$, where (x,t) and (x',t') are the coordinates of two frames, and frame (x',t') is moving towards the positive direction of the x axis with velocity v, and c is the speed of light, we can find the inverse transformation $\begin{cases} t = \gamma\left(t' + \frac{vx'}{c^2}\right) \\ x = \gamma(x'+vt') \end{cases}$ which makes perfect sense since relatively speaking, frame (x,t) is moving towards the negative direction of the x' axis with velocity v or rather moving towards the positive direction of the x' axis with velocity $-v$



More generally, if we consider $(\vec{x},t), (\vec{x}',t')$ are the coordinates of two frames, with frame $(\vec{x}',t')$ moving with velocity $\vec{v}$, then the Lorentz transformation will be

Deduction 32.5 (Time dilation) A frame moving (x',t') is at a constant speed v, then $\Delta t = \gamma \Delta t'$. Suppose you are on the train with constant speed v and height h, and let light bouncing up and down perpendicularly, then we have

$$2\sqrt{h^2 + \left(\frac{\Delta t}{2}v\right)^2} = c\Delta t, v\Delta t' = h$$
$$\Rightarrow \Delta t = \gamma \Delta t'$$

Things happen simultaneously in one frame may not be simultaneous in another frame

Deduction 32.6 (Length contraction) Suppose a train is moving with speed v, shed a beam light from one end to get to the other end

A in frame (x,t) send a signal when the left end of train passes, B in frame (x',t') on the right end of the train receives and return the signal, suppose the length of the train is l', and the length appears to be l in frame (x,t), then it takes time $\frac{l'}{c}$ for B to receive the signal in (x',t'), which takes time $\frac{l'\gamma}{c}$ in (x,t), when B should be in distance $l + \frac{vl'\gamma}{c}$ from A in (x,t) but distance $l' + \frac{vl'}{c}$ in (x',t') which take time $\frac{l+\frac{vl'\gamma}{c}}{c}$ and $\frac{l'+\frac{vl'}{c}}{c}$ to get back to A in (x,t) and (x',t'), hence we should have $\frac{l+\frac{vl'\gamma}{c}}{c} = \frac{l'+\frac{vl'}{c}}{c}\gamma \Rightarrow l = \gamma l'$

33 Quantum mechanics

Theorem 33.1 (Schrödinger's equation) $\Phi(x,t)$ is the wave function

$$= \frac{\partial^2 \Phi}{\partial x^2} + V\Phi$$

34 Examples

Example 34.1 Suppose $1 \mapsto k$ is an element in $Hom(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z})$, then $m \mid kn \Rightarrow \frac{m}{(n,m)}$ divides $k\frac{n}{(n,m)}$, thus $\frac{m}{(n,m)}$ divides k, thus $k = \frac{im}{(n,m)}, i = 0, \cdots, (n,m)-1$, thus $Hom(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/(n,m)\mathbb{Z}$

Consider $\mathbb{Z}/n\mathbb{Z} \otimes \mathbb{Z}/m\mathbb{Z}$, then $n(1 \otimes 1) = n \otimes 1 = 0$, $m(1 \otimes 1) = 1 \otimes m = 0$, thus $(n,m)(1 \otimes 1) = (rn + sm)(1 \otimes 1) = 0$

Apply functor $Hom(-, \mathbb{Z}/m\mathbb{Z})$ to short exact sequence $0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$, we get a left exact sequence $Hom(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \to \mathbb{Z}/m\mathbb{Z} \xrightarrow{\times n} \mathbb{Z}/m\mathbb{Z} \to 0$

Apply functor $- \otimes \mathbb{Z}/m\mathbb{Z}$ to short exact sequence $0 \to \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$, we get a left exact sequence $\mathbb{Z}/m\mathbb{Z} \xrightarrow{\times n} \mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \otimes \mathbb{Z}/m\mathbb{Z} \to 0$

And the kernel and cokernel of $\mathbb{Z}/m\mathbb{Z} \xrightarrow{\times n} \mathbb{Z}/m\mathbb{Z}$ are both $\mathbb{Z}/(n,m)\mathbb{Z}$

Example 34.2 $\mathbb{Q}/\mathbb{Z} \cong \bigoplus_p \mathbb{Z}\left[\frac{1}{p}\right]$

Example 34.3 $O(1,1) = \left\{ \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \right\}$

Example 34.4 F is a field, $R = \mathrm{End}(F^\infty) = \{\text{infinite dimensional matrices}\}$, Consider $R \hookrightarrow R$ by embedding into odd rows and even rows, we have $R^2 \cong R$ as right R modules

Example 34.5 $GL(2, \mathbb{F}_2) = SL(2, \mathbb{F}_2) \cong S_3$

Example 34.6 (A ring homomorphism between two local rings that isn't a local ring homomorphism) $A = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \text{ odd} \right\}$ with maximal ideal $2A$, $B = \mathbb{Q}$, inclusion is not a local ring homomorphism

Example 34.7 $V \subseteq \mathbb{A}^n$ is an affine variety, $m \in \mathrm{Spm} k[V]$, $k[V]_m$ is the stalk of the sheaf of regular functions. Tow representatives $\frac{f}{u}, \frac{g}{v}$ are of the same germ $\Leftrightarrow \frac{f}{u} = \frac{g}{v}$ on $D(wuv)$ for some $w(m) \neq 0 \Leftrightarrow w(fv - gu) = 0$

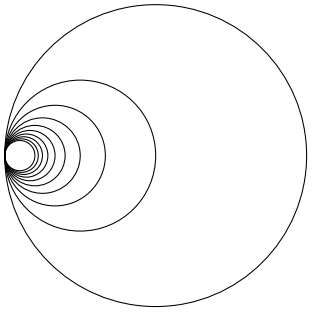
Example 34.8 (Multiplicative group scheme) The multiplicative group scheme over R is $\mathbb{G}_m = \mathrm{Spec}\ \mathbb{R}[x, x^{-1}]$, $\mathbb{G}_m(A) = \mathrm{Hom}(\mathbb{R}[x, x^{-1}], A) \cong A^\times, \phi_a \leftrightarrow a$, $\phi_a(x) = a$. The corresponding group multiplication $\mathbb{G}_m \times \mathbb{G}_m \xrightarrow{m} \mathbb{G}_m$ should be given by $\mathbb{G}_m(R) \times \mathbb{G}_m(R) \to \mathbb{G}_m(R)$, $(\phi_a, \phi_b) \mapsto \phi_{ab}$, which factors through $(\mathbb{G}_m \times \mathbb{G}_m)(R) \to \mathbb{G}_m(R)$, hence the corresponding comultiplication should be given by $\mathbb{Z}[x, x^{-1}] \xrightarrow{\Delta} \mathbb{Z}[x, x^{-1}] \otimes \mathbb{Z}[x, x^{-1}]$, $\Delta x = x \otimes x$, so that $\phi_a \otimes \phi_b(\Delta(x)) = \phi_{ab}(x)$. By Yoneda's Lemma, this is the unique choice

Example 34.9 (A surjective local homeomorphism may not be a covering) $p : \mathbb{R} \setminus \{0\} \to S^1$, or n sheeted cover with a point missing, p is discrete but not proper

Example 34.10 (Bundle with fiber isomorphic to vector space but not a vector bundle) $E \coloneqq \bigsqcup_{x \in X} \mathbb{R}^n$

Example 34.11 $H'_n = H_{k+n}$ also defines a homology theory where the dimension axiom fails

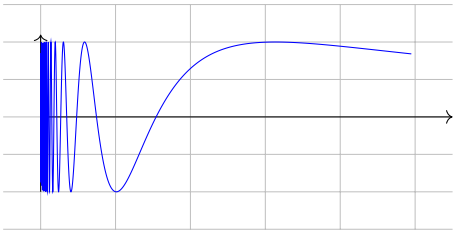
Example 34.12 (Hawaiian earring) The **Hawaiian earring** H is the union of circles with radius $\frac{1}{n}$ and centered at $(\frac{1}{n}, 0)$ with subspace topology in $\mathbb{R}^2$



Proposition 34.13 Hawaiian earring is not a CW complex since it is not locally contractible

Example 34.14 (Topologist's sine curve) The **topologist's sine curve** is

$$T = \left\{ \left(x, \sin\left(\frac{1}{x}\right) \right) \;\middle|\; x \in (0,1] \right\} \cup \{(0,0)\}$$



Proposition 34.15 The topologist's sine curve T is connected but not path connected

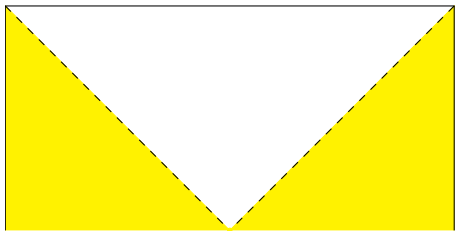
Example 34.16 (Warsaw circle) The **Warsaw circle** W is the topologist's sine curve enclosed

Bijective map $W \to [0,1)$ is not a homeomorphism, thus not a quotient map. W is weakly homotopic to a point but not homotopic

Example 34.17 $X = \mathbb{N}$ with discrete topology, $Y = \left\{0, 1, \frac{1}{2}, \frac{1}{3}, \cdots\right\}$ with subspace topology of $\mathbb{R}$, then $f : X \to Y, n \mapsto \frac{1}{n}$ is a weak homotopy equivalence, however, X, Y are not homotopy equivalent, otherwise suppose $g : X \to Y, h : Y \to X$ such that $hg \simeq 1_X$, $gh \simeq 1_Y$, suppose $F : Y \times I \to Y$ is a homotopy, then the restriction of F on $\{y\} \times I$ must be a constant map since the connected components of Y are just points, thus $F(y,0) = F(y,1)$, i.e. homotopic maps are in fact the same, for a similar argument on X, we have $hg = 1_X$, $gh = 1_Y$, thus h is injective which is impossible since $h^{-1}(h(0))$ consists of more then one point

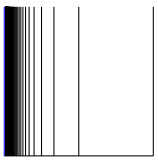
Example 34.18 $D^2 = S^2 \setminus \{N\} \subseteq S^2$ is not a cofibration. $D^2 \setminus \{0\} \subseteq D^2$ is not a cofibration

Example 34.19 $A = [-1,0) \cup (0,1]$, $X = [-1,1]$, then the mapping cylinder of the inclusion $A \overset{i}{\hookrightarrow} X$ has different topology from the subspace topology $X \times \{0\} \cup A \times I$ induced from $X \times I$

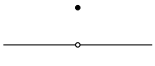


Example 34.20 $\{a, b\}$ with trivial topology, $\{a\} \subseteq \{a, b\}$ is a nonclosed cofibration since there is a retraction $I \sqcup I \to I \sqcup \{0\}$, $(s,t) \mapsto (s,0)$

Example 34.21 The **comb space** is $[(0,0),(1,0)] \bigcup \bigcup_{n=1}^{\infty} [(\frac{1}{n},0),(\frac{1}{n},1)]$

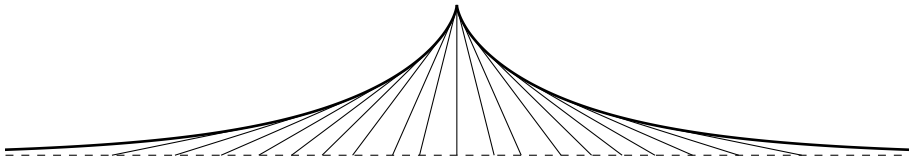


Example 34.22 A topological space with a cell decomposition may not be Hausdorff, consider $(-1,1)$ with two origins, which has $(-1,0),(0,1)$ as 1 cells and two origins as 0 cells



Example 34.23 $D \subseteq \mathbb{C}$ is the unit disc, $f(z) = \sum_{n=0}^{\infty} \frac{z^{2^n}}{n^2}$ is continuous on $\overline{D}$ and holomorphic on D but not on any point on ∂D

Example 34.24 (Tractrix) An interval I with one end point pushed or dragged along the x axis gives a **Tractrix**. The velocity has the same direction as I, i.e. $\frac{dx}{dy} = \pm\frac{\sqrt{a^2-y^2}}{y}$, which gives solution $x = \pm\left(\ln\frac{a+\sqrt{a^2-y^2}}{y} - \sqrt{a^2-y^2}\right)$



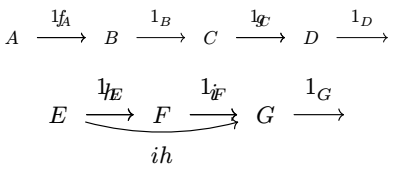
Example 34.25 R is a commutative ring. The category of R-modules is a monoidal category with

- $(A \otimes B) \otimes C \xrightarrow{\alpha} A \otimes (B \otimes C)$, $(a \otimes b) \otimes c \mapsto a \otimes (b \otimes c)$
- $R \otimes A \xrightarrow{\lambda} A$, $r \otimes a \mapsto ra$, $A \otimes R \xrightarrow{\rho} A$, $a \otimes r \mapsto ra$

Example 34.26 R is a commutative ring. R-algebra/coalgebra/bialgebra is a monoid/comonoid/ bimonoid of the category of R-modules,

Example 34.27 The image of a functor is not necessarily a category

Consider the following categories $\mathcal{C}$ and $\mathcal{D}$



Consider functor $F : \mathcal{C} \to \mathcal{D}$, $F(A) = E$, $F(B) = F$, $F(C) = F$, $F(D) = G$, $F(f) = h$, $F(g) = i$

Example 34.28 (Presheaves that are not sheaves)

1. G is a nontrivial abelian group. Consider presheaf $\mathcal{G}$ given by $\mathcal{G}(U) = G$ for any nonempty open subset and $\mathcal{G}(\varnothing) = 0$. For example $G = \mathbb{Z}/2$, $X = \{a, b\}$ with discrete topology, $\mathcal{G}(\{a, b\}) = \mathcal{G}(\{a\}) = \mathcal{G}(\{b\}) = \mathbb{Z}/2$, $\mathcal{G}(\varnothing) = 0$. Then 1 on $\{a\}$ and 0 on $\{b\}$ cannot be glued together.
2. Suppose $\phi : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves, im ϕ and coker ϕ are not generally sheaves. For example consider $\exp : \mathcal{O} \to \mathcal{O}^\times$, im exp is a presheaf but not a sheaf

Definition 34.29 ($\mathcal{O}(n)$ bundle over Riemann sphere $S^2 \cong \mathbb{C}P^1$) Suppose $(\mathbb{C}, z \mapsto z)$, $\left(S^2 \setminus 0, z \mapsto \frac{1}{z}\right)$ are the charts coordinate of S^2 with transition map $z \mapsto \frac{1}{z}$ both ways on $\mathbb{C} \setminus 0$, or equivalently $(U_0, [1, z] \mapsto z)$, $(U_1, [z, 1] \mapsto z)$ are the corresponding charts of $\mathbb{C}P^1$ with transition map $z \mapsto \frac{1}{z}$ both ways on $U_0 \cap U_1$, namely, $S^2 \to \mathbb{C}P^1$, $z \mapsto [1, z]$, $\infty \mapsto [0, 1]$ is and isomorphism because of isomorphisms on charts

Now define $\mathcal{O}(n)$ line bundle on S^2 by specify transition functions $g_{10}(z) = z^{-n}$, $g_{01}(z) = z^n$, $\forall z \in \mathbb{C} \setminus 0 \cong U_0 \cap U_1$

Definition 34.30 (Tautological line bundle over Riemann sphere) The tautological bundle is $\mathcal{O}(-1)$, tautological bundle is defined as a subspace E of $\mathbb{C}P^1 \times \mathbb{C}^2$ consists of (l, v) with $v \in l$ projects to the first factor, let's figure out the trivializations!

$\varphi_0 : U_0 \times \mathbb{C}^2 \cap E \to \mathbb{C} \times \mathbb{C}$, $([1,z], t(1,z)) \mapsto (z,t)$, and $\varphi_1 : U_1 \times \mathbb{C}^2 \cap E \to \mathbb{C} \times \mathbb{C}$, $([z,1], t(z,1)) \mapsto (z,t)$, since $\varphi_1 \circ \varphi_0^{-1} : (U_0 \cap U_1) \times \mathbb{C}^2 \cap E \to (U_0 \cap U_1) \times \mathbb{C}^2 \cap E$, $(z,t) \mapsto \left(\frac{1}{z}, zt\right)$, the transition function $g_{10}(z) = z$

Remark. $\mathcal{O}(-1)$ doesn't nonzero global section, suppose s is a global section of $\mathcal{O}(-1)$, then $s(x) = (x, f(x)) \in E \hookrightarrow \mathbb{C}P^1 \times \mathbb{C}^2$ is holomorphic, but then image of σ has to be a point, and this point must be zero

Example 34.31 We still use U_0, U_1 to denote coordinate charts, φ_0, φ_1 to denote corresponding trivializations

Global sections of $\mathcal{O} = \mathcal{O}(0)$ are exactly holomorphic functions which are just constants, suppose $s : S^2 \to \mathcal{O}$ is a section, and $\varphi_0 \circ s|_{U_0}(z) = (z, f_0(z))$, $\varphi_1 \circ s|_{U_1}\left(\frac{1}{z}\right) = \left(\frac{1}{z}, f_1\left(\frac{1}{z}\right)\right)$, then we have $(z, f_1(z)) = \varphi_1 \circ s|_{U_1}(z) = \varphi_1 \circ s|_{U_0}(z) = \varphi_1 \circ \varphi_0^{-1} \circ \varphi_0 \circ s|_{U_0}(z) = \varphi_1 \circ \varphi_0^{-1}(z, f_0(z)) = (z, g_{10}(z) f_0(z)), \forall z \in U_0 \cap U_1$, thus $f_1(z) = g_{10}(z) f_0(z) = f_0(z)$ which precisely means s correspond to holomorphic function f over X, $f|_{U_0} = f_0, f|_{U_1} = f_1$

Let's show that the canonical bundle(which in the case of a Riemann surface is the same as the cotangent bundle) is $\mathcal{O}(-2)$, since $d\left(\frac{1}{z}\right) = -\frac{1}{z^2} dz$, the transition function would be $g_{10}(z) = -z^2$, but using dz or $-dz$ as the a basis element would be isomorphic

Proposition 34.32 $H^0(CP^1, \mathcal{O}(n))$, the vector space of global sections of $\mathcal{O}(n) \to \mathbb{C}P^1, n \geq 0$ generated by homogeneous polynomials $z_0^n, z_0^{n-1} z_1, \cdots, z_0 z_1^{n-1}, z_1^n$

Proof. $z_0^k z_1^{n-k}$ have the forms z_1^{n-k} and z_0^k in U_0 and U_1

□

Example 34.33 (Line bundles on the projective space $\mathbb{C}P^n$) Suppose $(U_0, [1, z_1, \cdots, z_n] \mapsto (z_1, \cdots, z_n))$, $(U_n, [z_0, z_1, \cdots, z_{n-1}, 1] \mapsto (z_0, \cdots, z_{n-1}))$ be coordinate charts of $\mathbb{C}P^n$, with transition map $U_i \cap U_j \to U_i \cap U_j$, $\left(\frac{z_0}{z_i}, \cdots, \frac{\hat{z_i}}{z_i}, \cdots, \frac{z_n}{z_i}\right) \mapsto \left(\frac{z_0}{z_j}, \cdots, \frac{\hat{z_j}}{z_j}, \cdots, \frac{z_n}{z_j}\right)$, which is kind of like multiply by $\frac{z_i}{z_j}$, then the line bundle $\mathcal{O}(m)$ is defined by transition function $g_{ji} = \frac{z_j}{z_i}$ which satisfies the cocycle condition

Similarly, we can check that the tautological bundle $E = \{(l, v) \mid v \in l\} \subset \mathbb{C}P^n \times \mathbb{C}^{n+1}$ projects to $\mathbb{C}P^n$ is $\mathcal{O}(1)$

It is obvious that any degree n polynomial are global section of $\mathcal{O}(n)$

Example 34.34 X is topological space, $End(X)$ is a unital nonassociative $\mathbb{R}$ algebra which is not symmetric, antisymmetric, nor does it satisfy Jacobi identity

Example 34.35 Consider $C^\infty(M)$ where M is a smooth manifold, then $\mathcal{L}(M) = Der(C^\infty(M))$ consists of vector fields, it is a Lie algebra, hence we can think of derivations as linear differential operator of order 1, then we know that the commutator of two such operators is again a linear differential operator of order 1

Example 34.36 Let $\mathfrak{g}$ be a Lie algebra, then ideals of $\mathfrak{g}$ precisely the Lie algebra subrepresentations of the adjoint representation $(ad, \mathfrak{g})$

Example 34.37 (Lie algebra of $M_n(\mathbb{R})$) Suppose $X = \sum_{i,j} X_{ij} \frac{\partial}{\partial x_{ij}}$ is a left invariant

$$\begin{aligned} X_{kl}(A) &= \sum_{i,j} X_{ij}(A) \frac{\partial x_{kl}}{\partial x_{ij}}(A) \\ &= X_A(x_{kl}) = (L_A)_0 X_0(x_{kl}) \\ &= X_0(x_{kl} \circ L_A) \\ &= \sum_{i,j} X_{ij}(0) \frac{\partial (x_{kl} \circ L_A)}{\partial x_{ij}}(0) \\ &= X_{kl}(0) \end{aligned}$$

Thus X_{ij} are constants

$$\begin{aligned}
[X,Y] &= \left[\sum_{i,j} X_{ij}\frac{\partial}{\partial x_{ij}}, \sum_{k,l} Y_{kl}\frac{\partial}{\partial x_{kl}}\right] \\
&= \sum_{i,j,k,l} X_{ij}Y_{kl}\left[\frac{\partial}{\partial x_{ij}}, \frac{\partial}{\partial x_{kl}}\right] \\
&= \sum_{i,j} X_{ij}Y_{ij}\left[\frac{\partial}{\partial x_{ij}}, \frac{\partial}{\partial x_{kl}}\right] \\
&= 0
\end{aligned}$$

Therefore $\mathrm{Lie}(M_n(\mathbb{R})) = 0$

Example 34.38 (Lie algebra of $GL(n,\mathbb{R})$) Suppose $X = \sum_{i,j} c_{ij}\frac{\partial}{\partial x_{ij}}$ is a left invariant field

$$\begin{aligned}
c_{kl}(A) &= \sum_{i,j} c_{ij}(A)\frac{\partial x_{kl}}{\partial x_{ij}}(A) \\
&= X_A(x_{kl}) = (L_A)_I X_I(x_{kl}) \\
&= X_I(x_{kl}\circ L_A) \\
&= \sum_{i,j} c_{ij}(I)\frac{\partial(x_{kl}\circ L_A)}{\partial x_{ij}}(I) \\
&= \sum_i a_{ki}c_{il}(I)
\end{aligned}$$

Hence $C(A) = AC(I)$, $\frac{\partial c_{kl}}{\partial x_{ij}} = \delta_{ki}c_{jl}(I)$

$$
\begin{aligned}
[X,Y] &= \left[\sum_{i,j} c_{ij}\frac{\partial}{\partial x_{ij}}, \sum_{k,l} d_{kl}\frac{\partial}{\partial x_{kl}}\right] \\
&= \sum_{i,j,k,l}\left[c_{ij}\frac{\partial}{\partial x_{ij}}, d_{kl}\frac{\partial}{\partial x_{kl}}\right] \\
&= \sum_{i,j,k,l} c_{ij}\frac{\partial}{\partial x_{ij}}\left(d_{kl}\frac{\partial}{\partial x_{kl}}\right) - d_{kl}\frac{\partial}{\partial x_{kl}}\left(c_{ij}\frac{\partial}{\partial x_{ij}}\right) \\
&= \sum_{i,j,k,l} c_{ij}\left(\frac{\partial d_{kl}}{\partial x_{ij}}\frac{\partial}{\partial x_{kl}} + d_{kl}\frac{\partial^2}{\partial x_{ij}\partial x_{kl}}\right) - d_{kl}\left(\frac{\partial c_{ij}}{\partial x_{kl}}\frac{\partial}{\partial x_{ij}} + c_{ij}\frac{\partial^2}{\partial x_{ij}\partial x_{kl}}\right) \\
&= \sum_{i,j,k,l} c_{ij}\frac{\partial d_{kl}}{\partial x_{ij}}\frac{\partial}{\partial x_{kl}} - d_{kl}\frac{\partial c_{ij}}{\partial x_{kl}}\frac{\partial}{\partial x_{ij}} \\
&= \sum_{i,j,k,l} c_{ij}\frac{\partial d_{kl}}{\partial x_{ij}}\frac{\partial}{\partial x_{kl}} - \sum_{i,j,k,l} d_{kl}\frac{\partial c_{ij}}{\partial x_{kl}}\frac{\partial}{\partial x_{ij}} \\
&= \sum_{i,j,k,l} c_{ij}\frac{\partial d_{kl}}{\partial x_{ij}}\frac{\partial}{\partial x_{kl}} - \sum_{i,j,k,l} d_{ij}\frac{\partial c_{kl}}{\partial x_{ij}}\frac{\partial}{\partial x_{kl}} \\
&= \sum_{i,j,k,l}\left(c_{ij}\frac{\partial d_{kl}}{\partial x_{ij}} - d_{ij}\frac{\partial c_{kl}}{\partial x_{ij}}\right)\frac{\partial}{\partial x_{kl}} \\
&= \sum_{j,k,l}\left(c_{kj}d_{jl} - d_{kj}c_{jl}\right)\frac{\partial}{\partial x_{kl}} \\
&= \sum_{k,l}\left(\sum_j c_{kj}d_{jl} - d_{kj}c_{jl}\right)\frac{\partial}{\partial x_{kl}} \\
&= \sum_{k,l} b_{kl}\frac{\partial}{\partial x_{kl}}
\end{aligned}
$$

Here $B = [C, D]$. Therefore $\mathrm{Lie}(GL(n,\mathbb{R})) = \mathfrak{gl}(n,\mathbb{R})$

Example 34.39 Consider the $\Phi : GL(n,\mathbb{R}) \to M_n(\mathbb{R}), A \mapsto A^TA$ which is a smooth map, and level set $\Phi^{-1}(I) = O(n,\mathbb{R})$ is the orthogonal group, to show this is a Lie subgroup, thanks to Theorem 16.11, it suffices to show Φ is of constant rank, but Φ is equivariant assuming $GL(n,\mathbb{R})$ acts on itself by right multiplication and acts on $M_n(\mathbb{R})$ by $X \cdot A = A^TXA, X \in M_n(\mathbb{R}), A \in GL(n,\mathbb{R})$, since $\Phi(A)\cdot B = B^TA^TAB = \Phi(AB)$

$(d\Phi)_I(B) = B^T + B$, and $T_I(O(n,\mathbb{R})) = \ker(d\Phi)_I = \{B \in M(n,\mathbb{R}) | B^T + B = 0\}$

Example 34.40 Union of topologies may not be a topology

35 Exercises

Exercise 35.1 If R is a domain, so is $R[x]$

Solution. Suppose $f = ax^n + \cdots$, $g = bx^m + \cdots$ for some $a, b \neq 0$, then $fg = abx^{n+m} + \cdots \neq 0$

□

Exercise 35.2 If K/F is normal, L/F is the separable closure of F in K, then L is also normal

Solution. Suppose $a \in L$ has minimal polynomial p, then $p(x) = \prod_{j=1}^{n}(x - a_j)$ in K with a_j distinct since K is normal and L is separable, note that a_j has the same minimal polynomial p, hence $a_j \in L$, making L also normal

□

Exercise 35.3 If E/F is a Galois extension, then $Tr_{E/F}(\alpha)$ is the sum of all conjugates of α, $N_{E/F}(\alpha)$ is the product of all conjugates of α

Solution. Suppose the minimal polynomial of α is $m(x) = x^n + a_1 x^{n-1} + \cdots + a_n$

□

Exercise 35.4 If $F \subseteq E \subseteq L$ are field extensions, then $Tr_{L/F} = Tr_{E/F} \circ Tr_{L/E}$

Solution. Suppose $x_1, \cdots, x_n$ is a basis for L/E, $y_1, \cdots, y_m$ is a basis for E/F

□

Exercise 35.5 Suppose $T \in \mathrm{Hom}_{\mathbb{F}}(V, V)$ is a linear operator with $T(V) \leq W$, then $Tr(T) = Tr(T|_W)$

Exercise 35.6 R is a ring

1. $0x = 0$, $(-1)x = -x$

Solution.

1. $$0x = (0 + 0)x = 0x + 0x \Rightarrow 0x = 0$$

$$0x = (1 + (-1))x = 1x + (-1)x = x + (-1)x \Rightarrow (-1)x = -x$$

□

Exercise 35.7 Let R be a commutative ring, and $I_1, \cdots, I_n \leq R$ be pairwise coprime ideals, then $I_1 \cdots I_n = I_1 \cap \cdots \cap I_n$

Solution. By induction

□

Exercise 35.8 Every group G is naturally isomorphic to its opposite G^{op}

Solution. Consider $\phi : G \to G^{op}$, $g \mapsto g^{-1}$

□

Exercise 35.9 A morphism of G torsors is always an isomorphism

Exercise 35.10 X has a left G action and a right H action such that $(gx)h = g(xh)$

1. $X \times_G * \cong X/G$
2. $X \times_G G \cong X$
3. $(X \times_G Y) \times_H Z \cong X \times_G (Y \times_H Z)$
4. If $H \leq G$, then $X \times_G G \times_Y \cong X \times_H Y$
5. If $H \leq G$, $X \times_G (G/H) \cong X/H$

Exercise 35.11 $SL(n, F)$ is a perfect group for $n \geq 3$. $SL(2, F)$ is a perfect group if $|k| \geq 4$

Solution. Denote $G_n = SL(n, F)$. Elementary matrices generate G_n and are in $[G_n, G_n]$

□

Exercise 35.12 M is a finitely presented, then $N^* \otimes M \cong \mathrm{Hom}_R(M, N)^*$

R is a local ring, then flat, projective, free modules are equivalent notions

Solution. Finite presented and flat always imply projective

M has minimal generating set $m_1, \cdots, m_n$, $0 \to K \to R^n \to M \to 0$ is a split exact sequence, tensor with $k = R/m$, we have $0 \to k \otimes K \to k^n \to k \otimes M \to 0$, but $\dim k^n = \dim k \otimes M = n$, $K/mK = k \otimes K = 0$, by Nakayama's lemma 6.18, $K = 0$, hence $M = R^n$

□

Exercise 35.13 Let K be a field, and let n be a positive integer. Let $K(x_1, \cdots, x_n)$ be the field of rational functions over K with n variables, and let $L = K[x_1, \cdots, x_n, x_1^{-1}, \cdots, x_n^{-1}]$ be the subring of $K(x_1, \cdots, x_n)$. Let $R = K[x_1, \cdots, x_n, y_1, \cdots, y_n]$

1. For an element $p \in R$, let $\varphi(p)$ denote the element of L obtained by substituting x_i^{-1} into each variable y_i in p. This map $\varphi : R \to L$ is a ring homomorphism. Show that for an ideal J of L, $\varphi^{-1}(J)$ is an ideal of R
2. For $1 \leq i \leq n$, let $g_i = x_i y_i - 1$. Let

   $$R' = \{r \in R \mid \text{for } 1 \leq i \leq n \text{ , every monomial in } r \text{ does not involve } x_i \text{ and } y_i \text{ simultaneously}\}$$

   Show that for an arbitrary element $p \in R$, there exist $h_1, \cdots, h_n \in R$ and $r \in R'$ such that $p = h_1 g_1 + \cdots + g_n h_n + r$
3. Let I denote the ideal of R generated by $g_1, \cdots, g_n$. Show that $\ker \varphi = I$ and that L is isomorphic to the quotient ring R/I

Solution.

1. By definition
2. Suppose monomial q containing factor $(x_1 y_1)^k$ but not $(x_1 y_1)^{k+1}$, since $(x_1 y_1)^k = (g_1 + 1)^k$, q can be written as $ug_1 + v$, where every monomial in v does not involve x_1 and y_1 simultaneously. Repeat for $g_2, \cdots, g_n$, then we are done
3. $\varphi(g_i) = 0 \Rightarrow I \leq \ker \varphi$, conversely, if $p \in \ker \varphi$, then $0 = \varphi(r) \Rightarrow r = 0$, thus $\ker \varphi = I$. Since φ is surjective, by first isomorphism theorem, we have $L \cong R/I$

□

Exercise 35.14 A is an Abelian group, then V is an irreducible representation of A iff $\dim V = 1$

Solution. left multiplication by a induces an FA-module homomorphism, hence by Schur's lemma 6.13, a acts as scalar multiplication, thus and subspace would be a subrepresentation, so $\dim V$ must be 1

□

Exercise 35.15 Spec R is a disconnected $\Leftrightarrow$ R has a non trivial idempotent

Solution. Suppose $\mathrm{Spec}\ R = V(I) \sqcup V(J) = V(IJ)$, which implies that $IJ = \mathrm{Nil}(R)$, then $\varnothing = V(I) \cap V(J) = V(I + J)$, thus there are $e \in I, f \in J$ such that $e + f = 1$. Assume $(ef)^n = 0$. Then $(e^n) + (f^n) = I' + J' = R$ by expanding $(e + f)^{2n} = e^n * + * f^n$, and $((ef)^n) = I'J' = 0$, it is easy to show it has an idempotent

□

Exercise 35.16 An integral domain R has only trivial idempotents $0, 1$

Solution. Suppose $e \in R$ is an idempotent, then so is $1-e$ since $(1-e)^2 = 1-2e+e = 1-e$, also, $e(1-e) = 0$, by integrability, $e = 0$ or $e = 1$

□

Exercise 35.17 M, N are R-modules, then $(f \otimes g)(a \otimes b) = f(a) \otimes g(b)$ defines an injective map $M^\vee \otimes N^\vee \to (M \otimes N)^\vee$, which is an isomorphism when either M or N is finitely generated and projective

Solution.

□

Exercise 35.18 Suppose $\{A_i\}_{i \in I} \subseteq M_n(F)$ commute pairwise and diagonalizable, then they can be diagonalized simultaneously

Solution. Suppose $V_\lambda \subseteq F^n$ is the λ eigenspace of A, then $B(Ax) = ABx = \lambda Ax$, i.e. $B(V_\lambda) \subseteq V_\lambda$, thus V_λ will be an invariant space for all A_i's

□

Exercise 35.19 A is a Noetherian commutative ring, $\mathfrak{p}$ is a prime ideal, M is a finitely generated module, and $M_\mathfrak{p} = 0$, then $M_f = 0$ for some $f \in A - \mathfrak{p}$

Proof. Suppose $\{b_1, \cdots, b_n\}$ is a generating set of M, then $M_\mathfrak{p} = 0$ means that there exist $t_1, \cdots, t_n$ such that $t_i m_i = 0$, let $f = t_1 \cdots t_n$

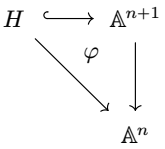
□

Exercise 35.20 A is a Noetherian commutative ring, $\mathfrak{p}$ is a prime ideal, M is a finitely generated module, and $M_\mathfrak{p}$ is a free module over $A_\mathfrak{p}$, show that there exists $f \in A$ such that M_f is a free module over A_f

Proof. Suppose $\left\{\frac{m_1}{s_1}, \cdots, \frac{m_n}{s_n}\right\}$ is a basis of M_p, then so is $\left\{\frac{m_1}{1}, \cdots, \frac{m_n}{1}\right\}$ since $\frac{m}{s} = \sum \frac{a_i}{t_i}\frac{m_i}{s_i} = \frac{a_i}{s_i t_i}\frac{m_i}{1}$. Consider morphism $A^n \to M$, $e_i \mapsto m_i$, suppose $0 \to K \to A^n \to M \to L \to 0$ is exact, we can localize at $\mathfrak{p}$, we get exact sequence $0 \to K_\mathfrak{p} \to A^n_\mathfrak{p} \to M_\mathfrak{p} \to L_\mathfrak{p} \to 0$, here $A^n_\mathfrak{p} \to M_\mathfrak{p}$, $\frac{e_i}{1} \mapsto \frac{m_i}{1}$ is an isomorphism, hence $K_\mathfrak{p} = L_\mathfrak{p} = 0$, and K as a submodule of A^n, L as a quotient of M are both finitely generated, thus there exists $f \in A - \mathfrak{p}$ such that $K_f = L_f = 0$

□

Exercise 35.21 $H = V(f)$ is a hypersurface, $f(t_1, \cdots, t_n, x) = a_0(t_1, \cdots, t_n)x^m + \cdots + a_m(t_1, \cdots, t_n)$



φ is finite iff $a_0 \neq 0$ is a constant. φ is quasifinite $\Rightarrow a_0, \cdots, a_m$ don't have common zeros

Exercise 35.22 R is a commutative ring, M is a finitely generated R-module. Show that $\mathrm{Supp}(M) \subseteq V(\mathrm{Ann}_R(M))$

Solution. The stacks project.

□

Exercise 35.23 Show that direct image of take presheaves to presheaves and take sheaves to sheaves

Proof.

□

Exercise 35.24 If X is Noetherian, then so is $Y \subseteq X$

Exercise 35.25 M is a locally Euclidean, Hausdorff and connected manifold, then paracompactness imples second countable

Proof. An open cover by precompact coordinate charts has a locally finite open refinement $\{U_i\}$, each U_i is precompact and second countable

Define $S_0 = \{U_0\}$ for some U_0, since $\{U_i\}$ is locally finite, define S_1 to be the union of S_0 and those intersects U_0, repeating this process, we get $S_2, \cdots, S_n, \cdots$, define $S = \bigcup_{n=0}^{\infty} S_n$

M is connected thus path connected, pick any $x_0 \in U_0$, for any $x \in M$, there is a path γ connecting x_0 and x, since γ is compact, it can be covered by S. Hence S is an open cover of M, thus M is second countable

□

Exercise 35.26 If G is a discrete group, P is connected, $P \xrightarrow{p} X$ is a principal G bundle iff it is a regular cover with $\mathrm{Aut}(p) = G$

Solution. $P \xrightarrow{p} X$ is a fiber bundle thus a cover, G acts regularly on fibers and $G \leq \mathrm{Aut}(p)$

□

Exercise 35.27 Use Theorem 2.108 to prove homotopy invariance of maps on homology

Solution. Suppose $F : X \times I \to Y$ is a homotopy between f and g, we only need to prove i_0, i_1 are naturally chain homotopic since $Fi_0 = f, Fi_1 = g$

$$\begin{array}{ccccc}
C_{n+1}(X) & \longrightarrow & C_n(X) & \longrightarrow & C_{n-1}(X) \\
{\scriptstyle i_{0\sharp}, i_{1\sharp}}\downarrow\downarrow & & {\scriptstyle i_{0\sharp}, i_{1\sharp}}\downarrow\downarrow & & {\scriptstyle i_{0\sharp}, i_{1\sharp}}\downarrow\downarrow \\
C_{n+1}(X \times I) & \longrightarrow & C_n(X \times I) & \longrightarrow & C_{n-1}(X \times I) \\
{\scriptstyle F}\downarrow & & {\scriptstyle F}\downarrow & & {\scriptstyle F}\downarrow \\
C_{n+1}(Y) & \longrightarrow & C_n(Y) & \longrightarrow & C_{n-1}(Y)
\end{array}$$

Consider Top with model $\mathcal{M} = \{\Delta^n\}$, $F, G : Top \to Ch_{\geq 0}$, $F(X) = C_*(X)$, $G(X) = C_*(X \times I)$, $H_i(\Delta^n \times I) = 0$ for $i \neq 0$, $F_k(X) = \left\{\Delta^k \xrightarrow{\mathrm{id}} \Delta^k \xrightarrow{\sigma} X\right\}$, there is an obvious natural equivalence $\phi_0 : H_0F \to H_0G$, then lifts i_0, i_1 are naturally chain homotopic

□

Exercise 35.28 K is a CW complex, $X \xrightarrow{f} Y$ is a weak equivalence, then $[K, X] \to [K, Y]$ is a bijection

Exercise 35.29 Quotient map $X \xrightarrow{q} Y$ is a homeomorphim iff q is bijective

Solution. If q is bijective, then for any open subset $U \subseteq X$, $U = q^{-1}(q(U))$, by definition, $q(U)$ is open, i.e. q^{-1} is continuous

□

Exercise 35.30 If X is Hausdorff, then cofibration $A \xrightarrow{i} X$ is closed. This is not true if X is not Hausdorff as showed in Example 34.20

Solution. Suppose $A \xrightarrow{i} X$ is a not closed, $X \times I \xrightarrow{r} X \times \{0\} \cup A \times I$ is the retraction, pick any $x \in \overline{A} \setminus A$ with x_n converging to x, then $A \times \{1\} \ni r(x, 1) = r(\lim x_n, 1) = \lim r(x_n, 1) = \lim(x_n, 1) = (x, 1)$ which is a contradiction

□

Exercise 35.31 $\mathbb{R} \times \mathbb{R} \xrightarrow{\wedge} \mathbb{R}$, $\mathbb{R} \times \mathbb{R} \xrightarrow{\vee} \mathbb{R}$ are continuous

Solution. $x \wedge y = \frac{x+y-|x-y|}{2}$, $x \vee y = \frac{x+y+|x-y|}{2}$

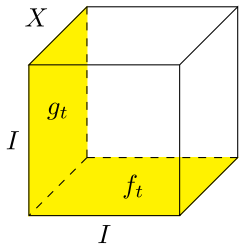
□

Exercise 35.32 $Y^I \to Y$, $\gamma \mapsto \gamma(0)$ and $Y^I \to Y \times Y$, $\gamma \mapsto (\gamma(0), \gamma(1))$ are Hurewicz fibrations

Solution. Need $g(x,s) = H(x,0,s)$, $f(x,t) = H(x,t,0)$ so that $g(x,0) = f(x,0)$

$$\begin{array}{ccc} X & \xrightarrow{g} & Y^I \\ \downarrow & \overset{H}{\nearrow} & \downarrow \\ X \times I & \xrightarrow{f_t} & Y \end{array}$$

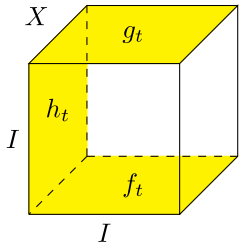
$X \times I^2$ can be deformed onto $X \times I \cup X \times I = X \times (I \cup I)$



Need $h(x,s) = H(x,0,s)$, $f(x,t) = H(x,t,0)$, $g(x,t) = H(x,t,1)$ so that $h(x,0) = f(x,0)$, $h(x,1) = g(x,0)$

$$\begin{array}{ccc} X & \xrightarrow{h} & Y^I \\ \downarrow & \overset{H}{\nearrow} & \downarrow \\ X \times I & \xrightarrow{(f_t, g_t)} & Y \times Y \end{array}$$

$X \times I^2$ can be deformed onto $X \times I \cup X \times I \cup X \times I = X \times (I \cup I \cup I)$



□

Exercise 35.33 $U \subsetneq \mathbb{R}^n$ is a nonempty open set, $x \in \partial U$, then there exists a discrete sequence $\{x_i\} \subseteq U$ converges to x

Solution. x is necessarily an accumulation point since $\partial U \cap U = \varnothing$. Pick $x_0 \in U$, then we can find $\epsilon > 0$ such that $x_0 \notin B(x, \epsilon)$, then pick $x_1 \in B(x, \epsilon/2) \cap U$, and so on

□

Exercise 35.34 f is analytic near 0, by rotation of coordinates, we can always make f has terms only involve one variable

Exercise 35.35 Evaluate $\int_0^\infty e^{-s^2 - \frac{1}{s^2}} ds$

Solution. $\left(s - \frac{1}{s}\right)^2 = s^2 + \frac{1}{s^2} - 2$, let $x = s - \frac{1}{s}$ which is increasing on $(0, \infty)$ since $0 < s < \infty$, $-\infty < x < \infty$, then $s = \frac{x + \sqrt{x^2+4}}{2}$ and

$$\int_0^\infty e^{-s^2-\frac{1}{s^2}}ds = e^{-2}\int_{-\infty}^{+\infty} e^{-x^2}\left(\frac{1}{2}+\frac{x}{2\sqrt{x^2+4}}\right)dx = e^{-2}\int_0^\infty e^{-x^2}dx = \frac{e^{-2}\sqrt{\pi}}{2}$$

□

Exercise 35.36 f is holomorphic on the punctured unit disc, $p>0$, $\int_D |f(z)|^p\ dz<\infty$. What can we say about the singularity?

Solution. $|f(z)|^p = e^{p\log|f(z)|}$ is subharmonic by Example 17.8, thus essential singularity is impossible

$$|f(z)|^p \le \frac{4}{\pi|z|^2}\int_{|w-z|<|z|/2}|f(w)|^p\ dw \le \frac{C}{|z|^2}$$

Thus $|z|^{\frac{2}{p}}|f(z)|<\infty$

□

Exercise 35.37 $U\subseteq\Omega\subseteq\mathbb{C}$ are open, f is holomorphic on U, $\hat{U}_\Omega$ be the union of U and compact connected components of $\Omega\setminus U$. There exist $\{f_n\}$ holomorphic on Ω converging uniformly to f on compact subsets of U iff there exists g holomorphic on $H(\hat{U}_\Omega)$ such that $g|_U=f$

Solution. Assume $\hat{U}_\Omega = U\cup K_1\cup\cdots$, where K_i's are compact

Suppose $\{f_n\}$ holomorphic on Ω converging uniformly to f on compact subsets of U, by maximum principle, $\{f_n\}$ would be uniformly bounded around K_i, by Montel's theorem 19.14, there exists a subsequence of $\{f_n\}$ converges uniformly on K_i, thus converging to g holomorphic on $H(\hat{U}_\Omega)$, hence $g|_U=f$

Conversely, suppose g holomorphic on $H(\hat{U}_\Omega)$ such that $g|_U=f$, $\hat{U}_\Omega$ is simply connected, by Riemann mapping theorem 19.22, we can think of $\hat{U}_\Omega$ as the unit disc or $\mathbb{C}$, by Runge's theorem, there exist $\{f_n\}$ holomorphic on Ω uniformly converging to g on each disc. Thus there exist a subsequence of $\{f_n\}$ converging uniformly to g on compact subsets of $\hat{U}_\Omega$

□

Exercise 35.38 Let Ω be an open subset of $\mathbb{C}$, $\mathcal{D}=\{D_i\}$ be an open cover of Ω with disks. Given meromorphic functions h_i on D_i, not identically zero. Assume $g_{ij}=\frac{h_i}{h_j}$ are holomorphic on $D_i\cap D_j$, then there exist holomorphic function f_i with no zeros on D_i such that $f_i = g_{ij}f_j$

Solution. It suffices to prove $H^1(\Omega,\mathcal{O}^*)=0$, since then $H^1(\mathcal{D},\mathcal{O}^*)=0$, $(g_{ij})\in Z^1(\mathcal{D},\mathcal{O}^*)=B^1(\mathcal{D},\mathcal{O}^*)$, i.e. there exists $(f_i)\in C^0(\mathcal{D},\mathcal{O}^*)$ such that $f_i=g_{ij}f_j$

Consider exact sequence of sheaves $0\to\mathbb{Z}\hookrightarrow\mathcal{O}\overset{\exp}{\to}\mathcal{O}^*\to 0$, then we get a long exact sequence $\cdots\to H^1(\Omega,\mathcal{O})\to H^1(\Omega,\mathcal{O}^*)\to H^2(\Omega,\mathbb{Z})\to\cdots$, $H^1(\Omega,\mathcal{O})=0$ by Mittag-Leffler theorem

□

Exercise 35.39 For each real r such that $0<|r|<1$, prove that there exists at most one real s with $0<s<1$ for which $\Omega := D\setminus\{0,r,s\}$ admits an analytic automorphism different from the identity

Solution. Suppose $\Omega\overset{\phi}{\to}\Omega$ is an analytic automorphism, then $0,r,s$ are all removable singularities, by continuity, ϕ can be extended to $D\overset{\phi}{\to}D$, so is ϕ^{-1}, by continuity, we know ϕ is an automorphism of D, sending $\{0,r,s\}$ to itself bijectively

By Schwarz lemma, we know that an automorphism ϕ of D with $\phi(\alpha)=0$ iff $\phi=e^{i\theta}\frac{z-\alpha}{1-\overline{\alpha}z}$

Now suppose ϕ is an automorphism different from the identity, if $0<r=s<1$, then $\phi=-\frac{z-r}{1-rz}$ is a choice, now we assumen $r\neq s$

1. $\phi(0)=0$

   $\phi=e^{i\theta}z$, $\phi(r)=s$, but $0<s<1$, thus $s=|r|$

2. $\phi(r) = 0$

   $\phi = e^{i\theta}\frac{z-r}{1-\overline{r}z}$, $\phi(0) = -re^{i\theta}$

   1. $\theta = \pi$, $\phi(0) = r$, then $s = \phi(s) \Rightarrow \overline{r}s^2 - 2s + r = 0 \Rightarrow s = \frac{1+\sqrt{1-|r|^2}}{\overline{r}}$ or $\frac{1-\sqrt{1-|r|^2}}{\overline{r}}$, r has to be a positive real number and $s = \frac{1-\sqrt{1-r^2}}{r}$

   2. $s = \phi(0) = |r|$

3. $\phi(s) = 0$

   $\phi = e^{i\theta}\frac{z-s}{1-sz}$, $\phi(0) = -se^{i\theta}$

   1. $\theta = \pi$, $\phi(0) = s$, then $r = \phi(r) \Rightarrow sr^2 - 2r + s = 0 \Rightarrow s = \frac{2r}{1+r^2}$.

   2. $r = \phi(0) = -se^{i\theta}$, $s = \phi(r) \Rightarrow s^2 = 1$ which is impossible

□

Exercise 35.40 $F \subseteq \mathbb{C}$ is closed, connected and noncompact, $\Omega = \mathbb{C} \setminus F$, then every $f \in \mathcal{O}(\Omega)$ has a primitive

Solution. It suffices to show that every connected component U of Ω is simply connected

Suppose U is not simply connected, then $\pi_1(U, z_0) \neq 0$, i.e. there is a simple(self non-intersecting) loop $\gamma \subseteq U$ with $\gamma(0) = \gamma(1)$ cannot be deform to z_0, by Jordan curve theorem 11.45, γ divides $\mathbb{C}$ into the exterior and the interior which is homeomorphic to the unit disc D, suppose $F \cap D$ is empty, then $\overline{D} \subseteq U$, γ can be deformed to z_0, giving a contradiction, hence $F \cap \overline{D}$ is a compact connected component of F which is also a contradiction

□

Exercise 35.41 Consider an open set $\Omega \subseteq \mathbb{C}^2$ such that

$$\{(z, w) \in \mathbb{C}^2 || z| \leq R_1, |w| \leq R_2\} \subseteq \Omega$$

for some positive reals R_1 and R_2. Let $f \in \mathcal{H}(\Omega)$ be such that $f(z, w) \neq 0$ for every z and w for which $|z| \leq R_1$, $|w| = R_2$

1. Prove that the number (counted with multiplicities) of zeros of $w \mapsto f(z, w)$ in $D(0, R_2)$ is the same for every $|z| \leq R_1$
2. Let $w_1(z), \cdots, w_m(z)$ denote the zeros of $w \mapsto f(z, w)$ (counted with multiplicities). Prove that for each $n \in \mathbb{N}$ the function

   $$z \mapsto w_1(z)^n + \cdots + w_m(z)^n$$

   is holomorphic for $z \in D(0, R_1)$
3. Deduce that nth elementary symmetric function σ_n of $w_1(z), \cdots, w_m(z)$ is holomorphic.
4. Prove that there exists a function h that is holomorphic and without any zeros on $\{(z, w) \in \mathbb{C}^2 || z| < R_1, |w| < R_2\}$ such that

   $$f(z, w) = h(z, w)[w^m + \sigma_1(z)w^{m-1} + \cdots + \sigma_{m-1}(z)w + \sigma_m(z)]$$

   for every z and w such that $|z| < R_1$ and $|w| < R_2$

Solution.

1. By Lemma 19.18, $\frac{1}{2\pi i}\int_{\partial D(0,R_2)} \frac{f_w(z,w)}{f(z,w)} dw$ is the number of zeros in $D(0, R_2)$ which is continuous, hence the same for every $|z| \leq R_1$
2. By Lemma 19.18, $\frac{1}{2\pi i}\int_{\partial D(0,R_2)} w^n \frac{f_w(z,w)}{f(z,w)} dw = w_1(z)^n + \cdots + w_m(z)^n$ is holomorphic
3. Directly follows from (2) thanks to Newton's identities

4. Since $\prod_{i=1}^{m}(w-w_i(z)) = w^m + \sigma_1(z)w^{m-1} + \cdots + \sigma_{m-1}(z)w + \sigma_m(z)$ is holomorphic

$$\frac{f(z,w)}{w^m + \sigma_1(z)w^{m-1} + \cdots + \sigma_{m-1}(z)w + \sigma_m(z)}$$

has no zeros on D and holomorphic on $\{R_2 - \varepsilon < |w| < R_2\}$, hence by Hartogs's extension theorem 19.25, can be extended to a holomorphic function $h(z,w)$, then $f(z,w) = h(z,w)[w^m + \sigma_1(z)w^{m-1} + \cdots + \sigma_{m-1}(z)w + \sigma_m(z)]$ on $\{R_2 - \varepsilon < |w| < R_2\}$, by identity theorem, this holds for all $|z| < R_1$ and $|w| < R_2$

□

Exercise 35.42 Suppose $p_1, \cdots, p_n$ are points on the compact Riemann surface X and $X' = X \setminus \{p_1, \cdots, p_n\}$. Suppose $f : X' \to \mathbb{C}$ is a non-constant holomorphic function. Show that the image of f comes arbitrarily close to every $c \in \mathbb{C}$

Solution. Suppose there exists $c \in \mathbb{C}$ such that $|f - c| \geq \varepsilon$ for some $\varepsilon > 0$, then $\frac{1}{f-c}$ would be a bounded holomorphic function on X', by Riemann's Removable singularity theorem, $\frac{1}{f-c}$ can be extended to a holomorphic function on X, but since X is compact, $\frac{1}{f-c}$ is a constant which is impossible

□

Exercise 35.43 Let X be a compact Riemann surface and let $X \xrightarrow{\sigma} X$ be a biholomorphic map of X onto itself, different from the identity. Let $a \in X$ be a point with $\sigma(a) \neq a$, and suppose that there is a non-constant meromorphic function f on X, holomorphic on $X \setminus \{a\}$, with a pole of order k at a. Prove that σ can have at most $2k$ fixed points on X

Solution. Suppose there are more than $2k$ fixed points of σ, then consider $f - f \circ \sigma^{-1} : X \to \mathbb{P}^1$ is holomorphic on $X \setminus \{a, \sigma^{-1}(a)\}$ with at least $2k+1$ zeros and with poles of order k at $a, \sigma^{-1}(a)$, but it should have as many poles as zeros which is a contradiction

□

Exercise 35.44 $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ and $\Lambda' = \mathbb{Z}\omega_1' + \mathbb{Z}\omega_2'$ are lattices in $\mathbb{C}$. Show that $\Lambda = \Lambda'$ iff there exists a matrix $A \in GL(2,\mathbb{Z})$ such that

$$\begin{pmatrix}\omega_1' \\ \omega_2'\end{pmatrix} = A\begin{pmatrix}\omega_1 \\ \omega_2\end{pmatrix}$$

Solution. First not that

$$\Lambda \subseteq \Lambda' \quad \Leftrightarrow \quad \begin{pmatrix}\omega_1 \\ \omega_2\end{pmatrix} = A\begin{pmatrix}\omega_1' \\ \omega_2'\end{pmatrix} \text{ for some } A \in \mathrm{M}(2,\mathbb{Z})$$

Hence we have

$$\Lambda = \Lambda' \quad \Leftrightarrow \quad \begin{pmatrix}\omega_1 \\ \omega_2\end{pmatrix} = A\begin{pmatrix}\omega_1' \\ \omega_2'\end{pmatrix}, \begin{pmatrix}\omega_1' \\ \omega_2'\end{pmatrix} = B\begin{pmatrix}\omega_1 \\ \omega_2\end{pmatrix} \text{ for some } A, B \in \mathrm{M}(2,\mathbb{Z})$$

Which is equivalent to $A \in \mathrm{GL}(2,\mathbb{Z})$

□

Exercise 35.45 $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$, $\Lambda' = \mathbb{Z}\omega_1' + \mathbb{Z}\omega_2'$ are lattices in $\mathbb{C}$ and $X = \mathbb{C}/\Lambda$, $X' = \mathbb{C}/\Lambda'$ are the corresponding complex tori

1. Prove that any holomorphic map $X \xrightarrow{f} X'$ is induced by a linear map $\mathbb{C} \xrightarrow{g} \mathbb{C}$ of the form $g(z) = \alpha z + \beta$, where $\alpha \in \mathbb{C}$ is such that $\alpha\Lambda \subseteq \Lambda'$. f is biholomorphic if and only if $\alpha\Lambda = \Lambda'$
2. Show that every torus $X = \mathbb{C}/\Lambda$ is isomorphic to a torus of the form $X(\tau) = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$, where $\tau \in \mathbb{C}$ satisfies $\mathrm{Im}(\tau) > 0$

3. Assume that $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{Z})$ and $\mathrm{Im}(\tau) > 0$. Let $\tau' := \frac{a\tau+b}{c\tau+d}$. Show that the tori $X(\tau)$ and $X(\tau')$ are biholomorphic

Solution.

1. Since $\mathbb{C}$ is the universal cover of $\mathbb{C}/\Lambda'$, $f \circ \pi : \mathbb{C} \to \mathbb{C}/\Lambda'$ has a lift $F : \mathbb{C} \to \mathbb{C}$, and locally we have $F = \pi'|_V^{-1} \circ f \circ \pi|_U$, thus F is holomorphic

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{F} & \mathbb{C} \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi'} \\ \mathbb{C}/\Lambda & \xrightarrow{f} & \mathbb{C}/\Lambda' \end{array}$$

   Fix $\omega \in \Lambda$, since $\pi(z+\omega) = \pi(z)$ for any $z \in \mathbb{C}$, we have $F(z+\omega) - F(z) \in \Lambda'$, hence $F(z+\omega) - F(z)$ is a continuous function of z but Λ' is discrete, thus $F(z+\omega) - F(z) \equiv C_\omega$, where $C_\omega \in \Lambda'$ is a constant. Then $F'(z+\omega) = F'(z)$ which shows $F' : \mathbb{C} \to \mathbb{C}$ is doubly periodic function, thus induces $G : \mathbb{C}/\Lambda \to \mathbb{C}$ with $F = G \circ \pi$. Thus G must be a constant, so is F', therefore F has the form $F(z) = \alpha z + \beta$. Then for any $\omega \in \Lambda$, we have $F(\omega) - F(0) = \alpha\omega \in \Lambda'$, thus $\alpha\Lambda \subset \Lambda'$. If f is biholomorphic, then $\pi' \circ F = f \circ \pi \Rightarrow \pi \circ F^{-1} = f^{-1} \circ \pi'$, which implies $\{ {\scriptstyle \alpha\Lambda \subset \Lambda' \atop \scriptstyle \alpha^{-1}\Lambda' \subset \Lambda}} \Rightarrow \alpha\Lambda = \Lambda'$

$$\begin{array}{ccc} \mathbb{C} & \xleftarrow{F^{-1}} & \mathbb{C} \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi'} \\ \mathbb{C}/\Lambda & \xleftarrow{f^{-1}} & \mathbb{C}/\Lambda' \end{array}$$

   Conversely, if $\alpha\Lambda = \Lambda'$, $\pi \circ F^{-1}$ is doubly periodic and induce f^{-1}, hence f is biholomorphic

2. Suppose $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2, \mathrm{Im}\left(\frac{\omega_2}{\omega_1}\right) > 0$, define $\Lambda' = \mathbb{Z} + \mathbb{Z}\tau$, where $\tau = \frac{\omega_2}{\omega_1}$, we have $\omega_1\Lambda' = \Lambda$, thus X and $X(\tau)$ are biholomorphic

3. $X(\tau)$ and $X(\tau')$ are biholomorphic iff $\begin{pmatrix} \tau' \\ 1 \end{pmatrix} = \alpha A \begin{pmatrix} \tau \\ 1 \end{pmatrix}, \alpha \in \mathbb{C} - \{0\}, A \in \mathrm{SL}(2,\mathbb{Z})$ If $X(\tau)$ and $X(\tau')$ are biholomorphic, then $\mathbb{Z} + \mathbb{Z}\tau' = \Lambda' = \alpha\Lambda = \mathbb{Z}\alpha + \mathbb{Z}\alpha\tau$ for some $\alpha \in \mathbb{C} - \{0\}$, thus $\begin{pmatrix} \tau' \\ 1 \end{pmatrix} = A \begin{pmatrix} \alpha\tau \\ \alpha \end{pmatrix} = \alpha A \begin{pmatrix} \tau \\ 1 \end{pmatrix}$, for some $A \in \mathrm{SL}(2,\mathbb{Z})$, the other direction is easy

□

Exercise 35.46 Determine the branch points(or ramification points) of the map $f : \mathbb{C} \to \mathbb{P}^1$ with

$$f(z) = \frac{1}{2}\left(z + \frac{1}{z}\right)$$

Solution. $f'(z) = \frac{1}{2}\left(1 - \frac{1}{z^2}\right)$ when $z \neq 0$, thus $1, -1$ are branch points.

Consider the chart $(P^1 - \{0\}, \varphi)$ with $\varphi(z) = \frac{1}{z}$

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{f} & \mathbb{P}^1 - \{0\} \\ & \searrow & \downarrow{\scriptstyle\varphi} \\ & & \mathbb{C} \end{array}$$

Thus $g(z) = \varphi \circ f(z) = \frac{z}{2(z^2+1)}$, $g'(z) = \frac{1-z^2}{2(z^2+1)}$, hence 0 is not a branch point

□

Exercise 35.47 If f and g are two elliptic functions with respect to the same lattice $\Omega \subseteq \mathbb{C}$, prove that there exists an irreducible polynomial $P(x,y) \in \mathbb{C}[x,y]$ such that $P(f,g) = 0$

Solution. If $f \equiv c$ is a constant, then $P(x,y) = x - c$ is an irreducible polynomial such that $P(f,g) = 0$, so we can assume f, g are not constants, Since $\mathcal{M}(X)$ is a finite algebraic extension of $\mathbb{C}(f)$, there exists rational functions $R_0, \cdots, R_n$ such that $R_0(f) + R_1(f)g + \cdots + R_n(f)g^n = 0$, then after multiplying denominators, we get a polynomial $P(x,y) \in \mathbb{C}[x,y]$ such that $P(f,g) = 0$, since $\mathbb{C}[x,y]$ is a UFD, $P = P_1 \cdots P_k$, where P_i are prime hence irreducible, then $0 = P_1(f,g) \cdots P_k(f,g) \in \mathcal{M}(X)$ which is a field, thus $P_j(f,g) = 0$ for some irreducible polynomial $P_j \in \mathbb{C}[x,y]$

□

Exercise 35.48 f is an elliptic function of order $n > 0$, then f' is an elliptic function of order m such that $n + 1 \leq m \leq 2n$. Both bounds can be attained

Solution. f' is elliptic since $f(z+\omega) = f(z) \Rightarrow f'(z+\omega) = f'(z)$ for all $\omega \in \Omega$. Suppose f has poles $[P_1], \cdots, [P_k]$ with multiplicities $r_1, \cdots, r_k$, $\sum r_i = n$, then f' also has poles $[P_1], \cdots, [P_k]$ with multiplicities $r_1 + 1, \cdots, r_k + 1$, $\sum r_i = n + k = m$, since $1 \leq k \leq n$, $n + 1 \leq m \leq 2n$

We can find an elliptic function f of order n which has $[P_1], \cdots, [P_{n-m}]$ as its poles with multiplicities $1, \cdots, 1, 2n + 1 - m$, then we get f' is another elliptic function which also has $[P_1], \cdots, [P_{n-m}]$ as its poles with multiplicities $2, \cdots, 2, 2n + 2 - m$, thus f' is of order m

□

Exercise 35.49 Prove that

$$\wp'(z) = \frac{2\,\sigma(z - \frac{\omega_1}{2})\,\sigma(z - \frac{\omega_2}{2})\,\sigma(z - \frac{\omega_3}{2})}{\sigma(\frac{\omega_1}{2})\,\sigma(\frac{\omega_2}{2})\,\sigma(\frac{\omega_3}{2})\,\sigma(z)^3}\,.$$

Solution. $\wp'(z)$ has a pole at $z = 0$ of order 3 and $\frac{\omega_1}{2}, \frac{\omega_2}{2}, \frac{\omega_3}{2}$ as simple roots, thus

$$\wp'(z) = \lambda \frac{\sigma(z - \frac{\omega_1}{2})\sigma(z - \frac{\omega_2}{2})\sigma(z - \frac{\omega_3}{2})}{\sigma(z)^3}$$

for some $\lambda \in \mathbb{C}$, multiply by z^3 on both sides, and let $z \to 0$, since $\lim_{z\to 0} \frac{z}{\sigma(z)} = 1, \lim_{z\to 0} z^3 \wp'(z) = -2$, we have

$$-2 = -\lambda \sigma\Big(\frac{\omega_1}{2}\Big)\sigma\Big(\frac{\omega_2}{2}\Big)\sigma\Big(\frac{\omega_3}{2}\Big) \Rightarrow \lambda = \frac{2}{\sigma(\frac{\omega_1}{2})\sigma(\frac{\omega_2}{2})\sigma(\frac{\omega_3}{2})}$$

Hence

$$\wp'(z) = \frac{2\sigma(z - \frac{\omega_1}{2})\sigma(z - \frac{\omega_2}{2})\sigma(z - \frac{\omega_3}{2})}{\sigma(\frac{\omega_1}{2})\sigma(\frac{\omega_2}{2})\sigma(\frac{\omega_3}{2})\sigma(z)^3}$$

□

Let $\Omega \subseteq \mathbb{C}$ be a lattice and $\wp(z)$ the associated Weierstrass $\wp$-function. We have seen that $\wp(z)$ satisfies the differential equation $(\wp'(z))^2 = p(\wp(z))$, where $p(x) = 4x^3 - g_2 x - g_3$. The following three problems examine the conditions under which the coefficients g_2 and g_3 of $p(x)$ are real numbers

Exercise 35.50 Prove that the following conditions are equivalent

1. $g_2, g_3 \in \mathbb{R}$
2. $G_k \in \mathbb{R}$ for all $k \geq 3$
3. $\wp(\bar{z}) = \overline{\wp(z)}$ for all $z \in \mathbb{C}$
4. $\overline{\Omega} = \Omega$ (the last condition says that Ω is a *real lattice*)

Solution. $(i) \Rightarrow (ii)$

$g_2 = 60G_4, g_3 = 140G_6 \in \mathbb{R} \Rightarrow G_4, G_6 \in \mathbb{R}$

Since

$$\begin{aligned}\wp(z) &= \frac{1}{z^2} + \sum_{n=2}^{\infty}(2n-1)G_{2n}z^{2n-2} \\ &= \frac{1}{z^2} + 3G_4z^2 + 5G_6z^4 + 7G_8z^6 + 9G_{10}z^8 + \cdots \\ \wp'(z) &= -\frac{2}{z^3} + \sum_{n=2}^{\infty}(2n-1)(2n-2)G_{2n}z^{2n-3} \\ &= -\frac{2}{z^3} + 6G_4z + 20G_6z^3 + 42G_8z^5 + 72G_{10}z^7 + \cdots \\ \wp''(z) &= \frac{6}{z^4} + \sum_{n=2}^{\infty}(2n-1)(2n-2)(2n-3)G_{2n}z^{2n-4} \\ &= \frac{6}{z^4} + 6G_4 + 60G_6z^2 + 210G_8z^4 + 504G_{10}z^6 + \cdots\end{aligned}$$

So we can conclude $\wp''(z) - 6\wp(z)^2 + 30G_4 = z\varphi(z)$, where $\varphi(z)$ is a holomorphic elliptic function, hence $\wp''(z) - 6\wp(z)^2 + 30G_4 = 0$, then the coefficients of $z^{2n}(n \geq 1)$ would be $(2n+1)(2n+2)(2n+3)(2n+4)G_{2n+4} - 6(2n+3)G_{2n+4}$ minus terms only involving $G_4, G_6, \cdots, G_{2n+2}$ and real numbers, thus by induction, we know $G_{2n+4} \in \mathbb{R}(n \geq 1)$

$(ii) \Rightarrow (iii)$

Since $\wp(z) = \frac{1}{z^2} + \sum_{n=2}^{\infty}(2n-1)G_{2n}z^{2n-2}$, if $G_k \in \mathbb{R}(k \geq 3)$, then $\wp(\bar{z}) = \overline{\wp(z)}$

$(iii) \Rightarrow (iv)$

The poles of $\overline{\wp(\bar{z})} = \wp(z)$ are exactly $\overline{\Omega}$, thus $\overline{\Omega} = \Omega$

$(iv) \Rightarrow (i)$

$g_2 = 60G_4 = 60\sum_{\omega\in\Omega^*}\frac{1}{\omega^4} = 60\sum_{\omega\in\overline{\Omega}^*}\frac{1}{\omega^4} = \overline{g_2} \Rightarrow g_2 \in \mathbb{R}$, similarly, $g_6 \in \mathbb{R}$

□

Exercise 35.51 We say that Ω is *real rectangular* if $\Omega = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ where $\omega_1 \in \mathbb{R}$ and $\omega_2 \in i\mathbb{R}$, and that Ω is *real rhombic* if $\Omega = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ where $\omega_2 = \overline{\omega}_1$. Prove that a lattice Ω is real if and only if it is real rectangular or real rhombic

Solution. If Ω is real rectangular or real rhombic, Ω is obviously a real lattice

Conversely, if Ω is a real lattice, suppose $\Omega = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$, then there exists $\omega \in \Omega \setminus (\mathbb{R} \cup i\mathbb{R})$, otherwise, $\omega_1 \in \mathbb{R}^*, \omega_2 \in i\mathbb{R}^*$ or $\omega_2 \in \mathbb{R}^*, \omega_1 \in i\mathbb{R}^*$, since ω_1, ω_2 are linear independent, but then $\omega = \omega_1 + \omega_2 \in \Omega \setminus (\mathbb{R} \cup i\mathbb{R})$ which is a contradiction

Since $\omega \in \Omega \setminus (\mathbb{R} \cup i\mathbb{R})$, $\omega + \overline{\omega} \in \mathbb{R}^*, \omega - \overline{\omega} \in i\mathbb{R}^*$, thus $\Omega \cap \mathbb{R}^* \neq \varnothing, \Omega \cap i\mathbb{R}^* \neq \varnothing$, let $\eta_1 = \min_{\eta\in\Omega\cap(0,\infty)}\eta$, then $\Omega \cap \mathbb{R} = \mathbb{Z}\eta_1$, otherwise $\exists\eta \in \mathbb{R} \setminus \mathbb{Z}\eta_1$, then $\eta - \left\lfloor\frac{\eta}{\eta_1}\right\rfloor\eta_1 \in \Omega \cap (0,\infty)$ which is a contradiction

Similarly, $\Omega \cap i\mathbb{R} = \mathbb{Z}\eta_2$ for some $\eta_2 \in i(0,\infty)$. If $\Omega = \mathbb{Z}\eta_1 + \mathbb{Z}\eta_2$, then Ω is real rectangular, if not, $\exists\gamma \in \Omega \setminus (\mathbb{R} \cup i\mathbb{R})$, such that $|\gamma| = \min_{\omega\in\Omega\setminus(\mathbb{R}\cup i\mathbb{R})}|\omega|$, then $\gamma + \overline{\gamma} = \eta_1$ or $-\eta_1$, otherwise $\gamma + \overline{\gamma} = k\eta_1$ for some $|k| \geq 2$

If $k = 2$, then $\gamma - \eta_1 = \eta_1 - \overline{\gamma} = -\overline{(\gamma - \eta_1)} \Rightarrow \gamma - \eta_1 \in i\mathbb{R} \Rightarrow \gamma \in \mathbb{Z}\eta_1 + \mathbb{Z}(\gamma - \eta_1) \subseteq \mathbb{Z}\eta_1 + \mathbb{Z}\eta_2$

If $k > 2$, then $\gamma - \eta_1 \notin \mathbb{R} \cup i\mathbb{R}$ and $|\gamma - \eta_1| < |\gamma|$, similarly for $k \leq -2$, these are all contradictions

Similarly, we know that $\gamma - \overline{\gamma} = \eta_2$ or $-\eta_2$

Now, for any $\omega \in \Omega \setminus (\mathbb{R} \cup i\mathbb{R})$, $\omega + \overline{\omega} = k\eta_1 = k(\gamma + \overline{\gamma})$ for some $k \neq 0$, then $\omega - k\gamma = k\overline{\gamma} - \overline{\omega} = -\overline{(\omega - k\gamma)} \Rightarrow \omega - k\gamma \in i\mathbb{R}$, if $\omega \neq k\gamma$, then $\omega - k\gamma = l\eta_2 = l(\gamma - \overline{\gamma}) \Rightarrow \omega \in \mathbb{Z}\gamma + \mathbb{Z}\overline{\gamma}$, therefore, we have $\Omega = \mathbb{Z}\gamma + \mathbb{Z}\overline{\gamma}$, Ω is real rhombic

□

Exercise 35.52 Let Ω be a real lattice. Define the real elliptic curve $E_{\mathbb{R}}$ to be the set $\{(x, y) \in \mathbb{R}^2 \mid y^2 = p(x)\}$. Prove that $E_{\mathbb{R}}$ has one or two connected components as Ω is real rhombic or real rectangular, respectively

Solution. The number of connected components of $E_{\mathbb{R}}$ is one or two if $p(x) = 0$ has one real root and two nonreal conjugate complex roots or three distinct real roots correspondingly

Since $\frac{\omega_1}{2}, \frac{\omega_2}{2}, \frac{\omega_3}{2}$ are simple roots of $\wp'(z)$, the three simple roots of $p(x)$ are $\wp(\frac{\omega_1}{2}), \wp(\frac{\omega_2}{2}), \wp(\frac{\omega_3}{2})$, since Ω is a real lattice, $G_k \in \mathbb{R}$ and $\wp(z) = \frac{1}{z^2} + \sum_{n=2}^{\infty}(2n-1)G_{2n}z^{2n-2}$

If Ω is real rectangular, then $\wp(\frac{\omega_1}{2}), \wp(\frac{\omega_2}{2})$ are both real, thus $E_{\mathbb{R}}$ has two connected components

If Ω is real rhombic, then $\wp(\frac{\omega_3}{2})$ is real $\wp(\frac{\omega_1}{2}) \neq \wp(\frac{\omega_2}{2})$ are nonreal conjugate, thus $E_{\mathbb{R}}$ has only one connected component

□

Exercise 35.53 $A(r, R) = \{r < |z| < R\}$ is biholomorphic to $\{s < |z| < S\}$ iff $R/r = S/s$, r can be 0, R can be ∞, but not at the same time

Solution. By scaling or inversion we can assume $r = s = 1$ and $|f(z)| \to 1$ as $|z| \to 1$. Suppose $f : A(r, R) \to A(s, S)$ is a biholomorphism, then consider the Laurent series $f = \sum_{k=-\infty}^{\infty} c_k z^k$, for $1 < t < R$, by Stokes theorem we have

$$A(t) = \frac{1}{2i}\int_{f(\{|z|=t\})} \bar{z}dz = \frac{1}{2i}\int_{|z|=t} \overline{f(z)}df(z) = \frac{1}{2i}\int_{|z|=t} \overline{f(z)}f'(z)dz = \pi\sum_{k\in\mathbb{Z}} k|c_k|^2\ t^{2k}$$

As $t \to 1$, we have $A(t) \to \pi \Rightarrow \sum k|c_k|^2 = 1$, thus

$$A(t) - \pi t^2 = \pi t^2 \sum_{k\in\mathbb{Z}} k|c_k|^2\ \left(t^{2k-2} - 1\right) \geq 0$$

Thus $A(t) \geq \pi t^2$, as $t \to R$, $A(t) \to \pi S^2 \geq \pi R^2 \Rightarrow S \geq R$. Therefore we have $S = R$

□

Exercise 35.54 Let $\mu : \mathcal{B}(\mathbb{R}) \to \mathbb{R}^+$ be a σ-additive set function defined on the Borel σ-algebra on $\mathbb{R}$ and let $D \subseteq \mathbb{R}$ be a discrete set with the property $x \in D$ if and only if there exists an open set U such that $x \in U$ and $\mu(U) > 0$. Show that μ can be expressed as a countable linear combination of measures of the form

$$\delta_x(A) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

Where $x \in D$ and $A \in \mathcal{B}(\mathbb{R})$

Proof. For each $x \in D$, denote $\mu(\{x\}) = c_x$. Since D is discrete, it is countable and any subset of D is closed. For any open set V in $\mathbb{R}$, by definition, $\mu(V - D) = 0$ since $V - D$ is an open set which doesn't intersect D. Hence

$$\mu(V) = \mu(V \cap D) + \mu(V - D) = \mu(V \cap D) = \sum_{x\in V\cap D} c_x = \sum_{x\in D} c_x\delta_x(V)$$

i.e. μ can be expressed as a countable linear combination of Dirac measures

□

Exercise 35.55 Let $c : [0,1] \to [0,1]$ denote the Cantor (ternary) function (known also as the Devil's staircase, or the Cantor-Lebesgue function). For each continuous real-valued function $f : [0,1] \to \mathbb{R}$, let $l(f)$ denote the Riemann integral

$$l(f) = \int_0^1 f(c(x))dx$$

Find explicitly a Borel measure $\mu : \mathcal{B}([0,1]) \to \mathbb{R}$ so that

$$l(f) = \int_{[0,1]} f d\mu$$

where $\mathcal{B}([0,1])$ denotes the Borel σ-algebra over $[0,1]$

Proof. Write m as the Lebesgue measure. Since c is a continuous function, the pushforward measure $\mu(E) = m(c^{-1}(E))$ is a Borel measure on $\mathcal{B}([0,1])$. Note that

$$\int_{[0,1]} \chi_E d\mu = \mu(E) = m(c^{-1}(E)) = \int_0^1 \chi_E(c(x))dx$$

and continuous functions are approximated by simple functions, thus

$$\int_{[0,1]} f d\mu = \int_0^1 f(c(x))dx$$

□

Exercise 35.56 Suppose $f(z)$ is a meromorphic function on $\mathbb{C}$ with finitely many zeros and poles at $\{z_1, \cdots, z_N\}$, i.e. e. $f(z)$ is holomorphic and nowhere vanishing on $\mathbb{C} \setminus \{z_1, \cdots, z_N\}$. Let $m_i \in \mathbb{Z}$ be the order of $f(z)$ at z_i. Furthermore, assume there is $A \in \mathbb{C} \setminus \{0\}$ and a real number $C > 0$, such that

$$|f(z) - A| \leq \frac{C}{|z|^2}$$

for all z with $|z|$ sufficiently large. Prove that

$$\sum_{i=1}^N m_i z_i = 0$$

Proof. Let $w = 1/z$, $g(w) = f(1/w) = f(z)$, then $|g(w) - A| \leq C|w|^2$, then $\frac{g(w)-A}{w^2}$ holomorphic and bounded around 0, denote as $h(w)$, we have $g(w) = A + w^2 h(w)$, hence for R large enough, $\epsilon = 1/R$ small enough, we have

$$\sum_{i=1}^N m_i z_i = \int_{|z|=R} z\frac{f'(z)}{f(z)}dz = \int_{|w|=\epsilon} \frac{g'(w)}{wg(w)}dw = 0$$

Since $\frac{g'(w)}{wg(w)}$ is holomorphic around 0

□

Exercise 35.57 Evaluate $\int_0^\infty \frac{\log x}{x^2+1}dx$

Solution.

$$\begin{aligned}\int_0^\infty \frac{\log x}{x^2+1}dx &= \int_{-\infty}^\infty \frac{ye^y}{e^{2y}+1}dy \\ &= \int_{-\infty}^\infty \frac{y}{e^y+e^{-y}}dy \\ &= 0\end{aligned}$$

□

Exercise 35.58 Let $\{f_n\}, n \geq 1$ be a sequence in $L^2([0,1])$ and f be a Lebesgue measurable function such that for every Lebesgue measurable set E in $[0,1]$

$$\int_E f_n dx \to \int_E f dx$$

as $n \to \infty$. Assume also that $\sup_n \int_0^1 |f_n|^2 \, dx < \infty$

1. Prove that $f \in L^2([0,1])$
2. Prove that for any function $g \in L^2([0,1])$

$$\int_0^1 f_n g dx \to \int_0^1 f g dx$$

Solution.

1. f_n converge to f in measure since otherwise there would exists $\epsilon > 0$ and E with $m(E) > 0$ such that $f_n - f \geq \epsilon$ on E, then

$$\int_E (f_n - f) dx \geq \epsilon m(E)$$

thus there is a subsequence $f_{n_k} \to f$ almost everywhere, by Fatou's lemma

$$\int_0^1 |f|^2 \, dx \leq \liminf_{k\to\infty} \int_0^1 |f_{n_k}|^2 \, dx \leq \sup_n \int_0^1 |f_n|^2 \, dx < \infty$$

2. Converges in measure $\Rightarrow$ converges weakly

$$\begin{aligned}\int_0^1 |(f_n - f)g| dx &= \int_{|f_n - f| \geq \epsilon} |f_n - f||g| dx + \int_{|f_n - f| < \epsilon} |f_n - f||g| dx \\ &\leq \| f_n - f \|_{L^2} \int_{|f_n - f| \geq \epsilon} |g| dx + \epsilon \| g \|_{L^2}\end{aligned}$$

□

Exercise 35.59 Suppose $f(z)$ is a holomorphic function on the unit disk with $|f(z)| \leq 3$ for all $|z| < 1$, and $f(1/2) = 2$. Show that $f(z)$ has no zeros in the disk $|z| < 1/8$. (Hint: first show $f(0) \neq 0$)

Solution. Consider $\phi = \frac{z - \frac{1}{2}}{1 - \frac{1}{2}z}, \psi = \frac{z - \frac{2}{3}}{1 - \frac{2}{3}z}$, $g = \psi \circ f/3 \circ \phi^{-1} \in \mathrm{Aut}(\mathbb{D})$,then $g(0) = 0$, by Schwartz lemma, $|g(z)| \leq |z|$, thus $g(-1/2) \neq -2/3 \Leftrightarrow f(0) \neq 0$, also it is easy to show that $\phi(\{|z| < 1/8\}) \subseteq \{|z| < 10/17\}$, and $10/17 < 2, 3$, f has no zeros in the disk $|z| < 1/8$

□

Exercise 35.60 (umd analysis qual 2021 January, problem 1) Does there exist a Lebesgue integrable function $f : [0, +\infty) \to \mathbb{R}$ which is nowhere differentiable and such that the following function $F : [0, +\infty) \to \mathbb{R}$

$$F(x) = \int_0^x f^{2021} dm$$

where m represents Lebesgue measure on $[0, +\infty)$, is well defined and infinitely differentiable? Provide an example or disprove its existence

Solution. Suppose such an f exists, then f is smooth since F is, assume f is not constantly zero, then $f = \sqrt[2021]{F}$ at where f is nonzero which is smooth, giving a contradiction

□

Exercise 35.61 (umd analysis qual 2021 January, problem 2) Let $f(z)$ be an entire function on $\mathbb{C}$, and suppose there is a constant $\lambda > 0$ such that for all $z \in \mathbb{C}$, $|\Re f(z)| \geq \lambda |\Im f(z)|$. Show that f must be constant

Solution. Pick $w_0 \notin \{|\Re z| \geq \lambda |\Im z|\}$, say i, then $\frac{1}{f - w_0}$ is bounded. $|f - i|^2 \geq \frac{\lambda^2}{1+\lambda^2} > 0$

□

Exercise 35.62 (umd analysis qual 2021 January, problem 3) Let $0 < \alpha < 1$. Verify the existence of a Lebesgue measurable set $A \subseteq [0, 1]$, with the property that for every open interval $(a, b) \subset [0, 1]$, we have $m(A \cap (a, b)) = \alpha m((a, b))$? Provide an example or disprove its existence

Solution. Suppose such an A exists, Let $G \supseteq A$ be a G_δ set with $m(G_\delta \setminus A) < \delta$, since $G_\delta = \bigcup I_i$ then we have

$$m(G_\delta) - \delta < m(A) = \sum m(A \cap I_i) = \sum \alpha m(I_i) = \alpha m(G_\delta) \Rightarrow m(A) \leq m(G_\delta) < \frac{\delta}{1 - \alpha}$$

Since δ can be arbitrarily small, so $m(A) = 0$, giving a contradiction

□

Exercise 35.63 (umd analysis qual 2021 January, problem 5) Let $\{f_n\} \subseteq L^1_m(\mathbb{R})$. Assume that the sequence f_n converges to f in $L^1_m(\mathbb{R})$. Furthermore, assume that there exists $M > 0$ such that for all $n = 1, \cdots$, we have $\| f_n \|_2 \leq M$. Verify whether or not f_n converges to f in $L^p_m(\mathbb{R})$ for $1 < p \leq 2$

Solution. Since $\{f_n\}$ converges to f in $L^1(\mathbb{R})$, there is a subsequence f_{n_k} such that $f_{n_k} \to f$ almost everywhere, thus

$$\int_{\mathbb{R}} |f|^2 \, dm = \int_{\mathbb{R}} \liminf_k |f_{n_k}|^2 \, dm \leq \liminf_k \int_{\mathbb{R}} |f_{n_k}|^2 \, dm \leq M^2$$

hence $f \in L^2(\mathbb{R})$, by interpolation, $f, f_n \in L^p(\mathbb{R})$ for $1 \leq p \leq 2$, since f_n is uniformally bounded, we may assume f_{n_k} also weakly converges to f, then $f_n \to f$ in $L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, again by intepolation, $f_n \to f$ in $L^p(\mathbb{R})$ for $1 \leq p \leq 2$

□

Exercise 35.64 (umd analysis qual 2021 January, problem 6) Define the following function of a complex variable z

$$F(z) = \int_0^\infty \frac{x^{z-1}}{e^x - 1} dx$$

Show that $F(z)$ is a holomorphic function for $\Re z > 2$. Prove that $F(z)$ admits an analytic continuation as a meromorphic function for $\Re z > 0$ whose only pole is simple at $z = 1$. Compute the residue of $F(z)$ at $z = 1$

Solution. Consider $\int_\epsilon^N \frac{x^{z-1}}{e^x - 1} dx$, we know that $F(z)$ is holomorphic for $\Re z > 2$. Define

$$G(z)=\int_0^\infty \frac{x^{z-1}}{e^x+1}dx$$

Then

$$F(z)-G(z)=\frac{1}{2^{z-1}}F(z)\Rightarrow\left(1-\frac{1}{2^{z-1}}\right)F(z)=G(z)$$

Note that $G(z)$ is holomorphic on $\Re z>0$ and $G(1)=\ln 2$. Note that

$$2^{z-1}=e^{(z-1)\ln 2}=1\Leftrightarrow z-1=\frac{2k\pi}{\ln 2}i$$

$(1-2^{1-z})'(1+2k\pi i/\ln 2)=-\ln 2$, hence $\operatorname{Res}(F,1)=-1$

□

Remark. Note that $F(z)=\zeta(z)\Gamma(z)$, $G(z)=\eta(z)\Gamma(z)$, here $\eta(z)$ is Dirichlet's eta function

Exercise 35.65 $E\overset{p}{\to}B$ is a Serre fibration, $A\overset{i}{\hookrightarrow}X$ is a subcomplex, if either p or i is a weak equivalence, then we have

$$\begin{array}{ccc} A & \xrightarrow{f} & E \\ {\scriptstyle i}\big\downarrow & \overset{\exists h}{\nearrow} & \big\downarrow{\scriptstyle p} \\ X & \xrightarrow{g} & B \end{array}$$

Solution. If p is a weak equivalence, then fibers are weak contractible

If i is a weak equivalence, then X deformation retracts onto A

□

Exercise 35.66 In category $\mathcal{C}$, if $X\overset{\phi_X}{\to}X'$, $Y\overset{\phi_Y}{\to}Y'$ are isomorphisms, then $Hom(X,Y)$, $Hom(X',Y')$ are in bijective correspondence

Solution. Consider $Hom(X,Y)\to Hom(X',Y')$, $f\mapsto\phi_Y f\phi_X^{-1}$ and $Hom(X',Y')\to Hom(X,Y)$, $f'\mapsto\phi_Y^{-1}f'\phi_X$ which are inverses to each other

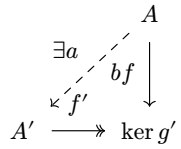
$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle \phi_X}\big\downarrow & & \big\downarrow{\scriptstyle \phi_Y} \\ X' & \xrightarrow{f'} & Y' \end{array}$$

□

Exercise 35.67 Suppose the bottom row of the following commutative diagram is exact, $gf=0$, then there exists a such that the following diagram commutes

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ {\scriptstyle \exists a}\,\vdots\downarrow & & {\scriptstyle b}\big\downarrow & & {\scriptstyle c}\big\downarrow \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array}$$

Solution. Since $0=cgf=g'bf$ and bottom row is exact, we have



□

Exercise 35.68 $F, G : \mathcal{C} \times \mathcal{D} \to \mathcal{E}$ are functors, $F \xrightarrow{\eta} G$ is a natural transformation iff η is natural on each factor

Solution. We have commutative diagram

$$\begin{array}{ccccc} & & F(f,1) & & F(1,g) & \\ F(A,B) & \longrightarrow & F(A',B) & \longrightarrow & F(A',B') \\ \eta_{A,B}\downarrow & & \eta_{A',B}\downarrow & & \eta_{A',B'}\downarrow \\ & & G(f,1) & & G(1,g) & \\ G(A,B) & \longrightarrow & G(A',B) & \longrightarrow & G(A',B') \end{array}$$

□

Exercise 35.69 A fully faithful functor $F : \mathcal{C} \to \mathcal{D}$ is injective on objects up to isomorphism

Solution. Suppose $F(X) = F(Y) = T$, let $f : X \to Y$ be the map corresponds to 1_T in $Hom(F(X), F(Y))$, then $f : X \to Y$ is an isomorphism because we can also let $g : Y \to X$ be the map corresponds to 1_T as in $Hom(F(Y), F(X))$, then $F(g \circ f) = F(g) \circ F(f) = 1_T \circ 1_T = 1_T$, thus $g \circ f$ corresponds to 1_T in $Hom(F(X), F(X))$, but $F(1_X) = 1_{F(X)} = 1_T$, thus $g \circ f = 1_X$, similarly, $f \circ g = 1_Y$

□

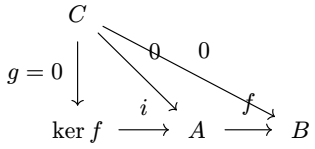
Exercise 35.70 Suppose $\mathcal{A}$ is an abelian category, show $\mathcal{A}$ is balanced. For any $A \xrightarrow{f} B$, $\ker f \xrightarrow{i} A$ is a monomorphism, $B \xrightarrow{\pi} \operatorname{coker} f$ is an epimorphism, and $\operatorname{im} f \coloneqq \ker \operatorname{coker} f$, $\operatorname{coim} f \coloneqq \operatorname{coker} \ker f$ are isomorphic

Solution. Suppose $A \xrightarrow{f} B$ is a bimorphism, it is the equaliser of

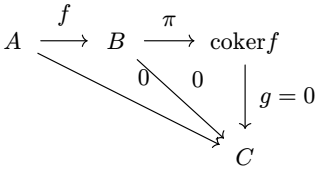
$$B \rightrightarrows \operatorname{coker} f$$

, then $\pi = 0$, $\operatorname{coker} f = 0$, but $A \xrightarrow{1_A} A$ is the kernel of $A \to 0$, hence A, B are isomorphic

$\ker f \xrightarrow{i} A$ is a monomorphism due to the following diagram

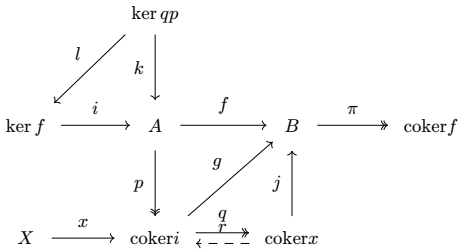


$B \xrightarrow{\pi} \operatorname{coker} f$ is a monomorphism due to the following diagram



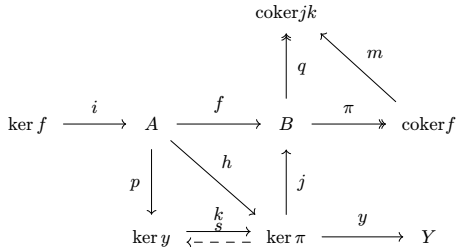
Now let's show coimage and image are isomorphic. $fi = 0$ induces $\operatorname{coker} i \xrightarrow{g} B$, claim that g is monic

Suppose $X \xrightarrow{x} \operatorname{coker} i$ is morphism such that $gx = 0$, it induces $\operatorname{coker} x \xrightarrow{j} B$, since $qpk = 0$, $fk = jqpk = 0$ induces $\ker qp \xrightarrow{l} \ker f$, since qp is epi, $pk = pil = 0$ induces $\operatorname{coker} x \xrightarrow{r} \operatorname{coker} i$, since p is epi, $p = rqp \Rightarrow rq = 1_{\operatorname{coker} i}$, hence q is monic, $qx = 0 \Rightarrow x = 0$

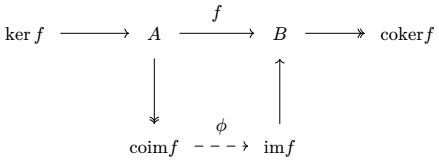


$\pi f = 0$ induces $A \overset{h}{\to} \ker \pi$, claim that h is epi

Suppose $\ker \pi \overset{y}{\to} Y$ is morphism such that $yh = 0$, it induces $A \overset{p}{\to} \ker y$, since $qjk = 0$, $qf = qjkp = 0$ induces $\operatorname{coker} f \overset{m}{\to} \operatorname{coker} jk$, since jk is monic, $qj = m\pi j = 0$ induces $\ker \pi \overset{s}{\to} \ker y$, since j is monic, $j = jks \Rightarrow ks = 1_{\ker \pi}$, hence k is epi, $yk = 0 \Rightarrow y = 0$



Since $\operatorname{im} f \to B$ is monic, $A \to \operatorname{coim} f$ is epi, g, h both induce $\operatorname{coim} f \overset{\phi}{\to} \operatorname{im} f$, then ϕ is monic and epi hence iso



□

Exercise 35.71 C is a bounded double complex with exact rows or exact columns, then $Tot(C)$ is exact

Solution. Without loss of generality, we may assume C is bounded in the first quadrant and has exact rows, use d', d'', d to denote row, column and total differentials

$Tot(C)$ is exact for all $n < 0$ since $Tot(C)_n = 0$ for all $n < 0$. now suppose $n \geq 0$, $d\left(\sum_{k=0}^{n} x_{k,n-k}\right) = 0$, i.e. $d' x_{k+1,n-k-1} + d'' x_{k,n-k} = 0$ for $0 \leq k < n$. Let $x_{0,n+1} = 0$, we can construct $x_{k,n+1-k}$ for $k > 0$ inductively such that $d'' x_{k,n-k+1} + d' x_{k+1,n-k} = x_{k,n-k}$ for $0 \leq k \leq n$ as follow:

For $k \geq -1$

$$\begin{aligned} d'(x_{k+1,n-k-1} - d'' x_{k+1,n-k}) &= d' x_{k+1,n-k-1} - d' d'' x_{k+1,n-k} \\ &= d' x_{k+1,n-k-1} + d'' d' x_{k+1,n-k} \\ &= d' x_{k+1,n-k-1} + d''(d'' x_{k,n-k+1} + d' x_{k+1,n-k}) \\ &= d' x_{k+1,n-k-1} + d'' x_{k,n-k} \\ &= 0 \end{aligned}$$

By exactness of rows, there exists $x_{k+2,n-k-1}$ such that

$$d' x_{k+2,n-k-1} = x_{k+1,n-k-1} - d'' x_{k+1,n-k} \Leftrightarrow d'' x_{k+1,n-k} + d' x_{k+2,n-k-1} = x_{k+1,n-k-1}$$

Therefore

$$
\begin{aligned}
d\left(\sum_{k=0}^{n+1} x_{k,n+1-k}\right) &= \sum_{k=1}^{n+1}(d' x_{k,n+1-k} + d'' x_{k,n+1-k}) \\
&= \sum_{k=1}^{n+1}(x_{k-1,n-k+1} - d'' x_{k-1,n-k+2} + d'' x_{k,n+1-k}) \\
&= \sum_{k=0}^{n}(x_{k,n-k} - d'' x_{k,n-k+1}) + \sum_{k=1}^{n+1} d'' x_{k,n+1-k} \\
&= \sum_{k=0}^{n} x_{k,n-k}
\end{aligned}
$$

□

Exercise 35.72 C, D are chain complexes with negative degree terms zeros, $H_n(C) = H_n(D) = 0$ for $n \neq 0$, then so is $C \otimes D$

Solution. Apply Exercise 35.71

□

Exercise 35.73 f is a retract of g in the arrow category, if g is an isomorphism, so is f

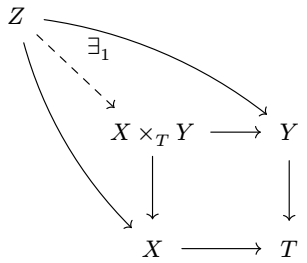
$$
\begin{array}{ccccc}
X & \xrightarrow{i} & Y & \xrightarrow{r} & X \\
{\scriptstyle f}\downarrow & & {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\
X' & \xrightarrow{i'} & Y' & \xrightarrow{r'} & X'
\end{array}
$$

Proof. $i' g^{-1} r$ is the inverse to f

□

Exercise 35.74 Suppose T is a terminal object, then the pullback $X \times_T Y$ is just the product $X \times Y$

Proof. For any $Z \to X$, $Z \to Y$, since T is final, there is a unique $Z \to T$, hence $Z \to X \to T$ is same as $Z \to Y \to T$, which then gives $Z \to X \times_T Y$, hence $X \times_T Y = X \times Y$



□

Exercise 35.75 In an abelian category $\mathcal{A}$, the equaliser of

$$
X \underset{g}{\overset{f}{\rightrightarrows}} Y
$$

is the isomorphic to the kernel of $f - g$

Exercise 35.76 $[n] = \{1, \cdots, n\}$, what is the cardinality of $\{f \in Aut([n]) | f(i) \neq i, \forall i \in [n]\}$

Solution. Consider $A_k = \{f \in Aut([n]) | f(k) = k\}$, by Inclusion-exclusion principle 1.4, we have

$$\begin{aligned} n! = \left|\bigcup_{i=1}^{n} A_n\right| &= \sum_{k=1}^{n}(-1)^k \sum_{i_1<\cdots<i_k} \left|A_{i_1} \cap \cdots \cap A_{i_k}\right| \\ &= \sum_{k=1}^{n}(-1)^k \binom{n}{k}(n-k)! \\ &= \sum_{k=1}^{n}(-1)^k \frac{n!}{k!} \end{aligned}$$

Thus the probability of picking such an auto morphism is $\sum_{i=1}^{n} \frac{(-1)^k}{k!}$ which approaches e^{-1} as n approaches infinity

□

Exercise 35.77 $E \xrightarrow{p} X$ is trivial iff there exist global sections $s_1, \cdots, s_n$ that they are linearly independent on each fiber

Exercise 35.78 $\phi \in \mathrm{Aut}(E)$, $\nabla^\phi = \phi^{-1} \circ \nabla \circ \phi$ is again a connection. Furthermore, ∇ is flat $\Rightarrow \nabla^\phi$ is flat

Solution.

$$\nabla^\phi(f \otimes s) = \phi^{-1} \circ \nabla(f \otimes \phi s) = \phi^{-1}(df \otimes \phi s + f \wedge \nabla(\phi s)) = df \otimes s + f \wedge \nabla^\phi s$$

Over a local trivialization, $\phi s = As$, then

$$A^{-1}(\nabla(As)) = A^{-1}(d(As) + \omega \wedge As) = A^{-1}(dAs + Ads + \omega \wedge As) = (d + A^{-1}dA + A^{-1}\omega A)s$$

Hence the connection form for ∇^ϕ is $A^{-1}dA + A^{-1}\omega A$. If $d\omega + \omega \wedge \omega = 0$, then

$$\begin{aligned} &d(A^{-1}dA + A^{-1}\omega A) + (A^{-1}dA + A^{-1}\omega A) \wedge (A^{-1}dA + A^{-1}\omega A) \\ &= dA^{-1}dA + dA^{-1} \wedge \omega A + A^{-1}d\omega A - A^{-1}\omega \wedge dA \\ &+A^{-1}dA \wedge A^{-1}dA + A^{-1}dA \wedge A^{-1}\omega A + A^{-1}\omega A \wedge A^{-1}dA + A^{-1}\omega A \wedge A^{-1}\omega A \\ &= dA^{-1}dA + dA^{-1} \wedge \omega A - A^{-1}\omega \wedge \omega A - A^{-1}\omega \wedge dA \\ &+A^{-1}dA \wedge A^{-1}dA + A^{-1}dA \wedge A^{-1}\omega A + A^{-1}\omega \wedge dA + A^{-1}\omega \wedge \omega A \\ &= dA^{-1} \wedge (dA + \omega A) + A^{-1}dA \wedge A^{-1}(dA + \omega A) \\ &= (dA^{-1} + A^{-1}dAA^{-1}) \wedge (dA + \omega A) \\ &= (dA^{-1} - A^{-1}AdA^{-1}) \wedge (dA + \omega A) \\ &= 0 \end{aligned}$$

□

Exercise 35.79 Suppose M, N are smooth manifolds of dimension m, n, $f : M \to N$ is a smooth map, $(x^1, \cdots, x^m)$, $(y^1, \cdots, y^n)$ are local coordinates around $p \in M$, $q = f(p) \in N$, then the corresponding matrix of df with respect to basis $\left(\frac{\partial}{\partial x^1}, \cdots, \frac{\partial}{\partial x^m}\right)$, $\left(\frac{\partial}{\partial y^1}, \cdots, \frac{\partial}{\partial y^n}\right)$ is $\left(\frac{\partial y^i}{\partial x^j}\right)$. In particular, this gives the change of coordinates formula

Solution.

$$df\left(\frac{\partial}{\partial x^i}\right)(g) = \frac{\partial(g \circ f)}{\partial x^i} = \sum_j \frac{\partial g}{\partial y^j}\frac{\partial y^j}{\partial x^i} = \sum_j \frac{\partial y^j}{\partial x^i}\frac{\partial}{\partial y^j}(g)$$

According to Exercise 35.96, $df = \sum_{i,j} \frac{\partial y^j}{\partial x^i} dx^i \otimes \frac{\partial}{\partial y^j}$, we can define higher differential $d^k f = \sum_{i_1,\cdots,i_k,j} \frac{\partial y^j}{\partial x^{i_1}\cdots\partial x^{i_k}} dx^{i_1}\cdots dx^{i_k} \otimes \frac{\partial}{\partial y^j}$

□

Exercise 35.80 Suppose $\omega \in \Omega^1(M)$, $X, Y \in TM$, then $d\omega(X, Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y])$

Solution. By linearity, we can assume $\omega = udv$, then

$$\begin{aligned} d\omega(X,Y) &= d(udv)(X,Y) \\ &= du \wedge dv(X,Y) \\ &= du(X)dv(Y) - du(Y)dv(X) \\ &= XuYv - YuXv \end{aligned}$$

And

$$\begin{aligned} & X(\omega(Y)) - Y(\omega(X)) - \omega([X,Y]) \\ &= X(udv(Y)) - Y(udv(X)) - udv([X,Y]) \\ &= Xudv(Y) + uX(dv(Y)) - Yudv(X) - uY(dv(X)) - u[X,Y]v \\ &= XuYv + uXYv - YuXv - uYXv - uXYv + uYXv \\ &= XuYv - YuXv \end{aligned}$$

□

Exercise 35.81 Suppose $\phi : M \to N$ is a map of smooth manifolds, then $X(f \circ \phi) = ((\phi_* X)f) \circ \phi$

Solution. $X(f \circ \phi)(p) = X_p(f \circ \phi) = \phi_p X_p(f) = (\phi_* X)_{\phi(p)}(f) = ((\phi_* X)f)(\phi(p))$

□

Exercise 35.82 Suppose X, Y are vector fields on M, $\phi : M \to N$ is a smooth map, then $\phi_*[X,Y] = [\phi_* X, \phi_* Y]$

Solution. Apply Exercise 35.81

$$\begin{aligned} \phi_*[X,Y](f) &= [X,Y](f \circ \phi) \\ &= X(Y(f \circ \phi)) - Y(X(f \circ \phi)) \\ &= X(((\phi_* Y)f) \circ \phi) - Y(((\phi_* X)f) \circ \phi) \\ &= ((\phi_* X)(\phi_* Y)f) \circ \phi - ((\phi_* Y)(\phi_* X)f) \circ \phi \\ &= ([\phi_* X, \phi_* Y]f) \circ \phi \\ &= [\phi_* X, \phi_* Y]f \end{aligned}$$

□

Exercise 35.83 Suppose $\mathfrak{g}$ is a real semisimple Lie algebra with a negative definite Killing form, then $\mathfrak{g}$ is the Lie algebra of some compact Lie group G

Exercise 35.84 Suppose A is a nonassociative algebra, $A = I_1 \oplus \cdots \oplus I_n$ is a direct sum of ideals, then $I_i I_j \subseteq I_i \cap I_j = 0$, hence $A = I_1 \times \cdots \times I_n$ can be viewed as product of ideals

Remark. If $A = I_1 \oplus \cdots \oplus I_n$ is just a direct sum of subalgebras, Exercise 35.84 may not hold true

Exercise 35.85 Let $S \subseteq M(n, \mathbb{F}), (\overline{\mathbb{F}} = \mathbb{F})$ be a set such that $[X,Y] = 0, \forall X, Y \in S$, then elements in S can be diagonalized simultaneously

Solution. Suppose $0 \neq V_\lambda$ is the λ-eigenspace of $X \in S$, then for any $Y \in S$, $XYv = YXv = \lambda Yv$ for $v \in V_\lambda$, thus $YV_\lambda \subseteq V_\lambda$, thus V_λ is an invariant subspace for all $Y \in \mathfrak{g}$, since Y are all semisimple, by induction we can write V as a direct sum of V_λ's, and they are all invariant under any $Y \in \mathfrak{g}$, we only need to show that all elements of $\mathfrak{g}$ can be diagonalized simultaneously on each V_λ, if $X|_{V_\lambda} = 1_{V_\lambda}$, then we are done, otherwise we can decompose it into smaller eigenspaces

□

Exercise 35.86 Let V be a finite dimensional $\mathbb{F}$ vector space with $\overline{\mathbb{F}} = \mathbb{F}$, and $\mathfrak{g} \leq \mathfrak{gl}(V)$ be a toral Lie algebra, then $\mathfrak{g}$ is abelian. Moreover, $X \in \mathfrak{g}$ can be diagonalized simultaneously

Solution. Suppose $ad(X)|_{\mathfrak{g}} \neq 0$ for some $X \in \mathfrak{g}$, since $\overline{\mathbb{F}} = \mathbb{F}$, there exists $0 \neq Y \in \mathfrak{g}$ and $\lambda \neq 0$ such that $ad(X)(Y) = \lambda Y$, by Proposition 8.53, $ad(Y)$ is semisimple, suppose λ_j, X_j are the eigenvalues

and linearly independent eigenvectors, then we have $X = \sum c_j X_j$ with $c_j \neq 0$, and $0 = ad(Y)(\lambda Y) = ad(Y)ad(X)(Y) = -ad(Y)^2(X) = -ad(Y)^2\big(\sum c_j X_j\big) = -\sum c_j \lambda_j^2 X_j$, thus $c_j\lambda_j^2 = 0 \Rightarrow \lambda_j = 0$, but $0 \neq \lambda Y = ad(X)(Y) = -ad(Y)(X) = -ad(Y)\big(\sum c_j X_j\big) = -\sum c_j\lambda_j X_j = 0$ which is a contradiction
Now that we know $[\mathfrak{g}, \mathfrak{g}] = 0$, use Lemma 35.85, we know all elements of $\mathfrak{g}$ are diagonalizable simultaneously

□

Exercise 35.87 Lie group homomorphism has constant rank

Solution. Let $\phi : G \to G'$ be a Lie group homomorphism, for any $g \in G$, it suffices to show $\text{rank}(d\phi)_g = \text{rank}(d\phi)_1$, since $\phi(gh) = \phi(g)\phi(h)$, thus $\phi \circ L_g = L_{\phi(g)} \circ \phi$, $(d\phi)_g(dL_g)_1 = d(L_{\phi(g)})_1(d\phi)_1$, and left multiplications are isomorphisms, we have $\text{rank}(d\phi)_g = \text{rank}(d\phi)_1$

□

Exercise 35.88 Let G be a Lie group, M, N be smooth manifolds with a G action, and G acts transitively on M, for any equivariant map $f : M \to N$, f has constant rank

Solution. For any $x \in M$, denote $y = f(x)$, it suffices to show $\text{rank}(df)_x = \text{rank}(df)_{gx}$ since G acts transitively on M, note that $f(gx) = gf(x)$, thus $f \circ L_g = L_g \circ f$, $(df)_{gx}(dL_g)_x = d(L_g)_y(df)_x$, and group actions are isomorphisms, we have $\text{rank}(df)_x = \text{rank}(df)_{gx}$

□

Exercise 35.89 If $\phi : G \to H$ be a bijective Lie group homomorphism, then it is an isomorphism

Solution. Apply Exercise 35.87 and Theorem 16.10

□

Exercise 35.90 Compact semisimple Lie group G has finite center

Solution. Since $\mathfrak{g} = \text{Lie}(G)$ is semisimple, $\text{Lie}(Z(G)) \leq Z(\mathfrak{g}) = 0$, thus $Z(G)$ is discrete, but G is compact, so $Z(G)$ is finite

□

Exercise 35.91 G is a topological group, A is called **symmetric** if $A = A^{-1}$

1. Topology of G is translation invariant, U is open $\Rightarrow xU, Ux$ are open
2. $e \in U$ is a neighborhood, then $e \in V \subseteq U$ a symmetric neighborhood
3. $e \in U$ is a neighborhood, then $e \in V \subseteq VV \subseteq U$ with V being a symmetric neighborhood
4. $H \leq G$ is a subgroup, then so is $\bar{H}$
5. Open subgroups of G are also closed(closed groups are not necessarily open, consider $\{e\}$)
6. $K_1, K_2 \subseteq G$ are compact sets, so is K_1K_2
7. Suppose G is a connected, U is a neighborhood of 1, then $G = \langle U \rangle$

Solution.

1. Multiplication by x is an isomorphism with x^{-1} being its inverse
2. Take $U \cap U^{-1}$
3. Since the multiplication $G \times G \to G$ is continuous, consider the preimage of U which contains $V_1 \times V_2$, take $V \subseteq V_1 \cap V_2$ symmetric
4. If $x_\alpha \to x$, $y_\beta \to y$, then $x_\alpha^{-1} \to x^{-1}$, $x_\alpha y_\beta \to xy$, since these maps are continuous. From this we know that $\bar{H} = \bigcap F$ wehre F runs over all closed subgroup containing H
5. Suppose $H \leq G$ is open, then $H = G \setminus \bigcup_{x \neq e} xH$ is closed, thus if G is connected, then $H = G$

6. K_1K_2 is the image of $K_1 \times K_2$ under multiplication
7. By b, we there is a symmetric neighborhood $1 \in V \subseteq U$, let V_k be the subset of elements can be written in the product of no more the k elements in V, then $V_1 = V$, $V_k = V_1V_{k-1}$ is open by induction, $\langle V \rangle = \bigcup_{k=1}^{\infty} V_k$ is also open, by e, G is generated by V hence by U, and if G is not connected, $G_0 = \langle V \rangle$ is called the identity component of G

□

Exercise 35.92 G is a topological group, if G is T_1, then G is Hausdorff, if G is not T_1, then $H := \overline{\{e\}}$ is normal subgroup, G/H is a Hausdorff topological group

Solution. If G is T_1, according to Exercise 35.91, for $x \neq y$, $\exists e \in VV \subseteq U$ with V a symmetric neighborhood of e disjoint from $y^{-1}x$, then $xV \cap yV = \emptyset$, suppose $z = xv_1 = yv_2$, then $y^{-1}x = v_2^{-1}v_1 \in VV$ thus reaches a contradiction

According to Proposiontion 35.91, $H = \bigcap H_i$, H_i runs over closed subgroups of G, thus H is the smallest closed subgroup, if H is normal, otherwise $xHx^{-1} \cap H$ is a smaller closed subgroup for some x

In G/H, identity is closed, by invariance of topology under translation, every point is closed, meaning G/H is T_1 thus Hausdorff

Checking G/H is still a topological group: $g \in \bigcup_x xH$ open in G, then $g^{-1} \in \left(\bigcup_x xH\right)^{-1} = \bigcup_x H^{-1}x^{-1} = \bigcup_x Hx^{-1} = \bigcup_x x^{-1}H$

If $V \times W \to VW \subseteq \bigcup_x xH$, then $vw \in \bigcup_x xH$, $\forall v \in V, w \in W$, then $\forall h \in H$, $vhw = vww^{-1}hw \in \bigcup_x xH$, therefore, $VH \times WH \to VHWH \subseteq \bigcup_x xH$, notice that VH is open as long as V is open since $VH = \bigcup_{h \in H} Vh$

□

Exercise 35.93 $(,)_B$ is the bilinear form given by matrix B, $O(B) = \{X \in GL_n(\mathbb{C}) | X^TBX = 1\}$, the Lie algebra is $\mathfrak{o}(B) = \{X \in M_n(\mathbb{C}) | X^TB + BX = 0\}$

Solution. $\frac{d}{dX}|_{X=0}(e^{X^T}Be^X) = X^TB + BX = 0$

□

Exercise 35.94 $T = \left\{ \begin{pmatrix} a & \\ & a^{-1} \end{pmatrix} \middle| a \in \mathbb{C}^\times \right\} \subseteq SL(2,\mathbb{C}) = G$ is the torus, the Weyl group $W(T) = N_G(T)/Z(T) = N/T \cong \mathbb{Z}/2\mathbb{Z}$ is generated by $\begin{pmatrix} & -1 \\ 1 & \end{pmatrix}$

Solution. Consider $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$,$ad - bc = 1$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x & \\ & x^{-1} \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} adx - bcx^{-1} & ab(x^{-1} - x) \\ cd(x - x^{-1}) & adx^{-1} - bcx \end{pmatrix} \in T$$

For any $x \in \mathbb{C}^\times$, which implies that $ab = cd = 0 \Rightarrow a = d = 0$ or $b = c = 0$ and

$$\begin{pmatrix} b & \\ & b^{-1} \end{pmatrix} \begin{pmatrix} & -b^{-1} \\ b & \end{pmatrix} = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix}$$

□

Exercise 35.95 $GL(n,F)$ acts transitively on flags

Exercise 35.96 $Hom(V,W) \to V^* \otimes W$, $A \mapsto \sum_{i,j} a_{ji} v_i^* \otimes w_j$ is an isomorphism where $A = (a_{ij})$ is the matrix with respect to basis $\{v_1^*, \cdots, v_m^*\}$, $\{w_1, \cdots, w_n\}$

Solution. $A(v_i) = \sum_j a_{ji} w_j$

□

Exercise 35.97 Let n be a positive integer. Let A be a real square matrix of size n, and let B be a real symmetric positive-definite matrix of size n

1. Show that there exists a unique real square matrix C of size n satisfying
$$BC + CB = A \tag{35.1}$$
In the following, this matrix C is denoted by $C_{A,B}$
2. Show that $BC_{A,B} = C_{A,B}B$ iff $AB = BA$

Solution.

1. $B = PDP^T$ for some orthogonal matrix P and digonal matrix $D = \text{diag}(d_1, \cdots, d_n)$ with $d_i > 0$, then 35.1 becomes
$$DP^TCP + P^TCPD = P^TAP$$
Denote as
$$DC' + C'D = A'$$
Then we have
$$(d_i + d_j)c'_{ij} = a'_{ij} \Rightarrow c'_{ij} = \frac{a'_{ij}}{d_i + d_j}$$
Therefore C' is uniquely determined, so is $C = PC'P^T$
2. We only need to prove $DC' = CD'$ iff $A'D = DA'$. But $A'D = DA'$ is equivalent to $D^2C' = C'D^2$. Therefore it suffices to prove $d_ic'_{ij} = d_jc'_{ij}$ iff $d_i^2c'_{ij} = d_j^2c'_{ij}$, i.e. $(d_i - d_j)c'_{ij} = (d_i + d_j)(d_i - d_j)c'_{ij}$ which is obviously true since $d_i + d_j > 0$

□

Exercise 35.98 Let n be a positive integer, and all matrices are supposed to be over the real numbers

1. Let A be a symmetric positive definite matrix of order n. Show that there exists a unique symmetric positive definite matrix R such that $R^2 = A$. We denote such R by $\sqrt{A}$
2. Let B be a nonsingular matrix of order n. Show that an orthogonal matrix Q that maximizes $f(Q) = \text{tr}(QB)$ satisfies
$$Q = \sqrt{B^TB}^{-1}B^T = B^T\sqrt{BB^T}^{-1}$$
3. Let G and H be symmetric positive definite matrices of order n. Find a square matrix L that minimizes
$$g(L) = \text{tr}\left\{(I - L)G(I - L)^T\right\}$$
subject to $LGL^T = H$

Solution.

1. $A = PDP^T$, here P is an orthogonal matrix, $D = \text{diag}(\lambda_1, \cdots, \lambda_n)$, $\lambda_i > 0$, take $R = P\sqrt{D}P^T$ where $\sqrt{D} = \text{diag}(\sqrt{\lambda_1}, \cdots, \sqrt{\lambda_n})$, and the uniqueness of R is obvious
2. Consider polar decomposition $B = \sqrt{BB^T}(\sqrt{BB^T}^{-1}B) = (B\sqrt{B^TB}^{-1})\sqrt{B^TB}$, $\sqrt{BB^T}, \sqrt{B^TB}$ are symmetric and positive definite, $\sqrt{BB^T}^{-1}B, B\sqrt{B^TB}^{-1}$ are orthogonal, then $\text{tr}(QB) = \text{tr}((QB\sqrt{B^TB}^{-1})\sqrt{B^TB}) = \text{tr}((\sqrt{BB^T}^{-1}BQ)\sqrt{BB^T})$. Hence the problem can be rewritten as given a symmetric positive definite matrix B, the orthogonal maximizer of $f(Q) = \text{tr}(QB)$ is $Q = I$, and by writing $B = PDP^T$, $\text{tr}(QB) = \text{tr}(P^TQPD)$, we may even assume $B = D$ is diagonal, but then $\text{tr}(QD) = \sum q_{kk}d_k$ which obtains maximum iff $q_{kk} = 1$ since $|q_{kk}| \leq 1$, this justifies our simplified version

3. $LGL^T = H$ can be rewritten as $Q^TQ = I$ with orthogonal matrix $Q = \sqrt{H}^{-1}L\sqrt{G}$, note that

$$\operatorname{tr}((I-L)G(I-L)^T) = \operatorname{tr}(G+H) - \operatorname{tr}(LG) - \operatorname{tr}(GL^T) = \operatorname{tr}(G+H) - 2\operatorname{tr}(LG)$$

Hence we only need to maximize $\operatorname{tr}(LG) = \operatorname{tr}(\sqrt{H}Q\sqrt{G}) = \operatorname{tr}(Q\sqrt{G}\sqrt{H})$, by 2. we know

$$Q = \sqrt{H}\sqrt{G}\sqrt{\sqrt{G}H\sqrt{G}}^{-1} \Rightarrow L = H\sqrt{G}\sqrt{\sqrt{G}H\sqrt{G}}^{-1}\sqrt{G}^{-1}$$

□

Exercise 35.99 Consider the heat equation with Neumann's boundary condition:

$$\begin{cases} u_t - \Delta u = 0, & \text{in } \Omega \times \mathbb{R}^+ \\ \frac{\partial u}{\partial n} = 0, & \text{on } \Gamma \times \mathbb{R}^+ \\ u(x,0) = v(x), & \text{in } \Omega \end{cases}$$

(a) Show that $\overline{u(t)} = \overline{v}$ for $t \geq 0$, where $\overline{v} = \frac{1}{|\Omega|}\int_\Omega v(x)dx$ denotes the average of v

(b) Show that $\| u(t) - \overline{v} \| \to 0$ as $t \to \infty$

Solution. **(a)** By divergence theorem, we have

$$0 = \int_\Omega u_t - \Delta u = \int_\Omega u_t + \nabla 1 \cdot \nabla u - \int_{\partial\Omega} \frac{\partial u}{\partial n} = \int_\Omega u_t = \left(\int_\Omega u\right)_t$$

Hence $\int_\Omega u = \int_\Omega v \Rightarrow \overline{u} = \overline{v}$

(b) By divergence theorem, we have

$$0 = \int_\Omega uu_t - u\Delta u = \frac{1}{2}\left(\int_\Omega u^2\right)_t + \int_\Omega |\nabla u|^2 \Rightarrow \frac{1}{2}\left(\int_\Omega u^2\right)_t = -\int_\Omega |\nabla u|^2 \leq 0$$

Hence $\int_\Omega u^2 \leq \int_\Omega v^2$. On the other hand, we have

$$\begin{aligned} 0 &= \int_\Omega (u_t - \Delta u)^2 \\ &= \int_\Omega u_t^2 - 2u_t\Delta u + (\Delta u)^2 \\ &= \int_\Omega u_t^2 - 2\nabla u_t \cdot \nabla u + (\Delta u)^2 \\ &= \int_\Omega 2(\Delta u)^2 + \left(\int_\Omega |\nabla u|^2\right)_t \end{aligned}$$

Which implies $\int_\Omega (\Delta u)^2 = -\frac{1}{2}\left(\int_\Omega |\nabla u|^2\right)_t$, thus

$$\left(\int_\Omega |\nabla u|^2\right)^2 = \left(\int_\Omega u\Delta u\right)^2 \leq \int_\Omega u^2 \cdot \int_\Omega (\Delta u)^2 \leq \int_\Omega v^2 \cdot \int_\Omega (\Delta u)^2 = -\frac{1}{2}\int_\Omega v^2 \cdot \left(\int_\Omega |\nabla u|^2\right)_t$$

Denote $\phi := \int_\Omega |\nabla u|^2$ which is a function of t, $C := \frac{1}{2}\int_\Omega v^2$, then the above equation becomes

$$\phi^2 \leq -C\phi' \Rightarrow 0 \geq \phi^2 + C\phi' \Rightarrow 0 \geq 1 + C\frac{\phi'}{\phi^2} = \left(t - \frac{C}{\phi}\right)'$$

Which implies

$$t - \frac{C}{\phi(t)} \leq -\frac{C}{\phi(0)} \Rightarrow \frac{C}{\phi(t)} \geq t + \frac{C}{\phi(0)} \geq t \Rightarrow \phi(t) \leq \frac{C}{t}$$

Thus $\phi(t) \to 0$ as $t \to \infty$

Now apply Poincaré's lemma, we get

$$\| u - \overline{v} \|_{L^2} = \| u - \overline{u} \|_{L^2} \leq C \| \nabla u \|_{L^2} \to 0, t \to \infty$$

□

Exercise 35.100

$$\begin{aligned}
\frac{d}{dr}\left(\frac{1}{|\partial B(x,r)|}\int_{\partial B(x,r)} u(y)dS\right) &= \frac{d}{dr}\left(\frac{1}{|\partial B(0,1)|}\int_{\partial B(0,1)} u(x+rz)dS\right) \\
&= \frac{1}{|\partial B(0,1)|}\int_{\partial B(0,1)} \frac{d}{dr}u(x+rz)dS \\
&= \frac{1}{|\partial B(0,1)|}\int_{\partial B(0,1)} z\cdot\nabla u(x+rz)dS \\
&= \frac{1}{|\partial B(x,r)|}\int_{\partial B(x,r)} \nu\cdot\nabla u(y)dS \\
&= \frac{1}{|\partial B(x,r)|}\int_{B(x,r)} \Delta u(y)dy \\
&= \frac{r}{n}\frac{1}{|B(x,r)|}\int_{B(x,r)} \Delta u(y)dy
\end{aligned}$$

$$\begin{aligned}
\frac{d}{dr}\left(\int_{B(x,r)} u(y)dy\right) &= \frac{d}{dr}\int_0^r\left(\int_{\partial B(x,s)} u(y)dS\right)ds \\
&= \int_{\partial B(x,r)} u(y)dS
\end{aligned}$$

Exercise 35.101 $\Box u = 0$ in $\mathbb{R}^{3+1}$, $u(x,0) = 0$, $u_t(x,0) = f(x) \in C^2(\mathbb{R}^3)$, show that $\int_0^\infty |u(0,t)|^2\, dt \leq C \| f \|_{L^2(\mathbb{R}^3)}$

Proof. Hint: $u(x,t) = \frac{t}{4\pi}\int_{S^2} f(x+tw)dS_w = -\frac{1}{4\pi}\int_{S^2}\int_t^\infty \frac{d}{d\lambda}f(x+\lambda t)d\lambda$

$u(0,t) = \frac{t}{4\pi}\int_{S^2} f(tw)dS_w$, $|u(x,t)| \leq \frac{C}{t}\int_{\mathbb{R}^3}|\nabla f|dx$

□

Exercise 35.102 Consider the differential equation

$$\frac{\partial}{\partial t}x(n,t) = x(n-1,t) + x(n+1,t) - 2x(n,t) \tag{35.2}$$

for a function $x(n,t)$ of an integer n and real number $t \geq 0$. Assume that the function $x(n,t)$ satisfies

$$x(n+N,t) = x(n,t) \tag{35.3}$$

for any integer n, where N is an integer larger than or equal to 3. Furthermore, let $e(m,n) = \exp\left(i\frac{2\pi mn}{N}\right)$ for integers m and n

1. Let $f_m(t)$ be a function of a real number $t \geq 0$ for an integer m with $f_m(0) = c_m$, where c_m is a complex number. Assume that the function of the form $x(n,t) = e(m,n)f_m(t)$ satisfies differential equation 35.2 and 35.3. Find $f_m(t)$
2. Let $(g_0, \cdots, g_{N-1})$ be an N-dimensional complex vector. Under the initial condition $x(n,0) = g_n$, $n = 0, \cdots, N-1$, find the solution of the differential equation 35.2 with condition 35.3
3. Find $\lim_{t\to\infty} x(n,t)$ for the solution $x(n,t)$ found in 2.

Solution.

1. Note that $e(m,n+N)=e(m,n)$, hence 35.3 is justified. Plug $x(n,t)=e(m,n)f_m(t)$ in 35.2, we have

$$e^{i\frac{2\pi mn}{N}}f'_m(t)=\Big(e^{i\frac{2\pi m(n-1)}{N}}+e^{i\frac{2\pi m(n+1)}{N}}-2e^{i\frac{2\pi mn}{N}}\Big)f_m(t)$$

   Which can be simplified as

$$f'_m(t)=\Big(e^{-i\frac{2\pi m}{N}}+e^{i\frac{2\pi m}{N}}-2\Big)f_m(t)=-4\sin^2\Big(\frac{\pi m}{N}\Big)f_m(t)=K_mf_m(t)$$

   Solve this with initial condition $f_m(0)=c_m$ we get $f_m(t)=c_me^{K_mt}$

2. Consider $x(n,t)=\sum_{m=0}^{N-1}e(m,n)f_m(t)$, then $x(n,0)=\sum_{m=0}^{N-1}e(m,n)f_m(0)=g_n$ which can be uniquely solved since $E=\{e(m,n)\}_{0\le m,n<N}$ is a Vandermonde matrix, suppose the solutions are $f_m(0)=c_m$, then use 1. to solve $f_m(t)$

3. Note that $K_m\le 0$ for $0\le m<N$ and $K_m=0$ iff $m=0$. Let $\omega=e^{\frac{2\pi i}{N}}$, then $\omega^N=1$ and the determinant of

$$\begin{bmatrix}1&1&\cdots&1\\ \omega&\omega^2&\cdots&\omega^{N-1}\\ \omega^2&\omega^4&\cdots&\omega^{2(N-1)}\\ \vdots&\vdots&\ddots&\vdots\\ \omega^{N-1}&\omega^{2(N-1)}&\cdots&\omega^{(N-1)^2}\end{bmatrix}$$

   without the $j+1$-th row would be

$$\begin{aligned}&\omega\cdots\hat{\omega^j}\cdots\omega^{N-1}\prod_{\substack{0\le p<q\le N-1\\ p,q\ne j}}(\omega^q-\omega^p)\\ &=(-1)^j\omega^{\frac{N(N-1)}{2}-j}\frac{\prod_{0\le p<q\le N-1}(\omega^q-\omega^p)}{\prod_{k\ne j}(\omega^k-\omega^j)}\\ &=(-1)^j\omega^{\frac{N(N-1)}{2}-j}\frac{\det E}{\omega^{j(N-1)}\prod_{k\ne 0}(\omega^k-1)}\\ &=\frac{(-1)^j\omega^{\frac{N(N-1)}{2}}}{N}\det E\end{aligned}$$

   thus by Cramer's rule we have

$$\begin{aligned}\lim_{t\to\infty}x(n,t)=c_0&=\frac{\begin{vmatrix}g_0&1&\cdots&1\\ g_1&\omega&\cdots&\omega^{N-1}\\ \vdots&\vdots&\ddots&\vdots\\ g_{N-1}&\omega^{N-1}&\cdots&\omega^{(N-1)^2}\end{vmatrix}}{\begin{vmatrix}1&1&\cdots&1\\ 1&\omega&\cdots&\omega^{N-1}\\ \vdots&\vdots&\ddots&\vdots\\ 1&\omega^{N-1}&\cdots&\omega^{(N-1)^2}\end{vmatrix}}\\ &=\frac{\sum_{j=0}^{N-1}(-1)^jg_j\frac{(-1)^j\omega^{\frac{N(N-1)}{2}}}{N}\det E}{\sum_{j=0}^{N-1}(-1)^j\frac{(-1)^j\omega^{\frac{N(N-1)}{2}}}{N}\det E}\\ &=\frac{g_0+\cdots+g_{N-1}}{N}\end{aligned}$$

□

36 Polylogarithms

36.1 Bloch Wigner polylogarithm

Definition 36.1 The *Bloch-Wigner polylogarithm* is defined as

$$\mathcal{L}_n(z) = \mathfrak{R}_n\left(\sum_{r=0}^{n-1} \frac{2^r B_r}{r!} \mathrm{Li}_{n-r}(z) \log^r|\ z\ |\right)$$

Here $\mathfrak{R}_n$ is Re if n is odd and Im if n is even. B_n are Bernoulli numbers. For instance, $\mathcal{L}_1(z) = \log|1-z|$, $\mathcal{L}_2(z) = \mathrm{Im}(\mathrm{Li}_2(z)) + \mathrm{Im}(\log(1-z))\log|z|$

Lemma 36.2 $\mathcal{L}_n(z) + (-1)^n \mathcal{L}_n(z^{-1}) = 0$. $\mathcal{L}_3(z) + \mathcal{L}_3(\frac{1}{1-z}) + \mathcal{L}_3(1-z^{-1}) = \zeta(3)$. $\mathcal{L}_2(z) - \mathcal{L}_2(\frac{1}{1-z}) = 0$

36.2 Exercises in polylogarithm

Exercise 36.3 (Derivatives of polylogarithms) If $m_i > 1$ for $1 \le i \le d$, then

$$\begin{aligned}
\frac{\partial}{\partial z_i}\mathrm{Li}_{m_1,\cdots,m_d}(z_1,\cdots,z_d) &= \sum_{k_1>\cdots>k_d\ge 1} \frac{z_1^{k_1}\cdots z_i^{k_i-1}\cdots z_d^{k_d}}{k_1^{m_1}\cdots k_i^{m_i-1}\cdots k_d^{m_d}} \\
&= \frac{1}{z_i}\mathrm{Li}_{m_1,\cdots,m_i-1,\cdots,m_d}(z_1,\cdots,z_d)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial z_1}\mathrm{Li}_{1,m_2,\cdots,m_d}(z_1,\cdots,z_d) &= \sum_{k_1>\cdots>k_d\ge 1} \frac{z_1^{k_1-1} z_2^{k_2}\cdots z_d^{k_d}}{k_2^{m_2}\cdots k_d^{m_d}} \\
&= \sum_{k_2>\cdots>k_d\ge 1} \frac{z_2^{k_2}\cdots z_d^{k_d}}{k_2^{m_2}\cdots k_d^{m_d}} \sum_{k_1=k_2+1}^{\infty} z_1^{k_1-1} \\
&= \sum_{k_2>\cdots>k_d\ge 1} \frac{z_2^{k_2}\cdots z_d^{k_d}}{k_2^{m_2}\cdots k_d^{m_d}} \frac{z_1^{k_2}}{1-z_1} \\
&= \frac{1}{1-z_1} \sum_{k_2>\cdots>k_d\ge 1} \frac{(z_1z_2)^{k_2}\cdots z_d^{k_d}}{k_2^{m_2}\cdots k_d^{m_d}} \\
&= \frac{1}{1-z_1}\mathrm{Li}_{m_2,\cdots,m_d}(z_1z_2,\cdots,z_d)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial z_d}\mathrm{Li}_{m_1,\cdots,m_{d-1},1}(z_1,\cdots,z_d) &= \sum_{k_1>\cdots>k_d\geq 1}\frac{z_1^{k_1}\cdots z_{d-1}^{k_{d-1}}z_d^{k_d-1}}{k_2^{m_2}\cdots k_{d-1}^{m_{d-1}}}\\
&= \sum_{k_1>\cdots>k_{d-1}\geq 2}\frac{z_1^{k_1}\cdots z_{d-1}^{k_{d-1}}}{k_1^{m_1}\cdots k_{d-1}^{m_{d-1}}}\sum_{k_d=1}^{k_{d-1}-1}z_d^{k_d-1}\\
&= \sum_{k_1>\cdots>k_{d-1}\geq 2}\frac{z_1^{k_1}\cdots z_{d-1}^{k_{d-1}}}{k_1^{m_1}\cdots k_{d-1}^{m_{d-1}}}\frac{1-z_d^{k_{d-1}-1}}{1-z_d}\\
&= \sum_{k_1>\cdots>k_{d-1}\geq 1}\frac{z_1^{k_1}\cdots z_{d-1}^{k_{d-1}}}{k_1^{m_1}\cdots k_{d-1}^{m_{d-1}}}\frac{1-z_d^{k_{d-1}-1}}{1-z_d}\\
&= \frac{1}{1-z_d}\sum_{k_1>\cdots>k_{d-1}\geq 1}\frac{z_1^{k_1}\cdots z_{d-1}^{k_{d-1}}}{k_1^{m_1}\cdots k_{d-1}^{m_{d-1}}}\\
&-\frac{1}{z_d(1-z_d)}\sum_{k_1>\cdots>k_{d-1}\geq 1}\frac{z_1^{k_1}\cdots(z_{d-1}z_d)^{k_{d-1}}}{k_1^{m_1}\cdots k_{d-1}^{m_{d-1}}}\\
&= \frac{1}{1-z_d}\mathrm{Li}_{m_1,\cdots,m_{d-1}}(z_1,\cdots,z_{d-1})\\
&-\frac{1}{z_d(1-z_d)}\mathrm{Li}_{m_1,\cdots,m_{d-1}}(z_1,\cdots,z_{d-1}z_d)
\end{aligned}$$

If $1<i<d$, then

$$\begin{aligned}
&\frac{\partial}{\partial z_i}\mathrm{Li}_{m_1,\cdots,1,\cdots,m_d}(z_1,\cdots,z_d)\\
&= \sum_{k_1>\cdots>k_d\geq 1}\frac{z_1^{k_1}\cdots z_i^{m_i-1}\cdots z_d^{k_d}}{k_2^{m_2}\cdots k_d^{m_d}}\\
&= \sum_{\substack{k_1>\cdots>k_{i-1}>k_{i+1}>\cdots>k_d\geq 1\\ k_{i-1}>k_{i+1}+1}}\frac{z_1^{k_1}\cdots z_{i-1}^{k_{i-1}}z_{i+1}^{k_{i+1}}\cdots z_d^{k_d}}{k_1^{m_1}\cdots k_{d-1}^{m_{d-1}}}\sum_{k_i=k_{i+1}+1}^{k_{i-1}-1}z_i^{k_i-1}\\
&= \sum_{\substack{k_1>\cdots>k_{i-1}>k_{i+1}>\cdots>k_d\geq 1\\ k_{i-1}>k_{i+1}+1}}\frac{z_1^{k_1}\cdots z_{i-1}^{k_{i-1}}z_{i+1}^{k_{i+1}}\cdots z_d^{k_d}}{k_1^{m_1}\cdots k_{d-1}^{m_{d-1}}}\frac{z_i^{k_{i+1}}-z_i^{k_{i-1}-1}}{1-z_i}\\
&= \sum_{k_1>\cdots>k_{i-1}>k_{i+1}>\cdots>k_d\geq 1}\frac{z_1^{k_1}\cdots z_{i-1}^{k_{i-1}}z_{i+1}^{k_{i+1}}\cdots z_d^{k_d}}{k_1^{m_1}\cdots k_{d-1}^{m_{d-1}}}\frac{z_i^{k_{i+1}}-z_i^{k_{i-1}-1}}{1-z_i}\\
&= \frac{1}{1-z_i}\sum_{k_1>\cdots>k_{i-1}>k_{i+1}>\cdots>k_d\geq 1}\frac{z_1^{k_1}\cdots z_{i-1}^{k_{i-1}}(z_iz_{i+1})^{k_{i+1}}\cdots z_d^{k_d}}{k_1^{m_1}\cdots k_{d-1}^{m_{d-1}}}\\
&-\frac{1}{z_i(1-z_i)}\sum_{k_1>\cdots>k_{i-1}>k_{i+1}>\cdots>k_d\geq 1}\frac{z_1^{k_1}\cdots(z_{i-1}z_i)^{k_{i-1}}z_{i+1}^{k_{i+1}}\cdots z_d^{k_d}}{k_1^{m_1}\cdots k_{d-1}^{m_{d-1}}}\\
&= \frac{1}{1-z_i}\mathrm{Li}_{m_1,\cdots,m_{i-1},m_{i+1},\cdots,m_d}(z_1,\cdots,z_{i-1},z_iz_{i+1},\cdots,z_d)\\
&-\frac{1}{z_i(1-z_i)}\mathrm{Li}_{m_1,\cdots,m_{i-1},m_{i+1},\cdots,m_{d-1}}(z_1,\cdots,z_{i-1}z_i,z_{i+1},\cdots,z_d)
\end{aligned}$$

36.3 Hodge structures

Definition 36.4 D is a simple normal crossing divisor of X, $Y=X-D$, $Y\overset{j}{\hookrightarrow}X$ is the embedding, the *log de Rham commplex* of X along D is $\Omega_X^*(\log D)$, which is the smallest chain complex of $j_*\Omega_Y^*$

closed under wedge product such that for any $f \in j_*\mathcal{O}_X^*(U)$ meromorphic along D, $\frac{df}{f} \in \Omega_X^*(\log D)(U)$. A section of $j_*\Omega_Y^*$ has *logarithmic poles* if it is a section of $\Omega_X^*(\log D)$

Proposition 36.5

1. Section ω of $j_*\Omega_Y^*$ has logarithmic poles along D iff both $\omega, d\omega$ have at most simple poles along D
2. $\Omega_X^1(\log D)$ is locally free and $\Omega_X^p(\log D) = \bigwedge^p \Omega_X^1(\log D)$
3. For $(X, D) = (X_1, D_1) \times (X_2, D_2) = (X_1 \times X_2, X_1 \times D_2 \cup X_2 \times D_1)$, isomorphism $\Omega_{Y_1}^* \boxtimes \Omega_{Y_2}^* \to \mathrm{pr}_{X_1}^*\Omega_{X_1}^* \otimes \mathrm{pr}_{X_2}^*\Omega_{X_2}^*$ induces isomorphism $\Omega_{X_1}^*(\log D_1) \boxtimes \Omega_{X_2}^*(\log D_2) \to \Omega_X^*(\log D)$
4. For $f : X_1 \to X_2$, $f^{-1}(D_2) = D_1$, $f^* : j_{2_*}\Omega_{Y_2}^* \to j_{1_*}\Omega_{Y_1}^*$ induces $f^* : \Omega_{X_2}^*(\log D_2) \to \Omega_{X_1}^*(\log D_1)$

Lemma 36.6 $X = D^n$, $D = \bigcup_{1\le i\le k} D_i$ with $D_i = \mathrm{pr}_i^{-1}(0)$, $Y = D^{(*)^k} \cup D^{n-k}$. Then $\Omega_X^1(\log D)$ is a free sheaf with base $\left\{\frac{dz_i}{z_i}\right\}_{1\le i\le k}$ and $\{dz_i\}_{k\le i\le n}$. In fact, any section of $j_*\mathcal{O}_Y^*$ meromorphic along D can be written locally as $f = g\prod_{i=1}^k z_i^{n_i}$, then

$$\frac{df}{f} = \frac{dg}{g} + \sum_{i=1}^k \frac{n_i}{z_i} dz_i$$

36.4 Hopf algebra structure

Definition 36.7 Iterated integrals form a Hopf algebra H with coproduct

$$\Delta I(a_0; a_1, \cdots; a_n; a_{n+1}) = \sum_{0=i_0<i_1<\cdots<i_k<i_{k+1}=n+1} I(a_{i_0}; a_{i_1}, \cdots, a_{i_k}; a_{ik+1}) \otimes \prod_{p=1}^k I(a_{i_p}; a_{i_p+1}, \cdots, a_{i_{p+1}-1}; a_{i_{p+1}})$$

The product is the just shuffle product, $\Delta_{i_1,\cdots,i_k}$ means those in grading $(i_1, \cdots, i_k)$. $\Delta'(x) = \Delta(x) - 1 \otimes x - x \otimes 1$ is the reduced coproduct. The space of indecomposables $Q(H) = H/(H_{>0} \cdot H_{>0})$ is mod products. The projection $\frac{1}{n}R = P : H \to Q(H)$, where R is defined inductively as $R(x) = nx - \mu(1 \otimes R)\Delta'(x)$, μ is multiplication. The cobracket is defined as $\delta(x) = (P \otimes P)(1 - \tau)\Delta(x)$, $\tau(x \otimes y) = y \otimes x$ Symbol of a multiple polylogarithm is defined to be $\Delta_{1,\cdots,1}(x)$, and omit log sign

36.5 Iterated integral

Definition 36.8 Chen's *iterated integral* is defined to be

$$\int_\gamma \omega_1(t)\cdots\omega_n(t) = \int_{0\le t_1\le\cdots\le t_n\le 1} \gamma^*\omega_1(t_1) \wedge \cdots \wedge \gamma^*\omega_n(t_n)$$

Theorem 36.9 [Unresolved source reference: Zhao: Multiple Zeta Functions, Multiple Polylogarithms and Their Special Values]

1. Inversion

$$\int_\gamma \omega_1(t)\cdots\omega_n(t) = (-1)^n \int_\gamma \omega_n(t)\cdots\omega_1(t)$$

2. Path composition ($\gamma_1(1) = \gamma_2(0)$)

$$\int_{\gamma_2\circ\gamma_1} \omega_1(t)\cdots\omega_n(t) = \sum_{i=0}^n \int_{\gamma_1} \omega_1(t)\cdots\omega_i(t) \int_{\gamma_2} \omega_{i+1}(t)\cdots\omega_n(t)$$

3. Shuffle Product

$$\textstyle\int_\gamma \omega_1(t)\cdots\omega_m(t) \int_\gamma \omega_{m+1}(t)\cdots\omega_{m+n}(t) = \sum_{\sigma \text{ runs over all } (m,n) \text{ shuffles}} \int_\gamma \omega_{\sigma^{-1}(1)}(t)\cdots\omega_{\sigma^{-1}(1)}(t)$$

4. Homotopy invariance
   $\int_\gamma \sum_{j=1}^n \omega_1^{(j)}\cdots\omega_m^{(j)}$ is homotopy invariant if $\omega_i^{(j)}$ are closed and $\sum_j \omega_{i-1}^{(j)} \wedge \omega_i^{(j)} = 0$

under the standard choice of path

$$I(0;a_1,\cdots,a_n;1)=\int_{0<t_1<\cdots<t_n<1}\frac{dt_1}{t_1-a_1}\wedge\cdots\wedge\frac{dt_n}{t_n-a_n}$$

Theorem 36.10 (Differential of iterated integrals)

$$dI(a_0;a_1,\cdots,a_n;a_{n+1})=\sum_{i=1}^{n}I(a_0;a_1,\cdots,\hat{a_i},\cdots,a_n;a_{n+1})d\log\frac{(a_{i+1}-a_i)}{a_{i+1}-a_i}$$

Proof. $dI(a_0;a_1,\cdots,a_n;a_{n+1})$ is

$$\begin{aligned}
&\sum_{i=1}^{n}\int_{a_0<t_1<\cdots<t_n<a_{n+1}}\frac{dt_1}{t_1-a_1}\wedge\cdots\wedge\frac{dt_i}{(t_i-a_i)^2}\wedge\cdots\wedge\frac{dt_n}{t_n-a_n}da_i\\
&=\sum_{i=1}^{n}\int_{a_0<t_1<\cdots<t_{i-1}<t_{i+1}<\cdots<t_n<a_{n+1}}\frac{dt_1}{t_1-a_1}\wedge\cdots\wedge\left(\frac{1}{t_{i-1}-a_i}-\frac{1}{t_{i+1}-a_i}\right)\wedge\cdots\wedge\frac{dt_n}{t_n-a_n}da_i\\
&=\sum_{i=1}^{n}\int_{a_0<t_1<\cdots<t_{i-1}<t_{i+1}<\cdots<t_n<a_{n+1}}\frac{dt_1}{t_1-a_1}\wedge\cdots\wedge\frac{dt_{i-1}}{(t_{i-1}-a_{i-1})(t_{i-1}-a_i)}\wedge\cdots\wedge\frac{dt_n}{t_n-a_n}da_i\\
&-\sum_{i=1}^{n}\int_{a_0<t_1<\cdots<t_{i-1}<t_{i+1}<\cdots<t_n<a_{n+1}}\frac{dt_1}{t_1-a_1}\wedge\cdots\wedge\frac{dt_{i+1}}{(t_{i+1}-a_i)(t_{i+1}-a_{i+1})}\wedge\cdots\wedge\frac{dt_n}{t_n-a_n}da_i\\
&=
\end{aligned}$$

$dI(0;a_1,\cdots,a_n;1)$ with respect to a_i is

$$\int_{0<t_1<\cdots<t_n<1}\frac{dt_1}{t_1-a_1}\wedge\cdots\wedge\frac{dt_i}{(t_i-a_i)^2}\wedge\cdots\wedge\frac{dt_n}{t_n-a_n}$$

By Fubini, we can first do integration

$$\int_{t_{i-1}}^{t_{i+1}}\frac{dt_i}{(t_i-a_i)^2}=\frac{1}{t_{i-1}-a_i}-\frac{1}{t_{i+1}-a_i}$$

On the other hand

□

36.6 $\mathrm{Li}_{1,1,1,1}$

$d\hat{L}_{1,1,1,1}-w_{1,1,1,1}+(w_2(x_2)-w_2(x_3)+w_{1,1}(x_2,x_3)-w_{1,1}(x_3,x_4))\hat{L}_{1,1}(x_1,x_2x_3x_4)+(w_2(x_1)-w_2(x_2)+w_{1,1}(x_1,x_2)-w_{1,1}(x_2,x_3))\hat{L}_{1,1}(x_1x_2x_3,x_4)-w_{1,1}(x_1,x_2)\hat{L}_{1,1}(x_3,x_4)$

36.7 Li 1,1,1

$$\begin{aligned}
\mathrm{Li}_{1,1,1}(x,y,z)=\int&-(dv_3dv_2+dv_{23}dw_{2,3})dv_1\\
&-(dv_3dv_{12}+dv_{123}dw_{12,3})dw_{1,2}\\
&-(dv_{23}dv_1+dv_{123}dw_{1,23})dw_{2,3}
\end{aligned}$$

$$\Lambda=\begin{bmatrix}
1 & & & & & & & \\
\mathrm{Li}_1(z) & 1 & & & & & & \\
\mathrm{Li}_1(yz) & & 1 & & & & & \\
\mathrm{Li}_1(xyz) & & & 1 & & & & \\
\mathrm{Li}_{1,1}(y,z) & \mathrm{Li}_1(y) & \mathrm{Li}_1(\frac{1-yz}{1-y}) & & 1 & & & \\
\mathrm{Li}_{1,1}(xy,z) & \mathrm{Li}_1(xy) & & \mathrm{Li}_1(\frac{1-xyz}{1-xy}) & & 1 & & \\
\mathrm{Li}_{1,1}(x,yz) & & \mathrm{Li}_1(x) & \mathrm{Li}_1(\frac{1-xyz}{1-x}) & & & 1 & \\
\mathrm{Li}_{1,1,1}(x,y,z) & \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x)\mathrm{Li}_1(\frac{1-yz}{1-y}) & \mathrm{Li}_{1,1}(\frac{1-xy}{1-x},\frac{1-xyz}{1-xy}) & \mathrm{Li}_1(x) & \mathrm{Li}_1(\frac{1-xy}{1-x}) & \mathrm{Li}_1(\frac{1-yz}{1-y}) & 1
\end{bmatrix}\tau_{1,1,1}(2\pi i)$$

Where

$$\mathrm{Li}_{1,1}\left(\frac{1-xy}{1-x},\frac{1-xyz}{1-xy}\right)=\mathrm{Li}_{1,1}(y,1)-\mathrm{Li}_{1,1}(y,x^{-1}y^{-1})+\mathrm{Li}_1(x^{-1}y^{-1})\mathrm{Li}_1(x^{-1})$$

$$\omega=\begin{bmatrix}0&&&&&&&\\-dv_3&0&&&&&&\\-dv_{23}&0&0&&&&&\\-dv_{123}&0&0&0&&&&\\0&-dv_2&-dw_{2,3}&0&0&&&\\0&-dv_{12}&0&-dw_{12,3}&0&0&&\\0&0&-dv_1&-dw_{1,23}&0&0&0&\\0&0&0&0&-dv_1&-dw_{1,2}&-dw_{2,3}&0\end{bmatrix}$$

To compute the monodromy around $x=0$, take $q(\epsilon)$ to be the loop $(x=\epsilon e^{it},y=z=\epsilon)$

$$\lim_{\epsilon\to 0}\int_{q(\epsilon)p(\epsilon)}-\int_{p(\epsilon)}=0$$

To compute the monodromy around $y=0$, take $q(\epsilon)$ to be the loop $(x=z=\epsilon,y=\epsilon e^{it})$

$$\lim_{\epsilon\to 0}\int_{q(\epsilon)p(\epsilon)}-\int_{p(\epsilon)}=0$$

To compute the monodromy around $z=0$, take $q(\epsilon)$ to be the loop $(x=y=\epsilon,z=\epsilon e^{it})$

$$\lim_{\epsilon\to 0}\int_{q(\epsilon)p(\epsilon)}-\int_{p(\epsilon)}=-\int_q dv_3\int_p(dv_2dv_1+dv_{12}dw_{1,2})=0$$

To compute the monodromy around $x=1$, take $q(\epsilon)$ to be the composition of $(x=(1-t)\epsilon+t(1-\epsilon),y=z=\epsilon)$, $(x=1-\epsilon e^{it},y=z=\epsilon)$ and $(x=(1-t)(1-\epsilon)+t\epsilon,y=z=\epsilon)$

$$\lim_{\epsilon\to 0}\int_{q(\epsilon)p(\epsilon)}-\int_{p(\epsilon)}=0$$

To compute the monodromy around $y=1$, take $q(\epsilon)$ to be the composition of $(x=z=\epsilon,y=(1-t)\epsilon+t(1-\epsilon))$, $(x=z=\epsilon,y=1-\epsilon e^{it})$ and $(x=z=\epsilon,y=(1-t)(1-\epsilon)+t\epsilon)$

$$\lim_{\epsilon\to 0}\int_{q(\epsilon)p(\epsilon)}-\int_{p(\epsilon)}=0$$

To compute the monodromy around $z=1$

$$\begin{aligned}\lim_{\epsilon\to 0}\int_{q(\epsilon)p(\epsilon)}-\int_{p(\epsilon)}&=-\int_q dv_3\int_p(dv_2dv_1+dv_{12}dw_{1,2})\\&=-2\pi i\mathrm{Li}_{1,1}(x,y)\end{aligned}$$

To compute the monodromy around $xy=1$, take $z=0$, x to be constant

$$\int_{qp}-\int_p=0$$

To compute the monodromy around $yz=1$, take x,y to be constants

$$\begin{aligned}\int_{q(\epsilon)p(\epsilon)} - \int_{p(\epsilon)} &= -\int_{q(\epsilon)} dv_{23}dv_3 \int_{p(\epsilon)} dv_1 - \int_{q(\epsilon)} dv_{123}dv_{23} \int_{p(\epsilon)} dw_{2,3} \\ &- \int_{q(\epsilon)} dv_{23} \int_{p(\epsilon)} dw_{2,3}dv_1 - \int_{q(\epsilon)} dv_{23} \int_{p(\epsilon)} dv_1 dw_{2,3} \\ &= -\int_{q(\epsilon)^{-1}} dv_3 dv_{23} \int_{p(\epsilon)} dv_1 - \int_{q(\epsilon)} dv_{123}dv_{23} \int_{p(\epsilon)} dw_{2,3} \\ &- \int_{q(\epsilon)} dv_{23} \int_{p(\epsilon)} dw_{2,3} \int_{p(\epsilon)} dv_1 \\ &= 2\pi i \log \frac{1-\epsilon^{-1}}{1-\epsilon} \log \frac{1-x}{1-\epsilon} \\ &-2\pi i \left(\log \frac{1-\epsilon}{1-\epsilon^3} + \log \frac{1-x}{1-\epsilon} \right) \left(\log \frac{y}{\epsilon} + \log \frac{1-z}{1-y} \right)\end{aligned}$$

Let $\epsilon \to 0$, we get

$$2\pi i \log(1-x) \log \frac{y-1}{y(1-z)} = -2\pi i \mathrm{Li}_1(x) \mathrm{Li}_1 \left(\frac{1-yz}{1-y} \right)$$

To compute the monodromy around $xyz = 1$

$$\begin{aligned}\int_{pq} - \int_p &= -\int_q dv_{123}dv_{23}dv_3 = \int_{q^{-1}} dv_3 dv_{23} dv_{123} \\ &= -2\pi i \int_z^{x^{-1}y^{-1}} \frac{dt}{1-t} \frac{dyt}{1-yt} \\ &= -2\pi i \int_{xyz}^1 \frac{dt}{xy-t} \frac{dt}{x-t} \\ &= -2\pi i \int_1^{xyz} \frac{dt}{x-t} \frac{dt}{xy-t} \\ &= -2\pi i \int_0^{xyz-1} \frac{dt}{(x-1)-t} \frac{dt}{(xy-1)-t} \\ &= -2\pi i \int_0^1 \frac{dt}{\frac{1-x}{1-xyz}-t} \frac{dt}{\frac{1-xy}{1-xyz}-t} \\ &= -2\pi i \mathrm{Li}_{1,1} \left(\frac{1-xy}{1-x}, \frac{1-xyz}{1-xy} \right)\end{aligned}$$

The monodromy representation ρ is as follows
For monodromy around $x = 0$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

For monodromy around $y = 0$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

For monodromy around $z = 0$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

For monodromy around $x = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

For monodromy around $y = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

For monodromy around $z = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

For monodromy around $xy = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

For monodromy around $yz = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

For monodromy around $xyz = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

36.8 Li 1,1

The following identity is true

$$\mathrm{Li}_{1,1}(x,y) = \mathrm{Li}_2\left(\frac{xy-y}{1-y}\right) - \mathrm{Li}_2\left(\frac{y}{y-1}\right) - \mathrm{Li}_2(xy)$$

$$\mathrm{Li}_2(1-x) = -\operatorname{Li}\,2(x) - \log x \log(1-x) + \frac{\pi^2}{6}$$

Note that $\log x \log(1-x) \to 0$ as $x \to 1$

$$\begin{aligned}\mathrm{Li}_{1,1}(x,y) &= \int \frac{dy}{1-y}\frac{dx}{1-x} + \frac{d(xy)}{1-xy}\left(\frac{dy}{1-y} - \frac{dx}{x(1-x)}\right) \\ &= \int d\log(1-y)d\log(1-x) + d\log(1-xy)d\log\frac{x(1-y)}{1-x} \\ &= \int dv_2 dv_1 + dv_{12}dw_1\end{aligned}$$

To compute the monodromy around $x=0$, take $q(\epsilon)$ to be the loop $(x=\epsilon e^{it}, y=\epsilon)$, we get 0.
To compute the monodromy around $y=0$, take $q(\epsilon)$ to be the loop $(x=\epsilon, y=\epsilon e^{it})$, we get 0.
To compute the monodromy around $x=1$, take $q(\epsilon)$ to be the composition of $(x=(1-t)\epsilon + t(1-\epsilon), y=\epsilon)$, $(x=1-\epsilon e^{it}, y=\epsilon)$ and $(x=(1-t)(1-\epsilon)+t\epsilon, y=\epsilon)$, we get 0.
To compute the monodromy around $y=1$, take $q(\epsilon)$ to be the composition of $(x=\epsilon, y=(1-t)\epsilon + t(1-\epsilon))$, $(x=\epsilon, y=1-\epsilon e^{it})$ and $(x=\epsilon, y=(1-t)(1-\epsilon)+t\epsilon)$, we get $-2\pi i \mathrm{Li}_1(x)$.
To compute the monodromy around $xy=1$, take q to be the loop $(x=x^0, y$ such that $\int_q d\log(1-xy) = 2\pi i)$, we get $-2\pi i \mathrm{Li}_1(\frac{1-xy}{1-x})$

The variation matrix is

$$\Lambda = \begin{bmatrix} 1 & & & \\ \mathrm{Li}_1(y) & 1 & & \\ \mathrm{Li}_1(xy) & & 1 & \\ \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x) & \mathrm{Li}_1\left(\frac{1-xy}{1-x}\right) & 1 \end{bmatrix} \tau_{1,1}(2\pi i)$$

$$\omega = \begin{bmatrix} 0 & & & \\ -dv_2 & 0 & & \\ -dv_{12} & 0 & 0 & \\ 0 & -dv_1 & -dw_1 & 0 \end{bmatrix}$$

Note that $\mathrm{Li}_1\left(\frac{1-xy}{1-x}\right) = -\log\left(\frac{x(y-1)}{1-x}\right) = \mathrm{Li}_1(y) - \mathrm{Li}_1(x^{-1}) = \mathrm{Li}_1(y) - \mathrm{Li}_1(x) - \log x - i\pi$
The monodromy representation ρ is as follows
For monodromy around $x=0$

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & -1 & 1 \end{bmatrix}$$

For monodromy around $y=0$, identity.
For monodromy around $x=1$

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & -1 & 1 & 1 \end{bmatrix}$$

For monodromy around $y=1$

$$\begin{bmatrix} 1 & & & \\ -1 & 1 & & \\ & & 1 & \\ & & -1 & 1 \end{bmatrix}$$

For monodromy around $xy=1$

$$\begin{bmatrix} 1 & & \\ & 1 & \\ -1 & 1 & \\ & & 1 \end{bmatrix}$$

$\hat{\mathcal{L}}_{1,1}(x) = \mathrm{Li}_{1,1}(x,y) - \frac{1}{2}v_{12}v_2 - \frac{1}{2}v_1v_2 + \frac{1}{2}v_1v_{12} - \frac{1}{2}u_1v_{12}$

36.9 Li 1,2

$$\mathrm{Li}_{1,2}(x,y) = dv_{12}du_{12}(du_1 - dv_1) + dv_2du_2dv_1 - (dv_2dv_1 + dv_{12}dw_{1,2})du_2$$

$$\Lambda = \begin{bmatrix} 1 & & & & & & \\ \mathrm{Li}_1(y) & 1 & & & & & \\ \mathrm{Li}_1(xy) & 0 & 1 & & & & \\ \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x) & \mathrm{Li}_1(y) - \mathrm{Li}_1(x^{-1}) & 1 & & & \\ \mathrm{Li}_2(y) & \log y & & & 1 & & \\ \mathrm{Li}_2(xy) & 0 & \log(xy) & & & 1 & \\ \mathrm{Li}_{1,2}(x,y) & \mathrm{Li}_1(x)\log y & g(x,y) & \log y & \mathrm{Li}_1(x) & -\mathrm{Li}_1(x^{-1}) & 1 \end{bmatrix} \tau_{1,2}(2\pi i)$$

Where $g(x,y) = \mathrm{Li}_2(y) - \log y\mathrm{Li}_1(x^{-1}) - \frac{1}{2}\log^2 x - \log x\mathrm{Li}_1(x) + \mathrm{Li}_2(x)$ $g(x,y) = -I((xy)^{-1}; y^{-1}, 0; 1)$

$$\omega = \begin{bmatrix} 0 & & & & & & \\ -dv_2 & 0 & & & & & \\ -dv_{12} & 0 & 0 & & & & \\ 0 & -dv_1 & d(-u_1 + v_1 - v_2) & 0 & & & \\ 0 & du_2 & 0 & 0 & 0 & & \\ 0 & 0 & d(u_1 + u_2) & 0 & 0 & 0 & \\ 0 & 0 & 0 & du_2 & -dv_1 & d(v_1 - u_1) & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & & & & & & \\ -I(0;y^{-1};1) & 1 & & & & & \\ -I(0;x^{-1}y^{-1};1) & 0 & 1 & & & & \\ -I(0;y^{-1},0;1) & I(y^{-1};0;1) & 0 & 1 & & & \\ -I(0;x^{-1}y^{-1},0;1) & 0 & I(x^{-1}y^{-1};0;1) & 0 & 1 & & \\ I(0;x^{-1}y^{-1},y^{-1};1) & -I(0;x^{-1}y^{-1};y^{-1}) & -I(x^{-1}y^{-1};y^{-1};1) & 0 & 0 & 1 & \\ I(0;x^{-1}y^{-1},y^{-1},0;1) & -I(0;x^{-1}y^{-1};y^{-1})I(y^{-1};0;1) & -I(x^{-1}y^{-1};y^{-1},0;1) & -I(0;x^{-1}y^{-1};y^{-1}) & -I(x^{-1}y^{-1};y^{-1};0) & I(y^{-1};0;1) & 1 \end{bmatrix}$$

For $x = 0$

$$\begin{bmatrix} 1 & & & & & & \\ 0 & 1 & & & & & \\ 0 & 0 & 1 & & & & \\ 0 & 0 & 0 & 1 & & & \\ 0 & 0 & -1 & 0 & 1 & & \\ 0 & 0 & 0 & 0 & 0 & 1 & \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix}$$

For $y = 0$

$$\begin{bmatrix} 1 & & & & & & \\ 0 & 1 & & & & & \\ 0 & 0 & 1 & & & & \\ 0 & 0 & 0 & 1 & & & \\ 0 & 0 & 0 & 0 & 1 & & \\ 0 & 0 & 0 & 0 & 0 & 1 & \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

For $x = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ 0 & 1 & & & & & & \\ 0 & 0 & 1 & & & & & \\ 0 & 0 & 0 & 1 & & & & \\ 0 & 0 & 0 & 0 & 1 & & & \\ 0 & 0 & 0 & 0 & 0 & 1 & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & \end{bmatrix}$$

For $y = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ 0 & 1 & & & & & & \\ 0 & 0 & 1 & & & & & \\ 0 & 0 & 0 & 1 & & & & \\ 0 & 0 & 0 & 0 & 1 & & & \\ 0 & 0 & 0 & 0 & 0 & 1 & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & \end{bmatrix}$$

For $xy = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ 0 & 1 & & & & & & \\ 0 & 0 & 1 & & & & & \\ 0 & 0 & 0 & 1 & & & & \\ 0 & 0 & 0 & 0 & 1 & & & \\ 0 & 0 & 0 & 0 & 0 & 1 & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & \end{bmatrix}$$

36.10 Li 2,1,1

$$\begin{bmatrix} 1 & & & & & & & & & & & \\ \mathrm{Li}_1(z) & 1 & & & & & & & & & & \\ \mathrm{Li}_1(yz) & 0 & 1 & & & & & & & & & \\ \mathrm{Li}_1(xyz) & 0 & 0 & 1 & & & & & & & & \\ \mathrm{Li}_{1,1}(y,z) & \mathrm{Li}_1(y) & \mathrm{Li}_1(\frac{1-yz}{1-y}) & 0 & 1 & & & & & & & \\ \mathrm{Li}_{1,1}(xy,z) & \mathrm{Li}_1(xy) & 0 & \mathrm{Li}_1(\frac{1-xyz}{1-xy}) & 0 & 1 & & & & & & \\ \mathrm{Li}_{1,1}(x,yz) & 0 & \mathrm{Li}_1(x) & \mathrm{Li}_1(\frac{1-xyz}{1-x}) & 0 & 0 & 1 & & & & & \\ \mathrm{Li}_2(xyz) & 0 & 0 & \log(xyz) & 0 & 0 & 0 & 1 & & & & \\ \mathrm{Li}_{1,1,1}(x,y,z) & \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x)\mathrm{Li}_1(\frac{1-yz}{1-y}) & \mathrm{Li}_{1,1}(\frac{1-xy}{1-x},\frac{1-xyz}{1-xy}) & \mathrm{Li}_1(x) & \mathrm{Li}_1(\frac{1-xy}{1-x}) & \mathrm{Li}_1(\frac{1-yz}{1-y}) & 0 & 1 & & & \\ \mathrm{Li}_{2,1}(xy,z) & \mathrm{Li}_2(xy) & 0 & \mathrm{Li}_2(x^{-1}y^{-1}) - \mathrm{Li}_2(z) + \mathrm{Li}_1(z)\log(xyz) & 0 & \log(xy) & 0 & \mathrm{Li}_1(z) & 0 & 1 & & \\ \mathrm{Li}_{2,1}(x,yz) & 0 & \mathrm{Li}_2(x) & \mathrm{Li}_2(x^{-1}) - \mathrm{Li}_2(yz) + \mathrm{Li}_1(yz)\log(xyz) & 0 & 0 & \log x & \mathrm{Li}_1(yz) & 0 & 0 & 1 & \\ \mathrm{Li}_{2,1,1}(x,y,z) & \mathrm{Li}_{2,1}(x,y) & \mathrm{Li}_2(x)\mathrm{Li}_1(\frac{1-yz}{1-y}) & \mathrm{Li}_{1,1,1}(\frac{1-y^{-1}z^{-1}}{1-z^{-1}},\frac{1}{1-y^{-1}z^{-1}},1-x^{-1}y^{-1}z^{-1}) & \mathrm{Li}_2(x) & -\mathrm{Li}_{1,1}(1-x,\frac{1-xy}{1-x}) & \log x \mathrm{Li}_1(\frac{1-yz}{1-y}) & \mathrm{Li}_{1,1}(y,z) & \log x & \mathrm{Li}_1(y) & \mathrm{Li}_1(\frac{1-yz}{1-y}) & 1 \end{bmatrix}$$

$$\int_{(xyz)^{-1}}^{1} \frac{dt}{(yz)^{-1}-t}\frac{dt}{z^{-1}-t} = \mathrm{Li}_{1,1}(y,z) + (u_1 - v_1 - \pi i)(u_{12} - v_{12} - \pi i) - \int_0^1 \frac{dt}{x-t}\frac{dt}{xy-t}$$

$$\omega = \begin{bmatrix} 0 & & & & & & & & & & \\ -dv_3 & 0 & & & & & & & & & \\ -dv_{23} & 0 & 0 & & & & & & & & \\ -dv_{123} & 0 & 0 & 0 & & & & & & & \\ & -dv_2 & -w_{2,3} & 0 & 0 & & & & & & \\ & -dv_{12} & 0 & -dw_{12,3} & 0 & 0 & & & & & \\ & & -dv_1 & -dw_{1,23} & 0 & 0 & 0 & & & & \\ & & & du_{123} & 0 & 0 & 0 & 0 & & & \\ & & & & -dv_1 & -dw_{1,2} & -dw_{2,3} & 0 & 0 & & \\ & & & & & du_{12} & 0 & -dv_3 & 0 & 0 & \\ & & & & & & du_1 & -dv_{23} & 0 & 0 & 0 \\ & & & & & & & & du_1 & -dv_2 & dw_{2,3} & 0 \end{bmatrix}$$

Monodromy around $x = 0, 1$, $y = 0, 1$, $xy = 1$, $z = 0$ are 0
Monodromy around $z = 1$ is $-2\pi i \mathrm{Li}_{2,1}(x, y)$
Monodromy around $yz = 1$ is $-2\pi i \mathrm{Li}_2(x)\mathrm{Li}_{1,1}(\frac{1-yz}{1-y})$
Monodromy around $xyz = 1$ is $-2\pi i I((xyz)^{-1}; 0, (yz)^{-1}, z^{-1}; 1)$
The monodromy representation ρ is as follows
For monodromy around $x = 0$

$$\begin{bmatrix}
1 & & & & & & & & & \\
 & 1 & & & & & & & & \\
 & & 1 & & & & & & & \\
 & & & 1 & & & & & & \\
 & & & & 1 & & & & & \\
 & & & -1 & & 1 & & & & \\
 & & & -1 & & & 1 & & & \\
 & & & 1 & & & & 1 & & \\
 & & & & -1 & & & & 1 & \\
 & & & & 1 & & & & & 1 \\
 & & & & & 1 & & & & \\
 & & & & & & 1 & & & \\
 & & & & & & & 1 & & \\
 & & & & & & & & 1 &
\end{bmatrix}$$

For monodromy around $y = 0$

$$\begin{bmatrix}
1 & & & & & & & & & \\
 & 1 & & & & & & & & \\
 & & 1 & & & & & & & \\
 & & & 1 & & & & & & \\
 & & -1 & & 1 & & & & & \\
 & & & -1 & & 1 & & & & \\
 & & & & & & 1 & & & \\
 & & & 1 & & & & 1 & & \\
 & & & & & -1 & & & 1 & \\
 & & & & 1 & & & & & 1 \\
 & & & & & & & & 1 & \\
 & & & & & & & & -1 & 1
\end{bmatrix}$$

For monodromy around $z = 0$

$$\begin{bmatrix}
1 & & & & & & & & & \\
 & 1 & & & & & & & & \\
 & & 1 & & & & & & & \\
 & & & 1 & & & & & & \\
 & & & & 1 & & & & & \\
 & & & & & 1 & & & & \\
 & & & & & & 1 & & & \\
 & & & 1 & & & & 1 & & \\
 & & & & & & & & 1 & \\
 & & & & & & & & & 1
\end{bmatrix}$$

For monodromy around $x = 1$

$$\begin{bmatrix}
1 & & & & & & & & & & \\
 & 1 & & & & & & & & & \\
 & & 1 & & & & & & & & \\
 & & & 1 & & & & & & & \\
 & & & & 1 & & & & & & \\
 & & & & & 1 & & & & & \\
 & -1 & 1 & & & & 1 & & & & \\
 & & & & & & & 1 & & & \\
 & & & -1 & 1 & & & & 1 & & \\
 & & & & & & & & & 1 & \\
 & & & & & & & & & & 1 \\
 & & & & & & & & & & \\
\end{bmatrix}$$

For monodromy around $y = 1$

$$\begin{bmatrix}
1 & & & & & & & & & & \\
 & 1 & & & & & & & & & \\
 & & 1 & & & & & & & & \\
 & & & 1 & & & & & & & \\
 & -1 & 1 & & 1 & & & & & & \\
 & & & & & 1 & & & & & \\
 & & & & & & 1 & & & & \\
 & & & & & & & 1 & & & \\
 & & & & -1 & 1 & & 1 & & & \\
 & & & & & & & & 1 & & \\
 & & & & & & & & & 1 & \\
 & & & & & & & -1 & 1 & 1 & \\
\end{bmatrix}$$

For monodromy around $z = 1$

$$\begin{bmatrix}
1 & & & & & & & & & & \\
-1 & 1 & & & & & & & & & \\
 & & 1 & & & & & & & & \\
 & & & 1 & & & & & & & \\
 & & -1 & & 1 & & & & & & \\
 & & & -1 & & 1 & & & & & \\
 & & & & & & 1 & & & & \\
 & & & & & & & 1 & & & \\
 & & & & & -1 & & & 1 & & \\
 & & & & & & -1 & & & 1 & \\
 & & & & & & & & & & 1 \\
 & & & & & & & & & -1 & 1 \\
\end{bmatrix}$$

For monodromy around $xy = 1$

$$\begin{bmatrix}
1 & & & & & & & & & & \\
 & 1 & & & & & & & & & \\
 & & 1 & & & & & & & & \\
 & & & 1 & & & & & & & \\
 & -1 & & & 1 & & & & & & \\
 & & & 1 & & 1 & & & & & \\
 & & & & & & 1 & & & & \\
 & & & & & & & 1 & & & \\
 & & & & & & & & 1 & & \\
 & & & & & & & & & 1 & \\
 & & & & & & & & & & 1
\end{bmatrix}$$

For monodromy around $yz = 1$

$$\begin{bmatrix}
1 & & & & & & & & & & \\
 & 1 & & & & & & & & & \\
-1 & & 1 & & & & & & & & \\
 & & & 1 & & & & & & & \\
 & & & & 1 & & & & & & \\
 & & & & & 1 & & & & & \\
 & & & & & & 1 & & & & \\
 & & & & & & & 1 & & & \\
 & & & & & & & & 1 & & \\
 & & & & & & -1 & & & 1 & \\
 & & & & & & & & & & 1
\end{bmatrix}$$

For monodromy around $xyz = 1$

$$\begin{bmatrix}
1 & & & & & & & & & & \\
 & 1 & & & & & & & & & \\
 & & 1 & & & & & & & & \\
-1 & & & 1 & & & & & & & \\
 & & & & 1 & & & & & & \\
 & & & & & 1 & & & & & \\
 & & & & & & 1 & & & & \\
 & & & & & & & 1 & & & \\
 & & & & & & & & 1 & & \\
 & & & & & & & & & 1 & \\
 & & & & & & & & & & 1
\end{bmatrix}$$

$d\omega_{2,1,1}(x,y,z) + (\omega_2(xyz) - w_2(x)) \wedge \omega_{1,1}(y,z) + (\omega_2(x) + \omega_2(y)) \wedge \omega_{1,1}(xy,z) = 0$, we have $d\boldsymbol{\omega} - \boldsymbol{\omega} \wedge \boldsymbol{\omega} = 0$ for

$$\boldsymbol{\omega} = \begin{bmatrix}
0 & & & & \\
\omega_{1,1}(y,z) & 0 & & & \\
\omega_{1,1}(xy,z) & 0 & 0 & & \\
\omega_2(xyz) & 0 & 0 & 0 & \\
\omega_{2,1,1}(x,y,z) & \omega_2(x) & -\omega_2(x) - \omega_2(y) & \omega_{1,1}(y,z) & 0
\end{bmatrix}$$

And the variation matrix should be

$$\begin{bmatrix} 1 & & & & \\ \hat{\mathcal{L}}_{1,1}(y,z) & 1 & & & \\ \hat{\mathcal{L}}_{1,1}(xy,z) & 0 & 1 & & \\ \hat{\mathcal{L}}_2(xyz) & 0 & 0 & 1 & \\ \hat{\mathcal{L}}_{2,1,1}(x,y,z) & \hat{\mathcal{L}}_2(x) & -\hat{\mathcal{L}}_2(x)-\hat{\mathcal{L}}_2(y) & \hat{\mathcal{L}}_{1,1}(y,z) & 1 \end{bmatrix}$$

Here

$$\hat{\mathcal{L}}_{1,1}(x,y) = \mathrm{Li}_{1,1}(x,y) - \frac{1}{2}(v_{12}v_2 - v_1v_2 + v_1v_{12} - u_1v_{12})$$

$$\hat{\mathcal{L}}_2(x) = \mathrm{Li}_2(x) + \frac{1}{2}v_1u_1$$

$$\begin{aligned}\hat{\mathcal{L}}_{2,1,1}(x,y,z) = \mathrm{Li}_{2,1,1}(x,y,z) &+ (v_3 - v_2 + u_2)\mathrm{Li}_{2,1}(x,yz) + v_2\mathrm{Li}_{2,1}(xy,z) - u_1\mathrm{Li}_{1,1,1}(x,y,z)\\ &+\tfrac{1}{2}(v_{23}(v_3+u_2-v_2)+v_2v_3)\mathrm{Li}_2(xyz) - (u_1(v_3+u_2-v_2))\mathrm{Li}_{1,1}(x,yz)\\ &+\big(-u_1v_2 - \tfrac{1}{2}u_1^2 + \tfrac{1}{2}u_1v_1 - \tfrac{1}{2}u_2v_2\big)\mathrm{Li}_{1,1}(xy,z) - \tfrac{1}{2}u_1v_1\mathrm{Li}_{1,1}(y,z)\end{aligned}$$

The unipotent monodromy representation would be $x = 0, xyz = 1$

$$\begin{bmatrix} 1 & & & & \\ & 1 & & & \\ 1 & & 1 & & \\ -1 & & & 1 & \\ -\frac{7}{12} & & & & 1 \end{bmatrix}$$

$y = 0, xyz = 1$

$$\begin{bmatrix} 1 & & & & \\ & 1 & & & \\ 1 & & 1 & & \\ -1 & & & 1 & \\ -\frac{1}{12} & & & & 1 \end{bmatrix}$$

$y = 0, yz = 1$

$$\begin{bmatrix} 1 & & & & \\ 1 & 1 & & & \\ & & 1 & & \\ & & & 1 & \\ & & & 1 & 1 \end{bmatrix}$$

$z = 0, xyz = 1$

$$\begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ -1 & & & 1 & \\ & & & & 1 \end{bmatrix}$$

$xy = 1, xyz = 1$

$$\begin{bmatrix} 1 & & & & \\ & 1 & & & \\ -1 & & 1 & & \\ & & & 1 & \\ \frac{1}{12} & & & & 1 \end{bmatrix}$$

$xy = 1, z = 1$

$$\begin{bmatrix} 1 & & & & \\ & 1 & & & \\ 1 & & 1 & & \\ & & & 1 & \\ -\frac{1}{12} & & & & 1 \end{bmatrix}$$

$xyz = 1, z = 1$

$$\begin{bmatrix} 1 & & & & \\ & 1 & & & \\ -1 & & 1 & & \\ & & & 1 & \\ \frac{1}{12} & & & & 1 \end{bmatrix}$$

$y = 1, yz = 1$

$$\begin{bmatrix} 1 & & & & \\ -1 & 1 & & & \\ & & 1 & & \\ & & & 1 & \\ & & & & 1 \end{bmatrix}$$

$y = 1, z = 1$

$$\begin{bmatrix} 1 & & & & \\ 1 & 1 & & & \\ & & 1 & & \\ & & & 1 & \\ & & & & 1 \end{bmatrix}$$

$yz = 1, z = 1$

$$\begin{bmatrix} 1 & & & & \\ -1 & 1 & & & \\ & & 1 & & \\ & & & 1 & \\ & & & & 1 \end{bmatrix}$$

36.11 Li 2,1

$$\begin{aligned} \mathrm{Li}_{2,1}(x,y) &= \int \frac{dy}{1-y}\frac{dx}{1-x}\frac{dx}{x} + \frac{d(xy)}{1-xy}\left(\frac{dy}{1-y} - \frac{dx}{x(1-x)}\right)\frac{dx}{x} + \frac{d(xy)}{1-xy}\frac{d(xy)}{xy}\frac{dy}{1-y} \\ &= \int dv_2 dv_1 du_1 + dv_{12} dw_{1,2} du_1 + dv_{12} du_{12} dv_2 \end{aligned}$$

To compute the monodromy around $x = 0$, take $q(\epsilon)$ to be the loop $(x = \epsilon e^{it}, y = \epsilon)$, we get 0
To compute the monodromy around $y = 0$, take $q(\epsilon)$ to be the loop $(x = \epsilon, y = \epsilon e^{it})$, we get 0
To compute the monodromy around $y = 1$, take $q(\epsilon)$ to be the composition of $(x = \epsilon, y = (1-t)\epsilon + t(1-$

$\epsilon)$), $(x=\epsilon, y=1-\epsilon e^{it})$ and $(x=\epsilon, y=(1-t)(1-\epsilon)+t\epsilon)$, we get $-2\pi i \mathrm{Li}_2(x)$
To compute the monodromy around $x=1$, take $q(\epsilon)$ to be the composition of $(x=(1-t)\epsilon+t(1-\epsilon), y=\epsilon)$, $(x=1-\epsilon e^{it}, y=\epsilon)$ and $(x=(1-t)(1-\epsilon)+t\epsilon, y=\epsilon)$, we get 0
To compute the monodromy around $xy=1$, take q to be the loop $(x=x^0, y$ such that $\int_q d\log(1-xy)=2\pi i)$

The variation matrix is

$$\Lambda = \begin{bmatrix} 1 & & & & & \\ \mathrm{Li}_1(y) & 1 & & & & \\ \mathrm{Li}_1(xy) & & 1 & & & \\ \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x) & \mathrm{Li}_1(y)-\mathrm{Li}_1(x^{-1}) & 1 & & \\ \mathrm{Li}_2(xy) & & \log(xy) & & 1 & \\ \mathrm{Li}_{2,1}(x,y) & \mathrm{Li}_2(x) & \log(xy)\mathrm{Li}_1(y)-\mathrm{Li}_2(y)+\mathrm{Li}_2(x^{-1}) & \log x & \mathrm{Li}_1(y) & 1 \end{bmatrix} \tau_{2,1}(2\pi i)$$

$-\mathrm{Li}_{1,1}(1-x, \frac{1-xy}{1-x}) = \log(xy)\mathrm{Li}_1(y)-\mathrm{Li}_2(y)+\mathrm{Li}_2(x^{-1})$

$$\omega = \begin{bmatrix} 0 & & & & & \\ -dv_2 & 0 & & & & \\ -dv_{12} & 0 & 0 & & & \\ 0 & -dv_1 & -w_1 & 0 & & \\ 0 & 0 & du_{12} & 0 & 0 & \\ 0 & 0 & 0 & du_1 & -dv_2 & 0 \end{bmatrix}$$

The monodromy representation ρ is as follows
For monodromy around $x=0$

$$\begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & -1 & 1 & & \\ & & 1 & & 1 & \\ & & & 1 & & 1 \end{bmatrix}$$

For monodromy around $y=0$

$$\begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & 1 & & 1 & \\ & & & & & 1 \end{bmatrix}$$

For monodromy around $x=1$

$$\begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & & & \\ & -1 & 1 & 1 & & \\ & & & & 1 & \\ & & & & & 1 \end{bmatrix}$$

For monodromy around $y=1$

$$\begin{bmatrix} 1 & & & & & \\ -1 & 1 & & & & \\ & & 1 & & & \\ & & -1 & 1 & & \\ & & & & 1 & \\ & & & & -1 & 1 \end{bmatrix}$$

For monodromy around $xy = 1$

$$\begin{bmatrix} 1 & & & & & \\ & 1 & & & & \\ -1 & & 1 & & & \\ & & & 1 & & \\ & & & & 1 & \\ & & & & & 1 \end{bmatrix}$$

$\mathrm{Li}_{2,1}(x,y) - \mathrm{Li}_2(xy)\mathrm{Li}_1(y) - \mathrm{Li}_{1,1}(x,y)\log x$ is well defined on the universal abelian cover, and

$$d\hat{L}_{2,1} = -u_1v_{12}du_1 + u_1v_{12}dv_1 - u_1v_{12}dv_2 - u_1v_2dv_1 - v_2v_{12}du_{12}$$

36.12 Li 2,2

$$\begin{aligned}\mathrm{Li}_{2,2} = &(dv_{12}du_{12}(du_1 - dv_1) + dv_2du_2dv_1 - (dv_2dv_1 + dv_{12}dw_{1,2})dv_2)du_1 \\ &+(dv_2dv_1du_1 + dv_{12}dw_{1,2}du_1 + dv_{12}du_{12}dv_2)du_2\end{aligned}$$

Monodromy around $x = 0$ Monodromy around $y = 0$ Monodromy around $x = 1$ Monodromy around $y = 1$ Monodromy around $xy = 1$

36.13 Li 3,1

$$\begin{aligned}\mathrm{Li}_{3,1}(x,y) = &\int \frac{dy}{1-y}\frac{dx}{1-x}\left(\frac{dx}{x}\right)^2 + \frac{d(xy)}{1-xy}\left(\frac{dy}{1-y} - \frac{dx}{x(1-x)}\right)\left(\frac{dx}{x}\right)^2 \\ &+\frac{d(xy)}{1-xy}\frac{d(xy)}{xy}\frac{dy}{1-y}\frac{dx}{x} + \frac{d(xy)}{1-xy}\left(\frac{d(xy)}{xy}\right)^2\frac{dy}{1-y} \\ &= \int dv_2dv_1(du_1)^2 + dv_{12}dw_1(du_1)^2 + dv_{12}du_{12}dv_2du_1 + dv_{12}(du_{12})^2dv_2\end{aligned}$$

Here we write $w_1 = u_1 + v_2 - v_1 = \log\frac{x(1-y)}{1-x}$

To compute the monodromy around $x = 0$, take $q(\epsilon)$ to be the loop $(x = \epsilon e^{it}, y = \epsilon)$, we get 0.
To compute the monodromy around $y = 0$, take $q(\epsilon)$ to be the loop $(x = \epsilon, y = \epsilon e^{it})$, we get 0.
To compute the monodromy around $x = 1$, take $q(\epsilon)$ to be the composition of $(x = (1-t)\epsilon + t(1-\epsilon), y = \epsilon)$, $(x = 1 - \epsilon e^{it}, y = \epsilon)$ and $(x = (1-t)(1-\epsilon) + t\epsilon, y = \epsilon)$, we get 0.
To compute the monodromy around $y = 1$, take $q(\epsilon)$ to be the composition of $(x = \epsilon, y = (1-t)\epsilon + t(1-\epsilon))$, $(x = \epsilon, y = 1 - \epsilon e^{it})$ and $(x = \epsilon, y = (1-t)(1-\epsilon) + t\epsilon)$, we get $-2\pi i\mathrm{Li}_3(x)$.
To compute the monodromy around $xy = 1$, take q to be the loop $(x = x^0, y$ such that $\int_q d\log(1-xy) = 2\pi i)$, we get $-2\pi i\mathrm{Li}_1(\frac{1-xy}{1-x})$

The variation matrix is

$$\Lambda = \begin{bmatrix} 1 & & & & & & & & & \\ \mathrm{Li}_1(y) & 1 & & & & & & & & \\ \mathrm{Li}_1(xy) & & 1 & & & & & & & \\ \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x) & \mathrm{Li}_1(y)-\mathrm{Li}_1(x^{-1}) & 1 & & & & & & \\ \mathrm{Li}_2(xy) & & \log(xy) & & 1 & & & & & \\ \mathrm{Li}_{2,1}(x,y) & \mathrm{Li}_2(x) & \log(xy)\mathrm{Li}_1(y)-\mathrm{Li}_2(y)+\mathrm{Li}_2(x^{-1}) & \log x & \mathrm{Li}_1(y) & 1 & & & & \\ \mathrm{Li}_3(xy) & & \log^2(xy)/2 & & \log(xy) & & 1 & & & \\ \mathrm{Li}_{3,1}(x,y) & \mathrm{Li}_3(x) & \frac{1}{2}\log^2(xy)\mathrm{Li}_1(y)-\log(xy)\mathrm{Li}_2(y)+\mathrm{Li}_3(y)-\mathrm{Li}_3(x^{-1}) & \log^2 x/2 & \log(xy)\mathrm{Li}_1(y)-\mathrm{Li}_2(y) & \log x & \mathrm{Li}_1(y) & 1 \end{bmatrix} \tau_{3,1}(2\pi i)$$

$$\begin{bmatrix}
1 & & & & & & & & & & & & & & & \\
-I(0;y^{-1};1) & 1 & & & & & & & & & & & & & & \\
-I(0;(xy)^{-1};1) & 0 & 1 & & & & & & & & & & & & & \\
I(0;(xy)^{-1},y^{-1};1) & -I(0;(xy)^{-1};y^{-1}) & -I((xy)^{-1};y^{-1};1) & 1 & & & & & & & & & & & & \\
-I(0;(xy)^{-1},0;1) & 0 & -I((xy)^{-1};0;y^{-1}) & 0 & 1 & & & & & & & & & & & \\
I(0;(xy)^{-1},0,y^{-1};1) & -I(0;(xy)^{-1},0;y^{-1}) & -I((xy)^{-1};0,y^{-1};1) & I((xy)^{-1};0;y^{-1}) & -I(0;y^{-1};1) & 1 & & & & & & & & & & \\
-I(0;(xy)^{-1},0,0;1) & 0 & I((xy)^{-1};0,0;1) & 0 & I((xy)^{-1};0;0) & 0 & 1 & & & & & & & & & \\
I(0;(xy)^{-1},0,0,y^{-1};1) & -I(0;(xy)^{-1},0,0;y^{-1}) & -I((xy)^{-1};0,0,y^{-1};1) & I((xy)^{-1};0,0;y^{-1}) & -I((xy)^{-1};0;0)I(0;y^{-1};1)-I(0;0,y^{-1};1) & I((xy)^{-1};0;0)+I(0;0;y^{-1}) & -I(0;y^{-1};1) & 1 & & & & & & & \\
\end{bmatrix}$$

$$\omega = \begin{bmatrix}
0 & & & & & & & & & & & & & & & \\
-dv_2 & 0 & & & & & & & & & & & & & & \\
-dv_{12} & 0 & 0 & & & & & & & & & & & & & \\
 & -dv_1 & -dw_1 & 0 & & & & & & & & & & & & \\
 & & du_{12} & 0 & 0 & & & & & & & & & & & \\
 & & & du_1 & -dv_2 & 0 & & & & & & & & & & \\
 & & & & du_{12} & 0 & 0 & & & & & & & & & \\
 & & & & & du_1 & -dv_2 & 0 & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
\end{bmatrix}$$

The monodromy representation ρ is as follows
For monodromy around $x = 0$

$$\begin{bmatrix}
1 & & & & & & & \\
 & 1 & & & & & & \\
 & & 1 & & & & & \\
 & & -1 & 1 & & & & \\
 & & 1 & & 1 & & & \\
 & & & 1 & & 1 & & \\
 & & \frac{1}{2} & & & & 1 & \\
 & & & & & 1 & & 1
\end{bmatrix}$$

For monodromy around $y = 0$

$$\begin{bmatrix}
1 & & & & & & & \\
 & 1 & & & & & & \\
 & & 1 & & & & & \\
 & & & 1 & & & & \\
 & & & & 1 & & & \\
 & & & & & 1 & & \\
 & & & & & & 1 & \\
 & & & & & & & 1
\end{bmatrix}$$

For monodromy around $x = 1$

$$\begin{bmatrix}
1 & & & & & & & \\
 & 1 & & & & & & \\
 & & 1 & & & & & \\
 & & & 1 & & & & \\
 & & & & 1 & & & \\
 & & & & & 1 & & \\
 & & & & & & 1 & \\
 & & & & & & & 1
\end{bmatrix}$$

For monodromy around $y = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

For monodromy around $xy = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

$dw_{3,1} - w_2(y) \wedge w_2(xy) = 0$. So $\xi = w_{3,1} - \frac{1}{2}(L_2(y) \wedge w_2(xy) - w_2(y) \wedge L_2(xy))$ should be closed. So we want $\hat{L}_{3,1}$ such that $\hat{d}L_{3,1} = \xi$

$d\omega_{3,1}(x,y) + \omega_2(y) \wedge \omega_2(xy) = 0$, we have $d\boldsymbol{\omega} - \boldsymbol{\omega} \wedge \boldsymbol{\omega} = 0$ for

$$\boldsymbol{\omega} = \begin{bmatrix} 0 & & \\ \omega_2(xy) & 0 & \\ \omega_{3,1}(x,y) & -\omega_2(y) & 0 \end{bmatrix}$$

And the variation matrix should be

$$\begin{bmatrix} 1 & & \\ \hat{\mathcal{L}}_2(xy) & 1 & \\ \hat{\mathcal{L}}_{3,1}(x,y) & -\hat{\mathcal{L}}_2(y) & 1 \\ & & \end{bmatrix}$$

Here

$$\hat{\mathcal{L}}_2(x) = \mathrm{Li}_2(x) - \frac{1}{2}\mathrm{Li}_1(x)\log x$$

$$\begin{aligned} \hat{\mathcal{L}}_{3,1}(x,y) = \mathrm{Li}_{3,1}(x,y) + \mathrm{Li}_3(xy)v_2 - \mathrm{Li}_{2,1}(x,y)u_1 + \tfrac{1}{2}\mathrm{Li}_{1,1}(x,y)u_1^2 - \tfrac{1}{2}\mathrm{Li}_2(xy)v_2(2u_1+v_1) \\ +\tfrac{1}{4}(-\tfrac{1}{2}v_1u_1^2v_2 + \tfrac{1}{2}v_1u_1^2v_{12} - \tfrac{1}{2}u_1^3v_{12} - \tfrac{3}{2}u_1^2v_2v_{12} - \tfrac{3}{2}u_1v_2u_2v_{12} - \tfrac{1}{2}v_2u_2^2v_{12}) \end{aligned}$$

The unipotent monodromy representation would be

Monodromy along the commutator of $x = 0, xy = 1$

$$\begin{bmatrix} 1 & & \\ -1 & 1 & \\ \frac{1}{2} & & 1 \end{bmatrix}$$

Monodromy along the commutator of $y = 0, y = 1$

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & -1 & 1 \end{bmatrix}$$

Monodromy along the commutator of $y=0, xy=1$

$$\begin{bmatrix} 1 & & \\ -1 & 1 & \\ & & 1 \end{bmatrix}$$

Monodromy along all other commutators

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}$$

36.14 Li 4,1

$$\mathrm{Li}_{n,1}(x,y) = \int (dv_2 dv_1 + dv_{12} dw_1)(du_1)^{n-1} + dv_{12}\left(\sum_{k=1}^{n-1} du_{12}^k dv_2 (du_1)^{n-1-k}\right)$$

Here $w_1 = u_1 + v_2 - v_1$

To compute the monodromy around $x=0$, take $q(\epsilon)$ to be the loop $(x = \epsilon e^{it}, y = \epsilon)$

$$\lim_{\epsilon\to 0} \int_{q(\epsilon)p(\epsilon)} - \int_{p(\epsilon)} = 0$$

To compute the monodromy around $y=0$, take $q(\epsilon)$ to be the loop $(x = \epsilon, y = \epsilon e^{it})$

$$\lim_{\epsilon\to 0} \int_{q(\epsilon)p(\epsilon)} - \int_{p(\epsilon)} = \int_q dv_2 \int_p dv_1 \cdots (du_1)^{n-1} = 0$$

To compute the monodromy around $x=1$, take $q(\epsilon)$ to be the composition of $(x = (1-t)\epsilon + t(1-\epsilon), y = \epsilon)$, $(x = 1 - \epsilon e^{it}, y = \epsilon)$ and $(x = (1-t)(1-\epsilon) + t\epsilon, y = \epsilon)$

$$\lim_{\epsilon\to 0} \int_{q(\epsilon)p(\epsilon)} - \int_{p(\epsilon)} = 0$$

To compute the monodromy around $y=1$, take $q(\epsilon)$ to be the composition of $(x = \epsilon, y = (1-t)\epsilon + t(1-\epsilon))$, $(x = \epsilon, y = 1 - \epsilon e^{it})$ and $(x = \epsilon, y = (1-t)(1-\epsilon) + t\epsilon)$

$$\lim_{\epsilon\to 0} \int_{q(\epsilon)p(\epsilon)} - \int_{p(\epsilon)} = \int_q dv_2 \int_p dv_1 (du_1)^{n-1} = -2\pi i \mathrm{Li}_n(x)$$

To compute the monodromy around $xy=1$, take q to be the loop $(x = x^0, y$ such that $\int_q d\log(1-xy) = 2\pi i)$

$$\begin{aligned}
\lim_{\epsilon\to 0} \int_{pq} - \int_p &= \sum_{k=0}^{n-1} \int_p dv_{12} (du_{12})^k \int_q (du_{12})^{n-1-k} dv_2 + \int_q dv_{12} du_{12}^{n-1} dv_2 \\
&= (-1)^{n+1} \int_{q^{-1}} dv_2 du_2^{n-1} dv_{12} \\
&= (-1)^{n+1} \int_{q^{-1}} \frac{\mathrm{Li}_n(y) - \mathrm{Li}_n(y^0)}{y - 1/x^0} \\
&= (-1)^n 2\pi i (\mathrm{Li}_n(y^0) - \mathrm{Li}_n(1/x^0))
\end{aligned}$$

The variation matrix is

$$\omega = \begin{bmatrix} 0 & & & & & & & & & \\ -dv_2 & 0 & & & & & & & & \\ -dv_{12} & & 0 & & & & & & & \\ & -dv_1 & -dw_1 & 0 & & & & & & \\ & & du_{12} & & 0 & & & & & \\ & & & du_1 & -dv_2 & 0 & & & & \\ & & & & du_{12} & & 0 & & & \\ & & & & & du_1 & -dv_2 & 0 & & \\ & & & & & & du_{12} & & 0 & \\ & & & & & & & du_1 & -dv_2 & 0 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 1 & & & & & & & & & & \\ \mathrm{Li}_1(y) & 1 & & & & & & & & & \\ \mathrm{Li}_1(xy) & & 1 & & & & & & & & \\ \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x) & \mathrm{Li}_1(y)-\mathrm{Li}_1(x^{-1}) & 1 & & & & & & & \\ \mathrm{Li}_2(xy) & & \log(xy) & & 1 & & & & & & \\ \mathrm{Li}_{2,1}(x,y) & \mathrm{Li}_2(x) & \log(xy)\mathrm{Li}_1(y)-\mathrm{Li}_2(y)+\mathrm{Li}_2(x^{-1}) & \log x & \mathrm{Li}_1(y) & 1 & & & & \\ \mathrm{Li}_3(xy) & & \log^2(xy)/2 & & \log(xy) & & 1 & & & \\ \mathrm{Li}_{3,1}(x,y) & \mathrm{Li}_3(x) & \frac{1}{2}\log^2(xy)\mathrm{Li}_1(y)-\log(xy)\mathrm{Li}_2(y)+\mathrm{Li}_3(y)-\mathrm{Li}_3(x^{-1}) & \log^2 x/2 & \log(xy)\mathrm{Li}_1(y)-\mathrm{Li}_2(y) & \log x & \mathrm{Li}_1(y) & 1 & \\ \mathrm{Li}_4(xy) & & \log^3(xy)/3! & & \log^2(xy)/2 & & \log(xy) & & 1 & \\ \mathrm{Li}_{4,1}(x,y) & \mathrm{Li}_4(x) & \frac{1}{3!}\log^3(xy)\mathrm{Li}_1(y)-\frac{1}{2}\log^2(xy)\mathrm{Li}_2(y)+\log(xy)\mathrm{Li}_3(y)-\mathrm{Li}_4(y)+\mathrm{Li}_4(x^{-1}) & \log^3 x/3! & \frac{1}{2}\log^2(xy)\mathrm{Li}_1(y)-\log(xy)\mathrm{Li}_2(y)+\mathrm{Li}_3(y) & \log^2 x/2 & \log(xy)\mathrm{Li}_1(y)-\mathrm{Li}_2(y) & \log x & \mathrm{Li}_1(y) & 1 \end{bmatrix} \tau_{1,1}(2\pi i)$$

The monodromy representation ρ is as follows
For monodromy around $x = 0$

$$\begin{bmatrix} 1 & & & & & & & & \\ & 1 & & & & & & & \\ & & 1 & & & & & & \\ & & -1 & 1 & & & & & \\ & & 1 & & 1 & & & & \\ & & & 1 & & 1 & & & \\ & & \frac{1}{2} & & 1 & & 1 & & \\ & & & \frac{1}{2} & & 1 & & 1 & \\ & & \frac{1}{3!} & & \frac{1}{2} & & 1 & & 1 \\ & & & \frac{1}{3!} & & \frac{1}{2} & & 1 & 1 \end{bmatrix}$$

For monodromy around $y = 0$

$$\begin{bmatrix} 1 & & & & & & & & \\ & 1 & & & & & & & \\ & & 1 & & & & & & \\ & & & 1 & & & & & \\ & & 1 & & 1 & & & & \\ & & & & & 1 & & & \\ & & \frac{1}{2} & & 1 & & 1 & & \\ & & & & & & & 1 & \\ & & \frac{1}{3!} & & \frac{1}{2} & & 1 & & 1 \end{bmatrix}$$

For monodromy around $x = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & -1 & 1 & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

For monodromy around $y = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ -1 & 1 & & & & & & \\ & & 1 & & & & & \\ & & -1 & 1 & & & & \\ & & & & 1 & & & \\ & & & & -1 & 1 & & \\ & & & & & & 1 & \\ & & & & & & -1 & 1 \end{bmatrix}$$

For monodromy around $xy = 1$

$$\begin{bmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ -1 & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ & & & & & & & 1 \end{bmatrix}$$

$d\omega_{4,1}(x,y) + \omega_2(y) \wedge \omega_3(xy) + \omega_2(xy) \wedge \omega_3(y) = 0$, we have $d\boldsymbol{\omega} - \boldsymbol{\omega} \wedge \boldsymbol{\omega} = 0$ for

$$\boldsymbol{\omega} = \begin{bmatrix} 0 & & & \\ \omega_2(xy) & 0 & & \\ \omega_3(xy) & 0 & 0 & \\ \omega_{4,1}(x,y) & \omega_3(y) & -\omega_2(y) & 0 \end{bmatrix}$$

And the variation matrix should be

$$\begin{bmatrix} 1 & & & \\ \hat{\mathcal{L}}_2(xy) & 1 & & \\ \hat{\mathcal{L}}_3(xy) & 0 & 1 & \\ \hat{\mathcal{L}}_{4,1}(x,y) & \hat{\mathcal{L}}_3(y) & -\hat{\mathcal{L}}_2(y) & 1 \end{bmatrix}$$

Here

$$\hat{\mathcal{L}}_2(x) = \mathrm{Li}_2(x) + \frac{1}{2}v_1u_1$$

$$\hat{\mathcal{L}}_3(x) = \mathrm{Li}_3(x) - \mathrm{Li}_2(x)u_1 - \frac{1}{3}u_1^2v_1$$

$$\begin{aligned}\hat{\mathcal{L}}_{4,1}(x,y) = {} & \mathrm{Li}_{4,1}(x,y) + \mathrm{Li}_4(xy)v_2 - \mathrm{Li}_{3,1}(x,y)u_1 + \tfrac{1}{2}Li_{2,1}(x,y)u_1^2 - \tfrac{1}{6}u_1^3\mathrm{Li}_{1,1}(x,y) \\ & -u_1v_2\mathrm{Li}_3(xy) - \tfrac{1}{2}u_2v_2\mathrm{Li}_3(xy) + \big(\tfrac{1}{2}u_1^2v_2 + \tfrac{1}{6}u_2^2v_2 + \tfrac{1}{2}u_1u_2v_2\big)\mathrm{Li}_2(xy) \\ & +\tfrac{1}{30}u_1^4v_{12} - \tfrac{1}{30}u_1^3v_1v_{12} + \tfrac{2}{15}u_1^3v_2v_{12} + \tfrac{1}{30}u_2^3v_{12}v_2 + \tfrac{1}{30}u_1^3v_1v_2 + \tfrac{1}{5}u_1^2u_2v_2v_{12} + \tfrac{2}{15}u_1u_2^2v_2v_{12}\end{aligned}$$

The unipotent monodromy representation would be

36.15 Li 5,1

$d\omega_{5,1}(x,y) - \omega_4(xy)\omega_4(y) + \omega_3(xy)\wedge\omega_3(y) - \omega_2(xy)\wedge\omega_4(y) = 0$, we have $d\boldsymbol{\omega} - \boldsymbol{\omega}\wedge\boldsymbol{\omega} = 0$ for

$$\boldsymbol{\omega} = \begin{bmatrix} 0 & & & & \\ \omega_2(xy) & 0 & & & \\ \omega_3(xy) & 0 & 0 & & \\ \omega_4(xy) & 0 & 0 & 0 & \\ \omega_{5,1}(x,y) & -\omega_4(y) & \omega_3(y) & -\omega_2(y) & 0 \end{bmatrix}$$

And the variation matrix should be

$$\begin{bmatrix} 1 & & & & \\ \hat{\mathcal{L}}_2(xy) & 1 & & & \\ \hat{\mathcal{L}}_3(xy) & 0 & 1 & & \\ \hat{\mathcal{L}}_4(xy) & 0 & 0 & 1 & \\ \hat{\mathcal{L}}_{5,1}(x,y) & -\hat{\mathcal{L}}_4(y) & \hat{\mathcal{L}}_3(y) & -\hat{\mathcal{L}}_2(y) & 1 \\ & & & & \end{bmatrix}$$

Here

$$\hat{\mathcal{L}}_2(x) = \mathrm{Li}_2(x) + \frac{1}{2}v_1u_1$$

$$\hat{\mathcal{L}}_3(x) = \mathrm{Li}_3(x) - \mathrm{Li}_2(x)u_1 - \frac{1}{3}u_1^2v_1$$

$$\hat{\mathcal{L}}_4(x) = \mathrm{Li}_4(x) - \mathrm{Li}_3(x)u_1 + \frac{1}{2}u_1^2\mathrm{Li}_2(x) + \frac{1}{8}u_1^3v_1$$

$$\hat{\mathcal{L}}_{5,1}(x,y) = \mathrm{Li}_{5,1}(x,y) + v_2\mathcal{L}_{4,1}(xy) - u_1\mathcal{L}_{4,1}(x,y) + \big(-u_1v_1 - \tfrac{1}{2}u_2v_2\big)\mathcal{L}_4(xy) + \tfrac{1}{2}u_1^2\mathcal{L}_{3,1}(x,y) + \big(\tfrac{1}{2}u_1u_2v_2 + \tfrac{1}{2}u_1^2v_2 + \tfrac{1}{2}u_1^2v_1v_2\big)\mathcal{L}_3(xy) - \tfrac{1}{6}u_1^3\mathcal{L}_{2,1}(x,y) + \big(-\tfrac{1}{24}u_1^4v_2 - \tfrac{1}{6}u_1u_2^2v_2 - \tfrac{1}{6}u_1^2u_2v_2 - \tfrac{1}{6}u_1^3v_2\mathcal{L}_{2,1}(xy) + \tfrac{1}{30}u_1^4\mathcal{L}_2(1,1,u_1,u_2) - (u_1^5v_{12})/144 + 1/144u_1^4v_1v_{12} - 5/144u_1^4v_2v_{12} - 5/72u_1^3u_2v_{12}v_2 - 5/72u_1^3u_2^2v_{12}v_2 - 5/144u_1u_2^2v_{12}v_2 - 1/144u_2^2v_{12}v_2$$

The unipotent monodromy representation would be

36.16 Li n,1

$$\mathrm{Li}_{1,1}(x,y) = \int_{(0,0)}^{(x,y)} dv_2dv_1 + dv_{12}dw_{1,2}$$

Here $w_{1,2} = u_1 + v_2 - v_1$, then

$$\mathrm{Li}_{n,1}(x,y) = \int_{(0,0)}^{(x,y)} (dv_2dv_1 + dv_{12}dw_{1,2})(du_1)^{n-1} + dv_{12}\left(\sum_{k=1}^{n-1} du_{12}^k dv_2(du_1)^{n-1-k}\right)$$

since inductively

$$\begin{aligned}\mathrm{Li}_{n,1}(x,y) &= \int_{(0,0)}^{(x,y)} \mathrm{Li}_{n-1,1}(x,y)du_1 - \mathrm{Li}_n(xy)dv_2 \\ &= \int_{(0,0)}^{(x,y)} \left[(dv_2dv_1 + dv_{12}dw_{1,2})(du_1)^{n-2} + dv_{12}\left(\sum_{k=1}^{n-2} du_{12}^k dv_2 (du_1)^{n-2-k}\right)\right]du_1 \\ &+ dv_{12}(du_{12})^{n-1}dv_2\end{aligned}$$

Suppose p is the path from $(0,0)$ to (x,y) that give the value of $\mathrm{Li}_{n,1}(x,y)$, q is a loop based at (x,y) that induces monodromy, we can take p to $p(\epsilon)$ to start at (ϵ,ϵ) and then take limit $\epsilon \to 0$, then pq can be homotopied to some $q(\epsilon)p(\epsilon)$

To compute the monodromy around $x=0$, take $q(\epsilon)$ to be the loop $(x = \epsilon e^{it}, y = \epsilon)$

$$\lim_{\epsilon\to 0} \int_{q(\epsilon)p(\epsilon)} - \int_{p(\epsilon)} = 0$$

To compute the monodromy around $y=0$, take $q(\epsilon)$ to be the loop $(x = \epsilon, y = \epsilon e^{it})$

$$\lim_{\epsilon\to 0} \int_{q(\epsilon)p(\epsilon)} - \int_{p(\epsilon)} = \int_q dv_2 \int_p dv_1 (du_1)^{n-1} = 0$$

To compute the monodromy around $x=1$, take $q(\epsilon)$ to be the composition of $(x = (1-t)\epsilon + t(1-\epsilon), y = \epsilon)$, $(x = 1 - \epsilon e^{it}, y = \epsilon)$ and $(x = (1-t)(1-\epsilon) + t\epsilon, y = \epsilon)$

$$\lim_{\epsilon\to 0} \int_{q(\epsilon)p(\epsilon)} - \int_{p(\epsilon)} = 0$$

To compute the monodromy around $y=1$, take $q(\epsilon)$ to be the composition of $(x = \epsilon, y = (1-t)\epsilon + t(1-\epsilon))$, $(x = \epsilon, y = 1 - \epsilon e^{it})$ and $(x = \epsilon, y = (1-t)(1-\epsilon) + t\epsilon)$

$$\lim_{\epsilon\to 0} \int_{q(\epsilon)p(\epsilon)} - \int_{p(\epsilon)} = \int_q dv_2 \int_p dv_1 (du_1)^{n-1} = -2\pi i \mathrm{Li}_n(x)$$

To compute the monodromy around $xy=1$, take q to be the loop $(x = x^0, y$ such that $\int_q d\log(1-xy) = 2\pi i)$ First it's easy to show inductively

$$\int_{(x^0,y^0)}^{(x,y)} dv_2 du_2^{n-1} = \mathrm{Li}_n(y) - \sum_{k=1}^{n} \frac{(\log y - \log y^0)^{n-k}}{(n-k)!} \mathrm{Li}_k(y^0)$$

Thus

$$\begin{aligned}\lim_{\epsilon\to 0} \int_{pq} - \int_p &= \sum_{k=0}^{n-1} \int_p dv_{12}(du_{12})^k \int_q (du_{12})^{n-1-k} dv_2 + \int_q dv_{12} du_{12}^{n-1} dv_2 \\ &= (-1)^{n+1} \int_{q^{-1}} dv_2 du_2^{n-1} dv_{12} \\ &= (-1)^{n+1} \int_{q^{-1}} \frac{\mathrm{Li}_n(y) - \sum_{k=1}^{n} \frac{(\log y - \log y^0)^{n-k}}{(n-k)!} \mathrm{Li}_k(y^0)}{y - 1/x^0} dy \\ &= (-1)^n 2\pi i (-g_n(x^0,y^0) - \mathrm{Li}_n(1/x^0))\end{aligned}$$

Where

$$g_m(x,y) = \sum_{k=1}^{m} (-1)^{k+1} \frac{\log^{m-k}(xy)}{(m-k)!} \mathrm{Li}_k(y)$$

The variation matrix is

$$\Lambda = \begin{bmatrix} 1 & & & & & & & & \\ \mathrm{Li}_1(y) & 1 & & & & & & & \\ \mathrm{Li}_1(xy) & 0 & 1 & & & & & & \\ \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x) & g_1(x,y)-\mathrm{Li}_1(x^{-1}) & 1 & & & & & \\ \mathrm{Li}_2(xy) & 0 & \log x & 0 & 1 & & & & \\ \mathrm{Li}_{2,1}(x,y) & \mathrm{Li}_2(x) & g_2(x,y)+\mathrm{Li}_2(x^{-1}) & \log x & g_1(x,y) & 1 & & & \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & & \\ \mathrm{Li}_n(xy) & 0 & \frac{\log^{n-1}(xy)}{(n-1)!} & 0 & \frac{\log^{n-2}(xy)}{(n-2)!} & 0 & \cdots & 1 & \\ \mathrm{Li}_{n,1}(x,y) & \mathrm{Li}_n(x) & g_n(x,y)+(-1)^n\mathrm{Li}_n(x^{-1}) & \frac{\log^{n-1}x}{(n-1)!} & g_{n-1}(x,y) & \frac{\log^{n-2}x}{(n-2)!} & \cdots & g_1(x,y) & 1 \end{bmatrix} \tau_{n,1}(2\pi i)$$

$$\omega = \begin{bmatrix} 0 & & & & & & & & \\ -dv_2 & 0 & & & & & & & \\ -dv_{12} & 0 & 0 & & & & & & \\ & -dv_2 & -dw_1 & 0 & & & & & \\ & & du_{12} & 0 & 0 & & & & \\ & & & du_1 & -dv_2 & 0 & & & \\ & & & & \ddots & \ddots & \ddots & & \\ & & & & & du_{12} & 0 & 0 & \\ & & & & & & du_1 & -dv_2 & 0 \end{bmatrix}$$

$$dg_m(x,y) = g_{m-1}(x,y)du_1 - \frac{\log^{m-1}(xy)}{(m-1)!}dv_2$$

Justifies the differential equation $d\Lambda = \omega\Lambda$
Monodromy of $\log^m x/m!$ around $x = 0$ is

$$\sum_{k=0}^{m-1} \frac{\log^k x}{k!}\frac{(2\pi i)^{m-k}}{(m-k)!}$$

Monodromy of $\log^m(xy)/m!$ around $x = 0$ is

$$\sum_{k=0}^{m-1} \frac{\log^k(xy)}{k!}\frac{(2\pi i)^{m-k}}{(m-k)!}$$

Monodromy of $\mathrm{Li}_n(x^{-1})$ around $x = 0$ is $2\pi i \log^{n-1}(x^{-1})/(n-1)! = (-1)^{n-1}2\pi i \log^{n-1}(x)/(n-1)!$
Monodromy of g_n around $x = 0$ or $y = 0$ is

$$\begin{aligned}\sum_{k=1}^{m}\sum_{l=1}^{m-k}(-1)^{k+1}\mathrm{Li}_k(y)\frac{\log^{m-k-l}(xy)}{(m-k-l)!}\frac{(2\pi i)^l}{l!} &= \sum_{l=1}^{m}\sum_{k=1}^{m-l}(-1)^{k+1}\mathrm{Li}_k(y)\frac{\log^{m-k-l}(xy)}{(m-k-l)!}\frac{(2\pi i)^l}{l!} \\ &= \sum_{l=1}^{m-1}\frac{(2\pi i)}{l!}g_{m-l}(x,y)\end{aligned}$$

Monodromy of g_n around $y = 1$ is

$$2\pi i\sum_{k=1}^{m}(-1)^k\frac{\log^{m-k}(xy)}{(m-k)!}$$

The monodromy representation ρ is as follows
For monodromy around $x = 0$

$$\begin{bmatrix}
1 & & & & & & & & & & \\
 & 1 & & & & & & & & & \\
 & & 1 & & & & & & & & \\
 & & -1 & 1 & & & & & & & \\
 & & 1 & 0 & 1 & & & & & & \\
 & & 0 & 1 & 0 & 1 & & & & & \\
 & & \frac{1}{2} & 0 & 1 & 0 & 1 & & & & \\
 & & 0 & \frac{1}{2} & 0 & 1 & 0 & 1 & & & \\
 & & \frac{1}{3!} & 0 & \frac{1}{2} & 0 & 1 & 0 & 1 & & \\
 & & \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \\
 & & \frac{1}{(n-1)!} & 0 & \frac{1}{(n-2)!} & \cdots & \frac{1}{2} & 0 & 1 & 0 & 1 \\
 & & 0 & \frac{1}{(n-1)!} & 0 & \cdots & 0 & \frac{1}{2} & 0 & 1 & 0 & 1
\end{bmatrix}$$

For monodromy around $y = 0$

$$\begin{bmatrix}
1 & & & & & & & & & \\
 & 1 & & & & & & & & \\
 & & 1 & & & & & & & \\
 & & 0 & 1 & & & & & & \\
 & & 1 & 0 & 1 & & & & & \\
 & & 0 & 0 & 0 & 1 & & & & \\
 & & \frac{1}{2} & 0 & 1 & 0 & 1 & & & \\
 & & 0 & 0 & 0 & 0 & 0 & 1 & & \\
 & & \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \ddots & \\
 & & \frac{1}{(n-1)!} & 0 & \frac{1}{(n-2)!} & \cdots & 0 & 1 & 0 & 1 & \\
 & & & & & & & & & & 1
\end{bmatrix}$$

For monodromy around $x = 1$

$$\begin{bmatrix}
1 & & & & & & \\
 & 1 & & & & & \\
 & & 1 & & & & \\
 & -1 & 1 & 1 & & & \\
 & & & & 1 & & \\
 & & & & & \ddots & \\
 & & & & & & 1
\end{bmatrix}$$

For monodromy around $y = 1$

$$\begin{bmatrix}
1 & & & & & & \\
-1 & 1 & & & & & \\
 & & 1 & & & & \\
 & & -1 & 1 & & & \\
 & & & \ddots & \ddots & & \\
 & & & & & 1 & \\
 & & & & & -1 & 1
\end{bmatrix}$$

For monodromy around $xy = 1$

$$\begin{bmatrix} 1 & & & & \\ & 1 & & & \\ -1 & & 1 & & \\ & & & 1 & \\ & & & & \ddots & \\ & & & & & 1 \end{bmatrix}$$

36.17 Lifted Hodge structures

Take $\mathrm{Li}_{2,1}(x,y)$ for an example, we have differential equation

$$dV = \omega V$$

Here ω is an integrable connection since $d\omega - \omega \wedge \omega = 0$. Let Ω be such that $d\Omega = \omega$

Then the lifted variation matrix should be $\hat{V} = e^{-\Omega}V$, and we should have $d\hat{V} = \hat{\omega}\hat{V}$, but we have

$$d\hat{V} = d(e^{-\Omega}V) = d(e^{-\Omega})e^{\Omega}e^{-\Omega}V + e^{-\Omega}\omega e^{\Omega}e^{-\Omega}V = (d(e^{-\Omega})e^{\Omega} + e^{-\Omega}\omega e^{\Omega})\hat{V}$$

Thus $\hat{\omega} = (de^{-\Omega})e^{\Omega} + e^{-\Omega}\omega e^{\Omega}$ gives the lifted connection form which is again integrable since

$$\begin{aligned} d\hat{\omega} - \hat{\omega}\wedge\hat{\omega} &= \big(-(de^{-\Omega})\wedge(de^{\Omega}) + (de^{-\Omega})\wedge\omega e^{\Omega} + e^{-\Omega}d\omega e^{\Omega} - e^{-\Omega}\omega\wedge(de^{\Omega})\big) - \\ &\quad \big((de^{-\Omega})e^{\Omega}\wedge(de^{-\Omega})e^{\Omega} + (de^{-\Omega})\wedge\omega e^{\Omega} + e^{-\Omega}\omega e^{\Omega}\wedge d(e^{-\Omega})e^{\Omega} + e^{-\Omega}\omega\wedge\omega e^{\Omega}\big) \\ &= -(de^{-\Omega})\wedge(de^{\Omega}) - e^{-\Omega}\omega\wedge(de^{\Omega}) - (de^{-\Omega})e^{\Omega}\wedge(de^{-\Omega})e^{\Omega} - e^{-\Omega}\omega e^{\Omega}\wedge(de^{-\Omega})e^{\Omega} \\ &\quad + e^{-\Omega}(d\omega - \omega\wedge\omega)e^{\Omega} \\ &= -\big(de^{-\Omega} + e^{-\Omega}\omega\big)\wedge\big(de^{\Omega} + e^{\Omega}(de^{-\Omega})e^{\Omega}\big) \\ &= -\big(de^{-\Omega} + e^{-\Omega}\omega\big)\wedge\big(de^{\Omega} - (de^{\Omega})e^{-\Omega}e^{\Omega}\big) \\ &= 0 \end{aligned}$$

But $de^{-\Omega} = e^{-\Omega}\frac{e^{\mathrm{ad}\Omega}-1}{\mathrm{ad}_\Omega}\omega$, thus $\hat{\omega} = e^{-\Omega}\frac{e^{\mathrm{ad}\Omega}+\mathrm{ad}_\Omega-1}{\mathrm{ad}_\Omega}\omega e^{\Omega}$

Conversely, $V = e^{\Omega}\hat{V}$, $dV = ((de^{\Omega})e^{-\Omega} + e^{\Omega}\hat{\omega}e^{-\Omega})V = \omega V$

Note that

$$\sum_{i=0}^{k}(-1)^i\binom{n+1}{i} = 1 + \sum_{i=1}^{k}(-1)^i\left(\binom{n}{i} + \binom{n}{i-1}\right) = (-1)^k\binom{n}{k}$$

$$\begin{aligned} &\sum_{\substack{p+q+r=n-1\\p,q,r\geq 0}}\frac{(-1)^{p+q+1}}{(p+q+1)!r!}\Omega^p\omega\Omega^{q+r} + \sum_{\substack{k+l=n-1\\k,l\geq 0}}\frac{(-1)^k}{k!l!}\Omega^k\omega\Omega^l \\ &= \sum_{k+l=n-1}\Omega^k\omega\Omega^l\left(\frac{(-1)^k}{k!l!} + \sum_{q+r=l}\frac{(-1)^{k+q+1}}{(k+q+1)!r!}\right) \\ &= \sum_{k+l=n-1}\Omega^k\omega\Omega^l\left(\frac{(-1)^k}{k!l!} + \frac{(-1)^n}{n!}\sum_{r=0}^{l}(-1)^r\binom{n}{r}\right) \\ &= \sum_{k+l=n-1}\Omega^k\omega\Omega^l\left(\frac{(-1)^k}{k!l!} + \frac{(-1)^n}{n!}(-1)^l\binom{n-1}{l}\right) \\ &= \frac{n-1}{n!}\sum_{k+l=n-1}(-1)^{k+1}\binom{n-1}{k}\Omega^k\omega\Omega^l \end{aligned}$$

Note that we also observe

$$
\begin{aligned}
&\left[\sum_{\substack{k+l=n-1\\k,l\geq 0}}(-1)^{k+1}\binom{n-1}{k}\Omega^k\omega\Omega^l,\Omega\right]\\
&=\sum_{\substack{k+l=n-1\\k,l\geq 0}}(-1)^{k+1}\binom{n-1}{k}(\Omega^k\omega\Omega^{l+1}-\Omega^{k+1}\omega\Omega^l)\\
&=\sum_{\substack{k+l=n\\k\geq 0,l\geq 1}}(-1)^{k+1}\binom{n-1}{k}\Omega^k\omega\Omega^l-\sum_{\substack{k+l=n\\l\geq 0,k\geq 1}}(-1)^{k}\binom{n-1}{k-1}\Omega^k\omega\Omega^l\\
&=\sum_{\substack{k+l=n\\k\geq 0,l\geq 1}}(-1)^{k+1}\binom{n-1}{k}\Omega^k\omega\Omega^l+\sum_{\substack{k+l=n\\l\geq 0,k\geq 1}}(-1)^{k+1}\binom{n-1}{k-1}\Omega^k\omega\Omega^l\\
&=\sum_{\substack{k+l=n\\k,l\geq 0}}(-1)^{k+1}\binom{n}{k}\Omega^k\omega\Omega^l
\end{aligned}
$$

A formula of Richard Hain says

$$V=\int I+\omega+\omega\omega+\omega\omega\omega+\cdots$$

So $\mathrm{SymbolMap}(V)=I+\Omega+\Omega\otimes\Omega+\Omega\otimes\Omega\otimes\Omega+\cdots$, and $\mathrm{SymbolToForms}(\mathrm{SymbolMap}(V))=\hat{\omega}$ which is symmetrized since

$$
\begin{aligned}
\text{Symmetrization}\left(\textstyle\sum_{k=0}^{n}(-1)^{k+1}\binom{n}{k}\Omega^k\omega\Omega^l\right)&=\textstyle\sum_{k=0}^{n}(-1)^{k+1}\binom{n}{k}\Omega^k\omega\Omega^l-\left(\frac{1}{n+1}\sum_{k=0}^{n}(-1)^{k+1}\binom{n}{k}\right)d(\Omega^{n+1})\\
&=\textstyle\sum_{k=0}^{n}(-1)^{k+1}\binom{n}{k}\Omega^k\omega\Omega^l
\end{aligned}
$$

36.18 Multiple polylogarithm

Definition 36.11 The *multiple polylogarithms* are

$$\mathrm{Li}_{\mathbf{n}}(\mathbf{x})=\sum_{\mathbf{k}}\frac{\mathbf{x}^{\mathbf{k}}}{\mathbf{k}^{\mathbf{n}}}=\int_0^1\frac{dt}{a_1-t}\left(\frac{dt}{t}\right)^{n_1-1}\cdots\frac{dt}{a_d-t}\left(\frac{dt}{t}\right)^{n_d-1}$$

Here $\mathbf{k}$ runs over $0<k_1<\cdots<k_d$, $a_j=a_j(\mathbf{x})=(x_j\cdots x_d)^{-1}$

Define $\mathrm{Li}_0(x)=\frac{x}{1-x}$

Note. For $\mathbf{k}$ runs over $(k_1,\cdots,k_d)\in\mathbb{Z}^d_{\geq 1}$

$$\sum_{\mathbf{k}}\frac{\mathbf{x}^{\mathbf{k}}}{\mathbf{k}^{\mathbf{n}}}=\left(\sum_{k_1}\frac{x_1^{k_1}}{k_1^{n_1}}\right)\cdots\left(\sum_{k_d}\frac{x_d^{k_d}}{k_d^{n_d}}\right)=\mathrm{Li}_{n_1}(x_1)\cdots\mathrm{Li}_{n_d}(x_d)$$

Can be written in terms of multiple polylogarithms

Note.

$$\begin{aligned}\mathrm{Li}_{n_1,\cdots,n_{i-1},0,n_{i+1},\cdots,n_d}(x_1,\cdots,x_d) &= \sum_{0<k_1<\cdots<k_d} \frac{x_1^{k-1}\cdots x_d^{k_d}}{k_1^{n_1}\cdots k_{i-1}^{n_{i-1}}k_{i+1}^{n_{i+1}}\cdots k_d^{n_d}} \\ &= \sum_{0<k_1<\cdots<k_d} \frac{x_1^{k-1}\cdots x_{i-1}^{k_{i-1}}x_{i+1}^{k_{i+1}}\cdots x_d^{k_d}}{k_1^{n_1}\cdots k_{i-1}^{n_{i-1}}k_{i+1}^{n_{i+1}}\cdots k_d^{n_d}} \frac{x_i^{k_{i-1}+1}-x_i^{k_{i+1}}}{1-x_i} \\ &= \sum_{0<k_1<\cdots<k_d} \frac{x_1^{k-1}(\cdots x_{i-1}x_i)^{k_{i-1}}x_{i+1}^{k_{i+1}}\cdots x_d^{k_d}}{k_1^{n_1}\cdots k_{i-1}^{n_{i-1}}k_{i+1}^{n_{i+1}}\cdots k_d^{n_d}} \frac{x_i}{1-x_i} \\ &- \sum_{0<k_1<\cdots<k_d} \frac{x_1^{k-1}\cdots x_{i-1}^{k_{i-1}}(x_ix_{i+1})^{k_{i+1}}\cdots x_d^{k_d}}{k_1^{n_1}\cdots k_{i-1}^{n_{i-1}}k_{i+1}^{n_{i+1}}\cdots k_d^{n_d}} \frac{1}{1-x_i} \\ &= \mathrm{Li}_{n_1,\cdots,n_{i-1},n_{i+1},\cdots,n_d}(x_1,\cdots,x_{i-1}x_i,x_{i+1},\cdots,x_d)\frac{x_i}{1-x_i} \\ &-\mathrm{Li}_{n_1,\cdots,n_{i-1},n_{i+1},\cdots,n_d}(x_1,\cdots,x_{i-1},x_ix_{i+1},\cdots,x_d)\frac{1}{1-x_i}\end{aligned}$$

$$\mathrm{Li}_{n_1,\cdots,n_{d-1},0}(x_1,\cdots,x_d) = \mathrm{Li}_{n_1,\cdots,n_{d-1}}(x_1,\cdots,x_{d-1}x_d)\frac{x_d}{1-x_d}$$

$$\mathrm{Li}_{0,n_2,\cdots,n_d}(x_1,\cdots,x_d) = \mathrm{Li}_{n_2,\cdots,n_{d-1}}(x_2,\cdots,x_d)\frac{x_1}{1-x_1} - \mathrm{Li}_{n_2,\cdots,n_{d-1}}(x_1x_2,\cdots,x_d)\frac{1}{1-x_1}$$

Exercise 36.12 (Derivatives of polylogarithms) Observe the following

$$\begin{aligned}\frac{\partial}{\partial x_i}\left(\sum_{\mathbf{k}} \frac{\cdots x_{i-1}^{k_{i-1}}x_i^{k_i}x_{i+1}^{k_{i+1}}\cdots}{\cdots k_{i-1}^{n_{i-1}}k_i^{n_i}k_{i+1}^{n_{i+1}}\cdots}\right) &= \sum_{\mathbf{k}} \frac{\cdots x_{i-1}^{k_{i-1}}x_i^{k_i-1}x_{i+1}^{k_{i+1}}\cdots}{\cdots k_{i-1}^{n_{i-1}}k_i^{n_i-1}k_{i+1}^{n_{i+1}}\cdots} \\ &= \sum_{\mathbf{k}} \frac{\cdots x_{i-1}^{k_{i-1}}x_i^{k_i}x_{i+1}^{k_{i+1}}\cdots}{\cdots k_{i-1}^{n_{i-1}}k_i^{n_i-1}k_{i+1}^{n_{i+1}}\cdots}\frac{1}{x_i}\end{aligned}$$

Write $u_i = \log(x_i)$, $v_i = \log(1-x_i)$, $u_{ij} = \log(x_i\cdots x_j)$, $v_{ij} = \log(1-x_i\cdots x_j)$. If $m_i > 1$, then

$$\begin{aligned}d_i\mathrm{Li}_{n_1,\cdots,n_d}(z_1,\cdots,z_d) &= \mathrm{Li}_{n_1,\cdots,n_i-1,\cdots,n_d}(z_1,\cdots,z_d)\frac{dx_i}{x_i} \\ &= \mathrm{Li}_{n_1,\cdots,n_i-1,\cdots,n_d}(z_1,\cdots,z_d)du_i\end{aligned}$$

$$\begin{aligned}d_d\mathrm{Li}_{n_1,\cdots,n_{d-1},1}(z_1,\cdots,z_d) &= \mathrm{Li}_{n_1,\cdots,n_{d-1}}(z_1,\cdots,z_{d-1}z_d)\frac{dx_d}{1-x_d} \\ &= -\mathrm{Li}_{n_1,\cdots,n_{d-1}}(z_1,\cdots,z_{d-1}z_d)dv_d\end{aligned}$$

$$\begin{aligned}d_1\mathrm{Li}_{1,n_2,\cdots,n_d}(z_1,\cdots,z_d) &= \mathrm{Li}_{n_2,\cdots,n_d}(z_2,\cdots,z_d)\frac{dx_1}{1-x_1} \\ &-\mathrm{Li}_{n_2,\cdots,n_d}(z_1z_2,\cdots,z_d)\frac{dx_1}{x_1(1-x_1)}\end{aligned}$$

$$\begin{aligned}d_i\mathrm{Li}_{n_1,\cdots,n_{i-1},1,n_{i+1},\cdots,n_d}(z_1,\cdots,z_d) &= \mathrm{Li}_{n_1,\cdots,n_{i-1},n_{i+1},\cdots,n_d}(z_1,\cdots,z_{i-1}z_i,z_{i+1},\cdots,z_d)\frac{dx_i}{1-x_i} \\ &-\mathrm{Li}_{n_1,\cdots,n_{i-1},n_{i+1},\cdots,n_d}(z_1,\cdots,z_{i-1},z_iz_{i+1},\cdots,z_d)\frac{dx_i}{x_i(1-x_i)}\end{aligned}$$

Remark.

Note that $\mathrm{Li}_{2n}(1) = \zeta(2n) = (-1)^{n+1}\frac{(2\pi)^{2n}B_{2n}}{2(2n)!}$, $\mathrm{Li}_{2n}(1) + \mathrm{Li}_{2n}(-1) = \frac{1}{2^{2n-1}}\mathrm{Li}_{2n}(1)$, which implies $\mathrm{Li}_{2n}(-1) = (\frac{1}{2^{2n-1}}-1)\mathrm{Li}_{2n}(1) = (-1)^{n+1}\frac{\pi^{2n}B_{2n}}{(2n)!}(1-2^{2n-1})$

Theorem 36.13 $\mathrm{Li}_{2n}(x)+\mathrm{Li}_{2n}(x^{-1})=-\sum_{k=0}^{k}\frac{\pi^{2(n-k)}}{(2k)!}\log^{2k}(-x)$ $\mathrm{Li}_{2n+1}(x)-\mathrm{Li}_{2n-1}(x^{-1})=$

Proof. $\mathrm{Li}_0(x)+\mathrm{Li}_0(x^{-1})=-1$, $\mathrm{Li}_1(x)+\mathrm{Li}_1(x^{-1})=\log(-x)$, $\mathrm{Li}_2(x)+\mathrm{Li}_2(x^{-1})=-\frac{1}{2}\log^2(-x)-\frac{\pi^2}{6}$

$$d(\mathrm{Li}_n(x)+(-1)^n\mathrm{Li}_n(x^{-1}))=(\mathrm{Li}_{n-1}(x)+(-1)^{n-1}\mathrm{Li}_{n-1}(x^{-1}))\frac{dx}{x}$$

□

Definition 36.14 The generating function of Bernoulli polynomials is $\frac{te^{xt}}{e^t-1}=\sum B(x)\frac{t^n}{n!}=\frac{d/dx}{e^{d/dx}-1}x^n$

We have

$$\mathrm{Li}_n(u)+(-1)^n\mathrm{Li}_n(-u)=-\frac{(2\pi i)^n}{n!}B_n(u)$$

36.19 Polylogarithm

Definition 36.15 The *Polylogarithms* are

$$\mathrm{Li}_n(z)=\sum_{k=1}^{\infty}\frac{z^k}{k^n}$$

Note that

$$\mathrm{Li}_{n+1}(z)=\int_0^z\frac{\mathrm{Li}_n(t)}{t}dt,\quad \mathrm{Li}_1(z)=-\ln(1-z)$$

Hence

$$\mathrm{Li}_n(z)=\int_0^z\left(\frac{dt}{t}\right)^{n-1}\frac{dt}{1-t}=\int_0^1\left(\frac{dt}{t}\right)^{n-1}\frac{dt}{z^{-1}-t}$$

Dilogarithm $\mathrm{Li}_2(z)=-\int_0^z\frac{\ln(1-u)}{u}du$ is the analytic continuation on $\mathbb{C}\setminus\{0,1\}$, avoiding the the cut $[1,\infty]$

Lemma 36.16 $\mathrm{Li}_k(z)$ satisfies differential equation

$$\left[(1-z)\frac{d}{dz}\right]\left(z\frac{d}{dz}\right)^{k-1}y=1$$

Other solutions are $\frac{\ln^j z}{j!}$, $1\le j\le k-1$

To compute the monodromy around $x=0$, take $q(\epsilon)$ to be the loop $x=\epsilon e^{it}$, we get 0.
To compute the monodromy around $x=1$, take $q(\epsilon)$ to be the composition of $x=(1-t)\epsilon+t(1-\epsilon)$, $x=1-\epsilon e^{it}$ and $x=(1-t)(1-\epsilon)+t\epsilon$, we get $-\frac{2\pi i}{(n-1)!}\log^{n-1}x$

The variation matrix is

$$\Lambda=\begin{bmatrix}1&&&&\\ \mathrm{Li}_1(x)&1&&&\\ \mathrm{Li}_2(x)&\log x&1&&\\ \vdots&\vdots&\ddots&\ddots&\\ \mathrm{Li}_{n-1}(x)&\frac{\log^{n-2}x}{(n-2)!}&\cdots&\log x&1\\ \mathrm{Li}_n(x)&\frac{\log^{n-1}x}{(n-1)!}&\cdots&\frac{\log^2 x}{2!}&\log x\ \ 1\end{bmatrix}\tau_n(2\pi i)$$

$$\omega = \begin{bmatrix} 0 & & & & & \\ -dv & 0 & & & & \\ 0 & du & 0 & & & \\ \vdots & \vdots & \ddots & \ddots & & \\ 0 & 0 & \cdots & du & 0 & \\ 0 & 0 & \cdots & 0 & du & 0 \\ & & & & & \end{bmatrix}$$

The monodromy representation ρ is as follows
For monodromy around $x = 0$

$$\begin{bmatrix} 1 & & & \\ 1 & & & \\ 1 & 1 & & \\ \vdots & \vdots & \ddots & \\ \frac{1}{n!} & \frac{1}{(n-1)!} & \cdots & 1 \end{bmatrix}$$

For monodromy around $x = 1$

$$\begin{bmatrix} 1 & & & \\ -1 & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$

$\hat{\mathbb{C}} = \{(u, v) \in \mathbb{C}^2 | e^u + e^v = 1\}$, $u = \log x, v = \log(1 - x)$

$\hat{\mathcal{L}_1} : \hat{\mathbb{C}} \to \mathbb{C}/(2\pi i)\mathbb{Z}$ is just Li_1

$\hat{\mathcal{L}_2} : \hat{\mathbb{C}} \to \mathbb{C}/(2\pi i)^2\mathbb{Z}$ is $\mathrm{Li}_2(x) - \frac{1}{2}\log x \mathrm{Li}_1(x)$, $[M_1, M_0]\hat{\mathcal{L}_2} = (2\pi i)^2$, $d\hat{\mathcal{L}_2} = \frac{1}{2}(udv - vdu)$

36.20 Symbols

Definition 36.17 The symbol mod product map is

$$\rho(a_1 \otimes \cdots \otimes a_k) = \frac{k-1}{k}[\rho(a_1 \otimes \cdots \otimes a_{k-1}) \otimes a_k - \rho(a_2 \otimes \cdots \otimes a_k) \otimes a_1]$$

Starting with $\rho(a_1) = a_1$

Note. Roughly ρ is given inductively by $\frac{k-1}{k}[\rho, 1]$

Theorem 36.18 $\ker \rho$ is precisely products(shuffle product) of symbols. And ρ is idempotent

36.21 Variation matrix

Theorem 36.19 Λ is the fundamental solution of the system of linear differential equations

$$d\Lambda = \omega\Lambda$$

Example 36.20 For

$$\begin{aligned} \mathrm{Li}_{1,1}(x, y) &= \int_{(0,0)}^{(x,y)} dv_1 dv_2 + dv_{12} d(u_1 - v_1 + v_2) \\ &= \int_{(0,0)}^{(x,y)} dv_1 dv_2 + dv_{12} du_1 - dv_{12} dv_1 + dv_{12} dv_2 \end{aligned}$$

$(0, 0) < (0, 1) < (1, 0) < (1, 1)$ in $\mathfrak{S}(1, 1)$

$$\Lambda = \begin{bmatrix} 1 & & & \\ \mathrm{Li}_1(y) & 1 & & \\ \mathrm{Li}_1(xy) & & 1 & \\ \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x) & \mathrm{Li}_1\left(\frac{1-xy}{1-x}\right) & 1 \end{bmatrix} \tau_{1,1}(2\pi i)$$

$$\omega = \begin{bmatrix} 0 & & & \\ -dv_2 & 0 & & \\ -dv_{12} & 0 & 0 & \\ 0 & -dv_1 & d(-u_1+v_1-v_2) & 0 \end{bmatrix}$$

Example 36.21 For

$$\begin{aligned} \mathrm{Li}_{2,1}(x,y) &= \int_{(0,0)}^{(x,y)} (dv_1 dv_2 + dv_{12} d(u_1 - v_1 + v_2)) du_1 + dv_{12} d(u_1 + u_2) dv_2 \\ &= \int_{(0,0)}^{(x,y)} dv_1 dv_2 du_1 + dv_{12} du_1 du_1 - dv_{12} dv_1 du_1 \\ &+ dv_{12} dv_2 du_1 + dv_{12} du_1 dv_2 + dv_{12} u_2 dv_2 \end{aligned}$$

$(0,0) < (0,1) < (1,0) < (1,1) < (2,0) < (2,1)$ in $\mathfrak{S}(2,1)$

$$\Lambda = \begin{bmatrix} 1 & & & & & \\ \mathrm{Li}_1(y) & 1 & & & & \\ \mathrm{Li}_1(xy) & & 1 & & & \\ \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x) & \log\frac{1-x}{(1-y)x} & 1 & & \\ \mathrm{Li}_2(xy) & & \log(xy) & & 1 & \\ \mathrm{Li}_{2,1}(x,y) & \mathrm{Li}_2(x) & g(x,y) & \log x & \mathrm{Li}_1(y) & 1 \end{bmatrix} \tau_{2,1}(2\pi i)$$

Where $dg = \log\frac{1-x}{(1-y)x}\frac{dx}{x} + \log(xy)\frac{dy}{1-y}$

$$\omega = \begin{bmatrix} 0 & & & & & \\ -dv_2 & 0 & & & & \\ -dv_{12} & 0 & 0 & & & \\ 0 & -dv_1 & d(-u_1+v_1-v_2) & 0 & & \\ 0 & 0 & d(u_1+u_2) & 0 & 0 & \\ 0 & 0 & 0 & du_1 & -dv_2 & 0 \end{bmatrix}$$

Example 36.22 For

$$\begin{aligned} \mathrm{Li}_{1,1,1}(x,y,z) &= \int_{(0,0,0)}^{(x,y,z)} \\ &= \int_{(0,0,0)}^{(x,y,z)} \end{aligned}$$

$(0,0,0) < (0,0,1) < (0,1,0) < (1,0,0) < (0,1,1) < (1,0,1) < (1,1,0) < (1,1,1)$ in $\mathfrak{S}(1,1,1)$

$$\Lambda = \begin{bmatrix} 1 & & & & & & & \\ \mathrm{Li}_1(z) & 1 & & & & & & \\ \mathrm{Li}_1(yz) & & 1 & & & & & \\ \mathrm{Li}_1(xyz) & & & 1 & & & & \\ \mathrm{Li}_{1,1}(y,z) & \mathrm{Li}_1(y) & \log\frac{1-y}{(1-z)y} & & 1 & & & \\ \mathrm{Li}_{1,1}(xy,z) & \mathrm{Li}_1(xy) & & \log\frac{1-xy}{(1-z)xy} & & 1 & & \\ \mathrm{Li}_{1,1}(x,yz) & & \mathrm{Li}_1(x) & \log\frac{1-x}{(1-yz)x} & & & 1 & \\ \mathrm{Li}_{1,1,1}(x,y,z) & g(x,y) & \mathrm{Li}_1(x)\log\frac{1-y}{(1-z)y} & h(x,y) & \mathrm{Li}_1(x) & \log\frac{1-y}{(1-x)x} & \log\frac{1-z}{(1-y)y} & 1 \end{bmatrix} \tau_{1,1,1}(2\pi i)$$

Where

$$\omega = \begin{bmatrix} 0 & & & & & & & \\ -dv_3 & 0 & & & & & & \\ -dv_{23} & 0 & 0 & & & & & \\ -dv_{13} & 0 & 0 & 0 & & & & \\ 0 & -dv_2 & d(v_2-u_2-v_3) & 0 & 0 & & & \\ 0 & -dv_{12} & 0 & d(v_{12}-u_1-u_2-v_3) & 0 & 0 & & \\ 0 & 0 & -dv_1 & d(v_1-u_1-v_{23}) & 0 & 0 & 0 & \\ 0 & 0 & 0 & 0 & -dv_1 & d(v_1-u_1-v_2) & d(v_2-u_2-v_3) & 0 \end{bmatrix}$$

Example 36.23 For

$$\begin{aligned} \mathrm{Li}_{1,2}(x,y) &= dv_{12}d(u_1+u_2)d(u_1-v_2) + (dv_1dv_2 + dv_{12}d(u_1-v_1+v_2))du_2 \\ &= dv_{12}du_1du_1 - dv_{12}du_1dv_2 + dv_{12}du_2du_1 - dv_{12}du_2dv_2 \\ &\quad + dv_1dv_2du_2 + dv_{12}du_1du_2 - dv_{12}dv_1du_2 + dv_{12}dv_2du_2 \end{aligned}$$

$(0,0) < (0,1) < (1,0) < (1,1) < (0,2) < (1,2)$ in $\mathfrak{S}(1,2)$

$$\Lambda = \begin{bmatrix} 1 & & & & & & \\ \mathrm{Li}_1(y) & 1 & & & & & \\ \mathrm{Li}_1(xy) & 0 & 1 & & & & \\ \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x) & \log\frac{1-x}{(1-y)x} & 1 & & & \\ \mathrm{Li}_2(y) & \log y & & & 1 & & \\ \mathrm{Li}_2(xy) & 0 & \log(xy) & & & 1 & \\ \mathrm{Li}_{1,2}(x,y) & \mathrm{Li}_1(x)\log y & g(x,y) & \log y & \mathrm{Li}_1(x) & -\mathrm{Li}_1(x^{-1}) & 1 \end{bmatrix} \tau_{1,2}(2\pi i)$$

Where $g(x,y) = -I((xy)^{-1}; y^{-1}, 0; 1)$

$$\omega = \begin{bmatrix} 0 & & & & & & \\ -dv_2 & 0 & & & & & \\ -dv_{12} & 0 & 0 & & & & \\ 0 & -dv_1 & d(-u_1+v_1-v_2) & 0 & & & \\ 0 & du_2 & 0 & 0 & 0 & & \\ 0 & 0 & d(u_1+u_2) & 0 & 0 & 0 & \\ 0 & 0 & 0 & du_2 & -dv_1 & d(v_1-u_1) & 0 \end{bmatrix}$$

Example 36.24 For

$$\begin{aligned}\mathrm{Li}_{2,2}(x,y) &= (dv_{12}du(u_1+u_2)d(u_1-v_2)+(dv_1dv_2+dv_{12}d(u_1-v_1+v_2))du_2)du_1\\ &+((dv_1dv_2+dv_{12}d(u_1-v_1+v_2))du_1+dv_{12}d(u_1+u_2)dv_2)du_2\\ &= dv_{12}du_1du_1du_1-dv_{12}du_1dv_2du_1+dv_{12}du_2du_1du_1-dv_{12}du_2dv_2du_1\\ &+dv_1dv_2du_2du_1+dv_{12}du_1du_2du_1-dv_{12}dv_1du_2du_1+dv_{12}dv_2du_2du_1\\ &+dv_1dv_2du_1du_2+dv_{12}du_1du_1du_2-dv_{12}dv_1du_1du_2\\ &+dv_{12}dv_2du_1du_2+dv_{12}du_1dv_2du_2+dv_{12}u_2dv_2du_2\end{aligned}$$

$(0,0)<(0,1)<(1,0)<(1,1)<(0,2)<(2,0)<(1,2)<(2,1)<(2,2)$ in $\mathfrak{S}(2,2)$

$$\Lambda=\begin{bmatrix}1\\ \mathrm{Li}_1(y) & 1\\ \mathrm{Li}_1(xy) & & 1\\ \mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(x) & \log\frac{1-x}{(1-y)x} & 1\\ \mathrm{Li}_2(y) & \log y & & & 1\\ \mathrm{Li}_2(xy) & & \log(xy) & & & 1\\ \mathrm{Li}_{1,2}(x,y) & \mathrm{Li}_1(x)\log y & g(x,y) & \log y & \mathrm{Li}_1(x) & \log\frac{1-x}{x} & 1\\ \mathrm{Li}_{2,1}(x,y) & \mathrm{Li}_2(x) & h(x,y) & \log x & & \mathrm{Li}_1(y) & & 1\\ \mathrm{Li}_{2,2}(x,y) & \mathrm{Li}_2(x)\log y & i(x,y) & \log x\log y & \mathrm{Li}_2(x) & \mathrm{Li}_2(y)-\mathrm{Li}_2(x)-\frac{1}{2}\log^2 x & \log x & \log y & 1\end{bmatrix}\tau_{2,2}(2\pi i)$$

$$\omega=\begin{bmatrix}0\\ -dv_2 & 0\\ -dv_{12} & 0 & 0\\ 0 & -dv_1 & d(-u_1+v_1-v_2) & 0\\ 0 & du_2 & 0 & 0 & 0\\ 0 & 0 & d(u_1+u_2) & 0 & 0 & 0\\ 0 & 0 & 0 & du_2 & -dv_1 & d(v_1-u_1) & 0\\ 0 & 0 & 0 & du_1 & 0 & -dv_2 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & du_1 & du_2 & 0\end{bmatrix}$$

36.22 Multiple polylogarithms (older draft)

Definition 36.25 The *multiple polylogarithms* are

$$\mathrm{Li}_{\mathbf{n}}(\mathbf{z})=\sum_{\mathbf{k}}\frac{\mathbf{z}^{\mathbf{k}}}{\mathbf{k}^{\mathbf{n}}}=\int_0^1\left(\frac{dt}{t}\right)^{n_1-1}\frac{dt}{a_1-t}\cdots\left(\frac{dt}{t}\right)^{n_d-1}\frac{dt}{a_d-t}$$

Here $\mathbf{k}$ runs over $k_1>\cdots>k_d\geq 1$, $a_j=a_j(\mathbf{z})=(z_1\cdots z_j)^{-1}$, $a_0=1$, $a_{n+1}=0$

Note. For $\mathbf{k}$ runs over $(k_1,\cdots,k_d)\in\mathbb{Z}^d_{\geq 1}$

$$\sum_{\mathbf{k}}\frac{\mathbf{z}^{\mathbf{k}}}{\mathbf{k}^{\mathbf{n}}}=\left(\sum_{k_1}\frac{z_1^{k_1}}{k_1^{n_1}}\right)\cdots\left(\sum_{k_d}\frac{z_d^{k_d}}{k_d^{n_d}}\right)=\mathrm{Li}_{n_1}(z_1)\cdots\mathrm{Li}_{n_d}(z_d)$$

Can be written in terms of multiple polylogarithms

Exercise 36.26 (Derivatives of polylogarithms) If $m_i>1$, then

$$\frac{\partial}{\partial z_i}\mathrm{Li}_{\mathbf{m}',m_i,\mathbf{m}''}(\mathbf{z}',z_i,\mathbf{z}'')=\frac{1}{z_i}\mathrm{Li}_{\mathbf{m}',m_i-1,\mathbf{m}''}(\mathbf{z}',z_i,\mathbf{z}'')$$

$$\frac{\partial}{\partial z_1}\mathrm{Li}_{1,\mathbf{m}'}(z_1,z_2,\mathbf{z}')=\frac{1}{1-z_1}\mathrm{Li}_{\mathbf{m}'}(z_1z_2,\mathbf{z}')$$

$$\frac{\partial}{\partial z_d}\mathrm{Li}_{\mathbf{m}',1}(\mathbf{z}',z_{d-1},z_d)=\frac{1}{1-z_d}\mathrm{Li}_{\mathbf{m}'}(\mathbf{z}',z_{d-1})-\frac{1}{z_d(1-z_d)}\mathrm{Li}_{\mathbf{m}'}(\mathbf{z}',z_{d-1}z_d)$$

If $1<i<d$, then

$$\frac{\partial}{\partial z_i}\mathrm{Li}_{\mathbf{m}',1,\mathbf{m}''}(\mathbf{z}',z_{i-1},z_i,z_{i+1},\mathbf{z}'')=\frac{1}{1-z_i}\mathrm{Li}_{\mathbf{m}',\mathbf{m}''}(\mathbf{z}',z_{i-1},z_iz_{i+1},\mathbf{z}'')$$
$$-\frac{1}{z_i(1-z_i)}\mathrm{Li}_{\mathbf{m}',\mathbf{m}''}(\mathbf{z}',z_{i-1}z_i,z_{i+1},\mathbf{z}'')$$

Lemma 36.27 Write $\mathfrak{S}_n=\{0,1\}^n$, $\mathbf{0}=(0,\cdots,0)$, $\mathbf{1}=(1,\cdots,1)$, $\mathbf{u}_s=(0,\cdots,\underset{s\text{-th}}{1\uparrow},\cdots,0)$, define *multiple logarithm* $\mathcal{L}_n=\mathrm{Li}_{\mathbf{1}}$, $\mathcal{L}_0=1$. Denote $\mathbf{x}_j=(x_1,\cdots,x_jx_{j+1},x_{j+2},\cdots,x_n)$, $\mathbf{x}_n=(x_1,\cdots,x_{n-1})$, we have

$$d_j\mathcal{L}_n(\mathbf{x})=\mathcal{L}_{n-1}(\mathbf{x}_{j-1})\frac{dx_j}{x_j(x_j-1)}-\mathcal{L}_{n-1}(\mathbf{x}_j)\frac{dx_j}{x_j-1}$$

For $2\leq j\leq n$, and

$$d_1\mathcal{L}_n(\mathbf{x})=-\mathcal{L}_{n-1}(\mathbf{x}_1)\frac{dx_1}{x_1-1}$$

Therefore

$$\begin{aligned}d\mathcal{L}_n(\mathbf{x})&=\sum_{j=1}^n d_j\mathcal{L}_n(\mathbf{x})\\&=\sum_{j=1}^{n-1}\left(\mathcal{L}_{n-1}(\mathbf{x}_j)\frac{dx_j}{1-x_j}+\mathcal{L}_{n-1}(\mathbf{x}_j)\frac{dx_{j+1}}{x_{j+1}(x_{j+1}-1)}\right)+\mathcal{L}_{n-1}(\mathbf{x}_n)\frac{dx_n}{x_n-1}\\&=\sum_{j=1}^{n-1}\mathcal{L}_{n-1}(\mathbf{x}_j)\left(-d\ln(1-x_j)+d\ln\left(\frac{x_{j+1}-1}{x_{j+1}}\right)\right)+\mathcal{L}_{n-1}(\mathbf{x}_n)\frac{dx_n}{x_n-1}\\&=\sum_{j=1}^{n}\mathcal{L}_{n-1}(\mathbf{x}_j)d\ln\left(\frac{1-x_{j+1}^{-1}}{1-x_j}\right)\end{aligned}$$

Here $x_{n+1}=\infty$, $\mathcal{L}_0=1$

Suppose $\mathbf{i}\in\mathfrak{S}_n$, $|\mathbf{i}|=k$ and $i_{\tau_1}=\cdots=i_{\tau_k}=1$ for some $1\leq\tau_1\leq\cdots\leq\tau_k\leq n$, set

$$\mathbf{x}(\mathbf{i})=\mathbf{y},\quad y_m=\prod_{j=\tau_{m-1}+1}^{\tau_m}x_j=\frac{a_{\tau_{m-1}}}{a_{\tau_m}}$$

$$w_j(\mathbf{x})=d\ln\left(\frac{1-x_{j+1}^{-1}}{1-x_j}\right)$$

With $\tau_0=0$, $w_0(\mathbf{x})=1$. A partial order $\preccurlyeq$ on $\mathfrak{S}_n$ is given by $\mathbf{i}\preccurlyeq\mathbf{j}$ if $i_k\leq j_k$

Define

$$X_n=\left\{\prod_{j\leq k}(1-x_j\cdots x_k)=0\right\},\quad S_n=\mathbb{C}^n\setminus X_n$$

$$X_n'=\left\{\prod_i x_i\prod_{j\leq k}(1-x_j\cdots x_k)=0\right\},\quad S_n'=\mathbb{C}^n\setminus X_n'$$

$$D_n=\bigcap_j\left\{\left|x_j-\frac{1}{2}\right|<\frac{1}{2}\right\}$$

Theorem 36.28 Multiple logarithm $\mathcal{L}_n(\mathbf{x})$ is a multi-valued holomorphic function on S_n

$$\mathcal{L}_n(\mathbf{x}) = \sum_{0\neq \mathbf{j}_1 \prec \cdots \prec \mathbf{j}_n} \int_0^{\mathbf{x}} w_{\mathbf{j}_n - \mathbf{j}_{n-1}}(\mathbf{x}(\mathbf{j}_n)) \cdots w_{\mathbf{j}_2 - \mathbf{j}_1}(\mathbf{x}(\mathbf{j}_2)) w_1(\mathbf{x}(\mathbf{j}_1))$$

Here $\mathbf{j} - \mathbf{i} = \begin{cases} s' & j_t = i_t + \delta_{st} \\ 0 & \text{otherwise} \end{cases}$ for $\mathbf{i} \prec \mathbf{j}$, j_s is the s'-th nonzero element in $\mathbf{j}$, and the integration is taken over $\alpha : I \to \mathbb{C}^n$

Proof. Use induction and Lemma 36.27

$$\begin{aligned}
\mathcal{L}_n(\mathbf{x}) &= \textstyle\int_{\mathbf{0}}^{\mathbf{x}} d\mathcal{L}_n(\mathbf{x}) \\
&= \textstyle\int_{\mathbf{0}}^{\mathbf{x}} \sum_{k=1}^{n-1} \mathcal{L}_{n-1}(\mathbf{x}_k) d\ln\left(\frac{1-x_{k+1}^{-1}}{1-x_k}\right) + \mathcal{L}_{n-1}(\mathbf{x}_n)\frac{dx_n}{1-x_n} \\
&= \textstyle\int_{\mathbf{0}}^{\mathbf{x}} \sum_{j=1}^{n} w_{\mathbf{1}-\mathbf{f}_k}(\mathbf{x}) \sum_{0\neq \mathbf{p}_1 \prec \cdots \prec \mathbf{p}_{n-1}} w_{\mathbf{p}_{n-1}-\mathbf{p}_{n-2}}(\mathbf{x}_k(\mathbf{p}_{n-1})) \cdots w_{\mathbf{p}_2-\mathbf{p}_1}(\mathbf{x}_k(\mathbf{p}_2)) w_1(\mathbf{x}_k(\mathbf{p}_1)) \\
&= \textstyle\int_{\mathbf{0}}^{\mathbf{x}} \sum_{j=1}^{n} w_{\mathbf{1}-\mathbf{f}_k}(\mathbf{x}) \sum_{0\neq \mathbf{q}_1 \prec \cdots \prec \mathbf{q}_{n-1}} w_{\mathbf{q}_{n-1}-\mathbf{q}_{n-2}}(\mathbf{x}(\mathbf{q}_{n-1})) \cdots w_{\mathbf{q}_2-\mathbf{q}_1}(\mathbf{x}(\mathbf{q}_2)) w_1(\mathbf{x}(\mathbf{q}_1)) \\
&= \textstyle\sum_{0\neq \mathbf{j}_1 \prec \cdots \prec \mathbf{j}_n} \int_0^{\mathbf{x}} w_{\mathbf{j}_n - \mathbf{j}_{n-1}}(\mathbf{x}(\mathbf{j}_n)) \cdots w_{\mathbf{j}_2 - \mathbf{j}_1}(\mathbf{x}(\mathbf{j}_2)) w_1(\mathbf{x}(\mathbf{j}_1))
\end{aligned}$$

Here $\mathbf{f}_k = (1, \cdots, \underset{\substack{\uparrow \\ k\text{-th}}}{0}, \cdots, 1)$, $\mathbf{q}_i$ is $\mathbf{p}_i$ with 0 inserted in as the k-th entry. Note that S_n is given so that $w_i(\mathbf{x}(\mathbf{j}))$ are defined

□

Example 36.29 When $n = 1$

$$\begin{aligned}
\mathcal{L}_1(x_1) &= \int_0^{x_1} w_1(\mathbf{x}(1)) \\
&= \int_0^{x_1} d\ln\left(\frac{1}{1-x_1}\right) \\
&= \int_0^{x_1} \frac{dx_1}{1-x_1}
\end{aligned}$$

When $n = 2$

$$\begin{aligned}
\mathcal{L}_2(\mathbf{x}) &= \int_{\mathbf{0}}^{\mathbf{x}} w_{(1,1)-(1,0)}(\mathbf{x}(\mathbf{1})) w_1(\mathbf{x}(1,0)) + w_{(1,1)-(0,1)}(\mathbf{x}(\mathbf{1})) w_1(\mathbf{x}(0,1)) \\
&= \int_{\mathbf{0}}^{\mathbf{x}} w_2(\mathbf{x}) w_1(x_1) + w_1(\mathbf{x}) w_1(x_1 x_2) \\
&= \int_{\mathbf{0}}^{\mathbf{x}} d\ln\left(\frac{1}{1-x_2}\right) d\ln\left(\frac{1}{1-x_1}\right) + d\ln\left(\frac{1-x_2^{-1}}{1-x_1}\right) d\ln\left(\frac{1}{1-x_1x_2}\right) \\
&= \int_{\mathbf{0}}^{\mathbf{x}} \frac{dx_2}{1-x_2}\frac{dx_1}{1-x_1} + \left(\frac{dx_2}{x_2(x_2-1)} + \frac{dx_1}{1-x_1}\right)\frac{d(x_1x_2)}{1-x_1x_2}
\end{aligned}$$

When $n = 3$

$$\begin{aligned}
\mathcal{L}_3(\mathbf{x}) &= \int_{\mathbf{0}}^{\mathbf{x}} w_{(1,1,1)-(1,1,0)}(\mathbf{x}(\mathbf{1}))w_{(1,1,0)-(1,0,0)}(\mathbf{x}(1,1,0))w_1(\mathbf{x}(1,0,0)) + \\
&\quad w_{(1,1,1)-(1,1,0)}(\mathbf{x}(\mathbf{1}))w_{(1,1,0)-(0,1,0)}(\mathbf{x}(1,1,0))w_1(\mathbf{x}(0,1,0)) + \\
&\quad w_{(1,1,1)-(1,0,1)}(\mathbf{x}(\mathbf{1}))w_{(1,0,1)-(1,0,0)}(\mathbf{x}(1,0,1))w_1(\mathbf{x}(1,0,0)) + \\
&\quad w_{(1,1,1)-(1,0,1)}(\mathbf{x}(\mathbf{1}))w_{(1,0,1)-(0,0,1)}(\mathbf{x}(1,0,1))w_1(\mathbf{x}(0,0,1)) + \\
&\quad w_{(1,1,1)-(0,1,1)}(\mathbf{x}(\mathbf{1}))w_{(0,1,1)-(0,1,0)}(\mathbf{x}(0,1,1))w_1(\mathbf{x}(0,1,0)) + \\
&\quad w_{(1,1,1)-(0,1,1)}(\mathbf{x}(\mathbf{1}))w_{(0,1,1)-(0,0,1)}(\mathbf{x}(0,1,1))w_1(\mathbf{x}(0,0,1)) \\
&= \int_{\mathbf{0}}^{\mathbf{x}} w_3(\mathbf{x})w_2(x_1,x_2)w_1(x_1) + w_3(\mathbf{x})w_1(x_1,x_2)w_1(x_1x_2) + \\
&\quad w_2(\mathbf{x})w_2(x_1,x_2x_3)w_1(x_1) + w_2(\mathbf{x})w_1(x_1,x_2x_3)w_1(x_1x_2x_3) + \\
&\quad w_1(\mathbf{x})w_2(x_1x_2,x_3)w_1(x_1x_2) + w_1(\mathbf{x})w_1(x_1x_2,x_3)w_1(x_1x_2x_3) \\
&= \int_{\mathbf{0}}^{\mathbf{x}} \frac{dx_3}{1-x_3}\frac{dx_2}{1-x_2}\frac{dx_1}{1-x_1} + \frac{dx_3}{1-x_3}\left(\frac{dx_2}{x_2(x_2-1)} + \frac{dx_1}{1-x_1}\right)\frac{d(x_1x_2)}{1-x_1x_2} + \\
&\quad \left(\frac{dx_3}{x_3(x_3-1)} + \frac{dx_2}{1-x_2}\right)\frac{d(x_2x_3)}{1-x_2x_3}\frac{dx_1}{1-x_1} + \\
&\quad \left(\frac{dx_3}{x_3(x_3-1)} + \frac{dx_2}{1-x_2}\right)\left(\frac{d(x_2x_3)}{x_2x_3(x_2x_3-1)} + \frac{dx_1}{1-x_1}\right)\frac{d(x_1x_2x_3)}{1-x_1x_2x_3} + \\
&\quad \left(\frac{dx_2}{x_2(x_2-1)} + \frac{dx_1}{1-x_1}\right)\frac{dx_3}{1-x_3}\frac{d(x_1x_2)}{1-x_1x_2} + \\
&\quad \left(\frac{dx_2}{x_2(x_2-1)} + \frac{dx_1}{1-x_1}\right)\left(\frac{dx_3}{x_3(x_3-1)} + \frac{d(x_1x_2)}{1-x_1x_2}\right)\frac{d(x_1x_2x_3)}{1-x_1x_2x_3}
\end{aligned}$$

Definition 36.30 $\mathbf{i},\mathbf{j} \in \mathfrak{S}_n$, $|\mathbf{i}| = k$, $|\mathbf{j}| = l$, the $\mathbf{i}$-th *retraction* map $\rho_{\mathbf{i}} : \mathfrak{S}_n \to \mathfrak{S}_k$ is defined by

- If $\mathbf{i} \not\succcurlyeq \mathbf{j}$, $\rho_{\mathbf{i}}(\mathbf{j}) = \mathbf{0}$
- If $\mathbf{i} \succcurlyeq \mathbf{j}$, assume $\tau_1,\cdots,\tau_k$ and $t_1,\cdots,t_l$ are the nonzero entries in $\mathbf{i}$ and $\mathbf{j}$, suppose $\tau_{\alpha_r} = t_r$, then $\alpha_1,\cdots,\alpha_l$ are the nonzero entries of $\rho_{\mathbf{i}}(\mathbf{j})$

Write $\theta_s = \theta_s(\mathbf{x}) = \frac{dt}{t-a_s}$, the $2^n \times 2^n$ *variation matrix* $\mathcal{M}_{\mathbf{1}}(\mathbf{x}) = (2\pi i)^l E_{\mathbf{i},\mathbf{j}}(\mathbf{x})$ associated with $\mathcal{L}_n(\mathbf{x})$ is defined by

$$\begin{aligned}
E_{\mathbf{i},\mathbf{j}} = \gamma^k_{\rho_{\mathbf{i}}(\mathbf{j})}(\mathbf{y}) &= (-1)^{k-l}\prod_{r=0}^{l}\int_{a_{\alpha_{r+1}}(\mathbf{y})}^{a_{\alpha_r}(\mathbf{y})} \theta_{\alpha_r+1}(\mathbf{y})\cdots\theta_{\alpha_{r+1}-1}(\mathbf{y}) \\
&= (-1)^{k-l}\prod_{r=0}^{l}\int_{a_{t_{r+1}}}^{a_{t_r}} \theta_{\tau_{\alpha_r+1}}(\mathbf{x})\cdots\theta_{\tau_{\alpha_{r+1}-1}}(\mathbf{x}) \\
&= (-1)^{k-l}\prod_{r=0}^{l}\int_{p_r} \theta_{\tau_{\alpha_r+1}}(\mathbf{x})\cdots\theta_{\tau_{\alpha_{r+1}-1}}(\mathbf{x})
\end{aligned}$$

$\tau_{k+1} = t_{l+1} = n+1$, $\alpha_{l+1} = k+1$. p_r are independent from $\mathbf{i}$ These thetas are very weird

Proposition 36.31

$$E_{\mathbf{i},\mathbf{j}} = \prod_{r=0}^{l} \mathcal{L}_{\alpha_{r+1}-\alpha_r-1}\left(\frac{a_{t_r}(\mathbf{x}) - a_{t_{r+1}}(\mathbf{x})}{a_{\tau_{\alpha_r+1}}(\mathbf{x}) - a_{t_{r+1}}(\mathbf{x})}, \cdots, \frac{a_{\tau_{\alpha_{r+1}-2}}(\mathbf{x}) - a_{t_{r+1}}(\mathbf{x})}{a_{\tau_{\alpha_{r+1}-1}}(\mathbf{x}) - a_{t_{r+1}}(\mathbf{x})}\right)$$
$$= \mathcal{L}_{k-\alpha_l}(x_{1+t_l}\cdots x_{\tau_{\alpha_l+1}}, \cdots, x_{1+\tau_{k-1}}\cdots x_{\tau_k}) \cdot$$
$$\prod_{r=0}^{l-1} \mathcal{L}_{\alpha_{r+1}-\alpha_r-1}\left(\frac{1 - x_{1+t_r}\cdots x_{t_{r+1}}}{1 - x_{1+\tau_{\alpha_r+1}}\cdots x_{t_{r+1}}}, \cdots, \frac{1 - x_{1+\tau_{\alpha_{r+1}-2}}\cdots x_{t_{r+1}}}{1 - x_{1+\tau_{\alpha_{r+1}-1}}\cdots x_{t_{r+1}}}\right)$$

Proof.

□

Example 36.32

$$E_{\mathbf{1},\mathbf{j}}(\mathbf{x}) = \gamma^n_{\mathbf{j}}(\mathbf{x}) = \prod_{r=0}^{l} \mathcal{L}_{t_{r+1}-t_r-1}\left(\frac{a_{t_r} - a_{t_{r+1}}}{a_{t_r+1} - a_{t_{r+1}}}, \cdots, \frac{a_{t_{r+1}-2} - a_{t_{r+1}}}{a_{t_{r+1}-1} - a_{t_{r+1}}}\right)$$
$$= \prod_{r=0}^{l} \mathcal{L}_{t_{r+1}-t_r-1}\left(\frac{1 - x_{1+t_r}\cdots x_{t_{r+1}}}{1 - x_{2+t_r}\cdots x_{t_{r+1}}}, \cdots, \frac{1 - x_{t_{r+1}-1}x_{t_{r+1}}}{1 - x_{t_{r+1}}}\right)$$
$$= \mathcal{L}_{k-\alpha_l}(x_{1+t_l}\cdots x_{t_l+1}, \cdots, x_n) \cdot$$
$$\prod_{r=0}^{l-1} \mathcal{L}_{t_{r+1}-t_r-1}\left(\frac{1 - x_{1+t_r}\cdots x_{t_{r+1}}}{1 - x_{2+t_r}\cdots x_{t_{r+1}}}, \cdots, \frac{1 - x_{t_{r+1}-1}x_{t_{r+1}}}{1 - x_{t_{r+1}}}\right)$$

In particular we have

$$E_{\mathbf{1},\mathbf{0}}(\mathbf{x}) = \gamma^n_{\mathbf{0}}(\mathbf{x}) = \mathcal{L}_n(\mathbf{x}), \quad E_{\mathbf{1},\mathbf{1}}(\mathbf{x}) = \gamma^n_{\mathbf{1}}(\mathbf{x}) = \prod_{r=0}^{n} \mathcal{L}_0 = 1$$

$E = E(\mathbf{x})$ has columns

$$C_{\mathbf{j}} = \sum_{\mathbf{i} \succcurlyeq \mathbf{j}} \gamma^{|\mathbf{i}|}_{\rho_{\mathbf{i}}(\mathbf{j})}(\mathbf{x}(\mathbf{i}))e_{\mathbf{i}}$$

$e_{\mathbf{i}}$ is the standard unit column vector, using the complete order $<$ on $\mathfrak{S}_n$: if $|\mathbf{i}| < |\mathbf{j}|$, then $\mathbf{i} < \mathbf{j}$, if $|\mathbf{i}| = |\mathbf{j}|$, then compare the lexicographic order from left to right with $1 < 0$. By definition, the $\mathbf{i}$-th row of $\mathcal{M}_{\mathbf{1}}$ is

$$R_{\mathbf{i}} = \sum_{\mathbf{i} \succcurlyeq \mathbf{j}} (2\pi i)^{|\mathbf{j}|} \gamma^{|\mathbf{i}|}_{\rho_{\mathbf{i}}(\mathbf{j})}(\mathbf{x}(\mathbf{i}))e^T_{\mathbf{j}}$$

And the $\mathbf{j}$-th column of $\mathcal{M}_{\mathbf{1}}$ is $(2\pi i)^{|\mathbf{j}|}C_{\mathbf{j}}$ Note that $\gamma^{|\mathbf{i}|}_{\rho_{\mathbf{i}}(\mathbf{i})}(\mathbf{x}(\mathbf{i})) = 1$, and the first entry of $R_{\mathbf{i}}$ is $\mathcal{L}_{|\mathbf{i}|}(\mathbf{x}(\mathbf{i}))$

Example 36.33 The variation matrix associated with double logarithm is

$$\mathcal{M}_{1,1} = \begin{bmatrix} \gamma^0_{\rho_{(0,0)}(0,0)} & (2\pi i)\gamma^0_{\rho_{(0,0)}(1,0)} & (2\pi i)\gamma^0_{\rho_{(0,0)}(0,1)} & (2\pi i)^2\gamma^0_{\rho_{(0,0)}(1,1)} \\ \gamma^1_{\rho_{(1,0)}(0,0)}(x_1) & (2\pi i)\gamma^1_{\rho_{(1,0)}(1,0)}(x_1) & (2\pi i)\gamma^1_{\rho_{(1,0)}(0,1)}(x_1) & (2\pi i)^2\gamma^1_{\rho_{(1,0)}(1,1)}(x_1) \\ \gamma^1_{\rho_{(0,1)}(0,0)}(x_1x_2) & (2\pi i)\gamma^1_{\rho_{(0,1)}(1,0)}(x_1x_2) & (2\pi i)\gamma^1_{\rho_{(0,1)}(0,1)}(x_1x_2) & (2\pi i)^2\gamma^1_{\rho_{(0,1)}(1,1)}(x_1x_2) \\ \gamma^2_{\rho_{(1,1)}(0,0)}(x_1,x_2) & (2\pi i)\gamma^2_{\rho_{(1,1)}(1,0)}(x_1,x_2) & (2\pi i)\gamma^2_{\rho_{(1,1)}(0,1)}(x_1,x_2) & (2\pi i)^2\gamma^2_{\rho_{(1,1)}(1,1)}(x_1,x_2) \end{bmatrix}$$
$$= \begin{bmatrix} 1 & & & \\ \mathcal{L}_1(x_1) & 2\pi i & & \\ \mathcal{L}_1(x_1x_2) & & 2\pi i & \\ \mathcal{L}_2(x_1,x_2) & (2\pi i)\mathcal{L}_1(x_2) & (2\pi i)\mathcal{L}_1\left(\frac{1-x_1x_2}{1-x_2}\right) & (2\pi i)^2 \end{bmatrix}$$

Lemma 36.34 The variation matrix is lower triangular. The principal submatrix of $\mathcal{M}_{\mathbf{1}}$ with $|\mathbf{i}| = |\mathbf{j}| = k$ is $(2\pi i)^k$ times the $\binom{n}{k} \times \binom{n}{k}$ identity matrix

Proof. By definition

□

Proposition 36.35 $\mathcal{M}_{\mathbf{1}}(\mathbf{x})$ is the fundamental matrix of linear differential equations

$$\begin{cases} dX_{\mathbf{i}} & = \sum_{|\mathbf{k}|=|\mathbf{i}|-1, \mathbf{k} \prec \mathbf{i}} d\rho_{\mathbf{i}}^{|\mathbf{i}|}(\mathbf{k})(\mathbf{x}(\mathbf{i}))X_{\mathbf{k}} \\ dX_{\mathbf{0}} & = \mathbf{0} \end{cases}$$

Theorem 36.36

Corollary 36.37 The monodromy representation $\rho_{\mathbf{x}} : \pi_1(S_n, \mathbf{x}) \to \mathrm{GL}_{2^n}(\mathbb{Z})$ is unipotent

Definition 36.38 Let $\mathcal{D}_n = X_n' \cup (\mathbb{CP}^n \setminus \mathbb{C}^n)$ and

$$\boldsymbol{\omega} = (c_{\mathbf{i},\mathbf{j}}) \in H^0(\mathbb{CP}^n, \Omega^1_{\mathbb{CP}^n}(\log(\mathcal{D}_n))) \otimes M_{2^n}(\mathbb{C})$$

Here

$$c_{\mathbf{i},\mathbf{j}} = \begin{cases} d\gamma^{|\mathbf{i}|}_{\rho_{\mathbf{i}}(\mathbf{j})}(\mathbf{x}(\mathbf{i})) & |\mathbf{j}| = |\mathbf{i}| - 1, \mathbf{j} \prec \mathbf{i} \\ \mathbf{0} & \text{otherwise} \end{cases}$$

All one forms in $\boldsymbol{\omega}$ has logarithmic singularity. By Proposition 36.35, $d\mathcal{M}_{\mathbf{1}} = \boldsymbol{\omega}\mathcal{M}_{\mathbf{1}}$, thus

$$\mathbf{0} = d^2\mathcal{M}_{\mathbf{1}} = d\boldsymbol{\omega} \wedge \mathcal{M}_{\mathbf{1}} - \boldsymbol{\omega} \wedge d\mathcal{M}_{\mathbf{1}} = d\boldsymbol{\omega} \wedge \mathcal{M}_{\mathbf{1}} - \boldsymbol{\omega} \wedge \boldsymbol{\omega}\mathcal{M}_{\mathbf{1}}$$

Since $\mathcal{M}_{\mathbf{1}}(\mathbf{x})$ is invertible and $\boldsymbol{\omega}$ is by definition closed, i.e. $d\boldsymbol{\omega} = 0$, hence $\boldsymbol{\omega} \wedge \boldsymbol{\omega} = 0$, which implies $\boldsymbol{\omega}$ is integrable. Define a meromorphic connection on trivial bundle $\mathbb{CP}^n \times \mathbb{C}^{2^n} \to \mathbb{CP}^n$

$$\nabla f = df - \boldsymbol{\omega} f$$

Here $f : S_n \to \mathbb{C}^{2^n}$ is a section

Example 36.39 If $n = 2$

$$\boldsymbol{\omega} = \begin{bmatrix} & & & \\ -d\ln(1-x_1) & & & \\ -d\ln(1-x_1x_2) & & & \\ & & -d\ln(1-x_2) & -d\ln\left(\frac{x_2(x_1-1)}{1-x_2}\right) \end{bmatrix}$$

Definition 36.40 Let $V_{\mathbf{1}}(\mathbf{x})$ be the locally constant sheaf of $\mathbb{Q}$ vector space generated by the column vectors in $\mathcal{M}_{\mathbf{1}}(\mathbf{x})$, define a weight filtration $W_\bullet$ by letting $W_{2k+1} = W_{2k}$ and W_{-2k} is generated by $(2\pi i)^{|\mathbf{j}|}C_{\mathbf{j}}(\mathbf{x})$, $|\mathbf{j}| \geq k$, and $W_{-2k} = 0$ for $k > n$, note that $W_{-2k} = V_{\mathbf{1}}(\mathbf{x})$ for $k \geq n$. Define filtration $\mathcal{F}^\bullet$ by $\mathcal{F}^{-k}V_{\mathbf{1}}(\mathbb{C}) = \langle e_{\mathbf{i}}, |\mathbf{i}| \leq k \rangle_{\mathbb{C}}$, $\mathcal{F}^{-k} = 0$ for $k < 0$, note that $\mathcal{F}^{-k} = V_{\mathbf{1}}(\mathbb{C})$ for $k \geq n$. By induction on n and Lemma 36.34

$$\mathcal{F}^{-p} \cap W_{-2k}(\mathbb{C}) = \begin{cases} 0 & p \leq k-1 \\ \langle (2\pi i)^{|\mathbf{j}|} e_{\mathbf{j}}, k \leq |\mathbf{j}| \leq p \rangle & k \leq p \leq n \\ \langle (2\pi i)^{|\mathbf{j}|} e_{\mathbf{j}}, k \leq |\mathbf{j}| \leq n \rangle & p > n \end{cases}$$

Which implies that

$$\mathcal{F}^{-p}(\mathrm{gr}^W_{-2k}V_{\mathbf{1}})(\mathbb{C}) = \begin{cases} 0 & p \leq k-1 \\ W_{-2k}V_{\mathbf{1}}(\mathbb{C})/W_{-2k-1}V_{\mathbf{1}}(\mathbb{C}) = \mathrm{gr}^W_{-2k}V_{\mathbf{1}}(\mathbb{C}) & k \leq p \end{cases}$$

$\mathrm{gr}^W_{-2k}V_{\mathbf{1}}(\mathbb{C})$ is isomorphic to a direct sum of $\binom{n}{k}$ copies of Tate structure $\mathbb{Z}(k)$

Let $\mathcal{D} \subseteq \mathbb{CP}^n$ be a subvariety, $T_{\mathcal{D}}$ be the restriction of $\mathcal{M}_{\mathbf{1}}$ on $\mathcal{D}$, $N_{\mathcal{D}} = \log T_{\mathcal{D}}/(2\pi i)$, note that $N_{\mathcal{D}}$ is well defined since $T_{\mathcal{D}}$ is unipotent

Definition 36.41 X is a complex manifold of dimension d, $X \overset{j}{\hookrightarrow} \tilde{X}$ is an embedding, $D = \tilde{X} - X$ is a divisor with normal crossings, let $\mathbb{V}$ is a unipotent local system of complex vector spaces over X, $\mathcal{V}$ is a corresponding vector bundle. By Theorem 37.32, there is a canonical extension of $\mathcal{V}$ to $\tilde{\mathcal{V}}$ over $\tilde{X}$

Definition 36.42 Variation matrix $\mathcal{M}_{\mathbf{n}}$

Example 36.43 $(1, 0, x^{-1}, x^{-1}y^{-1}, 0) = (1, 0, a_1, a_2, 0) = (b_0, b_1, b_2, b_3, b_4)$, we have

$$
\begin{aligned}
&\boxed{1}|0|a_1|a_2|\boxed{0}, \quad \mathbf{m} = (0,0)\\
&\boxed{1}|0|\boxed{a_1}|a_2|\boxed{0}, \quad \mathbf{m} = (1,0)\\
&\boxed{1}|0|a_1|\boxed{a_2}|\boxed{0}, \quad \mathbf{m} = (1,0)\\
&\boxed{1}|\boxed{0}|\boxed{a_1}|a_2|\boxed{0}, \quad \mathbf{m} = (2,0)\\
&\boxed{1}|0|\boxed{a_1}|\boxed{a_2}|\boxed{0}, \quad \mathbf{m} = (1,1)\\
&\boxed{1}|\boxed{0}|a_1|\boxed{a_2}|\boxed{0}, \quad \mathbf{m} = (2,0)\\
&\boxed{1}|\boxed{0}|\boxed{a_1}|\boxed{a_2}|\boxed{0}, \quad \mathbf{m} = (2,1)
\end{aligned}
$$

Therefore (columns tells how many pieces in multiplication, rows tells which can be chosen in the iterated integral)

$$
\mathcal{M}_{2,1} = \begin{bmatrix}
1 & & & & & \\
\mathrm{Li}_1(x) & 1 & & & & \\
\mathrm{Li}_1(xy) & 0 & 1 & & & \\
\mathrm{Li}_2(x) & \log(x) & 0 & 1 & & \\
\mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(y) & \mathrm{Li}_1(x) - \mathrm{Li}_1(y^{-1}) & 0 & 1 & \\
\mathrm{Li}_2(xy) & 0 & \log(xy) & 0 & 0 & 1 \\
\mathrm{Li}_{2,1}(x,y) & \mathrm{Li}_1(y)\log(x) & g(x,y) & \mathrm{Li}_1(y) & \log(x) & -\mathrm{Li}_1(y^{-1}) \quad 1
\end{bmatrix}
$$

Here

$$
\begin{aligned}
g(x,y) &= \int_{x^{-1}y^{-1}}^{1} \frac{dt}{t}\frac{dt}{x^{-1}-t}\\
&= \int_{x^{-1}y^{-1}}^{1} \frac{dt}{t}\int_{x^{-1}y^{-1}}^{t} \frac{d\tau}{x^{-1}-\tau}\\
&= \int_{x^{-1}y^{-1}}^{1} \frac{dt}{t}(\mathrm{Li}_1(xt) - \mathrm{Li}_1(y^{-1}))\\
&= \int_{x^{-1}y^{-1}}^{1} \frac{dt}{t}\mathrm{Li}_1(xt) - \mathrm{Li}_1(y^{-1})\int_{x^{-1}y^{-1}}^{1} \frac{dt}{t}\\
&= \int_{y^{-1}}^{x} \frac{dt}{t}\mathrm{Li}_1(t) - \mathrm{Li}_1(y^{-1})\log(xy)\\
&= \mathrm{Li}_2(x) - \mathrm{Li}_2(y^{-1}) - \mathrm{Li}_1(y^{-1})\log(xy)
\end{aligned}
$$

Note. Note that $\mathrm{Li}_1(x) - \mathrm{Li}_1(y^{-1}) = \mathrm{Li}_1\left(\frac{1-xy}{1-y}\right)$

Example 36.44 $(1, x^{-1}, 0, x^{-1}y^{-1}, 0) = (1, a_1, 0, a_2, 0) = (b_0, b_1, b_2, b_3, b_4)$, we have

$$
\begin{aligned}
&\boxed{1}|a_1|0|a_2|\boxed{0}, \quad \mathbf{m}=(0,0)\\
&\boxed{1}|\boxed{a_1}|0|a_2|\boxed{0}, \quad \mathbf{m}=(1,0)\\
&\boxed{1}|a_1|0|\boxed{a_2}|\boxed{0}, \quad \mathbf{m}=(1,0)\\
&\boxed{1}|\boxed{a_1}|0|\boxed{a_2}|\boxed{0}, \quad \mathbf{m}=(1,1)\\
&\boxed{1}|a_1|\boxed{0}|\boxed{a_2}|\boxed{0}, \quad \mathbf{m}=(2,0)\\
&\boxed{1}|\boxed{a_1}|\boxed{0}|\boxed{a_2}|\boxed{0}, \quad \mathbf{m}=(1,2)
\end{aligned}
$$

$$
\mathcal{M}_{1,2}=\begin{bmatrix}
1 & & & & & \\
\mathrm{Li}_1(x) & 1 & & & & \\
\mathrm{Li}_1(xy) & 0 & 1 & & & \\
\mathrm{Li}_{1,1}(x,y) & \mathrm{Li}_1(y) & \mathrm{Li}_1(x)-\mathrm{Li}_1(y^{-1}) & 1 & & \\
\mathrm{Li}_2(xy) & 0 & \log(xy) & 0 & 1 & \\
\mathrm{Li}_{1,2}(x,y) & \mathrm{Li}_2(y) & g(x,y) & \log(y) & \mathrm{Li}_1(x) & 1
\end{bmatrix}
$$

Here

$$
\begin{aligned}
g(x,y)&=\int_{x^{-1}y^{-1}}^1 \frac{dt}{x^{-1}-t}\frac{dt}{t}\\
&=\int_{x^{-1}y^{-1}}^1 \frac{dt}{x^{-1}-t}\int_{x^{-1}y^{-1}}^t \frac{d\tau}{\tau}\\
&=\int_{x^{-1}y^{-1}}^1 \frac{dt}{x^{-1}-t}(\log(t)+\log(xy))\\
&=\int_{x^{-1}y^{-1}}^1 \frac{dt}{x^{-1}-t}\log(t)+\log(xy)\int_{x^{-1}y^{-1}}^1 \frac{dt}{x^{-1}-t}\\
&=\int_{y^{-1}}^x \frac{dt}{1-t}(\log(t)-\log(x))+\log(xy)\int_{y^{-1}}^x \frac{dt}{1-t}\\
&=\int_{y^{-1}}^x \frac{dt}{1-t}\log(t)+\log(y)\mathrm{Li}_1\left(\frac{1-xy}{1-y}\right)\\
&=\log(y)\mathrm{Li}_1\left(\frac{1-xy}{1-y}\right)+\int_{y^{-1}}^x \log(t)d\mathrm{Li}_1(t)\\
&=\log(y)\mathrm{Li}_1\left(\frac{1-xy}{1-y}\right)+\log(t)\mathrm{Li}_1(t)|_{y^{-1}}^x-\int_{y^{-1}}^x \frac{dt}{t}\mathrm{Li}_1(t)\\
&=\log(y)\mathrm{Li}_1\left(\frac{1-xy}{1-y}\right)+\log(x)\mathrm{Li}_1(x)+\log(y)\mathrm{Li}_1(y^{-1})-\mathrm{Li}_2(x)+\mathrm{Li}_2(y^{-1})
\end{aligned}
$$

37 Miscellaneous

37.1 Algebraic K theory

Definition 37.1 The Grothendieck group of R is $K_0(R)$, the Grothendieck group of monoid of finitely generated projective modules over R

Theorem 37.2 (Swan's theorem) X is a compact Hausdorff space, $K(X) = K_0(C(X, \mathbb{R}))$

Proof. If $E \to X$ is a vector bundle, then it is the direct summand of some trivial vector bundle. Conversely, if P is a finitely generated module over $R = C(X, \mathbb{R})$, then P is image of some idempotent endomorphism of R^n which is a vector bundle

□

Definition 37.3 (Whitehead group) The **Whitehead group** of ring R is an abelian group $K_1(R)$ satisfying universal property

$$\begin{array}{ccc} GL(R) & & \\ {\scriptstyle\pi}\downarrow & \searrow & \\ K_1(R) & \overset{\exists_1}{\dashrightarrow} & A \end{array}$$

For any abelian group A

Construction 37.4 Thanks to Whitehead's lemma 3.9, $K_1(R) = GL(R)/[GL(R), GL(R)] = GL(R)/E(R)$

Definition 37.5 If R is commutative, $SL(R)$ is the kernel of $GL(R) \overset{\det}{\to} R^\times$, the special Whitehead group $SK_1(R) = SL(R)/E(R)$ is the kernel of $K_1(R) \overset{\det}{\to} R^\times$, $GL(R) \cong SL(R) \rtimes R^\times$, $K_1(R) \cong SK_1(R) \oplus R^\times$. $K_1(F) = F^\times$

Lemma 37.6 Since $GL(R_1 \times R_2) = GL(R_1) \times GL(R_2)$, $K_1(R_1 \times R_2) = K_1(R_1) \oplus K_1(R_2)$

37.2 Axiomatic sheaf cohomology theory

Definition 37.7 A *sheaf cohomology theory* H for M with coefficients in the sheaves of K-modules over M is a covariant cohomological δ functor that consists of

1. A family of covariant additive functors H^q from the category of sheaves of K-modules over M to the category of K-modules
2. For each short exact sequence $0 \to \mathcal{S}' \to \mathcal{S} \to \mathcal{S}'' \to 0$, a homomorphism $H^q(M, \mathcal{S}'') \to H^{q+1}(M, \mathcal{S}')$

such that

1. $H^q(M, \mathcal{S}) = 0$ for $q < 0$. $H^0(M, \mathcal{S}) \cong \Gamma(\mathcal{S})$, and for any homomorphism $\mathcal{S} \to \mathcal{S}'$

$$\begin{array}{ccc} H^0(M, \mathcal{S}) & \overset{\cong}{\longrightarrow} & \Gamma(\mathcal{S}) \\ \downarrow & & \downarrow \\ H^0(M, \mathcal{S}') & \overset{\cong}{\longrightarrow} & \Gamma(\mathcal{S}') \end{array}$$

   commutes
2. If $\mathcal{S}$ is a fine sheaf, then $H^q(M, \mathcal{S}) = 0$ for $q > 0$
3. For each short exact sequence $0 \to \mathcal{S}' \to \mathcal{S} \to \mathcal{S}'' \to 0$, we have long exact sequence

$$\cdots \to H^q(M, \mathcal{S}') \to H^q(M, \mathcal{S}) \to H^q(M, \mathcal{S}'') \to H^{q+1}(M, \mathcal{S}') \to \cdots$$

4. For commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathcal{S}' & \longrightarrow & \mathcal{S} & \longrightarrow & \mathcal{S}'' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{T}' & \longrightarrow & \mathcal{T} & \longrightarrow & \mathcal{T}'' & \longrightarrow & 0 \end{array}$$

we have commutative diagram

$$\begin{array}{ccccccccccc} \cdots & \longrightarrow & H^q(M,\mathcal{S}') & \longrightarrow & H^q(M,\mathcal{S}) & \longrightarrow & H^q(M,\mathcal{S}'') & \longrightarrow & H^{q+1}(M,\mathcal{S}') & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \cdots & \longrightarrow & H^q(M,\mathcal{T}') & \longrightarrow & H^q(M,\mathcal{T}) & \longrightarrow & H^q(M,\mathcal{T}'') & \longrightarrow & H^{q+1}(M,\mathcal{T}') & \longrightarrow & \cdots \end{array}$$

Existence of cohomology theories

Let $\mathcal{K} = M \times K$ be the constant sheaf over M, we shall show that any fine torsionless resolution of $\mathcal{K}$

$$0 \to \mathcal{K} \to \mathcal{C}^0 \to \mathcal{C}^1 \to \cdots$$

Give rise to a cohomology theory by defining $H^q(M,\mathcal{S}) = H^q(\Gamma(\mathcal{C}^* \otimes \mathcal{S}))$, since $\mathcal{C}_i$ are fine torsionless resolution, we have

$$0 \to \Gamma(C^* \otimes \mathcal{S}') \to \Gamma(C^* \otimes \mathcal{S}) \to \Gamma(C^* \otimes \mathcal{S}'') \to 0$$

is exact, which gives us the long exact sequence

Let $\mathcal{Z}^q = \ker(\mathcal{C}^q \to \mathcal{C}^{q+1})$, then we have short exact sequence $0 \to \mathcal{Z}^q \to \mathcal{C}^q \to \mathcal{Z}^{q+1} \to 0$, as a subsheaf $\mathcal{Z}^q$ is also torsionless, thus $0 \to \mathcal{Z}^q \otimes \mathcal{S} \to \mathcal{C}^q \otimes \mathcal{S} \to \mathcal{Z}^{q+1} \otimes \mathcal{S} \to 0$ is exact, so $0 \to \Gamma(\mathcal{Z}^q \otimes \mathcal{S}) \to \Gamma(\mathcal{C}^q \otimes \mathcal{S}) \to \Gamma(\mathcal{Z}^{q+1} \otimes \mathcal{S})$ is exact, and $\Gamma(\mathcal{C}^q \otimes \mathcal{S}) \to \Gamma(\mathcal{C}^{q+1} \otimes \mathcal{S})$ is the composition $\Gamma(\mathcal{C}^q \otimes \mathcal{S}) \to \Gamma(\mathcal{Z}^{q+1} \otimes \mathcal{S}) \to \Gamma(\mathcal{C}^{q+1} \otimes \mathcal{S})$, hence $H^q(M,\mathcal{S}) = H^q(\Gamma(\mathcal{C}^* \otimes \mathcal{S})) = \Gamma(\mathcal{Z}^q \otimes \mathcal{S})/\mathrm{im}(\Gamma(\mathcal{C}^{q-1} \otimes \mathcal{S}))$. If $\mathcal{S}$ is fine, then $0 \to \Gamma(\mathcal{Z}^{q-1} \otimes \mathcal{S}) \to \Gamma(\mathcal{C}^{q-1} \otimes \mathcal{S}) \to \Gamma(\mathcal{Z}^q \otimes \mathcal{S}) \to 0$ is exact, hence $H^q(M,\mathcal{S}) = 0$ for $q \geq 1$

Let $\mathcal{S}_0$ denote the sheaf of germs of discontinuous sections of $\mathcal{S}$ (which is the shefification of the sheaf of discontinuous sections of $\mathcal{S}$), we shall show that $\mathcal{S}_0$ is always a fine sheaf

Definition 37.8 $H, \tilde{H}$ are cohomology theories, a homomorphism $H \to \tilde{H}$ is a natural transformation such that

$$\begin{array}{ccc} H^0(M,\mathcal{S}) & \xrightarrow{\cong} & \Gamma(\mathcal{S}) \\ \downarrow & & \| \\ \tilde{H}^0(M,\mathcal{S}) & \xrightarrow{\cong} & \Gamma(\mathcal{S}) \end{array}$$

commutes

Theorem 37.9 $H, \tilde{H}$ are cohomology theories, then there is a unique homomorhpism $H \to \tilde{H}$

Corollary 37.10 Any two cohomology theories $H, \tilde{H}$ are uniquely isomorphic

Theorem 37.11 H is a cohomology theory

$$0 \to \mathcal{S} \to \mathcal{C}^0 \to \mathcal{C}^1 \to \cdots$$

is a fine resolution of $\mathcal{S}$, then there are canonical isomorphisms $H^q(M,\mathcal{S}) \xrightarrow{\cong} H^q(\Gamma(\mathcal{C}^*))$

37.3 Classical mechanics

Consider a system of N particles in $\mathbb{R}^n$ with masses m_i, with positions $\mathbf{r}_i = \mathbf{r}_i(t)$, $\dot{\mathbf{r}}_i, \ddot{\mathbf{r}}_i$ denote velocities and accelerations. Let $\mathbf{x}, t$ denote Cartesian coordinates and time, $\mathbf{r}_i$ may have some constraints(such as $f_j(\mathbf{r}_i, t) = 0$), consider another coordinates(generalized coordinates, preferably free, i.e. with no constraints) $\mathbf{y}$, particles will have generalized positions, generalized velocities and generalized accelerations $\mathbf{q}_i = \mathbf{q}_i(t)$, $\dot{\mathbf{q}}_i, \ddot{\mathbf{q}}_i$. The system itself should produce no work, i.e.

$$\begin{aligned} 0 &= \sum m_i \ddot{\mathbf{r}}_i \cdot \dot{\mathbf{r}}_i \\ &= \sum m_i \ddot{\mathbf{q}}_i \cdot \left[\frac{\partial \mathbf{x}}{\partial \mathbf{y}}(\mathbf{q}_i) \dot{\mathbf{q}}_i \right] \end{aligned}$$

Example 37.12

Definition 37.13

37.4 Differential Galois theory

Definition 37.14 Differential field extension E/F is a *Picard-Vessiot extension* if

1. E, F have the same ring of constants
2. E is by adjoining in solutions to a linear homogeneous differential equation

37.5 Electromagnetism

Theorem 37.15 (Curl-less(irrotational) fields) The following are equivalent

- curl $\mathbf{F} = \nabla \times \mathbf{F} = 0$
- $\int_a^b \mathbf{F} d\mathbf{l}$ is the independent of the path, for given end points
- $\oint \mathbf{F} d\mathbf{l} = 0$ for any closed loop
- $\mathbf{F} = -\nabla V$ for some potential V which is unique up to a constant

Theorem 37.16 (Divergence-less(solenoidal) fields) The following are equivalent

- div $\mathbf{F} = \nabla \mathbf{F} = 0$
- $\iint \mathbf{F} d\mathbf{a}$ is the independent of the path, for given boundary line
- $\oiint \mathbf{F} d\mathbf{a} = 0$ for any closed surface
- $\mathbf{F} =$ curl $\mathbf{A}$, unique up to a gradient of any scalar function

Theorem 37.17 (Helmholtz Theorem)

Theorem 37.18 (Maxwell's equations)

37.6 Elliptic curves

Consider ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the circumference is

$$4 \int_0^a \sqrt{1 + \frac{b^2 x^2}{a^2(a^2 - x^2)}} dx = 4a \int_0^{\frac{\pi}{2}} \sqrt{1 - e^2 \sin^2 \theta} d\theta$$

Definition 37.19 The elliptic integral of the *first* kind is

$$\int_0^{\varphi} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}}$$

Let $t = \sin \theta$, $x = \sin \varphi$, we have

$$\int_0^x \frac{dt}{\sqrt{(1 - t^2)(1 - k^2 t^2)}}$$

The elliptic integral of the *second* kind is

$$\int_0^{\varphi} \sqrt{1 - k^2 \sin^2 \theta} d\theta$$

Let $t = \sin \theta$, $x = \sin \varphi$, we have

$$\int_0^x \frac{\sqrt{1-k^2t^2}}{\sqrt{1-t^2}}dt$$

The elliptic integral of the *third* kind is

$$\int_0^\varphi \frac{d\theta}{(1-n\sin^2\theta)\sqrt{1-k^2\sin^2\theta}}$$

Let $t=\sin\theta$, $x=\sin\varphi$, we have

$$\int_0^x \frac{dt}{(1-nt^2)\sqrt{(1-t^2)(1-k^2t^2)}}$$

These elliptic integrals are called *incomplete*, they are *complete* if $\varphi=\frac{\pi}{2}$

Theorem 37.20 (Legendre's relation) For $k^2+k'^2=1$, E,E' are corresponding complete elliptic integrals of the second kind, K,K' are corresponding complete elliptic integrals of the first kind, then they satisfy the *Legendre's relation*

$$KE'+K'E-KK'=\frac{\pi}{2}$$

Equivalently

$$\omega_1\eta_2-\omega_2\eta_1=2\pi i$$

ω_1,ω_2 are the periods of Weierstrass $\wp$ function, η_1,η_2 are the quasiperiods of Weierstrass zeta function

Definition 37.21 An *elliptic integral* is of the form

$$\int_c^x R\big(x,\sqrt{P(x)}\big)dx$$

Here $R(x,w)$ is a rational function of x,w and $P(x)$ is a polynomial of degree 3 or 4. Every elliptic integral can be reduced into elliptic integrals of the first, second and third kinds

Definition 37.22 An *abelian integral* is of the form

$$\int_{z_0}^z R(x,w)dx$$

R is a rational function of x,w, and $F(x,w)=0$ for some

$$\varphi_n(x)w^n+\cdots+\varphi_0(x)=0$$

$\varphi_i(x)$ are rational functions of x. It is called a *hyperelliptic integral* if $F(x,w)=w^2-P(x)$ for some polynomial P, note that if degree of P is 3 or 4 than it is an elliptic integral

Definition 37.23 C is a compact algebraic curve of genus g, $H^0(X,K)=\mathbb{C}^g$ is generated by one forms $\omega_1,\cdots,\omega_g$, K is a canonical bundle, the *Abel-Jacobi map* is

$$\begin{aligned} J:C&\to J(C)=\mathbb{C}^g/\Lambda\\ P&\mapsto \left(\int_{P_0}^P \omega_1,\cdots,\int_{P_0}^P \omega_g\right) \bmod \Lambda \end{aligned}$$

Theorem 37.24 (Abel-Jacobi theorem) Abel-Jacobi map J is an isomorphism

37.7 Hodge structure

Use $H_{\mathbb{F}}$ or $H(\mathbb{F})$ to indicate coefficients in $\mathbb{F}$

Definition 37.25 An abelian group $H_{\mathbb{Z}}$ has a *pure Hodge structure* of weight n if

$$H_{\mathbb{C}} = \bigoplus_{p+q=n} H^{p,q}, \quad \overline{H^{p,q}} = H^{q,p}$$

Or equivalently, if there is a decreasing *Hodge filtration* F^p on $H_{\mathbb{C}}$ such that $H_{\mathbb{C}} = F^p \oplus \overline{F^{n+1-p}}$. The equivalence of the two definitions is given by $(p+q=n)$

$$F^p = \bigoplus_{i \geq p} H^{i,n-i}, \quad \overline{F^q} = \bigoplus_{j \leq p} H^{j,n-j}$$

$$H^{p,q} = F^p \cap \overline{F^q}, \quad F^p \cap \overline{F^{n+1-p}} = 0$$

Example 37.26 X is a complex manifold, $H_{\mathbb{Z}} = H^n(X;\mathbb{Z})$, then

$$H^n(X;\mathbb{C}) = \bigoplus_{p+q=n} H^{p,q} = \bigoplus_{p+q=n} H^p(X;\mathbb{C}) \wedge \overline{H^p(X;\mathbb{C})}$$

Example 37.27 (*Tate-Hodge structure*) $\mathbb{Z}(1) = (2\pi i)\mathbb{Z}$ has a pure Hodge structure of weight -2 such that $\mathbb{Z}(1)_{\mathbb{C}} = H^{-1,-1}$, this is the unique one dimensional weight -2 pure Hodge structure($H^{-2,0} = \overline{H^{0,-2}}$ should be 0 or nonzero at the same time). $\mathbb{Z}(n) \coloneqq \mathbb{Z}(1)^{\otimes n}$

Definition 37.28 A *polarization* over $\mathbb{Q}$ of a Hodge structure over $\mathbb{Q}$ of weight k is a $(-1)^k$ symmetric nondegenerate flat bilinear map $\beta : \mathbb{V}_{\mathbb{Q}} \times \mathbb{V}_{\mathbb{Q}} \to \mathbb{Q}$ such that the Hermitian form $\beta_x(C_x v, \bar{w})$ on each fiber $\mathcal{V}_x$ is positive definite, here C_x is the *Weil operator*, given as the direct sum of multiplication i^{p-q} on $H_x^{p,q}$

Definition 37.29 A *mixed Hodge structure* on $H_{\mathbb{Z}}$ consists of an increasing *weight filtration* $W_\bullet$ on $H_{\mathbb{Q}}$ and a decreasing filtration $F^\bullet$ on $H_{\mathbb{C}}$ that are compatible, i.e.

$$F^p(\mathrm{gr}_k W)_{\mathbb{C}} = \frac{F^p \cap W_{k+1}(\mathbb{C}) + W_k(\mathbb{C})}{W_k(\mathbb{C})}$$

is a pure Hodge structure of weight k of $\mathrm{gr}_k W$

Definition 37.30 A *variation* of Hodge structure of weight k over $\mathbb{Q}$ and a complex manifold X is $(\mathbb{V}_{\mathbb{Q}}, \mathcal{F}^\bullet)$, $\mathbb{V}_{\mathbb{Q}}$ is a locally constant sheaf of $\mathbb{Q}$ vector spaces, $\mathcal{F}^\bullet$ is a decreasing filtration of holomorphic subbundles of the locally free sheaf $\mathcal{V} = \mathcal{O}_X \otimes \mathbb{V}_{\mathbb{Q}}$ such that

- $(\mathcal{V}_x, \mathcal{F}_x^\bullet)$ has a pure Hodge structure of weight k, i.e. $\mathcal{V}_x = \mathcal{F}^p \oplus \overline{\mathcal{F}^{k+1-p}}$
- (Griffiths transversality) $\nabla \mathcal{F}^p \subseteq \Omega_X^1 \otimes_{\mathcal{O}_X} \mathcal{F}^{p-1}$

Definition 37.31 A variation of mixed Hodge structure over $\mathbb{Q}$ and a complex manifold X is $(\mathbb{V}_{\mathbb{Q}}, \mathcal{W}_\bullet, \mathcal{F}^\bullet)$, $\mathcal{W}_\bullet$ is an increasing filtration of $\mathbb{V}_{\mathbb{Q}}$ by locally constant subsheaves such that

- $(\mathcal{V}_x, (\mathcal{W}_\bullet)_x, \mathcal{F}_x^\bullet)$ has a mixed Hodge structure, i.e. () is a pure Hodge structure of weight k
- $\nabla \mathcal{F}^p \subseteq \Omega_X^1 \otimes_{\mathcal{O}_X} \mathcal{F}^{p-1}$

Remark. Given a locally constant sheaf is equivalent to given a monodromy representation $\rho_{\mathbf{x}} : \pi_1(X, \mathbf{x}) \to \mathrm{Aut}_{\mathbb{Q}}(\mathcal{V}_x)$. A variation is *unipotent* if the the monodromy representation is unipotent

Theorem 37.32 (Deligne) $\tilde{X}$ is a normalization of X, $(\mathbb{V}_{\mathbb{Q}}, \mathcal{W}_\bullet, \mathcal{F}^\bullet)$ is a unipotent variation of mixed Hodge structure of weight k, then there is a unique extension $\tilde{\mathcal{V}}$ over $\tilde{X}$ such that

- Inside every section of $\tilde{\mathcal{V}}$, flat sections increase at most at the rate of $O(\log(\| x \|^k))$ on each compact set of $\tilde{X} - X$
- Every flat section of $\mathcal{V}^\vee$ increases at most at the rate of $O(\log(\| x \|^k))$

These conditions are equivalent to

- In a local basis of $\tilde{\mathcal{V}}$, the connection matrix $\boldsymbol{\omega}$ of $\mathcal{V}$ has at most logarithmic singularities along $\tilde{X} - X$
- The residue of $\boldsymbol{\omega}$ along any irreducible component of $\tilde{X} - X$ is nilpotent

For smooth complex vector bundle E. Dolbeault operators are $\bar{\partial}_E : \Omega^{p,q}(E) \to \Omega^{p,q+1}(E)$ satisfying Leibniz. By Newlander-Nirenberg, E has a unique holomorphic structure iff $\bar{\partial}_E$ is the unique Dolbeault operator

For a holomorphic complex vector bundle E with a Hermitian metric h, the Chern connection is the unique metric connection such that the $(0,1)$ part of ∇s is $\bar{\partial}_E s$. The connection form is $h^{-1}\partial h$

Definition 37.33 A harmonic differential $\omega = Adx + Bdy$ is the real part of an analytic complex differential, then we can define its conjugate $*\,\omega = Ady - Bdx$, which should be the imaginary part, we may think of $*$ as the Hodge star operator. Note that the complex form should have been $(A - iB)(dx + idy)$. Thus ω is a harmonic differential iff $d\omega = d * \omega = 0$, $d * du = \Delta u dx \wedge dy$ gives the Laplacian

Let $\Delta = dd^* + d^*d$ be the Laplacian, then the following are equivalent(very easy to check): $\Delta\alpha = 0$, $d\alpha = d^*\alpha = 0$, $\alpha \in \ker d \cap (\operatorname{im} d)^\perp$, such forms are called harmonic forms

There is a unique harmonic representative for each cohomology class, which happen to be of the smallest norm, consider the isomorphism(very easy to check) $\ker d \cap (\operatorname{im} d)^\perp \to H^k$, suppose $\alpha \in \ker d \cap (\operatorname{im} d)^\perp$, then $|\alpha + \partial\beta|^2 = |\alpha|^2 + |\partial\beta|^2 \geq |\alpha|^2$, equality holds iff $\partial\beta = 0$

37.8 Hyperlogarithm and differential Galois

$X = \{x_0, \cdots, x_N\}$, $N \geq 1$, X^* is the free non-commutative monoid generated by X, $\mathbb{C}\langle\langle X\rangle\rangle$ is the ring of formal power series with non-commutative variables in X. Consider $j : X \to \mathbb{C}$, $\sigma_i = j(x_i)$, $\Sigma = j(X) \cup \{\infty\}$, $D = \mathbb{CP}^1 - \Sigma$ The differential equation

$$\frac{\partial}{\partial z}F(z) = \sum_{i=0}^{N} \frac{x_i}{z - \sigma_i}F(z) \tag{37.1}$$

is of Fuchs type. Where $F \in \mathcal{O}(D)\langle\langle X\rangle\rangle$, write $F(z) = \sum_{w\in X^*} F_w(z)w$, $F_w(z) \in \mathcal{O}(D)$, then the differential equation becomes

$$\frac{\partial}{\partial z}F_{x_k w}(z) = \frac{F_w(z)}{z - \sigma_k} \tag{37.2}$$

Find half line $\ell(\sigma_i)$ starting at σ_i such that they don't intersect, and $U = D - \bigcup \ell(\sigma_i)$ is a simply connected, pick a branch for $\log(z - \sigma_0)$ on $D - \ell(\sigma_0)$. $\hat{D} \overset{p}{\to} D$ is a the universal cover, when there is no confusion, we shall still use $L_w(z)$

Theorem 37.34 Equation 37.1 has a unique solution

$$L(z) = f_0(z)e^{x_0 \log(z-\sigma_0)}$$

such that f_0 is holomorphic on $D - \bigcup_{k\neq 0} \ell(\sigma_k)$ with $f_0(\sigma_0) = 1$ (We also write $L(z) \sim (z - \sigma_0)^{x_0}$ as $z \to \sigma_0$). And any other holomorphic solution can be uniquely written as $L(z)C$, where $C \in \mathbb{C}\langle\langle X\rangle\rangle$

Proof. For $w \in X^*$ not ending in x_0, write $w = x_0^{n_r-1}x_{i_r}\cdots x_0^{n_1-1}x_{i_1}$, define

$$L_w(z) = (-1)^r \sum_{1\leq m_1<\cdots<m_r} \frac{1}{m_1^{n_{i_1}}\cdots m_r^{n_{i_r}}} \left(\frac{z - \sigma_0}{\sigma_{i_1} - \sigma_0}\right)^{m_1} \cdots \left(\frac{z - \sigma_0}{\sigma_{i_r} - \sigma_0}\right)^{m_r - m_{r-1}}$$

in a neighborhood of σ_0. And analytically continue this by

$$L_{x_k w}(z) = \int_{\sigma_0}^{z} \frac{L_w(t)}{t - \sigma_k}dt$$

In particular

$$L_{x_i^n}(z) = \frac{1}{n!}\log^n\left(\frac{z - \sigma_i}{\sigma_0 - \sigma_i}\right)$$

$$L_{x_0^n}(z) = \frac{1}{n!}\log^n(z-\sigma_0)$$

The $f_0(z)$ can be computed explicitly by comparing the coefficients in the series expansion that satisfies equation 37.1. $L(z)$ is invertible since its leading coefficeint is 1, suppose $F(Z)$ is another solution, define $K(z) = L(z)^{-1}F(z)$, then differentiate $F(z) = L(z)K(z)$ we have

$$F(z)' = L(z)'K(z) + L(z)K(z)' = \sum_{i=0}^{N}\frac{x_i}{z-\sigma_i}L(z)K(z) + L(z)K(z)' = F(z)' + L(z)K(z)'$$

Thus $L(z)K(z)' = 0 \Rightarrow K(z)' = 0 \Rightarrow K(z) = C$

□

Remark. Heuristically, the solution might seem to be

$$\exp\left(x_0\log(z-\sigma_0) + \sum_{i=1}^{N} x_i \log\frac{z-\sigma_i}{\sigma_0-\sigma_i}\right)$$

But this violates the non-commutativity of variables x_i

Example 37.35 In the case $\sigma_0 = 0, \sigma_1 = 1$, $L_{x_0^m}(z) = \frac{1}{n!}\log^n z$ and $L_{x_1^{n-1}x_0}(z) = -\mathrm{Li}_n(z)$, $L_{x_0^{n_1-1}x_1\cdots x_0^{n_d-1}x_1}(z) = (-1)^{n_1+\cdots+n_d-d}\mathrm{Li}_{n_1,\cdots,n_d}(z)$

Corollary 37.36 By translation, there is a unique solution to equation 37.1

$$L^{\sigma_k}(z) = f_k(z)e^{x_k\log(z-\sigma_k)}$$

over $\mathbb{C} - \bigcup_{i\neq k}\ell(\sigma_i)$ with $f_k(\sigma_k) = 1$

Proof. Consider $z' - \sigma_0 = e^{i\theta}(z-\sigma_k)$, $L^{\sigma_k}(z) = L(z')$, then

$$\frac{\partial}{\partial z}L^{\sigma_k}(z) = \sum_{i=0}^{N}\frac{x_i}{z-\sigma_i}L^{\sigma_k}(z)$$

Becomes

$$\frac{\partial}{\partial z'}L(z') = \sum_{i\neq k}\frac{x_i}{z'-(\sigma_i - e^{i\theta}(\sigma_k-\sigma_i))}L(z') + \frac{x_k}{z'-\sigma_0}L(z')$$

Hence

$$L^{\sigma_k}(z) = L(z') = f_0(z')e^{x_k\log(z'-\sigma_0)} = f_k(z)e^{x_k\log(z-\sigma_k)}$$

□

Corollary 37.37

Proof. Consider $z' - \sigma_0 = z^{-1}$, $L^{\infty}(z) = L(z')$, then

$$\frac{\partial}{\partial z}L^{\infty}(z) = \sum_{i=0}^{N}\frac{x_i}{z-\sigma_i}L^{\infty}(z)$$

Becomes

$$\begin{aligned}\frac{\partial}{\partial z'}L(z') &= -\sum_{i=0}^{\infty}\frac{x_i}{(z'-\sigma_0)-(z'-\sigma_0)^2\sigma_i}L(z') \\ &= \frac{-x_\infty}{z-\sigma_0}L(z') + \sum_{i=0}^{\infty}\frac{x_i}{z-(\sigma_0+\sigma_i^{-1})}L(z')\end{aligned}$$

Here $x_\infty = \sum_{i=0}^{N}x_i$. Hence

$$L^{\sigma_k}(z) = L(z') = f_0(z')e^{x_k \log(z'-\sigma_0)} = f_\infty(z)e^{x_\infty \log z}$$

□

$\mathbb{C}\langle X\rangle$ is the free $\mathbb{C}$ algebra generated by non commuting variables of X, there is a shuffle product $\sqcup : \mathbb{C}\langle X\rangle \otimes \mathbb{C}\langle X\rangle \to \mathbb{C}\langle X\rangle$ defined inductively by

$$e \sqcup w = w \sqcup e = w$$

$$xw \sqcup yv = x(w \sqcup yv) + y(xw \sqcup v)$$

$\mathbb{C}\langle X\rangle \to \mathcal{O}(D)$, $w \mapsto L_w(z)$ is then a homomorphism, i.e. $L_{w\sqcup v}(z) = L_w(z)L_v(z)$. Let

$$\mathcal{O}_\Sigma = \mathbb{C}\left[z, \frac{1}{z-\sigma_0}, \cdots, \frac{1}{z-\sigma_N}\right]$$

be the ring of regular functions on D. Let L_Σ be the free $\mathcal{O}_\Sigma$ module generated by $L_w(z)$. L_Σ is multiplicatively closed due to the shuffle relation, it is a differential algebra with derivation $\frac{\partial}{\partial z}$, and it is independent of the choice of the choice of branch of log

For $0 \leq k \leq N$, define truncate operator $\partial_k : \mathbb{C}\langle X\rangle \to \mathbb{C}\langle X\rangle$, $\partial_k(x_j w) = \delta_{jk} w$, it is easy to show that ∂_k is a derivation for shuffle product by induction

$$\begin{aligned}\partial_k(x_i w \sqcup x_j v) &= \partial_k(x_i(w \sqcup x_j v) + x_j(x_i w \sqcup v)) \\ &= \delta_{ik} w \sqcup x_j v + \delta_{jk} x_i w \sqcup v \\ &= \partial_k(x_i w) \sqcup x_j v + x_i w \sqcup \partial_k(x_j v)\end{aligned}$$

The universal algebra of hyperlogarithm is the differential $\mathcal{O}_\Sigma$ algebra

$$\mathcal{HL}_\Sigma = \mathcal{O}_\Sigma \otimes_{\mathbb{C}} \mathbb{C}\langle X\rangle$$

With the induced shuffle product as multiplication and $\partial = \frac{\partial}{\partial z} \otimes 1 + \sum_{k=0}^N \frac{1}{z-\sigma_k} \otimes \partial_k$ as derivation

It is well known that shuffle product is associative and commutative. Define Lyndon words $\mathbf{Lyn}(X) \subseteq X^*$ by assuming lexicographical order $x_0 < \cdots < x_N$ and

$$w = x_{i_1} \cdots x_{i_n} \in \mathbf{Lyn}(X) \Leftrightarrow x_{i_1} \cdots x_{i_r} \leq x_{i_{r+1}} \cdots x_{i_n}, \forall r$$

Theorem 37.38 (Radford) $\mathbb{C}\langle X\rangle$ with shuffle product is the free polynomial algebra over $\mathbf{Lyn}(X)$

For $w \in X^*$, define weight $|w|$ to be the number of symbols in w. The induced filtration of $\mathcal{HL}_\Sigma$ is

$$\mathcal{HL}_\Sigma^{(m)} = \left\{ \sum_{|w|\leq m} f_w(z) \otimes w \;\middle|\; f_w(z) \in \mathcal{O}_\Sigma \right\}$$

Since $\partial \mathcal{HL}_\Sigma^{(m)} \subseteq \mathcal{HL}_\Sigma^{(m)}$, we can define $\mathrm{gr}_m \mathcal{HL}_\Sigma$, by Theorem 37.38, $\mathrm{gr}_m \mathcal{HL}_\Sigma$ is generated by Lyndon words of weight m which are polynomials in N variables of degree m. For example, if $m = 1$, the Lyndon words are x_i's, $0 \leq i \leq N$, if $m = 2$, the Lyndon words are $x_i x_j$, $0 \leq i < j \leq N$

$\mathbb{C}\langle X\rangle$ has two natural Hopf algebra structure. The first one has non-commutative multiplication given by concatenation, cocommutative coproduct uniquely given by that x_i are primitive, i.e. $\Delta(x_i) = x_i \otimes 1 + 1 \otimes x_i$, counit ϵ given the projection onto the coefficient of the trivial word e and antipode a given by $w \mapsto (-1)^{|w|}\tilde{w}$, $\tilde{w}$ is the reversal of w. The second one has commutative multiplication given by the shuffle product, coproduct given by $\Gamma(w) = \sum_{uv=w} u \otimes v$, same counit ϵ and antipode $w \mapsto (-1)^{|w|} w$. These two Hopf algebra structures are dual to each other

Corollary 37.39 The map $S : \mathbb{C}\langle X\rangle \to \mathbb{C}^\times$ is a homomorphism iff $S = \sum_{w\in X^*} S(w)w \in \mathbb{C}\langle\langle X\rangle\rangle$ is group like, i.e. $\Delta S = S \otimes S$

Remark. S may not be group like, but there exists $S^\times$ such that $\Delta S^\times = S^\times \otimes S^\times$ and $S^\times(w) = S(w)$ for all $w \in \mathbf{Lyn}(X)$

Recall. If (A, ∂) is a differential ring, A is differentially simple if A is a simple module over its ring of differential operators $A[\partial]$, or equivalently that A has no nontrivial differential ideal. A differential ideal $I \subseteq A$ is such that $\partial I \subseteq I$. The ring of constants in A is $\ker \partial$. An equivalent condition for differential simplicity is for any $\theta \in A$, there exists $D_\theta \in A[\partial]$ such that $D_\theta \theta = 1$, otherwise $A[\partial]\theta$ would be such a nontrivial ideal

Theorem 37.40 Every differential $\mathcal{O}_\Sigma$ subalgebra of $\mathcal{HL}_\Sigma$ is differentially simple. The ring of constants of $\mathcal{HL}_\Sigma$ is $\mathbb{C}$ and each element in $\mathcal{HL}_\Sigma$ has a primitive, i.e. the following sequence is exact

$$0 \to \mathbb{C} \to \mathcal{HL}_\Sigma \xrightarrow{\partial} \mathcal{HL}_\Sigma \to 0$$

Proof.

□

Definition 37.41 The differential Galois group of extension $\mathcal{HL}_\Sigma/\mathcal{O}_\Sigma$ is $\mathrm{Gal}(\mathcal{HL}_\Sigma/\mathcal{O}_\Sigma) = \mathrm{Aut}_{\mathcal{O}_\Sigma[\partial]}\mathcal{HL}_\Sigma$

Note. The differential Galois group is normally about the field of fractions of a differential ring, however this is fine here since $\mathrm{Gal}(\mathcal{HL}_\Sigma/\mathcal{O}_\Sigma) \cong \mathrm{Gal}(\mathrm{Frac}\ \mathcal{HL}_\Sigma/\mathbb{C}(z))$

Theorem 37.42 There is a canonical isomorphism of groups $\mathrm{Gal}(\mathcal{HL}_\Sigma/\mathcal{O}_\Sigma) \cong \{S \in \mathbb{C}\langle\langle X\rangle\rangle | \Delta S = S \otimes S\}$, the right hand side has the group low of concatenation. There are also natural bijections between both groups and $\mathrm{Hom}(\mathbf{Lyn}(X), \mathbb{C})$

Proof.

□

Both groups have natural filtrations induced from the filtration by weight. Let U_n be the group of endomorphisms of $\mathcal{HL}_\Sigma^{(n)}$ that commute with ∂ and $\sqcup\!\sqcup$, then $U_n \cong \{S \in \mathbb{C}\langle\langle X\rangle\rangle / X^{n+1}\mathbb{C}\langle\langle X\rangle\rangle | \Delta S = S \otimes S\}$. Since any group like S satisfies $(S-1)^{n+1} \in X^{n+1}\mathbb{C}\langle\langle X\rangle\rangle$, U_n is a unipotent matrix group and

$$\mathrm{Gal}(\mathcal{HL}_\Sigma/\mathcal{O}_\Sigma) = \lim_{\leftarrow} U_n$$

is pro-unipotent. Its Lie algebra is isomorphic to the primitive elements $\{T \in \mathbb{C}\langle\langle X\rangle\rangle | \Delta T = T \otimes 1 + 1 \otimes T\}$

Definition 37.43 A realization of hyperlogarithms is a differential $\mathcal{O}_\Sigma$ algebra A and a nonzero homorphism $\rho : \mathcal{HP}_\Sigma \to A$, since $\ker \rho$ is a differential ideal, ρ is injective, and therefore $\{\rho(w) | w \in X^*\}$ form an $\mathcal{O}_\Sigma$ basis of the image of ρ. The holomorphic realization of hyperlogarithms is

$$\mathcal{HP}_\Sigma \to L_\Sigma, w \mapsto L_w(z)$$

Corollary 37.44 $\{L_w(z)\}$ form an $\mathcal{O}_\Sigma$ basis of L_Σ, the only relations among $L_w(z)$ are the shuffle relations. Each element of L_Σ has a primitive in L_Σ which is unique up to a constant

Definition 37.45 Let M be an $\mathcal{O}_\Sigma$ module of finite type, equipped with a connection ∇, i.e. a linear map $M \to M$ such that $\nabla(fm) = f'm + f\nabla m$. Under some base $\nabla = \frac{\partial}{\partial z} + P$, where P is a matrix with entries in $\mathcal{O}_\Sigma$. Define the solutions of M to be the smallest $\mathcal{O}_\Sigma$ submodule $\mathrm{Sol}(M)$ such that closed under taking primitive, i.e. $\nabla\theta \in \mathrm{Sol}(M) \Rightarrow \theta \in \mathrm{Sol}(M)$. M is free over $\mathcal{O}_\Sigma$ since $\mathcal{O}_\Sigma$ is differentially simple

Lemma 37.46 The following are equivalent

1. There is a filtration

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

such that $\nabla M_i \subseteq M_{i-1}$ and $(M_i/M_{i-1}, \nabla) \cong (\mathcal{O}_\Sigma, \partial/\partial z)$

2. There is some basis with respect to which that P is strictly upper triangular. We say M is *unipotent*

3. M is spanned by its solutions, i.e. $M = \mathrm{Sol}(M)$

Definition 37.47 Suppose (A, ∂) is a finitely generated differential $\mathcal{O}_\Sigma$ algebra. A is generated by $\mathcal{O}_\Sigma$ submodule of finite type $M \subseteq A$ with $\partial M \subseteq M$. Then ∂ defines a connection on M which restricts to differentiation on $\mathcal{O}_\Sigma$, further we assume the ring of constants is $\mathbb{C}$. We say A is unipotent if M is unipotent

Lemma 37.48 Let (R, ∂) be a differentially simple extension of $\mathcal{O}_\Sigma$. $R[y]$ is an extension by adjoining an integral, i.e. $\partial y = r \in R$. If $\partial x = r$ has no solution $x \in R$, then y is transcendental over R and $R[y]$ is also differentially simple

Proof.

□

Theorem 37.49 The following are equivalent

1. A is unipotent

2. There is an $\mathcal{O}_\Sigma$ basis of M that can be annihilated by some $D \in \mathcal{O}_\Sigma[\partial]$ which is a product of ∂ and $(z - \sigma_k)$

3. $\mathrm{Gal}(A/\mathcal{O}_\Sigma)$ acts on M and is unipotent, and $A^{\mathrm{Gal}(A/\mathcal{O}_\Sigma)} = \mathcal{O}_\Sigma$

4. There is an embedding $A \overset{i}{\hookrightarrow} \mathcal{HL}_\Sigma$

In such case, Frac $A/\mathbb{C}(z)$ is an Picard-Vessiot extension, $\mathrm{Gal}(A/\mathcal{O}_\Sigma) \cong \mathrm{Gal}(\text{Frac } A/\mathbb{C}(z))$ and i in 4. is unique up to an element in this group

Corollary 37.50 $\mathcal{HL}_\Sigma$ is the unipotent closure of $\mathcal{O}_\Sigma$, i.e. $\mathcal{HL}_\Sigma = \lim_{\underset{A}{\leftarrow}} A$, where A ranges over all finitely generated unipotent differential algebra extensions over $\mathcal{O}_\Sigma$

Definition 37.51 The regularized zeta series at σ_k is $Z^{\sigma_k}(X) = L^{\sigma_k}(z)^{-1} L(z) \in \mathbb{C}\langle\langle X \rangle\rangle$ which is written as

$$Z^{\sigma_k}(z) = \sum_{w \in X^*} \zeta^{\sigma_k}(w) w$$

Lemma 37.52

$$Z^{\sigma_k}(z) = \lim_{z \to \sigma_k} e^{-x_k \log(z - \sigma_k)} L(z)$$

37.9 Isotopy

Definition 37.53 $f, g : X \to Y$ are topological embeddings, an **isotopy** from f to g is a homotopy H such that H_t are embeddings

Definition 37.54 g, h are embeddings of N in M, an **ambient isotopy** from g to h is $M \times I \overset{F}{\to} M$ such that F_t are homeomorphisms, and $F_0 = 1_M$, $F_1 \circ g = h$

Theorem 37.55 (Alexander's trick) $D \subseteq \mathbb{R}^n$ is the unit ball, homeomorphisms of D that are isotopic on ∂D are also isotopic on D

Proof. Suppose $f, g : D \to D$ are homeomorphisms with $f|_{\partial D}$, $g|_{\partial D}$ isotopic

□

37.10 Mathematical statistics

Definition 37.56 Suppose $X \sim N(\mu, \sigma^2)$, σ is known, $P\left(-1.96 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.96\right) = 0.95$ and

$$-1.96 < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < 1.96 \Leftrightarrow \bar{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}$$

$(\bar{x} - 1.96\sigma/\sqrt{n}, \bar{x} + 1.96\sigma/\sqrt{n})$ is called the 95% *confidence interval*

37.11 Moduli space

Consider a parametrized curve $C = \{(t, \mathbf{x}(t))\}_{t\in I}$, $\mathbf{x}(t) \in \mathbb{R}^n$, now we change I to some space X, $\mathbf{x}(t)$ to some algebro-geometric objects, then we have a parametrization of these objects by X

Definition 37.57 U is a family of some algebro-geometric objects. A parametrization of U by space X is a map $X \to U$, attaching some object U_x for each $x \in X$, we can also think of this map as a section of $X \times U \to X$

We say X is the parametrization space, U is parametrized over X

A moduli functor F is a contravariant functor $Space \to Set$ that takes a space X to the set of families of objects over X, and take a morphism f to the pullback f^* that taking section s to pullback section $f^*s(y) = (y, \mathrm{Pr}_U\, sf(y))$

$$\begin{array}{ccc} Y \times U & \longrightarrow & X \times U \\ \downarrow & & \downarrow \\ Y & \xrightarrow{f} & X \end{array}$$

The category of spaces can be the category of schemes, manifolds, topological spaces, etc.

M is a fine moduli space if F is corepresentable by M, i.e. there is a natural isomorphism $\tau : F \to \mathrm{Hom}(-, M)$. There is a universal family over M corresponds to $1_M \in \mathrm{Hom}(M, M)$. Then any family over X is the pullback along some $X \xrightarrow{f} M$ of the universal family. The universal family is essentially unique and "tautological"

M is a coarse moduli space if there is exists a natural transformation $\tau : F \to \mathrm{Hom}(-, M)$ and universal among these natural transformations, i.e. for any natural transformation $\tau' : F \to \mathrm{Hom}(-, M')$, there is a morphism $M' \xrightarrow{\phi} M$ such that the following diagram commutes

$$\begin{array}{ccc} & & F \\ & \overset{\tau'}{\swarrow} & \downarrow \tau \\ \mathrm{Hom}(-, M') & \xrightarrow{\exists_1 \phi} & \mathrm{Hom}(-, M) \end{array}$$

37.12 Ordinary differential equations

Theorem 37.58 Linear differential equations $y'(t) = A(t)y(t) + b(t)$ with initial condition $y(0) = y_0$, where A, b, y are smooth, then there exists unique local solution

Proof. Define $Ty(t) = \int_0^t A(s)y(s) + b(s)ds + y_0$, note that $\| Ty(t) - Tz(t) \| = \| \int_0^t A(s)(y(s) - z(s))ds \| \leq |t| \| A \| \| y - z \|$, then there exists $\delta > 0$ such that $|t| \| A \| < 1, \forall |t| \leq \delta$, here we use $\| \cdot \|$ to denote the supremum norm in $|t| \leq \delta$, by Banach fixed point theorem, we have a unique local solution

□

Example 37.59 $v'(t) = Av(t), v(0) = v_0$, $A \in M_n(\mathbb{C})$, the solution is $v(t) = e^{tA}v_0$ since $\frac{d}{dt}A^{tA} = Ae^{tA}$

Theorem 37.60 (**Picard-Lindelöf theorem**) Suppose $f(y, t)$ is uniformly Lipschitz continuous in y and continuous in t, then the ODE

$$\begin{cases} y'(t) = f(y(t), t) \\ y(0) = y_0 \end{cases}$$

Has a unique solution $y(t)$ on $[-\varepsilon, \varepsilon]$

Remark. $f(y,t)$ is Lipschitz continuous in y and continuous in t would implie loacl uniformly Lipschitz in y and f uniformly continuous

When you have a local solution, you can try to extend it to a maximal length, i.e. $y(t)$ is defined on $(a,b) \supset [-\varepsilon,\varepsilon]$, it is open precisely because of the theorem

Proof. Define $Ty(t) = \int_0^t f(y(s),s)ds$, then
$\| Ty - Tz \| = \| \int_0^t f(y(s),s) - f(z(s),s)ds \| \leq \| C\int_0^t | \ y(s) - z(s) \ | \ ds \| \leq C|t| \ \| y - z \|$, then there exists $\varepsilon > 0$ such that $C|t| < 1, \ \forall t \in [-\varepsilon,\varepsilon]$, then by Banach fixed point theorem, we have a unique local solution

□

Theorem 37.61 (**Peano existence theorem**) Let $f(y,t)$ be a continuous function around $(y_0,0)$, then the ODE

$$\begin{cases} y'(t) = f(y(t),t) \\ y(0) = y_0 \end{cases}$$

Has a local solution $y(t)$ on $[-\varepsilon,\varepsilon]$

Proof. Say $|f| \leq M$ around $(y_0,0)$, Define $\phi_n(t) = \begin{cases} y_0 & ,x \leq 0 \\ y_0 + \int_0^t \phi_n(s - \frac{\varepsilon}{n})ds & ,0 \leq x \leq \varepsilon \end{cases}$ for $n \geq 1$

By Arzelà-Ascoli theorem, we know that there is a subsequence ϕ_{n_k} converges on $[-\varepsilon,\varepsilon]$, and the limit $\phi(t)$ satisfies $\phi(t) = y_0 + \int_0^t \phi_n(s)ds$ which is a local solution to the problem

□

Remark. The uniqueness may fail without the Lipschitz condition in y, for example, consider $\frac{dy}{dt} = y^{\frac{1}{3}}$, $y(0) = 0$ has solutions $y(t) = 0$ or $y(t) = \pm\left(\frac{2}{3}t\right)^{\frac{3}{2}}$

37.13 Weil conjecture

Definition 37.62 X is a non-singular n dimensional projective algebraic variety over F_q, the *zeta function* is

$$\zeta(X,s) = \exp\left(\sum_{m=1}^{\infty} \frac{N_m}{m} q^{-ms}\right)$$

Where N_m are the number of points of X over F_{q_m}. The *Weil conjectures* are

1. Let $T = q^{-s}$

$$\zeta(X,s) = \frac{P_1(T)P_3(T)\cdots P_{2n-1}(T)}{P_0(T)P_2(T)\cdots P_{2n}(T)} = \prod_{i=0}^{2n} P_i(T)^{(-1)^{i+1}}$$

   Where $P_0(T) = 1 - T$, $P_{2n}(T) = 1 - q^n T$, $P_i(T)$ can be split into $\prod_j (1 - \alpha_{ij}T)$ over $\mathbb{C}$. In particular, $\zeta(X,s)$ is a rational function of T

2. $$\zeta(X, n-s) = \pm q^{\frac{nE}{2} - Es}\zeta(X,s)$$

   Or equivalently

$$\zeta(X, q^{-n}T^{-1}) = \pm q^{\frac{nE}{2}} T^E \zeta(X,T)$$

   E is the Euler characteristic. $\{\alpha_{2n-i,1}, \alpha_{2n-i,2}, \cdots\}$ coincide with $\left\{\frac{q^n}{\alpha_{i,1}}, \frac{q^n}{\alpha_{i,2}}, \cdots\right\}$ in some order

3. $|\alpha_{i,j}| = q^{i/2}$

4.

Example 37.63 If X is the n dimensional projective space, $N_m = 1 + q^m + \cdots + q^{nm}$, $\zeta(\mathbb{P}^n, s) = \frac{1}{(1-q^{-s})\cdots(1-q^{n-s})}$

References

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